ALTERNATING BUNDLES AND THE SPLITTING OF EXTERIOR POWERS

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October 2026

Abstract. The alternating-map bifunctor over a nontrivially normed field acts analytically on analytic bundles when precomposition is analytic. For normed E and D, precomposition is analytic for all Banach values exactly when one class vanishes: the class of the exact sequence of completed projective tensor and exterior powers of E, pushed along an evaluation map. When $k! = 0$, no splitting of this sequence for $E = K^k$ with the maximum norm commutes with the permutation matrix of a single p-cycle, where $p = \operatorname{char} K$. Source, target and parameter splittings, chiefly ordering splittings, make the class vanish; Ramsey forcing over finite residue fields obstructs it. Over subfields of $\kappa((X))$, κ finite, this decides several classes of pairs, and rigid Banach spaces give vanishing classes that no splitting decomposition explains. Seventy-six statements, listed in Section 13, have Lean proofs of corresponding statements.

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The original counterexample and its initial write-up were developed with Anthropic's Claude. Subsequent research and exposition used Claude and OpenAI's Codex, and the organization of the paper was prepared with Claude and Codex. AI assistants contributed to mathematical arguments and independent proof audits. The original Lean formalization used Codex and Autoform. The remaining claims and the per-claim ledger were formalized with Anthropic's Claude Code in October 2026. The splitting theory and the classification of good pairs, that is, the results of Sections 3–12 and Appendices J–O not listed in Section 13, and Proposition 10.4 were developed and written with Claude and are not formalized. Section 13 records the verification scope.

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1. Introduction

A functor on normed spaces extends fiberwise to C^n vector bundles, with its natural transformations, when C^n families of linear maps induce C^n families (Theorem 2.1). We ask when the bifunctor $(E,F)\mapsto \mathrm{Alt}^k(E;F)$ of alternating k-linear maps over a nontrivially normed field K carries analytic bundles to analytic bundles. For normed spaces E, D and F this concerns $A^k_{E,D;F}\colon \mathcal{L}(E,D)\to\mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Alt}^k(E;F))$, $A^k_{E,D;F}(u)(m)=m\circ(u,\dots,u)$. Let $j_{E,F}\colon \mathrm{Alt}^k(E;F)\to\mathrm{Mult}^k(E;F)$ be the isometric inclusion. Composed with the induced isometric inclusion $j_{E,F}\circ -\colon \mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Alt}^k(E;F))\to\mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Mult}^k(E;F))$, the map $A^k_{E,D;F}$ is the diagonal of a bounded k-linear map (Proposition 2.2 with $g=\mathrm{id}_F$), and Theorem 2.3 decides when such a diagonal descends analytically through $j_{E,F}\circ -$. By Proposition 2.5, $A^k_{E,D;F}$ is analytic at some point if and only if it has a *lift*: a bounded k-linear map $Q\colon \mathcal{L}(E,D)^k\to\mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Alt}^k(E;F))$ with $Q(u,\dots,u)=A^k_{E,D;F}(u)$.

The splitting question. Let $\mathcal{N}^k_E$ be the kernel of the quotient map $\widehat{\wedge}^k_E$ from the completed projective tensor power of E onto its completed projective exterior power (Section 3):

$$0\longrightarrow \mathcal{N}^k_E\longrightarrow \widehat{\otimes}^k_\pi E\xrightarrow{\widehat{\wedge}^k_E}\widehat{\Lambda}^k_\pi E\longrightarrow 0. \tag{1.1}$$

If $k!\neq 0$ in K, antisymmetrization divided by $k!$ splits (1.1) naturally in E and solves every lift problem (Propositions 4.1 and 5.1(1)). If $k!=0$ in K, one p-cycle of the coordinates of K^k, $p=\mathrm{char}\,K$, already forbids natural splittings (Theorem 4.2, Corollary 4.4). The question becomes when the class $\mathrm{ob}_k(E,D)$ obtained from (1.1) vanishes. It vanishes exactly when (E,D) is *good for Banach values in degree k*: $A^k_{E,D;F}$ is analytic for every Banach space F; the pair is *good in degree k* if this holds for every normed space F (Definition 3.3, Theorem A).

The class vanishes under a *source splitting* (S^k_E), a bounded section of $\widehat{\wedge}^k_E$; a *target splitting* (R^k_D), a bounded retraction of the antisymmetrization alt^k_D; or a *parameter splitting*, a bounded projection of $\widehat{\otimes}^k_\pi\mathcal{L}(E,D)$ onto the closed span of the powers $u^{\otimes k}$, which makes $\mathcal{L}(E,D)$ *polarizable in degree k*. Relative splittings, decompositions and block assemblies combine them. The class can be nonzero: *forced asymmetry* makes lifts asymmetric (Section 4), and over a finite residue field *Ramsey forcing* turns this asymmetry into a triangular pattern of coefficients that norm estimates forbid (Section 7).

Main results.

Theorem A (descent and one obstruction class)**.** *Let E and D be normed spaces and $k\geq 1$.*

1. *For every normed space F, $A^k_{E,D;F}$ is C^∞ on $\mathcal{L}(E,D)$; it is analytic at one point if and only if it has a lift, and then it is a continuous polynomial.*
2. *The following are equivalent:* (a) *(E,D) is good for Banach values in degree k;* (b) *$u\mapsto\Lambda^k_\pi u$ has a bounded k-linear lift $\mathcal{L}(E,D)^k\to\mathcal{L}(\Lambda^k_\pi E,\widehat{\Lambda}^k_\pi D)$;* (c) *$\mathrm{ob}_k(E,D)=0$, where $\mathrm{ob}_k(E,D)$ is the pushout of the class of (1.1) along the restriction $\mathrm{ev}^k_{E,D}|_{\mathcal{N}^k_E}\colon\mathcal{N}^k_E\to\mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ of the evaluation map $\mathrm{ev}^k_{E,D}(x_1\otimes\cdots\otimes x_k)(u_1,\dots,u_k)=u_1x_1\wedge\cdots\wedge u_kx_k$, with Mult^k_0 the bounded k-linear maps with zero diagonal;* (d) *some bounded k-linear map $\Psi\colon E^k\to\mathrm{Mult}^k_0(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ satisfies $\Psi(x_1,\dots,x_k)=\mathrm{ev}^k_{E,D}(x_1\otimes\cdots\otimes x_k)$ whenever two of the x_a coincide.*
3. *Let $(e_t)_{t\in\Sigma}$ be a linearly independent family in E with dense span V, indexed by a totally ordered set Σ; every normed space has one. The* ordering lift *$Q_<(u)$, for $u\in\mathcal{L}(E,D)^k$, is the linear map $\Lambda^k_K V\to\widehat{\Lambda}^k_\pi D$ sending $e_{t_1}\wedge\cdots\wedge e_{t_k}$, $t_1<\cdots<t_k$, to $u_1e_{t_1}\wedge\cdots\wedge u_ke_{t_k}$.*

Composed with the canonical map $\Lambda^k_K V \to \Lambda^k_\pi E$, every lift Q as in (b) *is $Q_< + \Phi$ for a correction Φ, k-linear in u with $\Phi(v,\dots,v) = 0$ for $v \in \mathcal{L}(E,D)$ (Theorem 3.18). The pair (E,D) is good for Banach values in degree k if and only if, for some such Φ and some C, $\|Q_<(u)y + \Phi(u)y\| \le C\|u_1\| \cdots \|u_k\|\|y\|_\pi$ for all u and $y \in \Lambda^k_K V$, where $\|y\|_\pi$ is the projective seminorm.*

Theorem A collects Lemma 2.4, Propositions 2.2 and 2.5, Lemma 3.4, Theorem 3.15, Corollary 3.20 and Theorem 3.18.

Theorem B (canonical splittings exist exactly when $k! \neq 0$)**.** *Let $k \ge 2$.*

1. *If $k! \neq 0$ in K, then $\mathrm{alt}^k_E/k!$, $\wedge^k_D/k!$ and the symmetrizer $(1/k!)\sum_{\sigma\in S_k}\sigma$ on $\widehat{\otimes}^k_\pi P$ split the source, target and parameter sequences naturally in E, D and P. Every pair is good in degree k, and Alt^k is analytic on every hom space.*
2. *Suppose $k! = 0$ in K. For every norm on K^k with bounded coordinate functionals, no section of $\wedge^k_{K^k}$, retraction of $\mathrm{alt}^k_{K^k}$ or projection onto the symmetric tensors commutes with the permutation matrix of one p-cycle of the coordinates. If E has k linearly independent bounded functionals and $\widehat{\Lambda}^k_\pi D \neq 0$, no lift of $u \mapsto \Lambda^k_\pi u$ is symmetric, no section of $\widehat{\wedge}^k_E$ takes sign-isotypic values and no retraction of alt^k_D is sign-equivariant.*
3. *If $k! = 0$ in K, then $A^k_{E,E;F}$ is analytic at no point for some Banach spaces E and F, so Alt^k is analytic on all hom spaces, or on those of Banach spaces, if and only if $k! \neq 0$. If K is moreover complete and not spherically complete, $A^k_{E,D;K}$ is analytic at no point for some nonarchimedean Banach spaces E and D.*
4. *If K is complete and nonarchimedean, $\mathrm{Alt}^k(-;K)$ is analytic on all hom spaces of normed spaces if and only if $k! \neq 0$ in K or K is spherically complete.*
5. *Let a group act on a set I, and on $K^{(I)}$ by permuting the unit vectors. On the algebraic powers of $K^{(I)}$, $\wedge^k$ has an equivariant section and alt^k an equivariant retraction if and only if the stabilizer of each k-subset of I induces on it a group of order nonzero in K; an equivariant projection onto the span of the pure powers exists if and only if the stabilizer of each k-multiset on I induces on its support a group of order nonzero in K. The obstruction to an equivariant section is a family of Tate classes. One element inducing a permutation of order p on some k-subset forbids all three equivariant splittings, also on completed powers of normed spaces with a compatible biorthogonal system of vectors and bounded functionals.*

Theorem B collects Propositions 4.1 and 5.1, Theorem 4.2, Proposition 4.6, Theorem 7.2, Corollary 7.3, Theorems 7.4 and 4.3, Corollary 4.4 and Proposition 4.5. Corollary 5.15 lists the bundle settings in which fiberwise alternating maps form an analytic bifunctor; Proposition 10.1, Corollary 10.2 and Proposition 10.3 give analytic atlases with nonanalytic induced alternating or cotangent atlases, the latter over complete fields of characteristic $p > 0$ that are not spherically complete, in degrees $k \ge p$; over complete fields, ambient analytic forms keep pullback, wedge products and exterior derivatives (Theorem 10.7). Over nonarchimedean fields, with bounded polynomial laws in place of bounded multilinear maps as Taylor coefficients, divided-power analytic maps compose (Theorem 10.11(2)), and in every characteristic the alternating and projective exterior power bundles of divided-power analytic vector bundles are divided-power analytic (Corollary 10.12). If K has characteristic $p > 0$, then in every degree at least p there are bounded polynomial laws that are not diagonals of bounded multilinear maps, even after an isometric embedding of their target (Propositions 10.13 and 10.14).

Theorem C (how the class vanishes)**.** *Let E and D be normed spaces and $k \ge 1$.*

(1) *The pair (E,D) is good for Banach values in degree k if (S^k_E) or (R^k_D) holds, if $\mathcal{L}(E,D)$ is polarizable in degree k, if (E,D) has a splitting decomposition of degree k, or if (E,D) is an ℓ^1 block assembly of pairs with uniformly bounded lifts in all degrees $j \le k$. For decompositions into $\mathrm{S}^{\le k}$-split and $\mathrm{R}^{\le k}$-split pieces, two pieces suffice.*

(2) *If E has a linearly independent family, indexed by a totally ordered set, with dense span V such that the strict triangle projection $\mathcal{P}_<$ is bounded on $V \otimes_\pi V$, for instance one with a lower r-estimate for some $r < 2$, then (S^n_E) and (R^n_E) hold and E is polarizable in degree n, for every $n \ge 1$.*

(3) *If K is nonarchimedean, E has a bounded linearly independent family indexed by a totally ordered set, with dense span on which E is ℓ^2-dominated, and D has an equivalent r-norm, $r \in (2,\infty)$, or an equivalent nonarchimedean norm, then (E,D) is good for Banach values in every degree.*

(4) *Suppose* $\operatorname{char} K = p > 0$. *The least degree, if any, in which (E,D) is not good, or not good for Banach values, is a power of p. If E^* and D^* are infinite-dimensional, the set of degrees in which (E,D) is good, or good for Banach values, is $\{1,\dots,p^h-1\}$ for some $h \ge 1$ or $\{1,2,\dots\}$. For every normed space G, each of (S^p_G), (R^p_G) and polarizability in degree p implies the same condition in every degree.*

(5) *If K is nonarchimedean and E has a bounded linearly independent family $(e_t)_{t\in\Sigma}$, indexed by a totally ordered set, with dense span V and coordinate functionals $x \mapsto x(t)$ uniformly bounded on V, then the class of (1.1) pushes out to zero along every bounded linear map from $\mathcal{N}^k_E$ to a Banach space with an equivalent nonarchimedean norm. In particular* $\mathrm{ob}_k(E,D) = 0$ *if $\widehat{\Lambda}^k_\pi D$ has such a norm.*

Theorem C collects Proposition 3.5, Corollary 3.17, Theorem 6.13, Propositions 6.15 and 6.27, Theorem 5.6, Proposition 5.7, Theorem 5.9, Corollaries 5.10 and 6.3, and Theorems 6.5, 6.6, 6.10 and 3.22. A bounded retraction of $j_{E,F}$ gives analyticity for that value space F only (Proposition 5.1, Theorems 5.2 and 5.4).

Theorem D (forcing)**.** *Assume condition* (L)*: κ is a finite field of characteristic p, K is a subfield of $\kappa((X))$ on which the absolute value $|a| = \varrho^{\operatorname{ord} a}$, $0 < \varrho < 1$, is nontrivial, and $k \ge p$.*

(1) *For $2 \le r \le \infty$, the pair $(\ell^r(\mathbb{N},K), \ell^r(\mathbb{N},K))$, with $c_0(\mathbb{N},K)$ for $r = \infty$, is not good for Banach values in degree k.*

(2) *If symmetric sequence spaces E and D form a pair good for Banach values in degree k, then for some C, the* product triangle bound (PTB)

$$\|\mathcal{P}_<\big((ax)\otimes(by)\big)\|_{D\otimes_\pi D} \le C\|\mu_a\|_{E\to D}\|x\|_E\|\mu_b\|_{E\to D}\|y\|_E$$

holds for all finitely supported a,x,b,y, where μ_a is multiplication by a. If moreover $\|\xi\|_D \le C'\|\xi\|_r$ for all finitely supported ξ and some $r \in (2,\infty]$, then E is ℓ^2-dominated.

(3) *If E has an equivalent nonarchimedean norm with $\dim_{\widehat{K}} \widehat{E} = \infty$, then the pairs $(E, c_0(\mathbb{N},\widehat{K}))$ and $(c_0(\mathbb{N},K), E)$ are not good for Banach values in degree k.*

Theorem D collects Theorems 7.7 and 7.13, Proposition 7.15, and Theorem 7.19 with its proof, all proved in Appendix K; Lemma 7.1 reduces condition (L) to fields dense in a Laurent series field. Over every nontrivially normed field K of characteristic 2, polynomial identities replace Ramsey's theorem: for $E = c_0(\mathbb{N},K)$, $\ell^\infty(\mathbb{N},K)$ and $\ell^r(\mathbb{N},K)$, $2 < r < \infty$, the pair (E,E) is not good for Banach values in any degree $k \ge 2$, and neither (S^k_E) nor (R^k_E) holds (Theorem 7.20). Over Laurent series fields over infinite fields and over densely valued fields, the case $E = c_0(\mathbb{N},K)$ for odd p and the case $E = \ell^2(\mathbb{N},K)$ for $p = 2$ are open (Question 12.9). In characteristic 2, Proposition 11.21 removes condition (L) from Theorems 11.14 and 11.19(4),

from Theorem 11.17(3) except its last sentence, and from Corollary 11.15(2) for $E = c_0(\mathbb{N}, K)$ and for $E = \ell^{r_2}(\mathbb{N}, K)$, $r_2 > 2$.

Theorem E (good pairs in classes)**.** *Let κ, ϱ and K be as in condition* (L)*, and let good mean good for Banach values.*

(1) *Let E be a symmetric sequence space in which c_{00} is dense. The following are equivalent: (E, E) is good in some degree $k \geq p$; (c_0, E) is good in some degree $k \geq p$; $\mathcal{P}_<$ is bounded on $E \otimes_\pi E$; (S^k_E) holds for some $k \geq p$; (R^k_E) holds for some $k \geq p$; E is polarizable in some degree $k \geq p$; (E, D) and (D, E) are good in every degree for every normed space D. If they hold, E is ℓ^2-dominated and $\varphi_E(m)^2 \geq c\, m \ln m$ for some $c > 0$ and all $m \geq 1$, with $\varphi_E(m)$ the norm of the indicator of $\{1, \dots, m\}$.*

(2) *Let E be as in* (1) *and D a symmetric sequence space with an equivalent r-norm, $r \in (2, \infty)$, or an equivalent nonarchimedean norm. Then (E, D) is good in some degree $k \geq p$ if and only if* (PTB) *holds, if and only if E is ℓ^2-dominated, if and only if (E, D) is good in every degree.*

(3) *For $1 \leq r_1, r_2 \leq \infty$, with c_0 in place of ℓ^∞, the pair (ℓ^{r_1}, ℓ^{r_2}) is good in some degree $k \geq p$ if and only if it is good in every degree, if and only if $\min(r_1, r_2) < 2$ or $r_1 = 2 < r_2$.*

(4) *If E has an equivalent nonarchimedean norm and $k \geq p$, then (E, D) is good in degree k for every normed space D if and only if (D, E) is good in degree k for every normed space D, if and only if $\dim_{\widehat{K}} \widehat{E} < \infty$. If K is complete and $k \geq p$, two spaces with equivalent nonarchimedean norms form a pair good in degree k if and only if one of them is finite-dimensional.*

For the pairs in (1)–(3) *and in the last sentence of* (4)*, the good degrees form $\{1, \dots, p-1\}$ or $\{1, 2, \dots\}$, and every good pair has a splitting decomposition by absolute or relative ordering splittings or by finite-coordinate splittings.*

Theorem E collects Theorems 8.1, 8.4 and 7.19, Corollaries 8.5 and 8.6, and Remark 8.7. Items (1) and (2), the last sentence for the pairs in them, and Theorem D(2) hold for spreading sequence spaces, whose norms need only be invariant under increasing relabelings of the unit vectors (Theorem 8.9); this class contains spaces with no equivalent symmetric norm (Example 8.10).

Theorem F (goodness without splitting decompositions)**.** (1) *Over every separable complete nonarchimedean field K there are nonzero infinite-dimensional Banach spaces $\mathfrak{X}_1, \mathfrak{X}_2, \dots$ with $\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\alpha) = K\,\mathrm{id}$ and $\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\beta) = 0$ for $\alpha \neq \beta$.*

(2) *Over $\kappa((X))$ with κ finite, suitable choices in* (1) *give $\mathfrak{X} = \mathfrak{X}_1$ with $(\mathfrak{X}, \mathfrak{X})$ good in every degree, while for every $k \geq p$ neither $(\mathrm{S}^k_{\mathfrak{X}})$ nor $(\mathrm{R}^k_{\mathfrak{X}})$ holds and $(\mathfrak{X}, \mathfrak{X})$ has no splitting decomposition of degree k.*

(3) *Over $\kappa((X))$ with κ finite and these choices, the completed ℓ^1-sum E of the spaces $\mathfrak{X}_\alpha$ gives an ℓ^1 block assembly (E, E) that is good for Banach values in every degree and has no splitting decomposition of any degree $k \geq p$, even with parameter pieces.*

Theorem F collects Theorems 9.2, 9.3 and 9.4, proved in Appendix M.

Bundles need goodness only for triples $(P; E, D)$, along analytic operator families on open subsets of the base model P (Section 11). Goodness of a triple reduces to homogeneous polynomial families and to familywise classes (Theorems 11.5 and 11.7). Under (L), $(\ell^2(\mathbb{N}, K); E, E)$ with $E = \ell^r(\mathbb{N}, K)$, $2 < r < \infty$, is good in every degree, although neither is $\ell^2(\mathbb{N}, K)$ polarizable nor (E, E) good for Banach values in degree k (Theorems 11.9 and 11.13, Proposition 11.11, Corollary 8.2, Theorem 7.7). For a structure group with an open subgroup that lies in a linear

space of operators and contains a ball of it around the identity, goodness is that of the linear space, not that of the closed linear span of the group (Theorem 11.19, Proposition 11.20).

Theorem G (what bundles need)**.** *Let P be a normed space and $k \geq 1$.*

(1) *Let $\mathcal{C}$ be a class of Banach spaces closed under finite direct sums. Fiberwise alternating maps with Banach value fibers preserve the analyticity of bundles with typical fibers in $\mathcal{C}$ over every analytic manifold modeled on P if and only if $x \mapsto \Lambda^k_\pi u(x)$ is analytic for every analytic family u of operators between spaces of $\mathcal{C}$ on an open subset of P. For a single Banach fiber, families of invertible operators give the same condition.*

(2) *The condition holds for all normed fibers if P is polarizable in every degree $n \leq k$, and over every P for fibers E, D with $(E, \widehat{D})$ good for Banach values in degree k. In characteristic p, the condition in all degrees $j \leq k$ is decided by the lifting constants of homogeneous polynomial families in the degrees $p^i \leq k$. Under* (L)*, it holds for all normed fibers over $P = \ell^r(\mathbb{N}, K)$, $1 \leq r < \infty$, if and only if $r < 2$, and it fails for some normed fibers over every P with an equivalent nonarchimedean norm and $\dim_{\widehat{K}} \widehat{P} = \infty$, such as $c_0(\mathbb{N}, K)$.*

(3) *Let K be complete and P a Banach space with $P \cong P^3$. The bundles $\widehat{\Lambda}^k_\pi TM$ are analytic for every analytic manifold M modeled on P if and only if the condition holds for $E = D = P$. Under* (L)*, $c_0(\mathbb{N}, K)$ and $\ell^r(\mathbb{N}, K)$, $4 < r < \infty$, model manifolds M with a nonanalytic bundle $\widehat{\Lambda}^k_\pi TM$.*

Theorem G collects Proposition 11.2, Corollary 11.3, Theorems 11.5, 11.9 and 11.10, Proposition 11.11 and Theorem 11.17.

Section 12 collects questions. Every theorem, proposition, lemma and corollary of the paper is proved in the text, except Theorem D.3 and Lemma M.9, which are cited; Section 13 lists the statements that also have Lean proofs of corresponding statements. Proofs that would interrupt the text are collected in Appendix P.

Related work. The bundle construction follows Lang's functors of class C^n [16, Ch. III, §4] and Bourbaki's vector functors [3, n° 7.6]. Bourbaki defines the alternating bifunctor when K has characteristic 0 or the bundles have finite rank, and states that it is not known whether this restriction can be removed [3, n° 7.8]; the paragraph *Bourbaki's question* in Section 10 shows that it cannot over complete fields of characteristic $p > 0$ that are not spherically complete. Gouëzel's construction [8, 9] covers characteristic zero; the finite-dimensional case in positive characteristic, noted in [8], is Proposition 5.1(2) over complete fields. Closed-range reflection of smoothness is formalized in [19]. The nonarchimedean extension mechanism is a norm-controlled form of spherical completeness [12, 24, 25, 27]. Completed nonarchimedean exterior powers go back to Serre [26] and Gruson [10], who do not study sections of the quotient map; we know no source on these splittings in positive characteristic. The rigid spaces of Section 9 transfer a construction of Kalton and Roberts [15]. Corollary 3.21 detects the class by the distance of a z-linear map from the linear maps, as for twisted sums [13, 14, 5].

Conventions. $\mathcal{L}(E, F)$ denotes bounded linear maps with the operator norm, $\mathrm{Mult}^k(E; F)$ bounded k-linear maps, and $\mathrm{Alt}^k(E; F)$ those vanishing whenever two arguments coincide; this condition, not antisymmetry, defines alternation in characteristic two. The degree-zero space is F. Products carry the maximum norm. E denotes a source fiber, D a target fiber, F a value space (a *target fiber* in Corollary 5.15 and Appendix C), G an intermediate space, and P a base model or parameter space. *Finite coordinates* means a bounded linear isomorphism with K^d; over a complete field this is finite dimension. Hats denote completions (Section 3). Exponents are written r; ℓ^r, lower r-estimates, r-norms and ℓ^2-domination of a family are defined in Section 5,

symmetric sequence spaces in Definition 7.10. A finite field κ has $|\kappa|$ elements. Condition (L) is fixed in Section 7. A norm is *nonarchimedean* if $\|x+y\| \le \max(\|x\|, \|y\|)$. Every normed field of characteristic $p > 0$ is nonarchimedean: Frobenius gives $|x+y|^{p^m} \le 2\max(|x|,|y|)^{p^m}$; take roots and let $m \to \infty$.

C^n ($n \le \infty$) means iterated Fréchet differentiability; analyticity means local expansion in bounded multilinear coefficients with a positive norm-controlled radius. On manifolds, C^ω also requires analytic iterated derivatives, as in Mathlib [17]. On open sets the two agree, even for incomplete ranges: the derivative series converges in the completion to the Fréchet derivative, which exists in the operator space since nearby points are points of analyticity.

2. Alternating bundles and analytic descent

Throughout, Vec_K is the category of normed K-spaces, not necessarily complete, and bounded linear maps, with hom spaces $\mathcal{L}(E,D)$. For $n \in \mathbb{N} \cup \{\infty, \omega\}$, C^n is as in the conventions of Section 1, and C^0 means continuous.

2.1. **The bundle construction.** Fix a C^n manifold M modeled on a normed space P. Write $\mathsf{VB}^n_K(M)$ for the category of C^n normed vector bundles over M, with morphisms over id_M whose local *operator-valued* expressions, not merely the maps of total spaces, are C^n. Thus a morphism from a bundle with fiber E to one with fiber F is locally a C^n map $x \mapsto T(x) \in \mathcal{L}(E,F)$. This condition is independent of the trivializations, since composition is bounded bilinear.

Let $\epsilon = (\epsilon_1, \dots, \epsilon_r)$ specify a variance in each variable, and put $\mathbf{C}^\epsilon = \mathsf{C}_1 \times \cdots \times \mathsf{C}_r$ with $\mathsf{C}_a = \mathsf{Vec}_K$ or $\mathsf{Vec}_K^{\mathrm{op}}$. A functor $\mathcal{F}\colon \mathbf{C}^\epsilon \to \mathsf{Vec}_K$ is C^n if all its maps on products of hom spaces are jointly C^n. Let $\mathbf{B}^\epsilon_M$ denote the corresponding product of bundle categories, with the same variances. Natural transformations between such functors have bounded linear components, since their codomain is Vec_K.

Theorem 2.1 (lifting linear constructions to bundles)**.** *For a fixed C^n manifold M, fiberwise application defines, up to canonical natural isomorphism, a functor*

$$\mathcal{B}^n_M\colon \mathrm{Fun}^{C^n}(\mathbf{C}^\epsilon, \mathsf{Vec}_K) \longrightarrow \mathrm{Fun}(\mathbf{B}^\epsilon_M, \mathsf{VB}^n_K(M)),$$
$$\mathcal{F} \longmapsto \mathcal{F}_M, \qquad \eta \longmapsto \eta_M.$$

Here the category on the left is the full subcategory of the ordinary functor category on the functors regular on hom spaces. The resulting bundles, bundle morphisms, and natural transformations are given fiberwise by $\mathcal{F}$, its action on maps, and the components of η, respectively.

The same conclusion holds if regularity on whole hom spaces is replaced by preservation of C^n families parametrized by open subsets of the model space P.

Proof. Step 1 (transitions). Over a common trivializing cover of the input bundles, let the C^n operator-valued maps g^a_{ij} convert coordinates of the ath bundle from chart j to chart i, so that $g^a_{ij} g^a_{j\ell} = g^a_{i\ell}$ and $g^a_{ii} = \mathrm{id}$. Let $\mathbf{g}_{ij}(x)$ be the tuple of arrows in $\mathbf{C}^\epsilon$ with entry $g^a_{ij}(x)$ in a covariant variable and $g^a_{ji}(x)$ in a contravariant one. By functoriality, $\theta_{ij}(x) = \mathcal{F}(\mathbf{g}_{ij}(x))$ satisfies $\theta_{ij}\theta_{j\ell} = \theta_{i\ell}$. The maps θ_{ij} are C^n by composition, so they glue the products with typical fiber $\mathcal{F}(E_1, \dots, E_r)$ into a C^n vector bundle; its coordinate changes are C^n because operator evaluation is bounded bilinear.

Step 2 (morphisms). Applied to the local operator families of a tuple of bundle morphisms, reversed in the contravariant variables, $\mathcal{F}$ gives families compatible with the θ_{ij} by functoriality and C^n by composition; this defines $\mathcal{F}_M$. The same argument for changes of trivialization identifies different choices canonically and preserves identities and composition.

Step 3 (natural transformations). For $\eta\colon \mathcal{F} \Rightarrow \mathcal{G}$ and $\mathbf{E} = (E_1, \dots, E_r)$, let η_M be locally the constant operator $\eta_{\mathbf{E}}$. Naturality gives $\eta_{\mathbf{E}}\, \mathcal{F}(\mathbf{g}_{ij}(x)) = \mathcal{G}(\mathbf{g}_{ij}(x))\, \eta_{\mathbf{E}}$, so η_M is a bundle

morphism, C^n because its local expression is constant. The same identity for arbitrary input morphisms proves naturality of η_M, and identities and compositions of transformations are preserved fiberwise.

Step 4 (familywise version). In a base chart, every transition and local morphism family is a C^n family on an open subset of P, so preservation of such families gives the regularity used in Steps 1 and 2; Step 3 uses none. □

The familywise hypothesis, that $\mathcal{F} \circ \gamma$ be C^n for each C^n map γ on an open subset of P into $\prod_a \operatorname{Hom}_{\mathsf{C}_a}(E_a, D_a)$, is weaker than regularity on whole hom spaces. The theorem also holds for restricted categories of fibers containing the bundle morphisms; compare [9, 16] and [3, n°s 7.6.1–7.6.4].

2.2. **Alternating maps inside multilinear maps.** For normed spaces E, F, let $\operatorname{Mult}^k(E;F)$ be the normed space of bounded k-linear maps $E^k \to F$, and $\operatorname{Alt}^k(E;F)$ its subspace of alternating maps, those vanishing whenever two arguments agree. The inclusion

$$j_{E,F}\colon \operatorname{Alt}^k(E;F) \longrightarrow \operatorname{Mult}^k(E;F)$$

is a linear isometry with closed range, since alternation is an intersection of closed conditions under evaluation at fixed tuples. Both constructions are contravariant in E and covariant in F, and the inclusions form a natural transformation. If F is complete, so are both spaces, since operator-norm Cauchy families have bounded pointwise limits satisfying the same identities; so the alternating bifunctor restricts to Banach spaces.

For $f \in \mathcal{L}(E, D)$ and $g \in \mathcal{L}(F, F')$, their joint action is

$$B(f,g)(m)(x_1,\dots,x_k) = g\big(m(fx_1,\dots,fx_k)\big).$$

On alternating maps, write

$$A = A^k_{E,D;F}\colon \mathcal{L}(E,D) \longrightarrow \mathcal{L}(\operatorname{Alt}^k(D;F), \operatorname{Alt}^k(E;F)), \qquad A(f) = B(f, \mathrm{id}_F),$$

for precomposition. Since $\|B(f,g)(m)\| \le \|g\|\,\|m\|\,\|f\|^k$, these actions are bounded operators.

Proposition 2.2 (ambient polynomial action)**.** *The joint action on multilinear maps*

$$B\colon \mathcal{L}(E,D) \times \mathcal{L}(F,F') \longrightarrow \mathcal{L}\big(\operatorname{Mult}^k(D;F), \operatorname{Mult}^k(E;F')\big)$$

is the diagonal of a bounded $(k+1)$-linear map, and hence is C^ω. Its restriction to alternating inputs is analytic as a map into

$$\mathcal{L}\big(\operatorname{Alt}^k(D;F), \operatorname{Mult}^k(E;F')\big),$$

and all its values belong to the closed subspace of operators taking values in $\operatorname{Alt}^k(E;F')$.

Proof. For $v_i = (f_i, g_i)$ in the product of hom spaces, define

$$\Phi(v_0,\dots,v_k)(m)(x_1,\dots,x_k) = g_0\big(m(f_1x_1,\dots,f_kx_k)\big).$$

This map is multilinear in the v_i, has norm at most 1, and has diagonal B. A diagonal of a bounded multilinear map has finite polynomial derivatives of every order, so it is C^ω even with incomplete range. Restricting m to alternating maps is bounded linear and preserves analyticity. On the diagonal, two equal x_i give two equal fx_i, so the output is alternating; the operator subspace is closed because $j_{E,F'}$ has closed range, applied pointwise. For $k = 0$, the formula is postcomposition by g_0. □

For bundles with analytic transition families g_{ij} and h_{ij}, the induced alternating transitions $m \mapsto h_{ij}(x) \circ m \circ (g_{ji}(x), \dots, g_{ji}(x))$ are therefore analytic as maps into the multilinear operator space. Whether their power-series coefficients can be chosen in the alternating operator space is the descent problem of Section 2.3.

2.3. **Analytic descent through closed subspaces.** Throughout this subsection, P, W, Z are normed K-spaces, not necessarily complete, and $j\colon W \to Z$ is a linear isometry with closed range; we identify W with its image when convenient. If $U \subseteq P$ is open and $f\colon U \to W$, an ambient expansion at $x_0 \in U$ means

$$jf(x_0 + h) = \sum_{n \geq 0} b_n(h, \dots, h), \qquad \sup_n \|b_n\| R^n < \infty \tag{2.1}$$

for some $R > 0$, with bounded n-linear $b_n\colon P^n \to Z$ (b_0 a constant) and convergence for $\|h\| < R$. For $j = \mathrm{id}_W$ this is, after shrinking R, the usual local definition of analyticity. For nonarchimedean spaces over a complete nonarchimedean field, this use of multilinear representatives agrees with the polynomial lifting norms of [2, 7.14–7.19].

Theorem 2.3 (the coefficient descent criterion)**.** *Suppose* (2.1) *holds. Then every diagonal $b_n(h, \dots, h)$ belongs to $j(W)$. The map f is analytic at x_0 if and only if there are bounded n-linear maps $q_n\colon P^n \to W$ and $r > 0$ such that*

$$jq_n(h, \dots, h) = b_n(h, \dots, h) \quad (n \geq 0,\ h \in P), \qquad \sup_n \|q_n\| r^n < \infty.$$

Only the diagonals must agree; the original mixed values of b_n need not lie in $j(W)$.

Proof. Step 1 (diagonals in $j(W)$). Fix $h \in P$. On the scalar line $x_0 + th$, the coefficients $b_n(h, \dots, h)$ of (2.1) satisfy a geometric bound, and $b_0 = jf(x_0)$. If the diagonals of degree below n lie in $j(W)$, then for small nonzero t

$$t^{-n}\Big(jf(x_0 + th) - \sum_{m<n} t^m b_m(h, \dots, h)\Big) \in j(W),$$

and this vector tends to $b_n(h, \dots, h)$ as $t \to 0$, by the tail estimate in the proof of Lemma A.1; since $j(W)$ is closed, $b_n(h, \dots, h) \in j(W)$.

Step 2 (the criterion). If f is analytic, apply j to one of its expansions; on every scalar line, Lemma A.1, applied to the difference of the two series, gives the diagonal identities. Conversely, shrink r below R and let $\|h\| < r$. The partial sums of $\sum_n q_n(h, \dots, h)$ in W have images converging to $jf(x_0 + h)$, so they converge to $f(x_0 + h)$ because j is an isometry, and the bound on $\|q_n\| r^n$ gives a positive radius. □

Lemma 2.4 (smooth descent)**.** *For $n \in \mathbb{N} \cup \{\infty\}$, a map $f\colon U \to W$ is C^n if and only if jf is C^n.*

Proof. Continuity reflects through an isometry. If jf has Fréchet derivative T at x, then $Th = \lim_{t \to 0,\, t \neq 0} t^{-1}(jf(x + th) - jf(x)) \in j(W)$, so $T = jS$ for a unique bounded linear $S\colon P \to W$ of the same norm; the isometry transfers the remainder estimate, so $Df(x) = S$. Postcomposition by j is a closed linear isometry $\mathcal{L}(P, W) \to \mathcal{L}(P, Z)$, since its image consists of the operators with values in $j(W)$, a closed condition under each evaluation. The same argument applied to the derivative maps proves every finite order by induction, hence C^∞. □

By Proposition 2.2 and Lemma 2.4 with postcomposition by $j_{E,F'}$ as j, the joint action on alternating maps is C^∞, and so is its case $g = \mathrm{id}_F$, the precomposition action $A^k_{E,D;F}$; this is the argument formalized in [19]. With Theorem 2.1, this constructs alternating bundles

and their morphisms at every finite order and at C^∞; analyticity is the additional coefficient problem.

We call a finite sum of diagonals of bounded multilinear maps a *continuous polynomial*.

Proposition 2.5 (one lift, every base point)**.** *Let $k \geq 1$, and suppose $a\colon H \to W$ satisfies $ja(h) = \beta(h,\dots,h)$ for a bounded k-linear map $\beta\colon H^k \to Z$. The following are equivalent:*

(1) *a is analytic at some point of H;*
(2) *there is a bounded k-linear $Q\colon H^k \to W$ with $Q(h,\dots,h) = a(h)$;*
(3) *a has a finite power-series expansion at every point, of infinite radius.*

Consequently each precomposition action $A^k_{E,D;F}$ is either a continuous polynomial on its whole domain or analytic nowhere.

Proof. An expansion at h_0, compared along $h_0 + th$ with the polynomial $\beta(h_0+th,\dots,h_0+th)$, has degree-$k$ diagonal $\beta(h,\dots,h)$ by Lemma A.1; its degree-k coefficient is the required Q. Conversely, expanding $Q(h_0+h,\dots,h_0+h)$ by multilinearity gives a finite expansion whose degree-i coefficient, the sum over the i occupied slots, has norm at most $\binom{k}{i}\|Q\|\|h_0\|^{k-i}$. The last assertion follows from Proposition 2.2. □

Retractions are splittings. A bounded linear retraction ρ of $j_{E,F}$ turns the representative $\beta(u_1,\dots,u_k)(m) = m\circ(u_1,\dots,u_k)$ of $j_{E,F}\circ A^k_{E,D;F}$ into the lift $\rho\circ\beta$ (Proposition 5.1). Section 3 treats all Banach F at once by one exact sequence and one class (Theorem 3.15).

3. Exterior powers and the obstruction class

3.1. **Projective tensor and exterior powers.** For $k \geq 1$ give the algebraic exterior power $\Lambda^k_K E$ the seminorm

$$\|z\|_\pi = \inf_{z=\sum_j x_{j1}\wedge\cdots\wedge x_{jk}} \sum_j \prod_{a=1}^{k} \|x_{ja}\|.$$

Its quotient by the zero-seminorm subspace is the normed space $\Lambda^k_\pi E$, with universal alternating map $\omega_E(x_1,\dots,x_k) = [x_1\wedge\cdots\wedge x_k]$. Put $\Lambda^0_\pi E = K$ and $\omega_E() = 1$.

Proposition 3.1 (representation)**.** *There is a natural linear isometry*

$$\mathrm{Alt}^k(E;F) \cong \mathcal{L}(\Lambda^k_\pi E, F)$$

for every normed F. Under it, precomposition by $u\colon E\to D$ is precomposition by $\Lambda^k_\pi u\colon \Lambda^k_\pi E \to \Lambda^k_\pi D$.

Proof. The algebraic linearization $\widetilde{m}$ of a bounded alternating map m satisfies $\|\widetilde{m}(z)\| \leq \|m\|\|z\|_\pi$, by the definition of $\|z\|_\pi$; so it kills the null subspace and descends contractively. Conversely, $\|\omega_E(x_1,\dots,x_k)\| \leq \prod_a \|x_a\|$, so composing a bounded linear map with ω_E is contractive. The two operations are inverse, hence isometric. Applying u to each wedge factor gives the intertwining identity and $\|\Lambda^k_\pi u\| \leq \|u\|^k$ for $k \geq 1$; in degree zero the action is the identity on K. □

If F is complete, the same correspondence holds with the completion $\widehat{\Lambda^k_\pi E}$.

The projective tensor product $X_1\otimes_\pi\cdots\otimes_\pi X_n$ of normed spaces, and $\otimes^k_\pi E$ with k factors E, are defined in the same way with $\otimes$ in place of $\wedge$. As in Proposition 3.1, bounded n-linear maps m on $X_1\times\cdots\times X_n$ correspond isometrically to bounded linear maps $m^\otimes$ on $X_1\otimes_\pi\cdots\otimes_\pi X_n$. For bounded operators u_i, $u_1\otimes\cdots\otimes u_n$ has norm at most $\prod_i\|u_i\|$ and is multilinear in $(u_1,\dots,u_n)$;

$u^{\otimes k} = u \otimes \cdots \otimes u$. Hats denote completions, maps extend to completions by continuity, and every normed space is a subspace of its completion. Let

$$\wedge_E^k \colon \otimes_\pi^k E \to \Lambda_\pi^k E, \qquad \mathrm{alt}_E^k \colon \Lambda_\pi^k E \to \otimes_\pi^k E$$

be the maps with $x_1 \otimes \cdots \otimes x_k \mapsto x_1 \wedge \cdots \wedge x_k$ and $x_1 \wedge \cdots \wedge x_k \mapsto \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\, x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(k)}$. The first is $\omega_E^{\otimes}$, a contraction. The second is well defined in every characteristic, since for $x_a = x_b$ the permutations σ and $(a\ b) \circ \sigma$ give equal tensors with opposite signs; its norm is at most the real number $k!$, and $\wedge_E^k \circ \mathrm{alt}_E^k = k!$. Both maps are natural: $\wedge_D^k \circ u^{\otimes k} = \Lambda_\pi^k u \circ \wedge_E^k$ and $u^{\otimes k} \circ \mathrm{alt}_E^k = \mathrm{alt}_D^k \circ \Lambda_\pi^k u$ for $u \in \mathcal{L}(E, D)$.

3.2. **Universal and scalar tests.** Write $Y^* = \mathcal{L}(Y, K)$ and $J_Y(y)(\lambda) = \lambda(y)$ for the canonical contraction into the continuous bidual; no point-separation or Hahn–Banach assumption is implicit.

Theorem 3.2 (universal and scalar tests)**.** *Let U be an open parameter domain and $W \colon U \to \mathcal{L}(X, Y)$ a family. Define*

$$R_F(t)(T) = T \circ W(t), \qquad R_F \colon U \to \mathcal{L}(\mathcal{L}(Y, F), \mathcal{L}(X, F)).$$

For power-series analyticity, or for any fixed class C^n, the following statements hold.

1. *W is regular if and only if R_F is regular for every normed F; the single target Y suffices.*
2. *R_K is regular if and only if $J_Y W$ is regular in $\mathcal{L}(X, Y^{**})$, and if and only if R_{G^*} is regular for every normed G. Any one nonzero continuous dual G^* suffices for the last test.*
3. *If J_Y has a bounded linear left inverse, regularity of R_K implies regularity of W. If J_F has such a left inverse, regularity of R_K implies regularity of R_F.*

Proof. All identities below use bounded linear operations, which preserve each regularity class. Precomposition depends bounded-linearly on W, and $W(t) = R_Y(t)(\mathrm{id}_Y)$; this proves (1). Exchanging the arguments defines a linear isometry

$$\Theta \colon \mathcal{L}(Y^*, X^*) \longrightarrow \mathcal{L}(X, Y^{**}), \qquad (\Theta(B)x)(\lambda) = (B\lambda)(x),$$

since both operator norms are the least C bounding the displayed scalar by $C\|x\|\|\lambda\|$; its inverse exchanges the arguments again, and $\Theta(R_K(t)) = J_Y W(t)$. The isometry $\mathcal{L}(X, G^*) \cong \mathcal{L}(G, X^*)$, $T \mapsto [g \mapsto (x \mapsto T(x)(g))]$, identifies $R_{G^*}(t)$ with $S \mapsto R_K(t) \circ S$, which gives regularity for every dual target. Conversely, choose $v \in G^*$ and $g \in G$ with $v(g) = 1$; the injection $\alpha \mapsto v\alpha$ and evaluation at g compress R_{G^*} to R_K. This proves (2). For (3), if $P_Y J_Y = \mathrm{id}_Y$, then $W = P_Y \Theta(R_K)$; if $P_F J_F = \mathrm{id}_F$, then $R_F(t)(T) = P_F \circ R_{F^{**}}(t)(J_F \circ T)$. □

3.3. **Good pairs and one test target.** A *Banach space* is a complete normed K-space.

Definition 3.3. A pair (E, D) of normed spaces is *good in degree k* if $A^k_{E,D;F}$ is analytic for every normed F, and *good for Banach values in degree k* if this holds for every Banach space F.

Lemma 3.4 (one test target)**.** *The pair (E, D) is good in degree k if and only if $u \mapsto \Lambda_\pi^k u$ is analytic at some point of $\mathcal{L}(E, D)$, and if and only if there is a bounded k-linear $Q \colon \mathcal{L}(E, D)^k \to \mathcal{L}(\Lambda_\pi^k E, \Lambda_\pi^k D)$ with $Q(u, \dots, u) = \Lambda_\pi^k u$. The same holds for Banach values, with $\mathcal{L}(\Lambda_\pi^k E, \widehat{\Lambda}_\pi^k D)$ in place of $\mathcal{L}(\Lambda_\pi^k E, \Lambda_\pi^k D)$, also as the space in which $u \mapsto \Lambda_\pi^k u$ is analytic.*

Proof. Apply Theorem 3.2(1) with $X = \Lambda_\pi^k E$, $W(u) = \Lambda_\pi^k u$, and $Y = \Lambda_\pi^k D$, respectively $Y = \widehat{\Lambda}_\pi^k D$. By Proposition 3.1 and its completed form, R_F is $A^k_{E,D;F}$ for every normed F, respectively every Banach F, and the target Y suffices. Proposition 2.5 applies to W, with the

closed isometry $T \mapsto T \circ \omega_E$ of $\mathcal{L}(\Lambda^k_\pi E, Y)$ onto $\mathrm{Alt}^k(E;Y) \subseteq \mathrm{Mult}^k(E;Y)$ and the ambient map $(u_1,\dots,u_k) \mapsto \omega_D \circ (u_1,\dots,u_k)$. □

We use two splitting conditions on normed spaces:

(S^k_E) there is a bounded linear $s\colon \Lambda^k_\pi E \to \widehat{\otimes}^k_\pi E$ with $\widehat{\wedge}^k_E \circ s = \mathrm{id}$;
(R^k_D) there is a bounded linear $\rho\colon \otimes^k_\pi D \to \widehat{\Lambda}^k_\pi D$ with $\rho \circ \mathrm{alt}^k_D = \mathrm{id}$.

They hold *without completion* if s takes values in $\otimes^k_\pi E$, respectively ρ in $\Lambda^k_\pi D$. Both hold without completion if $k! \neq 0$ in K, with $s = \mathrm{alt}^k_E/k!$ and $\rho = \wedge^k_D/k!$, and if the space has finite continuous coordinates (e_i, e_i^*), with

$$s(x_1 \wedge \cdots \wedge x_k) = \sum_{i_1<\cdots<i_k} \det\bigl(e^*_{i_a}(x_b)\bigr)_{a,b}\, e_{i_1} \otimes \cdots \otimes e_{i_k},$$

$$\rho(y_1 \otimes \cdots \otimes y_k) = \sum_{i_1<\cdots<i_k} \prod_a e^*_{i_a}(y_a)\, e_{i_1} \wedge \cdots \wedge e_{i_k}.$$

Proposition 3.5 (splittings)**.** *Let E and D be normed spaces.*

1. *If (S^k_E) holds, then (E,D) is good for Banach values in degree k for every normed D: $Q(u_1,\dots,u_k) = \widehat{\wedge}^k_D \circ (u_1 \otimes \cdots \otimes u_k) \circ s$ is a lift as in Lemma 3.4, of norm at most $\|s\|$.*
2. *If (R^k_D) holds, then (E,D) is good for Banach values in degree k for every normed E, with the lift $Q(u_1,\dots,u_k) = \rho \circ (u_1 \otimes \cdots \otimes u_k) \circ \mathrm{alt}^k_E$ of norm at most $k!\|\rho\|$.*
3. *If the condition holds without completion, the pairs in (1), respectively (2), are good in degree k.*
4. *(S^k_E) holds if and only if the inclusions $j_{E,F}\colon \mathrm{Alt}^k(E;F) \to \mathrm{Mult}^k(E;F)$, for all Banach spaces F, have bounded linear retractions that commute with postcomposition by bounded linear maps $F \to F'$. It holds without completion if and only if such retractions exist for all normed F.*

The proof is in Appendix P.

3.4. **Three exact sequences.** A sequence $\mathcal{Y}\colon 0 \to Y_1 \xrightarrow{i} Y_2 \xrightarrow{q} Y_3 \to 0$ of Banach spaces and bounded linear maps is *exact* if i is injective, $i(Y_1) = \ker q$, and for a constant C, $\|y\| \le C\|iy\|$ on Y_1 and every $z \in Y_3$ is qy for some y with $\|y\| \le C\|z\|$. It *splits* if q has a bounded linear right inverse. Exact sequences $\mathcal{Y}, \mathcal{Y}''$ with the same end terms are *equivalent* if a bounded linear $T\colon Y_2 \to Y_2''$ satisfies $Ti = i''$ and $q''T = q$; such a T is bijective with bounded inverse, since q is open and i'' is bounded below, so this is an equivalence relation, and it preserves splitting. $\mathrm{Ext}(Y_3, Y_1)$ is the set of classes $[\mathcal{Y}]$, and 0 is the class of $0 \to Y_1 \to Y_1 \oplus Y_3 \to Y_3 \to 0$, with the inclusion, the projection and the norm $\|y_1\| + \|y_3\|$. Then $[\mathcal{Y}] = 0$ if and only if $\mathcal{Y}$ splits: a left inverse λ of i (Lemma 3.6(2)) gives the equivalence $y \mapsto (\lambda y, qy)$. For a bounded linear $\alpha\colon Y_1 \to Y'$ into a Banach space, the *pushout* $\alpha_*\mathcal{Y}$ is $0 \to Y' \xrightarrow{i_\alpha} Y_\alpha \xrightarrow{q_\alpha} Y_3 \to 0$, where Y_α is the quotient of $Y' \oplus Y_2$, normed by $\|y'\| + \|y\|$, by $\{(\alpha y, -iy) : y \in Y_1\}$, which is closed because $y \mapsto (\alpha y, -iy)$ is bounded below on the complete space Y_1; $i_\alpha y' = [(y',0)]$ and $q_\alpha[(y',y)] = qy$. Put $\alpha_*[\mathcal{Y}] = [\alpha_*\mathcal{Y}]$; an equivalence T induces the equivalence $[(y',y)] \mapsto [(y',Ty)]$, so this depends only on $[\mathcal{Y}]$.

Lemma 3.6 (splitting criteria)**.** *Let $\mathcal{Y}$ and $\mathcal{Y}'\colon 0 \to Y_1' \xrightarrow{i'} Y_2' \xrightarrow{q'} Y_3' \to 0$ be exact.*

1. *Let $\alpha_a\colon Y_a \to Y_a'$, $a = 1,2,3$, be bounded linear maps with $\alpha_2 i = i'\alpha_1$ and $q'\alpha_2 = \alpha_3 q$. Then $\alpha_1 = \tilde\alpha_1 i$ for a bounded linear $\tilde\alpha_1\colon Y_2 \to Y_1'$ if and only if $\alpha_3 = q'\tilde\alpha_3$ for a bounded linear $\tilde\alpha_3\colon Y_3 \to Y_2'$, and the two can be chosen with $\alpha_2 = i'\tilde\alpha_1 + \tilde\alpha_3 q$.*
2. *$\mathcal{Y}$ splits if and only if i has a bounded linear left inverse.*

(3) *For a bounded linear* $\alpha\colon Y_1 \to Y'$ *into a Banach space,* $\alpha_*\mathcal{Y}$ *is exact, and* $\alpha_*[\mathcal{Y}] = 0$ *if and only if* α *extends to* Y_2*, that is,* $\alpha = \tilde{\alpha} i$ *for a bounded linear* $\tilde{\alpha}\colon Y_2 \to Y'$*.*

The proof is in Appendix P.

Lemma 3.7. *For a normed space* E *and* $k \geq 1$*,* $\wedge_E^k$ *is a metric surjection:* $\|y\| = \inf\{\|z\| : \wedge_E^k z = y\}$ *for* $y \in \Lambda_\pi^k E$*. Its extension* $\widehat{\wedge}_E^k\colon \widehat{\otimes}_\pi^k E \to \widehat{\Lambda}_\pi^k E$ *is surjective and again a metric surjection.*

The proof is in Appendix P.

Definition 3.8. Let E, D, P be normed spaces and $k, n \geq 1$.

(1) The *source sequence* of E in degree k is the exact sequence (Lemma 3.7)

$$\mathcal{S}_E^k\colon \quad 0 \to \mathcal{N}_E^k \to \widehat{\otimes}_\pi^k E \xrightarrow{\widehat{\wedge}_E^k} \widehat{\Lambda}_\pi^k E \to 0, \qquad \mathcal{N}_E^k = \ker \widehat{\wedge}_E^k, \tag{3.1}$$

whose first map is the inclusion.

(2) The *target sequence* of D in degree k is $\mathcal{R}_D^k\colon 0 \to \widehat{\Lambda}_\pi^k D \xrightarrow{\mathrm{alt}_D^k} \widehat{\otimes}_\pi^k D \to \widehat{\otimes}_\pi^k D/\overline{\mathrm{alt}_D^k(\widehat{\Lambda}_\pi^k D)} \to 0$, with the quotient map last. It is exact if and only if alt_D^k is bounded below on $\widehat{\Lambda}_\pi^k D$.

(3) Let $\widehat{\Gamma}_\pi^n P$ be the closed linear span in $\widehat{\otimes}_\pi^n P$ of the pure powers $h^{\otimes n} = h \otimes \cdots \otimes h$, $h \in P$. The *parameter sequence* of P in degree n is the exact sequence $\mathcal{G}_P^n\colon 0 \to \widehat{\Gamma}_\pi^n P \to \widehat{\otimes}_\pi^n P \to \widehat{\otimes}_\pi^n P/\widehat{\Gamma}_\pi^n P \to 0$. The space P is *polarizable in degree* n, written (Γ_P^n), if $\mathcal{G}_P^n$ splits, that is (Lemma 3.6(2)), if there is a bounded linear projection R_n of $\widehat{\otimes}_\pi^n P$ onto $\widehat{\Gamma}_\pi^n P$. Let $g_n(P)$ be the infimum of the norms of such projections.

(4) For a Banach space Y, $\mathrm{Mult}_0^n(P;Y)$ is the closed subspace of $\mathrm{Mult}^n(P;Y)$ of the maps Φ with zero diagonal, $\Phi(h,\dots,h) = 0$ for $h \in P$, and $\mathrm{Poly}^n(P;Y)$ is the quotient of $\mathrm{Mult}^n(P;Y)$ by $\mathrm{Mult}_0^n(P;Y)$, with the quotient norm; through $\Phi \mapsto (h \mapsto \Phi(h,\dots,h))$ its elements correspond to the diagonals of bounded n-linear maps $P^n \to Y$, the n-homogeneous continuous polynomials of Section 2.3. The resulting exact sequence is

$$\mathcal{M}^n(P;Y)\colon \quad 0 \to \mathrm{Mult}_0^n(P;Y) \to \mathrm{Mult}^n(P;Y) \to \mathrm{Poly}^n(P;Y) \to 0.$$

Lemma 3.9. *Condition* (S_E^k) *holds if and only if* $\mathcal{S}_E^k$ *in* (3.1) *splits, and if and only if* $\mathcal{N}_E^k$ *is complemented in* $\widehat{\otimes}_\pi^k E$*. A section* s *gives the projection* $\mathrm{id} - s\,\widehat{\wedge}_E^k$ *onto* $\mathcal{N}_E^k$*, of norm at most* $1 + \|s\|$*; a bounded projection* Π *onto* $\mathcal{N}_E^k$ *gives the section* $s(\widehat{\wedge}_E^k z) = (\mathrm{id} - \Pi)z$*, of norm at most* $\|\mathrm{id} - \Pi\|$*.*

Proof. A section on $\Lambda_\pi^k E$ extends by continuity to a right inverse of $\widehat{\wedge}_E^k$, and a right inverse restricts to a section. The map $\mathrm{id} - s\,\widehat{\wedge}_E^k$ is the identity on $\mathcal{N}_E^k$ and is killed by $\widehat{\wedge}_E^k$. Conversely, $\mathrm{id} - \Pi$ vanishes on $\mathcal{N}_E^k$, so the formula defines s, and Lemma 3.7 bounds its norm. □

Lemma 3.10. *For a normed space* D *and* $k \geq 1$ *the following are equivalent:* (1) (R_D^k)*;* (2) $\mathcal{R}_D^k$ *is exact and splits;* (3) $\omega_D = \mathcal{A}_D\eta$ *for some* $\eta \in \mathrm{Mult}^k(D; \widehat{\Lambda}_\pi^k D)$*, where* $(\mathcal{A}_D\eta)(y_1,\dots,y_k) = \sum_{\sigma\in S_k} \mathrm{sgn}(\sigma)\,\eta(y_{\sigma(1)},\dots,y_{\sigma(k)})$ *is unnormalized antisymmetrization.*

The proof is in Appendix P.

Lemma 3.11. *For a normed space* P *and* $n \geq 1$ *the following are equivalent.*

(1) P *is polarizable in degree* n*.*

(2) *The power map* $P \to \widehat{\Gamma}_\pi^n P$*,* $h \mapsto h^{\otimes n}$*, is the diagonal of a bounded* n*-linear map* $P^n \to \widehat{\Gamma}_\pi^n P$*.*

(3) *For every closed subspace W of a Banach space Z, every bounded n-linear $\Phi\colon P^n \to Z$ with $\Phi(h,\dots,h) \in W$ for all $h \in P$, and every $\delta > 0$, there is a bounded n-linear $Q\colon P^n \to W$ with $Q(h,\dots,h) = \Phi(h,\dots,h)$ for all h and $\|Q\| \le (g_n(P)+\delta)\|\Phi\|$.*

Under these conditions $\mathcal{M}^n(P;Y)$ splits for every Banach space Y, by right inverses that commute with postcomposition by bounded linear maps $Y \to Y'$.

The proof is in Appendix P.

Definition 3.12. Let $w\colon E \to G$ be a bounded linear map of normed spaces and $k \ge 1$. Then w is S^k*-split* if $\widehat{\wedge}^k_G \circ s = \Lambda^k_\pi w$ for a bounded linear $s\colon \Lambda^k_\pi E \to \widehat{\otimes}^k_\pi G$, and R^k*-split* if $\rho \circ \mathrm{alt}^k_E = \Lambda^k_\pi w$ for a bounded linear $\rho\colon \otimes^k_\pi E \to \widehat{\Lambda}^k_\pi G$; it is split *without completion* if s takes values in $\otimes^k_\pi G$, respectively ρ in $\Lambda^k_\pi G$. It is $\mathrm{S}^{\le k}$*-split* ($\mathrm{R}^{\le k}$*-split*) if it is S^j-split (R^j-split) for every $j \le k$.

Thus (S^k_E) says that id_E is S^k-split, and (R^k_D) that id_D is R^k-split. Every w is S^1- and R^1-split, and S^j- and R^j-split whenever $j! \ne 0$ in K, with $s = w^{\otimes j}\,\mathrm{alt}^j_E\,/j!$ and $\rho = \wedge^j_G w^{\otimes j}/j!$, since $\wedge^j_G w^{\otimes j}\,\mathrm{alt}^j_E = \Lambda^j_\pi w\ \wedge^j_E \mathrm{alt}^j_E = j!\,\Lambda^j_\pi w$.

Proposition 3.13 (split operators)**.** *Let $w\colon E \to G$ be a bounded linear map of normed spaces and $k \ge 1$.*

(1) *The map $w^{\otimes k}$ sends $\mathcal{N}^k_E$ into $\mathcal{N}^k_G$, and the following are equivalent: (a) w is S^k-split; (b) $\widehat{\Lambda}^k_\pi w = \widehat{\wedge}^k_G \circ \tilde{s}$ for a bounded linear $\tilde{s}\colon \widehat{\Lambda}^k_\pi E \to \widehat{\otimes}^k_\pi G$; (c) $w^{\otimes k}|_{\mathcal{N}^k_E}\colon \mathcal{N}^k_E \to \mathcal{N}^k_G$ extends to a bounded linear map $\widehat{\otimes}^k_\pi E \to \mathcal{N}^k_G$; (d) the* splitting class $\mathrm{sp}_k(w) = (w^{\otimes k}|_{\mathcal{N}^k_E})_*[\mathcal{S}^k_E] \in \mathrm{Ext}(\widehat{\Lambda}^k_\pi E, \mathcal{N}^k_G)$ *vanishes. In particular (S^k_E) holds if and only if $\mathrm{sp}_k(\mathrm{id}_E) = 0$, if and only if $[\mathcal{S}^k_E] = 0$ (Lemma 3.9).*

(2) *If w is S^k-split with s, and $v\colon G \to G_1$, $v'\colon G_0 \to E$ are bounded linear, then vwv' is S^k-split with $v^{\otimes k} s\, \Lambda^k_\pi v'$. If w is R^k-split with ρ, then vwv' is R^k-split with $\widehat{\Lambda}^k_\pi v\, \rho\, v'^{\otimes k}$. The same holds without completion.*

The proof is in Appendix P.

3.5. **The obstruction class.** Throughout this subsection E and D are normed spaces and $k \ge 1$; $\mathrm{Mult}^k(\mathcal{L}(E,D); \widehat{\Lambda}^k_\pi D)$ is a Banach space, and $\mathrm{Mult}^k_0(\mathcal{L}(E,D); \widehat{\Lambda}^k_\pi D)$ is its closed subspace of maps with zero diagonal.

Lemma 3.14 (evaluation)**.** *There is a unique bounded linear map* $\mathrm{ev} = \mathrm{ev}^k_{E,D}\colon \widehat{\otimes}^k_\pi E \to \mathrm{Mult}^k(\mathcal{L}(E,D); \widehat{\Lambda}^k_\pi D)$ *with*

$$\mathrm{ev}(x_1 \otimes \cdots \otimes x_k)(u_1,\dots,u_k) = u_1 x_1 \wedge \cdots \wedge u_k x_k, \tag{3.2}$$

and $\|\mathrm{ev}\| \le 1$. It satisfies $\mathrm{ev}(z)(u,\dots,u) = \widehat{\Lambda}^k_\pi u(\widehat{\wedge}^k_E z)$ for $z \in \widehat{\otimes}^k_\pi E$ and $u \in \mathcal{L}(E,D)$; in particular ev *maps $\mathcal{N}^k_E$ into $\mathrm{Mult}^k_0(\mathcal{L}(E,D); \widehat{\Lambda}^k_\pi D)$.*

Proof. For $x \in E^k$ the right side of (3.2) is k-linear in $(u_1,\dots,u_k)$, of norm at most $\prod_a \|x_a\|$, and k-linear in x. Its linearization in x is a contraction from $\otimes^k_\pi E$ into a Banach space and extends to $\widehat{\otimes}^k_\pi E$; pure tensors span a dense subspace, which gives uniqueness. For fixed u, both sides of the identity are bounded linear in z and agree on pure tensors. □

Let q be the quotient map of $\mathcal{M}^k(\mathcal{L}(E,D); \widehat{\Lambda}^k_\pi D)$. Since $q \circ \mathrm{ev}$ vanishes on $\mathcal{N}^k_E$, Lemma 3.7 gives a unique bounded linear $\overline{\mathrm{ev}}\colon \widehat{\Lambda}^k_\pi E \to \mathrm{Poly}^k(\mathcal{L}(E,D); \widehat{\Lambda}^k_\pi D)$ with $q \circ \mathrm{ev} = \overline{\mathrm{ev}} \circ \widehat{\wedge}^k_E$. By Lemma 3.14, $\overline{\mathrm{ev}}(y)$ is the class of every map in $\mathrm{Mult}^k(\mathcal{L}(E,D); \widehat{\Lambda}^k_\pi D)$ with diagonal $u \mapsto \widehat{\Lambda}^k_\pi u(y)$.

The *obstruction class* of the pair (E,D) in degree k is

$$\begin{aligned}\mathrm{ob}_k(E,D) &= (\mathrm{ev}\,|_{\mathcal{N}_E^k})_*[\mathcal{S}_E^k]\\ &\in \mathrm{Ext}\big(\widehat{\Lambda}_\pi^k E, \mathrm{Mult}_0^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D)\big).\end{aligned} \tag{3.3}$$

Theorem 3.15 (obstruction theorem)**.** *Let E and D be normed spaces and $k\geq 1$. The following are equivalent.*

(1) *The pair (E,D) is good for Banach values in degree k.*
(2) *The restriction $\mathrm{ev}\,|_{\mathcal{N}_E^k}$ extends to a bounded linear map $\widehat{\otimes}_\pi^k E\to \mathrm{Mult}_0^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D)$.*
(3) *$\mathrm{ob}_k(E,D)=0$ in (3.3).*

More precisely, a lift $Q\colon \mathcal{L}(E,D)^k\to\mathcal{L}(\Lambda_\pi^k E,\widehat{\Lambda}_\pi^k D)$ as in Lemma 3.4 gives an extension in (2) of norm at most $1+\|Q\|$, and an extension ξ in (2) gives such a lift of norm at most $1+\|\xi\|$.

Proof. By Lemma 3.4, (1) holds if and only if a lift Q exists.

Step 1 (lifts give extensions). Bounded k-linear maps $Q\colon \mathcal{L}(E,D)^k\to\mathcal{L}(\Lambda_\pi^k E,\widehat{\Lambda}_\pi^k D)$ correspond to bounded linear maps $Q^\sharp\colon \widehat{\Lambda}_\pi^k E\to \mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D)$, by $Q^\sharp(y)(u_1,\dots,u_k)=Q(u_1,\dots,u_k)y$ for $y\in\Lambda_\pi^k E$, with $\|Q^\sharp\|=\|Q\|$. Given a lift Q, put $\xi=\mathrm{ev}-Q^\sharp\widehat{\wedge}_E^k$; it equals ev on $\mathcal{N}_E^k=\ker\widehat{\wedge}_E^k$. By Lemma 3.14, $\xi(z)(u,\dots,u)=\Lambda_\pi^k u(\wedge_E^k z)-Q(u,\dots,u)(\wedge_E^k z)=0$ for $z\in\otimes_\pi^k E$, hence for all z by continuity. So ξ is an extension as in (2). Since $\|\mathrm{ev}\|\leq 1$ by Lemma 3.14 and $\widehat{\wedge}_E^k$ is contractive, $\|\xi\|\leq\|\mathrm{ev}\|+\|Q^\sharp\|\leq 1+\|Q\|$.

Step 2 (extensions give lifts). Given an extension ξ as in (2), the map $\mathrm{ev}-\xi$ vanishes on $\mathcal{N}_E^k$. Since $\widehat{\wedge}_E^k$ is a metric surjection with kernel $\mathcal{N}_E^k$ (Lemma 3.7), $\mathrm{ev}-\xi=Q^\sharp\widehat{\wedge}_E^k$ for a bounded linear $Q^\sharp$ with $\|Q^\sharp\|=\|\mathrm{ev}-\xi\|\leq 1+\|\xi\|$, using $\|\mathrm{ev}\|\leq 1$. The corresponding Q satisfies $Q(u,\dots,u)(\wedge_E^k z)=\mathrm{ev}(z)(u,\dots,u)-\xi(z)(u,\dots,u)=\Lambda_\pi^k u(\wedge_E^k z)$ for $z\in\otimes_\pi^k E$, by Lemma 3.14 and since $\xi(z)$ has zero diagonal; so $Q(u,\dots,u)=\Lambda_\pi^k u$, since $\wedge_E^k$ is surjective.

Step 3 (the class). Conditions (2) and (3) are equivalent by Lemma 3.6(3). □

If D is dense in a normed K-space D', the canonical maps $\otimes_\pi^k D\to\otimes_\pi^k D'$ and $\Lambda_\pi^k D\to\Lambda_\pi^k D'$ are isometric with dense range; so, with $\widehat{D}$ regarded as a normed K-space, $\widehat{\otimes}_\pi^k\widehat{D}=\widehat{\otimes}_\pi^k D$ and $\widehat{\Lambda}_\pi^k\widehat{D}=\widehat{\Lambda}_\pi^k D$. The maps are contractive. By continuity ω_D extends to a contractive map $\omega'\colon D'^k\to\widehat{\Lambda}_\pi^k D$, alternating because tuples with two equal entries in D'^k are limits of such tuples in D^k; its linearization (Proposition 3.1) is a contraction $\Lambda_\pi^k D'\to\widehat{\Lambda}_\pi^k D$ restricting to the inclusion on $\Lambda_\pi^k D$, so the maps are isometric. Density follows from $x_1\wedge\cdots\wedge x_k-y_1\wedge\cdots\wedge y_k=\sum_a y_1\wedge\cdots\wedge y_{a-1}\wedge(x_a-y_a)\wedge x_{a+1}\wedge\cdots\wedge x_k$. The tensor case is the same.

Theorem 3.16 (ways for the class to vanish)**.** *Let E and D be normed spaces and $k\geq 1$.*

(1) (Source.) *If (S_E^k) holds, then $\mathrm{ob}_k(E,D)=0$. More generally, $\mathrm{ob}_k(E,D)=0$ if $\mathrm{ev}\,|_{\mathcal{N}_E^k}=\tau\circ w^{\otimes k}|_{\mathcal{N}_E^k}$ for an S^k-split $w\colon E\to G$ and a bounded linear $\tau\colon\mathcal{N}_G^k\to\mathrm{Mult}_0^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D)$. This applies when $w\colon E\to G$ is S^k-split and $\varepsilon\colon\mathcal{L}(E,D)\to\mathcal{L}(G,\widehat{D})$ is bounded linear with $\varepsilon(u)\circ w=u$ for every u, with $\tau(\zeta)(u_1,\dots,u_k)=\widehat{\wedge}_{\widehat{D}}^k\big(\varepsilon(u_1)\otimes\cdots\otimes\varepsilon(u_k)\big)\zeta$.*

(2) (Parameter.) *If the sequence $\mathcal{M}^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D)$ splits, and in particular if $\mathcal{L}(E,D)$ is polarizable in degree k, then $\mathrm{ob}_k(E,D)=0$. More generally, $\mathrm{ob}_k(E,D)=0$ if there are a normed space P with (Γ_P^k), a bounded linear map $\psi\colon\mathcal{L}(E,D)\to P$ and a bounded k-linear map $\Phi\colon P^k\to\mathcal{L}(\widehat{\otimes}_\pi^k E,\widehat{\Lambda}_\pi^k D)$ such that $\Phi(h,\dots,h)$ vanishes on $\mathcal{N}_E^k$ for $h\in P$ and $\Phi(\psi u,\dots,\psi u)=\widehat{\Lambda}_\pi^k u\circ\widehat{\wedge}_E^k$ for $u\in\mathcal{L}(E,D)$.*

(3) (Target.) *Let* $\mathrm{alt}_*\colon \mathrm{Mult}_0^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D) \to \mathrm{Mult}_0^k(\mathcal{L}(E,D);\widehat{\otimes}_\pi^k D)$ *be postcomposition with* alt_D^k. *Then* $(\mathrm{alt}_* \circ \mathrm{ev}\,|_{\mathcal{N}_E^k})_*[\mathcal{S}_E^k] = 0$. *If* (R_D^k) *holds, then* $\mathrm{ob}_k(E,D) = 0$.

(3′) (Target, relative.) *If* $w'\colon G \to \widehat{D}$ *is* R^k*-split and* $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(E,G)$ *is bounded linear with* $w' \circ \varepsilon(u) = u$ *for every* u, *then* $\mathrm{ob}_k(E,D) = 0$.

The proof is in Appendix P.

Corollary 3.17. *Each of* (S_E^k), (R_D^k) *and* $(\Gamma^k_{\mathcal{L}(E,D)})$ *makes* (E,D) *good for Banach values in degree* k; *so does each of the data* (G,w,ε) *of Theorem 3.16(1) and* (G,w',ε) *of Theorem 3.16(3′).*

Proof. Theorems 3.16 and 3.15. □

3.6. **The ordering lift.** Throughout this subsection E and D are normed spaces, $k \geq 1$, and $(e_t)_{t\in\Sigma}$ is a linearly independent family in E, indexed by a totally ordered set Σ, whose span V is dense in E; a Hamel basis of E in any total order is one. Write $x = \sum_t x(t)e_t$ for $x \in V$, $\mathrm{Mult}^k = \mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D)$ and $\mathrm{Mult}_0^k = \mathrm{Mult}_0^k(\mathcal{L}(E,D);\widehat{\Lambda}_\pi^k D)$, and let q and $\overline{\mathrm{ev}}$ be the maps defined after Lemma 3.14. By the description of $\overline{\mathrm{ev}}$ given there, in the notation of the proof of Theorem 3.15, a bounded linear $T\colon \widehat{\Lambda}_\pi^k E \to \mathrm{Mult}^k$ is $Q^\sharp$ for a lift Q as in Lemma 3.4 if and only if $q \circ T = \overline{\mathrm{ev}}$. Let $c_k(E,D) \in [0,\infty]$ be the infimum of $\|Q\|$ over these lifts; by Lemma 3.4 it is finite if and only if (E,D) is good for Banach values in degree k.

Let $\Sigma_<^k \subseteq \Sigma^k$ be the set of strictly increasing t, and put $e_t^\wedge = e_{t_1} \wedge \cdots \wedge e_{t_k}$ and $e_t^\otimes = e_{t_1} \otimes \cdots \otimes e_{t_k}$ for $t \in \Sigma^k$; the $e_t^\wedge$ with $t \in \Sigma_<^k$ form a basis of $\Lambda_K^k V$. Let $\wedge_V\colon \otimes_K^k V \to \Lambda_K^k V$ be the quotient map and $j_\Lambda\colon \Lambda_K^k V \to \widehat{\Lambda}_\pi^k E$, $j_\otimes\colon \otimes_K^k V \to \widehat{\otimes}_\pi^k E$ the canonical maps, so that $\widehat{\wedge}_E^k \circ j_\otimes = j_\Lambda \circ \wedge_V$, and put $\|y\|_\pi = \|j_\Lambda y\|$. By the paragraph before Theorem 3.16, $\|y\|_\pi$ is the projective seminorm of y computed from representations with entries in V, and $j_\Lambda(\Lambda_K^k V)$ is dense in $\widehat{\Lambda}_\pi^k E$; the same holds for $j_\otimes$. Norms in $[0,\infty]$ of linear maps on $\Lambda_K^k V$, for $\|\cdot\|_\pi$, and of k-linear maps on V^k are best constants, as are the norms of lifts Q and of the maps $Q^\sharp$.

The *ordering section* is the linear map $s_<\colon \Lambda_K^k V \to \otimes_K^k V$ with $s_<(e_t^\wedge) = e_t^\otimes$ for $t \in \Sigma_<^k$. Then $\wedge_V \circ s_< = \mathrm{id}$, and $s_< = \mathcal{P}^{(k)} \circ \mathrm{alt}^k$ for the map $\mathcal{P}^{(k)}$ of Proposition 5.7 and the antisymmetrization $\mathrm{alt}^k\colon \Lambda_K^k V \to \otimes_K^k V$. The *ordering lift* assigns to $u_1,\dots,u_k \in \mathcal{L}(E,D)$ the linear map $Q_<(u_1,\dots,u_k)\colon \Lambda_K^k V \to \Lambda_\pi^k D$ with

$$Q_<(u_1,\dots,u_k)(e_t^\wedge) = u_1 e_{t_1} \wedge \cdots \wedge u_k e_{t_k} \qquad (t \in \Sigma_<^k); \tag{3.4}$$

on $x_1 \wedge \cdots \wedge x_k$ with $x_1,\dots,x_k \in V$ it is the right side of (J.6). Let $Q_<^\sharp\colon \Lambda_K^k V \to \mathrm{Mult}^k$ be the linear map with $Q_<^\sharp(y)(u_1,\dots,u_k) = Q_<(u_1,\dots,u_k)y$; it takes values in Mult^k by (3.4). A *zero-diagonal correction* is a linear map $L\colon \Lambda_K^k V \to \mathrm{Mult}_0^k$. The proofs of Theorems 3.18 and 3.22, Lemma 3.19 and Corollaries 3.20 and 3.21 are in Appendix P.

Theorem 3.18 (ordering representation). (1) $\mathrm{ev} \circ j_\otimes \circ s_< = Q_<^\sharp$ *and* $q \circ Q_<^\sharp = \overline{\mathrm{ev}} \circ j_\Lambda$.

(2) *For every lift* $Q\colon \mathcal{L}(E,D)^k \to \mathcal{L}(\Lambda_\pi^k E, \widehat{\Lambda}_\pi^k D)$ *as in Lemma 3.4,* $L = Q^\sharp \circ j_\Lambda - Q_<^\sharp$ *is a zero-diagonal correction; equivalently,* $Q(u_1,\dots,u_k) \circ j_\Lambda = Q_<(u_1,\dots,u_k) + \Phi(u_1,\dots,u_k)$, *where* $\Phi(u_1,\dots,u_k)y = L(y)(u_1,\dots,u_k)$ *is* k*-linear in* $(u_1,\dots,u_k)$ *and* $\Phi(u,\dots,u) = 0$ *for* $u \in \mathcal{L}(E,D)$.

(3) *For every zero-diagonal correction* L *with* $\|Q_<^\sharp + L\| < \infty$ *there is a unique lift* Q *with* $Q^\sharp \circ j_\Lambda = Q_<^\sharp + L$, *and* $\|Q\| = \|Q_<^\sharp + L\|$. *Hence* $c_k(E,D) = \inf_L \|Q_<^\sharp + L\|$ *over the zero-diagonal corrections* L, *and* (E,D) *is good for Banach values in degree* k *if and only if* $\|Q_<^\sharp + L\| < \infty$ *for some zero-diagonal correction* L.

(4) (Defect form.) *The formula* $\Omega_<(x) = \mathrm{ev}(x_1 \otimes \cdots \otimes x_k) - Q^\sharp_<(x_1 \wedge \cdots \wedge x_k)$, *for* $x \in V^k$, *defines a* k*-linear map* $\Omega_<\colon V^k \to \mathrm{Mult}^k_0$. *For every zero-diagonal correction* L, *with* $(L \circ \wedge)(x) = L(x_1 \wedge \cdots \wedge x_k)$,

$$\|Q^\sharp_< + L\| \le 1 + \|\Omega_< - L \circ \wedge\|, \qquad \|\Omega_< - L \circ \wedge\| \le 1 + \|Q^\sharp_< + L\|.$$

So (E, D) *is good for Banach values in degree* k *if and only if* $\Omega_<$ *is the sum of a bounded* k*-linear map* $V^k \to \mathrm{Mult}^k_0$ *and an alternating* k*-linear map* $V^k \to \mathrm{Mult}^k_0$. *The infima* $c_k(E, D)$ *and* $\inf_L \|\Omega_< - L \circ \wedge\|$ *are finite together, and then they differ by at most* 1.

If $\|\mathcal{P}_<(x \otimes y)\| \le C\|x\|\|y\|$ for $x, y \in V$, as in Proposition 5.7, no correction is needed: the lift Q of Proposition 3.5(1) built from the section $s = \mathcal{P}^{(k)} \circ \mathrm{alt}^k_E$ of Proposition 5.7(2) satisfies $Q(u_1, \dots, u_k) \circ j_\Lambda = Q_<(u_1, \dots, u_k)$, since $s \circ j_\Lambda = j_\otimes \circ s_<$.

Lemma 3.19. *The space* $\mathcal{N}^k_E$ *is the closure in* $\widehat{\otimes}^k_\pi E$ *of the span of the pure tensors* $x_1 \otimes \cdots \otimes x_k$ *with* $x_a = x_{a+1}$ *for some* $a < k$, *and also of the span of the pure tensors with two equal entries. The pure tensors of the first kind with all entries in* V *span a dense subspace of* $\mathcal{N}^k_E$.

Corollary 3.20 (adjacent diagonals)**.** *The pair* (E, D) *is good for Banach values in degree* k *if and only if there is a bounded* k*-linear* $\Psi\colon E^k \to \mathrm{Mult}^k_0$ *with*

$$\Psi(x_1, \dots, x_k)(u_1, \dots, u_k) = u_1 x_1 \wedge \cdots \wedge u_k x_k$$

whenever two of the x_a *coincide, equivalently whenever* $x_a = x_{a+1}$ *for some* $a < k$. *A lift* Q *gives such a* Ψ *with* $\|\Psi\| \le 1 + \|Q\|$, *and such a* Ψ *gives a lift* Q *with* $\|Q\| \le 1 + \|\Psi\|$.

Corollary 3.21 (z-linear form)**.** *Let* $\delta > 0$, *and let* $\nu\colon \widehat{\Lambda}^k_\pi E \to \widehat{\otimes}^k_\pi E$ *be a homogeneous map,* $\nu(cy) = c\,\nu(y)$ *for* $c \in K$, *with* $\widehat{\wedge}^k_E \circ \nu = \mathrm{id}$ *and* $\|\nu(y)\| \le (1 + \delta)\|y\|$; *it exists by Lemma 3.7, choosing* $\nu(y)$ *for one point* y *of each orbit* $K^\times y$, $y \ne 0$, *and extending homogeneously. Put* $\Omega^\nu_< = \mathrm{ev} \circ \nu \circ j_\Lambda - Q^\sharp_<\colon \Lambda^k_K V \to \mathrm{Mult}^k$. *Then* $\Omega^\nu_<$ *takes values in* Mult^k_0, *is homogeneous, and is* z-linear*:*

$$\Big\|\Omega^\nu_<\Big(\sum_i y_i\Big) - \sum_i \Omega^\nu_<(y_i)\Big\| \le 2(1 + \delta) \sum_i \|y_i\|_\pi$$

for every finite family (y_i) *in* $\Lambda^k_K V$. *Let* $\mathrm{dist}(\Omega^\nu_<)$ *be the infimum, over the linear maps* $L\colon \Lambda^k_K V \to \mathrm{Mult}^k_0$, *of the infimum of the* C *with* $\|\Omega^\nu_<(y) - L(y)\| \le C\|y\|_\pi$ *for all* y. *Then* $\mathrm{dist}(\Omega^\nu_<)$ *and* $c_k(E, D)$ *are finite together, and then they differ by at most* $1 + \delta$. *In particular* $\mathrm{ob}_k(E, D) = 0$ *if and only if* $\Omega^\nu_<$ *lies at finite distance from a linear map* $\Lambda^k_K V \to \mathrm{Mult}^k_0$.

Theorem 3.22 (nonarchimedean values)**.** *Let* K *be nonarchimedean, and suppose that* $C_e = \sup_t \|e_t\| < \infty$ *and that* $|x(t)| \le C_c\|x\|$ *for* $x \in V$, $t \in \Sigma$ *and a constant* C_c. *Let* Y *be a Banach space with a nonarchimedean norm* $|||\cdot|||_Y$ *and constants* $0 < c_- \le c_+$ *with* $c_-|||\cdot|||_Y \le \|\cdot\| \le c_+|||\cdot|||_Y$, *and let* $\alpha\colon \mathcal{N}^k_E \to Y$ *be bounded linear. Then there is a bounded linear* $\tilde{\alpha}\colon \widehat{\otimes}^k_\pi E \to Y$ *of norm at most* $2(c_+/c_-)\|\alpha\|(C_c C_e)^k$ *with* $\tilde{\alpha}(j_\otimes z) = \alpha\big(j_\otimes(z - s_<(\wedge_V z))\big)$ *for* $z \in \otimes^k_K V$ *and* $\tilde{\alpha} = \alpha$ *on* $\mathcal{N}^k_E$. *Hence* $\alpha_*[\mathcal{S}^k_E] = 0$. *In particular* $\mathrm{ob}_k(E, D) = 0$ *whenever* $\widehat{\Lambda}^k_\pi D$ *has an equivalent nonarchimedean norm.*

4. Canonical splittings and forced asymmetry

Throughout this section K is a nontrivially normed field and $[n] := \{1, \dots, n\}$. For $\sigma \in S_n$ let τ_σ be the isometry of the n-th projective tensor power of a normed space, and of its completion,

with $\tau_\sigma(x_1 \otimes \cdots \otimes x_n) = x_{\sigma^{-1}(1)} \otimes \cdots \otimes x_{\sigma^{-1}(n)}$; it commutes with $u^{\otimes n}$ for every bounded operator u, and $\tau_\sigma \tau_{\sigma'} = \tau_{\sigma\sigma'}$. Evaluation on pure tensors gives

$$\mathrm{alt}^k_E \circ \wedge^k_E = \sum_{\sigma \in S_k} \mathrm{sgn}(\sigma)\, \tau_\sigma, \qquad \tau_\sigma \circ \mathrm{alt}^k_E = \mathrm{sgn}(\sigma)\, \mathrm{alt}^k_E, \qquad \wedge^k_E \circ \tau_\sigma = \mathrm{sgn}(\sigma)\, \wedge^k_E, \tag{4.1}$$

and the first identity extends to $\widehat{\otimes}^k_\pi E$ by continuity. For vectors $x_1, \dots, x_n$ and $S \subseteq [n]$ put $v_S := \sum_{i \in S} x_i$. Expanding $v_S^{\otimes n}$ over the maps $f\colon [n] \to S$, and using that $\sum_{S \supseteq f([n])} (-1)^{n-|S|}$ is 1 if f is onto and 0 otherwise, gives in every characteristic

$$\sum_{S \subseteq [n]} (-1)^{n-|S|}\, v_S^{\otimes n} = \sum_{\sigma \in S_n} x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)}. \tag{4.2}$$

Proposition 4.1 (natural splittings)**.** *Let E, D and P be normed spaces.*

1. *If $k! \neq 0$ in K, then $\mathrm{alt}^k_E / k!$ is a section of $\wedge^k_E$ of norm at most $k!/|k!|$ and $\wedge^k_D / k!$ is a retraction of alt^k_D of norm at most $1/|k!|$; both are natural with respect to every $u \in \mathcal{L}(E, D)$ and satisfy the sign rules of (4.1).*
2. *If $n! \neq 0$ in K, then $R_n := \frac{1}{n!} \sum_{\sigma \in S_n} \tau_\sigma$ is a projection of $\widehat{\otimes}^n_\pi P$ onto $\widehat{\Gamma}^n_\pi P$ of norm at most $n!/|n!|$; it commutes with $u^{\otimes n}$ for bounded operators u and satisfies $\tau_\sigma R_n = R_n \tau_\sigma = R_n$.*

Proof. (1) holds by the properties of $\wedge^k_E$ and alt^k_E in Section 3. (2) The norm bound and the commutation rules hold since each τ_σ is an isometry commuting with $u^{\otimes n}$ and $\tau_\sigma \tau_{\sigma'} = \tau_{\sigma\sigma'}$. R_n fixes every pure power, hence $\widehat{\Gamma}^n_\pi P$ pointwise, and by (4.2) and continuity it maps $\widehat{\otimes}^n_\pi P$ into $\widehat{\Gamma}^n_\pi P$. □

So every pair is good in degree k when $k! \neq 0$ (Propositions 3.5(3) and 5.1(1)). A matrix in $GL_k(K)$ is *monomial* if each row and each column has exactly one nonzero entry.

Theorem 4.2 (no equivariant splitting)**.** *Let $k \geq 2$ with $k! = 0$ in K, and let E_0 be K^k with a norm for which the coordinate functionals $e_1^*, \dots, e_k^*$ of the standard basis $e_1, \dots, e_k$ are bounded; by Lemma A.3 every norm qualifies if K is complete. For a monomial g, extend $g^{\otimes k}$ and $\Lambda^k_\pi g$ to the completions.*

1. *No linear $s\colon \Lambda^k_\pi E_0 \to \widehat{\otimes}^k_\pi E_0$ with $\widehat{\wedge}^k_{E_0} \circ s = \mathrm{id}$ satisfies $s \circ \Lambda^k_\pi g = g^{\otimes k} \circ s$ for all monomial g.*
2. *No linear $\rho\colon \otimes^k_\pi E_0 \to \widehat{\Lambda}^k_\pi E_0$ with $\rho \circ \mathrm{alt}^k_{E_0} = \mathrm{id}$ satisfies $\rho \circ g^{\otimes k} = \Lambda^k_\pi g \circ \rho$ for all monomial g.*
3. *No linear projection R of $\widehat{\otimes}^k_\pi E_0$ onto $\widehat{\Gamma}^k_\pi E_0$ satisfies $R \circ g^{\otimes k} = g^{\otimes k} \circ R$ for all monomial g.*

Statements (1)–(3) also hold for the algebraic powers $\otimes^k_K K^k$ and $\Lambda^k_K K^k$.

The proof is in Appendix P.

Let a group H act on a set I, and let $V = K^{(I)}$ be the space of finitely supported functions $I \to K$, with unit vectors e_i. Each $h \in \mathsf{H}$ acts on V by $he_i = e_{h(i)}$, on $\otimes^k_K V$ by $h^{\otimes k}$ and on $\Lambda^k_K V$ by $\Lambda^k_K h$; a linear map between two of these spaces is H-*equivariant* if it intertwines the actions of every $h \in \mathsf{H}$. The maps $\wedge^k\colon \otimes^k_K V \to \Lambda^k_K V$, $\mathrm{alt}^k\colon \Lambda^k_K V \to \otimes^k_K V$ and τ_σ are given by the formulas of Section 3.1 and of this section. A *k-subset* is a subset with k elements; for a finite $T \subseteq I$, H_T is its setwise stabilizer and $\bar{\mathsf{H}}_T \subseteq \mathrm{Sym}(T)$ is the group of permutations of T induced by H_T. A *k-multiset* is a finitely supported $\mathbf{m}\colon I \to \mathbb{N}$ with $\sum_i \mathbf{m}(i) = k$; H acts on these by $(h\mathbf{m})(i) = \mathbf{m}(h^{-1}(i))$, $\mathsf{H}_\mathbf{m}$ is the stabilizer of $\mathbf{m}$, and $\bar{\mathsf{H}}_\mathbf{m} \subseteq \mathrm{Sym}(\mathrm{supp}\, \mathbf{m})$ is the group induced by $\mathsf{H}_\mathbf{m}$. Let $\Gamma^k_K V \subseteq \otimes^k_K V$ be the span of the pure powers $x^{\otimes k}$, $x \in V$; since K

is infinite, it is the space of tensors fixed by every τ_σ, by the paragraph at the beginning of Section L.3.

Theorem 4.3 (permutation groups)**.** *Let* H *act on* I*, and let* $k \geq 1$*.*

(1) *There is an* H*-equivariant linear* $s\colon \Lambda^k_K V \to \otimes^k_K V$ *with* $\wedge^k \circ s = \mathrm{id}$ *if and only if* $|\bar{\mathsf{H}}_T| \neq 0$ *in* K *for every* k*-subset* T *of* I*.*

(2) *There is an* H*-equivariant linear* $\rho\colon \otimes^k_K V \to \Lambda^k_K V$ *with* $\rho \circ \mathrm{alt}^k = \mathrm{id}$ *if and only if the condition in (1) holds.*

(3) *There is an* H*-equivariant linear projection of* $\otimes^k_K V$ *onto* $\Gamma^k_K V$ *if and only if* $|\bar{\mathsf{H}}_{\mathbf{m}}| \neq 0$ *in* K *for every* k*-multiset* $\mathbf{m}$ *on* I*.*

When the condition in (1), (2) or (3) holds, the corresponding map can be chosen to commute also with the operator $e_i \mapsto \lambda_i e_i$ *of* V *for every* $\lambda \in (K^\times)^I$*.*

(4) *Let* E *be a normed space with vectors* $e_i \in E$ *and bounded linear functionals* e_i^* *on* E*,* $i \in I$*, with* $e_i^*(e_j) = \delta_{ij}$*, and let* H *act on* I *and on* E *by bounded linear maps with*

$$he_i = e_{h(i)}, \qquad e_i^* \circ h = e^*_{h^{-1}(i)} \qquad (h \in \mathsf{H},\ i \in I); \tag{4.3}$$

extend $h^{\otimes k}$ *and* $\Lambda^k_\pi h$ *to the completions. If* $|\bar{\mathsf{H}}_T| = 0$ *in* K *for some* k*-subset* T*, no linear* $s\colon \Lambda^k_\pi E \to \widehat{\otimes}^k_\pi E$ *with* $\widehat{\wedge}^k_E \circ s = \mathrm{id}$ *satisfies* $s \circ \Lambda^k_\pi h = h^{\otimes k} \circ s$ *for all* $h \in \mathsf{H}_T$*, and no linear* $\rho\colon \otimes^k_\pi E \to \widehat{\Lambda}^k_\pi E$ *with* $\rho \circ \mathrm{alt}^k_E = \mathrm{id}$ *satisfies* $\rho \circ h^{\otimes k} = \Lambda^k_\pi h \circ \rho$ *for all* $h \in \mathsf{H}_T$*. If* $|\bar{\mathsf{H}}_{\mathbf{m}}| = 0$ *in* K *for some* k*-multiset* $\mathbf{m}$*, no linear projection* R *of* $\widehat{\otimes}^k_\pi E$ *onto* $\widehat{\Gamma}^k_\pi E$ *satisfies* $R \circ h^{\otimes k} = h^{\otimes k} \circ R$ *for all* $h \in \mathsf{H}_{\mathbf{m}}$*.*

The proof is in Appendix N.

Corollary 4.4 (one element)**.** (1) *In each of (1)–(3) of Theorem 4.3, the following are equivalent: the* H*-equivariant map exists; for every* $h \in \mathsf{H}$*, the map equivariant for the cyclic group generated by* h *exists; no* $h \in \mathsf{H}$ *maps a* k*-subset* T *onto itself, in (1) and (2), or fixes a* k*-multiset* $\mathbf{m}$*, in (3), and induces on* T*, respectively on* $\operatorname{supp}\mathbf{m}$*, a permutation of order* $\operatorname{char} K$*. If* $\operatorname{char} K = 0$*, the three statements hold.*

(2) *Let* $k! = 0$ *in* K*, so that* $p = \operatorname{char} K$ *satisfies* $2 \leq p \leq k$*. Let* E_0 *be as in Theorem 4.2, let* $\nu \in S_k$ *be a cycle of length* p*, and let* g *be the permutation matrix with* $ge_i = e_{\nu(i)}$*. No map* s*,* ρ *or* R *as in (1), (2) or (3) of Theorem 4.2 satisfies the corresponding equivariance identity for this single* g*.*

The proof is in Appendix P.

For a finite group G acting trivially on K, the zeroth Tate cohomology group is $\widehat{H}^0(\mathsf{G}; K) = K/|\mathsf{G}|K$ [4, Ch. VI]; it is K if $|\mathsf{G}| = 0$ in K and 0 otherwise.

Proposition 4.5 (Tate classes)**.** *Let* H *act on* I*, let* $k \geq 1$*, and let* $T \subseteq I$ *be a* k*-subset. For a bijection* $f\colon [k] \to T$ *put* $e_f := e_{f(1)} \otimes \cdots \otimes e_{f(k)}$*. Fix a bijection* $f_T\colon [k] \to T$*, put* $\eta_T := e_{f_T(1)} \wedge \cdots \wedge e_{f_T(k)}$*, let* $U_T \subseteq \otimes^k_K V$ *be the span of the tensors* e_f *over the bijections* $f\colon [k] \to T$*, put* $\mathcal{N}_T := U_T \cap \ker \wedge^k$*, and let* $\chi_T\colon \mathsf{H}_T \to \{1, -1\}$ *be the homomorphism with* $\Lambda^k_K h(\eta_T) = \chi_T(h)\eta_T$*. Let* H_T *act on* U_T *by* $h \cdot z := \chi_T(h) h^{\otimes k} z$*. Write* $H^1(\mathsf{H}_T; \mathcal{N}_T)$ *for the space of maps* $c\colon \mathsf{H}_T \to \mathcal{N}_T$ *with* $c(hh') = c(h) + h \cdot c(h')$ *for* $h, h' \in \mathsf{H}_T$*, modulo the maps* $h \mapsto h \cdot n - n$ *with* $n \in \mathcal{N}_T$*, and for* $a \in K$ *let* $\delta_T(a) \in H^1(\mathsf{H}_T; \mathcal{N}_T)$ *be the class of* $h \mapsto a(h \cdot e_{f_T} - e_{f_T})$*.*

(1) $\delta_T(1) = 0$ *if and only if some* $z \in U_T$ *satisfies* $\wedge^k z = \eta_T$ *and* $h^{\otimes k} z = \chi_T(h) z$ *for every* $h \in \mathsf{H}_T$*.*

(2) $\ker \delta_T = |\bar{\mathsf{H}}_T| K$*. Hence* δ_T *induces an injective linear map from* $\widehat{H}^0(\bar{\mathsf{H}}_T; K) = K/|\bar{\mathsf{H}}_T|K$ *into* $H^1(\mathsf{H}_T; \mathcal{N}_T)$ *that sends the class of* 1 *to* $\delta_T(1)$*.*

(3) *An* H*-equivariant linear section of* $\wedge^k\colon \otimes^k_K V \to \Lambda^k_K V$ *exists if and only if* $\delta_T(1) = 0$ *for every* k*-subset* T *of* I*.*

The proof is in Appendix N. By (2) and (3), an H-equivariant section exists if and only if the Tate classes $[1] \in \widehat{H}^0(\bar{\mathsf{H}}_T; K)$ vanish for all k-subsets T.

A *lift* is, as in Lemma 3.4, a bounded k-linear map $Q\colon \mathcal{L}(E,D)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ with $Q(u,\dots,u) = \Lambda^k_\pi u$. For $u_1,\dots,u_k \in \mathcal{L}(E,D)$ define $\Lambda_{\mathrm{sym}}(u_1,\dots,u_k) \in \mathcal{L}(\Lambda^k_\pi E, \Lambda^k_\pi D)$ by

$$\Lambda_{\mathrm{sym}}(u_1,\dots,u_k)(x_1\wedge\cdots\wedge x_k) := \sum_{\sigma\in S_k} u_{\sigma(1)}x_1 \wedge \cdots \wedge u_{\sigma(k)}x_k. \tag{4.4}$$

The right side is alternating in (x_b), since the terms for σ and $\sigma\circ(a\ b)$ cancel when $x_a = x_b$; by Proposition 3.1, Λ_{sym} is a bounded symmetric k-linear map with $\Lambda_{\mathrm{sym}}(u,\dots,u) = k!\,\Lambda^k_\pi u$. If $k! \neq 0$, $\Lambda_{\mathrm{sym}}/k!$ is a symmetric lift, the normalized alternatization of Remark B.15.

Proposition 4.6 (forced asymmetry)**.** *Let* E *and* D *be normed spaces.*

(1) *Every lift* Q *satisfies* $\sum_{\sigma\in S_k} Q(u_{\sigma(1)},\dots,u_{\sigma(k)}) = \Lambda_{\mathrm{sym}}(u_1,\dots,u_k)$.

Suppose now that $k! = 0$ *in* K*.*

(2) *If* E *has* k *linearly independent bounded functionals and* $\widehat{\Lambda}^k_\pi D \neq 0$*, no lift is symmetric.*

(3) *No linear retraction* $\rho\colon \otimes^k_\pi D \to \widehat{\Lambda}^k_\pi D$ *of* alt^k_D *satisfies* $\rho\circ\tau_\sigma = \mathrm{sgn}(\sigma)\rho$ *for all* $\sigma \in S_k$*, unless* $\Lambda^k_\pi D = 0$*.*

(4) *If* E *has* k *linearly independent bounded functionals, no linear section* $s\colon \Lambda^k_\pi E \to \widehat{\otimes}^k_\pi E$ *of* $\widehat{\wedge}^k_E$ *takes values in the sign-isotypic tensors* $\{y : \tau_\sigma y = \mathrm{sgn}(\sigma) y \text{ for all } \sigma\}$*.*

The proof is in Appendix P.

The functional hypothesis in (2) is needed: for $k = p$, $K = \mathbb{F}_p(t)$ and the space $E \subseteq A = \mathbb{F}_p((t))[\epsilon]/(\epsilon^p)$ of Lemma H.5, every bounded functional on E is zero and $\mathcal{L}(E,E) = K\cdot\mathrm{id}$, while $\Lambda^p_\pi E \neq 0$ by Proposition 3.1, since the determinant in the basis $(\epsilon^i)_{0\le i<p}$ of A is a bounded alternating map on E^p with value 1 at that basis. So $Q(c_1\mathrm{id},\dots,c_p\mathrm{id}) = c_1\cdots c_p\,\mathrm{id}$ is a symmetric lift.

5. Splittings that exist

5.1. Splittings for one value space.

Proposition 5.1 (splitting the alternating inclusion)**.** *If the inclusion* $j_{E,F}\colon \mathrm{Alt}^k(E;F) \hookrightarrow \mathrm{Mult}^k(E;F)$ *has a bounded linear retraction* ρ*, then* $A^k_{E,D;F}$ *has a bounded* k*-linear lift of norm at most* $\|\rho\|$*, for every normed* D*. Such a lift also exists in each of the following cases:*

(1) $k! \neq 0$ *in* K*;*

(2) *either* E *or* D *has a finite algebraic basis with continuous coordinate functionals.*

No completeness hypothesis is required.

Proof. For a retraction, $Q(u_1,\dots,u_k)(m) = \rho\big(m\circ(u_1,\dots,u_k)\big)$ is a lift: the mixed composition has norm at most $\|m\|\prod_i\|u_i\|$, and its diagonal is already alternating.

If $k! \neq 0$, the lift

$$Q(u_1,\dots,u_k)(m)(x) = \frac{1}{k!}\sum_{\sigma\in S_k} m(u_{\sigma(1)}x_1,\dots,u_{\sigma(k)}x_k)$$

vanishes when two x-arguments agree, by pairing the permutations that exchange their positions; on the diagonal all $k!$ summands coincide; and its norm is at most $k!/|k!|$, the numerator counting summands.

For continuous coordinates $(e_i, e_i^*)_{1\le i\le d}$ on E, sum over increasing words $i_1 < \cdots < i_k$:

$$Q(u_1,\dots,u_k)(m)(x) = \sum_{i_1<\cdots<i_k} \det\big(e_{i_a}^*(x_b)\big)_{a,b} m(u_1 e_{i_1},\dots,u_k e_{i_k}).$$

If the coordinates are on D, replace the summand by $\det(e_{i_a}^*(u_a x_b))_{a,b}\, m(e_{i_1},\dots,e_{i_k})$. Equal columns give alternation, and the exterior-basis expansion gives the diagonal identity. Each is a finite sum of norm at most $k!\sum_{i_1<\cdots<i_k}\prod_a \|e_{i_a}^*\|\|e_{i_a}\|$, empty if $k > d$. □

Over a complete field, finite-dimensional spaces have continuous coordinates (Lemma A.3), so cases (1) and (2) contain Bourbaki's hypotheses, characteristic zero and finite rank [3, n° 7.8]; over incomplete K finite rank does not suffice, since the a-coordinate on $K + Ka \subseteq \widehat{K}$, $a \in \widehat{K} \setminus K$, is discontinuous.

A nonarchimedean normed space is *spherically complete* if every nonempty family of pairwise-intersecting closed balls has a common point. In a nonarchimedean metric, intersecting balls are nested.

Theorem 5.2 (spherically complete targets)**.** *Suppose K is nonarchimedean and F is nonarchimedean and spherically complete. Then, for every normed E, the space $\mathrm{Alt}^k(E;F)$ is nonarchimedean and spherically complete, and its inclusion into $\mathrm{Mult}^k(E;F)$ has a contractive linear retraction. Thus $A^k_{E,D;F}$ is a continuous polynomial for every normed E, D and every k. Neither the field nor the source spaces need be complete.*

The proof is in Appendix P. An equivalent nonarchimedean spherically complete norm on F suffices, since equivalent norms preserve analyticity.

Corollary 5.3 (discretely valued fields)**.** *Suppose K is nonarchimedean with $|K^\times| = r^{\mathbb{Z}}$ for some $0 < r < 1$. Every Banach target F admitting an equivalent nonarchimedean norm makes $A^k_{E,D;F}$ analytic in every degree and for all normed E, D. The field itself need not be complete.*

Proof. Round the equivalent nonarchimedean norm upwards to $r^{\mathbb{Z}}$, with value zero at zero. The new norm lies between the old one and r^{-1} times it, and satisfies the norm axioms, since rounding is monotone and commutes with multiplication by $r^{\mathbb{Z}}$. It is complete with nonzero distances in $r^{\mathbb{Z}}$, hence spherically complete by Lemma D.1. Apply Theorem 5.2 and transport the resulting lift back through the equivalent norms. □

Changing the target category. Over nonarchimedean K, the maximum in place of the sum in the projective seminorm gives a completed exterior power $\widehat{\Lambda}^k_{\pi,\mathrm{na}}E$ representing $\mathrm{Alt}^k(E;F)$ for complete nonarchimedean F (the universal-property proof of Proposition 3.1; [26, §4]). Over a discretely valued field, $u \mapsto \widehat{\Lambda}^k_{\pi,\mathrm{na}}u$ is analytic: it is $A^k_{E,D;F}$ for $F = \widehat{\Lambda}^k_{\pi,\mathrm{na}}D$, evaluated at the universal alternating map, so Corollary 5.3 applies. The obstructions of Section 7 concern the sum construction, whose target has no equivalent nonarchimedean norm.

In the next theorem and its proof, K is complete and nonarchimedean, and q_Y, Y^u and $j_Y : Y \to Y^u$ are the seminorm, the ultrametric envelope and its canonical map of a normed space Y, constructed in the proof of Proposition I.1.

Theorem 5.4 (ultrametric values)**.** *Let F be a Banach space with a nonarchimedean norm, and suppose that E^u or D^u is a bounded linear retract of $c_0(I,K)$ for some set I: there are bounded linear maps ι into $c_0(I,K)$ and ι' out of it with $\iota'\iota = \mathrm{id}$. Then $A^k_{E,D;F}$ has a bounded lift of norm at most $(\|\iota\|\|\iota'\|)^k$, and is therefore a continuous polynomial. If E or D is of countable type (Appendix I), there are such lifts of norm at most t^{-k} for every $0 < t < 1$.*

The proof is in Appendix P. If $\operatorname{char} K = p > 0$ and $k \geq p$, both hypotheses matter: Theorem 7.2(2) gives a failure with $F = K$ if K is not spherically complete; for $K = \kappa((X))$ and $E = D = \ell^\infty(\mathbb{N}, K) \cong c_0(I, K)$ [26, §1, Proposition 1], Theorem C.1 gives one with the target F_1, which has no equivalent nonarchimedean norm, so this pair is not good for Banach values (Corollary 7.6).

5.2. **Source and target splittings.** By Proposition 3.5, a source splitting (S^k_E) or a target splitting (R^k_D) makes (E, D) good for Banach values in degree k. A Banach space Y is *projective* if, for every bounded linear surjection $\psi\colon Y_1 \to Y_2$ of Banach spaces, every bounded linear $T\colon Y \to Y_2$ equals $\psi \circ T'$ for a bounded linear $T'\colon Y \to Y_1$. For a totally ordered set J, let $J^k_<$ be the set of $i \in J^k$ with $i_1 < \cdots < i_k$, and let $(e_i)_{i\in J}$ and $(\delta_i)_{i\in J^k_<}$ be the unit vectors of $\ell^1(J, K)$ and of $\ell^1(J^k_<, \widehat{K})$.

Proposition 5.5 (projective sources)**.** *Let E be a normed space and $k \geq 1$.*

1. *For every totally ordered set J, the map $\delta_i \mapsto e_{i_1} \wedge \cdots \wedge e_{i_k}$ extends to an isometric isomorphism of $\ell^1(J^k_<, \widehat{K})$ onto $\widehat{\Lambda}^k_\pi \ell^1(J, K)$.*
2. *If $\widehat{\Lambda}^k_\pi E$ is projective, then (S^k_E) holds.*
3. *If E is a bounded linear retract of $\ell^1(J, K)$ for some set J, then (S^n_E) holds for every $n \geq 1$.*

The proof is in Appendix P.

Let Σ be a totally ordered set. An *interval* of Σ is a subset I containing every t with $t_1 < t < t_2$ for some $t_1, t_2 \in I$. For a linearly independent family $(e_t)_{t\in\Sigma}$ in E with span V, write $x(t)$ for the coefficients of $x \in V$ and $\mathcal{P}_I x = \sum_{t\in I} x(t) e_t$.

Theorem 5.6 (lower estimates)**.** *Let K be any nontrivially normed field and E a normed space with a linearly independent family $(e_t)_{t\in\Sigma}$ whose span V is dense. Suppose that for some $0 < r < 2$ and $C \geq 1$,*

$$\Big(\sum_{i=1}^{N} \|\mathcal{P}_{I_i} x\|^r\Big)^{1/r} \leq C\|x\| \qquad (x \in V,\ I_1, \ldots, I_N \text{ pairwise disjoint intervals}). \tag{5.1}$$

Put $C_\sharp = 2C^2/(1 - 2^{1-2/r})$.

1. *The linear map $\mathcal{P}_<$ on $V \otimes V$ with $\mathcal{P}_<(e_t \otimes e_{t'}) = e_t \otimes e_{t'}$ for $t < t'$ and $\mathcal{P}_<(e_t \otimes e_{t'}) = 0$ otherwise satisfies $\|\mathcal{P}_<(x \otimes y)\| \leq C_\sharp \|x\|\|y\|$ for $x, y \in V$. It therefore extends to a bounded linear map $E \otimes_\pi E \to E \widehat{\otimes}_\pi E$ of norm at most $C_\sharp$.*
2. *For every $k \geq 1$, (S^k_E) and (R^k_E) hold, with $\|s\| \leq k!\, C_\sharp^{k-1}$ and $\|\rho\| \leq C_\sharp^{k-1}$. If $E = V$, they hold without completion.*
3. *For every normed D and every k, the pairs (E, D) and (D, E) are good for Banach values, and they are good if $E = V$.*

The proof is in Appendix J.

A normed space E has a *lower r-estimate* if it has a family $(e_t)_{t\in\Sigma}$ as in Theorem 5.6 satisfying (5.1) for some $C \geq 1$. Examples with $C = 1$ are $\ell^r(\Sigma, K)$, $1 \leq r < 2$, with its unit vectors, its subspace of finitely supported vectors, and, for $r_1, r_2 \geq 1$ with $r = \max(r_1, r_2) < 2$, the space $\ell^{r_1}(\mathbb{N}; \ell^{r_2}(\mathbb{N}, K))$ ordered lexicographically, by Minkowski's inequality if $r_1 \leq r_2$ and by $\ell^{r_2} \subseteq \ell^{r_1}$ otherwise. Conclusions (2), with norms multiplied by $(\|\iota\|\|\iota'\|)^k$, and (3) pass to every Y with bounded $\iota\colon Y \to E$ and $\iota'\colon E \to Y$, $\iota'\iota = \mathrm{id}_Y$: the maps $\iota'^{\otimes k} \circ s \circ \Lambda^k_\pi \iota$ and $\widehat{\Lambda}^k_\pi \iota' \circ \rho \circ \iota^{\otimes k}$ satisfy (S^k_Y) and (R^k_Y), by naturality.

Proposition 5.7 (ordering splittings)**.** *Let E be a normed space with a linearly independent family $(e_t)_{t\in\Sigma}$ whose span V is dense, and suppose that the map $\mathcal{P}_<$ of Theorem 5.6(1) satisfies $\|\mathcal{P}_<(x\otimes y)\|\le C\|x\|\|y\|$ for $x,y\in V$. Let $n\ge 1$, and let $\mathcal{P}^{(n)}$ and R_n be the linear maps on $\otimes^n V$ that send $e_{t_1}\otimes\cdots\otimes e_{t_n}$ to itself if $t_1<\cdots<t_n$, respectively to the sum O_t of the tensors $e_{\tilde{t}_1}\otimes\cdots\otimes e_{\tilde{t}_n}$ over the distinct rearrangements $\tilde{t}$ of t if $t_1\le\cdots\le t_n$, and to 0 otherwise.*

1. *$\mathcal{P}^{(n)}$ extends to a bounded linear map $\otimes^n_\pi E\to\widehat{\otimes}^n_\pi E$ of norm at most C^{n-1}.*
2. *$s=\mathcal{P}^{(n)}\circ\operatorname{alt}^n_E$ and $\rho=\widehat{\wedge}^n_E\circ\mathcal{P}^{(n)}$ satisfy (S^n_E) and (R^n_E), with $\|s\|\le n!\,C^{n-1}$ and $\|\rho\|\le C^{n-1}$. If $E=V$, both hold without completion.*
3. *R_n extends to a projection of $\widehat{\otimes}^n_\pi E$ onto $\widehat{\Gamma}^n_\pi E$, so (Γ^n_E) holds. Its norm is at most the sum of $n!\,(c_1!\cdots c_m!)^{-1}C^{m-1}(1+2C)^{n-m}$ over the compositions $c=(c_1,\dots,c_m)$ of n into positive parts.*

The proof is in Appendix J.

By Theorem 5.6(1), the hypothesis holds with $C=C_\sharp$ for every family satisfying (5.1).

In the rest of this subsection K is nonarchimedean. A linearly independent family $(e_t)_{t\in\Sigma}$ in E whose span V is dense is *ℓ^2-dominated* if $(\sum_t|x(t)|^2)^{1/2}\le C_2\|x\|$ for $x\in V$ and some constant C_2. For $1\le r<\infty$, an *r-norm* is a norm ν with $\nu(y+z)^r\le\nu(y)^r+\nu(z)^r$; a normed space D has an *equivalent r-norm* if it carries an r-norm ν with $a\nu(y)\le\|y\|\le b\nu(y)$ for constants $0<a\le b$.

Proposition 5.8 (r-norms)**.** *Let $1\le r<\infty$. A normed space D has an equivalent r-norm if and only if there is a constant C with*

$$\Big\|\sum_t \xi_t y_t\Big\|\le C\Big(\sum_t|\xi_t|^r\Big)^{1/r}\max_t\|y_t\| \tag{5.2}$$

for all finite families (ξ_t) in K and (y_t) in D with a common index set. The proof is in Appendix J.

Theorem 5.9 (relative ordering splittings)**.** *Let E be a normed space with an ℓ^2-dominated linearly independent family $(e_t)_{t\in\Sigma}$ whose span V is dense and with $\sup_t\|e_t\|<\infty$.*

1. *For every $r\in(2,\infty)$, the coordinate map $V\to\ell^r(\Sigma,\widehat{K})$, $x\mapsto(x(t))_{t\in\Sigma}$, extends to a bounded linear map $w\colon E\to\ell^r(\Sigma,\widehat{K})$, which is S^n-split (Definition 3.12) for every $n\ge1$.*
2. *If D has an equivalent r-norm for some $r\in(2,\infty)$, then (E,D) is good for Banach values in every degree, and good in every degree if $E=V$.*

The proof is in Appendix J.

Corollary 5.10. *Let E be as in Theorem 5.9, and let D have an equivalent nonarchimedean norm. Then (E,D) is good for Banach values in every degree, and good in every degree if $E=V$. The extension $E\to c_0(\Sigma,\widehat{K})$ of the coordinate map is S^n-split for every $n\ge1$.*

The proof is in Appendix P. For example, the sources $\ell^r(\Sigma,K)$, $1\le r\le2$, are covered, with the targets c_0, ℓ^∞, spaces with an equivalent nonarchimedean norm, and ℓ^r, $r>2$, which is r-normed by its own norm, since $|\alpha+\beta|^r\le\max(|\alpha|,|\beta|)^r\le|\alpha|^r+|\beta|^r$.

5.3. **Parameter splittings.** A *parameter splitting* in degree n, a bounded projection of $\widehat{\otimes}^n_\pi P$ onto $\widehat{\Gamma}^n_\pi P$, exists if and only if P is polarizable in degree n (Definition 3.8).

Theorem 5.11 (finite-coordinate parameters)**.** *Let $j\colon W\to Z$ be a closed linear isometry. If P is boundedly linearly isomorphic to K^d and $U\subseteq P$ is open, then $f\colon U\to W$ is analytic if and*

only if jf is analytic. For the maximum norm on K^d, ambient coefficients b_n can be replaced by W-valued coefficients q_n with

$$\|q_n\| \le d^n \|b_n\| \quad (n \ge 1).$$

If Z is nonarchimedean, the factor d^n can be replaced by 1.

Proof. Theorem 2.3 puts every homogeneous diagonal in W. Expand it in coordinates as $b_n(h,\dots,h) = \sum_{|\alpha|=n} c_\alpha h^\alpha$. Every c_α lies in W, by polynomial uniqueness over the infinite field K in the algebraic quotient Z/W. For each α, let $i_1(\alpha) \le \cdots \le i_n(\alpha)$ be the coordinate word with these multiplicities, and set

$$q_n(h_1,\dots,h_n) = \sum_{|\alpha|=n} c_\alpha \prod_{a=1}^{n} (h_a)_{i_a(\alpha)}.$$

Before grouping there are exactly d^n ordered terms, so the triangle inequality gives the bound, and two maximum estimates give the nonarchimedean refinement. The diagonals agree, and the exponential bounds preserve a positive radius in Theorem 2.3. The case $d = 0$ is immediate. □

Over an incomplete field, finite dimension of the base does not suffice: for $K = \mathbb{F}_p(t)$ and $k \ge p$, the end of Appendix H gives a base of dimension $p+1$ without continuous coordinates and a nonanalytic alternating atlas.

Theorem 5.12 (summable parameter spaces)**.** *Let $j\colon W \to Z$ be a closed linear isometry.*

1. *Suppose K and Z are nonarchimedean and W is complete. For $U \subseteq c_0(I,K)$ open, analyticity of jf implies analyticity of $f\colon U \to W$. Coefficients can be lifted without increasing their norms.*
2. *Suppose Z is complete. Let $a\colon H \to W$ have an ambient bounded homogeneous polynomial representation of fixed degree d, that is, $ja(h) = B(h,\dots,h)$ for a bounded d-linear $B\colon H^d \to Z$ on a normed space H. For every analytic $\gamma\colon U \to H$, with $U \subseteq \ell^1(I,K)$ open, the composite $a\gamma$ is analytic.*

Here c_0 carries the supremum norm and ℓ^1 the ordinary sum norm. The field need not be complete. Each analyticity assertion remains valid for a parameter space P that is a bounded linear retract of the indicated space V, with bounded linear $i\colon P \to V$, $r\colon V \to P$ and $ri = \mathrm{id}_P$; in (1) the lifted coefficients then satisfy $\|q_n\| \le (\|r\|\|i\|)^n \|b_n\|$.

The proof is in Appendix P. Theorems 5.11 and 5.12 give analyticity at every point of an open set, hence the C^ω conclusion (see Conventions).

Proposition 5.13 (parameter splittings of ℓ^1)**.** *Let $n \ge 1$.*

1. *For every set J, $\ell^1(J,K)$ is polarizable in degree n, by a projection of norm at most $n!$.*
2. *Every normed space with finite continuous coordinates is polarizable in degree n.*

The proof is in Appendix P.

Proposition 5.14 (degree two)**.** *For every normed space P, $\mathcal{N}_P^2 = \widehat{\Gamma}_\pi^2 P$. Hence $\mathcal{S}_P^2$ and $\mathcal{G}_P^2$ coincide up to the isometry $\widehat{\Lambda}_\pi^2 P \cong \widehat{\otimes}_\pi^2 P/\mathcal{N}_P^2$, and (S_P^2) holds if and only if P is polarizable in degree 2. A section s gives a projection onto $\widehat{\Gamma}_\pi^2 P$ of norm at most $1 + \|s\|$, and a projection R_2 onto $\widehat{\Gamma}_\pi^2 P$ gives a section of norm at most $1 + \|R_2\|$.*

The proof is in Appendix P.

5.4. **The bundle theorem.**

Corollary 5.15. *Let M be an analytic manifold over K, with normed model space P. Fiberwise alternating maps give an analytic bifunctor on analytic normed vector bundles and their operator-valued analytic morphisms in each of the following settings:*

Base model	*Hypotheses on the typical fibers*
Arbitrary P	*$k! \neq 0$ in K; or K is nonarchimedean and the target fibers admit equivalent nonarchimedean spherically complete norms; or the source fibers have finite continuous coordinates.*
Finite continuous coordinates	*Arbitrary normed fibers.*
$c_0(I, K)$, or a bounded linear retract	*K nonarchimedean; complete nonarchimedean target fibers.*
$\ell^1(I, K)$, or a bounded linear retract	*Complete target fibers.*

The fiber hypotheses apply to both pairs of bundles at the endpoints of a morphism. Each alternative in the first row specifies a separate setting. At every finite smoothness order and at C^∞, no additional hypothesis on K, P or the fibers is needed.

Proof. The fiber results give a bounded k-linear lift Q of A, and $(u_1, \dots, u_k, g) \mapsto \big(m \mapsto g \circ Q(u_1, \dots, u_k)(m)\big)$ is a bounded $(k+1)$-linear lift of the joint action B, which is therefore analytic. The parameter results apply directly to the ambient polynomial along transition families and local morphism families; the ambient multilinear and operator spaces are complete when the final value-space fiber is complete, and nonarchimedean when it is nonarchimedean. Theorem 2.1 supplies the bundle construction in every case, and Lemma 2.4 the smooth assertion. □

Fields of characteristic zero, spherically complete or not, belong to the first row in every finite degree.

Good pairs of fibers. If (E, D) is good for Banach values in degree k for all E, D in a class $\mathcal{C}$, then Theorem 2.1 applies over every analytic manifold to bundles with source fibers in $\mathcal{C}$ and Banach value fibers: the joint action is polynomial in the precomposition variable (Lemma 3.4, Proposition 2.5) and linear in the other. Such classes are the spaces with a lower r-estimate, $r < 2$ (Theorem 5.6), and the symmetric spaces with c_{00} dense and $\mathcal{P}_<$ bounded (Propositions 5.7(2), 3.5(1); under (L), Theorem 8.1). Over complete nonarchimedean K, Theorem 5.4 gives the same for source fibers of countable type and nonarchimedean Banach value fibers.

6. Degrees, decompositions and permanence

6.1. **The degrees in which goodness can fail.**

For $a, b \geq 1$ and $n = a + b$, let $\wedge_E^{a,b} \colon \Lambda_\pi^a E \otimes_\pi \Lambda_\pi^b E \to \Lambda_\pi^n E$ be the wedge product, $z \otimes w \mapsto z \wedge w$, and define the shuffle coproduct by

$$\mathrm{sh}_E^{a,b}(x_1 \wedge \cdots \wedge x_n) = \sum_{|S|=a} \epsilon(S)\, x_S \otimes x_{S^c}, \tag{6.1}$$

where S runs over the a-element subsets of $\{1, \dots, n\}$, x_S is the wedge of the x_i, $i \in S$, in increasing order, and $\epsilon(S)$ is the sign of the permutation of $\{1, \dots, n\}$ that lists S in increasing order followed by S^c in increasing order. We write $\binom{n}{a}$ also for its image in K.

Lemma 6.1 (the Hopf identity)**.** *The map $\wedge_E^{a,b}$ is contractive, and $\mathrm{sh}_E^{a,b}$ is a well-defined linear map of norm at most $\binom{n}{a}$, in every characteristic. Both are natural in bounded operators, and*

$$\wedge_E^{a,b} \circ \mathrm{sh}_E^{a,b} = \tbinom{n}{a}\,\mathrm{id}. \tag{6.2}$$

Moreover $\wedge_E^n = \wedge_E^{a,b} \circ (\wedge_E^a \otimes \wedge_E^b)$ and $\mathrm{alt}_E^n = (\mathrm{alt}_E^a \otimes \mathrm{alt}_E^b) \circ \mathrm{sh}_E^{a,b}$, where $\otimes_\pi^n E$ is identified with $\otimes_\pi^a E \otimes_\pi \otimes_\pi^b E$.

The proof is in Appendix P.

Theorem 6.2 (Lucas step)**.** *Let $n = a + b$ with $a, b \geq 1$ and $\binom{n}{a} \neq 0$ in K, let F be normed, and let $G = \mathrm{Alt}^b(D; F)$. If $A^a_{E,D;G}$ and $A^b_{E,D;F}$ are analytic, so is $A^n_{E,D;F}$. In particular, if $n \neq 0$ in K and $A^{n-1}_{E,D;F}$ is analytic, so is $A^n_{E,D;F}$.*

The proof is in Appendix P.

Corollary 6.3 (the good degrees)**.** *Let $\operatorname{char} K = p > 0$, and let $\mathfrak{D}$ be one of the following sets of degrees $k \geq 1$: the set $\mathfrak{D}(E, D)$ of those in which (E, D) is good; the set $\mathfrak{D}_B(E, D)$ of those in which it is good for Banach values; or the set $\mathfrak{D}_K(E, D)$ of those for which $A^k_{E,D;K}$ is analytic.*

1. *$\{1, \dots, p-1\} \subseteq \mathfrak{D}$, and $a + b \in \mathfrak{D}$ whenever $a, b \in \mathfrak{D}$ and the addition of a and b in base p has no carry.*
2. *If $\mathfrak{D} \neq \{1, 2, \dots\}$, its least missing element is a power of p. Thus $\mathfrak{D} = \{1, 2, \dots\}$ if and only if $\mathfrak{D}$ contains every power of p.*
3. *Let F be normed, and suppose that $A^n_{E,D;F}$ is not analytic for some n; let n_F be the least such n. Then n_F is divisible by the least degree in which (E, D) is not good.*

The proof is in Appendix P.

Theorem 6.4 (hyperplane step)**.** *Give $E \oplus K$ and $D \oplus K$ the maximum norms, let $k \geq 2$, and let F be normed. If $A^k_{E\oplus K, D\oplus K;F}$ is analytic, then so is $A^{k-1}_{E,D;F}$.*

The proof is in Appendix P.

Theorem 6.5 (no gaps for large duals)**.** *Let $\operatorname{char} K = p > 0$, and let E and D be normed spaces whose continuous duals E^* and D^* are infinite-dimensional. Then each of $\mathfrak{D}(E, D)$ and $\mathfrak{D}_B(E, D)$ equals $\{1, \dots, p^h - 1\}$ for a unique $h \geq 1$, or equals $\{1, 2, \dots\}$.*

If $E \cong E \oplus K$ and $D \cong D \oplus K$ by bounded linear isomorphisms, the same holds for $\mathfrak{D}_K(E, D)$: composing with these isomorphisms on both sides shows that $A^k_{E,D;K}$ and $A^k_{E\oplus K, D\oplus K;K}$ are analytic together, so Theorem 6.4 makes $\mathfrak{D}_K(E, D)$ closed under $k \mapsto k - 1$ for $k \geq 2$, and Corollary 6.3(2) identifies its first missing element.

Theorem 6.6 (degree p for one space)**.** *Let $\operatorname{char} K = p > 0$, and let Y be a normed space. If (S^p_Y) holds, then (S^n_Y) holds for every $n \geq 1$. If (R^p_Y) holds, then (R^n_Y) holds for every $n \geq 1$. Both statements also hold without completion.*

The proofs of Theorems 6.5 and 6.6 are in Appendix L.

Corollary 6.7. *Let $\operatorname{char} K = p > 0$, and let E and D be normed spaces. If (S^p_E) holds, or if (R^p_D) holds, then (E, D) is good for Banach values in every degree. If that condition holds without completion, then (E, D) is good in every degree.*

Proof. By Theorem 6.6 the condition holds in every degree, and Proposition 3.5 turns it into goodness in every degree. □

Split operators. For bounded linear $w\colon E \to G$ and $n \geq 1$, a *section for w in degree n* is a map s as in Definition 3.12 with $\widehat{\wedge}^n_G \circ s = \Lambda^n_\pi w$, and a *retraction for w in degree n* is a map ρ with $\rho \circ \mathrm{alt}^n_E = \Lambda^n_\pi w$. In degree 0 both are id_K, $\otimes^0_\pi E = K$, and $\wedge^{0,b}$, $\wedge^{a,0}$, $\mathrm{sh}^{0,b}$, $\mathrm{sh}^{a,0}$ are the canonical identifications. For bounded linear $w_1, w_2\colon E \to G$ and $x_1, \dots, x_n \in E$, the term of $(w_1 + w_2)x_1 \wedge \cdots \wedge (w_1 + w_2)x_n$ with w_1 in the positions of $S \subseteq \{1, \dots, n\}$ is $\epsilon(S)\,(w_1 x)_S \wedge (w_2 x)_{S^c}$, with $(w_1 x)_S$ formed as x_S from the vectors $w_1 x_i$; so (6.1) gives

$$\Lambda^n_\pi(w_1 + w_2) = \sum_{a=0}^{n} \wedge^{a,n-a}_G \circ \left(\Lambda^a_\pi w_1 \otimes \Lambda^{n-a}_\pi w_2\right) \circ \mathrm{sh}^{a,n-a}_E \,. \tag{6.3}$$

By (6.3) and Lemma 6.1, sections $s^{(l)}_a$ for w_l in all degrees $a \leq n$ give the section $\sum_a (s^{(1)}_a \otimes s^{(2)}_{n-a}) \circ \mathrm{sh}^{a,n-a}_E$ for $w_1 + w_2$ in degree n, and retractions $\rho^{(l)}_a$ give the retraction $\sum_a \wedge^{a,n-a}_G \circ (\rho^{(1)}_a \otimes \rho^{(2)}_{n-a})$. By Lemma 6.1, if $a, b \geq 1$ and $\binom{a+b}{a} \neq 0$ in K, sections s_a, s_b for w in degrees a, b give the section $\binom{a+b}{a}^{-1}(s_a \otimes s_b) \circ \mathrm{sh}^{a,b}_E$ in degree $a + b$, and retractions ρ_a, ρ_b give the retraction $\binom{a+b}{a}^{-1} \wedge^{a,b}_G \circ (\rho_a \otimes \rho_b)$; this is the Lucas step for split operators. Every operator has sections and retractions in the degrees n with $n! \neq 0$ in K (the paragraph after Definition 3.12); so if $\mathrm{char}\, K = p > 0$, the degrees in which an operator is S-split, or R-split, contain $1, \dots, p-1$ and are closed under carry-free sums. The $\mathrm{S}^{\leq k}$-split maps $E \to G$ form a linear space, as do the $\mathrm{R}^{\leq k}$-split ones, and both are closed under composition with bounded linear maps on either side (Proposition 3.13(2)).

Theorem 6.8 (composition)**.** *Let $q, m \geq 1$ and $n = qm$, and suppose that the number $N_{q,m} = (qm)!/\big((q!)^m\, m!\big)$ of partitions of $\{1, \dots, n\}$ into m blocks of size q is nonzero in K; this holds, for instance, when q and m are powers of $p = \mathrm{char}\, K > 0$. Let $|N_{q,m}|$ be the absolute value of its image in K; elsewhere $N_{q,m}$ and $m!$ are real numbers.*

1. *If $w_1\colon E \to G_1$ has a section s_1 in degree q and $w_2\colon G_1 \to G_2$ has a section s_2 in degree m, then $w_2 w_1$ has a section s in degree n with $\|s\| \leq N_{q,m} \|s_1\|^m \|s_2\|^q / |N_{q,m}|$.*
2. *If $w_1\colon E \to G_1$ has a retraction ρ_1 in degree m and $w_2\colon G_1 \to G_2$ has a retraction ρ_2 in degree q, then $w_2 w_1$ has a retraction ρ in degree n with $\|\rho\| \leq (m!)^{q-1} \|\rho_2\|^m \|\rho_1\|^q / |N_{q,m}|$.*

If the given sections, respectively retractions, are without completion, so is the new one.

Corollary 6.9. *Let $\mathrm{char}\, K = p > 0$ and $h \geq 1$.*

1. *A composite of h operators that are S^p-split is S^n-split for every $n < p^{h+1}$, and a composite of h operators that are R^p-split is R^n-split for every $n < p^{h+1}$.*
2. *An operator that factors through a normed space Y with (S^p_Y) is S^n-split for every n, and an operator that factors through a normed space Y with (R^p_Y) is R^n-split for every n.*

Theorem 6.10 (degree p for polarizability)**.** *Let $\mathrm{char}\, K = p > 0$. If a normed space P is polarizable in degree p, then P is polarizable in every degree $n \geq 1$.*

The proofs of Theorems 6.8 and 6.10 and of Corollary 6.9 are in Appendix L.

6.2. **Relative splittings and decompositions.** Throughout this subsection E and D are normed spaces and $k \geq 1$. Since $\widehat{\Lambda}^n_\pi \widehat{D} = \widehat{\Lambda}^n_\pi D$ (the paragraph before Theorem 3.16), $\Lambda^n_\pi v$ maps $\Lambda^n_\pi E$ into $\widehat{\Lambda}^n_\pi D$ for every $v \in \mathcal{L}(E, \widehat{D})$.

Proposition 6.11 (relative splittings)**.** *Let G be a normed space.*

(1) *Let $w\colon E \to G$ have a section s in degree k, and let $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(G,\widehat{D})$ be bounded linear with $\varepsilon(u)\circ w = u$ for every $u \in \mathcal{L}(E,D)$. Then (E,D) is good for Banach values in degree k, with the lift*

$$Q(u_1,\dots,u_k) = \widehat{\wedge}^k_{\widehat{D}} \circ \big(\varepsilon(u_1)\otimes\cdots\otimes\varepsilon(u_k)\big)\circ s$$

of norm at most $\|\varepsilon\|^k\|s\|$.

(2) *Let $w'\colon G \to \widehat{D}$ have a retraction ρ in degree k, and let $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(E,G)$ be bounded linear with $w'\circ\varepsilon(u) = u$ for every $u \in \mathcal{L}(E,D)$. Then (E,D) is good for Banach values in degree k, with the lift*

$$Q(u_1,\dots,u_k) = \rho \circ \big(\varepsilon(u_1)\otimes\cdots\otimes\varepsilon(u_k)\big)\circ \mathrm{alt}^k_E$$

of norm at most $k!\,\|\varepsilon\|^k\|\rho\|$.

(3) *If in (1) the section s takes values in $\otimes^k_\pi G$ and ε takes values in $\mathcal{L}(G,D)$, or if in (2) the retraction ρ takes values in $\Lambda^k_\pi D$, then (E,D) is good in degree k.*

Proof. Both maps Q are bounded and k-linear, with the stated norms. In (1), naturality of $\wedge^k$, extended to completions, gives $Q(u,\dots,u) = \widehat{\Lambda}^k_\pi\varepsilon(u)\circ\widehat{\wedge}^k_G\circ s = \Lambda^k_\pi\varepsilon(u)\circ\Lambda^k_\pi w = \Lambda^k_\pi u$. In (2), naturality of alt^k gives $Q(u,\dots,u) = \rho\circ\mathrm{alt}^k_G\circ\Lambda^k_\pi\varepsilon(u) = \Lambda^k_\pi w'\circ\Lambda^k_\pi\varepsilon(u) = \Lambda^k_\pi u$. Under the hypotheses of (3) both lifts take values in $\mathcal{L}(\Lambda^k_\pi E,\Lambda^k_\pi D)$. Lemma 3.4 concludes. □

Definition 6.12. (1) A *source-side relative splitting* of degree k for (E,D) is a triple (G,w,ε) as in Proposition 6.11(1): $w\colon E\to G$ is S^k-split and $\varepsilon(u)\circ w = u$. A *target-side relative splitting* of degree k is a triple (G,w',ε) as in Proposition 6.11(2): $w'\colon G\to\widehat{D}$ is R^k-split and $w'\circ\varepsilon(u) = u$.

(2) A *piece* for (E,D) is a tuple (G,G',w,w',ε) of normed spaces G and G', bounded linear maps $w\colon E\to G$ and $w'\colon G'\to\widehat{D}$, and a bounded linear map $\varepsilon\colon\mathcal{L}(E,D)\to\mathcal{L}(G,G')$. It is of *source type* in degree k if w is $\mathrm{S}^{\le k}$-split, and of *target type* in degree k if w' is $\mathrm{R}^{\le k}$-split.

(3) A *splitting decomposition* of degree k is a finite family of pieces $(G_i,G_i',w_i,w_i',\varepsilon_i)$, $1\le i\le m$, each of source type or of target type in degree k, such that

$$u = \sum_{i=1}^m w_i'\,\varepsilon_i(u)\,w_i \qquad \text{for every } u\in\mathcal{L}(E,D). \tag{6.4}$$

A source-side relative splitting (G,w,ε), respectively a target-side relative splitting (G,w',ε), of degree k also counts as a splitting decomposition of degree k with the single piece $(G,\widehat{D},w,\mathrm{id}_{\widehat{D}},\varepsilon)$, respectively $(E,G,\mathrm{id}_E,w',\varepsilon)$.

(4) These structures are *without completion* if their sections take values in the uncompleted tensor powers, their retractions take values in the uncompleted exterior powers $\Lambda^n_\pi D$, the maps $\varepsilon(u)$ of a source-side relative splitting take values in D, and the maps w_i' of a decomposition take values in D.

A *linear family* is a bounded linear map $\theta\colon P\to\mathcal{L}(E,\widehat{D})$ on a normed space P. It is *liftable in degree $n\ge 1$* if there is a bounded n-linear map $Q\colon P^n\to\mathcal{L}(\Lambda^n_\pi E,\widehat{\Lambda}^n_\pi D)$ with $Q(h,\dots,h) = \Lambda^n_\pi\theta(h)$ for every $h\in P$. For $P = \mathcal{L}(E,D)$, Lemma 3.4 shows that (E,D) is good for Banach values in degree n if and only if the inclusion $\theta_{E,D}\colon\mathcal{L}(E,D)\to\mathcal{L}(E,\widehat{D})$ is liftable in degree n. The proofs of Theorem 6.13, Propositions 6.15 and 6.18 and Lemma 6.16 are in Appendix P.

Theorem 6.13 (decompositions)**.** (1) *For a piece (G,G',w,w',ε) of source type or of target type in degree k, the map $u\mapsto w'\varepsilon(u)w$ is liftable in every degree $n\le k$.*

(2) *Suppose that* $u = \sum_{i=1}^{m} \theta_i(u)$ *for every* $u \in \mathcal{L}(E,D)$*, where each* $\theta_i \colon \mathcal{L}(E,D) \to \mathcal{L}(E,\widehat{D})$ *is bounded linear and liftable in every degree* $n \le k$*. Then* (E,D) *is good for Banach values in degree* k*. If every* $\theta_i(u)$ *takes values in* D *and every lift can be chosen with values in* $\mathcal{L}(\Lambda^n_\pi E, \Lambda^n_\pi D)$*, then* (E,D) *is good in degree* k*.*
(3) *A splitting decomposition of degree* k *makes* (E,D) *good for Banach values in degree* k*, and good in degree* k *if it is without completion.*

Proposition 6.14 (transfer)**.** *Let* $w \colon E \to G$ *and* $w' \colon G' \to \widehat{D}$ *be bounded linear maps of normed spaces, and let* $\varepsilon \colon \mathcal{L}(E,D) \to \mathcal{L}(G,G')$ *be bounded linear with* $u = w'\varepsilon(u)w$ *for every* u*. If* (G,G') *is good for Banach values in degree* k*, with a lift* Q'*, then so is* (E,D)*, with the lift*

$$Q(u_1,\dots,u_k) = \widehat{\Lambda}^k_\pi w' \circ Q'\big(\varepsilon(u_1),\dots,\varepsilon(u_k)\big) \circ \Lambda^k_\pi w.$$

Proof. The map Q is bounded and k-linear, with values in $\mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi \widehat{D})$ and $\widehat{\Lambda}^k_\pi \widehat{D} = \widehat{\Lambda}^k_\pi D$; its diagonal is $\widehat{\Lambda}^k_\pi w' \circ \Lambda^k_\pi \varepsilon(u) \circ \Lambda^k_\pi w = \Lambda^k_\pi u$. Lemma 3.4 concludes. □

The same formula makes $u \mapsto w'\varepsilon(u)w$ liftable in degree n for every piece (G,G',w,w',ε) with (G,G') good for Banach values in degree n.

Proposition 6.15 (two pieces suffice)**.** *For a bounded bilinear map* $b \colon \mathcal{L}(E,D) \times E \to \widehat{D}$ *let* $w_b \colon E \to \mathcal{L}(\mathcal{L}(E,D),\widehat{D})$ *be the map* $w_b(x)(u) = b(u,x)$*, and let* $w'_b \colon \mathcal{L}(E,D) \otimes_\pi E \to \widehat{D}$ *be the linear map with* $w'_b(u \otimes x) = b(u,x)$*. The pair* (E,D) *has a splitting decomposition of degree* k *into pieces of source and target type if and only if there are bounded bilinear maps* $b_S, b_R \colon \mathcal{L}(E,D) \times E \to \widehat{D}$ *with* $b_S(u,x) + b_R(u,x) = ux$*,* w_{b_S} $\mathrm{S}^{\le k}$*-split and* w'_{b_R} $\mathrm{R}^{\le k}$*-split. In that case* (E,D) *has such a decomposition with at most two pieces.*

Lemma 6.16 (complemented sub-pairs)**.** *Let* E_0 *and* D_0 *be normed spaces,* $J \colon E_0 \to E$ *and* $\Pi \colon \widehat{D} \to \widehat{D}_0$ *bounded linear maps, and* $\Phi \colon \mathcal{L}(E_0,D_0) \to \mathcal{L}(E,D)$ *a bounded linear map with* $\Pi\,\Phi(v)\,J = v$ *for every* $v \in \mathcal{L}(E_0,D_0)$*. This holds, for instance, if* $\Pi' J = \mathrm{id}_{E_0}$ *and* $\Pi J' = \mathrm{id}_{D_0}$ *for bounded linear maps* $\Pi' \colon E \to E_0$ *and* $J' \colon D_0 \to D$*, and* $\Phi(v) = J' v \Pi'$*.*

(1) *If* (E,D) *has a splitting decomposition of degree* k*, so does* (E_0,D_0)*.*
(2) *If* (E,D) *is good for Banach values in degree* k*, so is* (E_0,D_0)*.*

Definition 6.17. A normed space P is *polarizable in degrees* $\le k$ if it is polarizable in degree n for every $n \le k$. A *parameter piece* of degree k for (E,D) is a piece (G,G',w,w',ε) with $\varepsilon = \varepsilon' \circ \psi$ for bounded linear maps $\psi \colon \mathcal{L}(E,D) \to P$ and $\varepsilon' \colon P \to \mathcal{L}(G,G')$, where P is polarizable in degrees $\le k$; the maps w and w' are arbitrary. A *generalized decomposition* of degree k is a finite family of pieces satisfying (6.4), each of source type or of target type in degree k, or a parameter piece of degree k. A source-side or target-side relative splitting of degree k also counts as a generalized decomposition of degree k, as in Definition 6.12(3).

Proposition 6.18 (parameter pieces)**.** (1) *For every set* I*, the space* $\ell^1(I,K)$ *is polarizable in every degree, and so is every normed space with finite continuous coordinates, for instance every finite-dimensional normed space over a complete field (Lemma A.3).*
(2) *If* P_1 *and* P_2 *are polarizable in degrees* $\le k$*, so is* $P_1 \oplus P_2$ *with the maximum norm.*
(3) *If* P *is polarizable in degree* n *and* P_0 *is a retract of* P*, then* P_0 *is polarizable in degree* n*.*
(4) *For a parameter piece* (G,G',w,w',ε) *of degree* k*, the map* $u \mapsto w'\varepsilon(u)w$ *is liftable in every degree* $n \le k$*. Hence a generalized decomposition of degree* k *makes* (E,D) *good for Banach values in degree* k*.*

Corollary 6.19. *Let* $\operatorname{char} K = p > 0$*.*

(1) *A parameter piece of degree p is a parameter piece of every degree, and a generalized decomposition of degree p into parameter pieces is one of every degree.*
(2) *If $\mathcal{L}(E,D)$, with the operator norm, is polarizable in degree p, then (E,D) is good for Banach values in every degree.*

Theorem 6.20 (decompositions of degree p)**.** *Let* $\operatorname{char} K = p > 0$*. A splitting decomposition of degree p is a splitting decomposition of degree p^2-1, with the same pieces. Hence (E,D) is then good for Banach values in every degree $n < p^2$, and good in every degree $n < p^2$ if the decomposition is without completion.*

Proposition 6.21 (target pieces)**.** *Suppose that $E^* \neq 0$ and that (E,D) has a splitting decomposition of degree k all of whose pieces are of target type in degree k. Then (R^k_D) holds, without completion if the decomposition is without completion. If moreover* $\operatorname{char} K = p > 0$ *and $k = p$, then (E,D) is good for Banach values in every degree, and good in every degree if the decomposition is without completion.*

The proofs of Corollary 6.19, Theorem 6.20 and Proposition 6.21 are in Appendix L.

6.3. **The class of an operator family.** Throughout this subsection E, D, F, P and P' are normed spaces and $k \geq 1$; linear families and liftability are as defined before Theorem 6.13. As in Lemma 3.14, with the same proof, a linear family $\theta\colon P \to \mathcal{L}(E,\widehat{D})$ gives a unique bounded linear map $\mathrm{ev}_\theta\colon \widehat{\otimes}^k_\pi E \to \mathrm{Mult}^k(P;\widehat{\Lambda}^k_\pi D)$ with

$$\mathrm{ev}_\theta(x_1\otimes\cdots\otimes x_k)(h_1,\dots,h_k) = \theta(h_1)x_1\wedge\cdots\wedge\theta(h_k)x_k \in \widehat{\Lambda}^k_\pi\widehat{D} = \widehat{\Lambda}^k_\pi D;$$

it has norm at most $\|\theta\|^k$, and $\mathrm{ev}_\theta(z)(h,\dots,h) = \widehat{\Lambda}^k_\pi\theta(h)(\widehat{\wedge}^k_E z)$ for $z \in \widehat{\otimes}^k_\pi E$ and $h \in P$, so it maps $\mathcal{N}^k_E$ into $\mathrm{Mult}^k_0(P;\widehat{\Lambda}^k_\pi D)$. Let q be the quotient map of the *coefficient sequence* $\mathcal{M}^k(P;\widehat{\Lambda}^k_\pi D)$ of P (Definition 3.8(4)). Since $q\circ\mathrm{ev}_\theta$ vanishes on $\mathcal{N}^k_E$, Lemma 3.7 gives a bounded linear map $\overline{\mathrm{ev}}_\theta\colon \widehat{\Lambda}^k_\pi E \to \mathrm{Poly}^k(P;\widehat{\Lambda}^k_\pi D)$ with $\overline{\mathrm{ev}}_\theta\circ\widehat{\wedge}^k_E = q\circ\mathrm{ev}_\theta$. The *class* of θ in degree k is

$$\mathrm{ob}_k(\theta) = (\mathrm{ev}_\theta|_{\mathcal{N}^k_E})_*[\mathcal{S}^k_E] \in \mathrm{Ext}\big(\widehat{\Lambda}^k_\pi E, \mathrm{Mult}^k_0(P;\widehat{\Lambda}^k_\pi D)\big), \tag{6.5}$$

and $\mathrm{ob}_k(\theta_{E,D}) = \mathrm{ob}_k(E,D)$, since $\mathrm{ev}_{\theta_{E,D}} = \mathrm{ev}^k_{E,D}$.

For exact sequences $\mathcal{Y}\colon 0\to Y_1 \xrightarrow{i} Y_2 \xrightarrow{q} Y_3 \to 0$ and $\mathcal{Y}'\colon 0\to Y_1' \xrightarrow{i'} Y_2' \xrightarrow{q'} Y_3' \to 0$, a *morphism* $\mathcal{Y}\to\mathcal{Y}'$ is a triple of bounded linear maps $\alpha_l\colon Y_l\to Y_l'$, $l=1,2,3$, with $\alpha_2 i = i'\alpha_1$ and $q'\alpha_2 = \alpha_3 q$. For a bounded linear $\gamma\colon Y'\to Y_3$ from a Banach space Y', the *pullback* $\gamma^*\mathcal{Y}\colon 0\to Y_1\to Y_\gamma\to Y'\to 0$ has $Y_\gamma = \{(y,y')\in Y_2\oplus Y' : qy = \gamma y'\}$, normed by $\|y\|+\|y'\|$, with the maps $y_1\mapsto(iy_1,0)$ and $(y,y')\mapsto y'$. It is exact, its class $\gamma^*[\mathcal{Y}]$ depends only on $[\mathcal{Y}]$, and $\gamma^*[\mathcal{Y}] = 0$ if and only if $\gamma = q\circ L$ for a bounded linear $L\colon Y'\to Y_2$ (Lemma N.1). Pushouts and pullbacks compose and commute, and a morphism $(\alpha_1,\alpha_2,\alpha_3)\colon\mathcal{Y}\to\mathcal{Y}'$ gives $\alpha_{1*}[\mathcal{Y}] = \alpha_3^*[\mathcal{Y}']$ (Lemma N.2, Corollary N.3).

Theorem 6.22 (the class of a family)**.** *Let $\theta\colon P\to\mathcal{L}(E,\widehat{D})$ be a linear family.*

(1) *The following are equivalent: θ is liftable in degree k; the restriction $\mathrm{ev}_\theta|_{\mathcal{N}^k_E}$ extends to a bounded linear map $\widehat{\otimes}^k_\pi E\to\mathrm{Mult}^k_0(P;\widehat{\Lambda}^k_\pi D)$; $\mathrm{ob}_k(\theta)=0$ in (6.5). A lift Q gives such an extension of norm at most $\|\theta\|^k+\|Q\|$, and such an extension ξ gives a lift of norm at most $\|\theta\|^k+\|\xi\|$.*
(2) *The triple $(\mathrm{ev}_\theta|_{\mathcal{N}^k_E}, \mathrm{ev}_\theta, \overline{\mathrm{ev}}_\theta)$ is a morphism $\mathcal{S}^k_E\to\mathcal{M}^k(P;\widehat{\Lambda}^k_\pi D)$, and $\mathrm{ob}_k(\theta) = \overline{\mathrm{ev}}_\theta^*[\mathcal{M}^k(P;\widehat{\Lambda}^k_\pi D)]$.*

Proof. (1) Steps 1–3 of the proof of Theorem 3.15, with its correspondence $Q \leftrightarrow Q^\sharp$, apply with P, $\theta(h)$, ev_θ and $\mathrm{Mult}^k(P;\widehat{\Lambda}^k_\pi D)$ in place of $\mathcal{L}(E,D)$, u, ev and $\mathrm{Mult}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$, using $\mathrm{ev}_\theta(z)(h,\dots,h) = \widehat{\Lambda}^k_\pi\theta(h)(\widehat{\wedge}^k_E z)$; the bound $\|\mathrm{ev}_\theta\| \le \|\theta\|^k$ replaces $\|\mathrm{ev}\| \le 1$ in the constants. The proof of (2) is in Appendix N. □

For $\psi \in \mathcal{L}(P',P)$ and a Banach space Y let $\psi^\sharp\colon \mathrm{Mult}^k(P;Y) \to \mathrm{Mult}^k(P';Y)$ be the map $\Phi \mapsto \Phi\circ(\psi\times\cdots\times\psi)$, and for a bounded linear map $\chi\colon Y \to Y'$ of Banach spaces let $\chi_\flat\colon \mathrm{Mult}^k(P;Y) \to \mathrm{Mult}^k(P;Y')$ be the map $\Phi \mapsto \chi\circ\Phi$. Both preserve zero diagonals, and their pushouts are taken along these restrictions.

Proposition 6.23 (naturality)**.** *Let $\theta\colon P \to \mathcal{L}(E,\widehat{D})$ be a linear family.*

(a) *For $\psi \in \mathcal{L}(P',P)$, $\mathrm{ob}_k(\theta\circ\psi) = (\psi^\sharp)_*\,\mathrm{ob}_k(\theta)$.*
(b) *Let E' be a normed space, $w \in \mathcal{L}(E',E)$, and $\theta_w(h) = \theta(h)\circ w$ for $h \in P$. Then $\mathrm{ob}_k(\theta_w) = (\widehat{\Lambda}^k_\pi w)^*\,\mathrm{ob}_k(\theta) = (\mathrm{ev}_\theta|_{\mathcal{N}^k_E})_*\,\mathrm{sp}_k(w)$, with $\mathrm{sp}_k(w)$ as in Proposition 3.13(1).*
(c) *Let D' be a normed space, $w'\colon \widehat{D} \to \widehat{D}'$ bounded linear, and $(w'\theta)(h) = w'\circ\theta(h)$ for $h \in P$. Then $\mathrm{ob}_k(w'\theta) = ((\widehat{\Lambda}^k_\pi w')_\flat)_*\,\mathrm{ob}_k(\theta)$.*
(d) *$((\mathrm{alt}^k_D)_\flat)_*\,\mathrm{ob}_k(\theta) = 0$, with alt^k_D extended to $\widehat{\Lambda}^k_\pi D$. If G is a normed space, $w'\colon G \to \widehat{D}$ is R^k-split and $\theta_0\colon P \to \mathcal{L}(E,G)$ is bounded linear, then the family $h \mapsto w'\circ\theta_0(h)$ is liftable in degree k.*

The proof is in Appendix N.

Corollary 6.24 (transport)**.** (1) *Let G and G' be normed spaces, and let $w\colon E \to G$, $w'\colon G' \to \widehat{D}$ and $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(G,G')$ be bounded linear maps with $u = w'\varepsilon(u)w$ for every $u \in \mathcal{L}(E,D)$. Then*

$$\mathrm{ob}_k(E,D) = ((\widehat{\Lambda}^k_\pi w')_\flat)_*(\varepsilon^\sharp)_*(\widehat{\Lambda}^k_\pi w)^*\,\mathrm{ob}_k(G,G'),$$

with w' extended to $\widehat{G}'$.

(2) *Let G be a normed space, and let $w\colon E \to G$ and $\varepsilon\colon \mathcal{L}(E,D) \to \mathcal{L}(G,\widehat{D})$ be bounded linear maps with $u = \varepsilon(u)\circ w$ for every $u \in \mathcal{L}(E,D)$. Then $\mathrm{ob}_k(E,D) = (\mathrm{ev}_\varepsilon|_{\mathcal{N}^k_G})_*\,\mathrm{sp}_k(w)$, which vanishes if w is S^k-split.*

The proof is in Appendix P.

Theorem 6.25 (sum formula)**.** *Let P_1 and P_2 be normed spaces, $\theta_1\colon P_1 \to \mathcal{L}(E,\widehat{D})$ and $\theta_2\colon P_2 \to \mathcal{L}(E,\widehat{D})$ linear families, $P = P_1 \oplus P_2$ with the maximum norm, $\theta_1 \boxplus \theta_2\colon P \to \mathcal{L}(E,\widehat{D})$ the family $(h_1,h_2) \mapsto \theta_1(h_1)+\theta_2(h_2)$, and $\iota_1\colon P_1 \to P$, $\iota_2\colon P_2 \to P$ the inclusions. Put $Y = \widehat{\Lambda}^k_\pi D$. For $S \subseteq \{1,\dots,k\}$ and $\Phi \in \mathrm{Mult}^k(P;Y)$ let $\Phi_S(h^1,\dots,h^k)$, where $h^j = (h^j_1,h^j_2) \in P$, be the value of Φ at the tuple whose j-th entry is $(h^j_1,0)$ for $j \in S$ and $(0,h^j_2)$ for $j \notin S$, and let $\Pi^{(a)}\Phi = \sum_{|S|=a}\Phi_S$. The maps $\Pi^{(a)}$, $0 \le a \le k$, are bounded commuting idempotents on $\mathrm{Mult}^k(P;Y)$ with sum* id *that map $\mathrm{Mult}^k_0(P;Y)$ into itself (Lemma N.4). Put $\mathrm{Mult}^{(a)}_0 = \Pi^{(a)}\,\mathrm{Mult}^k_0(P;Y)$ and $\mathrm{ob}^{(a)} = (\Pi^{(a)}|_{\mathrm{Mult}^k_0(P;Y)})_*\,\mathrm{ob}_k(\theta_1\boxplus\theta_2) \in \mathrm{Ext}(\widehat{\Lambda}^k_\pi E, \mathrm{Mult}^{(a)}_0)$.*

(1) *$\mathrm{ob}_k(\theta_1\boxplus\theta_2) = 0$ if and only if $\mathrm{ob}^{(a)} = 0$ for every a. The map $\iota_1^\sharp$ restricts to an isomorphism of $\mathrm{Mult}^{(k)}_0$ onto $\mathrm{Mult}^k_0(P_1;Y)$ with $(\iota_1^\sharp)_*\,\mathrm{ob}^{(k)} = \mathrm{ob}_k(\theta_1)$, and $\iota_2^\sharp$ restricts to an isomorphism of $\mathrm{Mult}^{(0)}_0$ onto $\mathrm{Mult}^k_0(P_2;Y)$ with $(\iota_2^\sharp)_*\,\mathrm{ob}^{(0)} = \mathrm{ob}_k(\theta_2)$.*

(2) *Let $0 < a < k$. Then $\mathrm{ob}^{(a)} = 0$ if and only if there is a bounded k-linear map $Q^{(a)}\colon P_1^a \times P_2^{k-a} \to \mathcal{L}(\widehat{\Lambda}_\pi^k E, \widehat{\Lambda}_\pi^k D)$ with*

$$Q^{(a)}(h_1,\dots,h_1;h_2,\dots,h_2) = \wedge_{\widehat{D}}^{a,k-a} \circ \big(\widehat{\Lambda}_\pi^a \theta_1(h_1) \otimes \widehat{\Lambda}_\pi^{k-a}\theta_2(h_2)\big) \circ \mathrm{sh}_E^{a,k-a}$$

for $h_1 \in P_1$ and $h_2 \in P_2$, the maps on the right extended to completions; the right side is the term of index a in (6.3) for $w_1 = \theta_1(h_1)$ and $w_2 = \theta_2(h_2)$.

(3) *Let $0 < a < k$. If θ_1 is liftable in degree a and θ_2 in degree $k-a$, then $\mathrm{ob}^{(a)} = 0$.*

(4) *Suppose that $P_1 = P_2$, and let $\Delta_{P_1}\colon P_1 \to P$ be the map $h \mapsto (h,h)$. If $\mathrm{ob}^{(a)} = 0$ for every a, then $\theta_1 + \theta_2 = (\theta_1 \boxplus \theta_2) \circ \Delta_{P_1}$ is liftable in degree k. In particular, $\theta_1 + \theta_2$ is liftable in degree k if θ_1 and θ_2 are liftable in every degree at most k.*

(5) *Suppose that $P_1 = P_2$ and $\theta_1 = \theta_2 = \theta$, let Δ_{P_1} be as in (4), and let $0 < a < k$. Then $\Delta_{P_1}^\sharp \circ \Pi^{(a)} \circ \mathrm{ev}_{\theta\boxplus\theta} = \binom{k}{a}\mathrm{ev}_\theta$. Hence, if $\binom{k}{a} \neq 0$ in K, then $\mathrm{ob}_k(\theta) = 0$ if and only if $(\Delta_{P_1}^\sharp|_{\mathrm{Mult}_0^{(a)}})_* \mathrm{ob}^{(a)} = 0$, and θ is liftable in degree k if it is liftable in degrees a and $k-a$.*

The proof is in Appendix N.

Proposition 6.26 (composite families)**.** *Let $\theta\colon P \to \mathcal{L}(E,\widehat{D})$ and $\theta'\colon P' \to \mathcal{L}(D,\widehat{F})$ be linear families. Extend each $\theta'(h')$ to $\widehat{D}$, and regard θ' also as a family $P' \to \mathcal{L}(\widehat{D},\widehat{F})$; its class is $\mathrm{ob}_k(\theta')$, since $\mathcal{S}^k_{\widehat{D}} = \mathcal{S}^k_D$ (the paragraph before Theorem 3.16). Let $\theta' \odot \theta\colon P' \otimes_\pi P \to \mathcal{L}(E,\widehat{F})$ be the bounded linear map on the uncompleted projective tensor product with $(\theta' \odot \theta)(h' \otimes h) = \theta'(h') \circ \theta(h)$.*

(1) *For $h \in P$ let $\psi_h\colon P' \to P' \otimes_\pi P$ be the map $h' \mapsto h' \otimes h$. Then $(\psi_h^\sharp)_* \mathrm{ob}_k(\theta' \odot \theta) = (\widehat{\Lambda}_\pi^k \theta(h))^* \mathrm{ob}_k(\theta')$. For $h' \in P'$ let $\psi'_{h'}\colon P \to P' \otimes_\pi P$ be the map $h \mapsto h' \otimes h$. Then $((\psi'_{h'})^\sharp)_* \mathrm{ob}_k(\theta' \odot \theta) = ((\widehat{\Lambda}_\pi^k \theta'(h'))_\flat)_* \mathrm{ob}_k(\theta)$.*

(2) *Let $\mathrm{comp}\colon \mathcal{L}(D,F) \otimes_\pi \mathcal{L}(E,D) \to \mathcal{L}(E,F)$ be the composition map, $v \otimes u \mapsto vu$. Then $\mathrm{ob}_k(\theta_{D,F} \odot \theta_{E,D}) = (\mathrm{comp}^\sharp)_* \mathrm{ob}_k(E,F)$. In particular, if (E,F) is good for Banach values in degree k, then $\theta_{D,F} \odot \theta_{E,D}$ is liftable in degree k.*

The proof is in Appendix N.

6.4. **Block assemblies and permanence.** The proofs of Propositions 6.27 and 6.28 are in Appendix P.

Proposition 6.27 (block assemblies)**.** *Let I be a totally ordered set, and let E be the ℓ^1-sum, algebraic or completed, of normed spaces E_i, $i \in I$, with the norm $\|x\| = \sum_i \|\Pi_i x\|$, inclusions $J_i\colon E_i \to E$ and block projections $\Pi_i\colon E \to E_i$. Let D_i be normed spaces and $J'_i\colon D_i \to D$ bounded linear maps with $\|J'_i\| \le 1$. Let $\mathcal{U} \subseteq \mathcal{L}(E,D)$ be a linear subspace with linear maps $\mathcal{U} \to \mathcal{L}(E_i,D_i)$, $u \mapsto u_i$, such that $\|u_i\| \le c\|u\|$ for a constant c, and $ux = \sum_i J'_i u_i \Pi_i x$ for every $x \in E$ with finitely many nonzero blocks. Suppose that there is a constant $C \ge 1$ such that for every $i \in I$ and every $n \le k$ there is a bounded n-linear map Q^i_n on $\mathcal{U}_i = \{u_i : u \in \mathcal{U}\}$ with values in $\mathcal{L}(\Lambda_\pi^n E_i, \widehat{\Lambda}_\pi^n D_i)$, with $Q^i_n(v,\dots,v) = \Lambda_\pi^n v$ and $\|Q^i_n\| \le C$. Then there is a bounded k-linear map $Q\colon \mathcal{U}^k \to \mathcal{L}(\Lambda_\pi^k E, \widehat{\Lambda}_\pi^k D)$ with $Q(u,\dots,u) = \Lambda_\pi^k u$ and $\|Q\| \le C^k c^k$. In particular, if $\mathcal{U} = \mathcal{L}(E,D)$, then (E,D) is good for Banach values in degree k.*

Proposition 6.28 (permanence)**.** *Let $k \ge 1$, and give finite direct sums the maximum norm.*

(1) (Retracts.) *Let Y_0 be a retract of a normed space Y. If Y satisfies (S^k_Y), or (R^k_Y), or is polarizable in degree k, then Y_0 satisfies the same condition. If (E,D) is good for Banach values in degree k and E_0, D_0 are retracts of E, D, then (E_0,D_0) is good for Banach values in degree k.*

(2) (Finite sums.) *For normed spaces Y_1 and Y_2, $(\mathrm{S}^n_{Y_1\oplus Y_2})$ holds for every $n \le k$ if and only if $(\mathrm{S}^n_{Y_1})$ and $(\mathrm{S}^n_{Y_2})$ hold for every $n \le k$; the same holds for (R^n) and for polarizability in degrees $\le k$. If (E_a, D_b) is good for Banach values in every degree $n \le k$ for all $a, b \in \{1, 2\}$, so is $(E_1 \oplus E_2, D_1 \oplus D_2)$.*

(3) (Completion of a space.) *Regard the completion $\widehat{Y}$ of a normed space Y as a normed space over K. Then Y satisfies (S^k_Y), (R^k_Y), or polarizability in degree k, if and only if $\widehat{Y}$ satisfies the same condition.*

(4) (Completion of a pair.) *If D is complete, then (E, D) is good for Banach values in degree k if and only if $(\widehat{E}, D)$ is. If $(E, \widehat{D})$ is good for Banach values in degree k, so is (E, D).*

Every normed space embeds isometrically in a space with all three splittings, and every Banach space is isomorphic to a quotient of such a space.

Theorem 6.29 (a space with all splittings)**.** *Let G be a normed space and $\Omega = \{0,1\}^{\mathbb{N}}$. For $m \ge 0$ and $\xi \in \{0,1\}^m$ let $C_\xi \subseteq \Omega$ be the set of sequences that begin with ξ, a dyadic cell of level m, of Haar measure 2^{-m}. A step function of level m is a function $f = \sum_{\xi\in\{0,1\}^m} 1_{C_\xi} g_\xi$ on Ω with $g_\xi \in G$; its norm $\|f\|_1 = \sum_\xi 2^{-m}\|g_\xi\|$ does not depend on the level used to write it. Let $L^1(\Omega; G)$ be the completion of the normed space of step functions. Then for every $k \ge 1$ the space $Y = L^1(\Omega; G)$ satisfies (S^k_Y) with a section of norm at most 1 and (R^k_Y) with a retraction of norm at most 1, and it is polarizable in degree k with a projection of norm at most $k!$.*

Corollary 6.30 (no hereditary permanence)**.** *Let E and D be Banach spaces, let $\mathrm{Ball}(E)$ be the closed unit ball of E, and let $\ell^1(\mathrm{Ball}(E), \widehat{K}) \to E$ be the bounded linear map that sends each unit vector e_x to x.*

(1) *The map $g \mapsto 1_\Omega g$ is an isometry of E onto a closed subspace of $L^1(\Omega; E)$, and $(L^1(\Omega; E), D)$ is good for Banach values in every degree. Likewise D is isometric to a closed subspace of $L^1(\Omega; D)$, and $(E, L^1(\Omega; D))$ is good for Banach values in every degree.*

(2) *The map $\ell^1(\mathrm{Ball}(E), \widehat{K}) \to E$ is surjective and open, so E is isomorphic to a quotient of $\ell^1(\mathrm{Ball}(E), \widehat{K})$, and $(\ell^1(\mathrm{Ball}(E), \widehat{K}), D)$ is good for Banach values in every degree. Likewise D is isomorphic to a quotient of $\ell^1(\mathrm{Ball}(D), \widehat{K})$, and $(E, \ell^1(\mathrm{Ball}(D), \widehat{K}))$ is good for Banach values in every degree.*

(3) *Assume condition* (L)*. Then $(c_0(\mathbb{N}, \widehat{K}), c_0(\mathbb{N}, \widehat{K}))$ is not good for Banach values in degree k, and $c_0(\mathbb{N}, \widehat{K})$ satisfies neither (S^k) nor (R^k) and is not polarizable in degree k (Theorem 7.7, Corollary 8.3 and Proposition 6.28). Hence goodness for Banach values in degree k passes neither to closed subspaces nor to quotients, in either slot of a pair, and none of (S^k), (R^k) and polarizability in degree k passes to closed subspaces or to quotients.*

The proofs of Theorem 6.29 and Corollary 6.30 are in Appendix L.

Theorem 6.31 (no three-space property in the source slot)**.** *Assume condition* (L)*, and let D be a Banach space with an equivalent nonarchimedean norm and $\dim_{\widehat{K}} D = \infty$, for example $D = c_0(\mathbb{N}, \widehat{K})$. Let $Y_1 = \ell^1(\mathrm{Ball}(D), \widehat{K})$, let $\varpi_D \colon Y_1 \to D$ be the map $e_x \mapsto x$, surjective and open by Corollary 6.30(2), and let $\eta \colon Y_1 \to L^1(\Omega; Y_1)$ be the isometry $g \mapsto 1_\Omega g$ of Corollary 6.30(1). Put $Y_2 = D \oplus L^1(\Omega; Y_1)$ with the maximum norm and $Y_3 = L^1(\Omega; Y_1)/\eta(\ker \varpi_D)$. The maps $y \mapsto (-\varpi_D y, \eta y)$ and $(d, v) \mapsto [v + \eta(y_d)]$, where $y_d \in Y_1$ is any element with $\varpi_D y_d = d$, form an exact sequence $0 \to Y_1 \to Y_2 \to Y_3 \to 0$ such that:*

(1) (Y_1, D') *is good for Banach values in every degree for every normed space* D'*;*
(2) $\mathcal{L}(Y_3, D) = 0$*, so* (Y_3, D) *is good in every degree;*
(3) (Y_2, D) *is not good for Banach values in degree* k*.*

The proof is in Appendix N.

7. Obstructions

7.1. **The field.** We say that *condition* (L) holds if κ is a finite field of characteristic p, $0 < \varrho < 1$, the Laurent series field $\kappa((X))$ carries the absolute value $|a| = \varrho^{\operatorname{ord} a}$, K is a subfield of $\kappa((X))$ on which this absolute value is nontrivial, and $k \geq p$; for example $K = \kappa((X))$ or $K = \kappa(X)$. Then K is nonarchimedean, and $\widehat{K}$ is the closure of K in the complete field $\kappa((X))$. For $a \in \kappa((X))$ and $n \in \mathbb{Z}$, $\operatorname{coeff}_n(a) \in \kappa$ is the X^n-coefficient of a, applied entrywise to vectors and matrices, and $M_\varrho = \max_{l \geq 0}(l+1)\varrho^l$, which exists since $(l+1)\varrho^l \to 0$. The group $|K^\times|$ is a nontrivial subgroup of $\varrho^{\mathbb{Z}}$, hence $|K^\times| = \varrho^{n_0\mathbb{Z}}$ for an integer $n_0 \geq 1$; under (L) we fix $\vartheta \in K$ with $|\vartheta| = \varrho^{n_0}$, so $|K^\times| = |\vartheta|^{\mathbb{Z}}$.

Lemma 7.1 (normalization of the field)**.** *Assume* (L)*. There are a finite field* κ' *of characteristic* p *and an isometric field isomorphism* σ *from* $\kappa'((X))$*, with the absolute value* $|g| = |\vartheta|^{\operatorname{ord} g}$*, onto* $\widehat{K}$ *with* $\sigma(X) = \vartheta$*. Hence* K *is isometrically isomorphic to a dense subfield of* $\kappa'((X))$*. Consequently, a statement whose hypotheses and conclusion depend only on* K *as a normed field holds for every* K *satisfying* (L) *once it holds for every subfield* K *dense in* $\kappa((X))$*, for every finite field* κ *of characteristic* p *and every* $\varrho \in (0,1)$*.*

The proof is in Appendix P. When K is dense in $\kappa((X))$, we have $\widehat{K} = \kappa((X))$ and $|K^\times| = \varrho^{\mathbb{Z}}$, so $|\vartheta| = \varrho$.

7.2. **Forcing and the two counterexamples.** The factorial hypothesis of Proposition 5.1(1) cannot be removed.

Theorem 7.2 (operator obstructions)**.** *Let* K *be a nontrivially normed field of characteristic* $p > 0$*, and let* $k \geq p$*.*

(1) *There are Banach* K*-spaces* E, F *for which*

$$A^k_{E,E;F} \colon \mathcal{L}(E,E) \longrightarrow \mathcal{L}\big(\operatorname{Alt}^k(E;F), \operatorname{Alt}^k(E;F)\big)$$

is analytic at no point. The target F *admits no equivalent nonarchimedean norm.*

(2) *If* K *is complete and not spherically complete, there are nonarchimedean Banach spaces* E, D *for which* $A^k_{E,D;K}$ *is analytic at no point. They can be chosen as closed subspaces of*

$$\ell^\infty(\Lambda, K^{k+1}) \quad \text{and} \quad \ell^\infty(\Lambda, K^k),$$

respectively, containing the corresponding c_0 *spaces, with* Λ *countable.*

Part (1) is assembled in Appendix E. For κ finite of characteristic p, Theorem C.1 excludes a bounded lift over $\kappa((X))$ for $E_1 = \ell^\infty(\mathbb{N}, \kappa((X)))$ and $F_1 = \widehat{\Lambda}^k_\pi E_1$ (Lemma C.2): through Proposition C.3 a lift gives the map Ψ of Lemma C.4, which Theorem B.9 excludes by Ramsey's theorem, clusters and a staircase matrix. Proposition D.9, which rests on Ingleton's Theorem D.3, and Lemma D.11 pass to every K; by Proposition E.1 the target F has no equivalent nonarchimedean norm. Part (2) is proved in Appendix F from Lemmas F.4 and F.2 and van der Put's description of the continuous dual of a bounded sequence space [20]. In both parts, Proposition 2.5 turns the absence of a lift into nonanalyticity at every point.

Corollary 7.3 (regularity on whole hom spaces)**.** *For every nontrivially normed field K, the bifunctor* $\mathrm{Alt}^k\colon \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K \to \mathsf{Vec}_K$ *is C^∞, and is analytic if and only if $k! \neq 0$ in K. Both assertions remain true on the full subcategory of Banach spaces.*

Proof. Smooth descent applies to the ambient polynomial action. An invertible factorial gives a finite multilinear lift, and bounded bilinear postcomposition supplies the joint bifunctor action. If the factorial vanishes, Theorem 7.2(1) excludes analyticity even on Banach spaces, after fixing the output morphism to be the identity. □

Theorem 7.4 (universal scalar classification)**.** *Let K be a complete nonarchimedean, nontrivially normed field. For any $k \geq 0$, the contravariant functor*

$$\mathrm{Alt}^k(-;K)\colon \mathsf{Vec}_K^{\mathrm{op}} \longrightarrow \mathsf{Vec}_K$$

is analytic on every hom space if and only if $k! \neq 0$ in K or K is spherically complete. The same equivalence holds when the source category is restricted to Banach spaces, or to nonarchimedean Banach spaces.

Proof. An invertible factorial gives a multilinear lift by normalized antisymmetrization. If K is spherically complete, Theorem 5.2 applies with target K. Both arguments permit arbitrary normed sources. If neither condition holds, then K has characteristic $p > 0$ with $k \geq p$, and Theorem 7.2(2) supplies nonarchimedean Banach witnesses against all three assertions. □

By Theorem 3.2(2), the same criterion holds for every nonzero continuous-dual target, and hence for every nonzero finite-dimensional target, which over complete K is isomorphic to a continuous dual (Lemma A.3).

Call a normed space F a *universal target* if $\mathrm{Alt}^k(-;F)$ is analytic on every hom space, that is, if $A^k_{E,D;F}$ is analytic on $\mathcal{L}(E,D)$ for all normed spaces E and D. Over a nonarchimedean field, every nonarchimedean spherically complete F is a universal target, by Theorem 5.2.

Proposition 7.5 (universal targets)**.** *If* $\operatorname{char} K = p > 0$*, $k \geq p$, and K is complete and not spherically complete, then every universal target has zero continuous dual.*

Proof. Let F be a normed space with a nonzero $\lambda \in F^*$, and choose $v \in F$ with $\lambda(v) = 1$. Let E and D be the spaces of Theorem 7.2(2), and for $a \in \mathrm{Alt}^k(D;K)$ let $va \in \mathrm{Alt}^k(D;F)$ be $y \mapsto a(y)v$. Then $A^k_{E,D;K}(u)(a) = \lambda \circ A^k_{E,D;F}(u)(va)$ for $u \in \mathcal{L}(E,D)$. Composing with the bounded linear maps $a \mapsto va$ and $m \mapsto \lambda \circ m$ is a bounded linear map $\mathcal{L}(\mathrm{Alt}^k(D;F), \mathrm{Alt}^k(E;F)) \to \mathcal{L}(\mathrm{Alt}^k(D;K), \mathrm{Alt}^k(E;K))$. So if $A^k_{E,D;F}$ were analytic at a point, so would be $A^k_{E,D;K}$, which Theorem 7.2(2) excludes. Hence F is not a universal target. □

Corollary 7.6 (bounded sequences)**.** *Let κ be a finite field of characteristic p, $K_1 = \kappa((X))$ with $|X| = \varrho$, $k \geq p$, and $E_1 = \ell^\infty(\mathbb{N}, K_1)$. Then (E_1, E_1) is not good for Banach values in degree k, and neither $(\mathrm{S}^k_{E_1})$ nor $(\mathrm{R}^k_{E_1})$ holds, although E_1 is isometrically isomorphic to $c_0(I, K_1)$ for a set I, and $A^k_{E_1,E_1;F}$ is a continuous polynomial for every Banach space F with an equivalent nonarchimedean norm.*

Proof. The space $F_1 = \widehat{\Lambda}^k_\pi E_1$ is a Banach space (Lemma C.2), and $A^k_{E_1,E_1;F_1}$ is analytic at no point by Theorem C.1; so (E_1, E_1) is not good for Banach values, and by Proposition 3.5(1),(2) neither $(\mathrm{S}^k_{E_1})$ nor $(\mathrm{R}^k_{E_1})$ holds. Since $|K_1^\times| = \varrho^{\mathbb{Z}}$, Corollary 5.3 and Proposition 2.5 give the assertion about nonarchimedean F. The norm of E_1 takes values in $|K_1|$, so E_1 has an orthonormal basis and is isometrically isomorphic to $c_0(I, K_1)$ [26, §1, Prop. 1]. □

7.3. **The ℓ^r threshold.** Throughout this subsection K satisfies (L). For a sequence a, $\mu_a\colon x \mapsto ax$ denotes multiplication by a. The proof of the next theorem follows the proof of Theorem B.9, with norm bounds on the staircase matrix in place of the support bound.

Theorem 7.7 (ℓ^r with $r \geq 2$)**.** *Let $2 \leq r \leq \infty$, and let E be a normed space with $c_{00}(\mathbb{N}, K) \subseteq E \subseteq \ell^r(\mathbb{N}, K)$ and the ℓ^r norm, the supremum norm if $r = \infty$. Suppose $Q\colon c_{00}(\mathbb{N}, K)^k \to \mathcal{L}(\Lambda_\pi^k E, \widehat{\Lambda}_\pi^k E)$ is k-linear, bounded for the supremum norm on $c_{00}(\mathbb{N}, K)$, and satisfies $Q(a, \dots, a) = \Lambda_\pi^k \mu_a$. Then for every $m \geq 1$, with $1/r = 0$ when $r = \infty$,*

$$\begin{aligned} \|Q\| &\geq \frac{(1-\varrho^2)\ln m}{k(k-1)2^k(1+\ln(2|\kappa|))} \quad \text{if } r = 2, \\ m - 1 &\leq M_\varrho k(k-1)\|Q\|(4m)^{2/r}4^{(k-2)/r} \quad \text{if } r > 2. \end{aligned} \tag{7.1}$$

Hence no such Q exists. Consequently (E, E) is not good for Banach values in degree k; $A^k_{E,E;F}$ is analytic at no point for $F = \Lambda_\pi^k E$ and for $F = \widehat{\Lambda}_\pi^k E$; (S^k_E) and (R^k_E) fail; and $\widehat{\Lambda}_\pi^k E$ has no equivalent nonarchimedean norm.

Corollary 7.8 (the ℓ^r threshold)**.** *Let $1 \leq r \leq \infty$, and let $E = \ell^r(\mathbb{N}, K)$, or $E = c_0(\mathbb{N}, K)$ if $r = \infty$. Then (E, E) is good for Banach values in degree k if and only if $r < 2$; in that case (E, D) and (D, E) are good for Banach values for every normed D. The same holds, with good in place of good for Banach values, for the finitely supported sequences with the ℓ^r norm.*

Corollary 7.9 (pairs of ℓ^r spaces)**.** *Let $1 \leq r_1, r_2 \leq \infty$, and let $E = \ell^{r_1}(\mathbb{N}, K)$ and $D = \ell^{r_2}(\mathbb{N}, K)$, with $c_0(\mathbb{N}, K)$ in place of ℓ^∞. The pair (E, D) is good for Banach values in degree k if $r_1 < 2$ or $r_2 < 2$, and it is not if $r_2 = 2 \leq r_1$ or if $r_1, r_2 > 2$.*

The proofs are in Appendix K. The pairs with $r_1 = 2 < r_2$ are good for Banach values in every degree (Corollary 8.5).

7.4. **The product triangle bound.** Throughout this subsection K satisfies (L). For $A \subseteq \mathbb{N}$ let 1_A be its indicator function, for $n \geq 1$ let $[n]$ be the set of the first n elements of $\mathbb{N}$, let e_t ($t \in \mathbb{N}$) be the unit vectors, and for a sequence x let $|x|$ be the sequence of absolute values $|x(t)|$.

Definition 7.10. A *sequence space* is a normed space D with $c_{00}(\mathbb{N}, K) \subseteq D \subseteq \ell^\infty(\mathbb{N}, K)$. It is *symmetric* if there is a norm on $c_{00}(\mathbb{N}, \mathbb{R})$, again written $\|\cdot\|_D$, with $\|f\|_D = \||f|\|_D$, with $\|f\|_D \leq \|g\|_D$ whenever $|f| \leq |g|$ pointwise, invariant under every permutation of $\mathbb{N}$ and equal to 1 at the unit vectors, such that $\|x\|_D = \||x|\|_D$ for $x \in c_{00}(\mathbb{N}, K)$ and $\|x\|_D = \sup_n \|x1_{[n]}\|_D$ for $x \in D$. A sequence space E is *ℓ^2-dominated* if there is C_2 with $\|x\|_2 \leq C_2\|x\|_E$ for all $x \in E$; for symmetric E in which $c_{00}(\mathbb{N}, K)$ is dense, this is the ℓ^2-domination of the unit vectors in the sense of Section 5, since $\|x\|_E = \sup_n \|x1_{[n]}\|_E$. The *fundamental function* of E is $\varphi_E(m) = \|1_{[m]}\|_E$. For a symmetric sequence space E, a sequence space D and $a \in c_{00}(\mathbb{N}, K)$, μ_a is bounded from E to D (Lemma 7.11(2)), and $\|\mu_a\|_{E\to D}$ denotes its norm.

For example, $\ell^r(\mathbb{N}, K)$ with $1 \leq r < \infty$ and $c_0(\mathbb{N}, K)$ are symmetric sequence spaces.

Lemma 7.11 (multiplier norms)**.** *Let E be a symmetric sequence space.*

1. *$\|x\|_\infty \leq \|x\|_E$ for $x \in E$, and $\|bx\|_E \leq \|x\|_E$ for $x \in E$ and every finitely supported b with $|b| \leq 1$.*
2. *For every sequence space D and $a \in c_{00}(\mathbb{N}, K)$, $\mu_a \in \mathcal{L}(E, D)$ and $\|\mu_a\|_{E\to D} = \sup\{\|ax\|_D : x \in c_{00}(\mathbb{N}, K),\ \operatorname{supp} x \subseteq \operatorname{supp} a,\ \|x\|_E \leq 1\}$.*

(3) *If D is symmetric, then $a \mapsto \|\mu_a\|_{E\to D}$ is a norm on $c_{00}(\mathbb{N},K)$ with $\|\mu_a\|_{E\to D} \geq \|a\|_\infty$ and $\|\mu_{e_t}\|_{E\to D} = 1$. It does not change when the coordinates of a are relabeled by an injective map $\mathbb{N}\to\mathbb{N}$ or when a is multiplied by a sequence of absolute value 1, and $\|\mu_a\|_{E\to D} \leq \|\mu_b\|_{E\to D}$ when $|a| \leq |b|$.*

(4) *If $\|y\|_D \leq C\|y\|_E$ for $y \in c_{00}(\mathbb{N},K)$, then $\|\mu_a\|_{E\to D} \leq C\|a\|_\infty$. If D is symmetric, then $\|\mu_a\|_{c_0\to D} = \|a\|_D$ and $\|\mu_a\|_{E\to c_0} = \|a\|_\infty$, where $c_0 = c_0(\mathbb{N},K)$.*

The proof is in Appendix P.

Definition 7.12. Let E be a symmetric sequence space, D a sequence space and $C \geq 0$, and let $\mathcal{P}_<$ be the strict triangle projection of the unit vectors: $\mathcal{P}_<(e_t \otimes e_{t'}) = e_t \otimes e_{t'}$ if $t < t'$ and 0 otherwise. The pair (E,D) satisfies the *product triangle bound* (PTB) with constant C if

$$\|\mathcal{P}_<((ax)\otimes(by))\|_{D\otimes_\pi D} \leq C\|\mu_a\|_{E\to D}\|x\|_E\|\mu_b\|_{E\to D}\|y\|_E \qquad (a,x,b,y \in c_{00}(\mathbb{N},K)). \tag{7.2}$$

We say that $\mathcal{P}_<$ is bounded on $E\otimes_\pi E$ if $\|\mathcal{P}_<(x\otimes y)\|_{E\otimes_\pi E} \leq C\|x\|_E\|y\|_E$ for some C and all $x,y \in c_{00}(\mathbb{N},K)$.

Theorem 7.13 (product triangle bound)**.** *Let E and D be symmetric sequence spaces, and suppose that (E,D) is good for Banach values in degree k, with a lift $Q\colon \mathcal{L}(E,D)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ as in Lemma 3.4. Then (E,D) satisfies* (PTB) *with a constant that depends only on k, K and $\|Q\|$. No inclusion between E and D is assumed, and $c_{00}(\mathbb{N},K)$ need not be dense in either.*

Lemma 7.14 (the triangle of one vector)**.** *Suppose that K is dense in $\kappa((X))$, and let D be a sequence space with $\|y\|_\infty \leq \|y\|_D$ for $y \in D$. Then for every $x \in c_{00}(\mathbb{N},K)$,*

$$\|x\|_2^2 \leq \frac{\|\mathcal{P}_<(x\otimes x)\|_{D\otimes_\pi D}}{1-\varrho} + \frac{\|x\|_\infty^2}{1-\varrho^2}.$$

Proposition 7.15 (ℓ^2-domination)**.** *Let E be a symmetric sequence space and D a sequence space with $\|y\|_\infty \leq \|y\|_D$ for $y \in D$. Suppose that $\|\xi\|_D \leq C'\|\xi\|_r$ for all $\xi \in c_{00}(\mathbb{N},K)$, for some $r \in (2,\infty]$ and $C' \geq 0$, and that (E,D) satisfies* (PTB) *with constant C. Put $\mathfrak{a} = CC'^2/(1-|\vartheta|)$ and $\mathfrak{b} = 1/(1-|\vartheta|^2)$. Then $\|x\|_2 \leq C_2\|x\|_E$ for all $x \in E$, where $C_2^2 = \mathfrak{a}+\mathfrak{b}$ if $r=\infty$ and $C_2^2 = \max\big((2\mathfrak{a})^{r/(r-2)}, 2\mathfrak{b}\big)$ if $r<\infty$. In particular E is ℓ^2-dominated.*

Proposition 7.16 (polarizable symmetric spaces)**.** *Let E be a symmetric sequence space in which $c_{00}(\mathbb{N},K)$ is dense. If (Γ^k_E) holds, then $\mathcal{P}_<$ is bounded on $E\otimes_\pi E$.*

The proofs of Theorem 7.13, Lemma 7.14 and Propositions 7.15 and 7.16 are in Appendix K.

7.5. **Nonarchimedean fibers.** Throughout this subsection K satisfies (L). The proofs, in Appendix K, take K dense in $\kappa((X))$; Lemma 7.1 extends them to every K satisfying (L).

Lemma 7.17 (marking)**.** *Let E be a normed space with an equivalent nonarchimedean norm and $\dim_{\widehat{K}} \widehat{E} = \infty$. There are $e_n \in E$, $n \in \mathbb{N}$, an equivalent nonarchimedean norm $\|\cdot\|'$ on $\widehat{E}$ with $\|\sum_n \xi_n e_n\|' = \max_n|\xi_n|$ for finitely supported (ξ_n) in $\widehat{K}$, and a $\widehat{K}$-linear map $\theta_E\colon \widehat{E} \to c_0(\mathbb{N},\widehat{K})$, contractive for $\|\cdot\|'$, that sends e_n to the nth unit vector. If E is complete, $\widehat{\psi}_E(\xi) = \sum_n \xi_n e_n$ defines an isometry of $c_0(\mathbb{N},\widehat{K})$ into $(E,\|\cdot\|')$ with $\theta_E\widehat{\psi}_E = \mathrm{id}$. If K is complete, θ_E takes values in $c_0(\mathbb{N},K)$.*

Theorem 7.18 (marking obstruction)**.** *Let E, E' be normed spaces, let $\psi\colon c_{00}(\mathbb{N},K) \to E$ and $\theta\colon E' \to \ell^\infty(\mathbb{N},\widehat{K})$ be bounded linear maps, with the supremum norm on $c_{00}(\mathbb{N},K)$, and let $a \mapsto u_a$ be a bounded linear map $c_{00}(\mathbb{N},K) \to \mathcal{L}(E,E')$ with*

$$\theta\big(u_a(\psi x)\big) = ax \qquad (a,x \in c_{00}(\mathbb{N},K)). \tag{7.3}$$

Let $\ell^\infty_{\widehat{K}} = \ell^\infty(\mathbb{N}, \widehat{K})$, regarded over K. Let F be normed, $\nu \in \mathrm{Alt}^k(E'; F)$, and $\theta_F \colon F \to \widehat{\Lambda}^k_\pi \ell^\infty_{\widehat{K}}$ bounded linear with $\theta_F \circ \nu = \omega_{\ell^\infty_{\widehat{K}}} \circ (\theta, \dots, \theta)$. Then no bounded k-linear $Q \colon c_{00}(\mathbb{N}, K)^k \to \mathrm{Alt}^k(E; F)$ satisfies $Q(a, \dots, a) = \nu \circ (u_a, \dots, u_a)$, and $A^k_{E,E';F}$ is analytic at no point. For $F = \Lambda^k_\pi E'$ and $F = \widehat{\Lambda}^k_\pi E'$, with $\nu = \omega_{E'}$ and θ_F induced by θ, the pair (E, E') is therefore not good, respectively not good for Banach values, in degree k.

Theorem 7.19 (nonarchimedean fibers over a local field)**.** *Let E be a normed space with an equivalent nonarchimedean norm.*

1. *The following are equivalent: $\dim_{\widehat{K}} \widehat{E} < \infty$; (E, E') is good for Banach values in degree k for every normed E'; (E', E) is good for Banach values in degree k for every normed E'.*
2. *If E is complete, these conditions are also equivalent to each of: (E, E) is good in degree k; (E, E) is good for Banach values in degree k.*
3. *If K is complete and E' also has an equivalent nonarchimedean norm, then (E, E') is good in degree k if and only if it is good for Banach values, if and only if $\dim E < \infty$ or $\dim E' < \infty$.*

Each failure is shown by an explicit pair whose action with values in the completed exterior power of its second space is analytic at no point; these values admit no equivalent nonarchimedean norm, while complete nonarchimedean values never fail, by Corollary 5.3.

By Theorem 5.6, $(c_0(\mathbb{N}, K), \ell^1(\mathbb{N}, K))$ is good for Banach values; so the hypothesis on E' in (3) cannot be dropped.

7.6. **Characteristic two.** Throughout this subsection condition (L) is not assumed. For a field $\Bbbk$, e_t are the unit vectors of $\Bbbk^n$, $[n] = \{1, \dots, n\}$, $1_S = \sum_{t \in S} e_t$ for $S \subseteq [n]$, also read in $c_{00}(\mathbb{N}, K)$, matrices are written in sans-serif, $\mathcal{P}_< \mathsf{Z}$ keeps the entries of Z above the diagonal and replaces the others by 0, and Ω_w is the alternating matrix of $w \in \Lambda^2 \Bbbk^n$. In characteristic 2, Ω_w is symmetric with zero diagonal and $\Omega_{x \wedge y} = xy^{\mathsf{T}} + yx^{\mathsf{T}}$. For $t \neq t'$, $\pi_{tt'}$ is the coordinate projection of $\Lambda^2 \Bbbk^n$ onto $\Bbbk\, e_t \wedge e_{t'}$, and for $a \in \Bbbk^n$, μ_a is multiplication by a, so that

$$\Lambda^2 \mu_a = \sum_{t<t'} a_t a_{t'}\, \pi_{tt'}. \tag{7.4}$$

For $L \subseteq [n]$ put $L^c = [n] \setminus L$, and for a vector a of indeterminates $a_L = \prod_{t \in L} a_t$; for a polynomial g, $[a_L b_{L^c} x_L y_{L^c}]g$ is the coefficient of the monomial $a_L b_{L^c} x_L y_{L^c}$ in g. The Pfaffian of an alternating matrix of even size over a commutative ring of characteristic 2 is the sum, over the partitions of the index set into pairs, of the products of the entries at the pairs; its square is the determinant, as for the generic alternating matrix over $\mathbb{Z}$. Sequence spaces are as in Definition 7.10, which does not use condition (L), and in this subsection a sequence space E is also assumed to satisfy $\|x\|_\infty \le \|x\|_E$ for $x \in E$; for finitely supported a, multiplication $\mu_a \colon x \mapsto ax$ then lies in $\mathcal{L}(E)$, as $\|ax\|_E \le \sum_t |a_t||x_t| \|e_t\|_E \le \big(\sum_t |a_t| \|e_t\|_E\big) \|x\|_E$. Put $\bar{\varphi}_E(n) = \max_{S \subseteq [n]} \|1_S\|_E$, which is nondecreasing in n. For $n \ge 1$, $c_0(n) = c_0([n], K)$.

Let $\Bbbk$ be a field of characteristic 2 and $n \ge 2$. A *multiplier lift* on $\Bbbk^n$ is a $\Bbbk$-bilinear map $\Psi \colon \Bbbk^n \times \Bbbk^n \to \mathrm{End}_\Bbbk(\Lambda^2 \Bbbk^n)$, $\Psi(a, b) = \sum_{t,t'} a_t b_{t'} \Psi_{tt'}$, with

$$\Psi_{tt'} + \Psi_{t't} = \pi_{tt'} \qquad (t \neq t'). \tag{7.5}$$

Its *output matrix* is $\mathsf{M}_\Psi(a, b, x, y) = \Omega_{\Psi(a,b)(x \wedge y)}$; regarding the entries of a, b, x, y as indeterminates, it is a matrix over $\Bbbk[a, b, x, y]$. The *ordering lift* $\Psi^<$ has $\Psi^<_{tt'} = \pi_{tt'}$ for $t < t'$ and $\Psi^<_{tt'} = 0$ for $t \ge t'$; thus $\Psi^<(a, b)$ is given by formula (3.4) with $k = 2$, $u_1 = \mu_a$, $u_2 = \mu_b$ and the natural order of $[n]$.

If $\Psi(a,a) = \Lambda^2\mu_a$ for all $a \in \Bbbk^n$, then Ψ is a multiplier lift: by (7.4), $\Psi_{tt} = \Lambda^2\mu_{e_t} = 0$ and $\Psi_{tt'} + \Psi_{t't} = \Lambda^2\mu_{e_t+e_{t'}} - \Lambda^2\mu_{e_t} - \Lambda^2\mu_{e_{t'}} = \pi_{tt'}$.

Theorem 7.20 (characteristic two)**.**

(1) (Identities.) *Let $\Bbbk$ be a field of characteristic 2 and $n \geq 2$. If n is even, every multiplier lift Ψ on $\Bbbk^n$ satisfies*

$$\sum_{L\subseteq[n],\ |L|=n/2} \left[a_L\, b_{L^c}\, x_L\, y_{L^c}\right] \operatorname{Pf} \mathsf{M}_\Psi(a,b,x,y) = 1, \tag{7.6}$$

and some $\mathsf{M}_\Psi(a,b,x,y)$ with $a,b,x,y \in \{0,1\}^n$ is invertible. For every $\Bbbk$-linear $\sigma\colon \Lambda^2\Bbbk^n \to \Bbbk^{n\times n}$ with $\wedge \circ \sigma = \mathrm{id}$, where $\wedge\mathsf{Z} = \sum_{t<t'}(\mathsf{Z}_{tt'} + \mathsf{Z}_{t't})\, e_t \wedge e_{t'}$, there are $S,T \subseteq [n]$ with $\operatorname{rank}\sigma(1_S \wedge 1_T) \geq n-1$, while $\sigma^<(x\wedge y) = \mathcal{P}_<(xy^{\mathsf{T}} + yx^{\mathsf{T}})$, $x,y \in \Bbbk^n$, defines such a map with strictly upper triangular values, of rank at most $n-1$. For every $\Bbbk$-linear $\tau\colon \Bbbk^{n\times n} \to \Lambda^2\Bbbk^n$ with $\tau(\Omega_w) = w$ for every w, and even n, there is $S \subseteq [n]$ with $\Omega_{\tau(1_S 1_S^{\mathsf{T}})}$ invertible.

(2) (Multiplier maps.) *Let K be a nontrivially normed field of characteristic 2, $k \geq 2$, and E a sequence space, with $\|e_t\|_E = 1$ for all t if $k \geq 3$. Every k-linear $Q\colon c_{00}(\mathbb{N},K)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi E)$ with $\|Q(a_1,\dots,a_k)\| \leq \|Q\| \prod_i \|a_i\|_\infty$ and $Q(a,\dots,a) = \Lambda^k_\pi \mu_a$ satisfies, for every even $n \geq 2$,*

$$n \leq k(k-1)\,\|Q\|\,\bar{\varphi}_E(n)^2. \tag{7.7}$$

(3) (Splittings.) *Let K be as in (2), E a sequence space and $n \geq 2$. Every section s of $\widehat{\wedge}^2_E$ has $\|s\| \geq (n-1)/\bar{\varphi}_E(n)^2$, every retraction ρ of alt^2_E has $\|\rho\| \geq \lfloor n/2\rfloor/\bar{\varphi}_E(n)^2$, and every bounded linear projection R_2 of $\widehat{\otimes}^2_\pi E$ onto $\widehat{\Gamma}^2_\pi E$ has $\|R_2\| \geq n/\bar{\varphi}_E(n)^2$. For the corresponding maps of $c_0(n)$ the bounds are $n-1$, $\lfloor n/2\rfloor$ and n.*

(4) (Consequences.) *Let K be as in (2), and let E be $c_0(\mathbb{N},K)$, $\ell^\infty(\mathbb{N},K)$, $\ell^r(\mathbb{N},K)$ with $2 < r < \infty$, or a symmetric sequence space with $\sup_n n/\varphi_E(n)^2 = \infty$. For every $k \geq 2$, (E,E) is not good for Banach values in degree k, $A^k_{E,E;F}$ is analytic at no point for $F = \Lambda^k_\pi E$ and for $F = \widehat{\Lambda}^k_\pi E$, neither (S^k_E) nor (R^k_E) holds, and $\widehat{\Lambda}^k_\pi E$ has no equivalent nonarchimedean norm; moreover (Γ^2_E) fails. For $2 < r_1, r_2 \leq \infty$, with $c_0(\mathbb{N},K)$ in place of $\ell^\infty(\mathbb{N},K)$, the pair $(\ell^{r_1}(\mathbb{N},K), \ell^{r_2}(\mathbb{N},K))$ is not good for Banach values in any degree $k \geq 2$.*

The proof is in Appendix N. In (7.7), $\bar{\varphi}_E(n) = 1$ for $c_0(\mathbb{N},K)$ and $\ell^\infty(\mathbb{N},K)$, and $\bar{\varphi}_E(n) = n^{1/r}$ for $\ell^r(\mathbb{N},K)$; so $\|Q\| \geq n/(k(k-1))$, respectively $\|Q\| \geq n^{1-2/r}/(k(k-1))$, and for $E = \ell^2(\mathbb{N},K)$ only $\|Q\| \geq 1/(k(k-1))$. The proof uses Q only on vectors supported in $[n+k-2]$, and for $k=2$ the same argument applies to $E = c_0(n)$: for even n, every lift for $(c_0(n), c_0(n))$ in degree 2, for Banach values, has norm at least $n/2$. Over Laurent series fields over infinite fields and over densely valued fields, the case $E = c_0(\mathbb{N},K)$ for odd p is Question 12.9(1), and the case $E = \ell^2(\mathbb{N},K)$ for $p=2$ is Question 12.9(2).

Remark 7.21 (exactness without splitting). Let K be nonarchimedean, of any characteristic, and let $D = c_0(\mathbb{N},K)$. Then

$$\|\operatorname{alt}^2_D w\|_\pi = 2\|w\|_\pi \qquad (w \in \widehat{\Lambda}^2_\pi D), \tag{7.8}$$

so the target sequence $\mathcal{R}^2_D$ of Definition 3.8 is exact. If K has characteristic 2, then $\mathcal{R}^2_D$ does not split: by Lemma 3.10 a splitting would give (R^2_D), which Theorem 7.20(4) excludes. So in characteristic 2 the failure of (R^2_D) is a failure of $\mathcal{R}^2_D$ to split, not of its exactness.

Proof. Let $\mathcal{O}$ be the closed unit ball of K, and let $\|\cdot\|_\infty$ be the supremum norm.

Step 1 (reduction). Since $\operatorname{alt}^2_D(x\wedge y) = x\otimes y - y\otimes x$, the left side of (7.8) is at most the right side. Both sides are continuous in w, and the union of the spaces $\Lambda^2_K K^m$, where $K^m \subseteq D$ is the span of $e_1,\dots,e_m$, is dense in $\widehat{\Lambda}^2_\pi D$. The coordinate projection of D onto K^m is contractive and fixes K^m, so for $w \in \Lambda^2_K K^m$ both norms may be computed in $c_0(m)$. Put $T = \operatorname{alt}^2_D w$, an alternating matrix in $K^{m\times m}$.

Step 2 (elimination). Let $a = T_{ij}$ be an entry of T of maximal absolute value; then $i \neq j$. Let B be the block of T on the rows and columns i,j, and C the part of the columns i,j outside these rows. As $|\det B| = |a|^2$, the matrix CB^{-1} has entries in $\mathcal{O}$. Let g and h be the unit vectors e_i and e_j plus the first and second columns of CB^{-1}, placed in the rows other than i,j; so $\|g\|_\infty = \|h\|_\infty = 1$. The Schur complement gives $T = a(gh^{\mathsf{T}} - hg^{\mathsf{T}}) + T'$, where $T' = \operatorname{alt}^2_D(w - a\, g\wedge h)$ vanishes on the rows and columns i,j. Repeating with T' gives $w = \sum_k a_k\, g_k \wedge h_k$ with $\|g_k\|_\infty = \|h_k\|_\infty = 1$, so $\|w\|_\pi \le \sum_k |a_k|$, and $T = U \operatorname{diag}(a_1 J, a_2 J, \dots) U^{\mathsf{T}}$ with $U \in GL_m(\mathcal{O})$, whose columns are the g_k, the h_k and unit vectors, and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

Step 3 (minors). Let $\mu_1 \ge \mu_2 \ge \dots \ge \mu_{2r}$ be the numbers $|a_k|$, each taken twice, and $\mu_{2r+1} = 0$. By the Cauchy–Binet formula the minors of T and of $\operatorname{diag}(a_k J)$ are $\mathcal{O}$-linear combinations of each other, so the largest absolute value of an $l\times l$ minor of T is $\mu_1\cdots\mu_l$. Let $T = \sum_q x_q y_q^{\mathsf{T}}$ be a finite representation, with costs $c_q = \|x_q\|_\infty \|y_q\|_\infty$ listed in decreasing order. The Cauchy–Binet formula and the ultrametric inequality bound every $l\times l$ minor of T by $c_1\cdots c_l$. So $\mu_1\cdots\mu_l \le c_1\cdots c_l$ for $l \le 2r$, and these c_l are positive. Since $c - \mu \ge \mu\log(c/\mu)$ for $c,\mu > 0$, Abel summation gives

$$\sum_{l\le 2r}(c_l - \mu_l) \ge \sum_{l\le 2r} \mu_l \log\frac{c_l}{\mu_l} = \sum_{l\le 2r}(\mu_l - \mu_{l+1}) \log\frac{c_1\cdots c_l}{\mu_1\cdots\mu_l} \ge 0.$$

Hence $2\|w\|_\pi \le 2\sum_k |a_k| = \sum_l \mu_l \le \sum_q c_q$, and taking the infimum over representations gives $2\|w\|_\pi \le \|\operatorname{alt}^2_D w\|_\pi$. □

8. Which pairs are good

Throughout this section κ, ϱ and K are as in condition (L), and each result states its degrees. Symmetric sequence spaces, their fundamental functions φ_E, ℓ^2-domination and the product triangle bound (PTB) are those of Definitions 7.10 and 7.12. For a sequence space, $\mathcal{P}_<$ is the strict triangle projection of the unit vectors $(e_t)_{t\in\mathbb{N}}$, and c_{00}, c_0 and ℓ^r stand for $c_{00}(\mathbb{N},K)$, $c_0(\mathbb{N},K)$ and $\ell^r(\mathbb{N},K)$.

8.1. **Symmetric spaces.**

Theorem 8.1 (symmetric spaces)**.** *Let E be a symmetric sequence space in which c_{00} is dense. The following are equivalent:*

1. *(E,E) is good for Banach values in some degree $k \ge p$;*
2. *(c_0,E) is good for Banach values in some degree $k \ge p$;*
3. *$\mathcal{P}_<$ is bounded on $E\otimes_\pi E$: there is C with $\|\mathcal{P}_<(x\otimes y)\|_{E\otimes_\pi E} \le C\|x\|_E\|y\|_E$ for all $x,y \in c_{00}$;*
4. *(S^k_E) holds for some $k \ge p$;*
5. *(R^k_E) holds for some $k \ge p$;*
6. *E is polarizable in some degree $k \ge p$;*
7. *for every normed D, the pairs (E,D) and (D,E) are good for Banach values in every degree.*

If these conditions hold, then (S^n_E), (R^n_E) *and* (Γ^n_E) *hold for every* $n \geq 1$, E *is* ℓ^2*-dominated, and there is* $c > 0$ *with* $\varphi_E(m)^2 \geq c\, m \ln m$ *for every* $m \geq 1$.

The proof is in Appendix K.

Corollary 8.2 (small fundamental functions)**.** *Let* E *be* ℓ^r *with* $2 \leq r < \infty$*, or* c_0*, or a symmetric sequence space in which* c_{00} *is dense and* $\varphi_E(m)^2 \leq Cm$ *for some* C *and every* $m \geq 1$*. Then for every* $k \geq p$ *neither* (E, E) *nor* (c_0, E) *is good for Banach values in degree* k*, and* (S^k_E), (R^k_E) *and* (Γ^k_E) *fail.*

Proof. For $E = \ell^r$ we have $\varphi_E(m) = m^{1/r}$, so $\varphi_E(m)^2 \leq m$ when $r \geq 2$, and for $E = c_0$ we have $\varphi_E(m) = 1$; in both cases c_{00} is dense. If one of the pairs were good in a degree $k \geq p$, or one of the splittings held, Theorem 8.1 would give $c\, m \ln m \leq \varphi_E(m)^2 \leq Cm$ for every m, which fails for large m. □

Corollary 8.3. *The space* c_0 *is not polarizable in any degree* $k \geq p$.

Proof. This is Corollary 8.2 for $E = c_0$. □

8.2. **Symmetric sources against r-normed targets.** For the targets of the next theorem, the necessary condition (PTB) of Theorem 7.13 is also sufficient.

Theorem 8.4 (r-normed targets)**.** *Let* E *be a symmetric sequence space in which* c_{00} *is dense, and let* D *be a symmetric sequence space with an equivalent* r*-norm for some* $r \in (2, \infty)$ *or with an equivalent nonarchimedean norm. The following are equivalent:*

(1) (E, D) *is good for Banach values in some degree* $k \geq p$*;*
(2) (E, D) *satisfies* (PTB) *with some constant;*
(3) E *is* ℓ^2*-dominated;*
(4) (E, D) *is good for Banach values in every degree.*

The proof is in Appendix K. Under (3), the coordinate map of E into $\ell^r(\mathbb{N}, \widehat{K})$, with $r = 3$ when D has an equivalent nonarchimedean norm, is a relative ordering splitting: it is S^n-split for every n, and every $u \in \mathcal{L}(E, D)$ factors through it (Theorem 5.9(1) and Step 1 of its proof, Corollary 5.10).

8.3. **The ℓ^r scale.**

Corollary 8.5 (the ℓ^r scale)**.** *Let* $1 \leq r_1, r_2 \leq \infty$*. For the pair* (ℓ^{r_1}, ℓ^{r_2})*, with* c_0 *in place of* ℓ^∞*, the following are equivalent: it is good for Banach values in some degree* $k \geq p$*; it is good for Banach values in every degree;* $\min(r_1, r_2) < 2$*, or* $r_1 = 2 < r_2$*. Hence* $\mathfrak{D}_B(\ell^{r_1}, \ell^{r_2})$ *is given by the following table, in which* $r_i = \infty$ *stands for* c_0*:*

$r_1 \backslash r_2$	$[1, 2)$	2	$(2, \infty)$	∞
$[1, 2)$	$\{1, 2, \dots\}$	$\{1, 2, \dots\}$	$\{1, 2, \dots\}$	$\{1, 2, \dots\}$
2	$\{1, 2, \dots\}$	$\{1, \dots, p-1\}$	$\{1, 2, \dots\}$	$\{1, 2, \dots\}$
$(2, \infty)$	$\{1, 2, \dots\}$	$\{1, \dots, p-1\}$	$\{1, \dots, p-1\}$	$\{1, \dots, p-1\}$
∞	$\{1, 2, \dots\}$	$\{1, \dots, p-1\}$	$\{1, \dots, p-1\}$	$\{1, \dots, p-1\}$

The proof is in Appendix K.

8.4. **Degrees and mechanisms.**

Corollary 8.6 (one critical degree gives all)**.** *For each of the following pairs* (E, D)*, the set* $\mathfrak{D}_B(E, D)$ *is* $\{1, \dots, p-1\}$ *or* $\{1, 2, \dots\}$*:*

(1) (E, E) *and* (c_0, E)*, for* E *as in Theorem 8.1;*
(2) (E, D) *as in Theorem 8.4;*

(3) (ℓ^{r_1}, ℓ^{r_2}) *for* $1 \le r_1, r_2 \le \infty$, *with* c_0 *in place of* ℓ^∞;
(4) (E, D) *for complete* K *and spaces* E, D *with equivalent nonarchimedean norms.*

Proof. By Corollary 6.3(1), $\{1, \dots, p-1\} \subseteq \mathfrak{D}_B(E, D)$. It remains to show that each listed pair that is good for Banach values in one degree $k \ge p$ is good for Banach values in every degree. For (1)–(3) this is the equivalence of conditions (1), (2) and (7) of Theorem 8.1, of (1) and (4) of Theorem 8.4, and Corollary 8.5. For (4), Theorem 7.19(3) shows that goodness in a degree $k \ge p$ forces $\dim E < \infty$ or $\dim D < \infty$; that space has continuous coordinates by Lemma A.3, and Proposition 5.1(2) gives a lift in every degree. □

Remark 8.7 (mechanisms). In every degree, each pair good for Banach values in Theorems 8.1, 8.4 and 7.19 and Corollary 8.5 has a splitting decomposition (Definition 6.12) with one piece of one of three kinds: an ordering splitting of the source or the target (Theorem 5.6, Proposition 5.7); the relative ordering splitting of the source described after Theorem 8.4; or a splitting of the source or the target by finite coordinates on its completion (proof of Theorem 7.19). In each of these results, goodness implies the stated criterion by forcing (Theorems 7.7, 7.13 and 7.18).

8.5. **Spreading spaces.**

Definition 8.8. A sequence space E is *spreading* if it satisfies Definition 7.10 for symmetric sequence spaces with invariance under permutations of $\mathbb{N}$ replaced by the following condition: for $d \ge 1$ and $f \in \mathbb{R}^d$, the norm $\|\sum_{i=1}^d f_i e_{t_i}\|_E$ is the same for all $t_1 < \dots < t_d$ in $\mathbb{N}$.

Symmetric sequence spaces are spreading. In Banach space terms, the unit vectors of a spreading sequence space are 1-unconditional and 1-spreading, that is, 1-subsymmetric. A map from a subset of $\mathbb{N}$ to $\mathbb{N}$ is *increasing* if it preserves the strict order.

Theorem 8.9 (spreading spaces)**.** *Let* E *and* D *be spreading sequence spaces.*

(1) *Lemma 7.11 holds for* E *and* D, *with relabeling by an injective map* $\mathbb{N} \to \mathbb{N}$ *replaced by relabeling along a map increasing on the support of* a; *so* (PTB) *is defined for* (E, D) *by Definition 7.12.*
(2) *Theorem 7.13, Propositions 7.15 and 7.16, Theorems 8.1 and 8.4, and Corollaries 8.2 and 8.6(1),(2) hold with spreading sequence spaces in place of symmetric ones, and so does Remark 8.7 for the pairs of Theorems 8.1 and 8.4.*
(3) *Over every nontrivially normed field* K *of characteristic* 2, *without condition* (L), *Theorem 7.20(4) holds for every spreading sequence space* E *over* K *with* $\sup_n n/\varphi_E(n)^2 = \infty$.

The proof is in Appendix K. The spreading spaces form a larger class than the symmetric ones, also up to equivalent norms.

Example 8.10 (Garling spaces)**.** In this example we identify $\mathbb{N}$ with $\{1, 2, 3, \dots\}$ through its order. Let $0 < \alpha < 1$, $1 \le q < \infty$ and $\beta = (1-\alpha)/q$. For $f \in c_{00}(\mathbb{N}, \mathbb{R})$ put

$$\|f\|_{\alpha,q} = \sup\Big\{ \Big(\sum_{j=1}^{L} j^{-\alpha} |f(t_j)|^q \Big)^{1/q} : L \ge 1,\ t_1 < \dots < t_L \Big\},$$

a norm of the kind introduced by Garling [7] and studied in [1], and let $E_{\alpha,q}$ be the closure of c_{00} in the space of all $x \in \ell^\infty(\mathbb{N}, K)$ with $\|x\| = \sup_n \| |x 1_{[n]}| \|_{\alpha,q} < \infty$.

(1) $E_{\alpha,q}$ is a spreading sequence space in which c_{00} is dense, and $m^{2\beta} \le \varphi_{E_{\alpha,q}}(m)^2 \le (1-\alpha)^{-2/q} m^{2\beta}$ for $m \ge 1$.
(2) No norm on c_{00} invariant under permutations of $\mathbb{N}$ is equivalent to that of $E_{\alpha,q}$, and the unit vectors of $E_{\alpha,q}$, in the order of $\mathbb{N}$, satisfy (5.1) for no $r \in (0, 2)$ and $C \ge 1$.

(3) If $2\beta \le 1$, then for every $k \ge p$ neither $(E_{\alpha,q}, E_{\alpha,q})$ nor $(c_0, E_{\alpha,q})$ is good for Banach values in degree k, and $(\mathrm{S}^k_{E_{\alpha,q}})$, $(\mathrm{R}^k_{E_{\alpha,q}})$ and $(\Gamma^k_{E_{\alpha,q}})$ fail. If $2\beta < 1$, then over every nontrivially normed field K of characteristic 2, with $E_{\alpha,q}$ formed over K in the same way, the pair $(E_{\alpha,q}, E_{\alpha,q})$ is good for Banach values in no degree $k \ge 2$.
(4) If $2\beta > 1$, then $\|x\|_2^2 \le \big(\sum_{m\ge1} m^{-2\beta}\big)\|x\|^2$ for $x \in E_{\alpha,q}$.

The proof is in Appendix K. For $2\beta > 1$ the necessary conditions of Theorem 8.1 hold: $E_{\alpha,q}$ is ℓ^2-dominated by (4), and $\varphi_{E_{\alpha,q}}(m)^2 \ge m^{2\beta}$ grows faster than $m \ln m$; by (2), Theorem 5.6 does not apply to the unit vectors in their natural order (Question 12.16).

9. Goodness without splitting decompositions

The good pairs of Remark 8.7 all have splitting decompositions; this section shows that goodness does not require one.

9.1. One-dimensional operator spaces.

Proposition 9.1. *Let E and D be normed spaces with $\mathcal{L}(E,D) = K\,u_0$ for some nonzero $u_0 \in \mathcal{L}(E,D)$.*

(1) *For every $k \ge 1$ the pair (E,D) is good in degree k, and the map*
$$Q(\lambda_1 u_0, \dots, \lambda_k u_0) = \lambda_1 \cdots \lambda_k \, \Lambda^k_\pi u_0$$
is a lift as in Lemma 3.4.
(2) *If D is complete, then for every $k \ge 1$ the pair (E,D) has a splitting decomposition of degree k if and only if u_0 is S^k-split or R^k-split.*
(3) *If $\dim D \ge 2$, then every bounded linear functional on E is zero.*

The proof is in Appendix P.

9.2. Rigid Banach spaces.

The next theorem transfers the rigid subspace of L_0 of Kalton and Roberts [15] to separable complete nonarchimedean fields.

Theorem 9.2. *Let K be a complete nonarchimedean field with nontrivial absolute value that is separable as a metric space. There are infinite-dimensional Banach spaces $\mathfrak{X}_1, \mathfrak{X}_2, \dots$ over K such that, for all $\alpha, \beta \ge 1$,*
$$\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\beta) = \begin{cases} K\,\mathrm{id}_{\mathfrak{X}_\alpha}, & \alpha = \beta, \\ 0, & \alpha \ne \beta. \end{cases}$$

The proof is in Appendix M. By Proposition 9.1(3) with $E = D = \mathfrak{X}_\alpha$ and $u_0 = \mathrm{id}_{\mathfrak{X}_\alpha}$, every bounded linear functional on $\mathfrak{X}_\alpha$ is zero.

Theorem 9.3. *Let κ be a finite field of characteristic p and $K = \kappa((X))$, with the absolute value of condition (L) of Section 7. For suitable choices in the construction of Theorem 9.2, the Banach space $\mathfrak{X} = \mathfrak{X}_1$ has the following properties.*

(1) $\mathcal{L}(\mathfrak{X}, \mathfrak{X}) = K\,\mathrm{id}_{\mathfrak{X}}$.
(2) *The pair $(\mathfrak{X}, \mathfrak{X})$ is good in every degree, hence good for Banach values in every degree.*
(3) *For every $k \ge p$, neither $(\mathrm{S}^k_{\mathfrak{X}})$ nor $(\mathrm{R}^k_{\mathfrak{X}})$ holds.*
(4) *For every $k \ge p$, the pair $(\mathfrak{X}, \mathfrak{X})$ has no splitting decomposition of degree k.*

The proof is in Appendix M. Since $\mathfrak{X}$ has no nonzero bounded functional, item (3) is proved by forcing through L^r-valued readouts (Corollary M.12) instead of the coordinates of Section 7.

Theorem 9.4. *Let κ be a finite field of characteristic p and $K = \kappa((X))$, with the absolute value of condition (L). Let $\mathfrak{X}_1, \mathfrak{X}_2, \dots$ be the spaces of Theorem 9.2, built with the choices of Theorem 9.3, and let E be the completed ℓ^1-sum of the $\mathfrak{X}_\alpha$, with the norm $\|x\| = \sum_\alpha \|x_\alpha\|$ for $x = (x_\alpha)_{\alpha \geq 1}$.*

1. *For a bounded sequence $\lambda = (\lambda_\alpha)_{\alpha \geq 1}$ in K put $u_\lambda x = (\lambda_\alpha x_\alpha)_\alpha$. Every bounded operator on E is some u_λ, and $\|u_\lambda\| = \sup_\alpha |\lambda_\alpha|$.*
2. *The pair (E, E) is good for Banach values in every degree.*
3. *For every $k \geq p$, the pair (E, E) has no generalized decomposition of degree k, that is, no decomposition into finitely many source, target and parameter pieces (Definition 6.17).*

The proof is in Appendix M. In Theorem 9.3 the lift Q of Proposition 9.1(1) is that of one parameter piece through K; in Theorem 9.4 the lift is a block assembly (Proposition 6.27) of infinitely many such lifts, which no finite family of pieces replaces. Question 12.4 asks whether these mechanisms and the operations of Section 6 yield every pair good for Banach values.

10. Bundles, tangent bundles and differential forms

10.1. **From a bad operator family to changes of frame.** Transition functions take values in invertible operators; for a single Banach fiber, invertible families impose the same condition as all operator families (Proposition 11.2(1)).

Proposition 10.1 (shear realization)**.** *Suppose $A = A^k_{E,D;F}$ is not analytic at $u_0 \in \mathcal{L}(E, D)$. Set $H = D \oplus E$, with the maximum norm, and*

$$g(u)(d, e) = (d + ue, e).$$

Both g and its inverse family are affine analytic. Nevertheless, the family of pullback operators $g(u)^*$ on $\mathrm{Alt}^k(H; F)$ is not analytic at u_0.

Proof. The inverse family is $g(-u)$. Let $\pi_D \colon H \to D$ be the projection and $i_E(e) = (0, e)$. Define bounded linear maps $J \colon \mathrm{Alt}^k(D; F) \to \mathrm{Alt}^k(H; F)$ by $Jm = m \circ \pi_D^{\times k}$ and $R \colon \mathrm{Alt}^k(H; F) \to \mathrm{Alt}^k(E; F)$ by restriction along i_E. Since $\pi_D g(u) i_E = u$,

$$R \circ g(u)^* \circ J = A(u). \tag{10.1}$$

Bounded linear compression preserves analyticity, so $u \mapsto g(u)^*$ is not analytic at u_0. □

Corollary 10.2 (shear bundle)**.** *Suppose $A = A^k_{E,D;F}$ is not analytic at $u_0 \in \mathcal{L}(E, D)$, and let H and g be as in Proposition 10.1. Take the base $M = \mathcal{L}(E, D)$ and the trivial bundle $M \times H$. Its product trivialization and the trivialization $(u, z) \mapsto (u, g(u)z)$ form an analytic bundle atlas, and the induced alternating atlas on the trivial bundle with fiber $\mathrm{Alt}^k(H; F)$ is not analytic. Equivalently, the analytic bundle automorphism $(u, z) \mapsto (u, g(u)z)$ does not induce an analytic morphism on the alternating product bundle.*

Proof. The transitions are $g(u)$ and $g(-u)$, affine analytic by Proposition 10.1. The induced alternating transitions are pullback by $g(u)$ and $g(-u)$, and the induced morphism is locally pullback by $g(u)$; by Proposition 10.1, $u \mapsto g(u)^*$ is not analytic at u_0. □

Thus every failure of the hom-space criterion has a bundle realization, and Theorem 7.2 gives fibers for which it occurs at every point of the base. Applied to analytic families $x \mapsto u(x)$, the shear also makes the familywise hypothesis of Theorem 2.1 necessary for all fibers and frames; Section 11 studies it (Corollary 11.3, Theorems 11.9 and 11.10).

10.2. **Tangent bundles and individual forms.**

Proposition 10.3 (tangent and pullback obstructions)**.** *Suppose K is complete, has characteristic $p > 0$, is not spherically complete, and $k \geq p$.*

(1) *There is an analytic Banach manifold with two global analytic charts whose induced transition on scalar alternating k-forms is analytic at no point.*
(2) *There are Banach spaces N, Y, a polynomial analytic map $h\colon N \to Y$, and an analytic map $\omega_0\colon Y \to \mathrm{Alt}^k(Y;K)$ such that $h^*\omega_0$, expressed in the product coordinates on N, is analytic at no point as an $\mathrm{Alt}^k(N;K)$-valued map. After inclusion in $\mathrm{Mult}^k(N;K)$ it is analytic everywhere.*

The proof is in Appendix P.

Assertion (2) separates intrinsic analyticity of individual forms from analyticity of bundle transitions; Section 10.3 builds a calculus on the ambient notion.

The chart ψ of the proof of Proposition 10.3 realizes any nowhere analytic action $A^k_{E,D;F}$; with Theorem 7.2(1) it works over every field of characteristic p.

Proposition 10.4 (tangent obstructions with other values)**.** *Let K be a nontrivially normed field of characteristic $p > 0$, complete or not, and let $k \geq p$.*

(1) *There are a Banach space F and an analytic Banach manifold with two global analytic charts whose induced transition on F-valued alternating k-forms is analytic at no point.*
(2) *There is an analytic Banach manifold with two global analytic charts whose induced transition on tangent-valued alternating k-forms is analytic at no point.*
(3) *If K is complete and not spherically complete, a manifold as in (2) exists with a nonarchimedean Banach model.*

The target F constructed for (1) admits no equivalent nonarchimedean norm; for $K = \mathbb{F}_q((X))$ as in Corollary C.8, the model space can be chosen nonarchimedean. The model space constructed for (2) admits no equivalent nonarchimedean norm.

The proof is in Appendix P.

Bourbaki's question. For $k \geq 1$, Alt^k on Banach spaces is the functor α_k of [3, n° 7.8.1], and $A^k_{E,D;F}(u) = \alpha_k(\mathrm{id}_F, u)$. There K is $\mathbb{R}$, $\mathbb{C}$ or complete nonarchimedean, norms over a nonarchimedean K are nonarchimedean, and class C^r means analytic if $K \neq \mathbb{R}$, in a sense that agrees with ours between Banach spaces. In that setting α_k is a vector functor of class C^r if $k! \neq 0$ in K (Corollary 7.3), and for every k if K is discretely valued (Corollary 5.3 and the proof of Corollary 5.15). It is not one if $k! = 0$ in K and K is not spherically complete, since Theorem 7.2(2) gives nonarchimedean Banach spaces E, D with $A^k_{E,D;K}$ analytic at no point; so the definition in [3, n° 7.8.1] needs a restriction. By Corollary 10.2 and Proposition 10.3, it is needed for the bundles themselves. Source spaces of countable type suffice in every degree, by the following corollary of Theorem 5.4.

Corollary 10.5 (countable type and Bourbaki's α_k)**.** *Let K be complete and nonarchimedean, let E, D be normed spaces, at least one of them of countable type, and let F be a nonarchimedean Banach space. For every normed F', the joint action B of Proposition 2.2 on $\mathcal{L}(E,D)\times\mathcal{L}(F,F')$ has a bounded $(k+1)$-linear lift. In the setting of the paragraph* Bourbaki's question*, α_k is therefore a vector functor of the class considered there on source spaces of countable type, in every degree.*

Proof. If Q lifts $A^k_{E,D;F}$ (Theorem 5.4), then $(u_1, \dots, u_k, g) \mapsto (m \mapsto g \circ Q(u_1, \dots, u_k)(m))$ lifts B, as in the proof of Corollary 5.15. □

Our results do not decide the case of a densely valued, spherically complete field with $k! = 0$ (Section 12).

A derivative adapted to positive characteristic. Yaël Dillies asked whether $A = A^k_{E,D;F}$ is analytic in the calculus of [2, Definition 4.7 and Theorem 5.4], where a C^k map f has Taylor coefficients $a_n(x,h)$ with $n!\,a_n(x,h) = d^n f(x)(h,\dots,h)$. There A is smooth [2, Remark 4.2 and Lemma 10.1] and $a_k(u_0,u) = A(u)$, while $d^k A(u_0)(u,\dots,u) = k!\,A(u)$ vanishes if $k! = 0$. Analyticity is not restored: it asks for a bounded k-linear lift (Proposition 2.5), and Theorem 7.2 gives spaces with none when $k! = 0$ in K. Section 10.4 changes the coefficients instead: over every nonarchimedean field, with bounded polynomial laws as Taylor coefficients, A is divided-power analytic on $\mathcal{L}(E,D)$ in every characteristic.

10.3. **Analytic differential forms.** Throughout this subsection K is complete and nontrivially normed, manifold models are normed spaces, and charts and maps are C^ω, so that their derivatives are analytic; on the Banach coefficient spaces this is power-series analyticity.

Definition 10.6. An *ambient analytic scalar k-form* on M is a family $\eta_x \in \mathrm{Alt}^k(T_xM;K)$ whose representative in each chart is analytic after the inclusion

$$j\colon\ \mathrm{Alt}^k(P;K) \hookrightarrow \mathrm{Mult}^k(P;K).$$

Write $\Omega^k_{\mathrm{amb}}(M)$ for the space of such forms. In degree zero this is the space of analytic scalar functions.

Under a coordinate change $\psi = \phi' \circ \phi^{-1}$ the representatives satisfy $j\eta_\phi(y) = (j\eta_{\phi'}(\psi(y))) \circ (D\psi(y),\dots,D\psi(y))$, which is analytic by bounded multilinear composition; so any analytic atlas may be used.

For an ambient k-form η, an ambient l-form ζ and $S \subseteq \{1,\dots,k+l\}$, let v_S list the arguments v_i, $i \in S$, in increasing order of i, and let $\epsilon(S)$ be the shuffle sign of Section 6.1. The product is the shuffle formula

$$(\eta \wedge \zeta)(v_1,\dots,v_{k+l}) = \sum_{|S|=k} \epsilon(S)\eta(v_S)\zeta(v_{S^c}), \tag{10.2}$$

a bounded bilinear operation on multilinear coefficient spaces, of norm at most $\binom{k+l}{k}$, which is alternating on alternating inputs. On an open subset of a model space P, the exterior derivative of an ambient coefficient map η of degree k is

$$(d\eta)_y(v_0,\dots,v_k) = \sum_{i=0}^{k} (-1)^i D\eta(y)(v_i)(v_0,\dots,\widehat{v_i},\dots,v_k). \tag{10.3}$$

Theorem 10.7 (ambient differential calculus)**.** *Ambient analytic forms define a sheaf of graded algebras on every analytic manifold. Analytic maps induce pullbacks, and there is an exterior derivative*

$$d\colon \Omega^k_{\mathrm{amb}}(M) \longrightarrow \Omega^{k+1}_{\mathrm{amb}}(M).$$

The product is associative and graded commutative, $d^2 = 0$, pullback commutes with both product and exterior derivative, and

$$d(\eta \wedge \zeta) = d\eta \wedge \zeta + (-1)^k \eta \wedge d\zeta \qquad (\eta \in \Omega^k_{\mathrm{amb}}(M)).$$

Thus $\Omega^\bullet_{\mathrm{amb}}$ is a contravariant differential graded algebra construction in every characteristic.

The proof is in Appendix P.

Proposition 10.8 (comparison with alternating-bundle sections)**.** *Every intrinsically analytic alternating-valued coefficient map is ambient analytic. If, for the model space P, the inclusion j has a bounded linear retraction, then the natural alternating cotangent atlas is analytic and its analytic sections are precisely the ambient analytic forms. The same conclusion holds when P has finite continuous coordinates.*

Proof. The first statement follows by composition with j. For a bounded retraction R, the identity $\eta = R(j\eta)$ gives the converse. Moreover, $(u_1, \dots, u_k) \mapsto \big[m \mapsto R\big(m \circ (u_1, \dots, u_k)\big)\big]$ is a bounded multilinear lift of the precomposition action, so the induced alternating cotangent transitions are analytic. For finite continuous coordinates, analytic descent through closed subspaces applies to the coefficient map itself, and Corollary 5.15 supplies the analytic bundle atlas. □

The retraction exists if $k!$ is invertible, by normalized antisymmetrization, and if K is spherically complete and nonarchimedean, by Theorem 5.2.
Exact constant forms in degree p. Here K is any nontrivially normed field of characteristic $p > 0$, Y and F are normed, and $\mathcal{A}_Y \colon \mathrm{Mult}^p(Y;F) \to \mathrm{Alt}^p(Y;F)$ is unnormalized antisymmetrization (Lemma 3.10). Define $(d\eta)_y \in \mathrm{Alt}^p(Y;F)$ by (10.3) for every map η from an open subset of Y to $\mathrm{Alt}^{p-1}(Y;F)$ differentiable at y. Constant forms are closed, and (R^p_Y) is the exactness of one of them.

Proposition 10.9 (exact constant forms)**.** *For $\omega \in \mathrm{Alt}^p(Y;F)$ the following are equivalent:*

1. *there are a map η from an open subset of Y to $\mathrm{Alt}^{p-1}(Y;F)$ and a point y at which η is differentiable with $(d\eta)_y = \omega$;*
2. *$\omega = d\eta$ on Y for a bounded linear map $\eta \colon Y \to \mathrm{Alt}^{p-1}(Y;F)$;*
3. *$\omega = \mathcal{A}_Y\beta$ for some $\beta \in \mathrm{Mult}^p(Y;F)$.*

For $F = \Lambda^p_\pi Y$ and $\omega = \omega_Y$ they hold if and only if alt^p_Y has a bounded linear left inverse, and then (E,Y) is good in degree p for every normed E. For $F = \widehat{\Lambda}^p_\pi Y$ and $\omega = \omega_Y$ they hold if and only if (R^p_Y) holds, and then (E,Y) is good for Banach values in degree p for every normed E.

The proof is in Appendix P.

Under condition (L) of Section 7.1, for $Y = c_0(\mathbb{N}, K)$ the closed constant form ω_Y with values in $\widehat{\Lambda}^p_\pi Y$ has no primitive near any point, by Theorem 7.7.

10.4. **Divided-power analyticity.** Throughout this subsection K is nonarchimedean, $\varpi \in K$ satisfies $0 < |\varpi| < 1$, and P, W and Y are normed spaces, not necessarily complete or nonarchimedean.

Definition 10.10. Let $n \geq 0$.

1. A map $a \colon P \to W$ is a *bounded polynomial law of degree n* if, for all $h_1, \dots, h_r \in P$, the map $s \mapsto a(s_1h_1 + \cdots + s_rh_r)$ on K^r is a homogeneous polynomial of degree n with coefficients in W, and $\|a\| = \sup_{h \neq 0} \|a(h)\|/\|h\|^n$ is finite. For $\alpha = (\alpha_1, \dots, \alpha_r)$ with integers $\alpha_i \geq 0$ and $|\alpha| = \alpha_1 + \cdots + \alpha_r = n$, $a^{[\alpha]}(h_1, \dots, h_r) \in W$ is the coefficient of $s_1^{\alpha_1} \cdots s_r^{\alpha_r}$, unique since K is infinite.
2. A map $a \colon P \to W$ is *ambient of degree n* if $ja(h) = \beta(h, \dots, h)$ for all $h \in P$, for a linear isometry j of W into a normed space Z and a bounded n-linear map $\beta \colon P^n \to Z$.
3. A map f from an open subset U of P to W is *divided-power analytic*, or *DP*, at $x_0 \in U$ if there are $R > 0$ and bounded polynomial laws $a_n \colon P \to W$ of degree n, $n \geq 0$, with $\sup_n \|a_n\| R^n < \infty$ and $f(x_0 + h) = \sum_n a_n(h)$ for $\|h\| < R$. It is *DP on U* if it is DP at every point of U.

Without the bound on $\|a\|$, the maps in Definition 10.10(1) are the maps $P \to W$ induced by the polynomial laws on (P, W) in the sense of Roby [22] that are homogeneous of degree n, and each of these maps determines its law, since K is an infinite field and the linear forms on the vector space W separate its points [22, Theorem I.1 and Propositions I.1, I.8 and IV.6].

Theorem 10.11 (divided-power calculus)**.**

(1) (Mixed terms.) *If $b\colon W \to Y$ is a bounded polynomial law of degree m, then*

$$\|b^{[\alpha]}(w_1, \dots, w_r)\| \le (8/|\varpi|)^m \|b\| \, (\|w_1\| + \cdots + \|w_r\|)^m \qquad (w_1, \dots, w_r \in W,\ |\alpha| = m).$$

Every bounded polynomial law is continuous and DP on its whole domain.

(2) (Composition.) *Let f map an open subset of P to W and g an open subset of W to Y. If f is DP at x_0 and g is DP at $f(x_0)$, then $g \circ f$, which is defined near x_0, is DP at x_0.*

(3) (Comparison.) *A map ambient of degree n, for instance the diagonal of a bounded n-linear map, is a bounded polynomial law of degree n, and a map analytic at a point is DP there. If P has finite coordinates, a map DP at a point is analytic there.*

(4) (Alternating maps.) *Let E, D, F and F' be normed spaces and $k \ge 1$. The maps $A^k_{E,D;F}\colon \mathcal{L}(E, D) \to \mathcal{L}(\mathrm{Alt}^k(D; F), \mathrm{Alt}^k(E; F))$, $\mathcal{L}(E, D) \to \mathcal{L}(\Lambda^k_\pi E, \Lambda^k_\pi D)$, $u \mapsto \Lambda^k_\pi u$, and $\mathcal{L}(E, D) \to \mathcal{L}(\widehat{\Lambda}^k_\pi E, \widehat{\Lambda}^k_\pi D)$, $u \mapsto \widehat{\Lambda}^k_\pi u$, are ambient of degree k, and the joint action B of Proposition 2.2, restricted to alternating maps, is ambient of degree $k + 1$ as a map from $\mathcal{L}(E, D) \times \mathcal{L}(F, F')$ to $\mathcal{L}(\mathrm{Alt}^k(D; F), \mathrm{Alt}^k(E; F'))$. As bounded polynomial laws, all four have norm at most 1.*

The proof is in Appendix P.

Corollary 10.12 (divided-power bundles)**.** *Theorem 2.1 holds with C^n replaced by DP on open sets, for manifolds and vector bundles whose transition maps are DP on open sets. The alternating bifunctor $(E, F) \mapsto \mathrm{Alt}^k(E; F)$ and the functors $E \mapsto \Lambda^k_\pi E$ and $E \mapsto \widehat{\Lambda}^k_\pi E$ satisfy its hypothesis. So in every characteristic and for all normed fibers, the alternating and the projective exterior power bundles of a DP vector bundle are DP vector bundles, with DP induced morphisms.*

Proof. Step 1 (the calculus). By Theorem 10.11(2), maps that are DP on open sets compose; by (3) and (1), bounded multilinear maps, in particular bounded linear maps and evaluation of operators, and constants are DP on their whole domains. A map into a product, with the maximum norm, is DP at a point when its components are, since a pair of bounded polynomial laws of degree n is a bounded polynomial law of degree n. Being DP at a point depends only on the map near that point.

Step 2 (the theorem). The proof of Theorem 2.1 uses of the class C^n only these properties: Step 1 there composes the transitions with the maps of the functor on hom spaces and evaluates operators, Step 2 composes local operator families, Step 3 uses constants and Step 4 restricts to open subsets of P.

Step 3 (the functors). By Theorem 10.11(4) and (1), the maps of the three functors on hom spaces are bounded polynomial laws and DP on their whole domains. □

The two notions differ: if $\operatorname{char} K = p > 0$ and $k \ge p$, then $A^k_{E,D;F}$ is DP on $\mathcal{L}(E, D)$ by Theorem 10.11(4) and (1), and for the spaces of Theorem 7.2 it is analytic at no point.

Bounded polynomial laws need not be ambient. Let $p = \operatorname{char} K > 0$ and $c_0 = c_0(\mathbb{N}, K)$, and identify K^m with the span of the first m unit vectors, $m \ge 1$. Let $\Gamma^p(K^m) \subseteq (K^m)^{\otimes p}$ be the space of symmetric tensors, put $Q_m = (K^m)^{\otimes p}/\Gamma^p(K^m)$, and let $\bar{z} \in Q_m$ be the image of $z \in (K^m)^{\otimes p}$. Let I_m be the set of the p^{pm} tuples $l = (y^1_l, \dots, y^p_l)$ of vectors in K^m with entries

in the prime field $\mathbb{F}_p$, and put $y_l = y_l^1 \otimes \cdots \otimes y_l^p$. Let $N_m \subseteq \ell^1(I_m, K)$ consist of the vectors $(\phi(\bar{y}_l))_{l \in I_m}$, ϕ a linear form on Q_m, let $G_m = \ell^1(I_m, K)/N_m$ with the quotient norm, and let $\varepsilon_l \in G_m$ be the image of the unit vector at l. Let G be the ℓ^1-sum of the spaces G_m, $m \geq 1$: its elements are the sequences whose m-th term lies in G_m and whose norm, the sum of the norms of the terms, is finite; we read G_m as the m-th summand of G. Let $H = K \oplus c_0 \oplus G$, with the maximum norm; it is a Banach space if K is complete. Let V be the span of the tensors $u^{\otimes p} \otimes g$, $u \in c_0$, $g \in G$, in $c_0^{\otimes p} \otimes G$, with the norm

$$\nu(t) = \inf\Big\{\sum_i |c_i| \|u_i\|^p \|g_i\| : t = \sum_i c_i\, u_i^{\otimes p} \otimes g_i\Big\},$$

which dominates the projective norm π of $c_0^{\otimes p} \otimes G$. Both are norms, since the coordinate forms of c_0, and the linear forms on the spaces G_m composed with the projections of G onto G_m, are bounded and separate points.

Proposition 10.13 (laws that are not ambient)**.** *Assume that either (a) K is discretely valued, that is, $|K^\times| = |\vartheta|^{\mathbb{Z}}$ for some $\vartheta \in K$ with $0 < |\vartheta| < 1$, or (b) $p = 2$. Put $t_m = \sum_{l \in I_m} y_l \otimes \varepsilon_l$.*

1. *$t_m \in V$ and $\pi(t_m) \leq p^{pm}$; under (a), $\nu(t_m) \geq m p^{pm} / \big(2^p M_{|\vartheta|}(1 + \lfloor \log_p m \rfloor)\big)$ with $M_{|\vartheta|} = \max_{j \geq 0}(j+1)|\vartheta|^j$, and under (b), $\nu(t_m) \geq m 4^{m-1}$.*
2. *For $n \geq p+1$, $R_n(\lambda, x, g) = \lambda^{n-p-1}\, x^{\otimes p} \otimes g$, with $\lambda^0 = 1$, defines a bounded polynomial law $R_n \colon H \to V$ of degree n and norm at most 1 that is not ambient of degree n, also as a map into the completion of V.*

The proof is in Appendix P. Under (a) it rests on Lemma P.1, by which every projection of $(\kappa^m)^{\otimes p}$ onto the symmetric tensors, κ the residue field of K, takes, on average over the tensors y_l, which have entries in $\mathbb{F}_p$, values whose $m \times m^{p-1}$ matrices of entries have large rank. Under (b) it rests on Proposition N.12, by which every projection Ψ of $(K^m)^{\otimes 2}$ onto the symmetric tensors has some $\Psi(y_l)$ whose determinant has absolute value at least 1, and on Lemma N.15(2), which turns this into a lower bound for its projective norm. Condition (a) holds under (L), with the element ϑ of Section 7.1, and then $M_{|\vartheta|} \leq M_\varrho$. In degrees below p every bounded polynomial law is ambient: a bounded polynomial law a of degree $n < p$ is the diagonal of $(h_1, \ldots, h_n) \mapsto a^{[(1,\ldots,1)]}(h_1, \ldots, h_n)/n!$. This map is n-linear, as comparing coefficients in $a(s_1 h_1 + s_1' h_1' + s_2 h_2 + \cdots + s_n h_n)$ after the substitutions $s_1' = s_1$, $s_1' = 0$ and $s_1 = 0$, and scaling s_1, shows, and likewise in every slot; it has diagonal a, since $a((s_1 + \cdots + s_n)h) = (s_1 + \cdots + s_n)^n a(h)$ gives $a^{[(1,\ldots,1)]}(h, \ldots, h) = n!\, a(h)$; and it is bounded by Theorem 10.11(1). A bounded polynomial law $a \colon P \to W$ of degree n whose composite ja with a linear isometry j of W into a normed space is analytic at one point h_0 is ambient of degree n: if $ja(h_0 + h) = \sum_i b_i(h, \ldots, h)$ near $h = 0$, Lemma A.1 on the line $s \mapsto h_0 + sh$ gives $b_n(h, \ldots, h) = ja(h)$, the coefficient of s^n in the polynomial $ja(h_0 + sh)$. So R_n is DP on H by Theorem 10.11(1), while by Proposition 10.13(2) jR_n is analytic at no point for every linear isometry j of V, or of its completion, into a normed space.

In degree p itself, with $p = \operatorname{char} K > 0$ as above, bounded polynomial laws need not be ambient either, over every such K. For a normed space P let $\Gamma^p(P) \subseteq P^{\otimes p}$ be the span of the powers $h^{\otimes p}$, $h \in P$; for $P = K^m$ it is the space of symmetric tensors, by the paragraph at the beginning of Section L.3, as K is infinite. Give $\Gamma^p(P)$ the seminorm

$$\|t\|_\Gamma = \inf\Big\{\sum_i |c_i| \|h_i\|^p : t = \sum_i c_i\, h_i^{\otimes p}\Big\}, \tag{10.4}$$

which dominates the projective seminorm π of $P^{\otimes p}$; both are norms if the bounded linear forms on P separate points. For integers $m \geq 1$ and $d \geq 0$ let $\mathbb{N}_d^m$ be the set of $\alpha \in \mathbb{N}^m$ with $|\alpha| = d$,

and let $e_1,\dots,e_m$ be the unit vectors of K^m. For a symmetric tensor $T\in(\mathbb{F}_p^m)^{\otimes p}$ and $\alpha\in\mathbb{N}_p^m$ let T_α be its coefficient at $e_{i_1}\otimes\cdots\otimes e_{i_p}$ for any $(i_1,\dots,i_p)$ in which each i occurs α_i times. For $\gamma\in\mathbb{N}_{p-1}^m$ let the *slice* $\partial^\gamma T\in\mathbb{F}_p^m$ be the vector whose i-th entry is $T_{\gamma+e_i}$, where $\gamma+e_i\in\mathbb{N}_p^m$ is γ with its i-th entry raised by 1. For a real number $\tau\ge1$ let $\mathcal{E}_{T,\tau}$ be K^m with the norm

$$\|h\|_{T,\tau}=\inf\Big\{\sum_{i=1}^m|z_i|+\tau\sum_{\gamma\in\mathbb{N}_{p-1}^m}|z_\gamma|:h=\sum_{i=1}^m z_ie_i+\sum_{\gamma\in\mathbb{N}_{p-1}^m}z_\gamma\,\partial^\gamma T\Big\},\tag{10.5}$$

the infimum over the scalars $z_i,z_\gamma\in K$. It is a norm, and the coordinate forms of $\mathcal{E}_{T,\tau}$ are bounded: as the entries of T lie in $\mathbb{F}_p$ and $\tau\ge1$, no entry of h has absolute value above $\|h\|_{T,\tau}$. The tensor T lies in $\Gamma^p(\mathcal{E}_{T,\tau})$.

Proposition 10.14 (laws that are not ambient, over every K)**.**

(1) *For every $m\ge1$ there are a nonzero symmetric tensor $T\in(\mathbb{F}_p^m)^{\otimes p}$ and a real number $\tau\ge1$ such that, in $\Gamma^p(\mathcal{E}_{T,\tau})$,*

$$\|T\|_\Gamma\ge\frac{(p-1)!}{3p^{3/2}(2p-1)!}\Big(\frac{m}{1+\log_p m}\Big)^{1/2}\pi(T).\tag{10.6}$$

(2) *Choose such $T^{(m)}$ and $\tau^{(m)}$ for every $m\ge1$, and let $\mathcal{E}$ be the ℓ^1-sum of the spaces $\mathcal{E}_m=\mathcal{E}_{T^{(m)},\tau^{(m)}}$, $m\ge1$, defined as for G; it is a Banach space if K is complete. For $n\ge p$, the map $(\lambda,h)\mapsto\lambda^{n-p}h^{\otimes p}$, with $\lambda^0=1$, from $K\oplus\mathcal{E}$, with the maximum norm, to $\Gamma^p(\mathcal{E})$, with the norm $\|\cdot\|_\Gamma$ of (10.4), is a bounded polynomial law of degree n and norm at most 1 that is not ambient of degree n, also as a map into the completion of $\Gamma^p(\mathcal{E})$.*

The proof is in Appendix P. A representation of T in $\Gamma^p(\mathcal{E}_{T,\tau})$ gives a symmetric tensor over $\mathbb{F}_p$ in $m+\binom{m+p-2}{p-1}$ symbols, which stand for the vectors e_i and $\partial^\gamma T$, that maps to T and whose nonzero coefficients have total weight at most the cost $\sum_i|c_i|\|h_i\|^p$ of the representation; a coefficient at two or more slices weighs at least τ^2, the others at least 1. The coefficients at exactly one slice act on T through a linear map $\mathsf{L}^\sharp$ of Lemma P.2, an identity for symmetric tensors in characteristic p by which $\mathrm{id}-\mathsf{L}^\sharp$ has rank at least $(p-1)!\,m^p/(2p-1)!$. So for most T every such tensor has many nonzero coefficients, or many at two or more slices. By Theorem 10.11(1) the law of (2) is DP on $K\oplus\mathcal{E}$, while, as for R_n, its composite with a linear isometry of $\Gamma^p(\mathcal{E})$, or of its completion, into a normed space is analytic at no point. For $n\ge p+1$ this gives laws with the properties of Proposition 10.13(2) over every K, on another space.

For $h_1,\dots,h_p\in P$ the tensor $\sum_{\sigma\in S_p}h_{\sigma(1)}\otimes\cdots\otimes h_{\sigma(p)}$ lies in $\Gamma^p(P)$, by (4.2); the elements of the span of these tensors are *hollow*. For $t\in\Gamma^p(P)$ let

$$\|t\|_{\mathrm{F}}=\inf\{\|t'\|_\Gamma:t'\in\Gamma^p(P),\ t-t'\text{ hollow}\},$$

the quotient seminorm of the seminorm $\|\cdot\|_\Gamma$ of (10.4) modulo the hollow tensors. For $C\ge0$, P satisfies (N_P) *with constant C* if $\|t\|_\Gamma\le C\pi(t)$ for every hollow t, and (F_P) *with constant C* if $\|t\|_{\mathrm{F}}\le C\pi(t)$ for every $t\in\Gamma^p(P)$. A *Frobenius law* is a map $a\colon P\to W$ with $a(h+h')=a(h)+a(h')$ and $a(\lambda h)=\lambda^p a(h)$ for $h,h'\in P$ and $\lambda\in K$, and with $\|a\|=\sup_{h\ne0}\|a(h)\|/\|h\|^p$ finite; it is a bounded polynomial law of degree p, as $a(s_1h_1+\cdots+s_rh_r)=s_1^pa(h_1)+\cdots+s_r^pa(h_r)$.

Theorem 10.15 (laws of degree p)**.** *Let $p=\mathrm{char}\,K>0$.*

(1) *P satisfies (F_P) with constant p, and with constant $\dim P$ if $\dim P<p$.*

(2) *Every Frobenius law $a\colon P\to W$ is ambient of degree p, with a p-linear map β of norm at most $p\|a\|$ in Definition 10.10(2).*

(3) *If P satisfies (N_P) with constant C, then $\|t\|_\Gamma \le (p+(p+1)C)\pi(t)$ for every $t \in \Gamma^p(P)$, and every bounded polynomial law $a\colon P \to W$ of degree p is ambient of degree p, with a p-linear map β of norm at most $(p+(p+1)C)\|a\|$. Conversely, if every bounded polynomial law of degree p on P, with values in any normed space, is ambient of degree p, then P satisfies (N_P) with some constant.*

The proof is in Appendix P. It rests on Lemma P.4: if $t \in \Gamma^p(P)$ is a sum of elementary tensors with nonzero factors, some of these factors form a basis of the span of all of them for which t is congruent, modulo the hollow tensors, to a combination of the p-th powers of the basis vectors in which the power of each basis vector h, taken as a factor of a term $h_1 \otimes \cdots \otimes h_p$, has a coefficient of absolute value at most $\|h_1\| \cdots \|h_p\|/\|h\|^p$. By (3), the space $K \oplus \mathcal{E}$ of Proposition 10.14(2) does not satisfy $(\mathrm{N}_{K\oplus\mathcal{E}})$ with any constant, as the law there with $n = p$ is not ambient of degree p.

For $n \ge 1$ let $\Gamma^n(P) \subseteq P^{\otimes n}$ be the span of the powers $h^{\otimes n}$, $h \in P$, with the seminorm $\|\cdot\|_\Gamma$ of (10.4) with n in place of p, and let π be the projective seminorm of $P^{\otimes n}$; for $n = p$ these are the space and the seminorms above. Let $\mathfrak{a}_n(P) \in [0,\infty]$ be the infimum of the $C \ge 0$ with $\|t\|_\Gamma \le C\pi(t)$ for all $t \in \Gamma^n(P)$; if it is finite, it is the least such C. For $m \ge 1$ let $[m] = \{1,\dots,m\}$, and let $c_0([m],K)$ be K^m with the maximum norm, read as the span of the first m unit vectors of $c_0(\mathbb{N},K)$. Put $\mathfrak{c}_K = |\vartheta|$ if K is discretely valued, with ϑ as in Proposition 10.13(a), and $\mathfrak{c}_K = 1$ otherwise.

Proposition 10.16 (ambience constants)**.** *Let $n \ge 1$.*

(1) *If $\mathfrak{a}_n(P) < \infty$, then every bounded polynomial law $a\colon P \to W$ of degree n is ambient of degree n, with an n-linear map β of norm at most $\mathfrak{a}_n(P)\|a\|$ in Definition 10.10(2). Conversely, if every bounded polynomial law of degree n on P, with values in any normed space, is ambient of degree n, then $\mathfrak{a}_n(P) < \infty$.*
(2) *$\mathfrak{a}_n(P) \le (n+1)\,\mathfrak{a}_{n+1}(K \oplus P)$, with the maximum norm on $K \oplus P$.*
(3) *If P is nonarchimedean, then $\mathfrak{a}_n(P) \le \mathfrak{c}_K^{-n} \sup_{m\ge1} \mathfrak{a}_n(c_0([m],K)) \le \mathfrak{c}_K^{-n}\,\mathfrak{a}_n(c_0(\mathbb{N},K))$.*
(4) *If P is nonarchimedean, then $\mathfrak{a}_2(P) \le \frac{3}{2}\,\mathfrak{c}_K^{-2}$.*

The proof is in Appendix P. By (1) and (3), if every bounded polynomial law of degree n on $c_0(\mathbb{N},K)$ is ambient of degree n, then so is every bounded polynomial law of degree n on every nonarchimedean normed space; by (1) and (4), every bounded polynomial law of degree 2 on a nonarchimedean normed space is ambient of degree 2.

11. Familywise analyticity and bundles

Throughout, P, E, D are normed spaces and $k \ge 1$; P is the model space of a base, since Theorem 2.1 needs only operator families on open subsets of P.

11.1. **Families of operators.**

Definition 11.1. The triple $(P;E,D)$ is *good in degree k* if for every open $U \subseteq P$ and every analytic $u\colon U \to \mathcal{L}(E,\widehat{D})$ the map $x \mapsto \Lambda^k_\pi u(x) \in \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ is analytic on U. We write $(P;E)$ for $(P;E,E)$. For a Banach space H and an open $U \subseteq P$, an *invertible family* on U is an analytic $g\colon U \to \mathcal{L}(H)$ whose values are invertible and for which $x \mapsto g(x)^{-1}$ is analytic.

Here $\Lambda^k_\pi u(x)$ maps $\Lambda^k_\pi E$ into $\Lambda^k_\pi \widehat{D} \subseteq \widehat{\Lambda}^k_\pi D$ (paragraph before Theorem 3.16). By Theorem 3.2(1) with $X = \Lambda^k_\pi E$, $Y = \widehat{\Lambda}^k_\pi D$, and the completed form of Proposition 3.1, $x \mapsto \Lambda^k_\pi u(x)$ is analytic at a point, or on U, if and only if $x \mapsto A^k_{E,\widehat{D};F}(u(x))$ is analytic there for every Banach space F. For a linear family $\theta\colon P \to \mathcal{L}(E,\widehat{D})$ (defined before Theorem 6.13), $h \mapsto \Lambda^k_\pi \theta(h)$

is analytic at some point if and only if θ is liftable in degree k, and then it is a continuous polynomial (Proposition 2.5, as in the proof of Lemma 3.4).

Proposition 11.2 (reductions). (1) *Let H be a Banach space. If $x \mapsto \Lambda^k_\pi g(x)$ is analytic for every invertible family g on every open subset of P, then $(P; H)$ is good in degree k. More precisely, for an analytic $u\colon U \to \mathcal{L}(H)$ and $x_0 \in U$ there are distinct $\lambda_0, \dots, \lambda_k \in K$ and a neighborhood $U_0 \subseteq U$ of x_0 such that every $1 + \lambda_i u$, with $1 = \mathrm{id}_H$, is an invertible family on U_0 and $\Lambda^k_\pi u$ is a fixed linear combination of the maps $\Lambda^k_\pi(1+\lambda_i u)$ on U_0.*

(2) *If $(P; E, D)$ is good in degree k, so is every triple of bounded linear retracts of P, E and D, respectively. The triple $(P; E \oplus E', D)$ is good in every degree $j \le k$ if and only if $(P; E, D)$ and $(P; E', D)$ are; the same holds in the target slot.*

(3) *Let E and D be Banach spaces and $H = D \oplus E$ with the maximum norm. Then $x \mapsto \Lambda^j_\pi g(x)$ is analytic for every invertible family g into $\mathcal{L}(H)$ over every open subset of P and every $j \le k$ if and only if the four triples $(P; E, E)$, $(P; E, D)$, $(P; D, E)$, $(P; D, D)$ are good in every degree $j \le k$. If $\mathcal{C}$ is a class of Banach spaces closed under finite direct sums, then, in the single degree k, $x \mapsto \Lambda^k_\pi g(x)$ is analytic for every $G \in \mathcal{C}$ and every invertible family g into $\mathcal{L}(G)$ over an open subset of P if and only if $(P; E, D)$ is good in degree k for all $E, D \in \mathcal{C}$.*

The proof is in Appendix O.

Corollary 11.3 (what bundles need). *Let $\mathcal{C}$ be a class of Banach spaces closed under finite direct sums. The following are equivalent.*

(1) *For every analytic manifold M modeled on P, every analytic vector bundle over M with typical fiber $E \in \mathcal{C}$ and every Banach space F, the atlas induced on the bundle with typical fiber $\mathrm{Alt}^k(E; F)$ is analytic.*

(2) *The triple $(P; E, D)$ is good in degree k for all $E, D \in \mathcal{C}$.*

If they hold, Theorem 2.1 applies to fiberwise alternating maps on analytic bundles with typical fibers in $\mathcal{C}$ and analytic bundles with Banach typical fibers, and to their operator-valued analytic morphisms, over every analytic manifold modeled on P.

The proof is in Appendix O.

11.2. **Homogeneous families.**

Definition 11.4. For $n \ge 0$ and $\beta \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D}))$ put $u_\beta(h) = \beta(h, \dots, h)$; for $n = 0$, β is a constant operator. For $j \ge 1$, a *lift of u_β in degree j* is a bounded jn-linear map $q\colon P^{jn} \to \mathcal{L}(\Lambda^j_\pi E, \widehat{\Lambda}^j_\pi D)$ with $q(h, \dots, h) = \Lambda^j_\pi u_\beta(h)$ for $h \in P$, and $c_j(\beta) \in [0, \infty]$ is the infimum of their norms.

For $n \ge 1$, $c_j(\beta) < \infty$ if and only if $h \mapsto \Lambda^j_\pi u_\beta(h)$ is analytic at some point of P (Lemma O.5). For $j \ge 1$ consider the condition

$$\begin{aligned} &(\mathrm{G}_j)\colon \text{ for every sequence } (\beta_n)_{n\ge1} \text{ with } \beta_n \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D})) \text{ and } \|\beta_n\| \le 1, \\ &\qquad \text{every } c_j(\beta_n) \text{ is finite and } \limsup_{n\to\infty} c_j(\beta_n)^{1/n} < \infty. \end{aligned} \tag{11.1}$$

Theorem 11.5 (reduction to homogeneous families). (1) *If $(P; E, D)$ is good in degree j, then (G_j) holds.*

(2) *If (G_j) holds for every $j \le k$, then $(P; E, D)$ is good in degree k.*

(3) *Condition (G_j) holds whenever $j! \ne 0$ in K. If $\mathrm{char}\, K = p > 0$, then $(P; E, D)$ is good in every degree $j \le k$ if and only if (G_{p^i}) holds for every $i \ge 1$ with $p^i \le k$.*

(4) *Let Y_n be the completed projective tensor product of n copies of P and one copy of E, and let $\beta_n^{\mathrm{u}} \in \mathrm{Mult}^n(P;\mathcal{L}(E,Y_n))$ be given by $\beta_n^{\mathrm{u}}(h_1,\dots,h_n)y = h_1\otimes\cdots\otimes h_n\otimes y$. For every normed D, $n\ge 1$ and $\beta\in\mathrm{Mult}^n(P;\mathcal{L}(E,\widehat{D}))$, $c_j(\beta)\le c_j(\beta_n^{\mathrm{u}})\|\beta\|^j$. Consequently $(P;E,D)$ is good in every degree $j\le k$ for every normed D if and only if, for every $j\le k$, every $c_j(\beta_n^{\mathrm{u}})$ is finite and $\limsup_n c_j(\beta_n^{\mathrm{u}})^{1/n}<\infty$; if $\operatorname{char}K=p>0$, the degrees $j=p^i\le k$ suffice. Equivalently, $(P;E,D_{\mathrm{u}})$ is good in every degree $j\le k$ for the ℓ^1-sum D_{u} of the spaces Y_n.*

Corollary 11.6 (degrees of triples)**.** *Let $\operatorname{char}K=p>0$, and let $\mathfrak{D}(P;E,D)$ be the set of $k\ge 1$ in which $(P;E,D)$ is good.*

(1) *$\{1,\dots,p-1\}\subseteq\mathfrak{D}(P;E,D)$, and $a+b\in\mathfrak{D}(P;E,D)$ whenever $a,b\in\mathfrak{D}(P;E,D)$ and the addition of a and b in base p has no carry. If $\mathfrak{D}(P;E,D)\ne\{1,2,\dots\}$, its least missing element is a power of p.*
(2) *If $E\cong E\oplus K$ and $D\cong D\oplus K$ by bounded linear isomorphisms, then $\mathfrak{D}(P;E,D)$ is $\{1,\dots,p^i-1\}$ for some $i\ge 1$ or $\{1,2,\dots\}$.*
(3) *$\mathfrak{D}(P;E,D)\supseteq\mathfrak{D}_B(E,\widehat{D})$, with equality for $P=\mathcal{L}(E,\widehat{D})$.*
(4) *The set of k for which $(P;E,D)$ is good in degree k for all normed E and D is $\{1,\dots,p^i-1\}$ for some $i\ge 1$, or $\{1,2,\dots\}$.*

The proofs are in Appendix O.

11.3. **The familywise class.** The tensor algebra $\mathcal{T}(P)$ is the Banach space of sequences $(w_n)_{n\ge0}$, $w_n\in\widehat{\otimes}_\pi^n P$, $\widehat{\otimes}_\pi^0 P=\widehat{K}$, with norm $\sum_n\|w_n\|<\infty$. The map $\mathrm{ver}_n\in\mathrm{Mult}^n(P;\mathcal{T}(P))$ places $h_1\otimes\cdots\otimes h_n$ in slot n, with ver_0 the unit of slot 0, and $\mathrm{ver}(h)=\sum_n\mathrm{ver}_n(h,\dots,h)=(h^{\otimes n})_{n\ge0}$ is analytic for $\|h\|<1$ (Lemma O.1). Every analytic germ factors through ver: if u is analytic at x_0 with values in $\mathcal{L}(E,\widehat{D})$, then $u(x_0+\lambda h)=\theta(\mathrm{ver}(h))$ for $\|h\|<1$, for some $\lambda\in K^\times$ and a linear family $\theta\colon\mathcal{T}(P)\to\mathcal{L}(E,\widehat{D})$ (Lemma O.8).

For a Banach space Y and $0<R<1$, $\mathrm{Mult}^\omega_R(P;Y)$ is the Banach space of sequences $(b_N)_{N\ge0}$, $b_N\in\mathrm{Mult}^N(P;Y)$, with norm $\sup_N R^N\|b_N\|<\infty$; $\mathrm{Mult}^\omega_{R,0}(P;Y)$ is its closed subspace where every b_N has zero diagonal, so that $b_0=0$; for $R'\le R$, $\iota_{R'R}$ is the contractive inclusion into $\mathrm{Mult}^\omega_{R'}(P;Y)$. With $I_1,\dots,I_k$ the consecutive blocks of $\{1,\dots,N\}$ of sizes $n_1,\dots,n_k$ and h_I the tuple of the h_i with $i\in I$, the map $\mathrm{ver}^\sharp\colon\mathrm{Mult}^k(\mathcal{T}(P);Y)\to\mathrm{Mult}^\omega_R(P;Y)$,

$$(\mathrm{ver}^\sharp\Phi)_N(h_1,\dots,h_N)=\sum_{n_1+\cdots+n_k=N}\Phi\big(\mathrm{ver}_{n_1}(h_{I_1}),\dots,\mathrm{ver}_{n_k}(h_{I_k})\big),$$

is bounded and maps $\mathrm{Mult}^k_0(\mathcal{T}(P);Y)$ into $\mathrm{Mult}^\omega_{R,0}(P;Y)$ (Lemma O.9). For a linear family $\theta\colon\mathcal{T}(P)\to\mathcal{L}(E,\widehat{D})$, with ev_θ and $\mathrm{ob}_k(\theta)$ as in Section 6.3, put

$$\begin{aligned}\mathrm{ob}^\omega_k(\theta;R)&=\big(\mathrm{ver}^\sharp\,\mathrm{ev}_\theta|_{\mathcal{N}_E^k}\big)_*[\mathcal{S}_E^k]=(\mathrm{ver}^\sharp)_*\,\mathrm{ob}_k(\theta)\\&\in\mathrm{Ext}\big(\widehat{\Lambda}^k_\pi E,\mathrm{Mult}^\omega_{R,0}(P;\widehat{\Lambda}^k_\pi D)\big),\end{aligned}\tag{11.2}$$

the second equality by Corollary N.3(b), and $\mathrm{ob}_{k,N}(\theta)=((\mathrm{pr}_N\,\mathrm{ver}^\sharp\,\mathrm{ev}_\theta)|_{\mathcal{N}_E^k})_*[\mathcal{S}_E^k]\in\mathrm{Ext}(\widehat{\Lambda}^k_\pi E,\mathrm{Mult}^N_0(P;\widehat{\Lambda}^k_\pi D))$, with pr_N the Nth term. The class $\mathrm{ob}^\omega_k(\theta)$ *vanishes after shrinking* if $(\iota_{R'R})_*\,\mathrm{ob}^\omega_k(\theta;R)=0$ for some $R'\in(0,R]$; this does not depend on R.

Theorem 11.7 (the familywise class)**.** (1) *For a linear family $\theta\colon\mathcal{T}(P)\to\mathcal{L}(E,\widehat{D})$ the following are equivalent: $x\mapsto\Lambda^k_\pi\theta(\mathrm{ver}(x))$ is analytic at 0; $\mathrm{ob}^\omega_k(\theta)$ vanishes after shrinking; there are $C,R'>0$ such that for every N the map $(\mathrm{pr}_N\,\mathrm{ver}^\sharp\,\mathrm{ev}_\theta)|_{\mathcal{N}_E^k}$ extends to a bounded linear map $\widehat{\otimes}^k_\pi E\to\mathrm{Mult}^N_0(P;\widehat{\Lambda}^k_\pi D)$ of norm at most CR'^{-N}.*

(2) $(P; E, D)$ *is good in degree* k *if and only if* $\mathrm{ob}_k^\omega(\theta)$ *vanishes after shrinking for every linear family* $\theta\colon \mathcal{T}(P) \to \mathcal{L}(E, \widehat{D})$.

(3) *Let* $\theta_{\mathrm{u}}\colon \mathcal{T}(P) \to \mathcal{L}(E, \mathcal{T}(P)\widehat{\otimes}_\pi E)$ *be given by* $\theta_{\mathrm{u}}(w)y = w \otimes y$. *For* θ *as in (1), let* $\tau_\theta\colon \mathcal{T}(P)\widehat{\otimes}_\pi E \to \widehat{D}$ *be the linearization of* $(w, y) \mapsto \theta(w)y$, *and let* $(\widehat{\Lambda}^k_\pi \tau_\theta)_\flat$ *apply* $\widehat{\Lambda}^k_\pi \tau_\theta$ *to every term. Then* $\mathrm{ob}_k^\omega(\theta; R) = ((\widehat{\Lambda}^k_\pi \tau_\theta)_\flat)_* \mathrm{ob}_k^\omega(\theta_{\mathrm{u}}; R)$. *Hence* $(P; E, D)$ *is good in degree* k *for every normed* D *if and only if* $\mathrm{ob}_k^\omega(\theta_{\mathrm{u}})$ *vanishes after shrinking, and if and only if* $x \mapsto \Lambda^k_\pi \mathcal{U}(x)$ *is analytic at* 0 *for the universal germ* $\mathcal{U}(x)y = \mathrm{ver}(x) \otimes y$.

The proof is in Appendix O.

11.4. **Base models good for all fibers.**

Definition 11.8. The space P is *universally good in degree* k, written (U^k_P), if $(P; E, D)$ is good in degree k for all normed E and D. It has *fixed-degree reflection in degree* k, written (FR^k_P), if for all Banach spaces G and Z, every closed subspace $W \subseteq Z$, every bounded k-linear $\Phi\colon G^k \to Z$ with $\Phi(y, \dots, y) \in W$ for $y \in G$, every open $U \subseteq P$ and every analytic $\gamma\colon U \to G$, the map $x \mapsto \Phi(\gamma(x), \dots, \gamma(x))$ is analytic as a W-valued map on U.

Recall $g_n(P)$ from Definition 3.8(3).

Theorem 11.9 (polarizable base models)**.** (1) *The following are equivalent:* (FR^k_P); (FR^k_P) *restricted to affine maps* $\gamma(x) = y_0 + Tx$ *on* $U = P$, *with* T *bounded linear;* P *is polarizable in every degree* $n \le k$.

(2) *If* P *is polarizable in every degree* $n \le k$, *then* (U^k_P) *holds. Quantitatively, let* $\bar{g} \ge \max(1, g_1(P), \dots, g_k(P))$, *with* $\bar{g} > g_n(P)$ *when no projection of norm* $g_n(P)$ *exists. If* $u\colon U \to \mathcal{L}(E, \widehat{D})$ *satisfies* $u(x_0+h) = \sum_n \beta_n(h, \dots, h)$ *for* $\|h\| < R$, *with* $\|\beta_n\| \le CR^{-n}$, *then* $\Lambda^k_\pi u$ *has at* x_0 *an expansion of radius at least* $R/\bar{g}$ *whose* N*th coefficient has norm at most* $k!\, C^k \binom{N+k-1}{k-1} (\bar{g}/R)^N$. *In particular, Theorem 2.1 applies to fiberwise alternating maps on analytic bundles with arbitrary normed typical fibers and analytic bundles with Banach typical fibers over every analytic manifold modeled on* P.

(3) *If* $\mathrm{char}\, K = 0$, *or* $\mathrm{char}\, K = p$ *and* $k < p$, *then* (U^k_P) *holds for every* P. *If* $\mathrm{char}\, K = p$ *and* P *is polarizable in degree* p, *then* (U^n_P) *holds for every* $n \ge 1$. *Every normed space* P *with a lower* r*-estimate for some* $r < 2$ *(Section 5.2) satisfies* (U^n_P) *for every* $n \ge 1$, *over every* K.

(4) *Let* K *be complete and* P *a Banach space. Then* P *has universal analytic reflection (Definition G.2) if and only if* $\sup_n g_n(P)^{1/n} < \infty$, *and* (FR^k_P) *holds for every* k *if and only if* $g_n(P) < \infty$ *for every* n. *If* K *is nonarchimedean and* $P = \ell^1(\mathbb{N}, K)$, *then* (FR^k_P) *and* (U^k_P) *hold for every* k, *but* P *does not have universal analytic reflection.*

Theorem 11.10 (base models under (L))**.** *Assume* (L).

(1) *Let* P *be a symmetric sequence space in which* $c_{00}(\mathbb{N}, K)$ *is dense, let* E_0 *be* $c_{00}(\mathbb{N}, K)$ *with the supremum norm, and let* $\theta_0\colon P \to \mathcal{L}(E_0, P)$ *be given by* $\theta_0(a)y = ay$. *The following are equivalent:* (U^k_P); $a \mapsto \Lambda^k_\pi \theta_0(a)$ *is analytic at some point of* P; $\mathcal{P}_<$ *is bounded on* $P \otimes_\pi P$; P *is polarizable in degree* p; (U^n_P) *holds for every* $n \ge 1$. *They are equivalent to the conditions of Theorem 8.1. In particular, for* $P = \ell^r(\mathbb{N}, K)$ *with* $1 \le r < \infty$, (U^k_P) *holds if and only if* $r < 2$; *and for* $P = \ell^r(\mathbb{N}, K)$ *with* $2 \le r < \infty$, *for* $P = c_0(\mathbb{N}, K)$, *and for every symmetric sequence space* P *in which* $c_{00}(\mathbb{N}, K)$ *is dense and* $\varphi_P(m)^2 \le Cm$ *for some* C, (U^n_P) *holds exactly for* $n \le p - 1$.

(2) *Let* P *be normed, and suppose there are bounded linear maps* $i\colon c_{00}(\mathbb{N}, K) \to P$, *with the supremum norm on* $c_{00}(\mathbb{N}, K)$, *and* $w\colon P \to \ell^\infty(\mathbb{N}, \widehat{K})$ *with* $w \circ i$ *the inclusion.*

Then for the linear family $x \mapsto \mu_{w(x)} \in \mathcal{L}(c_{00}(\mathbb{N}, K), \ell^\infty(\mathbb{N}, \widehat{K}))$, with $\ell^\infty(\mathbb{N}, \widehat{K})$ regarded over K, the map $x \mapsto \Lambda^k_\pi \mu_{w(x)}$ is analytic at no point, so (U^k_P) fails. Such maps exist for $c_0(\Sigma, K)$ and $\ell^\infty(\Sigma, K)$ with Σ infinite, and for every P with an equivalent nonarchimedean norm and infinite-dimensional completion, $\dim_{\widehat{K}} \widehat{P} = \infty$. For P with an equivalent nonarchimedean norm, (U^k_P) holds if and only if P is polarizable in degree p, and if and only if $\dim_{\widehat{K}} \widehat{P} < \infty$.

The proofs are in Appendix O.

11.5. **Fibers and mixed mechanisms.**

Proposition 11.11 (fibers good over every base)**.** (1) *If $(E, \widehat{D})$ is good for Banach values in degree k, with a lift Q as in Lemma 3.4, then $(P; E, D)$ is good in degree k for every normed P, and $c_k(\beta) \le \|Q\|\|\beta\|^k$ for every n and $\beta \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D}))$.*

(2) *$(\mathcal{L}(E, \widehat{D}); E, D)$ is good in degree k if and only if $(E, \widehat{D})$ is good for Banach values in degree k. Hence $(P; E, D)$ is good in degree k for every normed P if and only if $(E, \widehat{D})$ is good for Banach values in degree k.*

The proof is in Appendix O.

Definition 11.12. A bounded linear map $T\colon P \to P'$ is *Γ^k-split* if there is a bounded k-linear $\Pi\colon P^k \to \widehat{\Gamma}^k_\pi P'$ with $\Pi(a, \dots, a) = (Ta)^{\otimes k}$ for $a \in P$.

By Lemma 3.11, P is polarizable in degree k if and only if id_P is Γ^k-split, and T is Γ^k-split if P or P' is polarizable in degree k: compose $T^{\otimes k}$ with a projection onto $\widehat{\Gamma}^k_\pi P'$, or a projection onto $\widehat{\Gamma}^k_\pi P$ with the extension of $T^{\otimes k}$.

Theorem 11.13 (parameter-relative ordering)**.** *Let $2 < r < \infty$.*

(1) *For every totally ordered set Σ and every $k \ge 1$, the inclusion $\ell^2(\Sigma, \widehat{K}) \to \ell^r(\Sigma, \widehat{K})$, over K, is Γ^k-split.*

(2) *Let K be nonarchimedean, and let P have an ℓ^2-dominated linearly independent family $(e_t)_{t\in\Sigma}$ with dense span V and $\sup_t \|e_t\| < \infty$. Let G be a normed space with an equivalent r-norm, Z a Banach space, $W \subseteq Z$ a closed subspace and $\Phi\colon G^k \to Z$ a bounded k-linear map with $\Phi(y, \dots, y) \in W$ for $y \in G$. For every bounded linear $T\colon P \to G$ there is a bounded k-linear $Q\colon P^k \to W$ with $Q(a, \dots, a) = \Phi(Ta, \dots, Ta)$ for $a \in P$; and for every open $U \subseteq P$ and every analytic $\gamma\colon U \to G$, the map $x \mapsto \Phi(\gamma(x), \dots, \gamma(x))$ is analytic into W.*

(3) *For K and P as in (2), and for D with an equivalent r-norm or an equivalent nonarchimedean norm, $(P; E, D)$ is good in every degree for every normed E.*

Under (L) and for $2 < r < \infty$, (3) makes $(\ell^2(\mathbb{N}, K); \ell^r(\mathbb{N}, K))$ good in every degree, although neither space is polarizable in degree k (Corollary 8.2).

Theorem 11.14 (weighted staircase)**.** *Assume* (L). *Let E be a sequence space with $\|y\|_\infty \le \|y\|_E$ for $y \in E$, and let P be a normed space containing $c_{00}(\mathbb{N}, K)$ as a linear subspace. For $G \in \{E, P\}$ let*

$$\varphi^{\pm}_G(n) = \sup\big\{\|y\|_G : y \in c_{00}(\mathbb{N}, K),\ y(t) \in \{0, 1, -1\} \text{ for all } t,\ |\operatorname{supp} y| \le n\big\}.$$

If $\liminf_{m\to\infty} \varphi^{\pm}_P(2m)^2 \varphi^{\pm}_E(4m)^2/m = 0$, then no k-linear $Q\colon c_{00}(\mathbb{N}, K)^k \to \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi E)$ that is bounded for the norm of P satisfies $Q(a, \dots, a) = \Lambda^k_\pi \mu_a$ for $a \in c_{00}(\mathbb{N}, K)$. Consequently, if $T\colon P \to \mathcal{L}(E, \widehat{E})$ is bounded linear with $T(a) = \mu_a$ for $a \in c_{00}(\mathbb{N}, K)$, then $x \mapsto \Lambda^k_\pi T(x)$ is analytic at no point of P. For a symmetric sequence space G, $\varphi^{\pm}_G = \varphi_G$.

Corollary 11.15 (the table)**.** *Assume* (L). *Let P be $\ell^{r_1}(\mathbb{N},K)$ with $1\le r_1<\infty$, or $c_0(\mathbb{N},K)$. In (1) and (2), let E be $\ell^{r_2}(\mathbb{N},K)$ with $1\le r_2<\infty$, or $c_0(\mathbb{N},K)$, or, if K is complete, an infinite-dimensional Banach space with an equivalent nonarchimedean norm.*

(1) *$(P;E)$ is good in every degree if $r_1<2$; if $E=\ell^{r_2}(\mathbb{N},K)$ with $r_2<2$; or if $r_1=2$ and E is $\ell^{r_2}(\mathbb{N},K)$ with $r_2>2$, or $c_0(\mathbb{N},K)$, or has an equivalent nonarchimedean norm.*

(2) *There is a linear family $P\to\mathcal{L}(E,\widehat{E})$ along which Λ^k_π is analytic at no point, so $(P;E)$ is not good in degree k, if $P=c_0(\mathbb{N},K)$ and E is not $\ell^{r_2}(\mathbb{N},K)$ with $r_2<2$; or if $2<r_1<\infty$ and E is $c_0(\mathbb{N},K)$, or has an equivalent nonarchimedean norm, or is $\ell^{r_2}(\mathbb{N},K)$ with $1/r_1+1/r_2<1/2$.*

(3) *Let E be a symmetric sequence space in which $c_{00}(\mathbb{N},K)$ is dense. The multiplier family $c_0(\mathbb{N},K)\to\mathcal{L}(E,\widehat{E})$, $a\mapsto\mu_a$, is liftable in some degree $k\ge p$ if and only if (E,E) is good for Banach values in every degree, and if and only if $(c_0(\mathbb{N},K);E)$ is good in every degree.*

The proofs are in Appendix O.

11.6. **Tangent bundles.**

Definition 11.16. A normed space P_0 is *tangent-good in degree k* if for every open $U\subseteq P_0$ and every analytic $f\colon U\to P_0$ whose derivative $Df\colon U\to\mathcal{L}(P_0,P_0)$ is analytic, the map $x\mapsto\Lambda^k_\pi Df(x)\in\mathcal{L}(\Lambda^k_\pi P_0,\widehat{\Lambda}^k_\pi P_0)$ is analytic on U.

Let K be complete and P_0 a Banach space, so that analytic maps between open subsets of P_0 have analytic derivatives [3]. For a manifold M modeled on P_0 with chart transitions f_{ij}, the transitions of $\widehat{\Lambda}^k_\pi TM$ and $\mathrm{Alt}^k(TM;F)$ are $\Lambda^k_\pi Df_{ij}$ and $A^k_{P_0,P_0;F}(Df_{ji})$; so if P_0 is tangent-good in degree k, these bundles are analytic for every Banach space F, by the paragraph after Definition 11.1.

Theorem 11.17 (shear charts)**.** (1) *Let K be complete, let P_0 be a Banach space with a bounded linear isomorphism $J\colon P_1\times D\times E\to P_0$, let $U\subseteq P_1$ be open and let $u\colon U\to\mathcal{L}(E,D)$ be analytic. Then $\psi(x,d,e)=(x,d+u(x)e,e)$ is an analytic diffeomorphism of $U\times D\times E$ onto itself, with inverse $(x,d,e)\mapsto(x,d-u(x)e,e)$, and with $i_E(e)=(0,0,e)$ and $\pi_D(x,d,e)=d$,*

$$\Lambda^k_\pi u(x)=\widehat{\Lambda}^k_\pi\pi_D\circ\Lambda^k_\pi D\psi(x,d,e)\circ\Lambda^k_\pi i_E.$$

If $\Lambda^k_\pi u$ is not analytic at x_0, then neither $(x,d,e)\mapsto\Lambda^k_\pi D\psi(x,d,e)$ nor $(x,d,e)\mapsto\Lambda^k_\pi D(\psi^{-1})(x,d,e)$ is analytic at any point (x_0,d,e), and the manifold $M=J(U\times D\times E)$ with the charts id_M and $J\circ\psi\circ J^{-1}$ has bundles $\widehat{\Lambda}^k_\pi TM$ and $\mathrm{Alt}^k(TM;\widehat{\Lambda}^k_\pi D)$ whose induced atlases are not analytic.

(2) *Let K be complete and P_0 a Banach space with $P_0\cong P_0\times P_0\times P_0$. The following are equivalent: P_0 is tangent-good in degree k; for every analytic manifold M modeled on P_0 the induced atlases of $\widehat{\Lambda}^k_\pi TM$ and of $\mathrm{Alt}^k(TM;F)$, for every Banach space F, are analytic; $(P_0;P_0)$ is good in degree k.*

(3) *Assume* (L). *For $E=c_0(\mathbb{N},K)$ and for $E=\ell^r(\mathbb{N},K)$ with $4<r<\infty$, the polynomial map $\psi(a,d,e)=(a,d+a\cdot e,e)$ on E^3, with $a\cdot e$ the pointwise product, has the polynomial inverse $(a,d,e)\mapsto(a,d-a\cdot e,e)$, and $x\mapsto\Lambda^k_\pi D\psi(x)$ and $x\mapsto\Lambda^k_\pi D(\psi^{-1})(x)$ are analytic at no point of E^3; since $E\cong E^3$, E is not tangent-good in degree k. If K is complete, no infinite-dimensional Banach space with an equivalent nonarchimedean norm is tangent-good in degree k.*

The proof is in Appendix O.

11.7. **Structure groups.**

Definition 11.18. Let E be a normed space and $\mathrm{GL}(E)$ the group of invertible elements of $\mathcal{L}(E)$. For a subset $\mathsf{S} \subseteq \mathcal{L}(E)$, an S-*family* over P is an analytic map $u\colon U \to \mathcal{L}(E)$, with $U \subseteq P$ open, such that $u(U) \subseteq \mathsf{S}$; and S is *good over P in degree k* if $x \mapsto \Lambda^k_\pi u(x)$ is analytic on U for every S-family u over P. For a linear subspace $\mathsf{V} \subseteq \mathcal{L}(E)$ and $\varepsilon > 0$ put $\mathsf{V}(\varepsilon) = \{v \in \mathsf{V} : \|v\| < \varepsilon\}$ and $1 + \mathsf{V}(\varepsilon) = \{1 + v : v \in \mathsf{V}(\varepsilon)\}$, with $1 = \mathrm{id}_E$.

By Theorem 2.1, a bundle with transitions in a subgroup $\mathsf{H} \subseteq \mathrm{GL}(E)$ needs only H-families.

Theorem 11.19 (structure groups)**.** (1) *Let $\mathsf{H} \subseteq \mathrm{GL}(E)$ be a subgroup, $\mathsf{H}_0 \subseteq \mathsf{H}$ a subgroup that is open in H for the operator norm, and $\mathsf{V}_0 \subseteq \mathcal{L}(E)$ a linear subspace with $\mathsf{H}_0 \subseteq \mathsf{V}_0$ and $1 + \mathsf{V}_0(\varepsilon) \subseteq \mathsf{H}_0$ for some $\varepsilon > 0$. Then H is good over P in degree k if and only if V_0 is. Neither E nor K need be complete. If E is a Banach space, $\mathrm{GL}(E)$ is good over P in degree k if and only if $(P; E)$ is good in degree k.*

(2) *If E is a Banach space with a nonarchimedean norm, the group $\mathrm{Isom}(E)$ of surjective linear isometries of E is good over P in degree k if and only if $(P; E)$ is good in degree k.*

(3) *Let K be nonarchimedean and $E = c_0(\mathbb{N}, K)$ or $E = \ell^r(\mathbb{N}, K)$ with $1 \le r < \infty$. Let $\mathsf{V}_\mathrm{d} = \{\mu_a : a \in \ell^\infty(\mathbb{N}, K)\}$, let $\mathsf{H}_\mathrm{d} = \{\mu_a : a \in \ell^\infty(\mathbb{N}, K),\ \inf_t |a(t)| > 0\}$ be the diagonal group and $\mathsf{H}_\mathrm{m} = \{\sigma_\sharp \mu_a : \mu_a \in \mathsf{H}_\mathrm{d},\ \sigma$ a bijection of $\mathbb{N}\}$ the monomial group, where $\sigma_\sharp e_t = e_{\sigma(t)}$. Then H_m is good over P in degree k if and only if H_d is, and if and only if V_d is.*

(4) *Under* (L)*, the commutative group H_d of $E = c_0(\mathbb{N}, K)$ is not good over $P = c_0(\mathbb{N}, K)$ in degree k.*

Corollary C.7 gives isometric diagonal transitions with the same behavior over Laurent series fields.

Proposition 11.20 (the closed span does not decide)**.** *Assume* (L) *with K complete, and let κ' and ϑ be as in Lemma 7.1, with κ' identified with its image in K, so that the nonzero elements of κ' have absolute value* 1 *and every element of K is a convergent series $\sum_{n \ge n_0} \xi_n \vartheta^n$ with $\xi_n \in \kappa'$. Let $E = D = c_0(\mathbb{N}, K)$, let e_i^* be the ith coordinate functional, let $\mathsf{U} \subseteq \mathcal{L}(E)$ be the additive group of the operators u with $e_i^*(ue_j) \in \kappa'$ for all i, j, and let $\mathsf{H} = \{g(u) : u \in \mathsf{U}\} \subseteq \mathrm{GL}(D \oplus E)$, with g as in Proposition 10.1. Then H is a discrete subgroup, every H-family over every P is locally constant, and H is good over every P in every degree. But the closed linear span of H contains $g(\mu_x)$ for every $x \in c_0(\mathbb{N}, K)$, and $x \mapsto \Lambda^k_\pi g(\mu_x)$ is analytic at no point of $c_0(\mathbb{N}, K)$.*

The proofs are in Appendix O.

11.8. **Characteristic two.**

Proposition 11.21 (families in characteristic two)**.** *Let K be a nontrivially normed field of characteristic* 2*, not necessarily satisfying* (L)*, and let $k \ge 2$.*

(1) *Let E, P, $\varphi_E^\pm$ and $\varphi_P^\pm$ be as in Theorem 11.14. If $\sup_n n/(\varphi_P^\pm(n)^2 \varphi_E^\pm(n)^2) = \infty$, the conclusions of Theorem 11.14 hold.*

(2) *The conclusion of Corollary 11.15(2) holds for $P = c_0(\mathbb{N}, K)$ and E equal to $c_0(\mathbb{N}, K)$ or to $\ell^{r_2}(\mathbb{N}, K)$ with $2 < r_2 < \infty$, and for $P = \ell^{r_1}(\mathbb{N}, K)$ with $2 < r_1 < \infty$ and E equal to $c_0(\mathbb{N}, K)$ or to $\ell^{r_2}(\mathbb{N}, K)$ with $1/r_1 + 1/r_2 < 1/2$.*

(3) *For $E = c_0(\mathbb{N}, K)$ and for $E = \ell^r(\mathbb{N}, K)$ with $4 < r < \infty$, the maps $x \mapsto \Lambda^k_\pi D\psi(x)$ and $x \mapsto \Lambda^k_\pi D(\psi^{-1})(x)$ of Theorem 11.17(3) are analytic at no point of E^3, and E is not tangent-good in degree k.*

(4) *The commutative group* H_{d} *of Theorem 11.19(3), for* $E = c_0(\mathbb{N}, K)$*, is not good over* $P = c_0(\mathbb{N}, K)$ *in degree* k*.*

The proof is in Appendix O. Item (2) does not cover the pair $P = c_0(\mathbb{N}, K)$, $E = \ell^2(\mathbb{N}, K)$ of Corollary 11.15(2). Items (2) and (3) do not cover the spaces with an equivalent nonarchimedean norm in Corollary 11.15(2) and in the last sentence of Theorem 11.17(3), whose proofs use Lemma 7.17, which assumes (L).

12. Questions

The questions below concern $\operatorname{char} K = p > 0$ and $k \geq p$. For $1 < r < \infty$ and $1 \leq b \leq \infty$, the symmetric sequence space $\ell^{r,b}$ (Definition 7.10) is the closure of c_{00} in the space of all $x \in \ell^\infty(\mathbb{N}, K)$ with $\sup_n \|x1_{[n]}\|_{r,b} < \infty$, where on c_{00} $\|x\|_{r,b} = c_{r,b}\big(\sum_{n\geq 1} n^{b/r-1}(x_n^{**})^b\big)^{1/b}$ for $b < \infty$, $\|x\|_{r,\infty} = c_{r,\infty} \sup_{n\geq 1} n^{1/r} x_n^{**}$, x_n^{**} is the average of the n largest values of $|x(t)|$, and $c_{r,b} > 0$ makes $\|e_1\|_{r,b} = 1$.

Conjecture 12.1 (product triangle conjecture)**.** Let κ and K be as in (L), and let E, D be symmetric sequence spaces in which c_{00} is dense. If (E, D) satisfies (PTB) (Definition 7.12), then (E, D) is good for Banach values in every degree.

The converse holds (Theorem 7.13). The conjecture holds when $D = E$ or $E = c_0$, since indicator functions of supports in (7.2) bound $\mathcal{P}_<$ on $D \otimes_\pi D$ (Theorem 8.1), and when D has an equivalent r-norm, $2 < r < \infty$, or nonarchimedean norm (Theorem 8.4). The same holds for spreading sequence spaces (Theorem 8.9).

Conjecture 12.2. Let κ be a finite field of characteristic p, $K = \kappa((X))$ with the absolute value of (L), and $1 < r < 2 \leq b \leq \infty$. Then $\mathcal{P}_<$ is unbounded on $\ell^{r,b} \otimes_\pi \ell^{r,b}$.

By Theorem 8.1, the conjecture for given r, b is equivalent to the failure of $(\mathrm{S}^p_{\ell^{r,b}})$, of $(\mathrm{R}^p_{\ell^{r,b}})$, or of $(\Gamma^p_{\ell^{r,b}})$, and to $(\ell^{r,b}, \ell^{r,b})$ and $(c_0, \ell^{r,b})$ being good for Banach values in no degree $k \geq p$.

Conjecture 12.3 (sequence lattices)**.** Assume (L) with K complete. Let E and D be sequence lattices, that is, each is a Banach space Y with $c_{00}(\mathbb{N}, K) \subseteq Y \subseteq \ell^\infty(\mathbb{N}, K)$ in which c_{00} is dense, with $\|\cdot\|_\infty \leq \|\cdot\|_Y$, $\sup_t \|e_t\|_Y < \infty$, and $\|x\|_Y \leq \|y\|_Y$ whenever $|x(t)| \leq |y(t)|$ for every t. Then (E, D) is good for Banach values in degree k if and only if it has a splitting decomposition of degree k (Definition 6.12).

Question 12.4. Is every pair that is good for Banach values in all degrees $\leq k$ obtained from pairs (E, D) with (S^n_E) for all $n \leq k$, pairs with (R^n_D) for all $n \leq k$, relative splittings through $\mathrm{S}^{\leq k}$-split or $\mathrm{R}^{\leq k}$-split maps, and parameter pieces, by finite decompositions (Theorem 6.13, Proposition 6.18), transfer (Proposition 6.14) and block assemblies (Proposition 6.27)?

Question 12.5. Let κ and K be as in (L). Do (S^k_Y), (R^k_Y) and (Γ^k_Y) coincide for every Banach space Y and every $k \geq p$?

They coincide for spreading, in particular symmetric, sequence spaces in which c_{00} is dense (Theorems 8.1 and 8.9), and (S^2_Y) is equivalent to (Γ^2_Y) for every normed Y (Proposition 5.14).

Question 12.6. Let W be a closed subspace of a Banach space Z. If W and Z/W satisfy (S^k), does Z? The same question for (R^k) and for polarizability in degree k.

Under (L), goodness for Banach values with a fixed target has no three-space property in the source slot, even with an ℓ^1 kernel (Theorem 6.31).

Question 12.7. Let $k \geq p$. If (E, D) is good for Banach values in degree k for every normed D, does (S^k_E) hold?

Under (L) the answer is yes for spreading, in particular symmetric, sequence spaces E in which c_{00} is dense (Theorems 8.1 and 8.9).

Question 12.8. Assume (L). Does every S^k-split operator factor through a normed space G with (S^k_G)? Does the inclusion $\ell^2(\mathbb{N}, K) \to c_0(\mathbb{N}, \widehat{K})$, which is S^n-split for every n (Corollary 5.10), factor through such a space?

Question 12.9. Let K be a field of Laurent series in X over an infinite field of characteristic p, with an X-adic absolute value, or a densely valued field of characteristic p.

(1) If p is odd, is $(c_0(\mathbb{N}, K), c_0(\mathbb{N}, K))$ good for Banach values in degree p?
(2) If $p = 2$, is $(\ell^2(\mathbb{N}, K), \ell^2(\mathbb{N}, K))$ good for Banach values in degree 2?

For $p = 2$, $(c_0(\mathbb{N}, K), c_0(\mathbb{N}, K))$ is good for Banach values in no degree $k \geq 2$, over every nontrivially normed field of characteristic 2 (Theorem 7.20(4)); for $E = \ell^2(\mathbb{N}, K)$, $\bar{\varphi}_E(n)^2 = n$ and (7.7) gives no contradiction. For odd p, a positive answer to (1) needs a zero-diagonal correction making the ordering lift bounded (Theorem 3.18).

Question 12.10. If a pair is good in degree p, is it good in every degree? The same question for goodness for Banach values. Is every S^p-split operator S^n-split for every n, and every R^p-split operator R^n-split for every n?

The least degree in which a pair fails to be good is a power of p (Corollary 6.3). For the conditions (S^p_Y) and (R^p_Y) of one normed space Y, Theorem 6.6 answers the third question.

Question 12.11. Let $\operatorname{char} K = p$.

(1) If a normed space P satisfies (U^p_P) (Definition 11.8), is P polarizable in degree p?
(2) Assume (L). Is $(\ell^{r_1}(\mathbb{N}, K); \ell^{r_2}(\mathbb{N}, K))$ good in degree k when $r_1 = r_2 = 2$, when $2 < r_1 < \infty$ and $r_2 = 2$, or when $2 < r_1, r_2 < \infty$ and $1/r_1 + 1/r_2 \geq 1/2$?
(3) Do Theorems 11.5 and 11.7 hold for one Banach value space F, with $x \mapsto A^k_{E,\widehat{D};F}(u(x))$ in place of $x \mapsto \Lambda^k_\pi u(x)$?

Polarizability in the degrees $n \leq k$ gives (U^k_P) (Theorem 11.9(2)); under (L), (U^k_P) gives polarizability in degree p when P is a symmetric sequence space in which $c_{00}(\mathbb{N}, K)$ is dense or has an equivalent nonarchimedean norm (Theorem 11.10). The cases in (2) are those left open by Corollary 11.15; for $r_1 = r_2 = r$ and K complete they ask whether $\ell^r(\mathbb{N}, K)$ is tangent-good in degree k (Theorem 11.17(2)).

Question 12.12. Let P, E and D be normed spaces. If every linear family $P \to \mathcal{L}(E, \widehat{D})$ is liftable in degree k, is $(P; E, D)$ good in degree k?

The answer is yes for $P = \mathcal{L}(E, \widehat{D})$, whose identity is a linear family (Lemma 3.4, Proposition 11.11(2)); for fixed-degree reflection, affine families suffice (Theorem 11.9(1)).

Question 12.13. If $(P; E, D)$ is good in degree k, is it good in every degree $j < k$?

The answer is yes when $E \cong E \oplus K$ and $D \cong D \oplus K$, and for goodness for all normed E and D (Corollary 11.6(2), (4)).

Question 12.14. Is there a linear family $\theta\colon \mathcal{T}(P) \to \mathcal{L}(E, \widehat{D})$ for which every class $\mathrm{ob}_{k,N}(\theta)$ vanishes but $\mathrm{ob}^\omega_k(\theta)$ does not vanish after shrinking?

By Theorem 11.7(1), $\mathrm{ob}_k^{\omega}(\theta)$ vanishes after shrinking exactly when the classes $\mathrm{ob}_{k,N}(\theta)$ vanish with extension costs growing at most geometrically in N. For general closed subspaces, Taylor coefficients can have lifts without a common radius (Corollary G.5).

Question 12.15. Let E, D and F be normed spaces. If (E, D) and (D, F) are good for Banach values in degree k, is the composite family $\theta_{D,F} \odot \theta_{E,D}$ of Proposition 6.26 liftable in degree k?

The composite family is liftable in degree k when (E, F) is good for Banach values in that degree (Proposition 6.26(2)).

Question 12.16. Let κ and K be as in (L), $1 \leq q < 2$ and $0 < \alpha < 1 - q/2$. Is $(E_{\alpha,q}, E_{\alpha,q})$ (Example 8.10) good for Banach values in some degree $k \geq p$?

By Theorem 8.9, this holds if and only if $\mathcal{P}_<$ is bounded on $E_{\alpha,q} \otimes_\pi E_{\alpha,q}$, and then the pair is good for Banach values in every degree. The necessary conditions of Theorem 8.1 hold, and Theorem 5.6 does not apply to the unit vectors in their natural order (Example 8.10(2),(4)).

Question 12.17. Let K be nonarchimedean of characteristic $p > 0$. Is every bounded polynomial law on $c_0(\mathbb{N}, K)$ ambient of its degree?

Bounded polynomial laws of degree below p are ambient (Section 10.4), and for $p = 2$ so are those of degree 2 on nonarchimedean normed spaces (Proposition 10.16(1),(4)). By Proposition 10.16(1),(3), every bounded polynomial law of degree n on $c_0(\mathbb{N}, K)$ is ambient of degree n if and only if every bounded polynomial law of degree n on every nonarchimedean normed space over K is. As $K \oplus c_0(\mathbb{N}, K)$, with the maximum norm, is isometric to $c_0(\mathbb{N}, K)$, Proposition 10.16(1),(2) shows that the degrees with a positive answer form an initial segment of the positive integers: a negative answer in some degree gives a negative answer in every higher degree, and a positive answer in infinitely many degrees gives one in all. Frobenius laws are ambient, and every bounded polynomial law of degree p on a normed space P, with values in any normed space, is ambient of degree p if and only if P satisfies (N_P) with some constant (Theorem 10.15(2),(3)); so for $p \geq 3$, in degree p the question asks whether $c_0(\mathbb{N}, K)$ satisfies $(\mathrm{N}_{c_0(\mathbb{N},K)})$ with some constant. Over every K and in every degree $n \geq p$, Proposition 10.14(2) gives a law that is not ambient of degree n, on a space that is a Banach space if K is complete; if K is discretely valued or $p = 2$, Proposition 10.13(2) gives such laws in every degree $n \geq p+1$ also on the space $H = K \oplus c_0(\mathbb{N}, K) \oplus G$ of Section 10.4. These spaces have no equivalent nonarchimedean norm: they contain ℓ^1-sums of finite-dimensional quotients of weighted ℓ^1-spaces, and both proofs obtain their lower bounds, for ν and for $\|\cdot\|_\Gamma$, by adding up contributions of the coordinates of these ℓ^1-spaces.

Targets, bases and domains. An equivalent nonarchimedean, spherically complete norm makes a Banach space F a universal target (Section 7.2, Theorem 5.2), but the given norm of F need not be spherically complete (Remark I.3). Over complete, not spherically complete fields universal targets have zero dual (Proposition 7.5); we do not know whether that suffices, nor whether over densely valued, spherically complete fields every nonarchimedean Banach space is one. We do not know which P satisfy (U_P^k) (Question 12.11(1)); for Banach P over complete fields universal analytic reflection suffices (Theorem G.3) but, over nonarchimedean fields, is not necessary (Theorem 11.9(4)). We do not know whether compatibility with all analytic coordinate changes forces the ambient and intrinsic analytic forms to agree; one product structure does not (Proposition 10.3). Over the t-adic field $\mathbb{F}_p(t)$ a largest full analytic domain need not exist (Theorem H.4); we do not know whether this happens over complete fields. Under (L) with K complete, $\mathrm{Alt}^k(TM; K)$ is analytic on every analytic manifold M (Corollary 5.3); we do not know this over incomplete fields such as $\kappa(X)$, nor for tangent-valued forms on

$\ell^r(\mathbb{N}, K)$, $4 < r < \infty$ (Theorem 11.17(3)). Nor do we know whether Bourbaki's α_k is a vector functor of class C^r over densely valued, spherically complete fields with $k! = 0$.

13. Formal verification

The Lean proofs are in [18]. Its file `challenge.md` is a ledger of 76 claims of this paper. Each claim has a challenge file, stating it in Lean with `sorry` as proof, and a solution file proving the same statement. For a repository identifier `ID` such as `Thm4_4`, `python3 scripts/check_claim.py ID` builds both files and checks that the proof uses only the axioms `propext`, `Classical.choice` and `Quot.sound` and that the two statements agree when printed with all implicit arguments (`pp.all`). All 76 claims pass with Lean `4.34.0-rc2` and the Mathlib fork `Deicyde/mathlib4` at `2b73d98` (upstream `0383a80e64` plus the commit of [19]). The six statements of the comparator challenge in the same repository pass the comparator `https://github.com/leanprover/comparator` at revision `19e111e`.

Table 1 gives the identifiers of the 24 claims in Sections 2–10. The other 52 claims are the appendix statements A.1–A.3, B.2–B.13, B.15, C.1–C.4, C.6–C.8, D.1–D.9, D.11, E.1, E.2, F.1–F.5, G.1, G.3–G.5, H.3–H.7 and I.1–I.3; the identifier of each is formed from its type and number, for example `LemB_2` for Lemma B.2.

Table 1. Repository identifiers of the ledger claims in Sections 2–10

Statement	Identifier	Statement	Identifier
Theorem 2.1	`Thm2_1`	Theorem 5.12(2)	`Thm4_5b`
Proposition 2.2	`Prop2_2`	Corollary 5.15	`Cor4_6`
Theorem 2.3	`Thm3_1`	Theorem 7.2(1)	`Thm6_1a`
Lemma 2.4	`Lem3_2`	Theorem 7.2(2)	`Thm6_1b`
Proposition 2.5	`Prop3_3`	Corollary 7.3	`Cor6_2`
Proposition 3.1	`Prop5_1`	Theorem 7.4	`Thm6_3`
Theorem 3.2	`Thm5_2`	Proposition 7.5	`Sec9_zeroDual`
Proposition 5.1	`Prop4_1`	Proposition 10.1	`Prop6_4`
Theorem 5.2	`Thm4_2`	Corollary 10.2	`Prop6_4_bundle`
Corollary 5.3	`Cor4_3`	Proposition 10.3	`Prop6_5`
Theorem 5.11	`Thm4_4`	Theorem 10.7	`Thm7_2`
Theorem 5.12(1)	`Thm4_5a`	Proposition 10.8	`Prop7_3`

A numbered statement of this paper has a Lean proof of a corresponding statement exactly when it is one of the 24 claims of Table 1 or the 52 claims just listed. Excluded are the definitions, the conjectures and questions of Section 12, Remarks B.14, C.5 and D.10, Proposition 10.4, the coefficient bound on retracts in Theorem 5.12(1), and all other theorems, propositions, lemmas, corollaries and remarks of Sections 3–12 and Appendices J–O. The excluded results have written proofs only, except Lemma M.9, which is cited from [21], and apart from the Lean declarations named below.

`challenge.md` and the docstrings of the challenge files list where a ledger statement differs from the text. The main differences: Theorem 10.7 and Proposition 10.8 are stated on open subsets of a normed space, since Mathlib has no differential forms on manifolds; the Lean statement of Theorem 2.1 groups the variables by variance, uses the canonical trivializations and does not state independence of the trivializing cover; Corollary C.7 is stated for the explicit transition and section maps, and Corollary 5.15 for objects and morphisms without the functor laws.

In [18], `alternatingMapAction` is the joint action B of Section 2, and the alternating bundle construction rests on the family-preservation premise: for $n \in \mathbb{N} \cup \{\infty, \omega\}$, a C^n manifold M,

normed spaces E, D, F, F' and $B\colon \mathcal{L}(E,D)\times\mathcal{L}(F,F')\to\mathcal{L}(\mathrm{Alt}^k(D;F),\mathrm{Alt}^k(E;F'))$,

$$U\subseteq M \text{ open},\quad \gamma\colon M\to\mathcal{L}(E,D)\times\mathcal{L}(F,F'),\quad \gamma|_U \text{ is } C^n \quad\Longrightarrow\quad (B\circ\gamma)|_U \text{ is } C^n. \tag{13.1}$$

The predicate `PreservesAlternatingFamilies` states (13.1) and typechecks against [18]. The declarations

```
contMDiffOn_continuousAlternatingMapCoordChange_of_family
alternatingVectorPrebundle_isContMDiff_of_family
contMDiffVectorBundle_alternating_of_family
contMDiffVectorBundle_alternating_of_finiteCoordinates
alternatingBundleHom_of_family
```

of [18] index alternating maps by `Fin k`. The first three assume (13.1) with $D=E$ and $F'=F$ and show that the alternating coordinate changes, prebundle and bundle are C^n. The fourth shows that over an analytic manifold modeled on a space P with a continuous linear isomorphism $P\to K^d$ the alternating bundle of two analytic bundles is analytic, with the premise from Theorem 5.11. The fifth assumes (13.1) and turns C^n bundle morphisms into a C^n morphism of the induced alternating bundles when these are C^n.

Appendix A. Preliminaries for the obstruction proofs

The constructions below use three elementary facts about valued fields and power series. We record their proofs to make explicit the scope of the coefficient argument, including incomplete base fields.

Lemma A.1 (one-variable uniqueness)**.** *Let K be nontrivially normed and Z a normed K-space, not necessarily complete. Let $z_0, z_1, \ldots \in Z$, $\rho>0$ and $M\geq 0$ with $\|z_n\|\rho^n\leq M$ for all n. If $\sum_n t^n z_n$ converges to 0 for every $t\in K$ with $|t|<\rho$, then $z_n=0$ for all n.*

Proof. Otherwise let N be least with $z_N\neq 0$. For $0<|t|<\rho/2$ the series $\sum_{n>N} t^{n-N}z_n$ converges, being t^{-N} times the convergent tail $\sum_{n>N}t^n z_n$, and $t^N\bigl(z_N+\sum_{n>N}t^{n-N}z_n\bigr)=0$. Hence

$$\|z_N\| = \Bigl\|\sum_{n>N} t^{n-N}z_n\Bigr\| \leq \sum_{n>N} M\rho^{-n}|t|^{n-N} = M\rho^{-N}\frac{|t|/\rho}{1-|t|/\rho},$$

where the norm of a convergent series is bounded by the sum of the norms by continuity of the norm on partial sums. Since K is nontrivially normed, there are $t\neq 0$ with $|t|$ arbitrarily small, so $\|z_N\|=0$, a contradiction. □

Lemma A.2. *A normed field of characteristic $p>0$ is nonarchimedean.*

Proof. Frobenius and the triangle inequality give, for every $m\geq 0$,

$$|x+y|^{p^m} = |x^{p^m}+y^{p^m}| \leq 2\max(|x|,|y|)^{p^m}.$$

Taking p^m-th roots and letting $m\to\infty$ proves the assertion. □

Lemma A.3. *Let $\mathbb{K}$ be a complete nontrivially normed field and U a finite-dimensional normed $\mathbb{K}$-space. Then every linear functional on U is bounded, and U is complete.*

Proof. We prove both assertions by induction on $n=\dim U$, the case $n=0$ being immediate. Suppose $n\geq 1$ and let $g\neq 0$ be a linear functional on U. Its kernel H has dimension $n-1$, so the induction hypothesis makes H complete and therefore closed in U. Choose v with $g(v)=1$. Since $v\notin H$, $\delta:=\inf_{h\in H}\|v-h\|>0$. Write $x=g(x)v+h$ with $h\in H$. If $g(x)\neq 0$, then

$$\|x\| = |g(x)|\,\|v+g(x)^{-1}h\| \geq |g(x)|\delta.$$

Thus $|g(x)| \le \delta^{-1}\|x\|$ for every x, proving boundedness.

Choose a basis of U. Its coordinate functionals are now bounded, so the coordinate map $U \to \mathbb{K}^n$, with the max norm on $\mathbb{K}^n$, is bounded. The inverse map $(c_i) \mapsto \sum_i c_i e_i$ is bounded by $\sum_i \|e_i\|$. Hence U is bi-Lipschitz equivalent to the complete space $\mathbb{K}^n$, and is itself complete. □

When the scalar field needs to be displayed, $A^{k,L}$ denotes the degree-k precomposition action formed over L. For the precomposition action A, a *bounded lift* means a bounded k-linear map on its operator domain whose diagonal is A, as in Proposition 2.5. That proposition applies at every base point; Lemma A.1 provides the one-variable coefficient comparison used in its proof.

Appendix B. The multiplier theorem over a finite field

This appendix proves Theorem B.9: over a finite field with $k! = 0$, coordinatewise multiplication has no lift satisfying (Pol1) with uniformly bounded support dimension. Ramsey's theorem reduces the coefficients to cluster values (Lemmas B.10, B.11), (Pol1) makes them nonconstant (Lemma B.13), and the support bound forbids the resulting staircase (Lemma B.12).

B.1. **Exterior powers, supports and determinant arrays.** Let L be a field, V an L-vector space and $k \ge 1$. We write $\Lambda^k V = \Lambda^k_L V$ for the algebraic exterior power, in which $x_1 \wedge \cdots \wedge x_k = 0$ whenever two x_i coincide. We use the standard facts: for a subspace $W \subseteq V$ the map $\Lambda^k W \to \Lambda^k V$ is injective, and we regard $\Lambda^k W$ as a subspace of $\Lambda^k V$; if $u_1, \dots, u_d$ is a basis of a finite-dimensional space U, the wedges $u_I := u_{i_1} \wedge \cdots \wedge u_{i_k}$, $I = \{i_1 < \cdots < i_k\}$, form a basis of $\Lambda^k U$; and every $\omega \in \Lambda^k V$ lies in $\Lambda^k U$ for some finite-dimensional $U \subseteq V$.

Definition B.1. The *support dimension* of $\omega \in \Lambda^k V$ is

$$\operatorname{sdim}(\omega) := \min\{\dim W : \ W \subseteq V \text{ finite-dimensional}, \ \omega \in \Lambda^k W\}.$$

The definition gives $\operatorname{sdim}(0) = 0$, $\operatorname{sdim}(c\omega) = \operatorname{sdim}(\omega)$ for $c \neq 0$, $\operatorname{sdim}(\omega + \omega') \le \operatorname{sdim}(\omega) + \operatorname{sdim}(\omega')$ (use $W + W'$), and $\operatorname{sdim}(\omega) \le \dim \operatorname{span} S$ if ω is a sum of wedges of vectors from a finite set S. If $V \subseteq V'$, then sdim computed in V' is at most sdim computed in V.

For spaces of functions, exterior vectors can be detected by determinants of coordinate evaluations. These arrays also turn bounds on support dimension into ordinary matrix-rank bounds. Let S be a set and $V \subseteq L^S$ a space of functions on S. The *determinant array* is the linear map

$$\Omega\colon \Lambda^k V \to L^{S^k}, \qquad \Omega_{y_1 \wedge \cdots \wedge y_k}(c_1, \dots, c_k) := \det\big(y_b(c_a)\big)_{a,b=1}^{k}.$$

It is well defined because the right side is multilinear and alternating in $(y_1, \dots, y_k)$ (these are the columns). For each ω, the function Ω_ω is alternating in $(c_1, \dots, c_k)$ (the rows); in particular $\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)}) = \operatorname{sgn}(\sigma)\, \Omega_\omega(c_1, \dots, c_k)$.

Lemma B.2. *The determinant array $\Omega\colon \Lambda^k V \to L^{S^k}$ is injective.*

Proof. Let $\Omega_\omega = 0$ and choose a finite-dimensional $U \subseteq V$ with $\omega \in \Lambda^k U$. The evaluations $\mathrm{ev}_s|_U$, $s \in S$, separate the points of U (a function in U vanishing at every s is 0), so they span the dual space U^*. Choose $s_1, \dots, s_d \in S$ such that the $\mathrm{ev}_{s_i}|_U$ form a basis of U^*, and let $u_1, \dots, u_d$ be the dual basis of U, so that $u_j(s_i) = \delta_{ij}$. For increasing k-subsets I, J of $\{1, \dots, d\}$ we get $\Omega_{u_I}(s_{j_1}, \dots, s_{j_k}) = \det(u_{i_b}(s_{j_a}))_{a,b} = \delta_{IJ}$. Writing $\omega = \sum_I c_I u_I$ therefore gives $0 = \Omega_\omega(s_{j_1}, \dots, s_{j_k}) = c_J$ for every J, so $\omega = 0$. □

Lemma B.3 (support bounds flattening rank)**.** *Let $V \subseteq L^S$, $\omega \in \Lambda^k V$ and $\alpha \in \{1, \dots, k\}$. Let $R \subseteq S$ and $R' \subseteq S^{\{1,\dots,k\}\setminus\{\alpha\}}$ be finite. The $R \times R'$ matrix M with entries $M(c_\alpha, c') := \Omega_\omega(c)$, where c has c_α in slot α and the entries of c' in the other slots, has rank at most* $\operatorname{sdim}(\omega)$.

Proof. Write $\omega = \sum_j y_{j,1} \wedge \cdots \wedge y_{j,k}$ with all $y_{j,b}$ in a subspace W of dimension $\operatorname{sdim}(\omega)$. Expanding each determinant along row α, $\det(y_{j,b}(c_a))_{a,b} = \sum_b \pm y_{j,b}(c_\alpha) \cdot m_{j,b}(c')$, where the minor $m_{j,b}(c')$ does not involve c_α. So for fixed c' the column $c_\alpha \mapsto M(c_\alpha, c')$ is a linear combination of the restrictions $y_{j,b}|_R$. All columns lie in the restriction of W to R, a space of dimension at most $\operatorname{sdim}(\omega)$. □

A second way to obtain lower bounds on support is to contract all but one exterior factor. The resulting vectors must lie in every subspace supporting the original exterior vector.

Lemma B.4 (contraction)**.** *Let $k \geq 2$ and $\varphi = (\varphi_1, \ldots, \varphi_{k-1}) \in (V^*)^{k-1}$. There is a linear map $c_\varphi \colon \Lambda^k V \to V$ with*

$$c_\varphi(x_1 \wedge \cdots \wedge x_k) = \sum_{i=1}^{k} (-1)^{k+i} \det\bigl(\varphi_a(x_b)\bigr)_{1 \leq a \leq k-1,\ b \neq i}\, x_i,$$

and $c_\varphi(\Lambda^k W) \subseteq W$ for every subspace W. Consequently $\operatorname{sdim}(\omega) \geq \dim \operatorname{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$.

Proof. The right side is multilinear in (x_b). For $\psi \in V^*$, applying ψ to it gives, by Laplace expansion along the last row, the determinant of the $k \times k$ matrix with entries $\varphi_a(x_b)$ in the rows $a < k$ and $\psi(x_b)$ in the last row; it vanishes when two columns coincide. Since V^* separates the points of V, the right side vanishes whenever $x_a = x_b$ for some $a \neq b$, so it factors through $\Lambda^k V$. Each term is a multiple of some x_i, which gives $c_\varphi(\Lambda^k W) \subseteq W$. If $\omega \in \Lambda^k W$ with $\dim W = \operatorname{sdim}(\omega)$, all $c_\varphi(\omega)$ lie in W. □

Proposition B.5 (canonical exterior support)**.** *Let L be any field, V an L-vector space, $k \geq 2$, and $\omega \in \Lambda^k V$. Then*

$$S(\omega) := \operatorname{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$$

is the unique smallest subspace supporting ω. In particular, $\omega \in \Lambda^k S(\omega)$ and $\operatorname{sdim}(\omega) = \dim S(\omega)$.

Proof. Choose a finite-dimensional supporting subspace $U \subseteq V$. Lemma B.4 gives $S(\omega) \subseteq U$, so $S(\omega)$ is finite-dimensional. Choose a basis $e_1, \ldots, e_d$ of U whose first r vectors form a basis of $S(\omega)$, and extend its coordinate functionals algebraically to V. Write $\omega = \sum_{|I|=k} a_I e_I$ in the exterior basis of U. If $j \in I$ and $j > r$, contract against the $k-1$ coordinate functionals indexed by $I \setminus \{j\}$. The e_j-coordinate of this contraction is $\pm a_I$: no other basis wedge contributes to that coordinate. The contraction belongs to $S(\omega)$, so this coordinate is zero. Hence $a_I = 0$ whenever I meets $\{r+1, \ldots, d\}$, proving $\omega \in \Lambda^k S(\omega)$. Every supporting subspace contains all the contractions by Lemma B.4, and therefore contains $S(\omega)$. This proves minimality and the dimension formula. Only unit signs occur, so the argument is valid in every characteristic. □

The same conclusion holds in degree one: under $\Lambda^1 V \simeq V$, the empty-tuple contraction is the identity and $S(\omega) = L\omega$. In degree zero, every element of $\Lambda^0 V = L$ is supported by the zero subspace, so its support dimension is zero; no contraction to V is needed.

For matrix calculations, choose any finite-dimensional supporting subspace U and a basis of it. Form the finite matrix whose columns are the coordinates of the contractions against $(k-1)$-tuples of coordinate functionals, extended to V. Arbitrary functionals restrict to linear combinations of these coordinates on U; multilinearity in the covectors therefore shows that the column span is exactly $S(\omega)$. The matrix consequently has rank $\operatorname{sdim}(\omega)$. This uses ordinary matrix rank, and entails no choice of a rank convention for an infinite matrix.

B.2. **Lifts and three diagonal identities.** Fix a field L, an integer $k \geq 1$, L-vector spaces A and V, and a linear map $\mu\colon A \to \mathrm{End}_L(V)$, $a \mapsto \mu_a$. A *lift of μ in degree k* is a $2k$-linear map $\Psi\colon A^k \times V^k \to \Lambda^k V$ over L that is alternating in the V-slots; its diagonal identity will be specified separately. For every $n \geq 1$ write $[n] := \{1, \dots, n\}$, and for $\alpha \in \mathbb{N}^m$ put $|\alpha| := \sum_i \alpha_i$. Given $m \geq 1$, a map $f\colon [k] \to [m]$, a family $b = (b_1, \dots, b_m) \in A^m$ and $\lambda \in L^m$, put

$$b_f := (b_{f(1)}, \dots, b_{f(k)}), \quad \operatorname{type} f := \big(|f^{-1}(1)|, \dots, |f^{-1}(m)|\big) \in \mathbb{N}^m, \quad \lambda_f := \prod_{j=1}^{k} \lambda_{f(j)} = \lambda^{\operatorname{type} f}.$$

The *pointwise identity* is

$$\Psi(a, \dots, a; x_1, \dots, x_k) = \mu_a x_1 \wedge \cdots \wedge \mu_a x_k \qquad \text{for all } a \in A,\ x \in V^k. \tag{Pw}$$

The *polarized identity* is: for all $m \geq 1$, $b \in A^m$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in V^k$,

$$\sum_{\operatorname{type} f = \alpha} \Psi(b_f; x) = \sum_{\operatorname{type} f = \alpha} \mu_{b_{f(1)}} x_1 \wedge \cdots \wedge \mu_{b_{f(k)}} x_k. \tag{Pol}$$

Its *multilinear instance* is: for all $b_1, \dots, b_k \in A$ and $x \in V^k$,

$$\sum_{\sigma \in S_k} \Psi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x) = \sum_{\sigma \in S_k} \mu_{b_{\sigma(1)}} x_1 \wedge \cdots \wedge \mu_{b_{\sigma(k)}} x_k. \tag{Pol1}$$

Here $\sum_{\operatorname{type} f = \alpha}$ runs over the maps $f\colon [k] \to [m]$ of type α. (Pol1) is the instance $m = k$, $\alpha = (1, \dots, 1)$ of (Pol), since the maps of type $(1, \dots, 1)$ are the permutations. For $b \in A^m$ and $x \in V^k$ put $a_\lambda := \sum_i \lambda_i b_i$ and

$$\Delta_{b,x}(\lambda) := \Psi(a_\lambda, \dots, a_\lambda; x) - \mu_{a_\lambda} x_1 \wedge \cdots \wedge \mu_{a_\lambda} x_k = \sum_{|\alpha| = k} \lambda^\alpha\, \delta_\alpha(b, x),$$

$$\delta_\alpha(b, x) := \sum_{\operatorname{type} f = \alpha} \Big[\Psi(b_f; x) - \mu_{b_{f(1)}} x_1 \wedge \cdots \wedge \mu_{b_{f(k)}} x_k\Big].$$

Multilinearity of Ψ in the A-slots, linearity of μ and multilinearity of the wedge give the displayed expansion. Thus $\Delta_{b,x}$ is a homogeneous polynomial of degree k with coefficients in $\Lambda^k V$. Identity (Pol) says that every coefficient $\delta_\alpha(b, x)$ is zero. Identity (Pw) says instead that $\Delta_{b,x}$ vanishes at every point of L^m; conversely, this pointwise vanishing gives (Pw) by taking $m = 1$ and $\lambda = 1$. The following two identity principles identify the cases in which pointwise vanishing forces all coefficients to vanish.

Lemma B.6 (identity principle)**.** *Let $\mathbb{K}$ be an infinite field, Y a $\mathbb{K}$-vector space and $g(\lambda) = \sum_\alpha \lambda^\alpha y_\alpha$ a polynomial in $\lambda \in \mathbb{K}^m$ with finitely many nonzero coefficients $y_\alpha \in Y$. If $g(\lambda) = 0$ for all $\lambda \in \mathbb{K}^m$, then every $y_\alpha = 0$.*

Proof. The finitely many coefficients y_α span a finite-dimensional subspace of Y. Applying its coordinate functionals reduces the claim to scalar polynomials. For these, induction on m reduces to the fact that a nonzero polynomial in one variable has only finitely many roots. □

Lemma B.7 (homogeneous forms over a finite field)**.** *Let q be a prime power, $L' \supseteq \mathbb{F}_q$ a field, $1 \leq e \leq q$, and $g \in L'[\lambda_1, \dots, \lambda_m]$ homogeneous of degree e with $g(\lambda) = 0$ for every $\lambda \in \mathbb{F}_q^m$. Then $g = 0$.*

Proof. Call a monomial λ^β *reduced* if every $\beta_i \leq q - 1$. The indicator function of a point $a \in \mathbb{F}_q^m$ is $\prod_i \big(1 - (\lambda_i - a_i)^{q-1}\big)$, which expands into reduced monomials. So the q^m reduced monomials span the q^m-dimensional L'-space of functions $\mathbb{F}_q^m \to L'$, and they are linearly independent as functions. A monomial of degree $e \leq q$ that is not reduced is λ_i^q (and then $e = q \geq 2$); as a

function on $\mathbb{F}_q^m$ it agrees with the reduced monomial λ_i, of degree $1 < e$. If $e = q$, replace each λ_i^q in g by λ_i. These degree-one images are distinct, and homogeneity separates them from all the remaining monomials, which still have degree q. The replacement therefore sends the monomials of g injectively to reduced monomials without changing the associated function. Linear independence of the reduced monomials forces every coefficient to vanish. If $e < q$, no replacement is needed. □

Proposition B.8 (pointwise versus polarized)**.** *Let Ψ be a lift of μ in degree k over a field L.*

(1) (Pol)⇒(Pw)⇒(Pol1)*, over every field.*
(2) *If L is infinite, or if $L = \mathbb{F}_q$ and $k \le q$, then* (Pw)⇒(Pol)*.*
(3) *If $L = \mathbb{F}_q$ and $k \ge q + 1$, then* (Pw) *does not imply* (Pol) *in general.*
(4) *If $k! = 0$ in L, then* (Pol1) *does not imply* (Pw) *in general.*

Proof. (1) Inclusion–exclusion. The implication (Pol)⇒(Pw) is the case $m = 1$ and $\alpha = (k)$. To prove (Pw)⇒(Pol1), fix $b_1, \dots, b_k \in A$ and $x \in V^k$, and put $b_S := \sum_{i \in S} b_i$ for $S \subseteq [k]$. Multilinearity gives

$$\sum_{S \subseteq [k]} (-1)^{k-|S|} \Psi(b_S, \dots, b_S; x) = \sum_{S} (-1)^{k-|S|} \sum_{f\colon [k] \to S} \Psi(b_f; x)$$
$$= \sum_{f\colon [k] \to [k]} \Psi(b_f; x) \sum_{S \supseteq f([k])} (-1)^{k-|S|} = \sum_{f \text{ bijective}} \Psi(b_f; x),$$

because $\sum_{S \supseteq T} (-1)^{k-|S|} = (1-1)^{k-|T|}$ is 1 if $T = [k]$ and 0 otherwise. The same computation applies to the multilinear map $(a_1, \dots, a_k) \mapsto \mu_{a_1} x_1 \wedge \cdots \wedge \mu_{a_k} x_k$, whose diagonal is the right side of (Pw). Evaluating (Pw) at the 2^k subset sums b_S therefore gives (Pol1). The computation is valid in every characteristic and involves no division by $k!$.

(2) The identity principles. The coefficients of $\Delta_{b,x}$ span a finite-dimensional subspace of $\Lambda^k V$; in a basis of it each coordinate of $\Delta_{b,x}$ is a scalar form of degree k vanishing on L^m. It is zero by Lemma B.6 if L is infinite, and by Lemma B.7 if $L = \mathbb{F}_q$ and $k \le q$.

(3) Pointwise vanishing without full polarization. Let $\dim A \ge 3$, with $b_1, b_2, b_3 \in A$ and functionals g_1, g_2, g_3 on A such that $g_i(b_j) = \delta_{ij}$, and let $y_1, \dots, y_k \in V$ be linearly independent. Put

$$\theta(a_1, \dots, a_k) := \big[g_1(a_1) \cdots g_1(a_q)\, g_2(a_{q+1}) - g_1(a_1)\, g_2(a_2) \cdots g_2(a_{q+1})\big] \cdot g_3(a_{q+2}) \cdots g_3(a_k)$$

(for $k = q + 1$ the g_3-factor is empty and $\dim A \ge 2$ suffices), and $\Theta(a; x) := \theta(a)\, x_1 \wedge \cdots \wedge x_k$. Then Θ is a lift of $\mu = 0$. It satisfies (Pw), because $\theta(a, \dots, a) = (g_1(a)^q g_2(a) - g_1(a) g_2(a)^q)\, g_3(a)^{k-q-1} = 0$ since $c^q = c$ for $c \in \mathbb{F}_q$. It violates (Pol) for $\alpha = (q, 1, k - q - 1)$ and $b = (b_1, b_2, b_3)$: the first product contributes to $\sum_{\text{type } f = \alpha} \theta(b_f)$ exactly for $f = (1, \dots, 1, 2, 3, \dots, 3)$, with value 1; the second product would need $f(2) = \cdots = f(q+1) = 2$, that is, at least $q \ge 2$ values equal to 2, which is impossible for type α. So the coefficient $\delta_\alpha(b, y)$ of Θ is $y_1 \wedge \cdots \wedge y_k \ne 0$. When $k = q + 1$, the third label and its functional can be omitted throughout, with $\alpha = (q, 1)$.

(4) The multilinear identity without the pointwise identity. Take $\bar{a} \in A$ and a functional g_1 with $g_1(\bar{a}) = 1$, and $y_1, \dots, y_k$ linearly independent. The lift $\Theta'(a; x) := g_1(a_1) \cdots g_1(a_k)\, x_1 \wedge \cdots \wedge x_k$ of $\mu = 0$ satisfies (Pol1), since $\sum_\sigma \Theta'(b_\sigma; x) = k!\, g_1(b_1) \cdots g_1(b_k)\, x_1 \wedge \cdots \wedge x_k = 0$ for all $b_1, \dots, b_k \in A$ and $x \in V^k$, but $\Theta'(\bar{a}, \dots, \bar{a}; y) = y_1 \wedge \cdots \wedge y_k \ne 0$. □

The finite-field obstruction below requires only (Pol1), which follows from either (Pw) or (Pol). The distinction between the latter two remains important: over $\mathbb{F}_q$ they need not agree when $k > q$. The lift constructed in Appendix C nevertheless satisfies the full identity (Pol) in every

degree: the identity is polarized over an infinite Laurent-series field before constant coefficients are taken (Lemma C.4).

B.3. **Statement and proof strategy.** For the proof of the theorem, assume that L is a *finite* field with $k! = 0$ in L, that is, $\operatorname{char} L = p \le k$; in particular $k \ge 2$. Put $V := L^{\mathbb{N}}$ and let $V_{\mathrm{fin}} := L^{(\mathbb{N})} \subseteq V$ be the subspace of finitely supported sequences. For $u, x \in V$ we write ux for the coordinatewise product; so $\mu_u x = ux$ is the multiplier family.

Theorem B.9 (multipliers over a finite field)**.** *Let L be a finite field and $k \ge 1$ with $k! = 0$ in L.*

(1) *There is no $2k$-linear map $\Psi\colon V_{\mathrm{fin}}^k \times V_{\mathrm{fin}}^k \to \Lambda_L^k V$ over L such that*
   (a) *Ψ is antisymmetric in the last k slots;*
   (b) *Ψ satisfies (Pol1) for $\mu_u x = ux$: for all $u_1, \dots, u_k, x_1, \dots, x_k \in V_{\mathrm{fin}}$,*
   $$\sum_{\sigma \in S_k} \Psi(u_{\sigma(1)}, \dots, u_{\sigma(k)}; x_1, \dots, x_k) = \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k);$$
   (c) *for some $d \in \mathbb{N}$, $\operatorname{sdim}(\Psi(u; x)) \le d$ for all $u, x \in V_{\mathrm{fin}}^k$ with entries in $\{0, \pm 1\}$.*

(2) *Let $E_0 = V$ or $E_0 = V_{\mathrm{fin}}$. There is no $2k$-linear map $\Psi\colon E_0^k \times E_0^k \to \Lambda_L^k E_0$ over L that is antisymmetric in the last k slots, satisfies (Pol1) for all finitely supported u, x, and has $\operatorname{sdim}(\Psi(u; x)) \le d$ for all $u, x \in E_0^k$, for some d.*

Reduction to (1). Given a map as in (2), restrict its inputs to $V_{\mathrm{fin}}^k \times V_{\mathrm{fin}}^k$ and compose its output with the injective map $\Lambda^k E_0 \to \Lambda^k V$. This preserves antisymmetry and (Pol1). Support dimension can only decrease under the inclusion of the ambient space (§B.1), so the resulting map satisfies (a)–(c).

We therefore prove (1). Suppose that Ψ satisfies its hypotheses, and fix the bound d in (c). Write $e_i \in V_{\mathrm{fin}}$ for the unit vectors, $e_a := (e_{a_1}, \dots, e_{a_k})$ for $a \in \mathbb{N}^k$, and Ω for the determinant array of §B.1 with $S = \mathbb{N}$.

The proof has three steps. Ramsey's theorem first makes the coefficients of $\Omega\Psi$ depend only on the relative order and equalities among their indices. Special weights on four-point clusters then cancel every contribution from an occupied cluster containing no output index. This cancellation produces a two-valued staircase matrix from one exterior vector. Its bounded support forces the two values to agree, and hence makes all cluster order patterns have a common value. Finally, antisymmetry and (Pol1) say that the sum of these $k!$ equal values is 1, contradicting $k! = 0$ in L.

B.4. **Order-homogeneity from Ramsey's theorem.** We use Ramsey's theorem [21] in its infinite form: *for every $n \ge 1$ and every coloring of the n-element subsets of $\mathbb{N}$ with finitely many colors there is an infinite $H \subseteq \mathbb{N}$ all of whose n-element subsets have the same color.*

For a tuple $z \in \mathbb{N}^N$ with distinct values $y_1 < \cdots < y_r$, its *pattern* is the map $i \mapsto j$ determined by $z_i = y_j$. In particular, repeated entries are retained in the pattern. Two tuples z and z' have the same pattern if and only if $z_i < z_j \Leftrightarrow z'_i < z'_j$ and $z_i = z_j \Leftrightarrow z'_i = z'_j$ for all i, j.

Lemma B.10 (pattern homogeneity)**.** *For every $N \ge 1$ and every function $T\colon \mathbb{N}^N \to L$ there is an infinite $H \subseteq \mathbb{N}$ such that $T(z) = T(z')$ whenever $z, z' \in H^N$ have the same pattern.*

Proof. Color an N-element set $s = \{s_1 < \cdots < s_N\} \subseteq \mathbb{N}$ by the vector
$$\big(T(s_{\beta(1)}, \dots, s_{\beta(N)})\big)_{\beta\colon [N] \to [N]} \in L^{N^N}.$$
The coloring records all maps β, including those that repeat indices. It has finitely many values because L is finite; this is the only use of the finiteness of L in the obstruction proof. Ramsey's

theorem gives an infinite H on which the color is constant, say equal to $(t_\beta)_\beta$. Let $z \in H^N$ have value set $\{y_1 < \cdots < y_r\}$. Since H is unbounded, add $N - r$ elements of H above y_r to obtain an N-element set $s \subseteq H$. Padding above the largest value leaves the ranks unchanged: $s_j = y_j$ for $j \le r$, and $z = (s_{\beta(1)}, \dots, s_{\beta(N)})$, where $\beta(i)$ is the rank of z_i in $\{y_1, \dots, y_r\}$. The map β depends only on the pattern of z, and $T(z) = t_\beta$. □

Coefficients. For $(a; y; c) \in \mathbb{N}^k \times \mathbb{N}^k \times \mathbb{N}^k$ put

$$T(a; y; c) := \Omega_{\Psi(e_a; e_y)}(c_1, \dots, c_k) \in L.$$

Lemma B.10 with $N = 3k$ gives an infinite set $H = \{h_0 < h_1 < h_2 < \cdots\}$ on which T depends only on the pattern. For $u_j, v_j \in V_{\text{fin}}$ supported in H, multilinearity of Ψ and linearity of Ω give

$$\Omega_{\Psi(u;v)}(c) = \sum_{a,y \in H^k} \prod_{j=1}^{k} u_j(a_j) \prod_{j=1}^{k} v_j(y_j) \cdot T(a; y; c). \tag{B.1}$$

Although the sum is indexed by $H^k \times H^k$, only finitely many summands are nonzero, because all the input vectors have finite support. Every output tuple c used below also lies in H^k, so pattern homogeneity applies to all $3k$ indices in (B.1).

B.5. **Clusters and the cluster value.** A *cluster* is a set $C = \{h_n, h_{n+1}, h_{n+2}, h_{n+3}\}$ of four *consecutive* elements of H; we write $C(i) := h_{n+i-1}$ for $i = 1, \dots, 4$. Since a cluster is an interval of H, two disjoint clusters are *comparable*: every element of one is smaller than every element of the other; we write $C < C'$. Distinct clusters need not be adjacent in H.

We choose two weight vectors, one for each kind of input slot:

$$s = (s_1, s_2, s_3, s_4) := (1, 0, -1, 0) \qquad \text{on the operator slots,}$$
$$w = (w_1, w_2, w_3, w_4) := (1, -1, 1, -1) \qquad \text{on the vector slots.}$$

Over $\mathbb{Z}$, hence over every field,

$$\sum_i s_i = \sum_i w_i = \sum_i s_i w_i = \sum_{i<i'} s_i w_{i'} = \sum_{i>i'} s_i w_{i'} = 0, \qquad s_1 w_1 = 1. \tag{W}$$

Indeed $\sum_i s_i = 1 - 1$, $\sum_i w_i = 1 - 1 + 1 - 1$, $s_1 w_1 = 1$, and

$$\sum_i s_i w_i = s_1 w_1 + s_3 w_3 = 1 - 1 = 0,$$
$$\sum_{i<i'} s_i w_{i'} = s_1(w_2 + w_3 + w_4) + s_3 w_4 = -1 + 1 = 0,$$
$$\sum_{i>i'} s_i w_{i'} = s_3(w_1 + w_2) = -(1 - 1) = 0.$$

The first two zero sums will cancel a cluster occupied by only one input index. The other three will cancel a cluster occupied by an operator index and a vector index, according to their relative order. For a cluster C put

$$s_C := \sum_{i=1}^{4} s_i e_{C(i)}, \qquad w_C := \sum_{i=1}^{4} w_i e_{C(i)}.$$

These are vectors of V_{fin} with entries in $\{0, \pm 1\}$, and $s_C w_C = e_{C(1)} - e_{C(3)}$.

The cluster value. For pairwise disjoint clusters $C_1, \dots, C_k$, one for each slot, define

$$\chi(C_1, \dots, C_k) := \Omega_{\Psi(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})}\big(C_1(1), \dots, C_k(1)\big)$$
$$= \sum_{i, i' \in [4]^k} \prod_{j=1}^{k} s_{i_j} w_{i'_j} \cdot T\big(C_1(i_1), \dots, C_k(i_k);\, C_1(i'_1), \dots, C_k(i'_k);\, C_1(1), \dots, C_k(1)\big).$$

Let $\tau \in S_k$ be the *order pattern* of $(C_1, \dots, C_k)$: $\tau(j)$ is the position of C_j among $C_1, \dots, C_k$. The pattern of every $3k$-tuple in the sum is determined by (i, i') and τ: entries in different clusters compare as the clusters do, and the entries $C_j(i_j)$, $C_j(i'_j)$, $C_j(1)$ in the same cluster compare as their indices do, including equalities. Pattern homogeneity therefore shows that the displayed sum depends only on τ. Every permutation occurs as the order pattern of some disjoint clusters in H; we denote the resulting scalar by $\chi(\tau)$.

Lemma B.11 (cancellation of output-free clusters)**.** *Let $\mathcal{C}_1, \dots, \mathcal{C}_k$ be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in $\bigcup_j \mathcal{C}_j$ are pairwise disjoint. Put $u_j := \sum_{C \in \mathcal{C}_j} s_C$ and $v_j := \sum_{C \in \mathcal{C}_j} w_C$. Let $C_l \in \mathcal{C}_l$ for $l = 1, \dots, k$ and $c := (C_1(1), \dots, C_k(1))$. Then $\Omega_{\Psi(u;v)}(c) = \chi(C_1, \dots, C_k)$.*

Proof. We show that every cluster assignment except the one matching the output clusters has total contribution zero. Expand by (B.1). Since the clusters are disjoint, a nonzero term has, for each slot j, an operator index $a_j = C_j^{\mathrm{op}}(i_j)$ and a vector index $y_j = C_j^{\mathrm{vec}}(i'_j)$ with $C_j^{\mathrm{op}}, C_j^{\mathrm{vec}} \in \mathcal{C}_j$, and weight $\prod_j s_{i_j} w_{i'_j}$. Group the terms according to the *cluster assignment* $(C_j^{\mathrm{op}}, C_j^{\mathrm{vec}})_j$.

Call an occupied cluster *output-free* if it is none of $C_1, \dots, C_k$. Suppose a group has such a cluster Γ. Disjointness of the families assigns Γ to exactly one slot j, so the only tuple entries it can contain are a_j and y_j. Disjointness of the clusters excludes every output index from Γ.

Fix the positions of all indices outside Γ and sum over the positions inside it. Since Γ consists of consecutive elements of H, every outside entry lies either below the whole cluster or above it. Moving an index within Γ cannot change any comparison with an outside entry. By pattern homogeneity, the coefficient T therefore depends on the internal positions only as follows:

(i) if only $a_j \in \Gamma$, not at all, and the partial sum is $(\sum_i s_i) \cdot T = 0$;
(ii) if only $y_j \in \Gamma$, not at all, and the partial sum is $(\sum_i w_i) \cdot T = 0$;
(iii) if both lie in Γ, only through whether $i_j < i'_j$, $i_j = i'_j$ or $i_j > i'_j$; with the corresponding values $T_<, T_=, T_>$ of T the partial sum is

$$T_< \sum_{i < i'} s_i w_{i'} + T_= \sum_i s_i w_i + T_> \sum_{i > i'} s_i w_{i'} = 0.$$

In each case the weights of the other slots are a common factor, and the displayed partial sum vanishes by (W). Thus every assignment with an output-free cluster contributes zero. In any surviving assignment, every occupied cluster must be an output cluster. The only output cluster belonging to $\mathcal{C}_j$ is C_j, which forces $C_j^{\mathrm{op}} = C_j^{\mathrm{vec}} = C_j$ for every j. The contribution of this one assignment is exactly $\chi(C_1, \dots, C_k)$. □

B.6. **The staircase and the diagonal.** We now use the support bound to compare two order patterns that differ by an adjacent exchange. The key is to realize arbitrarily large staircase matrices as flattenings of a single value of Ψ.

Lemma B.12 (staircase)**.** *Let $\tau \in S_k$, let $\alpha \neq \beta$ be slots with $\tau(\alpha) = \ell$ and $\tau(\beta) = \ell + 1$, and let τ' be τ with these two positions exchanged, so $\tau'(\alpha) = \ell + 1$, $\tau'(\beta) = \ell$ and $\tau'(\gamma) = \tau(\gamma)$ otherwise. Then $\chi(\tau) = \chi(\tau')$.*

Proof. Let $m \geq 1$. Choose pairwise disjoint clusters, in increasing order in H: first a cluster C_γ for each slot γ with $\tau(\gamma) < \ell$, ordered as τ prescribes; then $2m$ clusters $A_1 < B_1 < A_2 < B_2 < \cdots < A_m < B_m$; then a cluster C_γ for each slot γ with $\tau(\gamma) > \ell + 1$, ordered as τ prescribes. Put $\mathcal{C}_\alpha := \{A_1, \dots, A_m\}$, $\mathcal{C}_\beta := \{B_1, \dots, B_m\}$ and $\mathcal{C}_\gamma := \{C_\gamma\}$ for the other slots, and let u, v be as in Lemma B.11 and $\omega := \Psi(u; v)$. Disjointness of the clusters ensures that the entries of u and v still lie in $\{0, \pm 1\}$, so $\operatorname{sdim}(\omega) \leq d$ by hypothesis (c) of Theorem B.9(1). For this fixed m, keep u, v and hence ω fixed while varying only the output indices.

Consider the $m \times m$ matrix M with rows indexed by $c_\alpha = A_n(1)$ and columns by $c_\beta = B_{n'}(1)$ $(n, n' = 1, \dots, m)$, all other output indices being fixed at $c_\gamma = C_\gamma(1)$, and entries $M_{nn'} := \Omega_\omega(c)$. By Lemma B.11, applied with the output clusters $A_n \in \mathcal{C}_\alpha$, $B_{n'} \in \mathcal{C}_\beta$ and $C_\gamma \in \mathcal{C}_\gamma$, $M_{nn'}$ is the cluster value of the tuple with A_n in slot α, $B_{n'}$ in slot β, and C_γ in every other slot. The interlacing order gives $A_n < B_{n'}$ exactly when $n \leq n'$. The tuple therefore has pattern τ on and above the matrix diagonal, and pattern τ' below it.

Put $\delta := \chi(\tau) - \chi(\tau')$, let $\mathbf{1} \in L^m$ be the all-ones column, and let U be the upper unitriangular matrix with $U_{nn'} = 1$ for $n \leq n'$ and $U_{nn'} = 0$ otherwise. Then

$$M = \begin{pmatrix} \chi(\tau) & \chi(\tau) & \cdots & \chi(\tau) \\ \chi(\tau') & \chi(\tau) & \cdots & \chi(\tau) \\ \vdots & \ddots & \ddots & \vdots \\ \chi(\tau') & \cdots & \chi(\tau') & \chi(\tau) \end{pmatrix} = \chi(\tau')\, \mathbf{1}\mathbf{1}^{\mathsf{T}} + \delta\, U,$$

The rows vary only slot α; each column fixes all the other slots, including the chosen value in slot β. Thus M is a submatrix of the flattening of this one array Ω_ω along slot α. Lemma B.3 gives $\operatorname{rank} M \leq \operatorname{sdim}(\omega) \leq d$.

If $\delta \neq 0$, the matrix δU is invertible, whereas the constant matrix $\chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}}$ has rank at most one. Applying rank subadditivity to $\delta U = M - \chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}}$ gives

$$m = \operatorname{rank}(\delta U) \leq \operatorname{rank} M + 1 \leq d + 1.$$

The construction is available for every $m \geq 1$; choosing $m = d + 2$ is a contradiction. Hence $\delta = 0$, as required. □

The support bound has now forced equality between adjacent order patterns. The remaining identity comes from (Pol1); it does not use the support bound.

Lemma B.13 (the diagonal)**.** $\sum_{\tau \in S_k} \chi(\tau) = 1$.

Proof. Take clusters $C_1 < \cdots < C_k$, and put $u_j := s_{C_j}$, $v_j := w_{C_j}$ and $c := (C_1(1), \dots, C_k(1))$. Apply $\omega \mapsto \Omega_\omega(c)$ to both sides of (Pol1), with the operator vectors $u_1, \dots, u_k$ and the vector arguments $v_1, \dots, v_k$.

Right side. If $\sigma \neq \mathrm{id}$, some slot j has $\sigma(j) \neq j$. Disjoint supports then give $u_{\sigma(j)} v_j = 0$, so the corresponding wedge vanishes. Only $\sigma = \mathrm{id}$ survives. It gives $\Omega_{(u_1 v_1) \wedge \cdots \wedge (u_k v_k)}(c) = \det\big((u_b v_b)(C_a(1))\big)_{a,b}$. Since $u_b v_b = e_{C_b(1)} - e_{C_b(3)}$ and $C_a(1) \neq C_b(3)$ for all a, b, this is the determinant of the identity matrix, so the right side is 1.

Left side. Fix $\sigma \in S_k$ and write $u_\sigma := (u_{\sigma(1)}, \dots, u_{\sigma(k)})$ and $v_\sigma := (v_{\sigma(1)}, \dots, v_{\sigma(k)})$. To identify the summand with a cluster value, we must permute both the vector arguments and the output coordinates to match the operator arguments. Each permutation contributes the same sign. More precisely, antisymmetry of Ψ in the vector slots gives $\Psi(u_\sigma; v) = \operatorname{sgn}(\sigma)\Psi(u_\sigma; v_\sigma)$. For every $\omega \in \Lambda^k V$ the function Ω_ω is alternating in its arguments, so $\Omega_\omega(c) = \operatorname{sgn}(\sigma)\, \Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)})$;

we apply this with $\omega = \Psi(u_\sigma; v_\sigma)$. Since $\operatorname{sgn}(\sigma)^2 = 1$,

$$\begin{aligned}\Omega_{\Psi(u_\sigma;\, v)}(c) &= \operatorname{sgn}(\sigma)\, \Omega_{\Psi(u_\sigma;\, v_\sigma)}(c) = \operatorname{sgn}(\sigma)^2\, \Omega_{\Psi(u_\sigma;\, v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}) \\ &= \Omega_{\Psi(u_\sigma;\, v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}).\end{aligned}$$

In the last expression slot j has operator $s_{C_{\sigma(j)}}$, vector $w_{C_{\sigma(j)}}$ and output $C_{\sigma(j)}(1)$, so it equals $\chi(C_{\sigma(1)}, \dots, C_{\sigma(k)})$ by definition. Its order pattern is σ, since $C_{\sigma(j)}$ is the $\sigma(j)$-th cluster in increasing order. Thus the convention for patterns gives σ, not σ^{-1}. No cluster cancellation is needed in this calculation.

As σ runs over S_k, each order pattern occurs exactly once. Comparing the two sides gives

$$\sum_{\sigma \in S_k} \Omega_{\Psi(u_\sigma;\, v)}(c) = \sum_{\tau \in S_k} \chi(\tau) = 1. \qquad \square$$

Proof of Theorem B.9. Exchanging adjacent ranks acts on an order pattern by $\tau \mapsto (\ell\ \ell{+}1) \circ \tau$, $1 \le \ell < k$. These transpositions generate S_k, and Lemma B.12 makes χ invariant under each exchange. Hence χ is constant; write its value as χ_0. Lemma B.13 now gives $k!\, \chi_0 = 1$ in L, contradicting $k! = 0$. This proves (1), and the preceding reduction gives (2). $\square$

Remark B.14 (where the hypotheses enter). Ramsey homogeneity is the only step in the obstruction proof that uses the finiteness of L. Multilinearity then gives the cluster expansion and cancellation. The support bound is used in Lemma B.12 only at the finitely many test inputs obtained by taking $m = d + 2$; all have entries in $\{0, \pm 1\}$. Antisymmetry and (Pol1) enter in Lemma B.13, which does not use that bound. Finally, $k! = 0$ is used only to turn the diagonal identity into a contradiction.

The inclusion–exclusion argument in Proposition B.8(1) uses only multilinearity in the operator slots. The theorem therefore remains true with the pointwise identity (Pw) at L-points in place of (Pol1), for this family of multipliers. In characteristic 2 antisymmetry is the same as symmetry, and the two signs in Lemma B.13 are trivial.

Remark B.15 (when $k!$ is invertible). The vanishing of $k!$ is essential. If instead $k! \neq 0$ in L, the *normalized alternatization*

$$\Psi_{\mathrm{alt}}(u; x) := \frac{1}{k!} \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k)$$

is alternating in x and satisfies (Pol1). Every summand is supported in the span of the k^2 vectors $u_i x_j$, so $\operatorname{sdim}(\Psi_{\mathrm{alt}}(u; x)) \le k^2$. The preceding homogeneity, cancellation, staircase and diagonal arguments apply, but now yield $\chi \equiv 1/k!$: for pairwise disjoint clusters only $\sigma = \mathrm{id}$ survives in $\Psi_{\mathrm{alt}}(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})$, and the determinant at $(C_1(1), \dots, C_k(1))$ is that of the identity. The final equation $k! \cdot (1/k!) = 1$ is then consistent. So the obstruction is exactly the vanishing of $k!$, not positive characteristic as such. For example, at $(p, k) = (5, 3)$ every step is consistent, since $3! = 6 = 1$ in $\mathbb{F}_5$ and $\chi_0 = 1$.

Appendix C. The Banach exterior target over a Laurent series field

We now realize the finite-field obstruction of Theorem B.9 in a Banach-space setting. The space F_1 of Theorem C.1 is $\widehat{\Lambda}^k_\pi E_1$, the test target of Lemma 3.4 for the pair (E_1, E_1), so Theorem C.1 says that (E_1, E_1) is not good for Banach values in degree k (Corollary 7.6).

Theorem C.1. *Let κ be a finite field of characteristic p, let $k \ge p$ and $\varrho \in (0, 1)$, and let $K_1 = \kappa((X))$ with $|X| = \varrho$. Let $E_1 = \ell^\infty(\mathbb{N}, K_1)$, and let F_1 be the completion of $\Lambda^k_{K_1} E_1$ under the projective exterior norm. Then $A^{k, K_1}_{E_1, E_1; F_1}$ admits no bounded k-linear lift over K_1. Consequently it is analytic at no point of $\mathcal{L}(E_1, E_1)$.*

C.1. **Setting.** Throughout this section $k \geq 1$, κ is a field, $\varrho \in (0,1)$, and $K_1 := \kappa((X))$ is the Laurent series field with absolute value $|a| := \varrho^{\operatorname{ord} a}$ ($|0| = 0$). This field is complete, nontrivially normed and nonarchimedean, and $|c| = 1$ for every $c \in \kappa^\times$. Write $\operatorname{coeff}_n(a) \in \kappa$ for the X^n-coefficient of $a \in K_1$. For a set S and $y \in \ell^\infty(S, K_1)$ we take the constant coefficient coordinatewise:

$$\operatorname{coeff}_0(y) := \big(\operatorname{coeff}_0(y(s))\big)_{s\in S} \in \kappa^S.$$

The resulting *constant-coefficient map* coeff_0 is κ-linear but not K_1-linear. A nonzero Laurent series of absolute value less than one has order at least one. Consequently,

$$\text{(C.1)}\quad \|y\|_\infty < 1 \implies \operatorname{coeff}_0(y) = 0, \qquad \text{hence} \qquad \|y - y'\|_\infty < 1 \implies \operatorname{coeff}_0(y) = \operatorname{coeff}_0(y').$$

Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$ with the sup norm, a nonarchimedean K_1-Banach space, and let $E_0 := \kappa^{\mathbb{N}} \subseteq E_1$ be the κ-subspace of sequences with entries in κ. Every nonzero $z \in E_0$ has $\|z\|_\infty = 1$. For $x \in E_1$ and $n \in \mathbb{Z}$ let $x_{[n]} := (\operatorname{coeff}_n(x(i)))_{i\in\mathbb{N}} \in E_0$ be the n-th Laurent component. For $x \neq 0$ the set $\{\operatorname{ord} x(i) : x(i) \neq 0\}$ is bounded below, because $\varrho^{\operatorname{ord} x(i)} \leq \|x\|_\infty$, so it has a minimum $\nu(x)$; then $\|x\|_\infty = \varrho^{\nu(x)}$ and $x_{[n]} = 0$ for $n < \nu(x)$. Finally put

$$M_\varrho := \max_{l\in\mathbb{N}} (l+1)\varrho^l.$$

The maximum exists because $(l+1)\varrho^l \to 0$. This constant will control the number of Laurent components that can contribute to a constant coefficient. For $\varrho \leq \frac{1}{2}$ one has $(l+1)\varrho^l \leq (l+1)2^{-l} \leq 1$, so $M_\varrho = 1$.

C.2. **The projective exterior norm.** On $\Lambda := \Lambda^k_{K_1} E_1$ put

$$\|\omega\|_\pi := \inf\Big\{\sum_j \prod_{i=1}^k \|x_{j,i}\|_\infty \;:\; \omega = \sum_j x_{j,1} \wedge \cdots \wedge x_{j,k}\Big\},$$

where the infimum is over finite decompositions and the cost is an ordinary sum. Let $\Omega^{K_1}\colon \Lambda \to \ell^\infty(\mathbb{N}^k, K_1)$ be the determinant array of §B.1 with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$, so that $\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)(i_1, \ldots, i_k) = \det(x_b(i_a))_{a,b}$; its values are bounded by the next lemma.

Lemma C.2. *$\|\cdot\|_\pi$ is a K_1-norm on Λ with $\|x_1 \wedge \cdots \wedge x_k\|_\pi \leq \prod_i \|x_i\|_\infty$, and Ω^{K_1} is K_1-linear and injective with $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. Let F_1 be the completion of $(\Lambda, \|\cdot\|_\pi)$, a K_1-Banach space, $J\colon F_1 \to \ell^\infty(\mathbb{N}^k, K_1)$ the continuous extension of Ω^{K_1} (so $\|J\| \leq 1$), and $\omega_{E_1}\colon E_1^k \to F_1$, $\omega_{E_1}(x) := x_1 \wedge \cdots \wedge x_k$. Then $\omega_{E_1} \in \operatorname{Alt}^k_{K_1}(E_1; F_1)$, $\|\omega_{E_1}\| \leq 1$, and $J(\omega_{E_1}(x)) = \Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)$.*

Proof. For a pure wedge, each entry of the determinant array is a signed sum of products $x_1(i_{\sigma(1)}) \cdots x_k(i_{\sigma(k)})$. Each product has absolute value at most $\prod_b \|x_b\|_\infty$, so the nonarchimedean inequality in K_1 gives the same bound for the determinant. For any finite decomposition of ω, it follows that

$$\|\Omega^{K_1}(\omega)\|_\infty \leq \max_j \prod_i \|x_{j,i}\|_\infty \leq \sum_j \prod_i \|x_{j,i}\|_\infty.$$

Thus the array is bounded, and taking the infimum gives $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. The map Ω^{K_1} is linear and injective by Lemma B.2, with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$. In particular, $\|\omega\|_\pi > 0$ whenever $\omega \neq 0$.

The remaining norm laws follow directly from decompositions. Concatenation gives the triangle inequality. Since $k \geq 1$, a scalar can be absorbed into the first factor of each wedge; for $\lambda \neq 0$ this gives $\|\lambda\omega\|_\pi \leq |\lambda|\|\omega\|_\pi$, and applying the same inequality to λ^{-1} gives equality. The pure-wedge bound follows by using a single summand.

The array space $\ell^\infty(\mathbb{N}^k, K_1)$ is complete, so Ω^{K_1} extends to the contraction J on F_1. The wedge map into F_1 is K_1-multilinear and vanishes whenever two arguments coincide. The pure-wedge bound makes it continuous with norm at most one, and the asserted identity with J holds on its values by construction. □

For $k \geq 2$, F_1 is not nonarchimedean (Proposition E.1).

C.3. **The support estimate.** We next show that the constant coefficient of every completed determinant array is represented by a finite algebraic exterior vector. The quantitative bound comes from counting the Laurent components needed as wedge *factors*; the number of wedge summands may be much larger.

Let $\Omega^\kappa\colon \Lambda^k_\kappa E_0 \to \kappa^{\mathbb{N}^k}$ be the determinant array over κ; it is injective by Lemma B.2 with $L = \kappa$ and $V = E_0$. For $y_1, \dots, y_k \in E_0$ we have $\Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^\kappa(y_1 \wedge \cdots \wedge y_k)$: the determinants are the same, with entries in $\kappa \subseteq K_1$. Support dimensions of elements of $\Lambda^k_\kappa E_0$ are computed in E_0 over κ.

Proposition C.3 (support estimate)**.** *For every $b \in F_1$ there is a unique $\eta(b) \in \Lambda^k_\kappa E_0$ with $\Omega^\kappa(\eta(b)) = \operatorname{coeff}_0(Jb)$. The map η is κ-linear, and*

$$\operatorname{sdim}\big(\eta(b)\big) \leq k M_\varrho \, \|b\|_{F_1} \qquad \text{for all } b \in F_1. \tag{C.2}$$

Proof. Step 1: one decomposable. Let $x_1, \dots, x_k \in E_1$. If some $x_b = 0$, the wedge is zero and we take $\eta = 0$. Otherwise put $\nu_b := \nu(x_b)$ and $l := -\sum_b \nu_b$, so that $\prod_b \|x_b\|_\infty = \varrho^{-l}$. The possible exponent tuples contributing to a constant coefficient form the set

$$\mathcal{N} := \Big\{ n \in \mathbb{Z}^k : \ n_b \geq \nu_b \text{ for all } b, \ \sum_b n_b = 0 \Big\}.$$

For such a tuple, the nonnegative integers $n_b - \nu_b$ sum to l. Hence $\mathcal{N}$ is empty if $l < 0$; if $l \geq 0$, every coordinate satisfies $\nu_b \leq n_b \leq \nu_b + l$, so $\mathcal{N}$ is finite.

Fix $(i_1, \dots, i_k) \in \mathbb{N}^k$ and expand each entry as $x_b(i_a) = \sum_{n \geq \nu_b} \operatorname{coeff}_n(x_b(i_a)) X^n$. In the determinant expansion

$$\det(x_b(i_a))_{a,b} = \sum_\sigma \operatorname{sgn}(\sigma) \prod_b x_b(i_{\sigma(b)}),$$

the constant coefficient of each product is its finite coefficient convolution over $\mathcal{N}$. Exchanging this finite sum with the permutation sum gives

$$\operatorname{coeff}_0 \det\big(x_b(i_a)\big)_{a,b} = \sum_{n \in \mathcal{N}} \det\big(x_{b,[n_b]}(i_a)\big)_{a,b}.$$

The same finite set $\mathcal{N}$ works for every array coordinate. Thus $\operatorname{coeff}_0\big(\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)\big) = \Omega^\kappa(\eta)$, where

$$\eta := \sum_{n \in \mathcal{N}} x_{1,[n_1]} \wedge \cdots \wedge x_{k,[n_k]} \in \Lambda^k_\kappa E_0.$$

This is a finite sum in the algebraic exterior power. When $l < 0$ it is zero. When $l \geq 0$, all its wedge factors belong to the set $\{x_{b,[n]} : 1 \leq b \leq k, \ \nu_b \leq n \leq \nu_b + l\}$, which has at most $k(l+1)$ elements. Taking their span gives

$$\operatorname{sdim}(\eta) \leq k(l+1) = k\,(l+1)\varrho^l \cdot \varrho^{-l} \leq k M_\varrho \prod_b \|x_b\|_\infty.$$

The final bound therefore holds in both cases.

Step 2: an element of Λ. Write $\omega = \sum_j w_j$, where $w_j = x_{j,1} \wedge \cdots \wedge x_{j,k}$, and take η_j from Step 1. Additivity gives

$$\operatorname{coeff}_0(\Omega^{K_1}(\omega)) = \sum_j \operatorname{coeff}_0(\Omega^{K_1}(w_j)) = \Omega^\kappa\Big(\sum_j \eta_j\Big).$$

Injectivity of Ω^κ shows that $\eta(\omega) := \sum_j \eta_j$ is independent of the chosen decomposition. Subadditivity of support dimension now yields

$$\operatorname{sdim}(\eta(\omega)) \le \sum_j \operatorname{sdim}(\eta_j) \le kM_\varrho \sum_j \prod_i \|x_{j,i}\|_\infty.$$

The left side is fixed while the decomposition varies. Taking the infimum therefore gives $\operatorname{sdim}(\eta(\omega)) \le kM_\varrho \|\omega\|_\pi$, without requiring a minimizing decomposition.

Step 3: completion. Choose $\omega_n \in \Lambda$ with $\|b - \omega_n\|_{F_1} \to 0$. Since J is a contraction,

$$\|Jb - \Omega^{K_1}(\omega_n)\|_\infty \le \|b - \omega_n\|_{F_1} < 1 \qquad (n \ge n_0).$$

Equation (C.1) therefore gives the *exact* identities

$$\operatorname{coeff}_0(Jb) = \operatorname{coeff}_0(\Omega^{K_1}(\omega_n)) = \Omega^\kappa(\eta(\omega_n)) \qquad (n \ge n_0).$$

By injectivity of Ω^κ, the algebraic vectors $\eta(\omega_n)$ are eventually equal. Define $\eta(b)$ to be this common value. Its array is $\operatorname{coeff}_0(Jb)$, so the same injectivity also proves independence of the approximating sequence and uniqueness. In particular, no limit in $\Lambda^k_\kappa E_0$ is involved.

For every $n \ge n_0$ the fixed vector $\eta(b)$ satisfies $\operatorname{sdim}(\eta(b)) \le kM_\varrho \|\omega_n\|_\pi$. Since $\|\omega_n\|_\pi \to \|b\|_{F_1}$, this gives (C.2). Finally, the array characterization proves κ-linearity: J is K_1-linear, coeff_0 is κ-linear, and Ω^κ is linear and injective. □

C.4. **The coefficient lift.** The support estimate converts norm control into an algebraic obstruction. We apply it to a hypothetical lift, evaluated on coordinatewise multiplication operators and on the universal wedge map ω_{E_1}.

Suppose that P is a bounded k-linear lift of $A^{k,K_1}_{E_1,E_1;F_1}$ over K_1. For $a \in E_1$ let $\mu_a \in \mathcal{L}(E_1, E_1)$ be coordinatewise multiplication, $\mu_a x = ax$. The map $a \mapsto \mu_a$ is K_1-linear and $\|\mu_a\| \le \|a\|_\infty$. Define

$$\Phi(a_1, \dots, a_k; x_1, \dots, x_k) := P(\mu_{a_1}, \dots, \mu_{a_k})(\omega_{E_1})(x_1, \dots, x_k) \in F_1,$$
$$\Psi := \eta \circ \Phi|_{E_0^{2k}} \colon E_0^k \times E_0^k \to \Lambda^k_\kappa E_0.$$

Since $\Omega^\kappa \circ \eta = \operatorname{coeff}_0 \circ J$, the map Ψ is $\operatorname{coeff}_0 \circ J \circ \Phi|_{E_0^{2k}}$, read in $\Lambda^k_\kappa E_0$ through the injective map Ω^κ.

Lemma C.4. *The map* Ψ *has the following properties.*

- (Ψ1) Ψ *is* $2k$*-linear over* κ*.*
- (Ψ2) Ψ *is alternating in the last* k *slots.*
- (Ψ3) $\operatorname{sdim}(\Psi(z)) \le d := \lfloor kM_\varrho \|P\| \rfloor$ *for* all $z \in E_0^{2k}$*.*
- (Ψ4) Ψ *satisfies* (Pol) *for the multiplier family* $\mu^\kappa \colon E_0 \to \operatorname{End}_\kappa(E_0)$, $\mu^\kappa_a x = ax$*: for all* $m \ge 1$, $b_1, \dots, b_m \in E_0$, $\alpha \in \mathbb{N}^m$ *with* $|\alpha| = k$*, and* $x \in E_0^k$*,*

$$\sum_{\text{type } f = \alpha} \Psi(b_f; x) = \sum_{\text{type } f = \alpha} (b_{f(1)} x_1) \wedge \cdots \wedge (b_{f(k)} x_k) \qquad \text{in } \Lambda^k_\kappa E_0.$$

In particular Ψ *satisfies* (Pol1).

Proof. The map Φ is K_1-multilinear, so its restriction to E_0^{2k} is κ-multilinear. Composing with the κ-linear map η proves (Ψ1). Since $P(\mu_{a_1},\dots,\mu_{a_k})(\omega_{E_1})$ belongs to $\mathrm{Alt}^k_{K_1}(E_1;F_1)$, it vanishes when two of the last k arguments coincide; the same holds after applying η, proving (Ψ2).

For (Ψ3), all inputs from E_0 have norm at most one. Thus

$$\|\Phi(a;x)\| \le \|P\| \prod_i \|\mu_{a_i}\| \, \|\omega_{E_1}\| \prod_i \|x_i\|_\infty \le \|P\|.$$

Apply (C.2) and use that support dimension is an integer to obtain the stated bound by d.

To prove (Ψ4), we first compare coefficients over the infinite field K_1, while the values still lie in F_1. Fix $b_1,\dots,b_m \in E_0$ and $x \in E_0^k$. For $\lambda \in K_1^m$ put $a_\lambda := \sum_i \lambda_i b_i \in E_1$. The diagonal identity for the lift gives

$$\Phi(a_\lambda,\dots,a_\lambda;x) = P(\mu_{a_\lambda},\dots,\mu_{a_\lambda})(\omega_{E_1})(x) = A(\mu_{a_\lambda})(\omega_{E_1})(x) = \omega_{E_1}(a_\lambda x_1,\dots,a_\lambda x_k).$$

Expanding both sides by multilinearity gives $\sum_\alpha \lambda^\alpha (L_\alpha - R_\alpha) = 0$ for all $\lambda \in K_1^m$, where

$$L_\alpha := \sum_{\text{type } f=\alpha} \Phi(b_f;x) \in F_1, \qquad R_\alpha := \sum_{\text{type } f=\alpha} \omega_{E_1}(b_{f(1)}x_1,\dots,b_{f(k)}x_k) \in F_1.$$

The field K_1 is infinite even when κ is finite. Lemma B.6, applied in the K_1-vector space F_1, therefore yields $L_\alpha = R_\alpha$ for every α.

We can now apply η. Additivity gives $\eta(L_\alpha) = \sum_{\text{type } f=\alpha} \Psi(b_f;x)$. To identify the other side, take $y_1,\dots,y_k \in E_0$. By Lemma C.2 and the determinant-array agreement preceding Proposition C.3,

$$J(\omega_{E_1}(y)) = \Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^{\kappa}(y_1 \wedge \cdots \wedge y_k).$$

This array has entries in κ, so taking its constant coefficient leaves it unchanged. Uniqueness in Proposition C.3 then gives $\eta(\omega_{E_1}(y)) = y_1 \wedge \cdots \wedge y_k \in \Lambda^k_\kappa E_0$. Using $y_j = b_{f(j)}x_j$ in each summand of R_α proves

$$\eta(R_\alpha) = \sum_{\text{type } f=\alpha} (b_{f(1)}x_1) \wedge \cdots \wedge (b_{f(k)}x_k),$$

which is the required identity. □

The order of these operations is essential for the full identity (Ψ4). Evaluating the diagonal only at κ-points would give (Pw) there, which for $\kappa = \mathbb{F}_q$ and $k > q$ is strictly weaker than (Pol) (Proposition B.8(3)). The proof below uses only the consequence (Pol1).

C.5. **No bounded lift over a Laurent series field.** We now combine the support estimate (Proposition C.3) and the coefficient lift (Lemma C.4) with the finite-field theorem of Appendix B (Theorem B.9) to prove Theorem C.1.

Proof of Theorem C.1. Suppose P exists, and form Ψ as in §C.4. By Lemma C.4, $\Psi\colon E_0^k \times E_0^k \to \Lambda^k_\kappa E_0$ with $E_0 = \kappa^{\mathbb{N}}$ is $2k$-linear over κ, alternating and hence antisymmetric in the last k slots, satisfies (Pol1), and has $\mathrm{sdim}(\Psi(z)) \le d$ for all inputs z. Moreover $k! = 0$ in κ, since $k \ge p = \mathrm{char}\,\kappa$. This contradicts Theorem B.9(2) with $L = \kappa$ and $E_0 = \kappa^{\mathbb{N}}$. The last sentence follows from Proposition 2.5. □

Remark C.5. Lemmas C.2 and C.4 and Proposition C.3 hold for every field κ, and nothing in them uses a normalization of ϱ: the constant M_ϱ makes the argument uniform in $\varrho \in (0,1)$. Only the final appeal to Theorem B.9 needs κ finite. We do not need, and do not discuss, the analogue of Theorem C.1 for infinite κ with this particular pair (E_1,F_1). For an infinite κ of

characteristic p the field $\kappa((X))$ is covered by Theorem 7.2(1), with the pair obtained by ascent from its subfield $\mathbb{F}_p((X))$.

C.6. **Multiplier families and c_0 bases.** The proof of Theorem C.1 uses the hypothetical lift only on coordinatewise multiplication operators and only at the single form ω_{E_1}. Moreover, Theorem B.9(1) needs only finitely supported test vectors with entries in $\{0, \pm 1\}$. Both observations give a much smaller bad parameter space. Throughout this subsection κ, k, ϱ, K_1, E_1, F_1 and ω_{E_1} are as in Theorem C.1, $c_0(\mathbb{N}, K_1) \subseteq E_1$ carries the supremum norm, and $\mu_a \in \mathcal{L}(E_1, E_1)$ denotes multiplication by a.

Proposition C.6 (multipliers on c_0)**.** *The homogeneous map*

$$\sigma\colon c_0(\mathbb{N}, K_1) \to \mathrm{Alt}^k_{K_1}(E_1; F_1), \qquad \sigma(a)(x_1, \dots, x_k) = \omega_{E_1}(ax_1, \dots, ax_k) = (ax_1) \wedge \cdots \wedge (ax_k),$$

has no bounded k-linear lift with values in $\mathrm{Alt}^k_{K_1}(E_1; F_1)$. Consequently, for every $u_0 \in \mathcal{L}(E_1, E_1)$, the map $a \mapsto A^{k,K_1}_{E_1,E_1;F_1}(u_0 + \mu_a)(\omega_{E_1})$ is analytic at no point of $c_0(\mathbb{N}, K_1)$. In particular $a \mapsto A^{k,K_1}_{E_1,E_1;F_1}(u_0 + \mu_a)$ is analytic at no point.

Proof of Proposition C.6. Suppose $p\colon c_0(\mathbb{N}, K_1)^k \to \mathrm{Alt}^k_{K_1}(E_1; F_1)$ is a bounded k-linear lift of σ. Let $V_{\mathrm{fin}} = \kappa^{(\mathbb{N})}$ be the finitely supported sequences with entries in κ; it lies in both $c_0(\mathbb{N}, K_1)$ and E_0. Define $\Psi(a; x) = \eta\big(p(a_1, \dots, a_k)(x_1, \dots, x_k)\big)$ for $a, x \in V^k_{\mathrm{fin}}$. The proof of Lemma C.4 applies verbatim. The map Ψ is $2k$-linear over κ and alternating in the last k slots. Its inputs have supremum norm at most one, so (C.2) gives $\operatorname{sdim} \Psi(a; x) \le \lfloor kM_\varrho \|p\| \rfloor$. The polarization points $a_\lambda = \sum_i \lambda_i b_i$, with $b_i \in V_{\mathrm{fin}}$, are finitely supported and hence lie in $c_0(\mathbb{N}, K_1)$, so the identity (Pol), and in particular (Pol1), follows from the diagonal identity as before. This contradicts Theorem B.9(1), since $k! = 0$ in κ.

For the second assertion, suppose that the map is analytic at a_0, with expansion $\sum_n p_n$ there. For $h \in c_0(\mathbb{N}, K_1)$ and $t \in K_1$, multilinearity gives, in $\mathrm{Mult}^k_{K_1}(E_1; F_1)$,

$$\omega_{E_1} \circ (u_0 + \mu_{a_0} + t\mu_h, \dots, u_0 + \mu_{a_0} + t\mu_h) = \sum_{r=0}^{k} t^r C_r(h), \qquad C_k(h) = \sigma(h).$$

Comparing this with the given expansion on scalar lines, Lemma A.1 shows, exactly as in Proposition 2.5, that p_k is a bounded lift of σ. This is impossible. The operator-valued statement follows by evaluation at ω_{E_1}. □

Corollary C.7 (diagonal transitions over a c_0 base)**.** *Let U be the open unit ball of $c_0(\mathbb{N}, K_1)$, an analytic manifold with one chart. Let $\mathcal{V} \to U$ be the trivial bundle with fiber E_1 and the two global trivializations $\tau_0(a, z) = (a, z)$ and $\tau_1(a, z) = (a, \mu_{1+a} z)$, and let $\mathcal{B} = U \times F_1$. Then $\mathcal{V}$ is an analytic normed vector bundle whose transitions are isometric diagonal automorphisms of E_1. The naturally induced atlas on $\mathrm{Alt}^k(\mathcal{V}; \mathcal{B})$ is not analytic: the section that is constantly ω_{E_1} in the chart induced by τ_1 is, in the chart induced by τ_0, the map $a \mapsto \omega_{E_1} \circ (\mu_{1+a}, \dots, \mu_{1+a})$, which is analytic at no point of U.*

Proof. For $a \in U$ every coordinate satisfies $|1 + a_i| = 1$, so μ_{1+a} is an isometry of E_1. Its inverse $\mu_{(1+a)^{-1}} = \sum_{n \ge 0} (-\mu_a)^n$ is a power series of radius one in a, since $\mu_a^n = \mu_{a^n}$ has norm at most $\|a\|^n$. Hence both transitions are analytic. If $\theta_a \in \mathrm{Alt}^k(\mathcal{V}_a; F_1)$ has coordinates $m_0(a)$ and $m_1(a)$ in the two induced charts, then $m_0(a) = m_1(a) \circ (\mu_{1+a}, \dots, \mu_{1+a})$. With $m_1 \equiv \omega_{E_1}$, this is the map of Proposition C.6 with $u_0 = \mathrm{id}$, which is analytic at no point. □

The obstruction concerns the atlas obtained from the two analytic frames. The underlying pointwise bundle is trivial and has an analytic product structure; the failure is compatibility

of that structure with the natural construction from all analytic frames. The complete target F_1 admits no equivalent nonarchimedean norm by Proposition E.1, so this example does not contradict Corollary 5.3.

Corollary C.8 (finite dimension is sharp for nonarchimedean bases)**.** *Let $K = \mathbb{F}_q((X))$, where q is a power of p, with an X-adic absolute value, and let $k \geq p$. Let P be a K-Banach space admitting an equivalent nonarchimedean norm. The alternating construction preserves analytic bundles and their operator-valued analytic morphisms over every analytic manifold modeled on P, with arbitrary normed fibers, if and only if P is finite-dimensional.*

Proof. If P is finite-dimensional, Lemma A.3 supplies continuous coordinates and Corollary 5.15 applies. Conversely, suppose that P is infinite-dimensional. Round an equivalent nonarchimedean norm upwards to the discrete value group $|K^\times|$; the result is still an equivalent nonarchimedean K-norm, and its nonzero values lie in $|K^\times|$. Serre's orthonormal-basis theorem [26, §1, Proposition 1] then identifies P topologically with $c_0(I, K)$ for an infinite set I. Choosing a countable subset of I gives bounded linear maps

$$\iota\colon c_0(\mathbb{N}, K) \longrightarrow P, \qquad \pi\colon P \longrightarrow c_0(\mathbb{N}, K), \qquad \pi\iota = \mathrm{id}.$$

Apply the Laurent construction with $\kappa = \mathbb{F}_q$ and $K_1 = K$. On the open set $V = \{x \in P : \|\pi x\| < 1\}$, take the trivial bundle with fiber E_1 and two analytic frames whose transition is $\mu_{1+\pi(x)}$, as in Corollary C.7, and the trivial target bundle with fiber F_1. If the induced alternating transition were analytic at $x_0 \in V$, restriction along $b \mapsto x_0 + \iota b$ would make

$$b \longmapsto A^k_{E_1,E_1;F_1}\big(\mu_{1+\pi(x_0)} + \mu_b\big)$$

analytic at zero. This contradicts Proposition C.6 with $u_0 = \mu_{1+\pi(x_0)}$. □

The dichotomy permits arbitrary normed target fibers. In the nonarchimedean Banach target category, Corollary 5.3 gives analyticity over every base for this discretely valued field.

Appendix D. Ascent to an arbitrary base field

Proposition D.9, applied in Appendix E with E_1 and F_1 as in Theorem C.1, carries analyticity at no point from a spherically complete Laurent subfield K_1 to the completion K' of the prescribed field, since a lift after scalar extension descends to K_1. Lemma D.11 then passes from K' to the field itself.

D.1. **The Laurent subfield.** We use spherical completeness in the sense used in Theorem 5.2: every nonempty family of pairwise-meeting closed balls has a common point. For the field case of the following lemma, see Schneider [25, Lemmas 1.3 and 1.6].

Lemma D.1 (discrete distances)**.** *Let Y be a complete nonarchimedean space whose distances lie in $r^{\mathbb{Z}} \cup \{0\}$ for some $r \in (0, 1)$. Then Y is spherically complete.*

Proof. Write $\mathcal{D}(a, \rho) = \{y \in Y : \mathrm{d}(y, a) \leq \rho\}$. In a nonarchimedean space, every point of a ball is a center. Consequently, whenever two closed balls meet, the one with the smaller radius lies in the other. The discrete distance assumption also gives

$$\mathcal{D}(a, \rho) = \mathcal{D}(a, \rho^\flat), \qquad \rho^\flat := \max\{s \in r^{\mathbb{Z}} \cup \{0\} : s \leq \rho\}.$$

We may therefore replace every radius by a value in $r^{\mathbb{Z}} \cup \{0\}$.

Let $(\mathcal{D}(a_i, \rho_i))_{i \in I}$ be a nonempty family of pairwise-meeting closed balls. If its radii have a least element ρ_{i_0}, then $\mathcal{D}(a_{i_0}, \rho_{i_0})$ lies in every ball of the family, so a_{i_0} is a common point.

Otherwise, discreteness implies that all the radii are positive and that their infimum is zero. Choose $i_1, i_2, \dots$ so that ρ_{i_n} decreases strictly to zero. The corresponding balls are nested, and

$$\mathrm{d}(a_{i_{n'}}, a_{i_n}) \le \rho_{i_n} \qquad (n' \ge n).$$

Their centers form a Cauchy sequence. By completeness it has a limit a, and closedness puts a in every ball of the sequence. For any $i \in I$, choose n with $\rho_{i_n} \le \rho_i$. The n-th ball then lies in $\mathcal{D}(a_i, \rho_i)$, so a belongs to every ball of the original family. □

Lemma D.2 (Laurent subfield)**.** *Let K' be a complete nontrivially normed field of characteristic p, and let $t \in K'$ with $0 < |t| = r < 1$.*

1. *K' is nonarchimedean, and $|c| = 1$ for every $c \in \mathbb{F}_p^\times$.*
2. *For $a_m, \dots, a_M \in \mathbb{F}_p$, not all zero, $\left|\sum_{i=m}^{M} a_i t^i\right| = r^{\min\{i : a_i \neq 0\}}$.*
3. *Give $\mathbb{F}_p((X))$ the absolute value with $|X| = r$. There is a continuous isometric ring homomorphism $\mathrm{ev}_t \colon \mathbb{F}_p((X)) \to K'$ such that $\mathrm{ev}_t\bigl(\sum_{i \ge m} a_i X^i\bigr) = \sum_{i \ge m} a_i t^i$. Its image K_1 is a closed subfield of K'; it is the closure of $\mathbb{F}_p(t)$, and it is isometrically isomorphic to $\mathbb{F}_p((X))$.*
4. *K_1 is spherically complete.*

Proof. (1) Lemma A.2 makes K' nonarchimedean. For $c \in \mathbb{F}_p^\times$, the identity $c^{p-1} = 1$ gives $|c| = 1$.

(2) The nonzero terms $a_i t^i$ have pairwise distinct absolute values r^i. In a nonarchimedean field, $|x + y| = |x|$ whenever $|y| < |x|$. Repeatedly applying this identity shows that the sum has the absolute value of its largest term, namely $r^{\min\{i : a_i \neq 0\}}$.

(3) Evaluation on Laurent polynomials defines a ring homomorphism $\mathrm{ev}_t \colon \mathbb{F}_p[X, X^{-1}] \to K'$, and (2) shows that $|\mathrm{ev}_t(g)| = r^{\operatorname{ord} g} = |g|$. Laurent polynomials are dense in $\mathbb{F}_p((X))$, since truncations converge. Completeness of K' therefore gives a unique continuous extension to $\mathbb{F}_p((X))$. Passing to limits preserves the isometry and the ring operations. It also gives the stated evaluation formula, because the partial sums of $\sum_{i \ge m} a_i X^i$ map to the partial sums of $\sum_{i \ge m} a_i t^i$.

The image is complete as the isometric image of a complete space, hence closed in K'. It is a subfield: for $g \neq 0$, the identity $\mathrm{ev}_t(g)\mathrm{ev}_t(g^{-1}) = 1$ provides the inverse. This subfield contains $\mathbb{F}_p(t)$, while the Laurent polynomials $\mathbb{F}_p[t, t^{-1}] \subseteq \mathbb{F}_p(t)$ are dense in it. The image is consequently exactly the closure of $\mathbb{F}_p(t)$.

(4) By (3), K_1 is complete and has value group $|K_1^\times| = r^{\mathbb{Z}}$. It is nonarchimedean by (1), so its distances lie in $r^{\mathbb{Z}} \cup \{0\}$. Lemma D.1 gives spherical completeness. □

D.2. **Ingleton's theorem and its consequences.** Throughout the rest of this section we fix fields $K_1 \subseteq K'$ with the following properties.

- (H1) K_1, with the restricted absolute value, is nontrivially normed, complete and spherically complete.
- (H2) K' is nonarchimedean.

By multiplication, K' is a nonarchimedean normed K_1-space. In our application $K' = \widehat{K}$ and K_1 is the field of Lemma D.2.

Theorem D.3 (Ingleton [12]; see also [23])**.** *Let K_1 be a spherically complete nonarchimedean normed field, V a K_1-vector space with a nonarchimedean norm, $V_0 \subseteq V$ a subspace, $C \ge 0$, and $T_0 \colon V_0 \to K_1$ linear with $|T_0 v| \le C\|v\|$ for $v \in V_0$. Then T_0 extends to a linear map $T \colon V \to K_1$ with $|Tx| \le C\|x\|$ for all $x \in V$.*

Schneider [25, Prop. 9.2] gives the more general formulation with a dominating nonarchimedean seminorm.

We will use Theorem D.3 on the nonarchimedean K_1-space K' in two ways: to obtain the scalar retraction below, and to extend finite-dimensional coordinate functionals in the proof of positivity of the projective tensor norm. Both applications have codomain K_1. The spaces E_1 and F_1 introduced later need not be nonarchimedean.

Corollary D.4 (Ingleton projection)**.** *There is a K_1-linear map $\varpi\colon K' \to K_1$ with $\varpi|_{K_1} = \mathrm{id}$ and $|\varpi(\lambda)| \le |\lambda|$ for all $\lambda \in K'$.*

Proof. Theorem D.3 with $V = K'$, $V_0 = K_1$, $T_0 = \mathrm{id}$ and $C = 1$; its hypotheses hold by (H1) and (H2). □

Lemma D.5 (extension)**.** *Let $\mathbb{K} \in \{K_1, K'\}$ and let $C \ge 0$.*

(1) *Let V_0 be a dense $\mathbb{K}$-subspace of a normed $\mathbb{K}$-space V, let Y be a $\mathbb{K}$-Banach space and $T_0\colon V_0 \to Y$ a $\mathbb{K}$-linear map with $\|T_0 x\| \le C\|x\|$. Then T_0 extends uniquely to a continuous map $T\colon V \to Y$, which is $\mathbb{K}$-linear with $\|Tx\| \le C\|x\|$.*

(2) *Let $V_1, \dots, V_k$ be normed K'-spaces with dense subspaces V_i^0, Y a K'-Banach space, and $M\colon \prod_i V_i^0 \to Y$ a K'-multilinear map with $\|M(d)\| \le C \prod_i \|d_i\|$. Then M extends uniquely to a continuous K'-multilinear map $\overline{M}\colon \prod_i V_i \to Y$ with $\|\overline{M}\| \le C$.*

Proof. (1) The bound makes T_0 Lipschitz, hence uniformly continuous. Density of V_0 and completeness of Y give a unique continuous extension. Linearity and the norm bound pass to limits by continuity of addition, scalar multiplication and the norm.

(2) The case $k = 0$ is immediate. For $k \ge 1$, multilinearity gives the telescoping estimate

$$\|M(d) - M(d')\| \le C\rho^{k-1} \sum_i \|d_i - d_i'\|$$

whenever $\|d_i\|, \|d_i'\| \le \rho$ for every i, with $\rho > 0$. For any $v \in \prod_i V_i$, choose tuples $d^{(n)} \in \prod_i V_i^0$ converging to v. Their coordinates are uniformly bounded, so the estimate makes $M(d^{(n)})$ Cauchy in Y. Define $\overline{M}(v)$ to be its limit. The same estimate, applied to two approximating sequences, proves that this definition is independent of the choices and is continuous on bounded sets. Multilinearity and the product norm bound pass to limits. Finally, density of $\prod_i V_i^0$ proves uniqueness. □

D.3. **Projective base change.** The next construction extends scalars while retaining the original norm and a contractive retraction whenever the original space is complete. The scalar retraction from Corollary D.4 is the essential ingredient; the construction applies to arbitrary normed spaces.

For a normed K_1-space V let $V_{K'} := V \otimes_{K_1} K'$, a K'-vector space through the second factor, and put

$$\|u\|_\pi := \inf\Big\{ \sum_j \|x_j\|\,|\lambda_j| \;:\; u = \sum_j x_j \otimes \lambda_j \Big\},$$

the infimum over finite decompositions, with an ordinary sum. For a bounded K_1-linear $\psi\colon K' \to K_1$ let $T_\psi\colon V_{K'} \to V$ be the K_1-linear map with $T_\psi(x \otimes \lambda) = \psi(\lambda)x$; it exists because $(x, \lambda) \mapsto \psi(\lambda)x$ is K_1-bilinear. Let $\iota_V\colon V \to V_{K'}$, $x \mapsto x \otimes 1$; it is K_1-linear, since $cx \otimes 1 = x \otimes c = c \cdot (x \otimes 1)$ for $c \in K_1$.

Lemma D.6 (projective base change)**.** *Let V be a normed K_1-space; no nonarchimedean hypothesis is made on V.*

(1) *$\|T_\psi u\| \le \|\psi\|\,\|u\|_\pi$ for every bounded K_1-linear $\psi\colon K' \to K_1$.*

(2) $\|\cdot\|_\pi$ *is a* K'*-norm on* $V_{K'}$.
(3) ι_V *is isometric.*
(4) *Let* $V\widehat{\otimes}_\pi K'$ *be the completion of* $(V_{K'}, \|\cdot\|_\pi)$. *If* V *is complete, then* T_ϖ, *with* ϖ *as in Corollary D.4, extends to a* K_1*-linear map* $\Pi_V \colon V\widehat{\otimes}_\pi K' \to V$ *with* $\|\Pi_V\| \le 1$ *and* $\Pi_V \circ \iota_V = \mathrm{id}_V$.

Proof. (1) For every decomposition $u = \sum_j x_j \otimes \lambda_j$,

$$\|T_\psi u\| = \Big\|\sum_j \psi(\lambda_j)x_j\Big\| \le \sum_j \|\psi\||\lambda_j|\|x_j\|.$$

Taking the infimum over decompositions proves the contraction estimate.

(2) Concatenating decompositions gives the triangle inequality. Multiplying their second factors by $\mu \in K'$ gives $\|\mu u\|_\pi \le |\mu|\|u\|_\pi$; for $\mu \ne 0$, applying the same inequality with μ^{-1} gives equality. The zero-scalar case is immediate.

It remains to prove positivity. Let $u \ne 0$. Choose a finite tensor representation and rewrite its second factors in a basis of their K_1-span. This gives

$$u = \sum_{j=1}^{n} x_j \otimes \lambda_j$$

with $\lambda_1, \dots, \lambda_n$ linearly independent over K_1. Some x_{j_0} is nonzero, since otherwise $u = 0$. On the finite-dimensional subspace $U := \mathrm{span}_{K_1}\{\lambda_1, \dots, \lambda_n\} \subseteq K'$, take the coordinate functional $\psi_0(\lambda_j) = \delta_{jj_0}$. It is bounded by some $C \ge 0$ by Lemma A.3, using completeness of K_1 in (H1). Apply Theorem D.3 to the nonarchimedean K_1-space K' to extend it to $\psi \colon K' \to K_1$ with $\|\psi\| \le C$. Then $T_\psi u = x_{j_0}$, so (1) gives

$$0 < \|x_{j_0}\| \le C\|u\|_\pi.$$

Thus $\|u\|_\pi > 0$. This argument detects a nonzero vector coefficient by a functional on the *scalar* factor K'; it requires no separating continuous dual on V.

(3) The elementary tensor representation gives $\|x \otimes 1\|_\pi \le \|x\|$. For the reverse inequality, use $\varpi(1) = 1$ and $\|\varpi\| \le 1$ in (1):

$$\|x\| = \|T_\varpi(x \otimes 1)\| \le \|x \otimes 1\|_\pi.$$

(4) When V is complete, the contraction T_ϖ on the dense tensor subspace extends by Lemma D.5(1), with $\mathbb{K} = K_1$, to a K_1-linear contraction Π_V with values in V. On the embedded copy of V it satisfies $\Pi_V \iota_V x = T_\varpi(x \otimes 1) = \varpi(1)x = x$. □

We now fix the spaces to which the descent argument will be applied. Let E_1 be a normed K_1-space and F_1 a K_1-Banach space, and put

$$E := E_1\widehat{\otimes}_\pi K', \qquad F := F_1\widehat{\otimes}_\pi K',$$

the completions of $((E_1)_{K'}, \|\cdot\|_\pi)$ and $((F_1)_{K'}, \|\cdot\|_\pi)$. They are K'-Banach spaces; $\iota_E \colon E_1 \to E$ and $\iota_F \colon F_1 \to F$ are K_1-linear isometries, and $\Pi_F \colon F \to F_1$ is K_1-linear with $\|\Pi_F\| \le 1$ and $\Pi_F \circ \iota_F = \mathrm{id}_{F_1}$ (Lemma D.6).

Lemma D.7 (operators)**.** *For* $f \in \mathcal{L}_{K_1}(E_1, E_1)$ *the map* $f \otimes \mathrm{id}$ *on* $(E_1)_{K'}$ *is* K'*-linear with* $\|(f \otimes \mathrm{id})u\|_\pi \le \|f\|\|u\|_\pi$. *Its extension* $f_{K'} \in \mathcal{L}_{K'}(E, E)$ *satisfies* $\|f_{K'}\| \le \|f\|$ *and* $f_{K'} \circ \iota_E = \iota_E \circ f$, *and* $f \mapsto f_{K'}$ *is* K_1*-linear.*

Proof. For any decomposition $u = \sum_j x_j \otimes \lambda_j$, the image under $f \otimes \mathrm{id}$ has the decomposition $\sum_j f(x_j) \otimes \lambda_j$, whose cost is at most $\|f\| \sum_j \|x_j\||\lambda_j|$. Taking the infimum proves the norm bound on the algebraic tensor product. Since E is complete, Lemma D.5(1), with $\mathbb{K} = K'$, gives

the continuous extension $f_{K'}$ with the same bound. Its action on $x \otimes 1$ proves the embedding identity. Additivity and K_1-homogeneity in f hold on elementary tensors, hence on their span, and then on E by continuity. □

Lemma D.8 (forms)**.** *For $m \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$ there is $m_{K'} \in \mathrm{Alt}^k_{K'}(E; F)$ with $\|m_{K'}\| \le \|m\|$ and*

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = \iota_F\big(m(x_1, \dots, x_k)\big) \qquad (x_i \in E_1),$$

and $m \mapsto m_{K'}$ is K_1-linear.

Proof. We first define the form on algebraic tensors, then establish a bound independent of their decompositions, and finally pass to the completions. Fix a K_1-basis $(\lambda_\beta)_{\beta \in \mathfrak{B}}$ of K'. Every $u \in (E_1)_{K'}$ has a unique finite expansion $u = \sum_\beta u^\beta \otimes \lambda_\beta$, with $u^\beta \in E_1$, and each coordinate map $u \mapsto u^\beta$ is K_1-linear. Define

$$m^{\mathrm{alg}}(u_1, \dots, u_k) := \sum_{\beta_1, \dots, \beta_k} m\big(u_1^{\beta_1}, \dots, u_k^{\beta_k}\big) \otimes \lambda_{\beta_1} \cdots \lambda_{\beta_k} \in (F_1)_{K'}.$$

The sum is finite, since each input has finitely many nonzero coordinates.

(i) Linearity of the coordinate maps and multilinearity of m make m^{alg} K_1-multilinear.

(ii) On elementary tensors, the definition gives

$$m^{\mathrm{alg}}(x_1 \otimes \mu_1, \dots, x_k \otimes \mu_k) = m(x_1, \dots, x_k) \otimes \mu_1 \cdots \mu_k \qquad (x_i \in E_1,\ \mu_i \in K').$$

Indeed, writing $\mu_i = \sum_\beta c_{i\beta} \lambda_\beta$ with $c_{i\beta} \in K_1$ gives $x_i \otimes \mu_i = \sum_\beta (c_{i\beta} x_i) \otimes \lambda_\beta$. Substitution in the defining sum yields

$$\sum_{\beta_1, \dots, \beta_k} \Big(\prod_i c_{i\beta_i}\Big) m(x) \otimes \prod_i \lambda_{\beta_i} = m(x) \otimes \prod_i \Big(\sum_\beta c_{i\beta} \lambda_\beta\Big),$$

which is the asserted formula.

(iii) The map m^{alg} is K'-multilinear. By additivity in each slot, it suffices to check K'-homogeneity when every argument is an elementary tensor; there it follows from (ii). The same formula determines the map on all tuples of algebraic tensors, so the initial basis choice does not affect it.

(iv) For arbitrary decompositions $u_i = \sum_j x_{ij} \otimes \mu_{ij}$, multilinearity and (ii) give

$$m^{\mathrm{alg}}(u) = \sum_{j_1, \dots, j_k} m(x_{1 j_1}, \dots, x_{k j_k}) \otimes \prod_i \mu_{i j_i}.$$

The cost of this tensor decomposition is at most $\|m\| \prod_i \big(\sum_j \|x_{ij}\| |\mu_{ij}|\big)$. Taking the infimum separately over the decompositions in each slot proves

$$\|m^{\mathrm{alg}}(u_1, \dots, u_k)\|_\pi \le \|m\| \prod_i \|u_i\|_\pi.$$

(v) To prove strong alternation, suppose two distinct slots a, b both contain $u = \sum_j x_j \otimes \mu_j$. Use this *same* decomposition in both slots and expand all arguments into elementary tensors. Terms using the same index j in slots a and b vanish because m has a repeated argument. For $j \ne j'$, the term using indices (j, j') in these slots pairs with the term using (j', j). Their scalar factors agree, while the values of m are negatives of one another, so the pair sums to zero. Thus m^{alg} vanishes whenever any two arguments coincide. Pair cancellation remains valid in characteristic 2; merely checking $m^{\mathrm{alg}}(u, \dots, u) = 0$ would not establish strong alternation for $k \ge 3$.

(vi) Regard m^{alg} as taking values in the completion F. By (iii), (iv) and Lemma D.5(2), it extends to a continuous K'-multilinear map $m_{K'} \colon E^k \to F$ with $\|m_{K'}\| \le \|m\|$. Strong

alternation also passes to the limit: if $v_a = v_b$, approximate this common vector by one sequence in $(E_1)_{K'}$, and approximate the other entries separately. Every approximating tuple has equal entries in slots a, b, so (v) and continuity give $m_{K'}(v) = 0$.

(vii) Taking every scalar factor equal to one in (ii) gives

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = m^{\mathrm{alg}}(x_1 \otimes 1, \dots, x_k \otimes 1) = m(x) \otimes 1 = \iota_F(m(x)).$$

The defining formula is K_1-linear in m. These linear identities pass to the continuous extensions by uniqueness, proving K_1-linearity of $m \mapsto m_{K'}$. □

Proposition D.9 (descent of a lift)**.** *Assume* (H1) *and* (H2)*. Let E_1 be a normed K_1-space, F_1 a K_1-Banach space, and E, F as above. If $A^{k,K'}_{E,E;F}$ admits a bounded k-linear lift P over K', then $A^{k,K_1}_{E_1,E_1;F_1}$ admits a bounded k-linear lift P_1 over K_1 with $\|P_1\| \le \|P\|$. Equivalently, by Proposition 2.5 over K_1 and over K': if $A^{k,K_1}_{E_1,E_1;F_1}$ is analytic at no point, then $A^{k,K'}_{E,E;F}$ is analytic at no point either.*

Proof. Extend the operator and form inputs to K', apply the given lift, and retract the value to F_1. Explicitly, for $f_1, \dots, f_k \in \mathcal{L}_{K_1}(E_1, E_1)$, $m \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$ and $x \in E_1^k$, set

$$P_1(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \Pi_F\Big(P\big((f_1)_{K'}, \dots, (f_k)_{K'}\big)(m_{K'})(\iota_E x_1, \dots, \iota_E x_k)\Big).$$

We check the norm bound and then the diagonal identity.

For fixed f and m, the map $P((f_i)_{K'})(m_{K'})$ is a continuous alternating K'-multilinear form on E. Precomposing each slot with the K_1-linear isometry ι_E and postcomposing with Π_F gives a K_1-multilinear map on E_1, still vanishing whenever two arguments coincide. Lemmas D.6–D.8 give

$$\|P_1(f)(m)\| \le \|\Pi_F\|\|P\| \prod_i \|(f_i)_{K'}\| \|m_{K'}\| \le \|P\| \prod_i \|f_i\| \|m\|.$$

In particular, $P_1(f)(m) \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$. The form extension is K_1-linear in m, the operator extension is K_1-linear in each f_i, and P is K'-multilinear with values in K'-linear operators. Consequently P_1 is K_1-linear in m and K_1-multilinear in $(f_1, \dots, f_k)$. The displayed estimate proves boundedness at both levels and yields $\|P_1\| \le \|P\|$.

On the diagonal, use $P(g, \dots, g) = A^{k,K'}(g)$ with $g = f_{K'}$, followed by the embedding identities in Lemmas D.7 and D.8 and the retraction identity $\Pi_F \iota_F = \mathrm{id}$:

$$\begin{aligned} P_1(f, \dots, f)(m)(x) &= \Pi_F\big(m_{K'}(f_{K'}\iota_E x_1, \dots, f_{K'}\iota_E x_k)\big) \\ &= \Pi_F\big(m_{K'}(\iota_E f x_1, \dots, \iota_E f x_k)\big) \\ &= \Pi_F\big(\iota_F(m(f x_1, \dots, f x_k))\big) \\ &= m(f x_1, \dots, f x_k) = A^{k,K_1}(f)(m)(x). \end{aligned}$$

□

Remark D.10. Proposition D.9 is characteristic-free and uses no hypothesis on k. It transports negative results upward only, from K_1 to K'. Ingleton's theorem is used only on K', with no nonarchimedean norm hypothesis on F_1. The completeness of F_1 is used essentially: it is what makes the retraction Π_F land in F_1.

D.4. **Incomplete base fields.** The preceding construction takes place over the completion $\widehat{K}$ of the prescribed field. To return to K, we restrict scalars on the resulting Banach spaces. Density of K in $\widehat{K}$ forces every continuous K-multilinear map between these spaces to be $\widehat{K}$-multilinear, so a lift cannot appear under this restriction.

Lemma D.11. *Let K be a nontrivially normed field with completion $\widehat{K}$, and let E, E', F be $\widehat{K}$-Banach spaces, regarded as K-Banach spaces by restriction of scalars (with the same norms). Then $\mathcal{L}_K(E,E') = \mathcal{L}_{\widehat{K}}(E,E')$, $\mathrm{Alt}^k_K(E;F) = \mathrm{Alt}^k_{\widehat{K}}(E;F)$ and $\mathrm{Alt}^k_K(E';F) = \mathrm{Alt}^k_{\widehat{K}}(E';F)$ with the same norms, $A^{k,K}_{E,E';F} = A^{k,\widehat{K}}_{E,E';F}$, and every bounded k-linear lift of $A^{k,K}_{E,E';F}$ over K is a bounded k-linear lift of $A^{k,\widehat{K}}_{E,E';F}$ over $\widehat{K}$.*

Proof. Let g be a continuous K-linear map between normed spaces that already carry compatible $\widehat{K}$-scalar actions. For $\lambda \in \widehat{K}$, choose $\lambda_n \in K$ with $\lambda_n \to \lambda$. Since $\|\lambda_n x - \lambda x\| = |\lambda_n - \lambda|\|x\|$, continuity gives

$$g(\lambda x) = \lim_n g(\lambda_n x) = \lim_n \lambda_n g(x) = \lambda g(x).$$

Thus g is $\widehat{K}$-linear. Applying the same argument with all but one slot fixed shows that every continuous K-multilinear map is $\widehat{K}$-multilinear. Alternation is the same equal-input vanishing condition over either field.

The spaces of bounded maps therefore coincide as sets. Their operator norms agree as well, since the norms are defined by the same pointwise bounds on the same vectors. They are $\widehat{K}$-Banach spaces, and $A^{k,K}_{E,E';F}$ and $A^{k,\widehat{K}}_{E,E';F}$ agree by their defining precomposition formula. Finally, apply the slotwise density argument once more to a bounded k-linear lift P between these map spaces. It becomes $\widehat{K}$-multilinear with the same norm and the same diagonal identity, and hence is a lift over $\widehat{K}$. □

Appendix E. Proof of the operator obstruction theorem

We show that the target spaces F_1 and F admit no equivalent nonarchimedean norm (Proposition E.1) and prove Theorem 7.2(1), which with Proposition 5.1(1) gives Corollary 7.3.

E.1. The target has no equivalent nonarchimedean norm.

Proposition E.1. *Let κ be any field, $k \geq 2$, $\varrho \in (0,1)$, and let $K_1 = \kappa((X))$, E_1, E_0 and F_1 be as in Appendix C. Let $e_i \in E_0$ be the unit vectors and $\omega_j := e_{kj+1} \wedge \cdots \wedge e_{kj+k} \in F_1$ $(j \geq 0)$, wedges on pairwise disjoint blocks of indices. Then $\|\omega_j\|_{F_1} = 1$ and*

$$\frac{N}{M_\varrho} \;\leq\; \Big\| \sum_{j<N} \omega_j \Big\|_{F_1} \;\leq\; N \qquad (N \geq 1).$$

Hence the norm of F_1 is not nonarchimedean, and F_1 admits no equivalent nonarchimedean norm. The same holds for $F = F_1 \widehat{\otimes}_\pi K'$ as in Appendix D, for any $K' \supseteq K_1$ satisfying (H1) *and* (H2)*, regarded as a normed space over K' or over any subfield of K'.*

Proof. Each ω_j is a wedge of unit vectors, so $\|\omega_j\|_{F_1} \leq 1$ by the defining projective norm. Evaluation of its determinant array at the block indices gives $\det I = 1$, hence $\|\omega_j\|_{F_1} \geq \|J\omega_j\|_\infty \geq 1$. The triangle inequality now gives the upper bound N.

For the lower bound, write ξ_N for the sum of the same block wedges in $\Lambda^k_\kappa E_0$, and put $\omega := \sum_{j<N} \omega_j \in F_1$. Its array $J\omega$ is the constant Laurent array $\Omega^\kappa(\xi_N)$. Thus $\mathrm{coeff}_0(J\omega) = \Omega^\kappa(\xi_N)$, and uniqueness in Proposition C.3 gives $\eta(\omega) = \xi_N$.

We apply Lemma B.4 to ξ_N. Fix one of the first N blocks and an index i_0 in it. Take for φ the $k-1$ coordinate functionals corresponding to the other indices of that block. On every other block all these functionals vanish, so its contraction is zero; here $k \geq 2$ ensures that the minors have positive size. On the chosen block, the coefficient of e_{i_0} is ± 1, while every other

cofactor contains the zero column obtained by evaluating the functionals at e_{i_0}. Consequently $c_\varphi(\xi_N) = \pm e_{i_0}$. These contractions span all kN unit vectors in the first N blocks, so

$$kN \le \operatorname{sdim}(\xi_N) = \operatorname{sdim}(\eta(\omega)) \le kM_\varrho \|\omega\|_{F_1}$$

by (C.2). This proves the lower bound.

If $\|\cdot\|_u$ were an equivalent nonarchimedean norm, choose positive constants c, c' with $\|\cdot\|_{F_1} \le c\|\cdot\|_u \le c'\|\cdot\|_{F_1}$. Then

$$\frac{N}{M_\varrho} \le \Big\|\sum_{j<N} \omega_j\Big\|_{F_1} \le c \max_{j<N} \|\omega_j\|_u \le c'$$

for every $N \ge 1$, a contradiction. In particular, the original norm is not nonarchimedean. This argument uses only the nonarchimedean inequality for $\|\cdot\|_u$, not its scalar homogeneity.

Finally, $\iota_F \colon F_1 \to F$ is a K_1-linear isometry by Lemma D.6(3). The images $\iota_F \omega_j$ satisfy the same unit-norm and sum-growth estimates, so the argument applies to F as well. Restriction to a subfield leaves these estimates unchanged. □

If $\varrho \le \frac{1}{2}$, then $M_\varrho = 1$ and $\|\sum_{j<N} \omega_j\|_{F_1} = N$ exactly.

E.2. **Proof of Theorem 7.2(1).**

Proof. Let K be a nontrivially normed field of characteristic $p > 0$, and let $k \ge p$.

(1) *Choose a Laurent subfield.* By Lemma A.2, K is nonarchimedean. Its completion $\widehat{K}$ is a complete, nontrivially normed, nonarchimedean field of characteristic p. Choose $t \in K$ with $0 < |t| = \varrho < 1$, and let K_1 be the closure of $\mathbb{F}_p(t)$ in $\widehat{K}$. Lemma D.2 identifies K_1 isometrically with $\mathbb{F}_p((X))$, with $|X| = \varrho$, by ev_t. In particular, K_1 is complete and spherically complete.

(2) *Obtain the obstruction over K_1.* Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$, and let F_1 be the completion of $\Lambda^k_{K_1} E_1$ under the projective exterior norm. Transport of scalars along ev_t preserves the norms, exterior powers and completions in this construction. It also identifies the operator and alternating-map spaces, their precomposition maps, and bounded lifts. Theorem C.1, with $\kappa = \mathbb{F}_p$, therefore shows that $A^{k,K_1}_{E_1,E_1;F_1}$ has no bounded k-linear lift.

(3) *Ascend to $\widehat{K}$.* The inclusion $K_1 \subseteq \widehat{K}$ satisfies (H1) and (H2), and F_1 is complete. Proposition D.9, with $K' = \widehat{K}$, gives Banach spaces

$$E := E_1 \widehat{\otimes}_\pi \widehat{K}, \qquad F := F_1 \widehat{\otimes}_\pi \widehat{K}$$

for which $A^{k,\widehat{K}}_{E,E;F}$ has no bounded k-linear lift.

(4) *Descend to K.* Regard the same E and F as K-Banach spaces. By Lemma D.11, a bounded lift of $A^{k,K}_{E,E;F}$ would extend to a bounded lift of $A^{k,\widehat{K}}_{E,E;F}$, which is impossible. Proposition 2.5 applies over the possibly incomplete field K: $A^{k,K}_{E,E;F}$ is analytic at no point. Reindexing the arguments by a bijection gives the same conclusion, with the same spaces E, F, for every finite index set with k elements.

(5) *Exclude an equivalent nonarchimedean target norm.* Since $k \ge p \ge 2$, Proposition E.1 applies to F_1 and F, after the scalar identification in step (2). Its conclusion persists when F is regarded over K. □

Remark E.2. If $K_1 = \widehat{K}$, as when $K = \mathbb{F}_p((X))$ and $t = X$, the completed tensor products identify isometrically with E_1 and F_1: use Lemma D.6(3) and $x \otimes \lambda = \lambda x \otimes 1$. Thus E is nonarchimedean in this case. For $K = \mathbb{F}_q((X))$, $q = p^n$, with its X-adic absolute value, one

can instead apply Theorem C.1 directly with $\kappa = \mathbb{F}_q$, obtaining $E = \ell^\infty(\mathbb{N}, K)$ and $F = F_1$. The proof over a general base field does not require, or assert, that E is nonarchimedean.

Appendix F. A counterexample with scalar values

We prove Theorem 7.2(2) in the form below: Lemma F.4 rules out k-linear lifts of the determinant vanishing on prescribed subspace pairs, and the chain spaces of (F.6) carry these conditions to Banach spaces. Its spaces E, E' are the pair (E, D) of Theorem 7.2(2).

Theorem F.1 (scalar counterexample)**.** *Let K be a complete, nontrivially normed field that is not spherically complete, and let $k \geq 1$ satisfy $k! = 0$ in K. There are nonarchimedean Banach spaces E, E' for which*

$$A^k_{E,E';K} \colon \mathcal{L}(E, E') \longrightarrow \mathcal{L}\big(\mathrm{Alt}^k(E'; K), \mathrm{Alt}^k(E; K)\big)$$

is analytic at no point. The spaces may be taken to be closed subspaces of $\ell^\infty(\Lambda; K^{k+1})$ and $\ell^\infty(\Lambda; K^k)$, respectively, containing the corresponding c_0 spaces, where Λ is countable. Both spaces are nonseparable.

The factorial hypothesis says precisely that K has characteristic $p > 0$ and $k \geq p$. In particular $k \geq 2$, and K is nonarchimedean by Lemma A.2. We first establish the three ingredients of the construction.

F.1. **The countable multilinear tail property.** For a set I, a family $(a_i)_{i \in I}$ is *null* if, for every $\varepsilon > 0$, only finitely many indices satisfy $|a_i| \geq \varepsilon$. Over a complete nonarchimedean field, every null scalar family is unconditionally summable: its finite partial sums form a Cauchy net, since the norm of a difference is at most the largest omitted term. Its sum has norm at most $\sup_i |a_i|$, and such sums may be regrouped.

We derive the multilinear tail property from van der Put's description of the dual of countable ℓ^∞. The induction is independent of the characteristic.

Lemma F.2 (multilinear tails)**.** *Let K be complete, nonarchimedean and not spherically complete, let I be countably infinite, and let $d \geq 1$. Every bounded d-linear map $\lambda \colon \ell^\infty(I, K)^d \to K$ has an expansion*

$$\lambda(x^1, \dots, x^d) = \sum_{(i_1, \dots, i_d) \in I^d} a_{i_1, \dots, i_d} x^1_{i_1} \cdots x^d_{i_d}, \qquad a_{i_1, \dots, i_d} = \lambda(e_{i_1}, \dots, e_{i_d}), \tag{F.1}$$

where $a \in c_0(I^d, K)$. Consequently, for every $\varepsilon > 0$, there is a finite $S \subseteq I$ such that $|\lambda(x^1, \dots, x^d)| \leq \varepsilon$ whenever all $\|x^r\| \leq 1$ and at least one x^r vanishes on S. In particular $\lambda(e_i, \dots, e_i) \to 0$ outside finite subsets of I.

Proof. After reindexing, take $I = \mathbb{N}$. Van der Put's theorem [20, Théorème 4.1], applied to countably many copies of K (a countable index set satisfies its nonmeasurability hypothesis), gives

$$\varphi(x) = \sum_i x_i \varphi(e_i), \qquad (\varphi(e_i))_i \in c_0(\mathbb{N}, K) \quad \big(\varphi \in \mathcal{L}(\ell^\infty, K)\big).$$

This is the case $d = 1$. Suppose the assertion about the array and expansion holds in smaller degrees. Fixing a coordinate vector in any slot of λ gives a form of degree $d - 1$. Hence, for every $\varepsilon > 0$, each slice of the set

$$W = \{(i_1, \dots, i_d) : |a_{i_1, \dots, i_d}| \geq \varepsilon\}$$

obtained by fixing one coordinate is finite.

If W were infinite, we could choose $w^{(0)}, w^{(1)}, \ldots \in W$ with the coordinates $w_r^{(n)}$ distinct as n varies, for each r: a finite list of forbidden coordinates excludes only finitely many elements of W. Define an isometric embedding $T_r\colon \ell^\infty(\mathbb{N}, K) \to \ell^\infty(\mathbb{N}, K)$ by putting $(T_r x)_{w_r^{(n)}} = x_n$ and zero elsewhere. The pulled-back form $\mu = \lambda \circ (T_1, \ldots, T_d)$ has array b with $|b_{n,\ldots,n}| \geq \varepsilon$ for all n. Its slices are null by induction. Thus

$$F_n = \{v \in \mathbb{N}^d : |b_v| \geq \varepsilon,\ v_r = n \text{ for some } r\}$$

is finite. Choose $n_1 < n_2 < \cdots$ so that n_s exceeds every coordinate occurring in $F_{n_1} \cup \cdots \cup F_{n_{s-1}}$. Every nonconstant tuple in $\{n_s : s \geq 1\}^d$ then has coefficient of norm $< \varepsilon$: its least coordinate puts a large tuple in the corresponding F_{n_r}, whereas its greatest coordinate violates the choice of the subsequence.

Let y be the indicator of $\{n_s : s \geq 1\}$ and set $\psi(x) = \mu(x, y, \ldots, y)$. The expansion of each fixed-coordinate slice gives

$$\psi(e_{n_s}) = b_{n_s,\ldots,n_s} + \sum_{v' \in \{n_r : r \geq 1\}^{d-1},\ v' \neq (n_s,\ldots,n_s)} b_{n_s, v'}.$$

The summands in the last sum form a null family and all have norm $< \varepsilon$. Their supremum is also $< \varepsilon$: outside a finite set they have norm $< \varepsilon/2$, and the remaining finite maximum is $< \varepsilon$. The nonarchimedean inequality therefore gives $|\psi(e_{n_s})| \geq \varepsilon$ for every s, contradicting the displayed dual formula. Thus a is null. Applying that formula in the first slot and the inductive expansion in the remaining slots gives (F.1); the null full array permits regrouping the iterated sums.

Finally choose a finite S containing all coordinates occurring in tuples with $|a_{i_1,\ldots,i_d}| > \varepsilon$. If one input vanishes on S, all these terms vanish. Every remaining term has norm at most ε on the product of unit balls, so the same holds for the sum. Apply this conclusion to coordinate vectors outside S to obtain the last assertion. □

F.2. **Chains and a finite-dimensional incompatibility.** Let L be a nonempty finite set and $\Lambda = L^{<\omega}$ the set of finite words in L, including the empty word. Write $s \preceq t$ when t extends s, and write sj for the word obtained by appending $j \in L$. A *j-chain* is a set $T = \{s_0, s_1, \ldots\}$ with $s_r j \preceq s_{r+1}$ for every r. Let Γ_j denote all such chains.

Lemma F.3 (chain gap)**.** *Chains with different labels have at most one common word. If sets $a_j \subseteq \Lambda$, $j \in L$, contain all but finitely many words of every chain in Γ_j, then $\bigcap_{j \in L} a_j \neq \varnothing$.*

Proof. If distinct words s, t belong to both a j-chain and an h-chain, take s shorter than t. Then t extends both sj and sh, forcing $j = h$.

For each word r and label j, there is an extension $s \succeq r$ such that every extension of sj belongs to a_j. Otherwise start with $s_0 = r$ and choose successively an extension $s_{m+1} \succeq s_m j$ outside a_j. This is a j-chain with infinitely many words outside a_j, a contradiction. Enumerate $L = \{j_1, \ldots, j_q\}$ and start at the empty word r_0. Choose $s_m \succeq r_{m-1}$ with the asserted property for j_m, and put $r_m = s_m j_m$. The word r_q extends every $s_m j_m$ and hence belongs to every a_j. □

The next lemma is purely algebraic; the alternating maps in its statement need no topology. Over a field κ, put $V = \kappa^{k+1}$ and $V' = \kappa^k$. Write $e_1, \ldots, e_{k+1}$ and $e'_1, \ldots, e'_k$ for their bases, ε^a for the coordinate forms on V, and δ' for the determinant on V'. Set

$$N = \{\varepsilon^a : 1 \leq a \leq k+1\} \cup \{\varepsilon^a + \varepsilon^b : 1 \leq a < b \leq k+1\},$$
$$\Sigma_\nu = \ker \nu, \qquad C'_s = \{z : z_s = 0\}, \qquad H'_{cd} = \{z : z_c = z_d\},$$

and consider the finite family of pairs

$$\mathcal{F} = \{(V, H'_{cd}) : c < d\} \cup \{(\Sigma_\nu, C'_s) : \nu \in N,\ 1 \le s \le k\}. \tag{F.2}$$

Let $g_0 \colon V \to V'$ be the coordinate projection onto the first k coordinates.

Lemma F.4 (finite fiber obstruction)**.** *Suppose $k! = 0$ in κ. There is no k-linear map*

$$\tau \colon \operatorname{Hom}_\kappa(V, V')^k \longrightarrow \operatorname{Alt}^k_\kappa(V; \kappa)$$

satisfying both

1. *$\tau(g_0, \dots, g_0) = \delta' \circ (g_0, \dots, g_0)$;*
2. *for every $(\Sigma, \Sigma') \in \mathcal{F}$, the restriction $\tau(g^1, \dots, g^k)|_{\Sigma^k}$ is zero whenever $g^r(\Sigma) \subseteq \Sigma'$ for every r.*

Proof. Let δ be the determinant on V. Every alternating k-form α has a unique expression $\alpha = \delta(w_\alpha, \cdot, \dots, \cdot)$: its evaluations on the basis tuples omitting one vector give

$$(w_\alpha)_a = (-1)^{a-1}\alpha(e_1, \dots, \widehat{e}_a, \dots, e_{k+1}).$$

For a nonzero ν, this description also gives

$$\alpha|_{(\ker \nu)^k} = 0 \quad \Longleftrightarrow \quad \nu(w_\alpha) = 0. \tag{F.3}$$

Indeed choose ζ with $\nu(\zeta) = 1$ and a basis $y_1, \dots, y_k$ of $\ker \nu$. Writing $w_\alpha = \nu(w_\alpha)\zeta + z$ with $z \in \ker \nu$ gives $\delta(w_\alpha, y_1, \dots, y_k) = \nu(w_\alpha)\delta(\zeta, y_1, \dots, y_k)$, whose last factor is nonzero.

Suppose τ exists. For a function $a \colon \{1, \dots, k\} \to \{1, \dots, k\}$ define

$$B_a(\varphi_1, \dots, \varphi_k) = \tau(\varphi_1 \otimes e'_{a(1)}, \dots, \varphi_k \otimes e'_{a(k)}), \qquad (\varphi \otimes e'_c)(x) = \varphi(x)e'_c.$$

If a omits s, every input map takes V into C'_s. Applying condition (2) to $(\ker \varepsilon^b, C'_s)$ for each b and using (F.3) makes every coordinate of $w_{B_a(\varphi)}$ zero. Hence $B_a = 0$ unless a is a permutation.

For a permutation a and distinct c, d, replace the target vector in slots $a^{-1}(c)$ and $a^{-1}(d)$ by $e'_c + e'_d$. All input maps then take V into H'_{cd}, so their image under τ is zero. Expansion in these two slots gives $B_a + B_{(c\ d)\circ a}$ and two terms with a missing target coordinate. The latter vanish by the preceding paragraph. Thus, with $B = B_{\mathrm{id}}$,

$$B_a = \operatorname{sgn}(a)B \qquad (a \in S_k). \tag{F.4}$$

Define the $(k+1)$-linear form

$$\beta(\varphi_1, \dots, \varphi_k; \chi) = \chi\big(w_{B(\varphi_1, \dots, \varphi_k)}\big).$$

If the sth argument is $\nu \in N$, its rank-one input vanishes on Σ_ν, and every other rank-one input takes values in C'_s. Conditions (2) and (F.3) give $\beta(\varphi_1, \dots, \nu, \dots, \varphi_k; \nu) = 0$. Use this identity successively for ε^a, ε^b and $\varepsilon^a + \varepsilon^b$ to obtain

$$\beta(\dots, \varepsilon^a, \dots; \varepsilon^b) = -\beta(\dots, \varepsilon^b, \dots; \varepsilon^a).$$

Consequently $t(\sigma) = \beta(\varepsilon^{\sigma(1)}, \dots, \varepsilon^{\sigma(k)}; \varepsilon^{\sigma(k+1)})$ changes sign under every transposition of a slot with the last slot. These transpositions generate S_{k+1}, so $t(\sigma) = \operatorname{sgn}(\sigma)t(\mathrm{id})$.

Expand $g_0 = \sum_{c=1}^k \varepsilon^c \otimes e'_c$ in condition (1). The missing-coordinate terms vanish, and (F.4) gives

$$\varepsilon^1 \wedge \cdots \wedge \varepsilon^k = \sum_{a \in S_k} \operatorname{sgn}(a) B(\varepsilon^{a(1)}, \dots, \varepsilon^{a(k)}).$$

Apply $\alpha \mapsto \varepsilon^{k+1}(w_\alpha)$. Its value on the left is $(-1)^k$; on the right it is $\sum_{a \in S_k} \operatorname{sgn}(a)^2 t(\mathrm{id}) = k!t(\mathrm{id}) = 0$. This is a contradiction. All identities remain valid in characteristic 2: repeated arguments vanish by alternation, and no step divides by 2 or by a factorial. □

We will need a quantitative consequence of this finite incompatibility. Give K^{k+1} and K^k their maximum norms and use the same family (F.2) over K.

Lemma F.5 (finite test certificate)**.** *Assume $k! = 0$ in K, as in Theorem F.1. There are finite sets of tests $\mathcal{T}_j$, one for each $j = (\Sigma^j, \Sigma'^j) \in \mathcal{F}$, consisting of tuples $(g^1, \dots, g^k; \xi_1, \dots, \xi_k)$ with $g^r\Sigma^j \subseteq \Sigma'^j$ and $\xi_r \in \Sigma^j$, such that every K-multilinear τ satisfying condition (1) of Lemma F.4 obeys*

$$\max_{j\in\mathcal{F}} \max_{(g;\xi)\in\mathcal{T}_j} |\tau(g^1, \dots, g^k)(\xi_1, \dots, \xi_k)| \geq 1. \tag{F.5}$$

Proof. Write $\operatorname{char} K = p$. All spaces in (F.2) are defined over $\mathbb{F}_p$. For each j, choose an $\mathbb{F}_p$-basis of the subspace Σ^j defined over $\mathbb{F}_p$, and one of

$$\{g \in \operatorname{Hom}_{\mathbb{F}_p}(\mathbb{F}_p^{k+1}, \mathbb{F}_p^k) : g\Sigma^j \subseteq \Sigma'^j\}.$$

Their scalar extensions are bases of the corresponding K-spaces: row reduction of the defining prime-field matrices has the same pivot positions over K. Take all k-tuples from each basis to form $\mathcal{T}_j$. Multilinearity in the maps and the vectors makes these tests equivalent to condition (2).

Choose coordinates θ for the finite-dimensional space of possible τ. Condition (1) is a system $A\theta = b$, and the tests for condition (2) form a system $C\theta = 0$, with all matrices and constants over $\mathbb{F}_p$. Lemma F.4 over $\mathbb{F}_p$ makes the combined system inconsistent. Row reduction of its augmented matrix therefore gives row vectors λ, μ over $\mathbb{F}_p$ with

$$\lambda A + \mu C = 0, \qquad \lambda b = 1.$$

For every K-valued solution of $A\theta = b$ it follows that $\mu C\theta = -1$. Every nonzero coefficient of μ has absolute value 1, since $\mathbb{F}_p^\times$ is finite. The nonarchimedean inequality now gives (F.5). □

F.3. **The Banach spaces defined by chain limits.** Set $L = \mathcal{F}$ and $\Lambda = L^{<\omega}$. Write $\Sigma^j \subseteq V = K^{k+1}$ and $\Sigma'^j \subseteq V' = K^k$ for the pair indexed by $j \in L$. Define

$$\begin{aligned} E &= \{x \in \ell^\infty(\Lambda; V) : \operatorname{dist}(x_i, \Sigma^j) \longrightarrow 0 \text{ along every } T \in \Gamma_j,\ j \in L\}, \\ E' &= \{z \in \ell^\infty(\Lambda; V') : \operatorname{dist}(z_i, \Sigma'^j) \longrightarrow 0 \text{ along every } T \in \Gamma_j,\ j \in L\}. \end{aligned} \tag{F.6}$$

Limits along a chain are taken outside its finite subsets. We verify the properties of these full solution spaces that will enter the proof.

Each subspace in the definition admits a projection R with $\|R\| \leq 1$ and $\|1 - R\| \leq 1$. For $\ker \varepsilon^a$, delete coordinate a; for $\ker(\varepsilon^a + \varepsilon^b)$, replace coordinate a by the negative of coordinate b; for V, use the identity. On V', delete coordinate s for C'_s, and replace coordinate c by coordinate d for H'_{cd}. These formulas are projections onto the stated spaces. Their norm bounds follow directly from the maximum norm and the nonarchimedean inequality. Moreover

$$\operatorname{dist}(v, \Sigma) = \|(1 - R)v\|. \tag{F.7}$$

Indeed $Rv \in \Sigma$ gives one inequality, and $(1 - R)(v - y) = (1 - R)v$ for every $y \in \Sigma$ gives the other.

For fixed j and $T \in \Gamma_j$, the corresponding condition in (F.6) is thus the inverse image of $c_0(T; V)$ under the bounded linear map $x \mapsto ((1 - R_j)x_i)_{i\in T}$, and similarly for V'. The space c_0 is closed in the sup norm: a uniform approximation within $\varepsilon/2$ by a null family leaves only finitely many coordinates of norm at least ε. Each condition is therefore closed and linear. Their intersection is a closed subspace of the Banach space $\ell^\infty(\Lambda; V)$, respectively $\ell^\infty(\Lambda; V')$, so E and E' are nonarchimedean Banach spaces.

Both contain their corresponding c_0 spaces. They are also modules over $\ell^\infty(\Lambda, K)$ under coordinatewise multiplication, with $\|tx\| \le \|t\|\|x\|$: each residual family in (F.7) is multiplied by the same bounded scalar family and remains null. Finally, if $T \in \Gamma_j$, every bounded Σ^j-valued family supported on T belongs to E. For chains with label j its residual is zero; for a chain with another label, its support meets that chain in at most one word by Lemma F.3. The identical assertion holds for E' and Σ'^j.

Since $k \ge 2$, the label $j = (V, H'_{12})$ occurs and both its spaces contain a vector of norm 1. Fix a j-chain T. Multiplying either such vector by arbitrary bounded scalar families on T gives an isometric embedding of $\ell^\infty(\mathbb{N}, K)$ into E, respectively E'. The binary-valued sequences form an uncountable set at pairwise distance 1, so both spaces are nonseparable.

F.4. **The global contradiction.**

Proof of Theorem F.1. Use the spaces (F.6). Suppose that $A = A^k_{E,E';K}$ has a bounded k-linear lift P. For $i \in \Lambda$, let $\xi_{[i]} \in E$ denote the family equal to $\xi \in V$ at i and zero elsewhere. Single-coordinate families belong to $c_0 \subseteq E$, and likewise in E'. For $g \in \mathrm{Hom}_K(V, V')$ let $g_{[i]} \in \mathcal{L}(E, E')$ send x to the family equal to $g(x_i)$ at i and zero elsewhere. Its norm is at most $\|g\|$. Set

$$\delta'_i(z_1, \dots, z_k) = \delta'(z_{1,i}, \dots, z_{k,i}) \in \mathrm{Alt}^k(E'; K).$$

The determinant estimate for the maximum norm gives $\|\delta'_i\| \le 1$. Define the fiber map

$$\pi_i(g^1, \dots, g^k)(\xi_1, \dots, \xi_k) = P(g^1_{[i]}, \dots, g^k_{[i]})(\delta'_i)(\xi_{1[i]}, \dots, \xi_{k[i]}). \tag{F.8}$$

It is k-linear in the maps and alternating in the vectors, and the lift identity gives

$$\pi_i(g, \dots, g) = \delta' \circ (g, \dots, g) \qquad \text{for every } g. \tag{F.9}$$

In particular every π_i satisfies condition (1) of Lemma F.4.

Fix $j = (\Sigma, \Sigma') \in L$, a chain $T \in \Gamma_j$, maps $g^1, \dots, g^k$ with $g^r\Sigma \subseteq \Sigma'$, and vectors $\xi_1, \dots, \xi_k \in \Sigma$. We claim that

$$\pi_i(g^1, \dots, g^k)(\xi_1, \dots, \xi_k) \longrightarrow 0 \qquad (i \in T). \tag{F.10}$$

For $t \in \ell^\infty(\Lambda, K)$ define u^r_t by

$$(u^r_t x)_i = \begin{cases} t_i g^r(x_i), & i \in T, \\ 0, & i \notin T. \end{cases}$$

This is a bounded map from E to E'. To see that its values lie in E', let R be the projection onto Σ above and decompose, on T,

$$t_i g^r(x_i) = t_i g^r(Rx_i) + t_i g^r((1 - R)x_i).$$

The first term is a bounded Σ'-valued family supported on T, hence belongs to E'. The second is null on T by (F.7), hence its zero extension belongs to $c_0 \subseteq E'$. The coordinate formula gives $\|u^r_t\| \le \|t\|\|g^r\|$, so $t \mapsto u^r_t$ is bounded linear with values in $\mathcal{L}(E, E')$.

For $c \in \ell^\infty(\Lambda, K)$ put

$$m_c(z_1, \dots, z_k) = \sum_{i \in T} c_i \delta'(z_{1,i}, \dots, z_{k,i}). \tag{F.11}$$

This sum converges. In fact, if R' projects onto Σ', write $z_{r,i} = a_{r,i} + b_{r,i}$ with $a_{r,i} = R'z_{r,i}$ and $b_{r,i} = (1 - R')z_{r,i}$. Both families are bounded by $\|z_r\|$, and $b_{r,i} \to 0$ along T. Expanding the

determinant, the term with only a's is zero because $\dim \Sigma' = k-1$. Every remaining term contains an error factor, giving

$$|\delta'(z_{1,i},\dots,z_{k,i})| \le \max_{1\le r\le k}\left(\|b_{r,i}\|\prod_{q\ne r}\|z_q\|\right)\longrightarrow 0.$$

Multiplication by bounded c preserves this limit. Completeness of K now gives the unconditional scalar sum in (F.11). Linearity in each argument and in c follows by taking limits of finite partial sums; repeated arguments make every summand zero. Moreover the determinant estimate gives

$$|m_c(z_1,\dots,z_k)| \le \|c\|\prod_{r=1}^{k}\|z_r\|.$$

Thus $c\mapsto m_c$ is a bounded linear map into $\mathrm{Alt}^k(E';K)$, of norm at most 1.

Let x^r be the family equal to ξ_r on T and zero elsewhere; it belongs to E by the chain-generator inclusion already proved. Coordinatewise multiplication defines a bounded linear map $\rho\mapsto \rho x^r$ from $\ell^\infty(\Lambda,K)$ to E. Consequently

$$\Phi(t^1,\dots,t^k,c,r^1,\dots,r^k) = P(u^1_{t^1},\dots,u^k_{t^k})(m_c)(r^1x^1,\dots,r^kx^k)$$

is a bounded $(2k+1)$-linear scalar form on $\ell^\infty(\Lambda,K)$, with bound

$$\|P\|\prod_{s=1}^{k}\big(\|g^s\|\|\xi_s\|\big)\prod_{s=1}^{k}\|t^s\|\,\|c\|\prod_{s=1}^{k}\|r^s\|.$$

The set Λ is countably infinite, so Lemma F.2 gives $\Phi(e_i,\dots,e_i)\to 0$ outside finite sets. For $i\in T$ we have $u^s_{e_i}=g^s_{[i]}$, $m_{e_i}=\delta'_i$ and $e_ix^s=\xi_{s[i]}$. Equation (F.8) therefore identifies this diagonal value with the expression in (F.10), proving the claim.

Choose the finite tests of Lemma F.5, and define

$$G_j(i)=\max_{(g;\xi)\in\mathcal{T}_j}|\pi_i(g^1,\dots,g^k)(\xi_1,\dots,\xi_k)|,\qquad a_j=\{i\in\Lambda: G_j(i)<1\}.$$

For fixed j and $T\in\Gamma_j$, equation (F.10) applies to every test in the finite set $\mathcal{T}_j$. Hence $G_j(i)\to 0$ along T, and $T\setminus a_j$ is finite. Lemma F.3 supplies an i belonging to every a_j. But (F.9) and Lemma F.5 force $G_j(i)\ge 1$ for some j, a contradiction.

There is therefore no bounded k-linear lift of A. Proposition 2.5 applies in this degree without dividing by $k!$: analyticity at any point would make the degree-k Taylor coefficient such a lift. Thus A is analytic at no point, as claimed. □

Appendix G. Parameter splittings and universal analytic reflection

A normed space P is polarizable in degree n (condition (Γ^n_P), Definition 3.8) if a bounded projection R_n of $\widehat{\otimes}^n_\pi P$ onto $\widehat{\Gamma}^n_\pi P$ exists; for Banach P over complete K this is reflection in degree n (Proposition G.1). Analytic descent (Section 2.3) of every Banach-valued analytic map through every closed inclusion is stronger; we characterize it and separate the two by an example.

Throughout this appendix, K is complete and nontrivially normed, and P is Banach. For $n\ge 1$, let $\widehat{\otimes}^n_\pi P$ be the completed projective tensor power as in Section 3.1; it represents bounded n-linear maps from P^n to Banach spaces. Recall from Definition 3.8 the closed subspace

$$\widehat{\Gamma}^n_\pi P=\overline{\mathrm{span}\{x^{\otimes n}: x\in P\}}\subseteq\widehat{\otimes}^n_\pi P.$$

Proposition G.1 (homogeneous reflection)**.** *For a fixed $n \geq 1$, the following conditions are equivalent.*

(1) *For every Banach space Z, closed subspace $W \subseteq Z$, and bounded n-linear map $B\colon P^n \to Z$ with diagonal values in W, there is a bounded W-valued n-linear map with the same diagonal.*
(2) *The subspace $\widehat{\Gamma}^n_\pi P$ is the range of a bounded projection in $\widehat{\otimes}^n_\pi P$.*

Proof. Suppose first that a projection R_n exists. Linearize B to a bounded map $\widetilde{B}\colon \widehat{\otimes}^n_\pi P \to Z$. The diagonal hypothesis and closedness of W imply $\widetilde{B}(\widehat{\Gamma}^n_\pi P) \subseteq W$. Composing $\widetilde{B}|_{\widehat{\Gamma}^n_\pi P}$ with R_n and the canonical tensor map gives the required multilinear representative, of norm at most $\|B\|\|R_n\|$. Conversely, take $Z = \widehat{\otimes}^n_\pi P$, $W = \widehat{\Gamma}^n_\pi P$, and B equal to the canonical tensor map. Linearizing the representative supplied by the first condition gives a bounded map into $\widehat{\Gamma}^n_\pi P$ fixing every pure power. It fixes their linear span and, by continuity, their closed span, so it is a projection. □

Definition G.2. The Banach parameter space P has *universal analytic reflection* if, for every Banach space Z, closed subspace $W \subseteq Z$, every open $U \subseteq P$, and every map $f\colon U \to W$, analyticity of the composite $U \to W \hookrightarrow Z$ implies analyticity of f.

Theorem G.3 (analytic tensor criterion)**.** *A Banach space P has universal analytic reflection if and only if there are projections $R_n\colon \widehat{\otimes}^n_\pi P \to \widehat{\Gamma}^n_\pi P$ for all $n \geq 1$ such that*

$$\limsup_{n\to\infty} \|R_n\|^{1/n} < \infty. \tag{G.1}$$

Proof. Assume first that the projections exist. Expand an ambient analytic map with values in W at a point. The scalar-line uniqueness argument in Theorem 2.3 puts every homogeneous diagonal in W. The construction in Proposition G.1 replaces its coefficient p_n by a W-valued coefficient of norm at most $\|p_n\|\|R_n\|$ and with the same diagonal. The exponential bound preserves a positive radius of convergence. The degree-zero coefficient is already in W, and the resulting expansion represents the given map.

Conversely, form the Banach spaces with supremum norms

$$Z = c_0\big(\{\widehat{\otimes}^n_\pi P\}_{n\geq 1}\big), \qquad W = c_0\big(\{\widehat{\Gamma}^n_\pi P\}_{n\geq 1}\big) \subseteq Z.$$

Here c_0 denotes families tending to zero in their respective Banach coordinate spaces; W is closed. On the open unit ball of P, define

$$f(x) = (x^{\otimes n})_{n\geq 1}.$$

Since $\|x^{\otimes n}\| \leq \|x\|^n$, this vector lies in W. As a Z-valued map it is analytic: its degree-n coefficient is the canonical tensor map placed in coordinate n, and has norm at most one. This gives an expansion at zero. To re-expand at a with $\|a\| < 1$, choose $r > 0$ with $\|a\| + r < 1$. Expanding each tensor power in $a + h$ gives translated coefficients, and the sum of all term bounds is at most $\sum_{n\geq 1}(\|a\| + r)^n < \infty$. Completeness of Z supplies the coefficients and the rearranged sum, proving analyticity throughout the unit ball. Universal reflection gives a W-valued expansion $\sum_m q_m(x, \ldots, x)$ near zero with $\|q_m\| \leq C\rho^{-m}$ for some $C, \rho > 0$. Compose this expansion with the projection $\pi_n\colon W \to \widehat{\Gamma}^n_\pi P$ and compare on scalar lines with $x \mapsto x^{\otimes n}$. Uniqueness of scalar power-series coefficients gives

$$\pi_n q_n(x, \ldots, x) = x^{\otimes n} \qquad (x \in P).$$

Linearizing $\pi_n q_n$ therefore gives a projection R_n onto $\widehat{\Gamma}^n_\pi P$ with $\|R_n\| \leq \|q_n\| \leq C\rho^{-n}$, proving (G.1). □

G.1. **Projection bounds and the special role of fixed degree.** For $P = K^d$ with the maximum norm, the coordinate construction in Theorem 5.11, applied to the canonical tensor map, gives $\|R_n\| \le d^n$ for $d \ge 1$. A bounded linear change of coordinates introduces only a fixed distortion factor to the n-th power, so it does not change the exponential-growth condition.

For $P = \ell^1(I, K)$, the proof of Proposition 5.13(1) identifies $\widehat{\otimes}^n_\pi P$ isometrically with $\ell^1(I^n, K)$, with unit vectors e_w for $w \in I^n$, puts the sum of the e_w over each permutation orbit in I^n into $\widehat{\Gamma}^n_\pi P$, and gives a projection onto $\widehat{\Gamma}^n_\pi P$ of norm at most $n!$. In these coordinates every element of $\widehat{\Gamma}^n_\pi P$ is constant on each permutation orbit, since every pure power is. Over a nonarchimedean field and for infinite I, the bound $n!$ is optimal even among projections that do not preserve the individual orbit blocks.

Proposition G.4. *If K is nonarchimedean and complete, and I is infinite, then*

$$\inf\{\|R\| : R\colon \widehat{\otimes}^n_\pi \ell^1(I,K) \to \widehat{\Gamma}^n_\pi \ell^1(I,K) \text{ is a bounded projection}\} = n!.$$

Consequently $\ell^1(I, K)$ does not have universal analytic reflection.

Proof. Proposition 5.13(1) gives the upper bound. For the lower bound, take an orbit $\mathcal{O}$ of a tuple with n distinct indices; it has $n!$ elements. For each $w \in \mathcal{O}$, let λ_w be the common coefficient on $\mathcal{O}$ of $R(e_w)$. The orbit sum $\sum_{w\in\mathcal{O}} e_w$ belongs to $\widehat{\Gamma}^n_\pi P$ and is fixed by R, hence

$$\sum_{w\in\mathcal{O}} \lambda_w = 1.$$

The nonarchimedean inequality implies that some $|\lambda_w| \ge 1$. For this w, the ordinary sum norm satisfies $\|R(e_w)\|_1 \ge n!|\lambda_w| \ge n!$. Thus $\|R\| \ge n!$. Finally, $(n!)^{1/n}$ is unbounded, so Theorem G.3 excludes universal analytic reflection. □

Corollary G.5 (entire polynomial approximation of a nonanalytic smooth map)**.** *Let K be a complete nonarchimedean nontrivially normed field and put $P = \ell^1(\mathbb{N}, K)$, with the ordinary sum norm. There are Banach spaces $W \subseteq Z$, with W closed, and a map f on the open unit ball of P such that f is analytic into Z and C^∞ into W, but is not analytic into W at zero. Moreover, f is a uniform limit on every smaller ball of entire W-valued polynomials.*

Proof. Take

$$Z = c_0\big(\{\widehat{\otimes}^n_\pi P\}_{n\ge1}\big), \qquad W = c_0\big(\{\widehat{\Gamma}^n_\pi P\}_{n\ge1}\big), \qquad f(x) = (x^{\otimes n})_{n\ge1}.$$

The spaces are Banach and W is closed in Z. As in the proof of Theorem G.3, the ambient expansion has coefficient norms at most one and defines an analytic map on the open unit ball. Smooth reflection through the closed isometric inclusion makes f C^∞ as a map into W.

Suppose a W-valued expansion at zero has coefficients q_n. Uniqueness on scalar lines gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n},$$

where $\pi_n\colon W \to \widehat{\Gamma}^n_\pi P$ is the coordinate projection. Linearizing $\pi_n q_n$ gives a projection $\widehat{\otimes}^n_\pi P \to \widehat{\Gamma}^n_\pi P$ of norm at most $\|q_n\|$. Proposition G.4 therefore gives $\|q_n\| \ge n!$. Since $(n!)^{1/n}$ is unbounded, these coefficients give a formal multilinear series of radius zero, contradicting the assumed analytic expansion.

For each n, choose the sorted-orbit projection $R_n\colon \widehat{\otimes}^n_\pi P \to \widehat{\Gamma}^n_\pi P$. Its composition with the canonical tensor map is a bounded multilinear representative of $x \mapsto x^{\otimes n}$ with values in $\widehat{\Gamma}^n_\pi P$. Consequently the truncation

$$f_N(x) = (x, x^{\otimes 2}, \dots, x^{\otimes N}, 0, \dots)$$

is an entire W-valued polynomial. For $0 < r < 1$ and $\|x\| \le r$,

$$\|f(x) - f_N(x)\|_W \le r^{N+1},$$

which proves the asserted uniform convergence. □

There is no conflict with Theorem 5.12(2): the ambient representation there has one fixed degree d, so factoring a general analytic parameter map through a free ℓ^1 space incurs a single factorial bound $d!$, rather than a sequence of factorial bounds across Taylor degrees. Over $\mathbb{R}$ and $\mathbb{C}$ every Banach space has universal analytic reflection, by Theorem G.3: the symmetrization $\frac{1}{n!}\sum_{\sigma\in S_n}\sigma$ has norm at most one on $\widehat{\otimes}^n_\pi P$ and, by polarization, is a projection onto $\widehat{\Gamma}^n_\pi P$.

Appendix H. Why a largest full analytic domain need not exist

In this appendix a pair (E, F) consists of a source space and a value space. It is split in degree k (Definition H.2) when $j_{E,F}$ has a bounded linear retraction, a splitting for the single value space F. Split pairs form an analytic domain (Definition H.1), since a splitting at the destination of a morphism suffices, with no compatibility between pairs (Lemma H.3).

There need not be a largest analytic domain: the endomorphism actions at two objects can each be analytic while their cross-action is not. As in Proposition 2.5, a bounded d-linear *lift* of a homogeneous map f is a bounded multilinear map B with $B(x, \dots, x) = f(x)$.

Definition H.1. A full subcategory $\mathcal{D} \subseteq \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K$ is an *analytic domain* for Alt^k if the restriction of Alt^k to $\mathcal{D}$ is analytic on every hom space.

Definition H.2. A pair (E, F) is *split in degree k* if the isometric inclusion

$$j_{E,F}\colon\ \mathrm{Alt}^k(E; F) \longrightarrow \mathrm{Mult}^k(E; F)$$

has a bounded linear retraction.

Lemma H.3. *Every morphism-space action whose destination is a split pair is analytic. In particular, the full subcategory of split pairs is an analytic domain.*

Proof. Let (E, F) be split, with retraction R, and let the other object be (D, G). For $u\colon E \to D$ and $v\colon G \to F$, the action is

$$m \longmapsto R\big(v \circ m \circ (u, \dots, u)\big).$$

Before applying R, replace the k copies of u by independent variables. This gives a bounded $(k+1)$-linear lift of the joint action, with values in $\mathcal{L}(\mathrm{Alt}^k(D; G), \mathrm{Mult}^k(E; F))$. Postcomposition with R gives a lift with the required alternating codomain. No compatibility between the retractions chosen for different objects is needed. □

Theorem H.4 (no largest full analytic domain)**.** *Let p be prime, let $k \ge p$, and equip $K = \mathbb{F}_p(t)$ with its t-adic absolute value. There is no largest full analytic domain for*

$$\mathrm{Alt}^k\colon \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K \longrightarrow \mathsf{Vec}_K.$$

The assertion remains true if the domains must be closed under isomorphisms, and if their objects are restricted to pairs of finite-dimensional, nonarchimedean normed K-spaces. In the latter case, isomorphism closure is understood inside that restricted ambient category.

More precisely, there are objects $X_k = (E_k, G)$ and $Y_k = (D_k, G)$ whose singleton full subcategories are analytic, such that Y_k is split and the action on morphisms from Y_k to X_k is not analytic.

Here finite-dimensional means finite *algebraic* dimension over the incomplete field K. These spaces need not admit continuous coordinates over K. The theorem does not give a counterexample over a complete field, or a counterexample with Banach objects.

We give the entire construction. Put $L = \mathbb{F}_p((t))$, the completion of K. We first work in degree p, then extend to every $k \geq p$ by adjoining coordinate summands.

H.1. **The rigid source and the split pair.** We use the following elementary completion principle. Let K be dense in a complete valued extension L, let (E_i) be a finite family of dense K-subspaces of Banach L-spaces H_i, and let Z be Banach over L. A bounded K-multilinear map $\prod_i E_i \to Z$ extends uniquely to a bounded L-multilinear map $\prod_i H_i \to Z$, with the same norm. Indeed, approximate each input by vectors in E_i. A telescoping expansion and the multilinear bound make the resulting values Cauchy, independently of the approximations. The bound persists in the limit, and density of K in L gives L-linearity in each slot. In the applications below the extended codomain is the complete space L or A, while membership of values on the original inputs in the smaller spaces is retained.

Put $L = \mathbb{F}_p((t))$, the completion of K, and consider the normed L-algebra

$$A = L[\epsilon]/(\epsilon^p), \qquad e_i = \epsilon^i \quad (0 \leq i < p),$$

with the maximum norm in this basis. Multiplication satisfies $\|xy\| \leq \|x\|\,\|y\|$. Choose elements $a_0, \dots, a_{p-1} \in L$, together with the finitely many auxiliary scalars τ_w introduced below, algebraically independent over K. Such a choice is possible because K is countable and L is uncountable: the elements of L algebraic over any finitely generated extension of K form a countable set. Set

$$a = \sum_{i=0}^{p-1} a_i e_i, \qquad E = K^p + Ka \subseteq A.$$

Every subspace used below carries the norm induced by A or L. In particular, E has K-dimension $p+1$ and completion A.

Lemma H.5. *Every bounded endomorphism of E is multiplication by a scalar in K, and every bounded linear map $E \to K$ is zero.*

Proof. A bounded K-linear map extends to completions, and the extension is L-linear by density of K in L. If it preserves E, its columns, the images of the e_i, give a matrix $M + ac^{\mathsf{T}}$, with $M \in M_p(K)$ and $c \in K^p$. Its value at a lies in $K^p + Ka$. Comparison of homogeneous degrees in the algebraically independent a_i shows first that $a(c^{\mathsf{T}}a) = 0$, hence $c = 0$, and then that $M = sI$ for some $s \in K$.

Similarly, the extension of a bounded map $E \to K$ is an L-linear functional on A. Its coefficients belong to K, since its values on the e_i do. Its value on a also lies in K, so algebraic independence makes all its coefficients zero. □

Multiplication by a has lower triangular Toeplitz matrix

$$(M_a)_{ij} = \begin{cases} a_{i-j}, & i \geq j, \\ 0, & i < j. \end{cases}$$

Define

$$D_0 = \operatorname{span}_K\{e_i, \epsilon^i a : 0 \leq i < p\} + Ka^2, \qquad \mathcal{W} = \{e_i : 0 \leq i < p\} \cup \{e_i + e_j : 0 \leq i < j < p\},$$

and put

$$D = D_0 + \sum_{w \in \mathcal{W}} K\tau_w w, \qquad G = \operatorname{span}_K\{\det(d_1, \dots, d_p) : d_i \in D\} \subseteq L.$$

The determinant is taken in the basis $(e_0, \dots, e_{p-1})$. These are finite-dimensional K-spaces, with completions A and L: D contains K^p, and G contains 1. By multilinearity, G is spanned by determinants of the displayed finite generating set for D. We may therefore identify G with a finite-dimensional subspace of $K[a_0, \dots, a_{p-1}, (\tau_w)_w]$. Every element has degree at most one in each individual τ_w, since a nonzero determinant cannot repeat the generator $\tau_w w$. Algebraic independence gives

$$\tau_w^2 G \cap G = \{0\} \qquad (w \in \mathcal{W}). \tag{H.1}$$

Lemma H.6. *Every bounded p-linear map $D^p \to G$ is alternating. Thus (D, G) is split in degree p.*

Proof. Compose such a map m with $G \hookrightarrow L$ and extend to a bounded L-multilinear map on A^p. Fix two argument positions. Put the same $w \in \mathcal{W}$ in those positions and arbitrary standard basis vectors in the others, and call the value $c \in G$. Replacing both copies of w by $\tau_w w \in D$ shows that $\tau_w^2 c \in G$, so (H.1) gives $c = 0$. Taking $w = e_i$ gives vanishing on repeated basis vectors. Taking $w = e_i + e_j$ then gives the signed interchange identity in the two positions. Expanding two equal arbitrary arguments in the basis proves that the completed map, and hence m, is alternating, including when $p = 2$. The inclusion of alternating maps into multilinear maps is therefore an isometric bijection, whose inverse is the desired retraction. □

H.2. **A missing coefficient.** The coefficient space of alternating maps on E is

$$C = \{\lambda \in G : a_i \lambda \in G \text{ for every } i\}.$$

Indeed, an alternating map extends to $\lambda \det$ on the p-dimensional completion A. Determinants of tuples from E span $\operatorname{span}_K\{1, a_0, \dots, a_{p-1}\}$: a determinant can contain at most one copy of a, and replacing one basis vector by a realizes each a_i up to sign. Consequently

$$\operatorname{Alt}^p(E; G) = C \det. \tag{H.2}$$

This identification is isometric, since the determinant bound gives one inequality and evaluation on the standard basis gives the reverse inequality.

Algebraically set every τ_w to zero. The image of G is

$$G_0 = \operatorname{span}_K\{\det(d_1, \dots, d_p) : d_i \in D_0\}.$$

Put $C_0 = \{\lambda \in G_0 : a_i \lambda \in G_0 \text{ for every } i\}$. Specialization sends C into C_0; no continuity of specialization is asserted. The displayed generators of D_0 are homogeneous polynomial vectors of degrees zero, one, and two. Their determinants are homogeneous, so G_0 is graded, and the defining conditions show that C_0 is graded as well.

Lemma H.7. *The homogeneous degree-one part of C_0 is zero.*

Proof. The degree-two part of G_0 is spanned by the 2×2 minors of M_a and the coordinates of a^2. These correspond respectively to choosing two degree-one columns, or one degree-two column, in a determinant of generators. On homogeneous quadratic polynomials define the K-linear map $\Phi\colon K[a_0, \dots, a_{p-1}]_2 \to K[T]$ by

$$\Phi(a_i a_j) = \begin{cases} T^{i+j}, & i + j \geq p - 1, \\ 0, & i + j < p - 1. \end{cases}$$

Every 2×2 minor of M_a is killed by Φ. When both product terms are nonzero, their index sums coincide. If only one is nonzero, write the rows as $r < s$ and columns as $i < j$. The surviving term has index sum $r + s - i - j$, and the missing term implies $r < j$. Hence $r + s - i - j \leq s - i - 1 \leq p - 2$, so its image is again zero.

The coefficient of ϵ^r in a^2 is $\sum_{i+j=r} a_i a_j$. Its image is zero for $r < p-1$; for $r = p-1$ there are exactly p summands and its image is $pT^{p-1} = 0$. Thus Φ kills the entire degree-two part of G_0. If $\ell = \sum_j c_j a_j$ lies in the degree-one part of C_0, then $a_{p-1}\ell \in G_0$, whereas

$$\Phi(a_{p-1}\ell) = \sum_j c_j T^{p-1+j}.$$

The powers are distinct, so all c_j vanish. □

H.3. **The nonanalytic cross-action.** Set $X = (E, G)$ and $Y = (D, G)$. Lemma H.5 identifies $\mathcal{L}(E, E)$ isometrically with K, and the self-action of X is

$$(s, v) \longmapsto s^p v_*.$$

This is a bounded polynomial, so the singleton full subcategory on X is analytic. The singleton on Y is analytic by Lemmas H.6 and H.3.

For $x \in E$, multiplication by x in A restricts to a bounded map $M_x \colon E \to D$. Indeed, writing $x = b + ha$ with $b \in K^p$ and $h \in K$, its values on the generators e_j, a lie in

$$K^p + \sum_i K\epsilon^i a + Ka^2 \subseteq D.$$

Submultiplicativity gives a bounded linear family $x \mapsto M_x$ from E to $\mathcal{L}(E, D)$. Its matrix is lower triangular with diagonal entry x_0, the e_0-coordinate of x in A, so $\det M_x = x_0^p$. In the cross-action from Y to X, fix the target morphism to id_G, evaluate at $\det \in \mathrm{Alt}^p(D; G)$, and use (H.2). The resulting map is

$$f \colon E \longrightarrow C, \qquad f(x) = x_0^p. \tag{H.3}$$

Its values lie in C because it was obtained by pulling back an alternating G-valued map.

Suppose that f were analytic at zero. Since $f(sx) = s^p f(x)$, uniqueness of power-series coefficients on scalar lines gives a bounded p-linear map $B \colon E^p \to C$ with diagonal f. This is the homogeneous coefficient argument used in Proposition 2.5; it requires no factorial division or completeness. Compose B with $C \hookrightarrow L$ and extend to an L-multilinear map on A^p. Put

$$b_i = B(e_0, \dots, e_0, e_i) \in C.$$

There are $p-1$ initial copies of e_0, and $b_0 = f(e_0) = 1$. Evaluation at a in the last position gives $\sum_i a_i b_i \in C$. Specializing all τ_w to zero yields elements $d_i \in C_0$ with

$$d_0 = 1, \qquad \sum_i a_i d_i \in C_0.$$

The degree-one part of this last expression is $\sum_i a_i (d_i)_0$, where $(d_i)_0 \in K$ is the constant term. Its a_0-coefficient equals one. This contradicts Lemma H.7. Thus (H.3), and therefore the cross-action from Y to X, is not analytic.

H.4. **Higher degrees and the categorical conclusion.** For $k = p + r$, let

$$E_k = E \oplus K^r, \qquad D_k = D \oplus K^r$$

with maximum norms. Their completions have dimension k over L. Determinants of tuples from E_k still span $\mathrm{span}_K\{1, a_0, \dots, a_{p-1}\}$, so extraction of the top-degree coefficient identifies

$$\mathrm{Alt}^k(E_k; G) = C \det_k \tag{H.4}$$

isometrically. In particular, the coefficient takes values in C, a fact needed to retain the preceding obstruction after increasing the degree.

By Lemma H.5, the extension of every bounded endomorphism of E_k to its L-completion $A \oplus L^r$ has block form

$$\begin{pmatrix} sI_p & B \\ 0 & T \end{pmatrix}, \qquad s \in K, \quad T \in M_r(K).$$

The scalar s and entries of T are bounded linear functions of the endomorphism, as is seen by evaluation on the standard basis. Its determinant is $s^p \det T$, so its action on (H.4) is multiplication by this bounded polynomial. Including postcomposition shows that the singleton full subcategory on $X_k = (E_k, G)$ is analytic.

The pair $Y_k = (D_k, G)$ is split. Given $m \in \mathrm{Mult}^k(D_k; G)$, restrict the first p arguments to D and fix the last r arguments at the standard auxiliary basis $e'_1, \dots, e'_r$ of K^r. The resulting p-linear map α_m is alternating by Lemma H.6. Write $\pi_D \colon D_k \to D$ and $\pi_r \colon D_k \to K^r$ for the two projections. Explicitly, set

$$(\mathcal{R}m)(v_1, \dots, v_k) = \sum_{\substack{S \subseteq \{1,\dots,k\} \\ |S|=p}} \varepsilon(S)\, \alpha_m(\pi_D v_{s_1}, \dots, \pi_D v_{s_p}) \det(\pi_r v_{t_1}, \dots, \pi_r v_{t_r}),$$

where $s_1 < \cdots < s_p$ list S, $t_1 < \cdots < t_r$ list its complement, and $\varepsilon(S)$ is the sign of the permutation $(s_1, \dots, s_p, t_1, \dots, t_r)$. The determinant and the sign are scalars multiplying the G-valued factor. In degree $r = 0$ the determinant is 1. This shuffle expression is alternating by the same repeated-argument cancellation as the wedge product, involves no factorial division, and gives a bounded linear map $\mathcal{R} \colon \mathrm{Mult}^k(D_k; G) \to \mathrm{Alt}^k(D_k; G)$ with norm at most $\binom{k}{p}$. It fixes every alternating map: after extension to the k-dimensional completion, both maps are determined by their value on the standard basis, where the shuffle formula has exactly one nonzero term. This is the required retraction. For $r > 0$, we do not need or assert that every k-linear map on D_k is alternating.

Finally, restrict the cross-action along the bounded affine family

$$E \longrightarrow \mathcal{L}(E_k, D_k), \qquad x \longmapsto M_x \oplus I_r.$$

Evaluation at $\det_k$, which is G-valued on D_k by Laplace expansion along the first p rows, and coefficient extraction through (H.4) give exactly the nonanalytic map (H.3). Thus the cross-action from Y_k to X_k is not analytic.

A largest analytic full subcategory would have to contain both individually analytic singleton subcategories, and fullness would then require this cross-action to be analytic. This is impossible. If closure under isomorphisms is required, replace each singleton by its full isomorphism closure. Every action between two copies is transported from the self-action by bounded linear changes of coordinates, so both closures remain analytic. The same contradiction proves Theorem H.4.

Finite-dimensional bases. The same spaces show that Theorem 5.11 needs continuous coordinates on the base, not only finite algebraic dimension. Regard E as an analytic manifold with one chart; by Lemma H.5 it has no continuous coordinates. Over it take the trivial bundle with fiber $D_k \oplus E_k$ and the two frames id and $S_x(d, e) = \big(d + (M_x \oplus I_r)e, e\big)$, together with the trivial bundle with fiber G; S_x and its inverse $(d, e) \mapsto \big(d - (M_x \oplus I_r)e, e\big)$ are affine in x, so these bundles are analytic. By (10.1), evaluation at $\det_k$ and (H.4), analyticity of the induced alternating atlas at $x = 0$ would make (H.3) analytic at zero, which was excluded above.

Appendix I. Further coordinate methods

The proof of Proposition I.1 gives a splitting for scalar values: it writes down an explicit section of unnormalized antisymmetrization.

I.1. **Countable-type scalar sources.** The scalar obstruction of Theorem 7.2(2) requires large source spaces in a stronger sense than infinite dimension. A normed space is *of countable type* if a countable subset has dense linear span.

Proposition I.1 (countable-type scalar sources)**.** *Let K be complete and nonarchimedean. If either E or D is of countable type, then $A^k_{E,D;K}$ is analytic in every degree. Neither completeness nor a nonarchimedean norm is required of E, D. In particular, the two spaces in Theorem 7.2(2) must both fail to be of countable type.*

Proof. The case $k = 0$ is constant. For a normed space V, set

$$q_V(x) = \inf_{x=\sum_i x_i} \max_i \|x_i\|,$$

where the decompositions use nonempty finite families, allowing a zero term. Scaling and concatenating them show that q_V is a nonarchimedean seminorm. Let V^u be the completion of its separated quotient; the canonical map $j_V \colon V \to V^u$ is contractive and has dense range. Expanding a bounded scalar multilinear form m along decompositions of each argument gives

$$|m(x_1, \dots, x_k)| \le \|m\| \prod_a q_V(x_a).$$

Thus scalar multilinear forms extend uniquely and isometrically to V^u, as do alternating forms. Extension preserves alternation by density of tuples with repeated entries, and commutes with permutations. Every bounded $u \colon V \to W$ induces $u^u \colon V^u \to W^u$ with $\|u^u\| \le \|u\|$. Uniqueness on the dense image makes $u \mapsto u^u$ bounded linear, and extension of forms intertwines precomposition by u and u^u.

If V is of countable type, so is V^u. By van der Put's basis theorem [20, Proposition 1.4], it has a t-orthogonal basis (b_i) for any fixed $0 < t < 1$. Writing f_i for its coordinate functionals, define

$$S(m)(x_1, \dots, x_k) = \sum_{i_1 < \cdots < i_k} m(b_{i_1}, \dots, b_{i_k}) \prod_{a=1}^{k} f_{i_a}(x_a).$$

The weighted coordinates $|f_i(x)|\|b_i\|$ form a null family bounded by $t^{-1}\|x\|$. Products of finitely many bounded null families are null, so the displayed sum converges unconditionally in K and $\|S(m)\| \le t^{-k}\|m\|$. Let $\mathcal{A}_V$ be unnormalized antisymmetrization, the signed sum over permutations of the arguments. It is bounded and alternating-valued in every characteristic. On finite-coordinate vectors, sorting distinct indices shows $\mathcal{A}_V S(m) = m$ for alternating m; repeated indices contribute zero. Density proves the same identity everywhere. Hence $R = \mathcal{A}_V S$ retracts all multilinear forms onto alternating forms, while $S|_{\mathrm{Alt}}$ is a bounded right inverse of $\mathcal{A}_V$. Transport through j_V gives both operators on the original V.

If E is of countable type, a lift of A is $R_E\big(m \circ (u_1, \dots, u_k)\big)$. If D is of countable type, use instead $\mathcal{A}_E\big(S_D(m) \circ (u_1, \dots, u_k)\big)$. Both are bounded multilinear in the u_i, linear in m, and have diagonal $A(u)(m)$; the second identity uses that antisymmetrization commutes with simultaneous precomposition. Proposition 2.5 concludes the proof. □

I.2. **Continuous algebraic polynomials on c_0.** Theorem 5.12(1) starts with bounded multilinear coefficients in an ambient space. For one homogeneous polynomial, interpolation supplies a lift even when only the diagonal is known to be continuous. The resulting bound is degreewise; it gives no assertion about a common radius across an infinite sequence of degrees.

Proposition I.2 (continuous algebraic polynomials on c_0)**.** *Let K be nontrivially normed and nonarchimedean, let Z be complete and nonarchimedean, and let $n \ge 1$. Suppose $p \colon c_0(I, K) \to Z$*

is continuous and is an algebraic homogeneous polynomial: there exists a K-multilinear map $b\colon c_0(I,K)^n \to Z$, not assumed continuous, with $p(x) = b(x,\dots,x)$. Then p has a bounded n-linear lift q. More precisely, there is $C_{K,n} < \infty$, independent of I, Z, p, such that

$$\|q\| \le C_{K,n}\|p\|_{\mathrm{diag}}, \qquad \|p\|_{\mathrm{diag}} := \inf\{D \ge 0 : \|p(x)\| \le D\|x\|^n \text{ for all } x\}.$$

No completeness assumption on K is needed.

Proof. Continuity at zero and $p(ax) = a^n p(x)$ imply $\|p\|_{\mathrm{diag}} < \infty$. Explicitly, choose $t \in K$ with $0 < |t| < 1$ and $\delta > 0$ such that $\|p(x)\| \le 1$ when $\|x\| < \delta$. For $x \neq 0$, choose an integer m with $|t|\delta \le |t^m|\|x\| < \delta$. Homogeneity gives $\|p(x)\| \le (|t|\delta)^{-n}\|x\|^n$.

Fix a total order on I. On each finite coordinate subspace, the algebraic formula for p is an ordinary homogeneous polynomial of degree n. Polynomial uniqueness over the infinite field K makes its coefficients compatible as the finite coordinate set varies. Denote the coefficient of x^α by c_α, where $\alpha\colon I \to \mathbb{N}$ is finitely supported and $\sum_i \alpha_i = n$. Its support has at most n elements.

Choose distinct $a_0, \dots, a_n$ in the valuation ring of K, and let $L_n \ge 1$ bound the absolute values of all entries of the inverse Vandermonde matrix $(a_i^j)_{0\le i,j\le n}$. If the support of α has $d \le n$ elements, interpolate the restriction of p to those coordinates on the grid $\{a_0,\dots,a_n\}^d$. Each grid vector has norm at most one. Applying the inverse Vandermonde matrix in each coordinate and using the nonarchimedean inequality in Z yields

$$\|c_\alpha\| \le L_n^d\|p\|_{\mathrm{diag}} \le L_n^n\|p\|_{\mathrm{diag}}.$$

For a nondecreasing word $i_1 \le \cdots \le i_n$, write $c_{i_1,\dots,i_n}$ for the coefficient with those multiplicities. Define

$$q(x_1,\dots,x_n) = \sum_{i_1\le\cdots\le i_n} c_{i_1,\dots,i_n}(x_1)_{i_1}\cdots(x_n)_{i_n}.$$

The coefficients are uniformly bounded and each input is a null family, so the summands tend to zero outside finite sets of words. Completeness and the nonarchimedean inequality give convergence, multilinearity, and the bound $\|q\| \le L_n^n\|p\|_{\mathrm{diag}}$. On finite supports its diagonal is p; density and continuity extend the identity to all of $c_0(I,K)$. Thus $C_{K,n} = L_n^n$ suffices. □

This bound is uniform in the number of coordinates, but the interpolation nodes and L_n depend on the degree. No exponential growth bound for $C_{K,n}$ is asserted. Consequently this proposition does not justify replacing lifting norms by diagonal norms in an arbitrary infinite analytic expansion. Theorem 5.12(1) obtains its stronger, radius-preserving conclusion from the already bounded ambient coefficients.

I.3. **A nonspherical target norm with analytic precomposition.**

Remark I.3. Spherical completeness of the given target norm is not necessary for analyticity, even in degrees at least the characteristic. Let $K = \mathbb{F}_p((t))$ with $|t| = 1/2$ and let $F = c_0(\mathbb{N}, K)$, the space of sequences tending to zero. Define

$$w_j = 1 + \frac{1}{j+2}, \qquad \|x\|_w = \sup_{j\ge 0} w_j|x_j|.$$

This is a nonarchimedean norm satisfying

$$\|x\|_\infty \le \|x\|_w \le \frac{3}{2}\|x\|_\infty.$$

Thus F is Banach for either norm. The usual supremum norm has nonzero values in $2^{\mathbb{Z}}$, since every nonzero null sequence attains its maximum absolute value. The discrete-distance argument in the proof of Corollary 5.3 therefore makes the supremum norm spherically complete.

Theorem 5.2 gives analytic precomposition in every finite degree for that norm, for all normed E, E'. Equivalent norms preserve the bounded maps and their analytic power-series expansions: the norms of coefficients change by a fixed factor and convergence is unchanged. Thus precomposition remains analytic for $\|\cdot\|_w$, also directly by Corollary 5.3.

Nevertheless, the weighted norm is not spherically complete. Write e_j for the coordinate vectors, set $a_n = \sum_{j<n} e_j$, and consider the closed balls

$$B_n = \overline{B}_w(a_n, w_n).$$

They are nested because $w_{n+1} < w_n$ and $\|a_{n+1} - a_n\|_w = w_n$; the nonarchimedean inequality gives $B_{n+1} \subseteq B_n$. If x belonged to all of them, then, for each j and any $n > j$,

$$|x_j - 1| \le \frac{w_n}{w_j} < 1.$$

Hence $|x_j| = 1$ for every j, contradicting $x_j \to 0$. Their intersection is empty. This example separates spherical completeness of a particular norm from analyticity, which is invariant under equivalent norms. It does not exclude a characterization using an equivalent spherically complete norm: this target has exactly such a norm.

Appendix J. Ordering splittings

Throughout, K is a nontrivially normed field, Σ is a totally ordered set, and $\vartheta \in K$ is the fixed element with $0 < |\vartheta| < 1$.

J.1. **Lower estimates.**

Proof of Theorem 5.6. (1) *Step 1 (the weight ν).* Let $x, y \in V$ be nonzero and $\Sigma_0 \subseteq \Sigma$ a finite set containing their supports. In this proof an interval is a set of consecutive elements of Σ_0; disjoint ones lie in disjoint intervals of Σ, so (5.1) applies to them. For an interval I let $\nu_x(I)$ be the supremum of $\sum_i \|\mathcal{P}_{I_i} x\|^r$ over the families of pairwise disjoint intervals $I_i \subseteq I$. Then ν_x is monotone and superadditive on disjoint intervals, $\|\mathcal{P}_I x\|^r \le \nu_x(I)$ and $\nu_x(\Sigma_0) \le C^r \|x\|^r$. Put $a_I = \nu_x(I)/\nu_x(\Sigma_0)$, $b_I = \nu_y(I)/\nu_y(\Sigma_0)$ and $\nu(I) = a_I + b_I$, so that $\nu(\Sigma_0) = 2$.

Step 2 (the binary tree). Build a binary tree of intervals with root Σ_0 of depth 0. A node I is a leaf if it has at most one element or $\nu(I) = 0$. Otherwise let t_I be the least $t \in I$ with $\nu(\{t' \in I : t' \le t\}) \ge \nu(I)/2$; the children of I, of depth one more, are $I_< = \{t \in I : t < t_I\}$ and $I_> = \{t \in I : t > t_I\}$. Then $\nu(I_<) < \nu(I)/2$ by minimality of t_I and $\nu(I_>) \le \nu(I)/2$ by superadditivity, so the nodes of depth d are pairwise disjoint and satisfy $\nu \le 2^{1-d}$.

Step 3 (the sum over the nodes). For an interval I let z_I be the sum of $x(t)y(t')\, e_t \otimes e_{t'}$ over $t < t'$ in I, so that $\mathcal{P}_<(x \otimes y) = z_{\Sigma_0}$. At an internal node

$$z_I = z_{I_<} + z_{I_>} + \mathcal{P}_{I_<} x \otimes \mathcal{P}_{I \setminus I_<} y + \mathcal{P}_{\{t_I\}} x \otimes \mathcal{P}_{I_>} y,$$

and $z_I = 0$ at a leaf I: it has at most one element, or $\nu(I) = 0$ and x vanishes on I. So $\mathcal{P}_<(x \otimes y)$ is the sum of the last two terms over the internal nodes. At a node I they have projective norm at most

$$2\nu_x(I)^{1/r} \nu_y(I)^{1/r} \le 2C^2 (a_I b_I)^{1/r} \|x\| \|y\|.$$

At depth d, $a_I b_I \le (\nu(I)/2)^2 \le 4^{-d}$, and $\sum_I (a_I b_I)^{1/2} \le 1$ by the Cauchy–Schwarz inequality, as $\sum_I a_I \le 1$ and $\sum_I b_I \le 1$; so $\sum_I (a_I b_I)^{1/r} \le 2^{-d(2/r-1)}$. Summing over d gives $\|\mathcal{P}_<(x \otimes y)\| \le C_\sharp \|x\| \|y\|$. The bounded bilinear map $(x, y) \mapsto \mathcal{P}_<(x \otimes y)$ extends from $V \times V$ to $E \times E$ with values in $E \widehat{\otimes}_\pi E$; its linearization is the asserted extension.

(2) For $1 \le a < k$ let $\mathcal{P}_<^{(a)} = \mathrm{id}^{\otimes(a-1)} \otimes \mathcal{P}_< \otimes \mathrm{id}^{\otimes(k-a-1)}$, a map $\otimes_\pi^k E \to \widehat{\otimes}_\pi^k E$ of norm at most $C_\sharp$, extended to the completion, and $\mathcal{P}^{(k)} = \mathcal{P}_<^{(k-1)} \circ \cdots \circ \mathcal{P}_<^{(1)}$. Then $\|\mathcal{P}^{(k)}\| \le C_\sharp^{k-1}$, and $\mathcal{P}^{(k)}$ fixes $e_{t_1} \otimes \cdots \otimes e_{t_k}$ if $t_1 < \cdots < t_k$ and maps it to 0 otherwise. Put $s = \mathcal{P}^{(k)} \circ \mathrm{alt}_E^k$ and $\rho = \widehat{\wedge}_E^k \circ \mathcal{P}^{(k)}$. For $x_1, \dots, x_k \in V$,

$$s(x_1 \wedge \cdots \wedge x_k) = \sum_{t_1 < \cdots < t_k} \det\big(x_b(t_a)\big)_{a,b}\, e_{t_1} \otimes \cdots \otimes e_{t_k},$$

whose image under $\widehat{\wedge}_E^k$ is $x_1 \wedge \cdots \wedge x_k$. Such wedges span a dense subspace of $\Lambda_\pi^k E$, as V is dense and $\|x_1 \wedge \cdots \wedge x_k\| \le \prod_i \|x_i\|$. Hence $\widehat{\wedge}_E^k \circ s = \mathrm{id}$ and $\rho \circ \mathrm{alt}_E^k = \widehat{\wedge}_E^k \circ s = \mathrm{id}$. If $E = V$, then $\mathcal{P}_<$ preserves $V \otimes V$, so s and ρ take values in the uncompleted spaces.

(3) follows from (2) and Proposition 3.5. □

J.2. **Ordering splittings in every degree.**

Proof of Proposition 5.7. Step 1 (source and target). As at the end of part (1) of the proof of Theorem 5.6, $\mathcal{P}_<$ extends to a map $E \otimes_\pi E \to E \widehat{\otimes}_\pi E$ of norm at most C, and part (2) of that proof, with C in place of $C_\sharp$ and n in place of k, proves (1) and (2).

Step 2 (diagonal operators). For $n = 1$ take $R_1 = \mathrm{id}$; let $n \ge 2$. Let $(1\,2)$ be the isometry exchanging the factors of $E \otimes_\pi E$, $\mathcal{P}_> = (1\,2) \circ \mathcal{P}_< \circ (1\,2)$ and $\Delta = \mathrm{id} - \mathcal{P}_< - \mathcal{P}_>$, so that $\|\Delta\| \le 1 + 2C$. For $T \in \{\mathcal{P}_<, \Delta\}$ and $1 \le a < n$ let $T^{(a)} = \mathrm{id}^{\otimes(a-1)} \otimes T \otimes \mathrm{id}^{\otimes(n-a-1)}$, extended to $\widehat{\otimes}_\pi^n E$, of norm at most $\|T\|$. For $\mathbf{t} \in \Sigma^n$ and $e_{\mathbf{t}} = e_{t_1} \otimes \cdots \otimes e_{t_n}$, the vector $\mathcal{P}_<^{(a)} e_{\mathbf{t}}$, respectively $\Delta^{(a)} e_{\mathbf{t}}$, is $e_{\mathbf{t}}$ if $t_a < t_{a+1}$, respectively $t_a = t_{a+1}$, and 0 otherwise. So these operators preserve $\otimes^n V$ and commute on it, hence on $\widehat{\otimes}_\pi^n E$.

Step 3 (the projection). For a composition $c = (c_1, \dots, c_m)$ of n let $J(c)$ be the set of partial sums $c_1 + \cdots + c_i$, $i < m$, and $\Pi_c = \prod_{a \in J(c)} \mathcal{P}_<^{(a)} \prod_{a \notin J(c)} \Delta^{(a)}$, with a ranging over $\{1, \dots, n-1\}$. Then $\Pi_c e_{\mathbf{t}} = e_{\mathbf{t}}$ if $\mathbf{t}$ is nondecreasing with constant blocks of lengths $c_1, \dots, c_m$, and $\Pi_c e_{\mathbf{t}} = 0$ otherwise. The stabilizer of such a $\mathbf{t}$ in S_n, acting by isometric permutations of the factors, is $S_{c_1} \times \cdots \times S_{c_m}$; if Ω_c is a set of representatives of its left cosets, then $O_{\mathbf{t}} = \sum_{\sigma \in \Omega_c} \sigma e_{\mathbf{t}}$. Hence $R_n = \sum_c \sum_{\sigma \in \Omega_c} \sigma \circ \Pi_c$, the first sum over all compositions of n, satisfies $R_n e_{\mathbf{t}} = O_{\mathbf{t}}$ for nondecreasing $\mathbf{t}$ and $R_n e_{\mathbf{t}} = 0$ otherwise, and $\|R_n\| \le \sum_c n!\,(c_1! \cdots c_m!)^{-1} C^{m-1} (1 + 2C)^{n-m}$.

Step 4 (the image). Let Γ_V be the span of the $O_{\mathbf{t}}$ with $\mathbf{t}$ nondecreasing. Then $R_n(\otimes^n V) = \Gamma_V$ and R_n fixes Γ_V, so by continuity R_n is a projection of $\widehat{\otimes}_\pi^n E$ onto $\overline{\Gamma_V}$. For $h \in V$, $h^{\otimes n} = \sum_{\mathbf{t}\ \text{nondecreasing}} h(t_1) \cdots h(t_n) O_{\mathbf{t}} \in \Gamma_V$; as $h \mapsto h^{\otimes n}$ is continuous and V is dense, $\widehat{\Gamma}_\pi^n E \subseteq \overline{\Gamma_V}$. Conversely, by the argument of Step 1 of the proof of Proposition 5.13, with the vectors e_t in place of the unit vectors, $O_{\mathbf{t}}$ is a linear combination of pure powers. Hence $\overline{\Gamma_V} = \widehat{\Gamma}_\pi^n E$, and (Γ_E^n) holds. □

J.3. **Boxes in chain sets.** Throughout this subsection and the next, $2 < r < \infty$ is fixed. For finite $T_1, \dots, T_k \subseteq \Sigma$ the *chain set* is $\Xi(T_1, \dots, T_k) = \{t \in T_1 \times \cdots \times T_k : t_1 < \cdots < t_k\}$, with $\Xi() = \{()\}$. A *box* is a product $J_1 \times \cdots \times J_k$ with $J_a \subseteq T_a$, of *cost* $\prod_a |J_a|^{1/r}$; $\chi(T_1, \dots, T_k)$ is the least total cost of a partition of $\Xi(T_1, \dots, T_k)$ into boxes contained in it. So $\chi = 0$ if the chain set is empty, $\chi() = 1$, and $\chi(T_1) = |T_1|^{1/r}$, as $y \mapsto y^{1/r}$ is subadditive.

For $y, y' > 0$ let $\psi(y, y') = \min(y, y')^{1-1/r} \max(y, y')^{1/r}$. A *matching* of a finite set S is a set $\mathfrak{m}$ of pairwise disjoint two-element subsets of S, *near-perfect* if it has $\lfloor |S|/2 \rfloor$ pairs. Let $\mathrm{Mat}(S)$ be the set of near-perfect matchings of S and $\mathrm{Mat}_q = \mathrm{Mat}(\{1, \dots, q\})$. For positive

reals $y = (y_a)_{a\in S}$ put

$$\langle y\rangle = \max_{\mathfrak{m}} \prod_{\{a,b\}\in\mathfrak{m}} \min(y_a, y_b), \qquad \Psi(y) = \prod_{a\in S} y_a^{1/r}\,\langle y\rangle^{1-2/r},$$

the maximum over all matchings of S, empty products being 1. For a matching $\mathfrak{m}$ let $\Psi_{\mathfrak{m}}(y)$ be the product of $\psi(y_a, y_b)$ over the pairs $\{a,b\}\in\mathfrak{m}$ and of $y_a^{1/r}$ over the unmatched a; as $\psi(y,y') = (yy')^{1/r}\min(y,y')^{1-2/r}$, it is $\Psi(y)$ with $\langle y\rangle$ replaced by the product of the minima over $\mathfrak{m}$. Write $y_{S'} = (y_a)_{a\in S'}$ for $S'\subseteq S$. Let $\lambda_1 = 1+(1-2^{-1/r})^{-1}$, $\lambda_2 = 1+2^{1-2/r}(2^{1-2/r}-1)^{-1}$ and $\lambda = \max(\lambda_1,\lambda_2)$. Four elementary facts are used.

- (F1) $\psi(y,y') \le y^{1-1/r}y'^{1/r}$ and $\psi(y,y') \le y^{1/r}y'^{1-1/r}$, with equality in the first if $y\le y'$: one bound equals $\psi(y,y')$, the other is $(\max/\min)^{1-2/r}$ times larger.
- (F2) $\Psi_{\mathfrak{m}}(y)\le\Psi(y)$ for every matching $\mathfrak{m}$ of S, as $1-2/r>0$. If all $y_a\ge 1$, then $\Psi(y)=\Psi_{\mathfrak{m}}(y)$ for some $\mathfrak{m}\in\mathrm{Mat}(S)$, as every matching extends to a near-perfect one and the added minima are ≥ 1; so $\Psi(y)\le\sum_{\mathfrak{m}\in\mathrm{Mat}(S)}\Psi_{\mathfrak{m}}(y)$.
- (F3) Ψ is nondecreasing in each variable.
- (F4) Let $N\ge 1$, $y_{b,i}\ge 0$ for b in a finite set S and $1\le i\le N$, $\alpha_b\ge 0$ and $\Theta=\sum_b\alpha_b$. Then, with $0^0=1$, $\sum_{i=1}^N\prod_b y_{b,i}^{\alpha_b}\le N^{(1-\Theta)_+}\prod_b(\sum_{i=1}^N y_{b,i})^{\alpha_b}$. For $\Theta\le 1$ this is Hölder's inequality, with the constant function 1 if $\Theta<1$; for $\Theta>1$, apply the case $\Theta=1$ to the exponents α_b/Θ and use $\sum_i z_i^\Theta\le(\sum_i z_i)^\Theta$ for $z_i=\prod_b y_{b,i}^{\alpha_b/\Theta}$.

The *hull* of a finite nonempty $Y\subseteq\Sigma$ is $\mathrm{hull}(Y)=\{t\in\Sigma:\min Y\le t\le\max Y\}$.

Lemma J.1 (pair tree)**.** *Let $T',T''\subseteq\Sigma$ be finite and nonempty, $m=\min(|T'|,|T''|)$ and $h=\lceil\log_2 m\rceil$; call T' the* tree side *if $|T'|\le|T''|$, and T'' otherwise. There are a finite set $\mathcal{T}$ with a partition $\mathcal{T}=\mathcal{T}_0\sqcup\cdots\sqcup\mathcal{T}_{h-1}\sqcup\mathcal{T}_*$ into* levels*, and nonempty sets $T'_i\subseteq T'$, $T''_i\subseteq T''$ with $\max T'_i<\min T''_i$ for $i\in\mathcal{T}$, such that*

$$\{(t',t'')\in T'\times T'' : t'<t''\} = \bigsqcup_{i\in\mathcal{T}} T'_i\times T''_i \tag{J.1}$$

and, if Y_i denotes the one of T'_i,T''_i that lies in the tree side: (T1) *$|\mathcal{T}_d|\le 2^d$ for $d<h$, and $|\mathcal{T}_*|\le m$;* (T2) *$|Y_i|\le m2^{-d}$ for $i\in\mathcal{T}_d$, and $|Y_i|=1$ for $i\in\mathcal{T}_*$;* (T3) *within each level, the sets T'_i have pairwise disjoint hulls, and so have the sets T''_i.*

Proof. Step 1 (the tree). Assume first $|T'|\le|T''|$. The nodes are pairs (U,I) of a nonempty set U of consecutive elements of T' and an interval I of Σ; the root (T',Σ) has depth 0. If $|U|\ge 2$, let U_L consist of the first $\lceil|U|/2\rceil$ elements of U and U_R of the others, $t_U=\max U_L$, $I_L=\{t\in I:t\le t_U\}$ and $I_R=\{t\in I:t>t_U\}$; the node has the children (U_L,I_L) and (U_R,I_R), of depth one more, and contributes the pair $(U_L,T''\cap I_R)$ to $\mathcal{T}_d$, d its depth, if $T''\cap I_R\ne\emptyset$. A node $(\{t_0\},I)$ is a leaf and contributes $(\{t_0\},\{t\in T''\cap I:t>t_0\})$ to $\mathcal{T}_*$ if this set is nonempty. Each pair satisfies $\max T'_i<\min T''_i$.

Step 2 (partition). Put $\Upsilon(U,I)=\{(t',t'')\in U\times(T''\cap I):t'<t''\}$. If $|U|\ge 2$, then $\Upsilon(U,I)=U_L\times(T''\cap I_R)\sqcup\Upsilon(U_L,I_L)\sqcup\Upsilon(U_R,I_R)$: for $t'\in U_L$ the cases $t''>t_U$ and $t''\le t_U$ give the first two pieces, and every $t'\in U_R$ exceeds t_U. At a leaf, Υ is the contributed pair or empty. Unwinding from $\Upsilon(T',\Sigma)$ gives (J.1).

Step 3 (counting and hulls). By induction a node of depth d has $|U|\le\lceil m/2^d\rceil$, as children have at most $\lceil|U|/2\rceil$ elements and $\lceil\lceil y\rceil/2\rceil=\lceil y/2\rceil$; there are at most 2^d such nodes. If $|U|\ge 2$, then $m>2^d$, so $d<h$, and $|U_L|\le\lceil m/2^{d+1}\rceil\le m/2^d$, as $\lceil y\rceil\le 2y$ for $y>1/2$. The leaves have distinct singletons $U\subseteq T'$. This proves (T1) and (T2). Children split the block U and the interval I of their parent into disjoint parts, so two nodes neither of which descends from

the other, such as two nodes of one depth or two leaves, have disjoint blocks U, one before the other, and disjoint intervals I. A pair contributed by a node has its first member in U and its second in I; this proves (T3).

Step 4 (the other case). If $|T''| < |T'|$, apply Steps 1–3 to T'', T' and the reversed order of Σ, which changes no hull. □

Lemma J.2 (level summation)**.** *Let $m \ge 1$, $h = \lceil \log_2 m \rceil$, and let $\mathcal{T} = \mathcal{T}_0 \sqcup \cdots \sqcup \mathcal{T}_{h-1} \sqcup \mathcal{T}_*$ be a finite set with $|\mathcal{T}_d| \le 2^d$ and $|\mathcal{T}_*| \le m$. Let $g\colon \mathcal{T} \to [0,\infty)$ satisfy $g \le m2^{-d}$ on $\mathcal{T}_d$ and $g \le 1$ on $\mathcal{T}_*$. For b in a finite set S let $y_b\colon \mathcal{T} \to [0,\infty)$ have sum at most Y_b over each $\mathcal{T}_d$ and over $\mathcal{T}_*$. Let $\alpha_0 \ge 1/r$, $\alpha_b \ge 0$, $\Theta = \sum_{b\in S} \alpha_b$, $Y = \prod_b Y_b^{\alpha_b}$ and $\Phi = \sum_{i\in\mathcal{T}} g(i)^{\alpha_0} \prod_b y_b(i)^{\alpha_b}$. (1) If $\Theta + \alpha_0 \ge 1 + 1/r$, then $\Phi \le \lambda_1 m^{\alpha_0} Y$. (2) If $\alpha_0 = \Theta = 1/r$, then $\Phi \le \lambda_2 m^{1-1/r} Y$.*

Proof. Put $\zeta = (1-\Theta)_+ - \alpha_0$. By (F4), the part of Φ over $\mathcal{T}_d$ is at most $(m2^{-d})^{\alpha_0} 2^{d(1-\Theta)_+} Y = m^{\alpha_0} 2^{d\zeta} Y$, and the part over $\mathcal{T}_*$ at most $m^{(1-\Theta)_+} Y$. (1) Here $\zeta \le -1/r$ and $(1-\Theta)_+ \le \alpha_0$, as $(1-\Theta)_+ = 0$ if $\Theta \ge 1$ and $1 - \Theta \le \alpha_0 - 1/r$ if $\Theta < 1$; so $\Phi \le m^{\alpha_0}(\sum_{d\ge 0} 2^{-d/r} + 1) Y = \lambda_1 m^{\alpha_0} Y$. (2) Here $(1-\Theta)_+ = 1 - 1/r$ and $\zeta = 1 - 2/r > 0$. For $m \ge 2$, $2^h < 2m$ gives $\sum_{d<h} 2^{d\zeta} \le 2^\zeta m^\zeta / (2^\zeta - 1)$, and $1/r + \zeta = 1 - 1/r$; for $m = 1$ there are no levels $\mathcal{T}_d$. Adding the two parts gives $\Phi \le \lambda_2 m^{1-1/r} Y$. □

Theorem J.3 (weighted boxes)**.** *Let $\eta_0 = \eta_1 = 1$ and $\eta_k = \lambda \sum_{l=2}^k |\mathrm{Mat}_{l-1}|\, \eta_{l-1}\eta_{k-l}$ for $k \ge 2$. For every $k \ge 1$ and all finite nonempty $T_1, \dots, T_k \subseteq \Sigma$ with $m_a = |T_a|$ elements,*

$$\chi(T_1, \dots, T_k) \le \eta_k \Psi(m_1, \dots, m_k) = \eta_k \prod_{a=1}^k m_a^{1/r} \, \langle m \rangle^{1-2/r}.$$

Proof. We induct on k; the cases $k \le 1$ are the values of $\chi()$ and $\chi(T_1)$ computed after the definition of χ. Let $k \ge 2$ and $m = (m_1, \dots, m_k)$.

Step 1 (decomposition). Apply Lemma J.1 to $T' = T_1$ and $T'' = T_2$. For $i \in \mathcal{T}$ and $T \subseteq \Sigma$ let $I_i = \mathrm{hull}(T_i'')$ and $T^{>i} = \{t \in T : t > \max T_i''\}$, and for $2 \le l \le k$ let

$$\Xi_{i,l} = T_i' \times \Xi(T_i'', T_3 \cap I_i, \dots, T_l \cap I_i) \times \Xi(T_{l+1}^{>i}, \dots, T_k^{>i}).$$

These sets partition $\Xi(T_1, \dots, T_k)$: for a chain t, (t_1, t_2) lies in exactly one $T_i' \times T_i''$ by (J.1), and if l is the largest a with $t_a \le \max T_i''$, then $t_a \in T_a \cap I_i$ for $3 \le a \le l$ and $t_a \in T_a^{>i}$ for $a > l$; conversely, elements of $\Xi_{i,l}$ are chains, as $\max T_i' < \min T_i''$ and $t_l \le \max T_i'' < t_{l+1}$. Products of box partitions of the three factors are box partitions of $\Xi_{i,l}$ with the product cost, so

$$\chi(T_1, \dots, T_k) \le \sum_{l=2}^k \sum_{i\in\mathcal{T}} |T_i'|^{1/r} \, \chi(T_i'', T_3 \cap I_i, \dots, T_l \cap I_i) \, \chi(T_{l+1}^{>i}, \dots, T_k^{>i}). \tag{J.2}$$

Step 2 (exponents). Fix l, and let $S_l = \{3, \dots, l\}$ and $S_l^+ = \{l+1, \dots, k\}$. By induction and (F3) the last factor in (J.2) is at most $\eta_{k-l} \Psi(m_{S_l^+})$. Put $m_{2,i} = |T_i''|$ and $m_{a,i} = |T_a \cap I_i|$ for $a \in S_l$. By induction and (F2), the middle factor is at most $\eta_{l-1} \sum_{\mathfrak{m}} \Psi_{\mathfrak{m}}(m_{2,i}, (m_{a,i})_{a\in S_l})$, the sum over $\mathfrak{m} \in \mathrm{Mat}(\{2\} \cup S_l)$, or zero if some $m_{a,i} = 0$. Fix $\mathfrak{m}$. For a pair $\{a, b\} \in \mathfrak{m}$ with $m_a \le m_b$, ties broken by the index, let $\alpha_a = 1 - 1/r$ and $\alpha_b = 1/r$; let $\alpha_a = 1/r$ for an unmatched a. By (F1),

$$\Psi_{\mathfrak{m}}\big(m_{2,i}, (m_{a,i})_{a\in S_l}\big) \le \prod_{a\in\{2\}\cup S_l} m_{a,i}^{\alpha_a}, \qquad \prod_{a\in\{2\}\cup S_l} m_a^{\alpha_a} = \Psi_{\mathfrak{m}}(m_{\{2\}\cup S_l}). \tag{J.3}$$

The sum $\Theta_{\mathfrak{m}}$ of the α_a is at least 1 if $S_l \ne \emptyset$, since $\mathfrak{m}$ then has a pair, and is $1/r$ if $S_l = \emptyset$.

Step 3 (level sums). Let $\Phi_{\mathfrak{m}} = \sum_{i\in\mathcal{T}} |T_i'|^{1/r} \prod_{a\in\{2\}\cup S_l} m_{a,i}^{\alpha_a}$. By (T3) the sums of $|T_i'|$, $m_{2,i}$ and $m_{a,i}$ over each level are at most m_1, m_2 and m_a. Apply Lemma J.2 with $\min(m_1,m_2)$ in the role of m, with $g(i) = |Y_i|$ for the set Y_i of Lemma J.1 and α_0 its exponent in $\Phi_{\mathfrak{m}}$, which is $1/r$ or $\alpha_2 \ge 1/r$, and with the other factors of $\Phi_{\mathfrak{m}}$ as the functions y_b; (T2) gives the bounds on g, and the exponents add up to $1/r + \Theta_{\mathfrak{m}}$. If $S_l \neq \emptyset$, part (1) and (J.3) give $\Phi_{\mathfrak{m}} \le \lambda m_1^{1/r} \Psi_{\mathfrak{m}}(m_{\{2\}\cup S_l})$; if $S_l = \emptyset$, part (2) gives $\Phi_{\mathfrak{m}} \le \lambda\psi(m_1,m_2)$.

Step 4 (assembly). By (F2), $\Psi(m_{S_l^+}) = \Psi_{\mathfrak{m}^+}(m_{S_l^+})$ for some $\mathfrak{m}^+ \in \mathrm{Mat}(S_l^+)$. Then $m_1^{1/r}\Psi_{\mathfrak{m}}(m_{\{2\}\cup S_l})\Psi_{\mathfrak{m}^+}(m_{S_l^+})$ is $\Psi_{\mathfrak{m}\cup\mathfrak{m}^+}(m)$, and $\psi(m_1,m_2)\Psi_{\mathfrak{m}^+}(m_{S_l^+})$ is $\Psi_{\{\{1,2\}\}\cup\mathfrak{m}^+}(m)$; both are at most $\Psi(m)$ by (F2). As $\{2\}\cup S_l$ has $l-1$ elements, the l-th term of (J.2) is at most $\lambda|\mathrm{Mat}_{l-1}|\eta_{l-1}\eta_{k-l}\Psi(m)$; summing over l proves the claim. □

J.4. **Triangular truncation from ℓ^2 into ℓ^r.** Let G_r be $c_{00}(\Sigma,K)$ with the norm $\|\cdot\|_r$; its completion is $\ell^r(\Sigma,\widehat{K})$. Let δ_t be the unit vectors, and for $g_1,\dots,g_k \in c_{00}(\Sigma,K)$ let $\mathcal{P}^{(k)}(g_1\otimes\cdots\otimes g_k) = \sum_{t_1<\cdots<t_k} g_1(t_1)\cdots g_k(t_k)\,\delta_{t_1}\otimes\cdots\otimes\delta_{t_k}$, the k-fold triangular truncation for the family (δ_t).

Lemma J.4 (flat vectors)**.** *If $g_1,\dots,g_k \in c_{00}(\Sigma,K)$ satisfy $\|g_a\|_\infty \le 1$ and have nonempty supports T_a with m_a elements, then $\|\mathcal{P}^{(k)}(g_1\otimes\cdots\otimes g_k)\|_{\otimes_\pi^k G_r} \le \chi(T_1,\dots,T_k) \le \eta_k\Psi(m_1,\dots,m_k)$.*

Proof. If boxes $J_1^i\times\cdots\times J_k^i$ partition $\Xi(T_1,\dots,T_k)$, then $\mathcal{P}^{(k)}(g_1\otimes\cdots\otimes g_k) = \sum_i (g_1 1_{J_1^i})\otimes\cdots\otimes(g_k 1_{J_k^i})$, as both sides have the coefficient $g_1(t_1)\cdots g_k(t_k)$ at each chain t and no other terms, and $\|g_a 1_J\|_r \le |J|^{1/r}$. Theorem J.3 gives the second inequality. □

Theorem J.5 (triangular truncation)**.** *Let $k\ge 1$, $\tau = |\vartheta|$, $\gamma = 1/2 - 1/r$ and*

$$C_{k,r} = \eta_k|\mathrm{Mat}_k|\Big(\frac{2\tau^{-2}}{1-\tau^\gamma}\Big)^{\lfloor k/2\rfloor}\Big(\frac{\tau^{-2/r}}{1-\tau^{2\gamma}}\Big)^{k-2\lfloor k/2\rfloor}.$$

For $x_1,\dots,x_k \in c_{00}(\Sigma,K)$,

$$\|\mathcal{P}^{(k)}(x_1\otimes\cdots\otimes x_k)\|_{\otimes_\pi^k G_r} \le C_{k,r}\tau^{-k}\prod_{a=1}^k \|x_a\|_2. \tag{J.4}$$

Consequently $\mathcal{P}^{(k)}$ extends to a bounded linear map from $\otimes_\pi^k\ell^2(\Sigma,K)$ into $\widehat{\otimes}_\pi^k\ell^r(\Sigma,\widehat{K})$ of norm at most $C_{k,r}\tau^{-k}$.

Proof. Step 1 (bands). For $x\in c_{00}(\Sigma,K)$ and $i\in\mathbb{Z}$ let $\Sigma_i(x) = \{t : \tau^{i+1} < |x(t)| \le \tau^i\}$, $m_i(x) = |\Sigma_i(x)|$ and $x^{(i)} = x1_{\Sigma_i(x)}$. Then $x = \sum_i x^{(i)}$ is a finite sum, $x^{(i)} = \vartheta^i g$ with $\|g\|_\infty \le 1$, and $\sum_i \tau^{2i} m_i(x) \le \tau^{-2}\|x\|_2^2$. Expanding each x_a in bands and applying Lemma J.4 and (F2), the left side of (J.4) is at most

$$\eta_k \sum_{\mathfrak{m}\in\mathrm{Mat}_k} \prod_{\{a,b\}\in\mathfrak{m}} \Phi_2(x_a,x_b) \prod_{a\text{ unmatched}} \Phi_1(x_a),$$

where $\Phi_1(x) = \sum_i \tau^i m_i(x)^{1/r}$ and $\Phi_2(x,y) = \sum_{i,l}\tau^{i+l}\psi(m_i(x),m_l(y))$, summed over the indices with $m_i(x), m_l(y)\ge 1$; for fixed $\mathfrak{m}$ the sum over the bands factors, as each slot occurs in one factor of $\Psi_{\mathfrak{m}}$.

Step 2 (single sums). If some $x_a = 0$ there is nothing to prove. Replacing x_a by $\vartheta^c x_a$, $c\in\mathbb{Z}$, multiplies both sides of (J.4) by τ^c, so we may assume $\tau < \|x_a\|_2 \le 1$ for every a. If $\|x\|_2 \le 1$, then $m_i(x)\ge 1$ forces $\tau^{i+1} < 1$, so $i\ge 0$, and $\tau^{2i}m_i(x) \le \tau^{-2}$; hence $\tau^i m_i(x)^{1/r} = (\tau^{2i}m_i(x))^{1/r}\tau^{2\gamma i} \le \tau^{-2/r}\tau^{2\gamma i}$ and $\Phi_1(x) \le \tau^{-2/r}/(1-\tau^{2\gamma})$.

Step 3 (pair sums). Let $\|x\|_2, \|y\|_2 \le 1$, $\alpha_i = \tau^i m_i(x)^{1/2}$ and $\alpha'_l = \tau^l m_l(y)^{1/2}$, so that $\sum_i \alpha_i^2$ and $\sum_l \alpha_l'^2$ are at most τ^{-2}. For $b', b'' > 0$, $\psi(b', b'') = (b'b'')^{1/2} \min(b'/b'', b''/b')^{\gamma} \le (b'b'')^{1/2} \min(b'/b'', b''/b')^{\gamma/2}$, as the minimum is at most 1. With $m_i(x)/m_l(y) = (\alpha_i/\alpha'_l)^2 \tau^{2(l-i)}$, the term of $\Phi_2(x,y)$ at (i,l) is at most $\alpha_i^{1+\gamma} \alpha_l'^{1-\gamma} \tau^{\gamma(l-i)}$ if $l \ge i$ and at most $\alpha_i^{1-\gamma} \alpha_l'^{1+\gamma} \tau^{\gamma(i-l)}$ if $l < i$. For fixed $l - i$, Hölder's inequality with exponents $2/(1 \pm \gamma)$ bounds the sum of these terms by $\tau^{-2} \tau^{\gamma|l-i|}$, so $\Phi_2(x,y) \le \tau^{-2}(1+\tau^{\gamma})/(1-\tau^{\gamma}) \le 2\tau^{-2}/(1-\tau^{\gamma})$.

Step 4 (conclusion). Each $\mathfrak{m} \in \mathrm{Mat}_k$ has $\lfloor k/2 \rfloor$ pairs and $k - 2\lfloor k/2 \rfloor$ unmatched slots, so the left side of (J.4) is at most $C_{k,r} < C_{k,r} \tau^{-k} \prod_a \|x_a\|_2$. As $c_{00}(\Sigma, K)$ is dense in $\ell^2(\Sigma, K)$, the bounded k-linear map $(x_1, \dots, x_k) \mapsto \mathcal{P}^{(k)}(x_1 \otimes \cdots \otimes x_k)$ extends to $\ell^2(\Sigma, K)^k$ with values in $\widehat{\otimes}_\pi^k G_r$; its linearization, followed by the contraction induced by the isometry $G_r \to \ell^r(\Sigma, \widehat{K})$, is the asserted extension. □

J.5. **Relative ordering splittings.**

Proof of Proposition 5.8. Let ν be an r-norm on D with $a\nu \le \|\cdot\| \le b\nu$, $0 < a \le b$. As $\nu(\sum_t z_t)^r \le \sum_t \nu(z_t)^r$ for finite families, the left side of (5.2) is at most $b(\sum_t |\xi_t|^r \nu(y_t)^r)^{1/r} \le (b/a)(\sum_t |\xi_t|^r)^{1/r} \max_t \|y_t\|$. Conversely, assume (5.2) with constant C, and let $\nu(y)$ be the infimum of $(\sum_i \|y_i\|^r)^{1/r}$ over the finite decompositions $y = \sum_i y_i$. Then $\nu \le \|\cdot\|$; $\nu(\xi y) = |\xi| \nu(y)$ for $\xi \in K^\times$, as $y_i \mapsto \xi y_i$ matches the decompositions of y and ξy; and $\nu(y+z)^r \le \nu(y)^r + \nu(z)^r$, by concatenating decompositions. Let $y = \sum_i y_i$ with all $y_i \ne 0$, and $\xi_i = \vartheta^{c_i}$ with c_i the largest integer such that $|\vartheta|^{c_i} \ge \|y_i\|$. Then $\|\xi_i^{-1} y_i\| \le 1$ and $|\xi_i| < \|y_i\|/|\vartheta|$, so (5.2) gives $\|y\| \le C(\sum_i |\xi_i|^r)^{1/r} \le (C/|\vartheta|)(\sum_i \|y_i\|^r)^{1/r}$, hence $\|y\| \le (C/|\vartheta|)\nu(y)$. So ν vanishes only at 0; it is a norm, since $r \ge 1$, and an equivalent r-norm. □

Proof of Theorem 5.9. Let $C_e = \sup_t \|e_t\|$, let C_2 be the constant of ℓ^2-domination, fix a degree $k \ge 1$ and $r \in (2, \infty)$, and let G_r, δ_t, τ and $C_{k,r}$ be as in Subsection J.4. In Steps 1 and 3 we assume, as in (2), that D has an r-norm ν with $c\nu \le \|\cdot\| \le c'\nu$.

Step 1 (factorization). For $u \in \mathcal{L}(E, D)$ let $\tilde{u} \colon G_r \to D$, $\tilde{u}(\xi) = \sum_t \xi(t)\, u e_t$. Then $\nu(\tilde{u}\xi)^r \le \sum_t |\xi(t)|^r \nu(u e_t)^r \le (C_e \|u\|/c)^r \|\xi\|_r^r$, so $\|\tilde{u}\| \le (c'/c) C_e \|u\|$; $\tilde{u}$ is linear in u, and $\tilde{u}\delta_t = u e_t$. For $x \in V$, $\tilde{u}(wx) = \sum_t x(t)\, u e_t = ux$, so the extension $\tilde{u} \colon \ell^r(\Sigma, \widehat{K}) \to \widehat{D}$ satisfies $\tilde{u} \circ w = u$, as V is dense; with Step 2, $(\ell^r(\Sigma, \widehat{K}), w, u \mapsto \tilde{u})$ is a source-side relative splitting of degree k (Definition 6.12).

Step 2 (relative section). For $x \in V$ the coordinate vector $wx = (x(t))_t$ satisfies $\|wx\|_r \le \|wx\|_2 \le C_2 \|x\|$, so w extends to a bounded map $w \colon E \to \ell^r(\Sigma, \widehat{K})$. By Theorem J.5 the k-linear map $(x_1, \dots, x_k) \mapsto \mathcal{P}^{(k)}(wx_1 \otimes \cdots \otimes wx_k)$ from V^k to $\otimes_\pi^k G_r$ has norm at most $C_{k,r} \tau^{-k} C_2^k$, so it extends to E^k with values in $\widehat{\otimes}_\pi^k G_r$. Let L be its linearization and $s = L \circ \mathrm{alt}_E^k$; then $\|s\| \le k!\, C_{k,r} \tau^{-k} C_2^k$ and, for $x_1, \dots, x_k \in V$,

$$s(x_1 \wedge \cdots \wedge x_k) = \sum_{t_1 < \cdots < t_k} \det\big(x_b(t_a)\big)_{a,b}\, \delta_{t_1} \otimes \cdots \otimes \delta_{t_k}. \tag{J.5}$$

Expanding $wx_b = \sum_t x_b(t) \delta_t$, dropping repeated indices and sorting shows that $\widehat{\wedge}^k$ maps the right side to $wx_1 \wedge \cdots \wedge wx_k$. The wedges $x_1 \wedge \cdots \wedge x_k$ with $x_a \in V$ span a dense subspace of $\Lambda_\pi^k E$, so $\widehat{\wedge}^k \circ s = \Lambda_\pi^k w$, with s read in $\widehat{\otimes}_\pi^k \ell^r(\Sigma, \widehat{K})$. Hence w is S^k-split, which proves (1). If $E = V$, s takes values in $\otimes_\pi^k G_r$.

Step 3 (lift). Put $Q(u_1, \dots, u_k) = \widehat{\wedge}_D^k \circ (\tilde{u}_1 \otimes \cdots \otimes \tilde{u}_k) \circ s$, with $\tilde{u}_1 \otimes \cdots \otimes \tilde{u}_k$ extended to $\widehat{\otimes}_\pi^k G_r$. Then Q is k-linear, $\|Q(u_1, \dots, u_k)\| \le \|s\|\, ((c'/c) C_e)^k \prod_a \|u_a\|$, and by (J.5), for $x_1, \dots, x_k \in V$,

Q is given by the *canonical lift formula*

$$Q(u_1,\dots,u_k)(x_1\wedge\cdots\wedge x_k)=\sum_{t_1<\cdots<t_k}\det\bigl(x_b(t_a)\bigr)_{a,b}\,u_1e_{t_1}\wedge\cdots\wedge u_ke_{t_k}. \tag{J.6}$$

For $u_1=\cdots=u_k=u$ the right side of (J.6) is $ux_1\wedge\cdots\wedge ux_k$, by expanding $ux_b=\sum_t x_b(t)\,ue_t$; by density, $Q(u,\dots,u)=\Lambda^k_\pi u$ as maps into $\widehat\Lambda^k_\pi D$. So (E,D) is good for Banach values in degree k by Lemma 3.4. If $E=V$, then Q takes values in $\mathcal{L}(\Lambda^k_\pi E,\Lambda^k_\pi D)$ by Step 2, and (E,D) is good in degree k by the same lemma. □

Appendix K. Forcing over Laurent series fields

Throughout this appendix K satisfies (L); from Section K.3 on, K is dense in $\kappa((X))$, so $\widehat K=\kappa((X))$ and $|\vartheta|=\varrho$. The results of Sections 7 and 8 proved from Section K.3 on, except Lemma 7.14, which assumes this density, depend only on K as a normed field; Lemma 7.1 extends them to every K satisfying (L). Theorem 8.9(3) and the last assertion of Example 8.10(3) are proved in Section K.8 over every nontrivially normed field of characteristic 2. Lemma K.2 holds, with the same proof, for every field κ of characteristic p in place of the finite field of (L); the proof of Proposition 10.13 uses it in that generality.

K.1. **Coefficients and costs.** For a normed space E with $c_{00}(\mathbb{N},K)\subseteq E\subseteq\ell^\infty(\mathbb{N},K)$ and $\|x\|_\infty\le\|x\|$, the determinant array Ω of §B.1, with $S=\mathbb{N}$, satisfies $|\Omega_z(c)|\le\|z\|$, as in the proof of Lemma C.2, so it extends to $\widehat\Lambda^k_\pi E$ with values in $\ell^\infty(\mathbb{N}^k,\kappa((X)))$. Likewise $z(t,t')$ denotes the continuous extension to $E\mathbin{\widehat\otimes_\pi}E$ of the entry $x(t)y(t')$ of $x\otimes y$.

For a matrix Z over κ with rows indexed by a finite set R and columns by a finite set R', $\mathrm{cost}(\mathsf{Z})$ is the least value of $\sum_i|\mathrm{supp}\,g_i|^{1/2}|\mathrm{supp}\,h_i|^{1/2}$ over the representations $\mathsf{Z}=\sum_i g_ih_i^{\mathsf T}$ with $g_i\in\kappa^R$ and $h_i\in\kappa^{R'}$; it is subadditive.

Lemma K.1 (tournament lemma)**.** *Let Z be an $m\times m$ matrix over κ such that every off-diagonal entry of $\mathsf{Z}-\mathsf{Z}^{\mathsf T}$ is nonzero. Then $\mathrm{cost}(\mathsf{Z})\ge m\ln m/(1+\ln(2|\kappa|))$.*

Proof. Step 1 (parts). Let $\mathsf{Z}=\sum_i g_ih_i^{\mathsf T}$ with all g_i,h_i nonzero, let $\mathcal{V}_i$ be the union of the supports of g_i and h_i, and for $t\in\mathcal{V}_i$ let the part $\mathcal{V}_i(t)$ be the set of points of $\mathcal{V}_i$ at which (g_i,h_i) is a multiple of $(g_i(t),h_i(t))$. The (t,t') entry $g_i(t)h_i(t')-h_i(t)g_i(t')$ of $g_ih_i^{\mathsf T}-h_ig_i^{\mathsf T}$ is nonzero exactly when $t,t'\in\mathcal{V}_i$ lie in different parts; so every pair $t\ne t'$ lies in some $\mathcal{V}_i$, in different parts.

Step 2 (deletion). Following Hansel's deletion argument [11] with weighted choices, choose independently for each i one part of $\mathcal{V}_i$, with probability proportional to its size, and delete the points of $\mathcal{V}_i$ outside it. At most one point survives, since two survivors would lie in different parts of some $\mathcal{V}_i$. The point t survives with probability $\exp(-L_t)$, where $L_t=\sum_{i:t\in\mathcal{V}_i}\ln(|\mathcal{V}_i|/|\mathcal{V}_i(t)|)$; so $\sum_t\exp(-L_t)\le1$, and convexity of the exponential gives $\sum_t L_t\ge m\ln m$. Moreover $\sum_t L_t=\sum_i\mathrm{ent}_i$ with $\mathrm{ent}_i=\sum_{t\in\mathcal{V}_i}\ln(|\mathcal{V}_i|/|\mathcal{V}_i(t)|)$.

Step 3 (one term). Exchanging g_i and h_i, which does not change the parts, we may assume that the sizes N_1 and N_2 of their supports satisfy $N_1\le N_2$, so $N=|\mathcal{V}_i|\le2N_2$. It remains to show $\mathrm{ent}_i\le(1+\ln(2|\kappa|))\sqrt{N_1N_2}$. The points of $\mathcal{V}_i$ outside the support of g_i form one part, which contributes $(N-N_1)\ln(N/(N-N_1))\le N_1$. The other parts divide the support of g_i by the value of h_i/g_i into at most $|\kappa|$ sets; since a distribution on at most $|\kappa|$ points has entropy at most $\ln|\kappa|$, they contribute at most $N_1\ln(N/N_1)+N_1\ln|\kappa|$. Hence

$$\mathrm{ent}_i\le N_1\Bigl(1+\ln\frac{2|\kappa|N_2}{N_1}\Bigr)\le(1+\ln(2|\kappa|))\sqrt{N_1N_2},$$

the second inequality with $x = N_2/N_1$, since $x^{-1/2}(1+\ln(2|\kappa|x))$ is decreasing for $x \geq 1$, as $\ln(2|\kappa|) > 1$. □

Lemma K.2 (constant coefficients)**.** *Let R, R' be finite sets and $r \in \{2, \infty\}$, let z belong to the completion of $\ell^r(R,K) \otimes_\pi \ell^r(R',K)$, and let $\operatorname{coeff}_0 z \in \kappa^{R\times R'}$ be the matrix of constant coefficients of its entries.*

(1) *If $r = 2$, then $\operatorname{cost}(\operatorname{coeff}_0 z) \leq \|z\|/(1-\varrho^2)$.*
(2) *If $r = \infty$, then $\operatorname{rank}(\operatorname{coeff}_0 z) \leq M_\varrho \|z\|$.*

Proof. For $x \in \ell^r(R,K)$ and $n \in \mathbb{Z}$ let $x_{[n]} \in \kappa^R$ have entries $\operatorname{coeff}_n(x(t))$. Then $\operatorname{coeff}_0(x\otimes y) = \sum_n x_{[n]} y_{[-n]}^{\mathsf{T}}$, a finite sum. (1) The vector $x_{[n]}$ is supported in $O_n(x) = \{t : x(t) \neq 0,\ \operatorname{ord} x(t) \leq n\}$, and $\sum_n |O_n(x)|\varrho^{2n} = \|x\|^2/(1-\varrho^2)$. By the Cauchy–Schwarz inequality, $\operatorname{cost}\operatorname{coeff}_0(x\otimes y) \leq \sum_n |O_n(x)|^{1/2}|O_{-n}(y)|^{1/2} \leq \|x\|\|y\|/(1-\varrho^2)$. (2) If $\|x\| = \varrho^\nu$ and $\|y\| = \varrho^{\nu'}$, only the $l+1$ values $\nu \leq n \leq -\nu'$ contribute, where $l = -\nu-\nu'$, and none if $l < 0$. Since $l+1 \leq M_\varrho \varrho^{-l}$, this gives $\operatorname{rank}\operatorname{coeff}_0(x\otimes y) \leq M_\varrho \|x\|\|y\|$.

For general z, write $z = \sum_j x_j \otimes y_j$ as an absolutely convergent series with $\sum_j \|x_j\|\|y_j\|$ as close to $\|z\|$ as we like, by approximating z quickly with algebraic tensors and decomposing the first approximation and the successive differences nearly optimally. The terms with $\|x_j\|\|y_j\| < 1$ have entries of absolute value at most ϱ, since $|x(t)| \leq \|x\|$ and absolute values lie in $\varrho^{\mathbb{Z}}$; by the nonarchimedean inequality so does their sum, which therefore does not change coeff_0. The finitely many remaining terms give the bounds with any number larger than $\|z\|$ in place of $\|z\|$, by subadditivity of cost and rank, and hence with $\|z\|$ itself. □

Lemma K.3 (two-slot slice)**.** *Let E be a normed space with $c_{00}(\mathbb{N},K) \subseteq E \subseteq \ell^\infty(\mathbb{N},K)$ and $\|x\|_\infty \leq \|x\|$, let $\gamma_1 \neq \gamma_2$ be slots in $\{1,\dots,k\}$, and fix $c_\gamma \in \mathbb{N}$ for the remaining slots γ. There is a bounded linear $\Upsilon\colon \widehat{\Lambda}^k_\pi E \to E\widehat{\otimes}_\pi E$ of norm at most $k(k-1)$ such that $(\Upsilon z)(t,t') = \Omega_z(c)$, where c has t in slot γ_1, t' in slot γ_2 and c_γ in each slot γ.*

Proof. For $x_1,\dots,x_k \in E$, expansion along the rows γ_1,γ_2 gives

$$\det(x_b(c_a))_{a,b} = \sum_{b\neq b'} x_b(c_{\gamma_1}) x_{b'}(c_{\gamma_2}) \operatorname*{det}_{bb'}(x),$$

where $\det_{bb'}(x)$, the sum of $\operatorname{sgn}(\sigma)\prod_{a\neq\gamma_1,\gamma_2} x_{\sigma(a)}(c_a)$ over the permutations with $\sigma(\gamma_1) = b$ and $\sigma(\gamma_2) = b'$, does not involve $c_{\gamma_1}, c_{\gamma_2}$, and $|\det_{bb'}(x)| \leq \prod_{b''\neq b,b'} \|x_{b''}\|$ by the nonarchimedean inequality. So the k-linear map $x \mapsto \sum_{b\neq b'} \det_{bb'}(x)\, x_b \otimes x_{b'}$ into $E\otimes_\pi E$ has norm at most $k(k-1)$, and its values have the stated determinants as entries. It is alternating, since these entries vanish when two x_b coincide and a tensor in $E\otimes E$ with all entries zero is zero. Proposition 3.1 gives Υ on $\Lambda^k_\pi E$, and continuity extends it and the entry formula to the completion. □

K.2. The ℓ^r threshold.

Proof of Theorem 7.7. Step 1 (a coefficient form). For finitely supported $a_1,\dots,a_k,x_1,\dots,x_k$ with entries in $\mathbb{F}_p$ and $c \in \mathbb{N}^k$ put

$$\Xi(a;x)(c) = \operatorname{coeff}_0\big(\Omega_{Q(a_1,\dots,a_k)(x_1\wedge\cdots\wedge x_k)}(c)\big) \in \kappa,$$

which is $\mathbb{F}_p$-multilinear in (a,x) and alternating in x and in c. For such $b_1,\dots,b_k,x_1,\dots,x_k$ and $\lambda \in K^k$, with $a_\lambda = \sum_i \lambda_i b_i$, the identity $Q(a_\lambda,\dots,a_\lambda)(x_1\wedge\cdots\wedge x_k) = (a_\lambda x_1)\wedge\cdots\wedge(a_\lambda x_k)$ is

an equality of polynomials in λ with coefficients in $\widehat{\Lambda}^k_\pi E$. Comparing the coefficients of $\lambda_1 \cdots \lambda_k$ (Lemma B.6) and applying $\mathrm{coeff}_0 \circ \Omega$ gives

$$\sum_{\sigma \in S_k} \Xi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x)(c) = \sum_{\sigma \in S_k} \det\big((b_{\sigma(j)} x_j)(c_i)\big)_{i,j}, \tag{K.1}$$

since these determinants lie in $\mathbb{F}_p$. The proofs of Lemmas B.10, B.11 and B.13 use only multilinearity, alternation and (K.1) of the map Ψ of Appendix B, and the expansion (B.1). For Ξ the coefficient function $T(d; y; c) = \Xi(e_{d_1}, \dots, e_{d_k}; e_{y_1}, \dots, e_{y_k})(c)$ takes values in the finite field κ and (B.1) follows from $\mathbb{F}_p$-multilinearity, the inputs there having entries in $\{0, \pm 1\}$; so those three lemmas hold for Ξ. Since $k! = 0$ in κ, Lemma B.13 shows that the cluster value χ is not constant; as adjacent rank exchanges generate S_k, there are τ and ℓ with $\delta = \chi(\tau) - \chi(\tau') \neq 0$, where $\tau' = (\ell\ \ell{+}1) \circ \tau$ as in Lemma B.12. Put $\gamma_1 = \tau^{-1}(\ell)$ and $\gamma_2 = \tau^{-1}(\ell + 1)$.

Step 2 (the quantitative bound). Fix $m \geq 1$, and let $a = (a_1, \dots, a_k)$ and $x = (x_1, \dots, x_k)$ be the inputs in the proof of Lemma B.12, with γ_1, γ_2 as the slots α, β there. By Lemma B.11, the $m \times m$ matrix Z of the values $\Xi(a; x)(c)$ with $A_n(1)$ in slot γ_1, $B_{n'}(1)$ in slot γ_2 and $C_\gamma(1)$ in each other slot γ is $\mathsf{Z} = \chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}} + \delta U$. All entries of a and x lie in $\{0, \pm 1\}$, and x_{γ_1}, x_{γ_2} have $4m$ nonzero entries and the other x_γ four, so $\zeta = Q(a_1, \dots, a_k)(x_1 \wedge \cdots \wedge x_k)$ has $\|\zeta\| \leq \|Q\|(4m)^{2/r} 4^{(k-2)/r}$. The slice Υ of Lemma K.3 for the slots γ_1, γ_2 and $c_\gamma = C_\gamma(1)$, followed by restriction of both tensor factors to the coordinates $\{A_n(1)\}$ and $\{B_{n'}(1)\}$, gives z with $\|z\| \leq k(k-1)\|\zeta\|$ and $\mathrm{coeff}_0 z = \mathsf{Z}$. If $r = 2$, Lemma K.2(1) gives $\mathrm{cost}(\mathsf{Z}) \leq k(k-1)2^k m \|Q\|/(1 - \varrho^2)$, while $\mathsf{Z} - \mathsf{Z}^{\mathsf{T}} = \delta(U - U^{\mathsf{T}})$ and Lemma K.1 give $\mathrm{cost}(\mathsf{Z}) \geq m \ln m/(1 + \ln(2|\kappa|))$. If $r > 2$, the ℓ^r norms dominate the supremum norms, and Lemma K.2(2) gives $m - 1 \leq \mathrm{rank}\, \mathsf{Z} \leq M_\varrho k(k-1)\|\zeta\|$, the first inequality since δU is invertible. This proves (7.1), which fails for large m.

Step 3 (lifts and splittings). A lift Q_0 as in Lemma 3.4, for Banach values, would give such a Q, namely $Q(a_1, \dots, a_k) = Q_0(\mu_{a_1}, \dots, \mu_{a_k})$, since $\|\mu_a\| \leq \|a\|_\infty$. Lemma 3.4, Theorem 3.2(1) and the dichotomy of Proposition 2.5 give the assertion about $A^k_{E,E;F}$, Proposition 3.5 the failure of the splittings, and Corollary 5.3 the last assertion, since an equivalent nonarchimedean norm on $\widehat{\Lambda}^k_\pi E$ would make $A^k_{E,E;\widehat{\Lambda}^k_\pi E}$ analytic. □

Proof of Corollary 7.8. Theorem 5.6, with the examples following it, and Theorem 7.7. □

Proof of Corollary 7.9. The positive cases are Theorem 5.6. In the negative cases, repeat the proof of Theorem 7.7 with $Q(a_1, \dots, a_k) = Q_0(\mu_{a_1}, \dots, \mu_{a_k})$ for a lift Q_0 on $\mathcal{L}(E, D)$ as in Lemma 3.4, with $\widehat{\Lambda}^k_\pi D$ in place of $\widehat{\Lambda}^k_\pi E$ and the slice formed in $D \widehat{\otimes}_\pi D$. If $r_1 \leq r_2$, then $\|\mu_a\| \leq \|a\|_\infty$ and $\|\zeta\| \leq \|Q_0\|(4m)^{2/r_1} 4^{(k-2)/r_1}$. If $r_1 > r_2$, Hölder's inequality gives $\|\mu_a\| \leq \|a\|_{r_0}$ with $1/r_0 = 1/r_2 - 1/r_1$; as a_{γ_1}, a_{γ_2} have $2m$ nonzero entries and the other a_γ two, $\|\zeta\| \leq \|Q_0\|(2m)^{2/r_0} 2^{(k-2)/r_0} (4m)^{2/r_1} 4^{(k-2)/r_1}$. In both cases $\|\zeta\| \leq 4^k \|Q_0\| m^{2/\min(r_1, r_2)}$. If $r_2 = 2 \leq r_1$, Lemmas K.2(1) and K.1 give $m \ln m/(1 + \ln(2|\kappa|)) \leq k(k-1)4^k m \|Q_0\|/(1 - \varrho^2)$; if $r_1, r_2 > 2$, Lemma K.2(2) gives $m - 1 \leq M_\varrho k(k-1)4^k \|Q_0\| m^{2/\min(r_1, r_2)}$. Both fail for large m. □

K.3. **A trace norm over $\kappa((X))$.** Let $\mathcal{O} = \kappa[[X]]$, the closed unit ball of $\kappa((X))$. For finite sets R, R' we identify $z \in \kappa((X))^{R \times R'}$ with $\sum_{t,t'} z(t, t')\, e_t \otimes e_{t'}$; its projective norm $\|z\|_\pi$ over $\kappa((X))$ for the supremum norms on $\kappa((X))^R$ and $\kappa((X))^{R'}$ is the infimum of $\sum_a \|g_a\|_\infty \|h_a\|_\infty$ over the finite representations $z = \sum_a g_a h_a^{\mathsf{T}}$. Fix enumerations $t_1, t_2, \dots$ of R and $t'_1, t'_2, \dots$ of R'. Since $\mathcal{O}$ is a principal ideal domain whose nonzero elements are units times powers of X,

the elementary divisor theorem, applied to $X^c z \in \mathcal{O}^{R\times R'}$ for a large integer c, writes every z as

$$z = U_1 \Delta U_2, \qquad U_1 \in GL_R(\mathcal{O}), \quad U_2 \in GL_{R'}(\mathcal{O}), \tag{K.2}$$

where Δ has the entries $X^{d_1}, \dots, X^{d_{\operatorname{rank} z}}$, with integers $d_1 \le \dots \le d_{\operatorname{rank} z}$, at the positions $(t_1, t'_1), \dots, (t_{\operatorname{rank} z}, t'_{\operatorname{rank} z})$ and zeros elsewhere. For $l \in \mathbb{Z}$ let $\operatorname{rank}_l(z)$ be the number of indices i with $d_i \le l$.

Lemma K.4 (a trace norm)**.** *Let R, R' be finite sets and $z \in \kappa((X))^{R\times R'}$.*

1. *The numbers* $\operatorname{rank}_l(z)$ *do not depend on the factorization* (K.2).
2. (Level formula.) $\|z\|_\pi = \sum_i \varrho^{d_i} = (1-\varrho)\sum_{l\in\mathbb{Z}} \varrho^l \operatorname{rank}_l(z)$.
3. (Submatrices.) *If z_0 is the submatrix of z with rows in $S \subseteq R$ and columns in $S' \subseteq R'$, then* $\operatorname{rank}_l(z_0) \le \operatorname{rank}_l(z)$ *for every $l \in \mathbb{Z}$.*
4. (Rank at a level.) *If $|R| = |R'|$, $e \in \mathbb{Z}$, and $z = X^e w$ for a matrix $w \in \mathcal{O}^{R\times R'}$ whose reduction modulo X is invertible over κ, then* $\operatorname{rank}_e(z) = |R|$.

Proof. For $l \ge 0$ let $\mathcal{O}_l = \mathcal{O}/X^{l+1}\mathcal{O}$, a local ring with residue field κ, and let $\operatorname{gen}(Y)$ be the least number of generators of a finitely generated $\mathcal{O}_l$-module Y. (a) If an $\mathcal{O}_l$-module is generated by N elements, so is each submodule: the module is a quotient of $\mathcal{O}^N$, and the preimage of the submodule is free of rank at most N, since $\mathcal{O}$ is a principal ideal domain. (b) $\operatorname{gen}(Y) = \dim_\kappa Y/XY$ by Nakayama's lemma; so $\bigoplus_{i=1}^N \mathcal{O}_l/X^{c_i}\mathcal{O}_l$ needs exactly N generators when every $c_i \ge 1$.

Step 1 (generators). Let $w \in \mathcal{O}^{R\times R'}$, $l \ge 0$, and let $\operatorname{Col}_l(w) \subseteq \mathcal{O}_l^R$ be the submodule generated by the columns of w reduced modulo X^{l+1}. If $w = U_1\Delta U_2$ is a factorization (K.2), then $\Delta = U_1^{-1} w U_2^{-1}$ has entries in $\mathcal{O}$, so all $d_i \ge 0$; we claim $\operatorname{gen}(\operatorname{Col}_l(w)) = \operatorname{rank}_l(w)$. The columns of w and of $U_1\Delta$ generate the same submodule, since U_2 is invertible over $\mathcal{O}$; multiplication by U_1 is an automorphism of $\mathcal{O}_l^R$; and the columns of Δ generate a module isomorphic to $\bigoplus_{d_i \le l} \mathcal{O}_l/X^{l+1-d_i}\mathcal{O}_l$, which needs exactly $\operatorname{rank}_l(w)$ generators by (b).

Step 2 (item (1)). Choose $c \in \mathbb{Z}$ with $X^c z \in \mathcal{O}^{R\times R'}$. Multiplying (K.2) by X^c gives a factorization of $X^c z$ with the exponents $d_i + c \ge 0$. So $\operatorname{rank}_l(z)$ is 0 if $l + c < 0$ and $\operatorname{gen}(\operatorname{Col}_{l+c}(X^c z))$ by Step 1 otherwise; neither depends on the factorization.

Step 3 (item (2)). The two right sides agree, since $(1-\varrho)\sum_{l\ge d} \varrho^l = \varrho^d$. Writing z as the sum over i of X^{d_i} times the product of the column t_i of U_1 and the row t'_i of U_2, vectors over $\mathcal{O}$, gives a representation of cost at most $\sum_i \varrho^{d_i}$. Conversely, let $z \ne 0$ and $z = \sum_{a=1}^N g_a h_a^{\mathsf{T}}$ with all g_a, h_a nonzero. Since $|\kappa((X))^\times| = \varrho^{\mathbb{Z}}$, $g_a = X^{\nu_a} g'_a$ and $h_a = X^{\nu'_a} h'_a$ with vectors g'_a, h'_a over $\mathcal{O}$ of supremum norm 1; with $m_a = \nu_a + \nu'_a$ the cost is $\sum_a \varrho^{m_a}$. Multiplying z and every g_a by X^c multiplies the cost and $\sum_i \varrho^{d_i}$ by ϱ^c, so we may assume every $m_a \ge 0$; then $z \in \mathcal{O}^{R\times R'}$. For $l \ge 0$, $z \equiv \sum_{m_a \le l} X^{m_a} g'_a h_a'^{\mathsf{T}}$ modulo X^{l+1}, so $\operatorname{Col}_l(z)$ lies in the submodule generated by the reductions of the g'_a with $m_a \le l$, and (a) with Step 1 gives $\operatorname{rank}_l(z) \le N_l = |\{a : m_a \le l\}|$; for $l < 0$ both are 0. Hence $\sum_a \varrho^{m_a} = (1-\varrho)\sum_l \varrho^l N_l \ge (1-\varrho)\sum_l \varrho^l \operatorname{rank}_l(z)$, and the infimum over representations gives the lower bound.

Step 4 (item (3)). Multiplying z and z_0 by the same X^c replaces l by $l + c$ on both sides, so we may assume $z \in \mathcal{O}^{R\times R'}$ and $l \ge 0$, both sides vanishing for $l < 0$. Deleting columns replaces $\operatorname{Col}_l(z)$ by a submodule, which needs no more generators by (a); deleting rows replaces it by its image under the coordinate projection $\mathcal{O}_l^R \to \mathcal{O}_l^S$, which is generated by the images of generators. Step 1 gives $\operatorname{rank}_l(z_0) \le \operatorname{rank}_l(z)$.

Step 5 (item (4)). Let $\Pi_0 \in \mathcal{O}^{R\times R'}$ have the entry 1 at each position (t_i, t'_i) and 0 elsewhere. The determinant of $\Pi_0^{-1} w$ reduces to a nonzero element of κ, so it is a unit and $\Pi_0^{-1} w \in GL_{R'}(\mathcal{O})$.

Then $z = I \cdot (X^e \Pi_0) \cdot \Pi_0^{-1} w$ is a factorization (K.2) with all $|R|$ exponents equal to e, so $\operatorname{rank}_e(z) = |R|$. □

K.4. **Coefficient functions of multiplier lifts.** Throughout this subsection G and D are sequence spaces (Definition 7.10) with $\|e_t\|_G \le 1$ and $\|y\|_\infty \le \|y\|_D$, Ω is the determinant array on $\widehat{\Lambda}^k_\pi D$ of Section K.1, and $\widetilde{Q}\colon c_{00}(\mathbb{N}, K)^k \to \mathcal{L}(\Lambda^k_\pi G, \widehat{\Lambda}^k_\pi D)$ is a *multiplier lift*: a k-linear map with

$$\widetilde{Q}(b, \dots, b)(x_1 \wedge \cdots \wedge x_k) = (bx_1) \wedge \cdots \wedge (bx_k) \qquad (b, x_1, \dots, x_k \in c_{00}(\mathbb{N}, K)) \tag{K.3}$$

and $\|\widetilde{Q}(e_{a_1}, \dots, e_{a_k})\| \le \varrho^{-l_0}$ for all $a \in \mathbb{N}^k$ and some integer $l_0 \ge 0$. For $l \in \mathbb{Z}$ and $a, y, c \in \mathbb{N}^k$ put

$$\Omega_{a;y}(c) = \Omega_{\widetilde{Q}(e_{a_1}, \dots, e_{a_k})(e_{y_1} \wedge \cdots \wedge e_{y_k})}(c), \qquad T_l(a; y; c) = \operatorname{coeff}_l \Omega_{a;y}(c) \in \kappa.$$

Since $|\Omega_z(c)| \le \|z\|$, $\|e_{y_1} \wedge \cdots \wedge e_{y_k}\| \le 1$ and the nonzero absolute values of $\kappa((X))$ lie in $\varrho^{\mathbb{Z}}$, a nonzero $\Omega_{a;y}(c)$ has order at least $-l_0$, so $T_l = 0$ for $l < -l_0$. For $g_1, \dots, g_k, h_1, \dots, h_k \in c_{00}(\mathbb{N}, K)$ and $c \in \mathbb{N}^k$, multilinearity gives the finite expansion

$$\Omega_{\widetilde{Q}(g_1, \dots, g_k)(h_1 \wedge \cdots \wedge h_k)}(c) = \sum_{a, y \in \mathbb{N}^k} \prod_{\gamma=1}^{k} g_\gamma(a_\gamma) h_\gamma(y_\gamma) \cdot \Omega_{a;y}(c). \tag{K.4}$$

Lemma K.5 (homogeneity)**.** *For every integer $l_1 \ge 0$ there is an infinite set $I \subseteq \mathbb{N}$ such that for every $l \le l_1$, $T_l(\mathbf{z}) = T_l(\mathbf{z}')$ whenever $\mathbf{z}, \mathbf{z}' \in I^{3k}$ have the same pattern in the sense of Appendix B.*

Proof. The map $\mathbf{z} \mapsto (T_l(\mathbf{z}))_{-l_0 \le l \le l_1}$ takes values in $\kappa^{l_0 + l_1 + 1}$, a finite set whose size is a power of p. Identifying it with a finite field of that size, Lemma B.10 with $N = 3k$ gives I; for $l < -l_0$, $T_l = 0$. □

For the rest of this subsection fix an integer $l_1 \ge 0$ and I as in Lemma K.5. As in Appendix B, clusters are the sets $C = \{C(1) < C(2) < C(3) < C(4)\}$ of four consecutive elements of I, disjoint clusters are compared by $C < C'$, and the operator and vector weights $o = (1, 0, -1, 0)$ and $w = (1, -1, 1, -1)$ satisfy (W), with o in place of s. Put $o_C = \sum_i o_i e_{C(i)}$ and $w_C = \sum_i w_i e_{C(i)}$, so that $o_C w_C = e_{C(1)} - e_{C(3)}$. For $l \le l_1$ and pairwise disjoint clusters $C_1, \dots, C_k$ put

$$\begin{aligned}\chi_l(C_1, \dots, C_k) &= \operatorname{coeff}_l \Omega_{\widetilde{Q}(o_{C_1}, \dots, o_{C_k})(w_{C_1} \wedge \cdots \wedge w_{C_k})}\big(C_1(1), \dots, C_k(1)\big) \\ &= \sum_{i, i' \in [4]^k} \prod_{\gamma=1}^{k} o_{i_\gamma} w_{i'_\gamma} \cdot T_l\big(C_1(i_1), \dots, C_k(i_k); C_1(i'_1), \dots, C_k(i'_k); C_1(1), \dots, C_k(1)\big),\end{aligned}$$

by (K.4), the weights being constants in $\{0, \pm 1\}$. As in Appendix B, the pattern of each $3k$-tuple in the sum is determined by (i, i') and the order pattern $\tau \in S_k$ of $(C_1, \dots, C_k)$, so by Lemma K.5 the value depends only on τ; we write it $\chi_l(\tau)$. Also $\chi_l = 0$ for $l < -l_0$.

Lemma K.6 (two-level cancellation)**.** *Let $\mathcal{C}_1, \dots, \mathcal{C}_k$ be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in $\bigcup_\gamma \mathcal{C}_\gamma$ are pairwise disjoint, and let $\operatorname{lev}^o_\gamma, \operatorname{lev}^w_\gamma\colon \mathcal{C}_\gamma \to \mathbb{Z}_{\ge 0}$ be functions. Put*

$$g_\gamma = \sum_{C \in \mathcal{C}_\gamma} \vartheta^{\operatorname{lev}^o_\gamma(C)} o_C, \qquad h_\gamma = \sum_{C \in \mathcal{C}_\gamma} \vartheta^{\operatorname{lev}^w_\gamma(C)} w_C \qquad (\gamma = 1, \dots, k).$$

Let $C_\gamma \in \mathcal{C}_\gamma$ for $\gamma = 1, \dots, k$, let $c = (C_1(1), \dots, C_k(1))$, and let $\lambda = \sum_\gamma \big(\mathrm{lev}^o_\gamma(C_\gamma) + \mathrm{lev}^w_\gamma(C_\gamma)\big)$. Then for every integer $l \le l_1$,

$$\mathrm{coeff}_l\, \Omega_{\widetilde{Q}(g_1,\dots,g_k)(h_1\wedge\cdots\wedge h_k)}(c) = \sum_{l'\ge 0} \mathrm{coeff}_{l'}(\vartheta^\lambda)\, \chi_{l-l'}(C_1, \dots, C_k),$$

a finite sum.

Proof. Step 1 (expansion). Expand by (K.4). As the clusters are disjoint and g_γ, h_γ are supported in the union of the clusters of $\mathcal{C}_\gamma$, a nonzero term has, in each slot γ, an operator index $a_\gamma = C^{\mathrm{op}}_\gamma(i_\gamma)$ and a vector index $y_\gamma = C^{\mathrm{vec}}_\gamma(i'_\gamma)$ with $C^{\mathrm{op}}_\gamma, C^{\mathrm{vec}}_\gamma \in \mathcal{C}_\gamma$, and the coefficient $\prod_\gamma o_{i_\gamma} w_{i'_\gamma} \cdot \vartheta^{\lambda'}$, where $\lambda' = \sum_\gamma \big(\mathrm{lev}^o_\gamma(C^{\mathrm{op}}_\gamma) + \mathrm{lev}^w_\gamma(C^{\mathrm{vec}}_\gamma)\big) \ge 0$ depends only on the *cluster assignment* $(C^{\mathrm{op}}_\gamma, C^{\mathrm{vec}}_\gamma)_\gamma$. Since $\vartheta^{\lambda'}$ has nonnegative order and $T_{l''} = 0$ for $l'' < -l_0$, $\mathrm{coeff}_l(\vartheta^{\lambda'}\Omega_{a;y}(c)) = \sum_{l'\ge 0} \mathrm{coeff}_{l'}(\vartheta^{\lambda'})\, T_{l-l'}(a; y; c)$, a finite sum.

Step 2 (output-free clusters). Group the terms by cluster assignment and by l'. Suppose an assignment has an occupied cluster C' that is none of $C_1, \dots, C_k$. Then C' belongs to exactly one family $\mathcal{C}_\gamma$, so only a_γ and y_γ can lie in C', and no output index does. Fix l' and all positions outside C', and sum over the positions inside C'. The factor $\mathrm{coeff}_{l'}(\vartheta^{\lambda'})$ and the weights of the other slots are common factors, and $T_{l-l'}$ is pattern-homogeneous on I since $l - l' \le l_1$. As C' consists of consecutive elements of I, cases (i)–(iii) of the proof of Lemma B.11 apply with $T_{l-l'}$ in place of T, and (W) makes the partial sum vanish.

Step 3 (the surviving assignment). In a surviving assignment every occupied cluster is an output cluster, and the only output cluster in $\mathcal{C}_\gamma$ is C_γ; so $C^{\mathrm{op}}_\gamma = C^{\mathrm{vec}}_\gamma = C_\gamma$ for every γ, $\lambda' = \lambda$, and the contribution is $\sum_{l'\ge 0} \mathrm{coeff}_{l'}(\vartheta^\lambda)\chi_{l-l'}(C_1, \dots, C_k)$. □

Lemma K.7 (the diagonal)**.** *For every integer $l \le l_1$, $\sum_{\tau\in S_k} \chi_l(\tau)$ equals 1 if $l = 0$ and 0 otherwise.*

Proof. Step 1 (polarization). Let $b_1, \dots, b_k, x_1, \dots, x_k \in c_{00}(\mathbb{N}, K)$ and $a_\alpha = \sum_i \alpha_i b_i$ for $\alpha \in K^k$. Both sides of (K.3) with $b = a_\alpha$ are polynomials in α, and comparing the coefficients of $\alpha_1 \cdots \alpha_k$ by Lemma B.6 gives

$$\sum_{\sigma\in S_k} \widetilde{Q}(b_{\sigma(1)}, \dots, b_{\sigma(k)})(x_1 \wedge \cdots \wedge x_k) = \sum_{\sigma\in S_k} (b_{\sigma(1)}x_1) \wedge \cdots \wedge (b_{\sigma(k)}x_k). \tag{K.5}$$

Step 2 (right side). Take clusters $C_1 < \cdots < C_k$, $b_\gamma = o_{C_\gamma}$, $x_\gamma = w_{C_\gamma}$ and $c = (C_1(1), \dots, C_k(1))$, and apply $\Omega_\cdot(c)$ to (K.5). On the right, a permutation $\sigma \ne \mathrm{id}$ has a slot γ with $\sigma(\gamma) \ne \gamma$, and $o_{C_{\sigma(\gamma)}} w_{C_\gamma} = 0$ by disjointness; the identity contributes $\det\big((o_{C_b} w_{C_b})(C_a(1))\big)_{a,b} = 1$, since $o_C w_C = e_{C(1)} - e_{C(3)}$.

Step 3 (left side). On the left, the sign manipulation in the proof of Lemma B.13, which uses only that the wedge and Ω_z are alternating, identifies the summand of σ with

$$\Omega_{\widetilde{Q}(o_{C_{\sigma(1)}},\dots,o_{C_{\sigma(k)}})(w_{C_{\sigma(1)}}\wedge\cdots\wedge w_{C_{\sigma(k)}})}\big(C_{\sigma(1)}(1), \dots, C_{\sigma(k)}(1)\big),$$

whose lth coefficient is $\chi_l(\sigma)$, since $(C_{\sigma(1)}, \dots, C_{\sigma(k)})$ has order pattern σ. Taking coeff_l gives $\sum_\sigma \chi_l(\sigma) = \mathrm{coeff}_l(1)$. □

Corollary K.8 (an adjacent transposition)**.** *There are $\tau \in S_k$ and $\ell \in \{1, \dots, k-1\}$ with $\chi_0(\tau) \ne \chi_0(\tau')$, where $\tau' = (\ell\ \ell{+}1) \circ \tau$. For such τ, ℓ put $\gamma_1 = \tau^{-1}(\ell)$, $\gamma_2 = \tau^{-1}(\ell+1)$, $\delta_l = \chi_l(\tau) - \chi_l(\tau')$ for $l \le l_1$, and let l_* be the least $l \le l_1$ with $\delta_l \ne 0$. Then $-l_0 \le l_* \le 0$.*

Proof. Otherwise χ_0 is invariant under the adjacent transpositions, which generate S_k, hence constant, and Lemma K.7 with $l = 0$ gives $k!\,\chi_0 = 1$, impossible since $k! = 0$ in κ. As $\delta_0 \ne 0$ and $\delta_l = 0$ for $l < -l_0$, $-l_0 \le l_* \le 0$. □

K.5. **The product triangle bound.** For a symmetric sequence space D and finite $J \subseteq \mathbb{N}$, D_J is the span of the e_t, $t \in J$; we use three facts.

(P1) For finite $J, J' \subseteq \mathbb{N}$, a tensor in $D_J \otimes D_{J'}$ has the same projective norm in $D_J \otimes_\pi D_{J'}$ as in $D \otimes_\pi D$, since multiplication by 1_J is a contractive projection of D onto D_J (Lemma 7.11(1)).

(P2) Relabeling the rows and the columns of finitely supported tensors by injective maps $\mathbb{N} \to \mathbb{N}$, and multiplying rows and columns by sequences in K of absolute value 1, are isometries for the norm of $D \otimes_\pi D$: by symmetry of the norm they map each D_J isometrically onto some $D_{J''}$, and (P1) applies.

(P3) For finite $J, J' \subseteq \mathbb{N}$, the completion of $D_J \otimes_\pi D_{J'}$ is $\widehat{K}^{J\times J'}$, with a norm between the maximum and the sum of the absolute values of the entries, since $\|\cdot\|_\infty \le \|\cdot\|_D$ and $\|e_t\|_D = 1$; as K is dense in $\widehat{K}$ and this norm is continuous for the maximum norm, $\|\lambda z\| = |\lambda|\|z\|$ for $\lambda \in \widehat{K}$. Multiplication by $1_J \otimes 1_{J'}$ extends to a contraction of $D\widehat{\otimes}_\pi D$ onto this completion whose value at z is the restriction of z to $J \times J'$ (Section K.1).

Proof of Theorem 7.13. Let $C_Q = \|Q\|$, $l_0 = \max(0, \lceil \log_{1/\varrho} C_Q \rceil)$ and $C_\star = 64 \cdot 8^{k-2} k(k-1)\, C_Q + \varrho^{-l_0} + 1$. We show (PTB) with constant $C_\star$. For general K satisfying (L), Lemma 7.1 then gives (PTB) with $|\vartheta|$ in place of ϱ in $C_\star$ and l_0.

Step 0 (a multiplier lift). For $a_1, \dots, a_k \in c_{00}(\mathbb{N}, K)$ put $\widetilde{Q}(a_1, \dots, a_k) = Q(\mu_{a_1}, \dots, \mu_{a_k})$, a k-linear map with

$$\|\widetilde{Q}(a_1, \dots, a_k)\| \le C_Q \prod_{\gamma=1}^{k} \|\mu_{a_\gamma}\|_{E\to D}. \tag{K.6}$$

Since $Q(u, \dots, u) = \Lambda^k_\pi u$, $\widetilde{Q}$ satisfies (K.3); since $\|e_t\|_E = 1$, $\|y\|_\infty \le \|y\|_D$ and $\|\mu_{e_t}\|_{E\to D} = 1$ (Lemma 7.11(1),(3)), it is a multiplier lift with $G = E$, as $\|\widetilde{Q}(e_{a_1}, \dots, e_{a_k})\| \le C_Q \le \varrho^{-l_0}$. The rest of the proof uses Q only through these properties of $\widetilde{Q}$ and (K.6).

Step 1 (configuration). Assume first:

(A) $\operatorname{supp} a = \operatorname{supp} x = J_1$ and $\operatorname{supp} b = \operatorname{supp} y = J_2$ with $J_1 \cap J_2 = \emptyset$, and every nonzero entry of a, x, b and y is a nonnegative power of ϑ.

If one of the four vectors is 0, both sides of (7.2) vanish; so let them be nonzero and put $N_0 = \|\mu_a\|_{E\to D}\|x\|_E\|\mu_b\|_{E\to D}\|y\|_E > 0$. List $J_1 \cup J_2$ as $f_1 < \dots < f_m$, and let $l_1 \ge 0$ be the least integer with $m^2\varrho^{l_1+1} \le N_0$. Take I as in Lemma K.5 for this l_1, and τ, ℓ, γ_1, γ_2, l_* as in Corollary K.8. Choose pairwise disjoint clusters in increasing order: a cluster C_γ for each slot γ with $\tau(\gamma) < \ell$, ordered as τ prescribes; then clusters $Y_1 < \dots < Y_m$; then a cluster C_γ for each slot γ with $\tau(\gamma) > \ell + 1$, ordered as τ prescribes. Let

$$\begin{aligned}
&\mathcal{C}_{\gamma_1} = \{Y_i : f_i \in J_1\}, && \operatorname{lev}^o_{\gamma_1}(Y_i) = \operatorname{ord} a(f_i), && \operatorname{lev}^w_{\gamma_1}(Y_i) = \operatorname{ord} x(f_i),\\
&\mathcal{C}_{\gamma_2} = \{Y_i : f_i \in J_2\}, && \operatorname{lev}^o_{\gamma_2}(Y_i) = \operatorname{ord} b(f_i), && \operatorname{lev}^w_{\gamma_2}(Y_i) = \operatorname{ord} y(f_i),
\end{aligned}$$

and $\mathcal{C}_\gamma = \{C_\gamma\}$ with both levels 0 for $\gamma \ne \gamma_1, \gamma_2$; these families are pairwise disjoint since $J_1 \cap J_2 = \emptyset$. Let g_γ and h_γ be as in Lemma K.6, and $\zeta = \widetilde{Q}(g_1, \dots, g_k)(h_1 \wedge \dots \wedge h_k) \in \widehat{\Lambda}^k_\pi D$.

Step 2 (a bounded slice). As ϑ has order 1, (A) gives $g_{\gamma_1} = g' - g''$ with $g' = \sum_{f_i \in J_1} a(f_i) e_{Y_i(1)}$ and g'' the same sum with $Y_i(3)$ in place of $Y_i(1)$, two relabelings of a; so $\|\mu_{g_{\gamma_1}}\|_{E\to D} \le 2\|\mu_a\|_{E\to D}$ by Lemma 7.11(3). Likewise $\|\mu_{g_{\gamma_2}}\|_{E\to D} \le 2\|\mu_b\|_{E\to D}$ and $\|\mu_{g_\gamma}\|_{E\to D} \le 2$ for the other slots, and since h_γ is a combination, with the coefficients $w_1, \dots, w_4$, of four relabelings of x, of y or of a unit vector, $\|h_{\gamma_1}\|_E \le 4\|x\|_E$, $\|h_{\gamma_2}\|_E \le 4\|y\|_E$ and $\|h_\gamma\|_E \le 4$. So (K.6) gives $\|\zeta\| \le 64 \cdot 8^{k-2} C_Q N_0$. Let $\Upsilon\colon \widehat{\Lambda}^k_\pi D \to D\widehat{\otimes}_\pi D$ be the slice of Lemma K.3 for the slots γ_1, γ_2

and $c_\gamma = C_\gamma(1)$, let $R = \{Y_i(1) : f_i \in J_1\}$ and $R' = \{Y_i(1) : f_i \in J_2\}$, and let $z \in \widehat{K}^{R\times R'}$ be the restriction of $\Upsilon\zeta$ to $R \times R'$ (fact (P3)). Then, in the completion of $D_R \otimes_\pi D_{R'}$,

$$\|z\| \le 64 \cdot 8^{k-2} k(k-1)\, C_Q N_0. \tag{K.7}$$

Step 3 (reading all coefficients). Let $t = Y_i(1) \in R$ and $t' = Y_{i'}(1) \in R'$. By Lemma K.3, $z(t,t') = \Omega_\zeta(c)$, where c has $Y_i(1)$ in slot γ_1, $Y_{i'}(1)$ in slot γ_2 and $C_\gamma(1)$ in every other slot γ. The clusters C_γ lie below or above all of $Y_1, \dots, Y_m$ as τ prescribes, so the output clusters have order pattern $\sigma = \tau$ if $f_i < f_{i'}$ and $\sigma = \tau'$ if $f_i > f_{i'}$. The level sum of Lemma K.6 is $\lambda = \operatorname{ord} a(f_i) + \operatorname{ord} x(f_i) + \operatorname{ord} b(f_{i'}) + \operatorname{ord} y(f_{i'})$, and $\vartheta^\lambda = (ax)(f_i)\,(by)(f_{i'})$ by (A). For $\sigma' \in S_k$ put $\hat\chi(\sigma') = \sum_{l=-l_0}^{l_1} \chi_l(\sigma') X^l \in \kappa((X))$. For $l \le l_1$, $\operatorname{coeff}_l(\vartheta^\lambda \hat\chi(\sigma)) = \sum_{l' \ge 0} \operatorname{coeff}_{l'}(\vartheta^\lambda) \chi_{l-l'}(\sigma)$, since ϑ^λ has nonnegative order and $\chi_{l''} = 0$ for $l'' < -l_0$. So by Lemma K.6, $z(t,t') - \vartheta^\lambda \hat\chi(\sigma)$ has no nonzero coefficient of index at most l_1, and

$$|z(t,t') - (ax)(f_i)\,(by)(f_{i'})\,\hat\chi(\sigma)| \le \varrho^{l_1+1}.$$

The relabeling $Y_i(1) \mapsto f_i$ of rows and columns is increasing on $R \cup R'$, and by (P2) and continuity it is an isometry onto the completion of $D_{J_1} \otimes_\pi D_{J_2}$; let $z^\sharp$ be the image of z. With $\tilde a = ax$ and $\tilde b = by$, as matrices indexed by $J_1 \times J_2$,

$$z^\sharp = \hat\chi(\tau')\,(\tilde a \otimes \tilde b) + \hat\delta \cdot \mathcal{P}_<(\tilde a \otimes \tilde b) + z_1, \qquad \hat\delta = \hat\chi(\tau) - \hat\chi(\tau') = \sum_{l=-l_0}^{l_1} \delta_l X^l, \tag{K.8}$$

where $z_1 \in \widehat{K}^{J_1\times J_2}$ has entries of absolute value at most ϱ^{l_1+1}. The lowest nonzero coefficient of $\hat\delta$ is δ_{l_*} with $l_* \le 0$, so $|\hat\delta| = \varrho^{l_*} \ge 1$, and $|\hat\chi(\tau')| \le \varrho^{-l_0}$.

Step 4 (conclusion under (A)). In the completion of $D_{J_1} \otimes_\pi D_{J_2}$, a $\widehat{K}$-normed space by (P3), $\|\tilde a \otimes \tilde b\| \le \|ax\|_D \|by\|_D \le N_0$ by Lemma 7.11(2), $\|z_1\| \le m^2 \varrho^{l_1+1} \le N_0$ by (P3) and the choice of l_1, and $\|z^\sharp\| = \|z\|$ is bounded by (K.7). By (K.8),

$$\|\mathcal{P}_<(\tilde a \otimes \tilde b)\| = |\hat\delta|^{-1} \|z^\sharp - \hat\chi(\tau')(\tilde a \otimes \tilde b) - z_1\| \le \big(64 \cdot 8^{k-2} k(k-1)\, C_Q + \varrho^{-l_0} + 1\big) N_0 = C_\star N_0.$$

By (P1), the left side is $\|\mathcal{P}_<((ax) \otimes (by))\|_{D \otimes_\pi D}$. The constant does not depend on I, τ, ℓ or l_1, which were chosen after a, x, b, y.

Step 5 (removing (A)). Both sides of (7.2) are homogeneous of degree one in each of a, x, b, y. The changes (a) and (b) below do not increase $\|\mu_a\|_{E\to D}$, $\|x\|_E$, $\|\mu_b\|_{E\to D}$ or $\|y\|_E$ (Lemma 7.11(1),(3) and symmetry of the norms), and they keep $\mathcal{P}_<((ax) \otimes (by))$ or change it by an isometry of (P2). (a) *Supports.* Replace (a, x) by $(a1_J, x1_J)$ with $J = \operatorname{supp}(ax)$, and likewise (b, y); then $\operatorname{supp} a = \operatorname{supp} x = J_1$ and $\operatorname{supp} b = \operatorname{supp} y = J_2$. If $J_1 \cap J_2 \ne \emptyset$, list $J_1 \cup J_2$ as $f_1 < \cdots < f_m$ and move the entries of a and x at $f_i \in J_1$ to the position $2i$ and those of b and y at $f_i \in J_2$ to the position $2i - 1$; since $2i < 2i' - 1$ exactly when $i < i'$, this relabels the rows and columns of $\mathcal{P}_<((ax) \otimes (by))$, and the new supports are disjoint. (b) *Units.* Since $|K^\times| = |\vartheta|^{\mathbb{Z}}$, each nonzero entry of a is $a(t) = \theta_a(t)\vartheta^{\operatorname{ord} a(t)}$ with $|\theta_a(t)| = 1$. Replace a by the vector a' with the entries $\vartheta^{\operatorname{ord} a(t)}$ on $\operatorname{supp} a$ and 0 elsewhere, and likewise x, b, y; this multiplies the rows of $\mathcal{P}_<((ax) \otimes (by))$ by $(\theta_a\theta_x)^{-1}$ and its columns by $(\theta_b\theta_y)^{-1}$. (c) *Levels.* By homogeneity we may multiply a, x, b, y by powers of ϑ that make all exponents nonnegative. After (a)–(c), (A) holds, and Steps 1–4 give (7.2) with constant $C_\star$. □

Proof of Lemma 7.14. We may assume $x \ne 0$. Let $R = \operatorname{supp} x$, and let $z \in K^{R\times R}$ have the entries $z(t,t') = x(t)x(t')$ for $t < t'$ and 0 otherwise; $\|\cdot\|_\pi$ is the norm of Section K.3.

Step 1 (reduction to matrices). Restricting to R the vectors of a representation $\mathcal{P}_<(x \otimes x) = \sum_a g_a \otimes h_a$ with $g_a, h_a \in D$ gives a representation $z = \sum_a (g_a|_R)(h_a|_R)^{\mathsf{T}}$ with $\|g_a|_R\|_\infty \le \|g_a\|_D$ and $\|h_a|_R\|_\infty \le \|h_a\|_D$. Hence $\|z\|_\pi \le \|\mathcal{P}_<(x \otimes x)\|_{D \otimes_\pi D}$.

Step 2 (normalization). Write $\|x\|_\infty = \varrho^c$ with $c \in \mathbb{Z}$ and put $x' = \vartheta^{-c}x$; the matrix $z' = \vartheta^{-2c}z$ is formed from x' in the same way, and $\|z\|_\pi = \varrho^{2c}\|z'\|_\pi$. Since $|K^\times| = |\vartheta|^{\mathbb{Z}}$, each nonzero entry is $x'(t) = \theta_t\vartheta^{\lambda_t}$ with an integer $\lambda_t \ge 0$ and $|\theta_t| = 1$, so $z' \in \mathcal{O}^{R\times R}$. For $i \ge 0$ let m_i be the number of $t \in R$ with $\lambda_t = i$.

Step 3 (one level). Fix i with $m_i \ge 2$, and let $t_1 < \cdots < t_{m_i}$ be the $t \in R$ with $\lambda_t = i$. The submatrix z'_i of z' with rows $t_1, \dots, t_{m_i-1}$ and columns $t_2, \dots, t_{m_i}$ has the entry $\theta_{t_a}\theta_{t_{b+1}}\vartheta^{2i}$ in position (a, b) if $a \le b$, and 0 otherwise. As $\vartheta = X\eta_0$ with a unit η_0 of $\mathcal{O}$, $z'_i = X^{2i}\eta_0^{2i}\Theta_1 U\Theta_2$, where Θ_1 and Θ_2 are diagonal with the units θ_{t_a} and $\theta_{t_{b+1}}$ and U is the upper unitriangular $(m_i - 1) \times (m_i - 1)$ matrix of ones; the reduction of $\eta_0^{2i}\Theta_1 U\Theta_2$ modulo X is invertible. Lemma K.4(4),(3) gives $\operatorname{rank}_{2i}(z') \ge \operatorname{rank}_{2i}(z'_i) = m_i - 1$.

Step 4 (conclusion). The indices $2i$ are distinct, so the level formula of Lemma K.4(2) gives $\|z'\|_\pi \ge (1 - \varrho)\sum_{i:m_i\ge1} \varrho^{2i}(m_i - 1)$, and

$$\|x'\|_2^2 = \sum_{i:m_i\ge1} (m_i - 1)\varrho^{2i} + \sum_{i:m_i\ge1} \varrho^{2i} \le \frac{\|z'\|_\pi}{1 - \varrho} + \frac{1}{1 - \varrho^2}.$$

Multiplying by $\varrho^{2c} = \|x\|_\infty^2$ and using Step 1 gives the lemma. □

Proof of Proposition 7.15. For $m \ge 1$ and $1 \le r' \le \infty$ put

$$\psi_{r'}(m) = \sup\{\|\xi\|_{r'}/\|\xi\|_E : \xi \in c_{00}(\mathbb{N}, K) \setminus \{0\},\ |\operatorname{supp}\xi| \le m\},$$

which is finite, since $\|\xi\|_{r'} \le m^{1/r'}\|\xi\|_\infty \le m^{1/r'}\|\xi\|_E$ by Lemma 7.11(1); and $\psi_\infty(m) \le 1$.

Step 1 (multipliers of indicators). For finite $J \subseteq \mathbb{N}$ and $x \in E$, $\mu_{1_J}x = x1_J$ and $\|x1_J\|_D \le C'\|x1_J\|_r \le C'\psi_r(|J|)\|x1_J\|_E \le C'\psi_r(|J|)\|x\|_E$, so $\|\mu_{1_J}\|_{E\to D} \le C'\psi_r(|J|)$.

Step 2 (one inequality for each support size). Let $x \in c_{00}(\mathbb{N}, K) \setminus \{0\}$ with $J = \operatorname{supp} x$ of size at most m. (PTB) with $a = b = 1_J$ and $y = x$, and Step 1, give $\|\mathcal{P}_<(x \otimes x)\|_{D\otimes_\pi D} \le CC'^2\psi_r(m)^2\|x\|_E^2$; Lemma 7.14 and $\|x\|_\infty \le \|x\|_E$ give $\|x\|_2^2 \le (\mathfrak{a}\psi_r(m)^2 + \mathfrak{b})\|x\|_E^2$. So

$$\psi_2(m)^2 \le \mathfrak{a}\,\psi_r(m)^2 + \mathfrak{b} \qquad (m \ge 1). \tag{K.9}$$

Step 3 (interpolation). If $r = \infty$, then $\psi_\infty \le 1$ and (K.9) give $\psi_2(m)^2 \le \mathfrak{a} + \mathfrak{b} = C_2^2$. Let $r < \infty$. For ξ with $|\operatorname{supp}\xi| \le m$, $\|\xi\|_r^r \le \|\xi\|_\infty^{r-2}\|\xi\|_2^2 \le \psi_2(m)^2\|\xi\|_E^r$, so $\psi_r(m)^2 \le \psi_2(m)^{4/r}$ and (K.9) gives $\psi_2(m)^2 \le \mathfrak{a}\,\psi_2(m)^{4/r} + \mathfrak{b}$. If $\psi_2(m)^2 > 2\mathfrak{b}$, then $\psi_2(m)^2 < 2\mathfrak{a}\,\psi_2(m)^{4/r}$, so $\psi_2(m)^2 < (2\mathfrak{a})^{r/(r-2)}$. In both cases $\psi_2(m)^2 \le C_2^2$.

Step 4 (all of E). So $\|x\|_2 \le C_2\|x\|_E$ for $x \in c_{00}(\mathbb{N}, K)$, and for $x \in E$, $\|x\|_2 = \sup_n\|x1_{[n]}\|_2 \le C_2 \sup_n\|x1_{[n]}\|_E = C_2\|x\|_E$ by Definition 7.10. □

Proof of Proposition 7.16. Write $c_0 = c_0(\mathbb{N}, K)$, a symmetric sequence space, and let $g_k(E)$, finite by (Γ_E^k), be the infimum of the norms of projections of $\widehat{\otimes}_\pi^k E$ onto $\widehat{\Gamma}_\pi^k E$ (Definition 3.8).

Step 1 (a multiplier lift for (c_0, E)). Let $Z = \mathcal{L}(\widehat{\otimes}_\pi^k c_0, \widehat{\Lambda}_\pi^k E)$ and let $W \subseteq Z$ be the closed subspace of operators vanishing on $\mathcal{N}_{c_0}^k = \ker\widehat{\wedge}_{c_0}^k$. Since $\|\mu_a\|_{c_0\to E} = \|a\|_E$ (Lemma 7.11(4)), $c_{00}(\mathbb{N}, K)$ is dense in E and Z is a Banach space, the map $\Phi(a_1, \dots, a_k) = \widehat{\wedge}_E^k \circ (\mu_{a_1} \otimes \cdots \otimes \mu_{a_k})$ extends from $c_{00}(\mathbb{N}, K)^k$ to a k-linear $\Phi\colon E^k \to Z$ of norm at most 1. Its diagonal lies in W, since $\Phi(a, \dots, a) = \widehat{\Lambda}_\pi^k\mu_a \circ \widehat{\wedge}_{c_0}^k$ for $a \in c_{00}(\mathbb{N}, K)$ by naturality of the wedge and W is closed. Lemma 3.11(3), with $P = E$, $n = k$ and $\delta = 1$, gives a bounded k-linear $\Phi'\colon E^k \to W$ with the diagonal of Φ and $\|\Phi'\| \le g_k(E) + 1$. By Lemma 3.7, $T \mapsto T \circ \widehat{\wedge}_{c_0}^k$ is a linear isometry of $\mathcal{L}(\widehat{\Lambda}_\pi^k c_0, \widehat{\Lambda}_\pi^k E)$ onto W. For $a_1, \dots, a_k \in c_{00}(\mathbb{N}, K)$ let $\widetilde{Q}(a_1, \dots, a_k)$ be the restriction to $\Lambda_\pi^k c_0$

of the preimage of $\Phi'(a_1,\dots,a_k)$. Then $\widetilde{Q}$ is k-linear, $\widetilde{Q}(a,\dots,a)=\Lambda_\pi^k\mu_a$, and

$$\|\widetilde{Q}(a_1,\dots,a_k)\|\le(g_k(E)+1)\prod_{\gamma=1}^{k}\|\mu_{a_\gamma}\|_{c_0\to E}.$$

Step 2 (the triangle bound). By Lemma 7.11(1),(3), $\widetilde{Q}$ is a multiplier lift with $G=c_0$, target E and $l_0=\max(0,\lceil\log_{1/\varrho}(g_k(E)+1)\rceil)$, and it satisfies (K.6) for (c_0,E) with $C_Q=g_k(E)+1$. So Steps 1–5 of the proof of Theorem 7.13 give (PTB) for (c_0,E), and the choice for $G=c_0$ in Step 1 of the proof of Theorem 8.1 bounds $\mathcal{P}_<$ on $E\otimes_\pi E$. □

K.6. **Nonarchimedean fibers.**

Proof of Lemma 7.17. Step 1 (a normalized basis). Extend an equivalent nonarchimedean norm of E to $\widehat{E}$ and round it up to $\varrho^{\mathbb{Z}}\cup\{0\}$, as in the proof of Corollary 5.3; this gives $\|\cdot\|'$. We construct $f_1,f_2,\dots\in\widehat{E}$ with $\|\sum_n\xi_nf_n\|'=\max_n|\xi_n|$. Given $f_1,\dots,f_n$, their span V_n is closed, being finite-dimensional over the complete field $\widehat{K}$ (Lemma A.3). For $v\in\widehat{E}\setminus V_n$ the distances $\|v-w\|'$, $w\in V_n$, lie in $\varrho^{\mathbb{Z}}$ and are at least $\operatorname{dist}(v,V_n)>0$, so some w_0 attains their minimum, and the nonarchimedean inequality gives $\|\lambda(v-w_0)+w\|'=\max(|\lambda|\|v-w_0\|',\|w\|')$ for $\lambda\in\widehat{K}$ and $w\in V_n$. Let f_{n+1} be a multiple of $v-w_0$ of norm one.

Step 2 (the projection). The closed span V_∞ of the f_n is isometric to $c_0(\mathbb{N},\widehat{K})$, complete with distances in $\varrho^{\mathbb{Z}}\cup\{0\}$, hence spherically complete (Lemma D.1); the extension argument in the proof of Theorem 5.2, over $\widehat{K}$, gives a contractive projection Π of $\widehat{E}$ onto V_∞. Choose $e_n\in E$ with $\|e_n-f_n\|'\le\varrho$. Then $\|\sum_n\xi_n(e_n-f_n)\|'<\max_n|\xi_n|$ for $\xi\ne0$, which gives the norm formula for the e_n. The operator Π' on V_∞ with $\Pi'(\sum_n\xi_nf_n)=\Pi(\sum_n\xi_ne_n)$ satisfies $\|\Pi'-\mathrm{id}_{V_\infty}\|\le\varrho<1$, so it is invertible with $\|\Pi'^{-1}\|\le1$. Let $\theta_E(x)$ be the coordinate sequence of $\Pi'^{-1}\Pi x$ in the basis (f_n); then $\theta_E(e_n)$ is the nth unit vector. If E is complete, then $E=\widehat{E}$ is a $\widehat{K}$-Banach space and the series defining $\widehat{\psi}_E$ converge; if K is complete, $K=\widehat{K}$. □

Proof of Theorem 7.18. Step 1 (the form Φ). Let F_1 and η be as in Appendix C for $K_1=\widehat{K}$, so that $E_1=\ell^\infty_{\widehat{K}}$, and let η' be η composed with the map $\widehat{\Lambda}^k_\pi\ell^\infty_{\widehat{K}}\to F_1$ induced by the identity of $\ell^\infty_{\widehat{K}}$, a contraction since a decomposition over K is one over $\widehat{K}$ with the same sum $\sum_j\prod_i\|x_{ji}\|$. Suppose Q exists, and for $a,x\in c_{00}(\mathbb{N},K)^k$ put

$$\Phi(a;x)=\theta_F\big(Q(a_1,\dots,a_k)(\psi x_1,\dots,\psi x_k)\big)\in\widehat{\Lambda}^k_\pi\ell^\infty_{\widehat{K}}.$$

It is k-linear in a and in x, alternating in x, of norm at most $\|\theta_F\|\|Q\|\|\psi\|^k$, and $\Phi(a,\dots,a;x)=\omega_{\ell^\infty_{\widehat{K}}}(ax_1,\dots,ax_k)$ by (7.3).

Step 2 (reduction to κ). For a,x with entries in $\mathbb{F}_p$, $\Psi(a;x)=\eta'(\Phi(a;x))\in\Lambda^k_\kappa(\kappa^{\mathbb{N}})$ is $\mathbb{F}_p$-multilinear and alternating in x, and by Proposition C.3, $\operatorname{sdim}\Psi(a;x)\le\lfloor kM_\varrho\|\theta_F\|\|Q\|\|\psi\|^k\rfloor$ when all entries lie in $\{0,\pm1\}$. As in Step 1 of the proof of Theorem 7.7, polarizing the diagonal identity over the infinite field K and applying η', which sends $\omega_{\ell^\infty_{\widehat{K}}}(y_1,\dots,y_k)$ to $y_1\wedge\cdots\wedge y_k$ when every y_j has entries in κ (proof of Lemma C.4), gives (Pol1) for Ψ on such inputs. Let Ψ^κ be the κ-multilinear map on $(\kappa^{(\mathbb{N})})^{2k}$ that agrees with Ψ on unit vectors. Antisymmetry in the last k slots and (Pol1) are identities between κ-multilinear expressions that hold on unit vectors, so they hold for Ψ^κ, and $\Psi^\kappa=\Psi$ on inputs with entries in $\{0,\pm1\}$. Since $k!=0$ in κ, this contradicts Theorem B.9(1).

Step 3 (consequences). If $A^k_{E,E';F}$ were analytic at a point, Proposition 2.5 would give a bounded lift Q_0, and $Q(a_1,\dots,a_k) = Q_0(u_{a_1},\dots,u_{a_k})(\nu)$ would be excluded. The last assertion follows from Lemma 3.4. □

Proof of Theorem 7.19. Step 1 (finite dimension). If $\dim_{\widehat{K}} \widehat{E} < \infty$, then $\widehat{E}$ has a $\widehat{K}$-basis (e_i) with continuous coordinates (ε_i) (Lemma A.3). For $u_1,\dots,u_k \in \mathcal{L}(E,E')$, with extensions $\widehat{u}_i$ to the completions, put

$$Q(u_1,\dots,u_k)(x_1\wedge\cdots\wedge x_k) = \sum_{i_1<\cdots<i_k} \det\big(\varepsilon_{i_a}(x_b)\big)_{a,b}\,\widehat{\omega}_{E'}(\widehat{u}_1 e_{i_1},\dots,\widehat{u}_k e_{i_k}),$$

where the continuous extension $\widehat{\omega}_{E'}$ of $\omega_{E'}$ is $\widehat{K}$-multilinear by density. Expanding $x_b = \sum_i \varepsilon_i(x_b)e_i$ shows, as in Proposition 5.1(2), that Q is a bounded lift of $u \mapsto \Lambda^k_\pi u$ with values in $\mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi E')$. For $u_i \in \mathcal{L}(E',E)$ use the terms $\det(\varepsilon_{i_a}(u_a x_b))_{a,b}\,\widehat{\omega}_E(e_{i_1},\dots,e_{i_k})$ instead. If E is complete, then $e_i \in E$, and moving the scalar into the first argument shows that these lifts take values in the uncompleted spaces. This proves the positive directions; in (3) with $\dim E' < \infty$, exchange the roles of E and E' in the second formula.

Step 2 (infinite dimension). Let $\dim_{\widehat{K}} \widehat{E} = \infty$. Take e_n, θ_E and, for complete E, $\widehat{\psi}_E$ from Lemma 7.17, and put $\psi_E(\xi) = \sum_n \xi_n e_n$ on $c_{00}(\mathbb{N},K)$. Theorem 7.18 applies to the following data, which satisfy (7.3):

(a) the pair $(E, c_0(\mathbb{N},\widehat{K}))$, with $\psi = \psi_E$, θ the inclusion, and $u_a(x) = a\,\theta_E(x)$;
(b) the pair $(c_0(\mathbb{N},K), E)$, with ψ the inclusion, $\theta = \theta_E$, and $u_a(x) = \psi_E(ax)$;
(c) for complete E, the pair (E,E), with $\psi = \psi_E$, $\theta = \theta_E$, and $u_a(x) = \widehat{\psi}_E(a\,\theta_E(x))$;
(d) for complete K and infinite-dimensional E,E', the pair (E,E'), with $\psi = \psi_E$, $\theta = \theta_{E'}$, and $u_a(x) = \psi_{E'}(a\,\theta_E(x))$, since θ_E then takes values in $c_0(\mathbb{N},K)$.

Over complete K, $\dim_{\widehat{K}} \widehat{E} = \dim E$, since finite-dimensional spaces are complete. An equivalent nonarchimedean norm on the values would make the action analytic by Corollary 5.3, since $|K^\times|$ is discrete. □

K.7. **Proofs of the classification results.** As in Section 8, c_{00}, c_0 and ℓ^r stand for $c_{00}(\mathbb{N},K)$, $c_0(\mathbb{N},K)$ and $\ell^r(\mathbb{N},K)$.

Proof of Theorem 8.1. Step 1 ((1)⇒(3) and (2)⇒(3)). Let $G \in \{E, c_0\}$ with (G,E) good for Banach values in a degree $k \ge p$; Theorem 7.13 gives (PTB) for (G,E) with a constant C. For $x,y \in c_{00}$, (7.2) with $a = 1_{\operatorname{supp} x}$ and $b = 1_{\operatorname{supp} y}$ if $G = E$, and with $(x, 1_{\operatorname{supp} x}, y, 1_{\operatorname{supp} y})$ in place of (a,x,b,y) if $G = c_0$, gives $\|\mathcal{P}_<(x\otimes y)\|_{E\otimes_\pi E} \le C\|x\|_E\|y\|_E$, since $\|\mu_{1_S}\|_{E\to E} \le 1$, $\|1_S\|_{c_0} = 1$ and $\|\mu_x\|_{c_0\to E} = \|x\|_E$ (Lemma 7.11(1),(4)).

Step 2 ((3)⇒(4), (5), (6), (7)). Condition (3) is the hypothesis of Proposition 5.7 for the unit vectors, a linearly independent family indexed by $\mathbb{N}$ with dense span c_{00}. So (S^n_E), (R^n_E) and (Γ^n_E) hold for every $n \ge 1$, which gives (4), (5) and (6) with $n = p$, and Proposition 3.5(1),(2) gives (7).

Step 3 ((4)⇒(1), (5)⇒(1), (6)⇒(3), (7)⇒(1), (2)). Proposition 3.5(1), respectively (2), with $D = E$ gives the first two implications, and Proposition 7.16 the third; for the last take $D = E$, respectively $D = c_0$, and the degree p.

Step 4 (ℓ^2-domination). Let C be the constant in (3). Lemma 7.14 with $D = E$ and $\|x\|_\infty \le \|x\|_E$ (Lemma 7.11(1)) gives $\|x\|_2 \le C_2\|x\|_E$ for $x \in c_{00}$, with $C_2^2 = C/(1-\varrho) + 1/(1-\varrho^2)$, hence for $x \in E$ by taking suprema over the $x1_{[n]}$ (Definition 7.10).

Step 5 (the fundamental function). We may assume $C > 0$. Let $m \ge 1$ and $z_m = \mathcal{P}_<(1_{[m]}\otimes 1_{[m]}) = \sum_{t<t'} e_t \otimes e_{t'}$, the sum over $t,t' \in [m]$. By (3) and fact (P1) of Section K.5, z_m has norm at most $C\varphi_E(m)^2$ in $E_{[m]} \otimes_\pi E_{[m]}$; the identity of $E_{[m]}$ onto $\ell^2([m],K)$ has norm

at most C_2 by Step 4, so z_m has norm at most $C_2^2 C\varphi_E(m)^2$ in $\ell^2([m],K)\otimes_\pi \ell^2([m],K)$. The matrix $\mathsf{Z} = \operatorname{coeff}_0 z_m$ is the strictly upper triangular matrix of ones, so every off-diagonal entry of $\mathsf{Z}-\mathsf{Z}^{\mathsf{T}}$ is ±1, and Lemmas K.1 and K.2(1) give

$$\frac{m\ln m}{1+\ln(2|\kappa|)} \le \operatorname{cost}(\mathsf{Z}) \le \frac{C_2^2 C\varphi_E(m)^2}{1-\varrho^2},$$

so $\varphi_E(m)^2 \ge c\,m\ln m$ with $c>0$ independent of m. □

Proof of Theorem 8.4. (1)⇒(2) is Theorem 7.13, and (4)⇒(1) follows with $k=p$.

(2)⇒(3). Let ν be an equivalent r-norm on D, or an equivalent nonarchimedean norm, in which case put $r=\infty$, scaled so that $\nu(y)\le\|y\|_D\le C''\nu(y)$ for $y\in D$. Since $\nu(e_t)\le 1$, iterating $\nu(y+y')^r\le\nu(y)^r+\nu(y')^r$, respectively $\nu(y+y')\le\max(\nu(y),\nu(y'))$, gives $\|\xi\|_D\le C''\nu(\xi)\le C''\|\xi\|_r$ for $\xi\in c_{00}$. As $\|y\|_\infty\le\|y\|_D$ (Lemma 7.11(1)), Proposition 7.15 with $C'=C''$ gives (3).

(3)⇒(4). The unit vectors of E are a linearly independent family indexed by $\mathbb{N}$, of norm one with dense span c_{00}, and ℓ^2-dominated by (3) (Definition 7.10). As K is nonarchimedean, Theorem 5.9(2) for $r<\infty$, and Corollary 5.10 for a nonarchimedean norm, give (4). □

Proof of Corollary 8.5. Positive cases. If $r_i<2$ for some i, then ℓ^{r_i} has a lower r_i-estimate with its unit vectors and $C=1$ (the examples after Theorem 5.6), and Theorem 5.6(3) makes the pair good for Banach values in every degree. If $r_1=2<r_2$, the unit vectors of ℓ^2 are ℓ^2-dominated, of norm one with dense span; ℓ^{r_2} is r_2-normed by its own norm if $r_2<\infty$, as noted after Corollary 5.10, and c_0 has a nonarchimedean norm, so Theorem 5.9(2), respectively Corollary 5.10, makes the pair good for Banach values in every degree.

Negative cases. For the remaining pairs, $r_2=2\le r_1$ or $r_1,r_2>2$, Corollary 7.9 excludes every degree $k\ge p$ and Corollary 6.3(1) gives every degree $k<p$. So $\mathfrak{D}_B(\ell^{r_1},\ell^{r_2})$ is $\{1,2,\dots\}$ in the positive cases and $\{1,\dots,p-1\}$ otherwise, which gives the equivalence and the table. □

K.8. **Spreading spaces.** For a finite set $S\subseteq\mathbb{N}$, a map $\psi\colon S\to\mathbb{N}$ that is increasing and $x\in c_{00}(\mathbb{N},K)$ supported in S, put $\psi_*x=\sum_{t\in S}x(t)e_{\psi(t)}$. If E is spreading, then $|\psi_*x|=\sum_{t\in S}|x(t)|e_{\psi(t)}$ and Definition 8.8 give $\|\psi_*x\|_E=\|x\|_E$; in particular $\|1_S\|_E=\varphi_E(|S|)$.

Proof of Theorem 8.9. Step 1 (item (1)). In the proof of Lemma 7.11, invariance under permutations is used only in (3), to see that a vector y supported in $\operatorname{supp}a$ and its relabeling y^γ have the same norm. For γ increasing on $\operatorname{supp}a$ this is the identity $\|\psi_*y\|=\|y\|$ above, with ψ the restriction of γ to $\operatorname{supp}a$. The rest of the proof uses only monotonicity, $\|e_t\|=1$ and $\|x\|=\sup_n\|x1_{[n]}\|$.

Step 2 (the facts (P1)–(P3)). They hold for spreading D, with (P2) for relabelings of the rows along a map increasing on the row support and of the columns along a map increasing on the column support: such relabelings map D_J and $D_{J'}$ isometrically onto $D_{\psi(J)}$ and $D_{\psi'(J')}$, so their tensor product is an isometry for the projective norms, and (P1) applies.

Step 3 (Theorem 7.13). Lemmas K.5, K.6 and K.7 and Corollary K.8 concern sequence spaces G and D with $\|e_t\|_G\le1$ and $\|\cdot\|_\infty\le\|\cdot\|_D$, and these properties hold for spreading spaces by Step 1. In the proof of Theorem 7.13, the norms of E and D enter through Lemma 7.11, the facts (P1)–(P3) and relabelings, and each relabeling is increasing on the support it is applied to: in Step 2, $f_i\mapsto Y_i(c)$ on J_1 and on J_2, $c=1,\dots,4$, since $Y_1<\dots<Y_m$; in Step 3, $Y_i(1)\mapsto f_i$ on R and on R'; in Step 5(a), $f_i\mapsto 2i$ on J_1 and $f_i\mapsto 2i-1$ on J_2. Steps 5(b) and (c) use only $\|x\|=\||x|\|$ and homogeneity. So Steps 1 and 2 give the theorem for spreading E and D.

Step 4 (the other results of item (2)). The proofs of Propositions 7.15 and 7.16 and of Theorems 8.1 and 8.4 use the symmetry of the spaces only through Lemma 7.11, Theorem 7.13

and fact (P1), together with $\|e_t\|_D = 1$ in Theorem 8.4; Lemma 7.14 concerns every sequence space D with $\|\cdot\|_\infty \le \|\cdot\|_D$, and c_0 is spreading. Corollaries 8.2 and 8.6(1),(2) follow from these results as before. Remark 8.7 rests, for these pairs, on Proposition 5.7 (Step 2 of the proof of Theorem 8.1) and on the relative ordering splitting described after Theorem 8.4 (Theorem 5.9(1) and Corollary 5.10), which concern arbitrary normed spaces.

Step 5 (item (3)). Here K is any nontrivially normed field of characteristic 2; condition (L) is not used. Steps 7 and 8 of the proof of Theorem 7.20 use of E only that it is a sequence space with $\|e_t\|_E = 1$, $\|x\|_\infty \le \|x\|_E$, $\|\mu_a\|_{E\to E} \le \|a\|_\infty$ and $\sup_n n/\bar\varphi_E(n)^2 = \infty$. For spreading E the first three follow from monotonicity, as there, and $\bar\varphi_E = \varphi_E$, since $\|1_S\|_E = \varphi_E(|S|)$ and φ_E is nondecreasing. □

Proof of the assertions of Example 8.10. Write $\|\cdot\|$ for $\|\cdot\|_{\alpha,q}$ on $c_{00}(\mathbb{N},\mathbb{R})$, φ for $\varphi_{E_{\alpha,q}}$, and put $\mathfrak{h}_n = \sum_{j\le n} 1/j$.

Step 1 (spreading). Minkowski's inequality along each increasing sequence makes $\|\cdot\|$ a norm. It depends only on $|f|$, is monotone in $|f|$ and dominates $\|f\|_\infty$, and $\|e_t\| = 1$. Let $f = \sum_{i\le d} f_i e_{t_i}$ with $t_1 < \cdots < t_d$. An increasing sequence (u_j) contributes only through its terms in $\{t_1,\dots,t_d\}$, say $u_{j_s} = t_{a_s}$ with $j_1 < j_2 < \cdots$ and $a_1 < a_2 < \cdots$; as $j_s \ge s$, its sum is at most $\sum_s s^{-\alpha}|f_{a_s}|^q$, the sum of the sequence $(t_{a_s})_s$. So $\|f\|^q$ is the largest value of $\sum_s s^{-\alpha}|f_{a_s}|^q$ over the increasing sequences (a_s) in $\{1,\dots,d\}$, which does not depend on $t_1,\dots,t_d$. The other conditions of Definition 8.8 hold by the definition of $E_{\alpha,q}$.

Step 2 (decreasing profiles; item (1)). If $|g(1)| \ge |g(2)| \ge \cdots$, then $|g(u_j)| \le |g(j)|$ for every increasing sequence (u_j), so $\|g\|^q = \sum_j j^{-\alpha}|g(j)|^q$. In particular $\varphi(m)^q = \sum_{j\le m} j^{-\alpha}$, which lies between $m^{1-\alpha}$ and $1 + \int_1^m \tau^{-\alpha}\,d\tau \le m^{1-\alpha}/(1-\alpha)$.

Step 3 (blocks). For $n, M \ge 1$ let $I_i = \{(i-1)n+1,\dots,in\}$ and $x = \sum_{i\le M}\sum_{j\le n} j^{-\beta} e_{(i-1)n+j}$, a real vector. By Steps 1 and 2 each $x1_{I_i}$ has norm $\mathfrak{h}_n^{1/q}$, since $j^{-\alpha}j^{-\beta q} = j^{-1}$. We show $\|x\|^q \le 1 + \ln(Mn) + (M-1)/\alpha$. It suffices to consider increasing sequences in the support of x, since other terms only raise the indices of the later ones. Fix one, let m_i be the number of its terms in I_i, and $\lambda_i = m_1 + \cdots + m_{i-1}$. The terms in I_i have indices $\lambda_i + s$, $s \le m_i$, at positions $(i-1)n + j_s$ with $j_s \ge s$, so they contribute at most $G_i = \sum_{s\le m_i}(\lambda_i + s)^{-\alpha} s^{\alpha-1}$. If $\lambda_i = 0$, then $G_i = \mathfrak{h}_{m_i} \le 1 + \ln m_i$. If $\lambda_i \ge 1$, the function $\tau \mapsto (\lambda_i + \tau)^{-\alpha}\tau^{\alpha-1}$ decreases on $(0,\infty)$, so the substitution $\tau = \lambda_i v$ gives

$$G_i \le \int_0^{m_i/\lambda_i} (1+v)^{-\alpha} v^{\alpha-1}\,dv \le \frac{1}{\alpha} + \ln\frac{\lambda_i + m_i}{\lambda_i},$$

using $(1+v)^{-\alpha} \le 1$ on $(0,1]$ and $(1+v)^{-\alpha}v^{\alpha-1} \le v^{-1}$ on $[1,\infty)$. Let i_0 be the first i with $m_i > 0$. At most $M-1$ later blocks are met, and their logarithms add up to the logarithm of the number of terms divided by m_{i_0}, at most $\ln(Mn/m_{i_0})$. So the sum of the sequence is at most $1 + \ln m_{i_0} + (M-1)/\alpha + \ln(Mn/m_{i_0})$.

Step 4 (item (2): symmetric norms). Let $x^* = x\circ\sigma$ be the decreasing rearrangement of x, for a permutation σ of $[Mn]$; it takes the value $j^{-\beta}$ at the M positions $(j-1)M+1,\dots,jM$. By Step 2,

$$\|x^*\|^q = \sum_{j\le n} j^{\alpha-1} \sum_{l\le M} ((j-1)M + l)^{-\alpha} \ge M^{1-\alpha}\mathfrak{h}_n,$$

so by Step 3 the ratio $\|x\circ\sigma\|^q/\|x\|^q$ is at least $M^{1-\alpha}\mathfrak{h}_n/(1 + \ln(Mn) + (M-1)/\alpha)$, which is unbounded as n and then M grow. For real $f \ge 0$ in $c_{00}(\mathbb{N},\mathbb{R})$ let $f' \in c_{00}(\mathbb{N},K)$ have $f'(t) = 0$ if $f(t) = 0$ and $f'(t) = \vartheta^{c(t)}$ otherwise, with $c(t)$ the largest integer such that $|\vartheta|^{c(t)} \ge f(t)$. Then $f \le |f'| < |\vartheta|^{-1} f$, so $\|f\| \le \|f'\| \le |\vartheta|^{-1}\|f\|$, and $(f\circ\sigma)' = f'\circ\sigma$. Hence

$\|x'\circ\sigma\|/\|x'\| \geq |\vartheta|\|x\circ\sigma\|/\|x\|$ is unbounded, which a norm on c_{00} invariant under permutations and equivalent to that of $E_{\alpha,q}$ would exclude.

Step 5 (item (2): lower estimates). Let $r \in (0,2)$ and $C \geq 1$, and let x' be as in Step 4. The I_i are intervals, $\|x'1_{I_i}\| \geq \|x1_{I_i}\| = \mathfrak{h}_n^{1/q}$ and $\|x'\| \leq |\vartheta|^{-1}\|x\|$, so (5.1) would give $M^{1/r}\mathfrak{h}_n^{1/q} \leq C|\vartheta|^{-1}(1+\ln(Mn)+(M-1)/\alpha)^{1/q}$. For $M = \lceil \ln n \rceil$ and large n the right side is at most $C|\vartheta|^{-1}((2+1/\alpha)\ln n)^{1/q}$ and the left side at least $(\ln n)^{1/r+1/q}$, so $(\ln n)^{1/r} \leq C|\vartheta|^{-1}(2+1/\alpha)^{1/q}$, which fails for large n.

Step 6 (items (3) and (4)). If $2\beta \leq 1$, then $\varphi(m)^2 \leq (1-\alpha)^{-2/q}m$ by (1), and Corollary 8.2, which holds for spreading spaces by Theorem 8.9(2), gives the first assertion of (3); if $2\beta < 1$, then $\sup_m m/\varphi(m)^2 = \infty$, and Theorem 8.9(3) gives the second, Steps 1 and 2 holding over every nontrivially normed field. For (4), let $y \in c_{00}$ and let y_m^* be the mth largest value of $|y|$. The positions of the m largest values, in increasing order, give $\|y\|^q \geq \sum_{s\leq m} s^{-\alpha}(y_m^*)^q \geq m^{1-\alpha}(y_m^*)^q$, so $y_m^* \leq m^{-\beta}\|y\|$ and $\|y\|_2^2 \leq (\sum_m m^{-2\beta})\|y\|^2$. For $y \in E_{\alpha,q}$ take the supremum over the $y1_{[n]}$. □

Appendix L. Degrees of splittings

This appendix proves the results of Section 6 on degrees, composition, polarizability and permanence. Since K is nontrivially normed, it is infinite, so the coefficients of a polynomial map from K^d to a vector space are linear combinations of its values. Degree 0 follows the paragraph on split operators in Section 6.1, with $\Lambda_\pi^0 u = \mathrm{id}_K$.

L.1. **Gaps in the degree sets.**

Lemma L.1 (retracts and lines)**.** *Let E and D be normed spaces and $k \geq 1$.*

1. *Let E_0 and D_0 be normed spaces, and let $\iota_1\colon E_0 \to E$, $\iota_1'\colon E \to E_0$, $\iota_2\colon D_0 \to D$ and $\iota_2'\colon D \to D_0$ be bounded linear maps with $\iota_1'\iota_1 = \mathrm{id}_{E_0}$ and $\iota_2'\iota_2 = \mathrm{id}_{D_0}$. If (E,D) is good, or good for Banach values, in degree k, then so is (E_0,D_0). In particular, isomorphic pairs have the same degree sets.*
2. *If $f\colon E \to K^d$ has as coordinates linearly independent bounded linear functionals $f_1,\dots,f_d$, then E is isomorphic to $\ker f \oplus K^d$ with the maximum norm.*
3. *Give $E\oplus K$ and $D\oplus K$ the maximum norms. If (E,D) is good, or good for Banach values, in every degree $n \leq k$, then so is $(E\oplus K, D\oplus K)$ in degree k.*

Proof. (1) For Banach values this is Proposition 6.28(1). For goodness, if Q is a lift for (E,D) as in Lemma 3.4, then $Q'(u_1,\dots,u_k) = \Lambda_\pi^k\iota_2' \circ Q(\iota_2u_1\iota_1',\dots,\iota_2u_k\iota_1')\circ\Lambda_\pi^k\iota_1$ is a lift for (E_0,D_0), since $Q'(u,\dots,u) = \Lambda_\pi^k(\iota_2'\iota_2 u\iota_1'\iota_1) = \Lambda_\pi^k u$ by Proposition 3.1. Isomorphic pairs are retracts of each other in this sense.

(2) The map f is onto, since a nonzero functional c on K^d vanishing on its image would give $\sum_a c_a f_a = 0$. For $x_1,\dots,x_d \in E$ with $f_a(x_b) = \delta_{ab}$, the maps $x \mapsto (x - \sum_a f_a(x)x_a, f(x))$ and $(y,\lambda) \mapsto y + \sum_a \lambda_a x_a$ are mutually inverse bounded linear maps between E and $\ker f \oplus K^d$.

(3) Let J_E,Π_E,J_K,Π_K be the inclusions and projections of $E\oplus K$, and J_D,Π_D,J_K',Π_K' those of $D\oplus K$. Since $J_E\Pi_E + J_K\Pi_K$ and $J_D\Pi_D + J_K'\Pi_K'$ are identities, every $u \in \mathcal{L}(E\oplus K, D\oplus K)$ is $\psi_1(u)+\psi_2(u)+\psi_3(u)$ with $\psi_1(u) = J_D\Pi_D u J_E\Pi_E$, $\psi_2(u) = J_K'\Pi_K' u J_E\Pi_E$ and $\psi_3(u) = uJ_K\Pi_K$. For $n \leq k$ and a lift Q_n for (E,D) in degree n, the map $\Lambda_\pi^n J_D \circ Q_n(\Pi_D u_1 J_E,\dots,\Pi_D u_n J_E)\circ \Lambda_\pi^n\Pi_E$, with $\widehat{\Lambda}_\pi^n J_D$ for Banach values, lifts ψ_1 in degree n. The operators $\psi_2(u)$ and $\psi_3(u)$ have rank at most one, so their exterior powers of degree $n \geq 2$ vanish, and ψ_2 and ψ_3 are liftable in every degree, with the lift 0 in degrees $n \geq 2$. Theorem 6.13(2) concludes. □

Theorem L.2 (no gap after the first failure)**.** *Let* $\operatorname{char} K = p > 0$*, let* E *and* D *each have* d *linearly independent bounded linear functionals, and let* $\mathfrak{D}$ *be* $\mathfrak{D}(E,D)$ *or* $\mathfrak{D}_B(E,D)$*. If* $\mathfrak{D} \neq \{1,2,\dots\}$ *and* n *is its least missing element, then none of* $n, n+1, \dots, n+d$ *lies in* $\mathfrak{D}$*.*

Proof. Step 1 (splitting off lines). Let $f\colon E \to K^d$ and $g\colon D \to K^d$ have the functionals as coordinates. For $0 \le l \le d$ let $E^{(l)} = \ker f \oplus K^{d-l}$ and $D^{(l)} = \ker g \oplus K^{d-l}$ with the maximum norms, so that $E^{(l)} = E^{(l+1)} \oplus K$ and $D^{(l)} = D^{(l+1)} \oplus K$. Let $\mathfrak{D}^{(l)}$ be the set of the same kind for $(E^{(l)}, D^{(l)})$, and $n^{(l)}$ its least missing element, or ∞ if there is none. By Lemma L.1(1),(2), $\mathfrak{D}^{(0)} = \mathfrak{D}$ and $n^{(0)} = n$.

Step 2 (the first failure does not move). The pair $(E^{(l+1)}, D^{(l+1)})$ is a retract of $(E^{(l)}, D^{(l)})$, so $\mathfrak{D}^{(l)} \subseteq \mathfrak{D}^{(l+1)}$ and $n^{(l)} \le n^{(l+1)}$ by Lemma L.1(1). If $i < n^{(l+1)}$, then $1, \dots, i \in \mathfrak{D}^{(l+1)}$, so $i \in \mathfrak{D}^{(l)}$ by Lemma L.1(3); hence $n^{(l)} \ge n^{(l+1)}$. By induction $n^{(l)} = n$ for every $l \le d$.

Step 3 (descent). Suppose that $k \in \mathfrak{D}$ with $n < k \le n + d$, and put $i = k - n$. By Corollary 6.3(1), $n \ge p \ge 2$, so $k, k-1, \dots, n+1 \ge 2$. Theorem 6.4, applied with every normed, respectively Banach, space F, turns a degree $k' \ge 2$ in $\mathfrak{D}^{(l)}$ into the degree $k' - 1$ in $\mathfrak{D}^{(l+1)}$. Applied i times, it gives $n \in \mathfrak{D}^{(i)}$, contradicting $n^{(i)} = n$. □

Proof of Theorem 6.5. Since E^* and D^* are infinite-dimensional, Theorem L.2 applies for every d. So if $\mathfrak{D}$, which is $\mathfrak{D}(E,D)$ or $\mathfrak{D}_B(E,D)$, has a least missing element n, then no degree $k \ge n$ lies in $\mathfrak{D}$, and $\mathfrak{D} = \{1, \dots, n-1\}$. By Corollary 6.3(2), $n = p^h$ for a unique $h \ge 1$. □

L.2. **Column constructions.** Throughout this subsection $q, m \ge 1$ and $n = qm$. For vectors $x_1, \dots, x_n$ and $C \subseteq \{1, \dots, n\}$, x_C is the wedge of the x_c, $c \in C$, in increasing order. For disjoint sets $C_1, \dots, C_l$, $\epsilon(C_1, \dots, C_l)$ is the sign of the listing of C_1, then C_2, and so on, each increasingly, relative to the increasing listing of their union; so $x_{C_1} \wedge \dots \wedge x_{C_l} = \epsilon(C_1, \dots, C_l)\, x_{C_1 \cup \dots \cup C_l}$. A listing $\lambda = (\lambda_1, \dots, \lambda_n)$ of $\{1, \dots, n\}$ has the sign $\operatorname{sgn}(\lambda)$ as a permutation, and $N_{q,m}$ is as in Theorem 6.8.

For a normed space G let $\wedge_G^{[q,m]}\colon (\Lambda_\pi^m G)^{\otimes_\pi q} \to \Lambda_\pi^n G$ be the iterated wedge product $\xi_1 \otimes \dots \otimes \xi_q \mapsto \xi_1 \wedge \dots \wedge \xi_q$, contractive by Lemma 6.1, and $\operatorname{sh}_G^{[q,m]} = (\mathrm{id} \otimes \dots \otimes \mathrm{id} \otimes \operatorname{sh}_G^{m,m}) \circ \dots \circ (\mathrm{id} \otimes \operatorname{sh}_G^{m,(q-2)m}) \circ \operatorname{sh}_G^{m,(q-1)m}$ from $\Lambda_\pi^n G$ to $(\Lambda_\pi^m G)^{\otimes_\pi q}$, so that

$$\operatorname{sh}_G^{[q,m]}(x_1 \wedge \dots \wedge x_n) = \sum_{(J_1, \dots, J_q)} \epsilon(J_1, \dots, J_q)\, x_{J_1} \otimes \dots \otimes x_{J_q},$$

the sum over the ordered partitions of $\{1, \dots, n\}$ into q blocks of size m. Both maps are natural in bounded operators, and iterating Lemma 6.1 gives, on $\otimes_\pi^n G = (\otimes_\pi^m G)^{\otimes_\pi q}$,

$$\wedge_G^n = \wedge_G^{[q,m]} \circ (\wedge_G^m)^{\otimes q}, \qquad \operatorname{alt}_G^n = (\operatorname{alt}_G^m)^{\otimes q} \circ \operatorname{sh}_G^{[q,m]}. \tag{L.1}$$

Their extensions to completions keep their names, with a hat on $\wedge_G^{[q,m]}$. In $\otimes_\pi^n G = (\otimes_\pi^q G)^{\otimes_\pi m}$, the tth block of q factors is column t, and the ith factors of the columns form row i; the isometry τ_ζ of Section 4, where $\zeta \in S_n$ sends the position $(t-1)q + i$ to $(i-1)m + t$, passes to the row-major order, in which row i is the ith block of m factors.

Lemma L.3. *If* q *and* m *are powers of a prime* p*, then* p *does not divide* $N_{q,m}$*. Hence, if* $\operatorname{char} K = p$*, the image of* $N_{q,m}$ *in* K *is a nonzero element of the prime field, and its absolute value is* 1*.*

Proof. By Legendre's formula, the exponent of p in $(p^c)!$ is $p^{c-1} + \dots + 1 = (p^c - 1)/(p-1)$. For $q = p^a$ and $m = p^b$, the exponent of p in $N_{q,m} = (qm)!/((q!)^m\, m!)$ is therefore $\big((p^{a+b} - 1) - p^b(p^a - 1) - (p^b - 1)\big)/(p-1) = 0$. Every nonzero element of $\mathbb{F}_p$ is a root of unity, so its absolute value is 1. □

Proposition L.4 (source column map)**.** *Let $w\colon E \to G$ have a section s in degree q. There is a bounded linear map col_s from $\Lambda^n_\pi E$ to the completion of $(\Lambda^m_\pi G)^{\otimes_\pi q}$, of norm at most $N_{q,m}\|s\|^m$ with $N_{q,m}$ read as a real number, such that*

$$\widehat{\wedge}^{[q,m]}_G \circ \mathrm{col}_s = \eta\, N_{q,m}\, \Lambda^n_\pi w,$$

where $\eta = \pm 1$ depends only on q and m. If s takes values in $\otimes^q_\pi G$, then col_s takes values in $(\Lambda^m_\pi G)^{\otimes_\pi q}$.

Proof. Step 1 (regrouping). Let $\Theta = (\wedge^m_G)^{\otimes q} \circ \tau_\zeta$, a contraction from $\otimes^n_\pi G$ to $(\Lambda^m_\pi G)^{\otimes_\pi q}$. It sends the tensor with columns $g_{1t} \otimes \cdots \otimes g_{qt}$ to $(g_{11} \wedge \cdots \wedge g_{1m}) \otimes \cdots \otimes (g_{q1} \wedge \cdots \wedge g_{qm})$. If $\gamma \in S_m$ and τ'_γ moves column t to column $\gamma(t)$, then $\Theta \circ \tau'_\gamma = \mathrm{sgn}(\gamma)^q\, \Theta$, since γ permutes the factors of each of the q row wedges. By (L.1) and Section 4, $\wedge^{[q,m]}_G \circ \Theta = \wedge^n_G \circ \tau_\zeta = \eta \wedge^n_G$ with $\eta = \mathrm{sgn}(\zeta)$.

Step 2 (the multilinear map). For $x_1, \dots, x_n \in E$ put

$$\Xi_s(x_1, \dots, x_n) = \sum_{\{C_1, \dots, C_m\}} \epsilon(C_1, \dots, C_m)\, \Theta\big(s(x_{C_1}) \otimes \cdots \otimes s(x_{C_m})\big),$$

the sum over the partitions of $\{1, \dots, n\}$ into m blocks of size q, each with some order of its blocks. Reordering the blocks by $\gamma \in S_m$ multiplies $\epsilon(C_1, \dots, C_m)$ by $\mathrm{sgn}(\gamma)^{q^2} = \mathrm{sgn}(\gamma)^q$ and, by Step 1, the Θ-term by $\mathrm{sgn}(\gamma)^q$; so each term depends only on the partition. The map Ξ_s is n-linear of norm at most $N_{q,m}\|s\|^m$.

Step 3 (Ξ_s is alternating). For a listing λ of $\{1, \dots, n\}$, cut into m consecutive groups of q entries, let $\Phi(\lambda) = \mathrm{sgn}(\lambda)\, \Theta\big(s(x_{\lambda_1} \wedge \cdots \wedge x_{\lambda_q}) \otimes s(x_{\lambda_{q+1}} \wedge \cdots \wedge x_{\lambda_{2q}}) \otimes \dots\big)$. Permuting entries inside a group changes $\mathrm{sgn}(\lambda)$ and that group's wedge by the same sign, so $\Phi(\lambda)$ is unchanged; and the term of a partition is $\Phi(\lambda)$ for λ listing C_1, then C_2, and so on. Let $x_a = x_b$ with $a \neq b$. The terms with a and b in one block vanish. Exchanging a and b is an involution without fixed points on the partitions that separate them, and it replaces λ by the listing λ' with the entries a and b exchanged; since $\mathrm{sgn}(\lambda') = -\mathrm{sgn}(\lambda)$ and $x_a = x_b$, $\Phi(\lambda') = -\Phi(\lambda)$, so these terms cancel in pairs. Proposition 3.1 gives col_s with $\mathrm{col}_s(x_1 \wedge \cdots \wedge x_n) = \Xi_s(x)$.

Step 4 (the identity). By Step 1, (L.1) for m blocks of size q, and $\widehat{\wedge}^q_G s = \Lambda^q_\pi w$,

$$\begin{aligned}\widehat{\wedge}^{[q,m]}_G \Theta\big(s(x_{C_1}) \otimes \cdots \otimes s(x_{C_m})\big) &= \eta\, \Lambda^q_\pi w(x_{C_1}) \wedge \cdots \wedge \Lambda^q_\pi w(x_{C_m}) \\ &= \eta\, \epsilon(C_1, \dots, C_m)\, \Lambda^n_\pi w(x_1 \wedge \cdots \wedge x_n),\end{aligned}$$

so each of the $N_{q,m}$ terms of Ξ_s contributes $\eta \Lambda^n_\pi w(x_1 \wedge \cdots \wedge x_n)$. If s takes values in $\otimes^q_\pi G$, every term lies in $(\Lambda^m_\pi G)^{\otimes_\pi q}$. □

Proposition L.5 (target column map)**.** *Let $w\colon G \to D$ have a retraction ρ in degree q. There is a bounded linear map col^ρ from the completion of $(\widehat{\Lambda}^m_\pi G)^{\otimes_\pi q}$ to $\widehat{\Lambda}^n_\pi D$, of norm at most $(m!)^{q-1}\|\rho\|^m$, such that*

$$\mathrm{col}^\rho \circ \mathrm{sh}^{[q,m]}_G = \eta'\, N_{q,m}\, \Lambda^n_\pi w,$$

where $\eta' = \pm 1$ depends only on q and m. If ρ takes values in $\Lambda^q_\pi D$, then col^ρ maps $(\Lambda^m_\pi G)^{\otimes_\pi q}$ into $\Lambda^n_\pi D$.

Proof. Step 1 (the map). For vectors $y_{i,t} \in G$ in rows i and columns t, put

$$\Psi_\rho(y) = \sum_{g_2, \dots, g_q \in S_m} \mathrm{sgn}(g_2) \cdots \mathrm{sgn}(g_q) \bigwedge_{t=1}^{m} \rho\big(y_{1,t} \otimes y_{2,g_2(t)} \otimes \cdots \otimes y_{q,g_q(t)}\big) \in \widehat{\Lambda}^n_\pi D,$$

the wedge taken in the order $t = 1, \dots, m$. It is multilinear of norm at most $(m!)^{q-1}\|\rho\|^m$.

Step 2 (alternation in each row). Let $y_{i,t} = y_{i,t'}$ with $t \neq t'$. If $i \geq 2$, replacing g_i by $(t\ t') \circ g_i$ keeps every factor and changes the sign. If $i = 1$, replacing every g_l by $g_l \circ (t\ t')$ exchanges the wedge factors in the positions t and t', which gives $(-1)^{q^2} = (-1)^q$, and multiplies the coefficient by $(-1)^{q-1}$; so the sign changes. Both replacements are involutions without fixed points, so the terms cancel in pairs. For $q = 1$, Ψ_ρ is alternating since $\rho = w$. By Proposition 3.1 in each row, Ψ_ρ induces a bounded linear map col^ρ on $(\Lambda^m_\pi G)^{\otimes_\pi q}$, which extends to the completion.

Step 3 (arrays). Let $x_1, \dots, x_n \in G$. For an ordered partition $(J_1, \dots, J_q)$ with $J_i = \{b_{i,1} < \cdots < b_{i,m}\}$, the corresponding term of $\mathrm{col}^\rho \mathrm{sh}^{[q,m]}_G(x_1 \wedge \cdots \wedge x_n)$ is $\epsilon(J_1, \dots, J_q)\Psi_\rho(y)$ with $y_{i,t} = x_{b_{i,t}}$. An *array* is a bijection $\alpha\colon \{1, \dots, q\} \times \{1, \dots, m\} \to \{1, \dots, n\}$; $\mathrm{sgn}_{\mathrm{row}}(\alpha)$ and $\mathrm{sgn}_{\mathrm{col}}(\alpha)$ are the signs of its readings row by row and column by column, and $\mathrm{sgn}_{\mathrm{row}} = \eta' \mathrm{sgn}_{\mathrm{col}}$ for a sign η' depending only on q and m. Through $\alpha(1,t) = b_{1,t}$ and $\alpha(i,t) = b_{i,g_i(t)}$ for $i \geq 2$, the pairs $((J_i),(g_i))$ correspond bijectively to the arrays with increasing first row, and $\epsilon(J_1, \dots, J_q) \prod_i \mathrm{sgn}(g_i) = \mathrm{sgn}_{\mathrm{row}}(\alpha)$. Hence

$$\mathrm{col}^\rho \mathrm{sh}^{[q,m]}_G(x_1 \wedge \cdots \wedge x_n) = \eta' \sum_\alpha \Phi(\alpha), \qquad \Phi(\alpha) = \mathrm{sgn}_{\mathrm{col}}(\alpha) \bigwedge_{t=1}^{m} \rho\big(x_{\alpha(1,t)} \otimes \cdots \otimes x_{\alpha(q,t)}\big),$$

the sum over the arrays with increasing first row.

Step 4 (columns). Permuting the columns of an array by $\gamma \in S_m$ leaves Φ unchanged, since $\mathrm{sgn}_{\mathrm{col}}$ and the wedge of m elements of degree q both change by $\mathrm{sgn}(\gamma)^{q^2}$. Group the arrays by the partition $\{C_1, \dots, C_m\}$ formed by their column sets. For a fixed partition, the arrays with increasing first row correspond bijectively to the choices of an ordering o_t of each C_t, since the first entries then fix the order of the columns. By the column invariance of Φ, the partition contributes

$$\epsilon(C_1, \dots, C_m) \sum_{o_1, \dots, o_m} \prod_t \mathrm{sgn}(o_t) \bigwedge_t \rho(x_{o_t}) = \epsilon(C_1, \dots, C_m) \bigwedge_t \rho\big(\mathrm{alt}^q_G(x_{C_t})\big),$$

where x_{o_t} is the tensor of the x_c, $c \in C_t$, in the order o_t, and $\mathrm{sgn}(o_t)$ is relative to the increasing order. Since $\rho\,\mathrm{alt}^q_G = \Lambda^q_\pi w$, this is $\epsilon(C_1, \dots, C_m)\Lambda^q_\pi w(x_{C_1}) \wedge \cdots \wedge \Lambda^q_\pi w(x_{C_m}) = \Lambda^n_\pi w(x_1 \wedge \cdots \wedge x_n)$. Summing over the $N_{q,m}$ partitions gives the identity on pure wedges, hence everywhere. □

Proof of Theorem 6.8. Put $N = N_{q,m}$, invertible in K. (1) Let $s = (\eta N)^{-1}(s_2)^{\otimes q} \circ \mathrm{col}_{s_1}$, with col_{s_1} the map of Proposition L.4 for w_1 and $(s_2)^{\otimes q}$ from the completion of $(\Lambda^m_\pi G_1)^{\otimes_\pi q}$ to $\widehat{\otimes}^n_\pi G_2$, rows in the order $1, \dots, q$. By (L.1) and naturality of $\wedge^{[q,m]}$, $\widehat{\wedge}^n_{G_2}(s_2)^{\otimes q} = \widehat{\wedge}^{[q,m]}_{G_2}(\Lambda^m_\pi w_2)^{\otimes q} = \widehat{\Lambda}^n_\pi w_2 \circ \widehat{\wedge}^{[q,m]}_{G_1}$, so Proposition L.4 gives $\widehat{\wedge}^n_{G_2} s = \Lambda^n_\pi(w_2 w_1)$, and $\|s\| \leq N_{q,m}\|s_1\|^m\|s_2\|^q/|N|$.

(2) Let $\rho = (\eta' N)^{-1}\,\mathrm{col}^{\rho_2} \circ (\rho_1)^{\otimes q}$ on $\otimes^n_\pi E = (\otimes^m_\pi E)^{\otimes_\pi q}$, with col^{ρ_2} the map of Proposition L.5 for w_2. By (L.1), $\rho_1\,\mathrm{alt}^m_E = \Lambda^m_\pi w_1$ and naturality of $\mathrm{sh}^{[q,m]}$, the map ρ satisfies $\rho\,\mathrm{alt}^n_E = (\eta' N)^{-1}\,\mathrm{col}^{\rho_2}\mathrm{sh}^{[q,m]}_{G_1}\Lambda^n_\pi w_1 = \Lambda^n_\pi(w_2 w_1)$, and $\|\rho\| \leq (m!)^{q-1}\|\rho_2\|^m\|\rho_1\|^q/|N|$. Uncompleted data give uncompleted maps, by the last assertions of Propositions L.4 and L.5. □

Proof of Theorem 6.6. Let s_p be a section for id_Y in degree p. If s_{p^j} is one in degree p^j, Theorem 6.8(1) with $w_1 = w_2 = \mathrm{id}_Y$, $q = p$ and $m = p^j$, which applies by Lemma L.3, gives one in degree p^{j+1}. So $(\mathrm{S}^{p^j}_Y)$ holds for every $j \geq 0$. By its base-p expansion, every $n \geq 1$ is a carry-free sum of powers of p, so (S^n_Y) holds by the Lucas step for split operators in Section 6.1. For retractions use Theorem 6.8(2) with $w_1 = w_2 = \mathrm{id}_Y$, $m = p^j$ and $q = p$. Uncompleted data stay uncompleted at each step. □

Proof of Corollary 6.9. (1) Let $w = w_h \cdots w_1$ with each w_l S^p-split, and put $\gamma_l = w_l \cdots w_1$. Then γ_l is S^{p^l}-split for $l \leq h$: for $l = 1$ by hypothesis, and for $l \geq 2$ by Theorem 6.8(1) applied

to γ_{l-1} and w_l with $q = p^{l-1}$ and $m = p$, by Lemma L.3. So $w = (w_h \cdots w_{l+1})\gamma_l$ is S^{p^l}-split for $l \le h$, by Proposition 3.13(2). Every $n < p^{h+1}$ is a carry-free sum of powers p^l with $l \le h$, so w is S^n-split by the Lucas step for split operators in Section 6.1. For retractions use Theorem 6.8(2) with the inner map γ_{l-1}, $m = p^{l-1}$, and the outer map w_l, $q = p$. (2) If $w = v_2 v_1$ with $v_1 \colon E \to Y$, $v_2 \colon Y \to G$ and (S^p_Y), then id_Y is S^n-split for every n by Theorem 6.6, hence so is $w = v_2 \,\mathrm{id}_Y\, v_1$ by Proposition 3.13(2). The same argument applies to (R^p_Y). □

L.3. **Polarizability in every degree.** A bounded linear map $v \colon P \to P'$ of normed spaces is *polarizable in degree n* if there is a bounded n-linear map $\Pi \colon P^n \to \widehat{\Gamma}^n_\pi P'$ with $\Pi(h, \dots, h) = (vh)^{\otimes n}$ for $h \in P$; by Lemma 3.11, id_P is polarizable in degree n if and only if P is. Each isometry τ_σ of Section 4 fixes every pure power, hence every point of $\widehat{\Gamma}^n_\pi P'$. Every tensor y in the algebraic tensor power $\otimes^n_K P'$ with $\tau_\sigma y = y$ for all $\sigma \in S_n$ is a linear combination of pure powers: if $e_1, \dots, e_d$ is a basis of a finite-dimensional subspace containing the factors of y, then y is a combination of the sums of the tensors $e_{i_1} \otimes \cdots \otimes e_{i_n}$ over the rearrangements of a fixed index tuple, and each such sum is a coefficient of the polynomial $\lambda \mapsto (\sum_i \lambda_i e_i)^{\otimes n}$ on K^d. For $a, b \ge 0$, $y \in \widehat{\otimes}^a_\pi P'$ and $z \in \widehat{\otimes}^b_\pi P'$, the *shuffle product* $y * z \in \widehat{\otimes}^{a+b}_\pi P'$ is the sum of the $\binom{a+b}{a}$ tensors obtained from $y \otimes z$ by distributing the factors of y and of z over the $a+b$ slots in their orders; it is bilinear with $\|y * z\| \le \binom{a+b}{a}\|y\|\|z\|$.

Lemma L.6 (polarizable maps)**.** *Let $v \colon P \to P'$ be a bounded linear map of normed spaces.*

(1) *The shuffle product maps $\widehat{\Gamma}^a_\pi P' \times \widehat{\Gamma}^b_\pi P'$ into $\widehat{\Gamma}^{a+b}_\pi P'$. If v is polarizable in degrees $a, b \ge 1$, with Π_a and Π_b, and $\binom{a+b}{a} \ne 0$ in K, then v is polarizable in degree $a+b$, with $(h_1, \dots, h_{a+b}) \mapsto \binom{a+b}{a}^{-1} \Pi_a(h_1, \dots, h_a) * \Pi_b(h_{a+1}, \dots, h_{a+b})$.*

(2) *If $n! \ne 0$ in K, then v is polarizable in degree n. Hence, if $\operatorname{char} K = p > 0$, the degrees in which v is polarizable contain $1, \dots, p-1$ and are closed under carry-free sums.*

Proof. (1) For $h, h' \in P'$, $h^{\otimes a} * h'^{\otimes b}$ is the coefficient of λ^a in $(\lambda h + h')^{\otimes(a+b)}$, hence a combination of pure powers; by bilinearity and continuity the shuffle product maps $\widehat{\Gamma}^a_\pi P' \times \widehat{\Gamma}^b_\pi P'$ into $\widehat{\Gamma}^{a+b}_\pi P'$. The stated map has diagonal $\binom{a+b}{a}^{-1}(vh)^{\otimes a} * (vh)^{\otimes b} = (vh)^{\otimes(a+b)}$. (2) By Proposition 4.1(2), $R_n = \frac{1}{n!}\sum_\sigma \tau_\sigma$ is a projection onto $\widehat{\Gamma}^n_\pi P'$, and $\Pi(h_1, \dots, h_n) = R_n(vh_1 \otimes \cdots \otimes vh_n)$ has diagonal $(vh)^{\otimes n}$. By Lucas's theorem, $\binom{a+b}{a} \ne 0$ in K when a and b add without carry in base p, and (1) gives the last assertion. □

Lemma L.7 (composition of polarizable maps)**.** *Let $q, m \ge 1$ and $n = qm$, and suppose that the number $N_{m,q}$ of partitions of $\{1, \dots, n\}$ into q blocks of size m is nonzero in K. If $v_1 \colon P \to P'$ is polarizable in degree q, with Π_1, and $v_2 \colon P' \to P''$ is polarizable in degree m, with Π_2, then $v_2 v_1$ is polarizable in degree n, with a map of norm at most $N_{m,q}\|\Pi_1\|^m\|\Pi_2\|^q/|N_{m,q}|$.*

Proof. Step 1 (the map). Use the rows, columns and τ_ζ of Subsection L.2, with P' and P'' in place of G. The stabilizer $S_{\mathrm{row}} \subseteq S_n$ of the partition into rows consists of the permutations inside rows composed with permutations of whole rows, and it has $N_{m,q}$ left cosets. For representatives $\sigma_1, \dots, \sigma_{N_{m,q}}$ put $T = \sum_l \tau_{\sigma_l}$ on $\widehat{\otimes}^n_\pi P''$, of norm at most $N_{m,q}$. If $\tau_\sigma Y = Y$ for all $\sigma \in S_{\mathrm{row}}$, then TY does not depend on the representatives, and $\tau_\sigma TY = TY$ for all $\sigma \in S_n$. For an array $(h_{i,t})$ in P let $c_t = \Pi_1(h_{1,t}, \dots, h_{q,t}) \in \widehat{\Gamma}^q_\pi P'$ and

$$Y(h) = (\Pi_2^{\otimes})^{\otimes q}\, \tau_\zeta(c_1 \otimes \cdots \otimes c_m), \qquad \Pi(h) = N_{m,q}^{-1}\, T\, Y(h),$$

where the linearization $\Pi_2^{\otimes} \colon \widehat{\otimes}^m_\pi P' \to \widehat{\Gamma}^m_\pi P''$ of Π_2 acts on each row. The map Π is n-linear with the stated bound.

Step 2 (invariance). A permutation of whole rows permutes the q factors of every column block in the same way, which fixes each c_t, so it fixes $\tau_\zeta(c_1 \otimes \cdots \otimes c_m)$. The operator $(\Pi_2^\otimes)^{\otimes q}$ commutes with these permutations, and its values are fixed by permutations inside rows, since $\Pi_2^\otimes$ takes values in $\widehat{\Gamma}_\pi^m P''$. Hence $\tau_\sigma Y(h) = Y(h)$ for $\sigma \in S_{\mathrm{row}}$.

Step 3 (the diagonal). If every $h_{i,t}$ is h, then $c_t = (v_1 h)^{\otimes q}$, $\tau_\zeta(c_1 \otimes \cdots \otimes c_m) = (v_1 h)^{\otimes n}$, and $\Pi_2^\otimes$ sends each row $(v_1 h)^{\otimes m}$ to $(v_2 v_1 h)^{\otimes m}$. So $Y(h) = (v_2 v_1 h)^{\otimes n}$, $TY(h) = N_{m,q}(v_2 v_1 h)^{\otimes n}$ and $\Pi(h, \dots, h) = (v_2 v_1 h)^{\otimes n}$.

Step 4 (values). Fix the array h, and choose $c_t^{(j)} \to c_t$ in the linear span of the pure powers in $\otimes_\pi^q P'$. Expanding, $\tau_\zeta(c_1^{(j)} \otimes \cdots \otimes c_m^{(j)})$ is a linear combination of tensors $\tau_\zeta(h_1'^{\otimes q} \otimes \cdots \otimes h_m'^{\otimes q}) = y^{\otimes q}$ with $y = h_1' \otimes \cdots \otimes h_m'$, and $(\Pi_2^\otimes)^{\otimes q}(y^{\otimes q}) = z^{\otimes q}$ with $z = \Pi_2^\otimes(y) \in \widehat{\Gamma}_\pi^m P''$. For $z_l \to z$ in the linear span of the pure powers, $z_l^{\otimes q}$ lies in the algebraic tensor power and is fixed by τ_σ for $\sigma \in S_{\mathrm{row}}$, so $T z_l^{\otimes q}$ is fixed by every τ_σ and is a linear combination of pure powers; hence $T z^{\otimes q} = \lim_l T z_l^{\otimes q}$ lies in $\widehat{\Gamma}_\pi^n P''$. By linearity and continuity, so does $TY(h)$. □

Proof of Theorem 6.10. If P is polarizable in degrees p and p^i, then Lemma L.7 with $v_1 = v_2 = \mathrm{id}_P$, $q = p$ and $m = p^i$, which applies by Lemma L.3, makes P polarizable in degree p^{i+1}. By induction P is polarizable in every degree p^i, and Lemma L.6 gives every degree, since every $n \geq 1$ is a carry-free sum of powers of p. □

Proof of Corollary 6.19. (1) The parameter space of a parameter piece of degree p is polarizable in degree p, hence in every degree by Theorem 6.10; so the piece, and every piece of a generalized decomposition of degree p into parameter pieces, is a parameter piece of every degree. (2) By Theorem 6.10, $\mathcal{L}(E, D)$ is polarizable in every degree, and Corollary 3.17 applies in each degree. □

L.4. **Decompositions of degree p.**

Proof of Theorem 6.20. Let w be $\mathrm{S}^{\leq p}$-split, as every S^p-split map is, since every map is S^n-split for $n < p$. Every $n < p^2$ is $c_1 p + c_0$ with base-p digits $c_0, c_1 < p$, a carry-free sum of copies of p and of 1, so w is S^n-split by the Lucas step for split operators in Section 6.1. Thus w is $\mathrm{S}^{\leq p^2-1}$-split, and the same holds for R. Hence every piece of the decomposition, including the single piece of a relative splitting, is of the same type in every degree $n < p^2$. Theorem 6.13(3) gives the goodness; the Lucas step keeps uncompleted data uncompleted. □

Proof of Proposition 6.21. Choose $f \in E^*$ and $x_0 \in E$ with $f(x_0) = 1$, and let the pieces be $(G_i, G_i', w_i, w_i', \varepsilon_i)$. For $y \in D$, the operator $u_y = f(\cdot)\, y \in \mathcal{L}(E, D)$ has $\|u_y\| \leq \|f\|\|y\|$, and (6.4) gives $y = u_y x_0 = \sum_i w_i' \chi_i(y)$ with the bounded linear maps $\chi_i \colon D \to G_i'$, $\chi_i(y) = \varepsilon_i(u_y) w_i x_0$. So the inclusion ι_D of D in $\widehat{D}$ is $\sum_i w_i' \chi_i$, which is $\mathrm{R}^{\leq k}$-split, since every w_i' is, by Proposition 3.13(2) and the paragraph on split operators in Section 6.1. A retraction for ι_D in degree k is a bounded linear $\rho \colon \otimes_\pi^k D \to \widehat{\Lambda}_\pi^k \widehat{D} = \widehat{\Lambda}_\pi^k D$ such that $\rho\, \mathrm{alt}_D^k$ is the inclusion of $\Lambda_\pi^k D$ in its completion; this is (R_D^k). Without completion, the maps w_i' take values in D and their retractions in the uncompleted exterior powers, and so does ρ. For $k = p$, Corollary 6.7 gives the last assertion. □

L.5. **Permanence and its failure.**

Proof of Theorem 6.29. Step 0 (notation). Order the words of each length m lexicographically. For a normed space U, a step function of level m on Ω^k is a function $z = \sum_\xi 1_{C_{\xi_1} \times \cdots \times C_{\xi_k}} u_\xi$, the sum over k-tuples ξ of words of length m, with $u_\xi \in U$; put $\|z\|_1 = \sum_\xi 2^{-mk} \|u_\xi\|$ and

$\|z\|_\infty = \max_\xi \|u_\xi\|$, and let $L^1(\Omega^k; U)$ be the completion of the step functions, in which the U_0-valued ones are dense if U_0 is dense in U. Let $\Delta_m \subseteq \Omega^k$ be the union of the cells $C_{\xi_1} \times \cdots \times C_{\xi_k}$ of level m with $\xi_1 < \cdots < \xi_k$, and T_m the union of those with two equal words; T_m has measure at most $\binom{k}{2}2^{-m}$, so $\|1_{T_m} z\|_1 \le \binom{k}{2}2^{-m}\|z\|_\infty$ for step functions z. Every extension of a word ξ precedes every extension of a later word of the same length, so $\Delta_m \subseteq \Delta_{m'}$ and $\Delta_{m'} \setminus \Delta_m \subseteq T_m$ for $m \le m'$.

Step 1 (tensor powers). For a finite set I with weights $c_t > 0$, the map $\sum_t e_t \otimes u_t \mapsto (u_t)_t$ is an isometry of $\ell^1(I, c) \otimes_\pi U$ onto U^I with the norm $\sum_t c_t\|u_t\|$, since this decomposition costs $\sum_t c_t\|u_t\|$ and $(\eta, u) \mapsto (\eta_t u)_t$ is contractive. With associativity and commutativity of $\otimes_\pi$, the contractive k-linear map $(f_1, \dots, f_k) \mapsto [\zeta \mapsto f_1(\zeta_1) \otimes \cdots \otimes f_k(\zeta_k)]$ therefore linearizes to an isometry of the kth projective tensor power of the step functions of level m onto the $\otimes_\pi^k G$-valued step functions of level m on Ω^k; since the tensor power at level m maps contractively into that of all step functions, it is an isometry of the latter onto all $\otimes_\pi^k G$-valued step functions. Passing to completions, $\widehat{\otimes}_\pi^k Y = L^1(\Omega^k; \widehat{\otimes}_\pi^k G)$; then $\widehat{\wedge}_Y^k$ sends $1_{C_1 \times \cdots \times C_k}(g_1 \otimes \cdots \otimes g_k)$ to $(1_{C_1} g_1) \wedge \cdots \wedge (1_{C_k} g_k)$, and alt_Y^k sends $f_1 \wedge \cdots \wedge f_k$ to $\zeta \mapsto \sum_\sigma \operatorname{sgn}(\sigma) f_{\sigma(1)}(\zeta_1) \otimes \cdots \otimes f_{\sigma(k)}(\zeta_k)$.

Step 2 (ordered antisymmetrization). For step functions $f_a = \sum_\xi 1_{C_\xi} g_{a,\xi}$ of level at most m, let $\Phi_m(f)$ be the step function equal to $\sum_\sigma \operatorname{sgn}(\sigma) f_{\sigma(1)}(\zeta_1) \otimes \cdots \otimes f_{\sigma(k)}(\zeta_k)$ on Δ_m and to 0 elsewhere. Since $(\xi, \sigma) \mapsto \upsilon$ with $\upsilon_{\sigma(a)} = \xi_a$ is a bijection from the increasing k-tuples of words times S_k onto the k-tuples of distinct words, $\|\Phi_m(f)\|_1 \le \sum_\upsilon 2^{-mk} \prod_b \|g_{b,\upsilon_b}\| \le \prod_b \|f_b\|_1$. For $m' \ge m$, $\Phi_{m'}(f)$ and $\Phi_m(f)$ agree outside T_m by Step 0, so $\|\Phi_{m'}(f) - \Phi_m(f)\|_1 \le k!\binom{k}{2}2^{-m} \prod_a \|f_a\|_\infty$. Hence $\Phi(f) = \lim_m \Phi_m(f)$ exists; it is k-linear, alternating, since the terms σ and $\sigma \circ (a\ b)$ cancel when $f_a = f_b$, and bounded by $\prod_a \|f_a\|_1$, so it extends to a contractive alternating map on Y^k.

Step 3 (sections). For the f_a of Step 2, $f_1 \wedge \cdots \wedge f_k$ is the sum over k-tuples of words of $\bigwedge_a 1_{C_{\xi_a}} g_{a,\xi_a}$. A tuple of distinct words is $(\xi_{\sigma^{-1}(1)}, \dots, \xi_{\sigma^{-1}(k)})$ for a unique increasing tuple ξ and a unique $\sigma \in S_k$, and its wedge is $\operatorname{sgn}(\sigma) \bigwedge_b 1_{C_{\xi_b}} g_{\sigma(b),\xi_b}$; by Step 1 these terms add up to $\widehat{\wedge}_Y^k \Phi_m(f)$. Since $\|1_{C_\xi} g\|_1 = 2^{-m}\|g\|$, the tuples with $\xi_a = \xi_b$ for some $a < b$ contribute at most $\sum_{a<b} 2^{-m}\|f_a\|_\infty \|f_b\|_1 \prod_{c \ne a,b} \|f_c\|_1$ in norm. Letting $m \to \infty$ gives $f_1 \wedge \cdots \wedge f_k = \widehat{\wedge}_Y^k \Phi(f)$. By Proposition 3.1, Φ induces $s\colon \Lambda_\pi^k Y \to \widehat{\otimes}_\pi^k Y$ with $\|s\| \le 1$, and $\widehat{\wedge}_Y^k s = \mathrm{id}$ on wedges of step functions, hence everywhere, since these span a dense subspace (the paragraph before Theorem 3.16).

Step 4 (retractions). For a step function z of level at most m and $m' \ge m$, $\|1_{\Delta_{m'}} z - 1_{\Delta_m} z\|_1 \le \binom{k}{2}2^{-m}\|z\|_\infty$ by Step 0. So $\mathcal{P}_\Delta z = \lim_m 1_{\Delta_m} z$ exists, with $\|\mathcal{P}_\Delta z\|_1 \le \|z\|_1$, and $\mathcal{P}_\Delta$ extends to a contraction of $L^1(\Omega^k; \widehat{\otimes}_\pi^k G) = \widehat{\otimes}_\pi^k Y$. Let $\rho = \widehat{\wedge}_Y^k \circ \mathcal{P}_\Delta$ on $\otimes_\pi^k Y$, so $\|\rho\| \le 1$. For step functions f_a, Step 1 gives $1_{\Delta_m} \operatorname{alt}_Y^k(f_1 \wedge \cdots \wedge f_k) = \Phi_m(f)$, so $\rho \operatorname{alt}_Y^k(f_1 \wedge \cdots \wedge f_k) = \widehat{\wedge}_Y^k \Phi(f) = f_1 \wedge \cdots \wedge f_k$ by Step 3; by density $\rho \operatorname{alt}_Y^k = \mathrm{id}$.

Step 5 (polarization). On step functions $\tau_\sigma z(\zeta) = \tau_\sigma\big(z(\zeta_{\sigma(1)}, \dots, \zeta_{\sigma(k)})\big)$, so $\tau_\sigma(1_S z) = 1_{\sigma S} \tau_\sigma z$ with $\sigma S = \{\zeta : (\zeta_{\sigma(1)}, \dots, \zeta_{\sigma(k)}) \in S\}$; the sets $\sigma\Delta_m$, $\sigma \in S_k$, are disjoint with union $\Omega^k \setminus T_m$. Put $R_k = \sum_\sigma \tau_\sigma \mathcal{P}_\Delta$, of norm at most $k!$. For a step function h, $h^{\otimes k}$ is fixed by every τ_σ, so $R_k h^{\otimes k} = \lim_m 1_{\Omega^k \setminus T_m} h^{\otimes k} = h^{\otimes k}$; by continuity R_k fixes every point of $\widehat{\Gamma}_\pi^k Y$. For cells $C_1 < \cdots < C_k$ of level m and $g_a \in G$, $\sum_\sigma \tau_\sigma(1_{C_1} g_1 \otimes \cdots \otimes 1_{C_k} g_k)$ is the coefficient of $\lambda_1 \cdots \lambda_k$ in $(\sum_a \lambda_a 1_{C_a} g_a)^{\otimes k}$, hence a linear combination of pure powers. For a $\otimes_\pi^k G$-valued step function z of level at most m, $1_{\Delta_m} z$ is a finite sum of such tensors, so $R_k z = \lim_m \sum_\sigma \tau_\sigma 1_{\Delta_m} z$ lies in $\widehat{\Gamma}_\pi^k Y$; by continuity R_k maps $\widehat{\otimes}_\pi^k Y$ onto $\widehat{\Gamma}_\pi^k Y$. □

Proof of Corollary 6.30. (1) The isometry $g \mapsto 1_\Omega g$ has complete, hence closed, image. By Theorem 6.29, $L^1(\Omega;E)$ and $L^1(\Omega;D)$ satisfy (S^n) and (R^n) for every n, and Proposition 3.5(1),(2) gives the goodness statements.

(2) The map $q_E\colon \ell^1(\mathrm{Ball}(E),\widehat{K}) \to E$, $e_x \mapsto x$, is contractive. Fix $c \in K$ with $0<|c|<1$. For $x \neq 0$ in E choose $j \in \mathbb{Z}$ with $|c|^{j+1} < \|x\| \le |c|^j$; then $x/c^j \in \mathrm{Ball}(E)$, $q_E(c^j e_{x/c^j}) = x$ and $\|c^j e_{x/c^j}\|_1 = |c|^j < |c|^{-1}\|x\|$. So q_E maps each open ball of radius δ onto a set containing the open ball of radius $|c|\delta$; it is surjective and open, and E is isomorphic to $\ell^1(\mathrm{Ball}(E),\widehat{K})/\ker q_E$. The unit vectors of $\ell^1(\mathrm{Ball}(E),\widehat{K})$ are linearly independent over K with dense K-span, on which $\|\mathcal{P}_<(x\otimes y)\| \le \|x\|_1\|y\|_1$ for every total order of $\mathrm{Ball}(E)$. So Proposition 5.7 gives (S^n), (R^n) and polarizability in degree n for every n, and Proposition 3.5(1),(2) gives the goodness statements. The same holds for D.

(3) By Theorem 7.7 with $r=\infty$ and Corollary 8.3, $(c_0(\mathbb{N},K), c_0(\mathbb{N},K))$ is not good for Banach values in degree k, and $c_0(\mathbb{N},K)$ satisfies neither (S^k) nor (R^k) and is not polarizable in degree k. Proposition 6.28(3) transfers the last three statements to the completion $Y_0 = c_0(\mathbb{N},\widehat{K})$. If (Y_0,Y_0) were good for Banach values in degree k, then Proposition 6.28(4), applied to the source with the complete target Y_0 and then to the target, would make $(c_0(\mathbb{N},K), c_0(\mathbb{N},K))$ good for Banach values in degree k. By (1) and (2) with $E=D=Y_0$, Y_0 is a closed subspace of $L^1(\Omega;Y_0)$ and isomorphic to a quotient of $\ell^1(\mathrm{Ball}(Y_0),\widehat{K})$; these two spaces satisfy the three conditions in degree k, and replacing either entry of (Y_0,Y_0) by one of them gives a pair that is good for Banach values in degree k. Since the three conditions and goodness for Banach values are invariant under isomorphisms (Proposition 6.28(1)), the last assertion follows. □

Appendix M. Rigid Banach spaces

This appendix proves Theorems 9.2, 9.3 and 9.4. It transfers the rigid subspace of L_0 of Kalton and Roberts [15, §3]: their quasi-norms become r-norms, their exponents increasing to 1 become exponents $r_i \uparrow \infty$, and forcing goes through L^r readouts.

M.1. **Blocks and the Nakano sum.** Throughout Subsections M.1–M.4, K is a complete nonarchimedean field with nontrivial absolute value that is separable as a metric space, and $\vartheta \in K$ is fixed with $0<|\vartheta|<1$.

For a finite-dimensional normed space Y and $1 \le r < \infty$, $L^r((0,1);Y)$ is the completion of the step functions $f\colon (0,1)\to Y$ modulo null functions for $\|f\|_r = (\int_0^1 \|f(t)\|^r\,dt)^{1/r}$, and $1\otimes y$ is the constant function with value y. By the Riesz–Fischer argument, Y being complete (Lemma A.3), each element is a measurable function, unique up to null sets, with the same formula for the norm. If the norm of Y is nonarchimedean, $\|\cdot\|_r$ is an r-norm (Section 5), since $\|f(t)+g(t)\|^r \le \max(\|f(t)\|,\|g(t)\|)^r \le \|f(t)\|^r + \|g(t)\|^r$. An r-norm is an r'-norm for $1 \le r' \le r$, since $(c^r + c'^r)^{1/r} \le (c^{r'} + c'^{r'})^{1/r'}$ for reals $c,c' \ge 0$. The space K^{1+N} carries the maximum norm, with unit vectors $e_1,\dots,e_{1+N}$.

Let $1 \le r_0 \le r_1 \le \cdots$ be real numbers. For a sequence $a=(a_i)_{i\ge0}$ of reals $a_i \ge 0$ put

$$\nu(a) = \inf\Big\{\varsigma>0 : \sum_{i\ge0}(a_i/\varsigma)^{r_i} \le 1\Big\} \in [0,\infty].$$

Convexity of $\varsigma \mapsto \varsigma^{r_i}$, with the weights $\varsigma/(\varsigma+\varsigma')$ and $\varsigma'/(\varsigma+\varsigma')$, gives $\nu(a+a') \le \nu(a)+\nu(a')$. Moreover ν is positively homogeneous and monotone, $\nu(a)=0$ only for $a=0$, $\nu(a)=a_i$ if a_i is the only nonzero entry, and $\nu(a) = \sup_h \nu(a^{\le h})$, where $a^{\le h}$ replaces the entries with index above h by zero. For finitely supported a the sum is continuous and decreasing in ς, so

$$\nu(a) \le \varsigma \iff \textstyle\sum_i (a_i/\varsigma)^{r_i} \le 1 \qquad (\varsigma>0). \tag{M.1}$$

Given Banach spaces L_i, $i \geq 0$, the *Nakano sum* $\mathcal{Z}$ is the normed space of the sequences $f = (f_i)_{i\geq 0}$, $f_i \in L_i$, with $\|f\| := \nu((\|f_i\|)_{i\geq 0}) < \infty$. It is complete: a Cauchy sequence converges in each L_i, since $\|f_i\| \leq \|f\|$, and to its coordinatewise limit in $\mathcal{Z}$, since $\nu(a) = \sup_h \nu(a^{\leq h})$. We identify L_i isometrically with the i-th block of $\mathcal{Z}$. The projections Π_i, $\Pi_{<i}$, $\Pi_{\leq i}$ and $\Pi_{>i}$ keep the coordinates with index equal to, less than, at most and greater than i and replace the others by zero; they are contractive, since ν is monotone, and $\mathrm{id} = \Pi_{<i} + \Pi_i + \Pi_{>i}$. Let $\mathcal{Z}_{<i}$, $\mathcal{Z}_{\leq i}$ and $\mathcal{Z}_{>i}$ be the ranges of the last three.

Lemma M.1. *If the norm of every L_m is an r_m-norm and $i \geq 0$, then $\|f^{(1)} + \cdots + f^{(d)}\|^{r_{i+1}} \leq \|f^{(1)}\|^{r_{i+1}} + \cdots + \|f^{(d)}\|^{r_{i+1}}$ for $f^{(1)}, \ldots, f^{(d)} \in \mathcal{Z}_{>i}$.*

Proof. By induction it suffices to take $d = 2$, $f = f^{(1)}$ and $g = f^{(2)}$. Put $r = r_{i+1}$ and, for $m > i$, $a_m = \|f_m\|$, $a'_m = \|g_m\|$ and $a''_m = \|f_m + g_m\|$; then $a''^r_m \leq a^r_m + a'^r_m$, since the norm of L_m is an r_m-norm and $r_m \geq r$. Let ν' be defined like ν on sequences indexed by $m > i$, with the exponents $r_m/r \geq 1$; like ν, it is monotone and subadditive. With $a^r = (a^r_m)_{m>i}$, the substitution $\varsigma = \varsigma'^r$ gives $\nu'(a^r) = \inf\{\varsigma'^r : \sum_{m>i}(a_m/\varsigma')^{r_m} \leq 1\} = \nu(a)^r$. Hence

$$\|f + g\|^r = \nu'(a''^r) \leq \nu'(a^r + a'^r) \leq \nu'(a^r) + \nu'(a'^r) = \|f\|^r + \|g\|^r.$$

□

M.2. **Operators between blocks.**

Lemma M.2 (compare [15, Lemma 2.2])**.** *Let $1 \leq r \leq r'$, let Y be a finite-dimensional normed space, $N' \geq 0$ an integer and $h \in L^r((0,1); Y)$ a step function. There is a bounded linear map $\Phi\colon L^{r'}((0,1); K^{1+N'}) \to L^r((0,1); Y)$ with $\Phi(1 \otimes e_1) = h$ and $\|\Phi g\|_r \leq \|h\|_r \|g\|_{r'}$ for every g.*

Proof. If $h = 0$, take $\Phi = 0$. Otherwise let $e_1^*\colon K^{1+N'} \to K$ be the first coordinate, so $e_1^*(e_1) = 1$ and $|e_1^*(y)| \leq \|y\|$. The function $\xi(t) = \|h\|_r^{-r} \int_0^t \|h(t')\|^r \, dt'$ maps $[0,1]$ onto $[0,1]$ and is continuous, nondecreasing and piecewise affine, with positive slope exactly where $h \neq 0$. For a step function g the step function $(\Phi g)(t) = e_1^*(g(\xi(t)))\, h(t)$ depends only on the class of g, since $\Phi g = 0$ where $h = 0$ and preimages of null sets under ξ are null where $h \neq 0$. On the intervals where $h \neq 0$, which ξ maps onto $(0,1)$ up to finitely many points, the substitution $t'' = \xi(t)$ has $\|h(t)\|^r \, dt = \|h\|_r^r \, dt''$. Hence

$$\|\Phi g\|_r^r = \|h\|_r^r \int_0^1 |e_1^*(g(t''))|^r \, dt'' \leq \|h\|_r^r \|g\|_r^r \leq \|h\|_r^r \|g\|_{r'}^r,$$

the last step by Jensen's inequality on the probability space $(0,1)$, since $r \leq r'$. Extend Φ by continuity; then $\Phi(1 \otimes e_1) = e_1^*(e_1)\, h = h$. □

From here to the end of Subsection M.4, $N_i \geq i$ are integers, $L_i = L^{r_i}((0,1); K^{1+N_i})$ and $\mathbf{e}_i = 1 \otimes e_1 \in L_i$ for $i \geq 0$, and $\mathcal{Z}$ is the Nakano sum of the L_i.

Lemma M.3 (compare [15, Lemma 3.2])**.** *Let $i \geq 1$, and let $f \in \mathcal{Z}_{<i}$ have step functions as coordinates and $\|f\| \leq 1$. There is a bounded linear map $\Phi\colon L_i \to \mathcal{Z}_{<i}$ with $\Phi\mathbf{e}_i = f$ and $\|\Phi\| \leq 1$.*

Proof. By (M.1), $\sum_{m<i} \|\Pi_m f\|^{r_m} \leq 1$. For $m < i$, since $r_m \leq r_i$, Lemma M.2 gives $\Phi_m\colon L_i \to L_m$ with $\Phi_m \mathbf{e}_i = \Pi_m f$ and $\|\Phi_m g\| \leq \|\Pi_m f\| \|g\|$. Put $\Phi g = \sum_{m<i} \Phi_m g$, with $\Phi_m g$ in block m. Then $\Phi \mathbf{e}_i = f$, and for $g \neq 0$, $\sum_{m<i} (\|\Phi_m g\|/\|g\|)^{r_m} \leq \sum_{m<i} \|\Pi_m f\|^{r_m} \leq 1$, so $\|\Phi g\| \leq \|g\|$ by (M.1). □

M.3. **The inductive choices.** Fix an integer $h_0 \geq 0$ with $\sum_{h\geq 0}|\vartheta|^{h+h_0} \leq 1/4$ and put $\zeta_h = \vartheta^{h+h_0}$ for $h \geq 0$. Given reals $\eta_i > 0$ and integers $d_i \geq 1$, let $\mathbf{e}_{i,l} \in L_i$, $1 \leq l \leq d_i$, equal e_1 on $[(l-1)/d_i, l/d_i)$ and 0 elsewhere; then $\sum_l \mathbf{e}_{i,l} = \mathbf{e}_i$ and $\|\mathbf{e}_{i,l}\| = d_i^{-1/r_i}$. Let $Y_i = \operatorname{span}(e_2, \dots, e_{1+N_i}) \subseteq K^{1+N_i}$ and

$$\mathcal{U}_i = \operatorname{span}\{\mathbf{e}_{i,1}, \dots, \mathbf{e}_{i,d_i}\} + \{1 \otimes y : y \in Y_i\} \subseteq L_i, \qquad \mathcal{U}_{<i} = \mathcal{U}_0 + \cdots + \mathcal{U}_{i-1} \subseteq \mathcal{Z}_{<i},$$

finite-dimensional spaces of step functions. For $i \geq 1$ let $C_i \geq 1$ satisfy $\|\sum_l x_l\| \leq C_i \max_l \|x_l\|$ for every finite family (x_l) in $\mathcal{U}_{<i}$; it exists because the norm of $\mathcal{U}_{<i}$ is equivalent to the maximum of the coordinates in a basis (Lemma A.3).

Lemma M.4 (compare [15, Lemma 3.1])**.** *Let real numbers $\underline{r}_i \geq 1$, $i \geq 0$, be given. One can choose r_i, η_i and d_i inductively so that for every $i \geq 0$:*

(1) $r_i \geq \max(r_{i-1}, \underline{r}_i, i)$, *where* $r_{-1} = 1$;
(2) $d_i^{-1/r_i} < \eta_i$, *that is,* $\|\mathbf{e}_{i,l}\| < \eta_i$ *for every* l;
(3) $(\sum_{l=1}^{d_i} \|\mathbf{e}_{i,l}\|^{r_{i+1}})^{1/r_{i+1}} < \eta_i$;
(4) $iC_i\eta_i < |\zeta_i|$ *if* $i \geq 1$.

Proof. At stage i the data with indices below i are fixed. If $i \geq 1$, then $(\sum_l \|\mathbf{e}_{i-1,l}\|^r)^{1/r}$, a finite sum, tends to $\max_l \|\mathbf{e}_{i-1,l}\| < \eta_{i-1}$ as $r \to \infty$, by (2) for $i-1$; choose r_i satisfying (1) and so large that (3) holds for $i-1$. This determines $\mathcal{U}_{<i}$ and its norm, which involves only $r_0, \dots, r_{i-1}$, hence C_i. Choose $\eta_i > 0$ satisfying (4), and then d_i satisfying (2). □

M.4. **The spaces $\mathfrak{X}_\alpha$.** Throughout this subsection the data of Subsection M.3 are chosen by Lemma M.4 for some lower bounds $\underline{r}_i$. The norm of each L_m is an r_m-norm, since K^{1+N_m} is nonarchimedean, so Lemma M.1 applies.

Index sets. Partition $\{0, 1, 2, \dots\} = \bigsqcup_{\alpha\geq 1} \mathcal{I}_\alpha$ into infinite sets, enumerate $\mathcal{I}_\alpha$ as $i_\alpha(0) < i_\alpha(1) < \cdots$, and put $\mathcal{J}_{i_\alpha(a)} = \{i_\alpha(2^a(2a'+1)) : a' \geq 0\}$ for $a \geq 0$. Since every positive integer is $2^a(2a'+1)$ for exactly one pair (a, a') and $2^a > a$, the sets $\mathcal{J}_i$, $i \in \mathcal{I}_\alpha$, are pairwise disjoint infinite subsets of $\mathcal{I}_\alpha$ with union $\mathcal{I}_\alpha \setminus \{\min \mathcal{I}_\alpha\}$, and $\min \mathcal{J}_i > i$.

Dense sequences. For $i \in \mathcal{I}_\alpha$, the set $\mathcal{F}_i$ of the $f \in \sum_{m\in\mathcal{I}_\alpha, m\leq i} \mathcal{U}_m$ with $|\vartheta| < \|f\| \leq 1$ contains $\mathbf{e}_i$ and lies in a finite-dimensional space over the separable field K; let $\mathcal{F}_i^0 \ni \mathbf{e}_i$ be a countable dense subset. Through a bijection of $\mathcal{J}_i$ onto $\mathcal{F}_i^0 \times \mathbb{N}$, choose $\gamma_h \in \mathcal{F}_i^0$, $h \in \mathcal{J}_i$, such that every element of $\mathcal{F}_i^0$ equals γ_h for infinitely many h. Since $\gamma_h \in \mathcal{Z}_{\leq i} \subseteq \mathcal{Z}_{<h}$ has step functions as coordinates and norm at most 1, Lemma M.3 gives $\Phi_h \colon L_h \to \mathcal{Z}_{<h}$ with $\Phi_h \mathbf{e}_h = \gamma_h$ and $\|\Phi_h\| \leq 1$. For $h = \min \mathcal{I}_\alpha$ put $\Phi_h = 0$.

The construction. For $\alpha \geq 1$ put $\Delta_\alpha = \sum_{h\in\mathcal{I}_\alpha} \zeta_h \Phi_h \Pi_h$; let $\mathcal{E}_\alpha$ and $\mathcal{Y}_\alpha$ be the closed spans of $\{\mathbf{e}_i : i \in \mathcal{I}_\alpha\}$ and of $\bigcup_{i\in\mathcal{I}_\alpha} \mathcal{U}_i$; put $\mathcal{Q}_\alpha = (\mathrm{id} - \Delta_\alpha)(\mathcal{E}_\alpha)$ and $\mathfrak{X}_\alpha = \mathcal{Y}_\alpha/\mathcal{Q}_\alpha$; and let $[f]_\alpha$ be the class of $f \in \mathcal{Z}$ in $\mathcal{Z}/\mathcal{Q}_\alpha$. The series Δ_α converges absolutely, with norm at most $\sum_h |\zeta_h| \leq 1/4$, so the Neumann series inverts $\mathrm{id} - \Delta_\alpha$, with $\|(\mathrm{id} - \Delta_\alpha)^{-1}\| \leq 4/3$. Since Φ_h maps into $\mathcal{Z}_{<h}$, Δ_α maps $\mathcal{Z}_{\leq i}$ into $\mathcal{Z}_{<i}$, so $\Pi_i \Delta_\alpha \Pi_{\leq i} = \Pi_{>i} \Delta_\alpha \Pi_{\leq i} = 0$, for every i. For $h \in \mathcal{I}_\alpha$ the vector $(\mathrm{id} - \Delta_\alpha)\mathbf{e}_h$ is $\mathbf{e}_h - \zeta_h\gamma_h$, or $\mathbf{e}_h$ if $h = \min \mathcal{I}_\alpha$, and lies in $\mathcal{Y}_\alpha$. So $\mathcal{Q}_\alpha$, closed as the image of a closed subspace under an isomorphism, lies in $\mathcal{Y}_\alpha$, and the Banach space $\mathfrak{X}_\alpha$ is the closed subspace of $\mathcal{Z}/\mathcal{Q}_\alpha$ of the classes $[f]_\alpha$, $f \in \mathcal{Y}_\alpha$. For $i \in \mathcal{I}_\alpha$ and $h \in \mathcal{J}_i$,

$$[\mathbf{e}_h]_\alpha = \zeta_h [\gamma_h]_\alpha, \tag{M.2}$$

and $\Pi_i \mathcal{E}_\alpha$ lies in the line $K\mathbf{e}_i$, which is closed by Lemma A.3.

Lemma M.5 (compare [15, Lemma 3.3])**.** *If $i \in \mathcal{I}_\alpha$ and $f \in \mathcal{Z}_{\leq i}$, then* $\inf_{\lambda\in K} \|\Pi_i f - \lambda \mathbf{e}_i\| \leq \frac{4}{3}\|[f]_\alpha\|$.

Proof. Let $t > \|[f]_\alpha\|$, choose $g \in \mathcal{E}_\alpha$ with $\|f - (\mathrm{id} - \Delta_\alpha)g\| < t$, and let $\Pi_i g = \lambda \mathbf{e}_i$. Since $\Pi_i \Delta_\alpha \Pi_{\le i} = 0$, the vector $\Pi_i f - \lambda \mathbf{e}_i = \Pi_i(f - (\mathrm{id} - \Delta_\alpha)g) - \Pi_i \Delta_\alpha \Pi_{>i} g$ has norm less than $t + \frac{1}{4}\|\Pi_{>i} g\|$. Since $\Pi_{>i} f = 0$ and $\Pi_{>i}\Delta_\alpha \Pi_{\le i} = 0$, the vector $\Pi_{>i}((\mathrm{id} - \Delta_\alpha)g - f) = \Pi_{>i} g - \Pi_{>i}\Delta_\alpha \Pi_{>i} g$ has norm less than t, so $\|\Pi_{>i} g\| < t + \frac{1}{4}\|\Pi_{>i} g\|$, that is, $\|\Pi_{>i} g\| < \frac{4}{3}t$. Hence $\|\Pi_i f - \lambda \mathbf{e}_i\| < \frac{4}{3}t$; let t decrease to $\|[f]_\alpha\|$. □

Lemma M.6 (compare [15, Lemma 3.5])**.** *The set $\bigcup_{i \in \mathcal{I}_\alpha} K[\mathbf{e}_i]_\alpha$ is dense in $\mathfrak{X}_\alpha$.*

Proof. Let $\mathcal{C}$ be the closure of this set; it is closed under scalar multiplication. Let $i \in \mathcal{I}_\alpha$. Each $g \in \mathcal{F}_i^0$ equals γ_h for some $h \in \mathcal{J}_i$, so $[g]_\alpha = \zeta_h^{-1}[\mathbf{e}_h]_\alpha \in \mathcal{C}$ by (M.2); by density, $[f]_\alpha \in \mathcal{C}$ for every $f \in \mathcal{F}_i$. A nonzero f in the span of $\bigcup_{m \in \mathcal{I}_\alpha} \mathcal{U}_m$, which is dense in $\mathcal{Y}_\alpha$, lies in $\sum_{m \in \mathcal{I}_\alpha, m \le i} \mathcal{U}_m$ for some $i \in \mathcal{I}_\alpha$, and a power of ϑ times f lies in $\mathcal{F}_i$; so $[f]_\alpha \in \mathcal{C}$. □

Lemma M.7. *Let $i \in \mathcal{I}_\alpha$, and let $\iota_i \colon Y_i \to L^{r_i}((0,1); Y_i)$, $y \mapsto 1 \otimes y$, be the constant embedding, which is isometric. There are bounded linear maps $J_i \colon Y_i \to \mathfrak{X}_\alpha$ and $\Xi_i \colon \mathfrak{X}_\alpha \to L^{r_i}((0,1); Y_i)$ with $\|J_i\| \le 1$, $\|\Xi_i\| \le 4/3$ and $\Xi_i J_i = \iota_i$.*

Proof. Put $J_i y = [1 \otimes y]_\alpha$, where $1 \otimes y \in \mathcal{U}_i \subseteq \mathcal{Y}_\alpha$ has norm $\|y\|$. The pointwise coordinate projection $\mathrm{pr}_{Y_i} \colon L_i \to L^{r_i}((0,1); Y_i)$ of K^{1+N_i} onto Y_i along Ke_1 is contractive and kills $\mathbf{e}_i$. So the map $z \mapsto \mathrm{pr}_{Y_i}(\Pi_i(\mathrm{id} - \Delta_\alpha)^{-1} z)$ on $\mathcal{Z}$, of norm at most $4/3$, vanishes on $\mathcal{Q}_\alpha$, since $(\mathrm{id} - \Delta_\alpha)^{-1}\mathcal{Q}_\alpha = \mathcal{E}_\alpha$ and $\Pi_i \mathcal{E}_\alpha \subseteq K\mathbf{e}_i$; let Ξ_i be the restriction to $\mathfrak{X}_\alpha$ of the induced map on $\mathcal{Z}/\mathcal{Q}_\alpha$. For $y \in Y_i$, $(\mathrm{id} - \Delta_\alpha)^{-1}(1 \otimes y) - 1 \otimes y = \sum_{c \ge 1} \Delta_\alpha^c (1 \otimes y)$ lies in $\mathcal{Z}_{<i}$, since Δ_α maps $\mathcal{Z}_{\le i}$ into $\mathcal{Z}_{<i}$. Hence $\Xi_i J_i y = \mathrm{pr}_{Y_i}(1 \otimes y) = 1 \otimes y$. □

Lemma M.8. *Each $\mathfrak{X}_\alpha$ is nonzero and infinite-dimensional.*

Proof. For $i \in \mathcal{I}_\alpha$ and $y \in Y_i$, Lemma M.7 gives $\|y\| = \|\Xi_i J_i y\| \le \frac{4}{3}\|J_i y\|$. So $\mathfrak{X}_\alpha$ contains the subspaces $J_i(Y_i)$ of dimension $N_i \ge i$, for the infinitely many $i \in \mathcal{I}_\alpha$. □

Proof of Theorem 9.2. The argument follows [15, Theorem 3.6]. Make the construction with any integers $N_i \ge i$ and any lower bounds $\underline{r}_i$; the Banach spaces $\mathfrak{X}_\alpha$ are infinite-dimensional by Lemma M.8. Let $u \in \mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\beta)$; multiplying u by a power of ϑ, we may assume $\|u\| < 1$.

Step 1 (approximation). Fix $i \in \mathcal{I}_\alpha$; the set $\mathcal{J}_i' = \{h \in \mathcal{J}_i : \gamma_h = \mathbf{e}_i\}$ is infinite. Let $h \in \mathcal{J}_i'$; then $h > i \ge 0$ and $[\mathbf{e}_h]_\alpha = \zeta_h[\mathbf{e}_i]_\alpha$ by (M.2). Since $\|u[\mathbf{e}_{h,l}]_\alpha\| \le \|u\|\|\mathbf{e}_{h,l}\| < \|\mathbf{e}_{h,l}\|$, there are $g_{h,l} \in \mathcal{Y}_\beta$, $1 \le l \le d_h$, with $[g_{h,l}]_\beta = u[\mathbf{e}_{h,l}]_\alpha$ and $\|g_{h,l}\| < \|\mathbf{e}_{h,l}\| < \eta_h$ (Lemma M.4(2)); put $g_h = \sum_l g_{h,l}$, so that $[g_h]_\beta = u[\mathbf{e}_h]_\alpha$. Since Π_m maps $\mathcal{Y}_\beta$ into the closed space $\mathcal{U}_m$ if $m \in \mathcal{I}_\beta$ and to zero otherwise, $\Pi_{<h} g_{h,l} \in \mathcal{U}_{<h}$ and $\|\Pi_{<h} g_h\| \le C_h \max_l \|\Pi_{<h} g_{h,l}\| < C_h \eta_h$. Lemmas M.1 and M.4(3) give $\|\Pi_{>h} g_h\| \le (\sum_l \|\mathbf{e}_{h,l}\|^{r_{h+1}})^{1/r_{h+1}} < \eta_h$. So $\|g_h - \Pi_h g_h\| < (C_h + 1)\eta_h \le 2C_h\eta_h < 2|\zeta_h|/h$ by Lemma M.4(4), and

$$\big\|u[\mathbf{e}_i]_\alpha - \zeta_h^{-1}[\Pi_h g_h]_\beta\big\| = |\zeta_h|^{-1}\|[g_h - \Pi_h g_h]_\beta\| < 2/h \qquad (h \in \mathcal{J}_i'). \tag{M.3}$$

Step 2 (the case $\alpha \ne \beta$). Then $h \notin \mathcal{I}_\beta$, so $\Pi_h g_h = 0$ and $\|u[\mathbf{e}_i]_\alpha\| < 2/h$ for infinitely many h. Hence $u[\mathbf{e}_i]_\alpha = 0$ for every $i \in \mathcal{I}_\alpha$, and $u = 0$ by Lemma M.6.

Step 3 (the case $\alpha = \beta$: eigenvectors). Let $h' < h$ be elements of $\mathcal{J}_i'$. The vector $f = \zeta_h^{-1}\Pi_h g_h - \zeta_{h'}^{-1}\Pi_{h'} g_{h'} \in \mathcal{Z}_{\le h}$ has $\Pi_h f = \zeta_h^{-1}\Pi_h g_h$ and, by (M.3) for h and h', $\|[f]_\alpha\| < 2/h + 2/h'$. Lemma M.5 gives $\xi \in K$ with $\|\zeta_h^{-1}\Pi_h g_h - \xi \mathbf{e}_h\| < \frac{4}{3}(2/h + 2/h')$. Passing to classes and using $[\mathbf{e}_h]_\alpha = \zeta_h[\mathbf{e}_i]_\alpha$ and (M.3), we get $\mathrm{dist}(u[\mathbf{e}_i]_\alpha, K[\mathbf{e}_i]_\alpha) < 2/h + \frac{4}{3}(2/h + 2/h')$. Letting $h \to \infty$ and then $h' \to \infty$ gives $u[\mathbf{e}_i]_\alpha = \lambda_i[\mathbf{e}_i]_\alpha$ for some $\lambda_i \in K$, since $K[\mathbf{e}_i]_\alpha$ is closed (Lemma A.3). If $[\mathbf{e}_i]_\alpha \ne 0$, then λ_i is unique and $|\lambda_i| \le \|u\|$.

Step 4 (one eigenvalue). Let $[\mathbf{e}_i]_\alpha$ and $[\mathbf{e}_{i'}]_\alpha$ be linearly independent and $x = [\mathbf{e}_i]_\alpha + [\mathbf{e}_{i'}]_\alpha$. By Lemma M.6, nonzero $y_\ell \in K[\mathbf{e}_{i_\ell}]_\alpha$, $i_\ell \in \mathcal{I}_\alpha$, converge to x. Since $uy_\ell = \lambda_{i_\ell} y_\ell$ and $|\lambda_{i_\ell}| \le \|u\|$,

$\|\lambda_{i_\ell}x - ux\| \le |\lambda_{i_\ell}|\|x - y_\ell\| + \|u\|\|y_\ell - x\| \to 0$. So ux lies in the closed line Kx, say $ux = \lambda x$, and comparing coefficients gives $\lambda_i = \lambda = \lambda_{i'}$. Nonzero dependent $[\mathbf{e}_i]_\alpha$, $[\mathbf{e}_{i'}]_\alpha$ have equal eigenvalues directly. So u is a scalar multiple of the identity on $\bigcup_i K[\mathbf{e}_i]_\alpha$, hence on $\mathfrak{X}_\alpha$ by Lemma M.6. □

M.5. **Forcing through an L^r readout.** Throughout this subsection condition (L) holds with $K = \kappa((X))$; so K is complete, and separable since the Laurent polynomials over κ are dense. Let $c_0(N) = K^N$, $N \ge 1$, carry the maximum norm, with unit vectors $e_1, \dots, e_N$, and let $\mu_a x = ax$ be the coordinatewise product of $a \in K^N$ and $x \in K^N$; then $\|\mu_a\| = \|a\|_\infty$. For $r \ge k$ put $G_{N,r} = L^r((0,1); c_0(N))$, with the constant embedding $\iota_{N,r}\colon c_0(N) \to G_{N,r}$, $x \mapsto 1 \otimes x$. A *readout lift* for $\iota_{N,r}$ is a k-linear map $Q\colon (K^N)^k \to \mathcal{L}(\Lambda^k_\pi c_0(N), \widehat{\Lambda}^k_\pi G_{N,r})$ with $Q(a, \dots, a) = \Lambda^k_\pi(\iota_{N,r}\mu_a)$ for $a \in K^N$; its norm is $\|Q\| = \sup\{\|Q(a_1, \dots, a_k)\| : \|a_j\|_\infty \le 1\}$.

Lemma M.9 (Ramsey [21, Theorem B])**.** *For integers $g, h, C \ge 1$ there is an integer $\mathrm{Ram}(g, h, C)$ such that for every coloring of the g-element subsets of $\{1, \dots, \mathrm{Ram}(g, h, C)\}$ with C colors there is an h-element set all of whose g-element subsets have the same color.*

Lemma M.10. *Let Y be a finite-dimensional normed space and $r \ge k$. The map sending $(g_1, \dots, g_k)$ to the function $t \mapsto g_1(t) \wedge \cdots \wedge g_k(t)$ is a k-linear alternating map from $L^r((0,1); Y)^k$ to $L^{r/k}((0,1); \Lambda^k_\pi Y)$ of norm at most 1. It induces a contraction*

$$\mathrm{pw}\colon \widehat{\Lambda}^k_\pi L^r((0,1); Y) \to L^{r/k}((0,1); \Lambda^k_\pi Y)$$

with $\mathrm{pw}(\Lambda^k_\pi \iota(\xi)) = 1 \otimes \xi$ for $\xi \in \Lambda^k_\pi Y$, where ι is the constant embedding of Y into $L^r((0,1); Y)$.

Proof. For step functions on a common partition the value is a step function with $\|g_1(t) \wedge \cdots \wedge g_k(t)\| \le \prod_j \|g_j(t)\|$, so its $L^{r/k}$-norm is at most $\prod_j \|g_j\|_r$, by Hölder's inequality with k exponents equal to k; it vanishes if two of the g_j coincide. Since step functions are dense and $r/k \ge 1$, the map extends to a k-linear alternating map of norm at most 1 into the Banach space $L^{r/k}((0,1); \Lambda^k_\pi Y)$, and Proposition 3.1, in its form for complete targets, gives pw. On constant functions $g_j = 1 \otimes x_j$ the value is $1 \otimes (x_1 \wedge \cdots \wedge x_k)$. □

Lemma M.11 (test lemma)**.** *For every $m \ge 1$ there are an integer $N(m) \ge m$ and a real number $r^*(m) \ge k$, depending only on m, k and $|\kappa|$, such that for all $N \ge N(m)$ and $r \ge r^*(m)$ every readout lift Q for $\iota_{N,r}$ satisfies $\|Q\| \ge m/(2M_\varrho\, k(k-1))$.*

Proof. Step 0 (parameters and reduction). Put $h = 4(2m + 2 + k) + 3k$ and $N(m) = \mathrm{Ram}(3k, h, |\kappa|^{(3k)^{3k}}) \ge h \ge m$, and choose $r^*(m) \ge k$ with $\binom{N(m)}{h} k!\,(k-1)\, 2^{-r^*(m)/k} < 1$. Let $N \ge N(m)$, $r \ge r^*(m)$, and Q a readout lift for $\iota_{N,r}$. The extension by zero $\mathrm{ext}\colon c_0(N(m)) \to c_0(N)$ and the pointwise restriction $\mathrm{res}\colon G_{N,r} \to G_{N(m),r}$ to the first $N(m)$ coordinates are contractive, and $\mathrm{res}\, \iota_{N,r}\, \mu_{\mathrm{ext}\, a}\, \mathrm{ext} = \iota_{N(m),r}\, \mu_a$. So $(a_1, \dots, a_k) \mapsto \widehat{\Lambda}^k_\pi \mathrm{res} \circ Q(\mathrm{ext}\, a_1, \dots, \mathrm{ext}\, a_k) \circ \Lambda^k_\pi \mathrm{ext}$ is a readout lift for $\iota_{N(m),r}$ of norm at most $\|Q\|$, and we may assume $N = N(m)$.

Step 1 (readout). For $a \in (K^N)^k$ and $\xi \in \Lambda^k_\pi c_0(N)$, Lemma M.10 gives an element $\widetilde{Q}(a; \xi) = \mathrm{pw}(Q(a)(\xi))$ of $L^{r/k}((0,1); \Lambda^k_\pi c_0(N))$, k-linear in a, linear in ξ, with $\|\widetilde{Q}(a; \xi)\|_{r/k} \le \|Q\| \prod_j \|a_j\|_\infty \|\xi\|$. The k-linear maps $a \mapsto Q(a)(x_1 \wedge \cdots \wedge x_k)$ and $a \mapsto \iota_{N,r}(a_1 x_1) \wedge \cdots \wedge \iota_{N,r}(a_k x_k)$ agree on the diagonal, so the inclusion–exclusion computation in the proof of Proposition B.8(1), followed by pw, gives for all $a_1, \dots, a_k, x_1, \dots, x_k \in K^N$

$$\sum_{\sigma \in S_k} \widetilde{Q}(a_{\sigma(1)}, \dots, a_{\sigma(k)}; x_1 \wedge \cdots \wedge x_k) = 1 \otimes \sum_{\sigma \in S_k} (a_{\sigma(1)} x_1) \wedge \cdots \wedge (a_{\sigma(k)} x_k). \tag{M.4}$$

Step 2 (representatives). For $i \in \{1,\dots,N\}^k$ write $e_i = (e_{i_1},\dots,e_{i_k})$. For each i and each strictly increasing $y \in \{1,\dots,N\}^k$ fix an everywhere defined representative $f_{i,y}\colon (0,1) \to \Lambda^k_\pi c_0(N)$ of $\widetilde{Q}(e_i; e_{y_1} \wedge \cdots \wedge e_{y_k})$ (Subsection M.1); put $f_{i,y} = \operatorname{sgn}(\sigma) f_{i,y\circ\sigma}$ if y has distinct entries and $y \circ \sigma$ is increasing, and $f_{i,y} = 0$ if y has a repeated entry. Then $f_{i,y}$ represents $\widetilde{Q}(e_i; e_{y_1} \wedge \cdots \wedge e_{y_k})$ for every y. For $t \in (0,1)$ and $a, x \in (\mathbb{F}_p^N)^k$ put

$$\Psi_t(a;x) = \sum_{i,y \in \{1,\dots,N\}^k} \prod_j a_j(i_j) \prod_j x_j(y_j)\, f_{i,y}(t) \in \Lambda^k_\pi c_0(N).$$

Then $t \mapsto \Psi_t(a;x)$ represents $\widetilde{Q}(a; x_1 \wedge \cdots \wedge x_k)$, and each Ψ_t is $\mathbb{F}_p$-multilinear and alternating in x: if $x_\gamma = x_{\gamma'}$ with $\gamma \neq \gamma'$, the terms of y and of $y \circ (\gamma\ \gamma')$ have equal weights and $f_{i,y\circ(\gamma\,\gamma')} = -f_{i,y}$, and the terms with $y_\gamma = y_{\gamma'}$ vanish.

Step 3 (test inputs and the point t_0). For an h-element set $\mathcal{H} \subseteq \{1,\dots,N\}$ let $\mathcal{H}'$ consist of its $h - 3k$ smallest elements; clusters of $\mathcal{H}'$ are sets of four consecutive elements of $\mathcal{H}'$, as in Appendix B. Let $(a^{\mathcal{H}}; x^{\mathcal{H}})$ be the inputs of the proof of Lemma B.13 for the first k clusters of $\mathcal{H}'$; by (M.4) and Step 2, the set $O_{\mathcal{H}}$ of the t with $\sum_\sigma \Psi_t(a^{\mathcal{H}}_{\sigma(1)},\dots,a^{\mathcal{H}}_{\sigma(k)}; x^{\mathcal{H}}) \neq \sum_\sigma (a^{\mathcal{H}}_{\sigma(1)} x^{\mathcal{H}}_1) \wedge \cdots \wedge (a^{\mathcal{H}}_{\sigma(k)} x^{\mathcal{H}}_k)$ is null. For $\tau \in S_k$ and $1 \le \ell < k$, let $(a^{\mathcal{H},\tau,\ell}; x^{\mathcal{H},\tau,\ell})$, with entries in $\{0,\pm 1\}$, be the inputs of the proof of Lemma B.12 for τ and ℓ, with $m+1$ pairs of clusters, all clusters taken consecutively from the smallest element of $\mathcal{H}'$; there are enough, since $4(2m+k) \le h - 3k$. For these inputs $(a;x)$, the set $O_{\mathcal{H},\tau,\ell}$ of the t with $\|\Psi_t(a;x)\| > 2\|\widetilde{Q}(a; x_1 \wedge \cdots \wedge x_k)\|_{r/k}$ has measure at most $2^{-r/k}$, by Chebyshev's inequality for $t \mapsto \|\Psi_t(a;x)\|^{r/k}$, and is null if $\widetilde{Q}(a; x_1 \wedge \cdots \wedge x_k) = 0$. There are at most $\binom{N}{h}\, k!\,(k-1)$ sets $O_{\mathcal{H},\tau,\ell}$, so the union of all of them and of the $O_{\mathcal{H}}$ has measure at most $\binom{N}{h}\, k!\,(k-1)\, 2^{-r/k} < 1$. Fix $t_0 \in (0,1)$ outside this union, and write $\Psi = \Psi_{t_0}$.

Step 4 (Ramsey at t_0). Let Ω be the determinant array of §B.1 on $\Lambda^k_\pi c_0(N)$, with $S = \{1,\dots,N\}$, and put $T(i;y;c) = \operatorname{coeff}_0 \Omega_{\Psi(e_i;e_y)}(c) \in \kappa$ for $i,y,c \in \{1,\dots,N\}^k$. Color each $3k$-element set $\{d_1 < \cdots < d_{3k}\} \subseteq \{1,\dots,N\}$ by the family $(T(d_{\eta(1)},\dots,d_{\eta(3k)}))_\eta$ over all maps $\eta\colon \{1,\dots,3k\} \to \{1,\dots,3k\}$, with at most $|\kappa|^{(3k)^{3k}}$ colors. Lemma M.9 gives a homogeneous h-element set $\mathcal{H}$. Padding with the $3k$ elements of $\mathcal{H} \setminus \mathcal{H}'$, which lie above $\mathcal{H}'$, as in the proof of Lemma B.10, shows that $T(i;y;c)$ depends only on the pattern of (i,y,c) when all its entries lie in $\mathcal{H}'$.

Step 5 (the argument of Appendix B at t_0). For inputs supported in $\mathcal{H}'$ with entries in $\{0,\pm 1\}$, $\operatorname{coeff}_0 \Omega_{\Psi(a;x)}(c) = \sum_{i,y \in \mathcal{H}'^k} \prod_j a_j(i_j) \prod_j x_j(y_j)\, T(i;y;c)$, as in (B.1), since Ψ is $\mathbb{F}_p$-multilinear, Ω is linear and coeff_0 is $\mathbb{F}_p$-linear. Define the cluster values $\chi(\tau) \in \kappa$ for clusters of $\mathcal{H}'$ as in Appendix B, with $\operatorname{coeff}_0 \Omega_{\Psi(\cdot)}$ in place of $\Omega_{\Psi(\cdot)}$. The proofs of Lemmas B.11 and B.13 use only this expansion, pattern homogeneity of T, that clusters are intervals of the set on which T is homogeneous, (W), alternation of Ψ in its vector slots and of Ω_z in its arguments, and (Pol1) at $(a^{\mathcal{H}}; x^{\mathcal{H}})$; the last holds since $t_0 \notin O_{\mathcal{H}}$ and the determinants on the right of (Pol1) lie in $\mathbb{F}_p$. So both lemmas hold, and $\sum_\tau \chi(\tau) = 1$ in κ. Since $k! = 0$ in κ, χ is not constant; since exchanges of adjacent ranks generate S_k, some τ and ℓ have $\delta = \chi(\tau) - \chi(\tau') \neq 0$, where $\tau' = (\ell\ \ell{+}1) \circ \tau$.

Let $(a;x) = (a^{\mathcal{H},\tau,\ell}; x^{\mathcal{H},\tau,\ell})$, let $\gamma_1 = \tau^{-1}(\ell)$ and $\gamma_2 = \tau^{-1}(\ell+1)$ be the exchanged slots, let I_1 and I_2 be the sets of first elements of the $m+1$ clusters given to γ_1 and γ_2 in the proof of Lemma B.12, and let c_γ be the first element of the cluster given to any other slot γ. By Lemma B.11, as in that proof, the $(m+1) \times (m+1)$ matrix St of the values $\operatorname{coeff}_0 \Omega_{\Psi(a;x)}(c)$, with an element of I_1 in slot γ_1, an element of I_2 in slot γ_2 and c_γ in each other slot γ of c, is $\chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}} + \delta U$ with U upper unitriangular, so $\operatorname{rank} \mathrm{St} \ge m$. The extension by zero $\operatorname{ext}_\infty\colon c_0(N) \to c_0(\mathbb{N}, K)$ makes $\Lambda^k_\pi \operatorname{ext}_\infty$ a contraction that keeps the determinant array on

$\{1,\dots,N\}^k$. Let Υ be the slice of Lemma K.3 for $c_0(\mathbb{N},K)$, the slots γ_1,γ_2 and the entries c_γ, and obtain z from $\Upsilon\Lambda^k_\pi \operatorname{ext}_\infty(\Psi(a;x))$ by restricting its two tensor factors to the coordinates in I_1 and in I_2. Then $\operatorname{coeff}_0 z = \operatorname{St}$ and $\|z\| \le k(k-1)\|\Psi(a;x)\|$, so Lemma K.2(2) gives $m \le \operatorname{rank}\operatorname{St} \le M_\varrho k(k-1)\|\Psi(a;x)\|$. Since $t_0 \notin O_{\mathcal{H},\tau,\ell}$ and the entries of a and x lie in $\{0,\pm1\}$, $\|\Psi(a;x)\| \le 2\|\widetilde{Q}(a;x_1\wedge\dots\wedge x_k)\|_{r/k} \le 2\|Q\|$, and $m \le 2M_\varrho k(k-1)\|Q\|$. □

Corollary M.12. *Let $m \ge 1$, $N \ge N(m)$ and $r \ge r^*(m)$. Let E be a normed space, and $J\colon c_0(N) \to E$ and $\Xi\colon E \to G_{N,r}$ bounded linear maps with $\Xi J = \iota_{N,r}$. Then every s as in (S^k_E) satisfies $\|s\| \ge m/(2M_\varrho k(k-1)(\|\Xi\|\|J\|)^k)$, and every ρ as in (R^k_E) satisfies $\|\rho\| \ge m/(2M_\varrho k(k-1)k!\,(\|\Xi\|\|J\|)^k)$.*

Proof. Let v_a be pointwise multiplication by $a \in K^N$ on $G_{N,r}$; then $\|v_a\| \le \|a\|_\infty$ and $v_a\iota_{N,r} = \iota_{N,r}\mu_a$. Given s, the k-linear map $Q(a_1,\dots,a_k) = \widehat{\wedge}^k_{G_{N,r}} \circ ((v_{a_1}\Xi)\otimes\dots\otimes(v_{a_k}\Xi)) \circ s \circ \Lambda^k_\pi J$ has norm at most $(\|\Xi\|\|J\|)^k\|s\|$ and, by naturality of $\wedge^k$, diagonal $\widehat{\Lambda}^k_\pi(v_a\Xi)\circ\widehat{\wedge}^k_E\circ s\circ\Lambda^k_\pi J = \Lambda^k_\pi(v_a\Xi J) = \Lambda^k_\pi(\iota_{N,r}\mu_a)$. Given ρ, the k-linear map $Q(a_1,\dots,a_k) = \widehat{\Lambda}^k_\pi\Xi\circ\rho\circ((J\mu_{a_1})\otimes\dots\otimes(J\mu_{a_k}))\circ\operatorname{alt}^k_{c_0(N)}$ has norm at most $k!\,(\|\Xi\|\|J\|)^k\|\rho\|$, since $\|\operatorname{alt}^k_{c_0(N)}\| \le k!$, and, by naturality of alt^k, diagonal $\widehat{\Lambda}^k_\pi\Xi\circ\rho\circ\operatorname{alt}^k_E\circ\Lambda^k_\pi(J\mu_a) = \Lambda^k_\pi(\Xi J\mu_a) = \Lambda^k_\pi(\iota_{N,r}\mu_a)$. Lemma M.11 applies to both. □

M.6. **Proofs of the goodness theorems.** Throughout this subsection κ, K and ϱ are as in Subsection M.5. For $k' \ge p$ and $m \ge 1$ let $N_{k'}(m)$ and $r^*_{k'}(m)$ be the numbers of Lemma M.11 in degree k', and make the construction of Subsections M.3 and M.4 with

$$N_i = \max_{p\le k'\le p+i} N_{k'}(i+1), \qquad \underline{r}_i = \max_{p\le k'\le p+i} r^*_{k'}(i+1) \qquad (i\ge 0); \tag{M.5}$$

then $N_i \ge i+1$, as Subsection M.2 requires.

Proof of Theorem 9.3. Step 1 (no splitting). Let $\alpha \ge 1$, $k \ge p$, and $i \in \mathcal{I}_\alpha$ with $i \ge k-p$. Identifying Y_i with $c_0(N_i)$ through the basis $e_2,\dots,e_{1+N_i}$ turns the map ι_i of Lemma M.7 into ι_{N_i,r_i}. By (M.5) and Lemma M.4(1), $N_i \ge N_k(i+1)$ and $r_i \ge r^*_k(i+1)$. Corollary M.12, with $m = i+1$, $J = J_i$ and $\Xi = \Xi_i$, shows that every s as in $(\mathrm{S}^k_{\mathfrak{X}_\alpha})$ and every ρ as in $(\mathrm{R}^k_{\mathfrak{X}_\alpha})$ have norm at least $(3/4)^k(i+1)/(2M_\varrho k(k-1)k!)$. Since $\mathcal{I}_\alpha$ is infinite, neither $(\mathrm{S}^k_{\mathfrak{X}_\alpha})$ nor $(\mathrm{R}^k_{\mathfrak{X}_\alpha})$ holds.

Step 2 (the four items). Let $\mathfrak{X} = \mathfrak{X}_1$. Item (1) is Theorem 9.2. Since $\mathrm{id}_{\mathfrak{X}} \ne 0$ (Lemma M.8), Proposition 9.1(1) with $E = D = \mathfrak{X}$ and $u_0 = \mathrm{id}_{\mathfrak{X}}$ makes $(\mathfrak{X},\mathfrak{X})$ good in every degree, hence good for Banach values by Lemma 3.4; this is (2). Item (3) is Step 1 for $\alpha = 1$. Since $\mathfrak{X}$ is complete, by Proposition 9.1(2) a splitting decomposition of degree k would make $\mathrm{id}_{\mathfrak{X}}$ S^k-split or R^k-split, that is, give $(\mathrm{S}^k_{\mathfrak{X}})$ or $(\mathrm{R}^k_{\mathfrak{X}})$, contrary to (3). This proves (4). □

Proof of Theorem 9.4. The inclusion $\mathrm{in}_\alpha\colon \mathfrak{X}_\alpha \to E$ of the α-th summand and the coordinate projection $\mathrm{pr}_\alpha\colon E \to \mathfrak{X}_\alpha$ are contractive, and $x = \sum_\alpha \mathrm{in}_\alpha\mathrm{pr}_\alpha x$ for $x \in E$.

Step 1 (item (1)). Let $u \in \mathcal{L}(E,E)$. By Theorem 9.2, $\mathrm{pr}_\beta u\,\mathrm{in}_\alpha = 0$ for $\beta \ne \alpha$ and $\mathrm{pr}_\alpha u\,\mathrm{in}_\alpha = \lambda_\alpha \mathrm{id}_{\mathfrak{X}_\alpha}$ for some $\lambda_\alpha \in K$, with $|\lambda_\alpha| \le \|u\|$ since $\mathfrak{X}_\alpha \ne 0$ (Lemma M.8). Hence $u\,\mathrm{in}_\alpha = \sum_\beta \mathrm{in}_\beta\mathrm{pr}_\beta u\,\mathrm{in}_\alpha = \lambda_\alpha\mathrm{in}_\alpha$ and $ux = \sum_\alpha \lambda_\alpha\mathrm{in}_\alpha\mathrm{pr}_\alpha x = u_\lambda x$. Conversely $\|u_\lambda x\| = \sum_\alpha|\lambda_\alpha|\|\mathrm{pr}_\alpha x\| \le \sup_\alpha|\lambda_\alpha|\|x\|$, with $\|u_\lambda x\| = |\lambda_\alpha|\|x\|$ on the summand $\mathfrak{X}_\alpha \ne 0$. So $\lambda \mapsto u_\lambda$ is an isometric isomorphism onto $\mathcal{L}(E,E)$ from the space ℓ^∞ of bounded sequences $(\lambda_\alpha)_{\alpha\ge1}$ in K.

Step 2 (item (2)). Let $k \ge 1$. Proposition 6.27 applies with $I = \{1,2,\dots\}$ in its natural order, $E_\alpha = D_\alpha = \mathfrak{X}_\alpha$, $D = E$, inclusions in_α, block projections pr_α, $J'_\alpha = \mathrm{in}_\alpha$, $\mathcal{U} = \mathcal{L}(E,E)$, $u_\alpha = \mathrm{pr}_\alpha u\,\mathrm{in}_\alpha$ and $c = C = 1$: $\|u_\alpha\| \le \|u\|$ and $ux = \sum_\alpha \mathrm{in}_\alpha u_\alpha \mathrm{pr}_\alpha x$ by Step 1, and for $n \le k$, since $\mathcal{L}(\mathfrak{X}_\alpha,\mathfrak{X}_\alpha) = K\mathrm{id}_{\mathfrak{X}_\alpha}$ with $\mathrm{id}_{\mathfrak{X}_\alpha} \ne 0$, Proposition 9.1(1) with $u_0 = \mathrm{id}_{\mathfrak{X}_\alpha}$ gives the lift $Q^\alpha_n(\theta_1\mathrm{id}_{\mathfrak{X}_\alpha},\dots,\theta_n\mathrm{id}_{\mathfrak{X}_\alpha}) = \theta_1\cdots\theta_n\Lambda^n_\pi\mathrm{id}_{\mathfrak{X}_\alpha}$, $\theta_j \in K$, which satisfies $Q^\alpha_n(v,\dots,v) = \Lambda^n_\pi v$ and $\|Q^\alpha_n\| \le 1$. So (E,E) is good for Banach values in degree k.

Step 3 (item (3)). Suppose that (E,E) has a generalized decomposition of degree $k \geq p$ (Definition 6.17) with pieces $(G_l, G_l', w_l, w_l', \varepsilon_l)$, so that $u = \sum_l w_l' \varepsilon_l(u) w_l$ for $u \in \mathcal{L}(E,E)$; the maps w_l' take values in the complete space E. Fix α and u; the compression $\mathrm{pr}_\alpha w_l' \varepsilon_l(u) w_l \mathrm{in}_\alpha$ lies in $\mathcal{L}(\mathfrak{X}_\alpha, \mathfrak{X}_\alpha) = K\mathrm{id}_{\mathfrak{X}_\alpha}$. For a piece of source or target type, and for the single piece of a relative splitting, w_l is S^k-split or w_l' is R^k-split, so the compression is S^k-split or R^k-split by the ideal property (Proposition 3.13(2)). If it were a nonzero multiple of $\mathrm{id}_{\mathfrak{X}_\alpha}$, the ideal property would make $\mathrm{id}_{\mathfrak{X}_\alpha}$ S^k-split or R^k-split, contrary to Step 1 of the proof of Theorem 9.3; so it is zero.

For a parameter piece, $\varepsilon_l = \varepsilon_l' \circ \psi_l$ with bounded linear $\psi_l \colon \mathcal{L}(E,E) \to P_l$ and $\varepsilon_l' \colon P_l \to \mathcal{L}(G_l, G_l')$, where P_l is polarizable in degrees $\leq k$. The linear maps $g_{l,\alpha} \colon P_l \to K$ given by $\mathrm{pr}_\alpha w_l' \varepsilon_l'(y) w_l \mathrm{in}_\alpha = g_{l,\alpha}(y)\mathrm{id}_{\mathfrak{X}_\alpha}$ satisfy $|g_{l,\alpha}(y)| \leq \|w_l'\|\|\varepsilon_l'\|\|w_l\|\|y\|$ for every α, since $\mathfrak{X}_\alpha \neq 0$. Compressing $u_\lambda = \sum_l w_l' \varepsilon_l(u_\lambda) w_l$ to the summand α gives $\lambda_\alpha = \sum_l g_{l,\alpha}(\psi_l(u_\lambda))$ for every $\lambda \in \ell^\infty$ and every α, the sum over the parameter pieces. Let P be the direct sum of their spaces P_l with the maximum norm. By Step 1 and the uniform bound, $\mathcal{A} \colon \ell^\infty \to P$, $\mathcal{A}\lambda = (\psi_l(u_\lambda))_l$, and $\mathcal{A}' \colon P \to \ell^\infty$, $\mathcal{A}'(y_l)_l = (\sum_l g_{l,\alpha}(y_l))_{\alpha \geq 1}$, are bounded linear, and $\mathcal{A}'\mathcal{A} = \mathrm{id}_{\ell^\infty}$.

The subspace $c_0 \subseteq \ell^\infty$ of null sequences, a copy of $c_0(\mathbb{N}, K)$, is complete with distances in $\varrho^{\mathbb{Z}} \cup \{0\}$, hence spherically complete by Lemma D.1. The extension argument behind Ingleton's Theorem D.3, in the form for spherically complete values given in the proof of Theorem 5.2, extends id_{c_0} to a contractive linear map $\mathcal{B} \colon \ell^\infty \to c_0$ on the nonarchimedean space ℓ^∞. Then $\mathcal{B}\mathcal{A}' \circ \mathcal{A}|_{c_0} = \mathrm{id}_{c_0}$, so c_0 is a retract of P, which is polarizable in degrees $\leq k$ by Proposition 6.18(2). By Proposition 6.18(3), c_0 is polarizable in degree k, contrary to Corollary 8.3, since K satisfies (L) and $k \geq p$. □

Appendix N. Properties of the obstruction class

This appendix contains the proofs to which Sections 4, 6 and 7 refer. Throughout, K is a nontrivially normed field, hence infinite.

N.1. **Pullbacks and morphisms.** Exact sequences, equivalences, classes and pushouts are as in Section 3.4, morphisms and pullbacks as in Section 6.3. Throughout this subsection $\mathcal{Y} \colon 0 \to Y_1 \xrightarrow{i} Y_2 \xrightarrow{q} Y_3 \to 0$ and $\mathcal{Y}' \colon 0 \to Y_1' \xrightarrow{i'} Y_2' \xrightarrow{q'} Y_3' \to 0$ are exact, with a constant C for $\mathcal{Y}$ as in Section 3.4.

Lemma N.1. *For a bounded linear map $\gamma \colon Y' \to Y_3$ from a Banach space Y', $\gamma^*\mathcal{Y}$ is exact, its class depends only on $[\mathcal{Y}]$, and $\gamma^*[\mathcal{Y}] = 0$ if and only if $\gamma = q \circ L$ for a bounded linear $L \colon Y' \to Y_2$.*

Proof. The space Y_γ is the kernel of $(y, y') \mapsto qy - \gamma y'$, hence complete. The first map has image $\{(y,0) : qy = 0\}$, the kernel of the second, and $\|(iy_1, 0)\| \geq \|y_1\|/C$. For $y' \in Y'$ some y with $qy = \gamma y'$ has $\|(y,y')\| \leq (1 + C\|\gamma\|)\|y'\|$. An equivalence T of $\mathcal{Y}$ with an exact sequence $\mathcal{Y}''$ induces the equivalence $(y,y') \mapsto (Ty, y')$ of the pullbacks. The splittings of $\gamma^*\mathcal{Y}$, the bounded linear right inverses of $(y,y') \mapsto y'$, are the maps $y' \mapsto (Ly', y')$ with L bounded linear and $qL = \gamma$. □

Lemma N.2. *For a morphism $(\alpha_1, \alpha_2, \alpha_3) \colon \mathcal{Y} \to \mathcal{Y}'$, $\alpha_{1*}[\mathcal{Y}] = \alpha_3^*[\mathcal{Y}']$ in $\mathrm{Ext}(Y_3, Y_1')$.*

Proof. The map $T[(y_1', y)] = (i'y_1' + \alpha_2 y, qy)$ from the middle space of $\alpha_{1*}\mathcal{Y}$, the quotient of $Y_1' \oplus Y_2$ by the vectors $(\alpha_1 y_1, -iy_1)$, to that of $\alpha_3^*\mathcal{Y}'$ is well defined, since $i'\alpha_1 = \alpha_2 i$ and $qi = 0$, bounded, and takes values in the pullback, since $q'(i'y_1' + \alpha_2 y) = \alpha_3 qy$. It sends $[(y_1', 0)]$ to $(i'y_1', 0)$ and has second coordinate qy, so it is an equivalence (Lemmas 3.6(3) and N.1). □

Corollary N.3. (a) *If* $(\alpha_1, \alpha_2, \mathrm{id})\colon \mathcal{Y} \to \mathcal{Y}'$ *is a morphism, then* $[\mathcal{Y}'] = \alpha_{1*}[\mathcal{Y}]$. *If* $\mathcal{Y}''\colon 0 \to Y_1 \xrightarrow{i''} Y_2'' \to Y_3'' \to 0$ *is exact and* $(\mathrm{id}, \alpha_2, \gamma)\colon \mathcal{Y}'' \to \mathcal{Y}$ *is a morphism, then* $[\mathcal{Y}''] = \gamma^*[\mathcal{Y}]$.

(b) *For bounded linear maps* $\alpha, \alpha', \gamma, \gamma'$ *of Banach spaces for which the expressions are defined,* $(\alpha'\alpha)_* = \alpha'_*\alpha_*$, $(\gamma\gamma')^* = \gamma'^*\gamma^*$, $\mathrm{id}_* = \mathrm{id}^* = \mathrm{id}$ *and* $\alpha_*\gamma^* = \gamma^*\alpha_*$.

(c) *For a bounded linear* $\alpha\colon Y_1 \to Y'$ *into a Banach space,* $\alpha_* 0 = 0$, *and* $(c\alpha)_*[\mathcal{Y}] = 0$ *if and only if* $\alpha_*[\mathcal{Y}] = 0$, *for* $c \in K$, $c \neq 0$.

Proof. (a) Lemma N.2 gives $\alpha_{1*}[\mathcal{Y}] = \mathrm{id}^*[\mathcal{Y}']$ and $\mathrm{id}_*[\mathcal{Y}''] = \gamma^*[\mathcal{Y}]$, and $(y, y_3) \mapsto y$ and $[(y_1, y)] \mapsto i''y_1 + y$ are equivalences of $\mathrm{id}^*\mathcal{Y}'$ with $\mathcal{Y}'$ and of $\mathrm{id}_*\mathcal{Y}''$ with $\mathcal{Y}''$. (b) The triples $(\alpha, y \mapsto [(0, y)], \mathrm{id})\colon \mathcal{Y} \to \alpha_*\mathcal{Y}$ and $(\mathrm{id}, (y, y') \mapsto y, \gamma)\colon \gamma^*\mathcal{Y} \to \mathcal{Y}$ are morphisms, since $[(0, iy_1)] = [(\alpha y_1, 0)]$. Composites of such morphisms, the identity morphism and (a) give the first three identities; the composite $\gamma^*\mathcal{Y} \to \mathcal{Y} \to \alpha_*\mathcal{Y}$ and Lemma N.2 give the last. (c) By Lemma 3.6(3), $\alpha_*[\mathcal{Y}] = 0$ if and only if $\alpha = \tilde{\alpha} i$ for a bounded linear $\tilde{\alpha}$, that is, $c\alpha = (c\tilde{\alpha})i$. If $[\mathcal{Y}] = 0$, then i has a bounded linear left inverse λ (Lemma 3.6(2)), and $\alpha = (\alpha\lambda)i$ gives $\alpha_* 0 = 0$. □

N.2. **Permutation groups.** The notation is that of Section 4, before Theorem 4.3 and in Proposition 4.5.

Proof of Theorem 4.3. Step 1 (notation). For a *tuple* $t\colon [k] \to I$ put $e_t = e_{t(1)} \otimes \cdots \otimes e_{t(k)}$; these form a basis of $\otimes_K^k V$, and $h^{\otimes k} e_t = e_{h\circ t}$. The *type* of t is the k-multiset $i \mapsto |t^{-1}(i)|$; if t has type $\mathbf{m}$, then $h \circ t$ has type $h\mathbf{m}$. For a k-subset T let $\mathrm{Bij}(T)$ be the set of bijections $[k] \to T$. Then $\wedge^k e_f = \mathrm{sgn}(f_T^{-1} f)\eta_T$ for $f \in \mathrm{Bij}(T)$, $\wedge^k e_t = 0$ for t not injective, the η_T form a basis of $\Lambda_K^k V$, and $\chi_T(h) = \mathrm{sgn}(f_T^{-1} h f_T)$ depends only on the permutation that h induces on T, so χ_T is also a homomorphism on $\bar{\mathsf{H}}_T$. Since $f_T^{-1} h f = (f_T^{-1} h f_T)(f_T^{-1} f)$ in S_k,

$$\mathrm{sgn}(f_T^{-1} h f) = \chi_T(h)\,\mathrm{sgn}(f_T^{-1} f) \qquad (h \in \mathsf{H}_T,\ f \in \mathrm{Bij}(T)). \tag{N.1}$$

The groups $\bar{\mathsf{H}}_T$ and $\bar{\mathsf{H}}_{\mathbf{m}}$ act freely, by $f \mapsto h \circ f$, on $\mathrm{Bij}(T)$ and on the tuples of type $\mathbf{m}$. This gives the *orbit count*: if $c_f \in \widehat{K}$ satisfy $c_{h\circ f} = \chi_T(h) c_f$ for $h \in \mathsf{H}_T$, then by (N.1) each orbit contributes $|\bar{\mathsf{H}}_T| c_f \,\mathrm{sgn}(f_T^{-1} f)$ to $\sum_{f \in \mathrm{Bij}(T)} c_f \,\mathrm{sgn}(f_T^{-1} f)$, which therefore lies in $|\bar{\mathsf{H}}_T|\widehat{K}$; if $c_t \in \widehat{K}$ satisfy $c_{h\circ t} = c_t$ for the tuples t of type $\mathbf{m}$ and $h \in \mathsf{H}_{\mathbf{m}}$, then $\sum_t c_t \in |\bar{\mathsf{H}}_{\mathbf{m}}|\widehat{K}$. Let $O_{\mathbf{m}}$ be the sum of the e_t over the tuples of type $\mathbf{m}$, as in Proposition 5.7. Then $h^{\otimes k} O_{\mathbf{m}} = O_{h\mathbf{m}}$, and the $O_{\mathbf{m}}$ form a basis of $\Gamma_K^k V$, the space of tensors fixed by every τ_σ, since $\tau_\sigma e_t = e_{t\circ\sigma^{-1}}$.

Step 2 (necessity). Assume the hypotheses of (4). For a tuple t let e_t^* be the bounded functional on $\widehat{\otimes}_\pi^k E$ linearizing $(x_j) \mapsto \prod_j e^*_{t(j)}(x_j)$; then $e_t^*(e_{t'}) = \delta_{tt'}$, and $e_t^* \circ h^{\otimes k} = e^*_{h^{-1}\circ t}$ by (4.3). Let ℓ_T be the bounded functional on $\widehat{\Lambda}_\pi^k E$ linearizing $(x_{j'}) \mapsto \det(e^*_{f_T(j)}(x_{j'}))_{j,j'}$ (Proposition 3.1). Then $\ell_T(\eta_T) = 1$; $\ell_T \circ \widehat{\Lambda}_\pi^k h = \chi_T(h)\ell_T$ for $h \in \mathsf{H}_T$, since (4.3) permutes the rows of the determinant by $f_T^{-1} h^{-1} f_T$; and $\ell_T \circ \widehat{\wedge}_E^k = \sum_{f \in \mathrm{Bij}(T)} \mathrm{sgn}(f_T^{-1} f) e_f^*$ by the Leibniz formula.

Let $|\bar{\mathsf{H}}_T| = 0$ in K. For s as in (4), $z = s(\eta_T)$ satisfies $z = \chi_T(h) h^{\otimes k} z$ for $h \in \mathsf{H}_T$, so $c_f = e_f^*(z)$ satisfies $c_{h\circ f} = \chi_T(h) c_f$, and $1 = \ell_T(\widehat{\wedge}_E^k z) = \sum_f c_f \,\mathrm{sgn}(f_T^{-1} f) = 0$ by the orbit count. For ρ as in (4), $a_f = \ell_T(\rho(e_f))$ satisfies $a_{h\circ f} = \chi_T(h) a_f$, and $\mathrm{alt}_E^k \eta_T = \sum_f \mathrm{sgn}(f_T^{-1} f) e_f$ gives $1 = \ell_T(\rho\,\mathrm{alt}_E^k \eta_T) = \sum_f \mathrm{sgn}(f_T^{-1} f) a_f = 0$.

Let $|\bar{\mathsf{H}}_{\mathbf{m}}| = 0$ in K, R be as in (4), and $t_{\mathbf{m}}$ a tuple of type $\mathbf{m}$. As $e_t^*(x^{\otimes k}) = \prod_j e^*_{t(j)}(x)$ depends only on the type of t, every e_t^* with t of type $\mathbf{m}$ agrees on $\widehat{\Gamma}_\pi^k E$ with $\varphi_{\mathbf{m}} = e^*_{t_{\mathbf{m}}}$. So for $h \in \mathsf{H}_{\mathbf{m}}$, $\varphi_{\mathbf{m}} \circ h^{\otimes k} = \varphi_{\mathbf{m}}$ on $\widehat{\Gamma}_\pi^k E$, and $c_t = \varphi_{\mathbf{m}}(R e_t)$ satisfies $c_{h\circ t} = c_t$. By Step 1,

$O_{\mathbf{m}}$ is a combination of pure powers of vectors in the span of the e_i, so $O_{\mathbf{m}} \in \widehat{\Gamma}^k_\pi E$, and $1 = \varphi_{\mathbf{m}}(RO_{\mathbf{m}}) = \sum_t c_t = 0$ by the orbit count. With the algebraic powers of V, its coordinate functionals and the determinant functional on $\Lambda^k_K V$, the same argument gives the necessity in (1)–(3).

Step 3 (sufficiency in (1)). Let $|\bar{\mathsf{H}}_T| \neq 0$ in K for all T, and let $\mathcal{T}$ be a set of representatives of the H-orbits of k-subsets. For $T \in \mathcal{T}$ put $z_T = |\bar{\mathsf{H}}_T|^{-1} \sum_{g \in \bar{\mathsf{H}}_T} \chi_T(g) e_{g \circ f_T} \in U_T$; then $\wedge^k z_T = \eta_T$ by (N.1), and $h'^{\otimes k} z_T = \chi_T(h') z_T$ for $h' \in \mathsf{H}_T$ by reindexing. The vectors $\Lambda^k_K g\, \eta_T = \pm \eta_{gT}$, one for each k-subset gT, form a basis of $\Lambda^k_K V$; put $s(\Lambda^k_K g\, \eta_T) = g^{\otimes k} z_T$. If $gT = g'T$, then $h' = g^{-1}g' \in \mathsf{H}_T$, and $\Lambda^k_K g' \eta_T$ and $g'^{\otimes k} z_T$ are $\chi_T(h')$ times $\Lambda^k_K g\, \eta_T$ and $g^{\otimes k} z_T$; so s is well defined. It is H-equivariant by construction and a section, since $\wedge^k g^{\otimes k} z_T = \Lambda^k_K g \wedge^k z_T$. For $\lambda \in (K^\times)^I$ let $d_\lambda e_i = \lambda_i e_i$. Then $d_\lambda^{\otimes k}$ and $\Lambda^k_K d_\lambda$ act on $U_{T'}$ and on $K\eta_{T'}$ by $\prod_{i \in T'} \lambda_i$, and $s(K\eta_{T'}) \subseteq U_{T'}$, so s commutes with d_λ.

Step 4 (sufficiency in (2) and (3)). For $T \in \mathcal{T}$ let φ_T be the functional on U_T with $\varphi_T(e_{g \circ f_T}) = |\bar{\mathsf{H}}_T|^{-1} \chi_T(g)$ for $g \in \bar{\mathsf{H}}_T$ that vanishes on the other e_f. Then $\varphi_T \circ h'^{\otimes k} = \chi_T(h') \varphi_T$ on U_T for $h' \in \mathsf{H}_T$, and $\varphi_T(\operatorname{alt}^k \eta_T) = 1$ by (N.1). Put $\rho(e_t) = 0$ for t not injective and $\rho(e_{g \circ f}) = \varphi_T(e_f) \Lambda^k_K g\, \eta_T$ for $g \in \mathsf{H}$, $T \in \mathcal{T}$ and $f \in \operatorname{Bij}(T)$; this is well defined, since $g \circ f = g' \circ f'$ implies $h' = g^{-1}g' \in \mathsf{H}_T$ and $f = h' \circ f'$. The map ρ is equivariant, $\rho(\operatorname{alt}^k \Lambda^k_K g\, \eta_T) = \rho(g^{\otimes k} \operatorname{alt}^k \eta_T) = \Lambda^k_K g\, \eta_T$, and $\rho(U_{T'}) \subseteq K\eta_{T'}$. For (3), choose representatives $\mathbf{m}$ of the H-orbits of k-multisets and tuples $t_{\mathbf{m}}$ of type $\mathbf{m}$. Let $\psi_{\mathbf{m}}(e_t)$ be $|\bar{\mathsf{H}}_{\mathbf{m}}|^{-1}$ for $t \in \bar{\mathsf{H}}_{\mathbf{m}} \circ t_{\mathbf{m}}$ and 0 for the other t of type $\mathbf{m}$; then $\psi_{\mathbf{m}}$ is $\mathsf{H}_{\mathbf{m}}$-invariant and $\psi_{\mathbf{m}}(O_{\mathbf{m}}) = 1$. Put $R(e_{g \circ t}) = \psi_{\mathbf{m}}(e_t) O_{g\mathbf{m}}$ for $g \in \mathsf{H}$ and t of type $\mathbf{m}$. As for ρ, R is well defined and equivariant; it maps into $\Gamma^k_K V$, and $R(O_{g\mathbf{m}}) = \psi_{\mathbf{m}}(O_{\mathbf{m}}) O_{g\mathbf{m}} = O_{g\mathbf{m}}$, so it is a projection onto $\Gamma^k_K V$. It maps e_t into $KO_{\mathbf{m}'}$, $\mathbf{m}'$ the type of t, on which $d_\lambda^{\otimes k}$ acts by the same factor as on e_t. □

Proof of Proposition 4.5. Step 1 (augmentation). Let $\operatorname{aug}_T \colon U_T \to K$ be linear with $\operatorname{aug}_T(e_f) = \operatorname{sgn}(f_T^{-1} f)$. Then $\wedge^k z = \operatorname{aug}_T(z) \eta_T$ for $z \in U_T$, so $\mathcal{N}_T = \ker \operatorname{aug}_T$; and $\operatorname{aug}_T(h \cdot z) = \operatorname{aug}_T(z)$ by (N.1), so $h \cdot e_{f_T} - e_{f_T} \in \mathcal{N}_T$. As $h \mapsto h \cdot y - y$ satisfies the cocycle identity for every $y \in U_T$, δ_T is a well-defined linear map. Call $z \in U_T$ *invariant* if $h \cdot z = z$ for all $h \in \mathsf{H}_T$, that is, $h^{\otimes k} z = \chi_T(h) z$.

Step 2 (item (1)). For $a \in K$, $\delta_T(a) = 0$ means that $z = a e_{f_T} - n$ is invariant for some $n \in \mathcal{N}_T$, and then $\operatorname{aug}_T(z) = a$; conversely, for an invariant z with $\operatorname{aug}_T(z) = a$, the vector $n = a e_{f_T} - z$ lies in $\mathcal{N}_T$ and gives $\delta_T(a) = 0$. For $a = 1$ this is (1).

Step 3 (items (2) and (3)). By Step 2, $\ker \delta_T$ is the image under aug_T of the invariant vectors. A vector $\sum_f c_f e_f$ is invariant if and only if $c_{h \circ f} = \chi_T(h) c_f$, so by the orbit count this image lies in $|\bar{\mathsf{H}}_T| K$; it contains $|\bar{\mathsf{H}}_T|$, the image of $\sum_{g \in \bar{\mathsf{H}}_T} \chi_T(g) e_{g \circ f_T}$. So $\ker \delta_T = |\bar{\mathsf{H}}_T| K$, which gives (2), since δ_T is linear. By (2), $\delta_T(1) = 0$ if and only if $|\bar{\mathsf{H}}_T| \neq 0$ in K, and Theorem 4.3(1) gives (3). □

N.3. **Operator families.** The notation is that of Section 6.3.

Proof of Theorem 6.22(2). The squares commute, since $\operatorname{ev}_\theta(\mathcal{N}^k_E) \subseteq \operatorname{Mult}^k_0(P; \widehat{\Lambda}^k_\pi D)$ and $\overline{\operatorname{ev}}_\theta \circ \widehat{\wedge}^k_E = q \circ \operatorname{ev}_\theta$ by the definition of $\overline{\operatorname{ev}}_\theta$ before Theorem 6.22. Lemma N.2 gives the formula. □

Proof of Proposition 6.23. Step 1 (items (a)–(c)). On pure tensors, hence everywhere, $\operatorname{ev}_{\theta \circ \psi} = \psi^\sharp \circ \operatorname{ev}_\theta$, $\operatorname{ev}_{\theta_w} = \operatorname{ev}_\theta \circ w^{\otimes k}$ and $\operatorname{ev}_{w'\theta} = (\widehat{\Lambda}^k_\pi w')_\flat \circ \operatorname{ev}_\theta$. Corollary N.3(b) gives (a) and (c). For (b), $(w^{\otimes k}|_{\mathcal{N}^k_{E'}}, w^{\otimes k}, \widehat{\Lambda}^k_\pi w)$ is a morphism $\mathcal{S}^k_{E'} \to \mathcal{S}^k_E$ (proof of Proposition 3.13), so Corollary N.3(b)

and Lemma N.2 give

$$\begin{aligned}\mathrm{ob}_k(\theta_w) &= (\mathrm{ev}_\theta|_{\mathcal{N}^k_E})_*(w^{\otimes k}|_{\mathcal{N}^k_{E'}})_*[\mathcal{S}^k_{E'}] = (\mathrm{ev}_\theta|_{\mathcal{N}^k_E})_*\,\mathrm{sp}_k(w)\\ &= (\mathrm{ev}_\theta|_{\mathcal{N}^k_E})_*(\widehat{\Lambda}^k_\pi w)^*[\mathcal{S}^k_E] = (\widehat{\Lambda}^k_\pi w)^*\,\mathrm{ob}_k(\theta).\end{aligned}$$

Step 2 (item (d)). The map $(\mathrm{alt}^k_D)_\flat$ preserves zero diagonals, so it induces a bounded linear $\overline{\mathrm{alt}}\colon \mathrm{Poly}^k(P;\widehat{\Lambda}^k_\pi D)\to\mathrm{Poly}^k(P;\widehat{\otimes}^k_\pi D)$ and a morphism $\mathcal{M}^k(P;\widehat{\Lambda}^k_\pi D)\to\mathcal{M}^k(P;\widehat{\otimes}^k_\pi D)$. By Theorem 6.22(2), Lemma N.2 and Corollary N.3(b), the class $((\mathrm{alt}^k_D)_\flat)_*\,\mathrm{ob}_k(\theta)$ equals $(\overline{\mathrm{alt}}\circ\overline{\mathrm{ev}}_\theta)^*[\mathcal{M}^k(P;\widehat{\otimes}^k_\pi D)]$. The bounded linear map $L\colon\widehat{\Lambda}^k_\pi E\to\mathrm{Mult}^k(P;\widehat{\otimes}^k_\pi D)$, $L(y)(h_1,\dots,h_k)=(\theta(h_1)\otimes\cdots\otimes\theta(h_k))\,\mathrm{alt}^k_E\,y$, has the diagonal $\theta(h)^{\otimes k}\,\mathrm{alt}^k_E\,y=\mathrm{alt}^k_D\,\widehat{\Lambda}^k_\pi\theta(h)y$, by naturality of alt^k; so its composite with the quotient map of $\mathcal{M}^k(P;\widehat{\otimes}^k_\pi D)$ is $\overline{\mathrm{alt}}\circ\overline{\mathrm{ev}}_\theta$, and Lemma N.1 gives the first assertion. For the second, extend w' to $\widehat{G}$, regard θ_0 as a family $P\to\mathcal{L}(E,\widehat{G})$, and let $\rho\colon\otimes^k_\pi G\to\widehat{\Lambda}^k_\pi D$ satisfy $\rho\circ\mathrm{alt}^k_G=\Lambda^k_\pi w'$, so that $\widehat{\Lambda}^k_\pi w'=\rho\circ\mathrm{alt}^k_G$ on the completions. By (c), Corollary N.3(b),(c) and the first assertion for θ_0, $\mathrm{ob}_k(w'\theta_0)=(\rho_\flat)_*((\mathrm{alt}^k_G)_\flat)_*\,\mathrm{ob}_k(\theta_0)=0$, and Theorem 6.22(1) concludes. □

Lemma N.4. *Let P_1 and P_2 be normed spaces, $P=P_1\oplus P_2$ with the maximum norm, Y a Banach space, and Φ_S, $\Pi^{(a)}$ as in Theorem 6.25. For $\Phi\in\mathrm{Mult}^k(P;Y)$, $\Phi=\sum_{S\subseteq[k]}\Phi_S$, $(\Phi_S)_{S'}$ is Φ_S if $S'=S$ and 0 otherwise, and $\|\Phi_S\|\le\|\Phi\|$. Hence the $\Pi^{(a)}$ are bounded commuting idempotents with sum* id. *Each maps $\mathrm{Mult}^k_0(P;Y)$ into itself.*

Proof. Expanding $h^j=(h^j_1,0)+(0,h^j_2)$ multilinearly gives $\Phi=\sum_S\Phi_S$. The map $(\Phi_S)_{S'}$ evaluates Φ at a tuple whose jth entry is 0 for j in exactly one of S and S', and equals Φ_S if $S'=S$. Since $(h^j_1,0)$ and $(0,h^j_2)$ have norm at most $\|h^j\|$, $\|\Phi_S\|\le\|\Phi\|$. For $\Phi\in\mathrm{Mult}^k_0(P;Y)$, $h=(h_1,h_2)\in P$ and $\lambda\in K$,

$$0=\Phi((\lambda h_1,h_2),\dots,(\lambda h_1,h_2))=\sum_S\lambda^{|S|}\Phi_S(h,\dots,h)=\sum_{a=0}^k\lambda^a(\Pi^{(a)}\Phi)(h,\dots,h).$$

Since K is infinite, the Vandermonde determinant for $k+1$ distinct values of λ shows that $(\Pi^{(a)}\Phi)(h,\dots,h)=0$ for every a. □

Proof of Theorem 6.25. Put $\mathrm{Mult}^k_0=\mathrm{Mult}^k_0(P;Y)$, $\mathrm{Mult}^{(a)}=\Pi^{(a)}\,\mathrm{Mult}^k(P;Y)$ and $\mathrm{ev}=\mathrm{ev}_{\theta_1\boxplus\theta_2}$. By Lemma N.4, $\mathrm{Mult}^{(a)}$ and $\mathrm{Mult}^{(a)}_0$ are closed and $\mathrm{Mult}^{(a)}\cap\mathrm{Mult}^k_0=\mathrm{Mult}^{(a)}_0$; by Corollary N.3(b), $\mathrm{ob}^{(a)}=(\Pi^{(a)}\circ\mathrm{ev}|_{\mathcal{N}^k_E})_*[\mathcal{S}^k_E]$.

Step 1 (item (1)). By Lemma 3.6(3), $\mathrm{ob}_k(\theta_1\boxplus\theta_2)=0$ if and only if $\mathrm{ev}|_{\mathcal{N}^k_E}$ extends to a bounded linear $\xi\colon\widehat{\otimes}^k_\pi E\to\mathrm{Mult}^k_0$, and $\mathrm{ob}^{(a)}=0$ if and only if $\Pi^{(a)}\,\mathrm{ev}|_{\mathcal{N}^k_E}$ extends to a bounded linear $\xi_a\colon\widehat{\otimes}^k_\pi E\to\mathrm{Mult}^{(a)}_0$. An extension ξ gives the extensions $\Pi^{(a)}\xi$, and extensions ξ_a give the extension $\sum_a\xi_a$. For $S\ne[k]$, $\Phi_S\circ\iota_1^{\times k}$ evaluates Φ at tuples with a zero entry, so $\iota_1^\sharp=\iota_1^\sharp\Pi^{(k)}$. With the projections $\mathrm{pr}_l\colon P\to P_l$, $l=1,2$, $\Phi_{[k]}=(\iota_1^\sharp\Phi)\circ\mathrm{pr}_1^{\times k}$, so $\iota_1^\sharp$ is an isometry of $\mathrm{Mult}^{(k)}$ onto $\mathrm{Mult}^k(P_1;Y)$ with inverse $\mathrm{pr}_1^\sharp$, and both preserve zero diagonals. By Corollary N.3(b) and Proposition 6.23(a), $(\iota_1^\sharp)_*\,\mathrm{ob}^{(k)}=(\iota_1^\sharp)_*\,\mathrm{ob}_k(\theta_1\boxplus\theta_2)=\mathrm{ob}_k((\theta_1\boxplus\theta_2)\circ\iota_1)=\mathrm{ob}_k(\theta_1)$. The case of ι_2, with $S=\emptyset$ and pr_2, is the same.

Step 2 (item (2)). Let $\Theta_a(h_1,h_2)$ be the right side of the identity in (2). By Lemma 3.7, the relation $\Pi^{(a)}\,\mathrm{ev}-\xi_a=\Psi\circ\widehat{\wedge}^k_E$ matches the extensions ξ_a of Step 1 with the bounded linear $\Psi\colon\widehat{\Lambda}^k_\pi E\to\mathrm{Mult}^{(a)}$ such that $\Psi(\widehat{\wedge}^k_E z)-\Pi^{(a)}\,\mathrm{ev}(z)\in\mathrm{Mult}^k_0$ for all z. For $z=x_1\otimes\cdots\otimes x_k$ and $h=(h_1,h_2)$, $\mathrm{ev}(z)_S(h,\dots,h)$ is the wedge, in the order of j, of the vectors $\theta_1(h_1)x_j$, $j\in S$,

and $\theta_2(h_2)x_j$, $j \notin S$; by the expansion before (6.3), the sum over $|S| = a$ is $\Theta_a(h_1,h_2)(x_1 \wedge \cdots \wedge x_k)$. By continuity, the condition on Ψ says $\Psi(y)(h,\dots,h) = \Theta_a(h_1,h_2)y$. Given $Q^{(a)}$, the map $\Psi(y)(h^1,\dots,h^k) = Q^{(a)}(h_1^1,\dots,h_1^a;h_2^{a+1},\dots,h_2^k)y$ satisfies it and lies in $\mathrm{Mult}^{(a)}$, since $\Psi(y) = \Psi(y)_{[a]}$ (Lemma N.4). Conversely, given Ψ, for $g \in P_1^a$, $g' \in P_2^{k-a}$ and $|S| = a$ let $(g;g')_S \in P^k$ have the entries $(g_1,0),\dots,(g_a,0)$ in the slots of S and $(0,g'_1),\dots,(0,g'_{k-a})$ in the others, in increasing order, and put $Q^{(a)}(g;g')y = \sum_{|S|=a}\Psi(y)((g;g')_S)$; its diagonal is $(\Pi^{(a)}\Psi(y))(h,\dots,h) = \Theta_a(h_1,h_2)y$.

Step 3 (items (3)–(5)). (3) For lifts Q_a of θ_1 and Q'_{k-a} of θ_2, the map $\wedge_{\widehat{D}}^{a,k-a}\circ(Q_a(g)\otimes Q'_{k-a}(g'))\circ \mathrm{sh}_E^{a,k-a}$, extended to completions, is bounded and k-linear by Lemma 6.1, with diagonal $\Theta_a(h_1,h_2)$; apply Step 2. (4) If all $\mathrm{ob}^{(a)}$ vanish, then $\mathrm{ob}_k(\theta_1 \boxplus \theta_2) = 0$ by (1), so $\mathrm{ob}_k(\theta_1+\theta_2) = (\Delta_{P_1}^\sharp)_*\,\mathrm{ob}_k(\theta_1\boxplus\theta_2) = 0$ by Proposition 6.23(a) and Corollary N.3(c), and Theorem 6.22(1) applies. If θ_1 and θ_2 are liftable in every degree at most k, then $\mathrm{ob}_k(\theta_1) = \mathrm{ob}_k(\theta_2) = 0$ by Theorem 6.22(1); applying $(\mathrm{pr}_1^\sharp)_*$ and $(\mathrm{pr}_2^\sharp)_*$ to the identities of (1) gives $\mathrm{ob}^{(k)} = \mathrm{ob}^{(0)} = 0$ by Corollary N.3(b),(c), and the other $\mathrm{ob}^{(a)}$ vanish by (3). (5) For $|S| = a$, $(\mathrm{ev}_{\theta\boxplus\theta}(z))_S \circ \Delta_{P_1}^{\times k} = \mathrm{ev}_\theta(z)$, on pure tensors since $(\theta\boxplus\theta)(h,0) = (\theta\boxplus\theta)(0,h) = \theta(h)$, hence everywhere; summing over the $\binom{k}{a}$ sets S gives the identity. By Corollary N.3(b), $(\Delta_{P_1}^\sharp|_{\mathrm{Mult}_0^{(a)}})_*\,\mathrm{ob}^{(a)} = (\binom{k}{a}\,\mathrm{ev}_\theta|_{\mathcal{N}_E^k})_*[\mathcal{S}_E^k]$, which by Corollary N.3(c) vanishes if and only if $\mathrm{ob}_k(\theta) = 0$. If θ is liftable in degrees a and $k-a$, then $\mathrm{ob}^{(a)} = 0$ by (3), so $\mathrm{ob}_k(\theta) = 0$, and Theorem 6.22(1) applies. □

Proof of Proposition 6.26. (1) As $(\theta' \odot \theta)(\psi_h h') = \theta'(h')\circ\theta(h)$, the family $(\theta'\odot\theta)\circ\psi_h$ is the family θ'_w of Proposition 6.23(b) for θ' on $\widehat{D}$ and $w = \theta(h) \in \mathcal{L}(E,\widehat{D})$, with $\widehat{\Lambda}_\pi^k w$ mapping into $\widehat{\Lambda}_\pi^k\widehat{D} = \widehat{\Lambda}_\pi^k D$; items (a) and (b) give the first identity. Likewise $(\theta'\odot\theta)\circ\psi'_{h'}$ is the family $w'\theta$ of Proposition 6.23(c) for $w' = \theta'(h')\colon \widehat{D}\to\widehat{F}$, and (a) and (c) give the second. (2) On pure tensors $(\theta_{D,F}\odot\theta_{E,D})(v\otimes u) = vu$, so $\theta_{D,F}\odot\theta_{E,D} = \theta_{E,F}\circ\mathrm{comp}$, and Proposition 6.23(a) gives the identity. If (E,F) is good for Banach values in degree k, then $\mathrm{ob}_k(E,F) = 0$ by Theorem 3.15, so the class of $\theta_{D,F}\odot\theta_{E,D}$ vanishes by Corollary N.3(c), and Theorem 6.22(1) applies. □

N.4. **A three-space counterexample.** The Cantor set Ω, its dyadic cells and the spaces $L^1(\Omega;U)$ are as in Theorem 6.29.

Lemma N.5. *Let K be nonarchimedean, U a Banach space, U_0 a closed subspace of $L^1(\Omega;U)$, and Y a normed space with an equivalent nonarchimedean norm. Then every bounded linear map $L^1(\Omega;U)/U_0 \to Y$ is zero.*

Proof. Let ψ be such a map, ψ_0 its composite with the quotient map, and ν a nonarchimedean norm on Y with $\nu \le c_\nu\|\cdot\|$. For a dyadic cell C_ξ of level m, $u \in U$ and $j \ge 0$, $1_{C_\xi}u$ is the sum of the 2^j functions $1_{C_{\xi'}}u$, $C_{\xi'} \subseteq C_\xi$ of level $m+j$, each of norm $2^{-m-j}\|u\|$. So $\nu(\psi_0(1_{C_\xi}u)) \le \max_{\xi'}\nu(\psi_0(1_{C_{\xi'}}u)) \le c_\nu\|\psi_0\|2^{-m-j}\|u\|$, and $j\to\infty$ gives $\psi_0(1_{C_\xi}u) = 0$. Step functions are dense, so $\psi_0 = 0$, and $\psi = 0$. □

Lemma N.6. *Let Y, Y', Y'' be Banach spaces, $T\colon Y\to Y'$ bounded linear such that every $y' \in Y'$ equals Ty for some y with $\|y\| \le C_T\|y'\|$, $R = \ker T$, and $\eta\colon Y\to Y''$ bounded linear and bounded below. Give $Y'\oplus Y''$ the maximum norm. Then $y \mapsto (-Ty,\eta y)$ and $(y',y'') \mapsto [y''+\eta(y_{y'})]$, where $Ty_{y'} = y'$, form an exact sequence $0\to Y\to Y'\oplus Y''\to Y''/\eta(R)\to 0$.*

Proof. The subspace $\eta(R)$ is complete, hence closed. Two choices of $y_{y'}$ differ by an element of R, so the second map is well defined and linear; it is bounded for the choice $\|y_{y'}\| \le C_T\|y'\|$, and it maps $(0, y'')$ to $[y'']$, where y'' can be chosen with $\|y''\| \le 2\|[y'']\|$. If $y'' + \eta y_{y'} = \eta r$ with $r \in R$, then $(y', y'') = (-Tw, \eta w)$ for $w = r - y_{y'}$; conversely $(-Tw, \eta w)$ maps to $[\eta w - \eta w] = 0$, with $y_{-Tw} = -w$. The first map is bounded below, as η is. □

Proof of Theorem 6.31. Step 1 (exactness). As ϖ_D is open (Corollary 6.30(2)), for some $\delta > 0$ the image of the open unit ball contains the open ball of radius δ. For $d \neq 0$ and $j \in \mathbb{Z}$ with $|\vartheta|^{j+1}\delta \le \|d\| < |\vartheta|^j\delta$, ϑ as in Section 7.1, some y with $\|y\| < 1$ has $\varpi_D y = \vartheta^{-j}d$, and $y_d = \vartheta^j y$ has $\|y_d\| < \|d\|/(|\vartheta|\delta)$. As η is isometric, Lemma N.6 applies with $T = \varpi_D$.

Step 2 (items (1) and (2)). By Proposition 5.5(3), $\ell^1(\mathrm{Ball}(D), K)$ satisfies (S^n) for every n, and it is dense in Y_1; so Proposition 6.28(3) gives $(\mathrm{S}^n_{Y_1})$, and Proposition 3.5(1) gives (1). Under (L), K is nonarchimedean, so Lemma N.5 with $U = Y_1$, the closed subspace $U_0 = \eta(\ker \varpi_D)$ and $Y = D$ gives $\mathcal{L}(Y_3, D) = 0$; the zero map on $\mathcal{L}(Y_3, D)^n$ is then a lift as in Lemma 3.4 for every $n \ge 1$.

Step 3 (item (3)). Let $J\colon D \to Y_2$ be $d \mapsto (d, 0)$ and $\Pi'\colon Y_2 \to D$ the first coordinate. If (Y_2, D) were good for Banach values in degree k, Lemma 6.16(2), with J, Π' and $J' = \Pi = \mathrm{id}_D$, would make (D, D) good for Banach values in degree k, contrary to Theorem 7.19(2), since D is complete, has an equivalent nonarchimedean norm and $\dim_{\widehat{K}} D = \infty$. □

N.5. **Characteristic two.** Throughout this subsection the notation is that of Section 7.6, $\Bbbk$ is a field of characteristic 2, and K is a nontrivially normed field of characteristic 2. By Lemma A.2, K is nonarchimedean, and so is its completion $\widehat{K}$; K is infinite, as a finite field has only the trivial absolute value. For z in $\Lambda^k K^n$ or in $K^n \otimes K^n$, with coordinates z_c,

$$\max_c |z_c| \le \|z\|_\pi \le \sum_c |z_c|, \tag{N.2}$$

since each coordinate of $x_1 \wedge \cdots \wedge x_k$ is a $k \times k$ minor, of absolute value at most $\prod_a \|x_a\|$ by the ultrametric inequality. So the completions of $\Lambda^k_\pi c_0(n)$ and $c_0(n) \otimes_\pi c_0(n)$ are coordinate spaces over $\widehat{K}$ with norms equivalent to the coordinate supremum norm, and their elements are identified with their coordinate vectors.

Definition N.7. Let $J \subseteq [n]$, let $x|_J$ be the restriction of $x \in \Bbbk^n$ to J, and let $\mathrm{Sym}_J(\Bbbk)$ be the space of symmetric $J \times J$ matrices. A *J-polarization* is a $\Bbbk$-bilinear map $\mathsf{N}\colon \Bbbk^n \times \Bbbk^n \to \mathrm{Sym}_J(\Bbbk)$ with $\mathsf{N}(x, x) = x|_J(x|_J)^{\mathsf{T}}$. Regarding the entries of x and y also as indeterminates, $\mathsf{N}(x, y) = \sum_{t,t'} x_t y_{t'} \mathsf{N}(e_t, e_{t'})$ is a matrix over $\Bbbk[x, y]$. For $L, L' \subseteq [n]$ put $x_L y_{L'} = \prod_{t\in L} x_t \prod_{t'\in L'} y_{t'}$, and for $g \in \Bbbk[x, y]$ let $c_g(L, L')$ be the coefficient of $x_L y_{L'}$ in g.

Lemma N.8. *Let J be finite, M a symmetric $J \times J$ matrix over a commutative ring of characteristic 2, λ a ring element, and $\mathsf{M}[J']$ the principal submatrix on $J' \subseteq J$, with $\det \mathsf{M}[\emptyset] = 1$. For $a \neq a'$ in J,*

$$\det(\mathsf{M} + \lambda(e_a e_{a'}^{\mathsf{T}} + e_{a'} e_a^{\mathsf{T}})) = \det \mathsf{M} + \lambda^2 \det \mathsf{M}[J \setminus \{a, a'\}],$$
$$\det(\mathsf{M} + \lambda e_a e_a^{\mathsf{T}}) = \det \mathsf{M} + \lambda \det \mathsf{M}[J \setminus \{a\}].$$

Proof. The second identity is the expansion along row a. For the first, expand multilinearly in the rows a and a', which change by $\lambda e_{a'}^{\mathsf{T}}$ and λe_a^{T}. The two first-order terms are λ times the cofactors of M at (a, a') and (a', a), which are equal as M is symmetric, so they cancel; the second-order term is $\pm\lambda^2 \det \mathsf{M}[J \setminus \{a, a'\}]$, and signs do not matter in characteristic 2. □

Lemma N.9. *Let $J \subseteq [n]$, and let N and N' be J-polarizations. Then $c_{\det \mathsf{N}}(L, L') = c_{\det \mathsf{N}}(L', L)$ and $c_{\det \mathsf{N}}(L, L) = c_{\det \mathsf{N}'}(L, L)$ for all $L, L' \subseteq [n]$.*

Proof. Step 1 (swap symmetry). Expanding $\mathsf{N}(x+y,x+y)$ at unit vectors gives $\mathsf{N}(y,x) = \mathsf{N}(x,y) + x|_J(y|_J)^{\mathsf{T}} + y|_J(x|_J)^{\mathsf{T}}$, and $x|_J(y|_J)^{\mathsf{T}} + y|_J(x|_J)^{\mathsf{T}}$, whose diagonal vanishes, is the sum of the matrices $(x_a y_{a'} + x_{a'} y_a)(e_a e_{a'}^{\mathsf{T}} + e_{a'} e_a^{\mathsf{T}})$ over $a < a'$ in J. Adding them one at a time, Lemma N.8 changes the determinant by terms $(x_a^2 y_{a'}^2 + x_{a'}^2 y_a^2)g$, whose monomials contain a squared variable. So the coefficient of $x_L y_{L'}$ in $\det \mathsf{N}(x,y)$ equals that in $\det \mathsf{N}(y,x)$, which is $c_{\det \mathsf{N}}(L', L)$.

Step 2 (invariance). The map $\mathsf{A} = \mathsf{N}' - \mathsf{N}$ is bilinear with symmetric values and $\mathsf{A}(x,x) = 0$, so $\mathsf{A}(e_t, e_t) = 0$, $\mathsf{A}(e_{t'}, e_t) = \mathsf{A}(e_t, e_{t'})$ and $\mathsf{A}(x,y) = \sum_{t<t'}(x_t y_{t'} + x_{t'} y_t)\mathsf{A}(e_t, e_{t'})$. Writing each $\mathsf{A}(e_t, e_{t'})$ as a combination of the matrices $e_a e_a^{\mathsf{T}}$ and $e_a e_{a'}^{\mathsf{T}} + e_{a'} e_a^{\mathsf{T}}$, we pass from N to N' through J-polarizations, each obtained from the previous one, N_1, by adding $\lambda(x_t y_{t'} + x_{t'} y_t)\mathsf{E}$ with $\lambda \in \Bbbk$ and E one of these matrices. For $\mathsf{E} = e_a e_{a'}^{\mathsf{T}} + e_{a'} e_a^{\mathsf{T}}$, Lemma N.8 changes the determinant by $\lambda^2(x_t^2 y_{t'}^2 + x_{t'}^2 y_t^2)g$, which has no term $x_L y_L$. For $\mathsf{E} = e_a e_a^{\mathsf{T}}$, the change is $\lambda(x_t y_{t'} + x_{t'} y_t)g$ with $g = \det \mathsf{N}_1(x,y)[J \setminus \{a\}]$; its coefficient of $x_L y_L$ is 0 unless $t, t' \in L$, when it is $\lambda(c_g(L \setminus \{t\}, L \setminus \{t'\}) + c_g(L \setminus \{t'\}, L \setminus \{t\})) = 0$ by Step 1 for the $(J \setminus \{a\})$-polarization $\mathsf{N}_1[J \setminus \{a\}]$. □

Lemma N.10. *The map with* $\mathsf{N}_<(x,y)_{tt'} = x_{\max(t,t')} y_{\min(t,t')}$ *is an* $[n]$*-polarization,*

$$\det \mathsf{N}_<(x,y) = y_1 x_n \prod_{t=1}^{n-1} (x_t y_{t+1} + x_{t+1} y_t),$$

and $c_{\det \mathsf{N}_<}([n],[n]) = 1$.

Proof. The map is bilinear with symmetric values, and $\mathsf{N}_<(x,x)_{tt'} = x_t x_{t'}$. For $n = 1$ the determinant is $x_1 y_1$. For $n \geq 2$, over the field of rational functions, subtracting x_n/x_{n-1} times row $n-1$ from row n leaves the last row $(0, \dots, 0, (x_n/x_{n-1})(x_{n-1} y_n + x_n y_{n-1}))$ below the matrix $\mathsf{N}_<$ for $n-1$. By induction $\det \mathsf{N}_< = x_1 y_1 \prod_{t=2}^{n} (x_t/x_{t-1})(x_{t-1} y_t + x_t y_{t-1})$, which telescopes to the formula. A term of the product equal to $x_{[n]} y_{[n]}$ takes $x_{n-1} y_n$ from the factor of index $n-1$, since x_n is used, then $x_{n-2} y_{n-1}$ from the factor of index $n-2$, and so on; so the coefficient is 1. □

Lemma N.11. *Let* g *be a polynomial in* x *and* y *over a commutative ring of characteristic* 2*, of degree at most* n *in* x *and at most* n *in* y*. Then* $\sum_{S,T \subseteq [n]} g(1_S, 1_T)$ *is the coefficient of* $x_{[n]} y_{[n]}$ *in* g*. Likewise, if* g *is a polynomial in* x *of degree at most* n*, then* $\sum_{S \subseteq [n]} g(1_S)$ *is the coefficient of* $x_1 \cdots x_n$*.*

Proof. A monomial $x^\alpha y^\beta$ takes the value 1 at $(1_S, 1_T)$ if $\operatorname{supp} \alpha \subseteq S$ and $\operatorname{supp} \beta \subseteq T$, and 0 otherwise, so its sum over S, T is $2^{2n - |\operatorname{supp} \alpha| - |\operatorname{supp} \beta|}$. This vanishes unless both supports are $[n]$, which under the degree bounds happens only for $x_{[n]} y_{[n]}$. The second statement is proved in the same way. □

Proposition N.12. *Let* $n \geq 2$ *and* $\sigma \colon \Lambda^2 \Bbbk^n \to \Bbbk^{n \times n}$ *be* $\Bbbk$*-linear with* $\wedge \circ \sigma = \mathrm{id}$*. Then* $\mathsf{N}(x,y) = \sigma(x \wedge y) + xy^{\mathsf{T}}$ *is an* $[n]$*-polarization,* $\sum_{S,T \subseteq [n]} \det \mathsf{N}(1_S, 1_T) = 1$*, and* $\operatorname{rank} \sigma(1_S \wedge 1_T) \geq n - 1$ *for some* $S, T \subseteq [n]$*.*

Proof. The map $\Phi(x,y) = \sigma(x \wedge y)$ is bilinear, $\Phi(x,x) = 0$, and $\Phi(x,y) + \Phi(x,y)^{\mathsf{T}} = \Omega_{x \wedge y} = xy^{\mathsf{T}} + yx^{\mathsf{T}}$; so N is bilinear, $\mathsf{N}(x,y)^{\mathsf{T}} = \mathsf{N}(x,y)$ and $\mathsf{N}(x,x) = xx^{\mathsf{T}}$. By Lemmas N.9 and N.10, $c_{\det \mathsf{N}}([n],[n]) = c_{\det \mathsf{N}_<}([n],[n]) = 1$, and $\det \mathsf{N}(x,y)$ has degree n in x and in y, so Lemma N.11 gives the identity. Hence some $\mathsf{N}(1_S, 1_T)$ is invertible, and $\sigma(1_S \wedge 1_T) = \mathsf{N}(1_S, 1_T) + 1_S 1_T^{\mathsf{T}}$ has rank at least $n-1$. □

Proposition N.13. *Let $n \geq 2$ and $\tau\colon \Bbbk^{n\times n} \to \Lambda^2\Bbbk^n$ be $\Bbbk$-linear with $\tau(\Omega_w) = w$ for $w \in \Lambda^2\Bbbk^n$. If n is even, $\sum_{S\subseteq[n]} \operatorname{Pf} \Omega_{\tau(1_S 1_S^{\mathsf{T}})} = 1$, and some $\Omega_{\tau(1_S 1_S^{\mathsf{T}})}$ is invertible. For every n, some $S \subseteq [n]$ has $\operatorname{rank} \Omega_{\tau(1_S 1_S^{\mathsf{T}})} \geq 2\lfloor n/2 \rfloor$.*

Proof. Step 1 (even n). Let n be even and $w_t = \tau(e_t e_t^{\mathsf{T}})$. Since $\tau(e_t e_{t'}^{\mathsf{T}}) + \tau(e_{t'} e_t^{\mathsf{T}}) = \tau(\Omega_{e_t \wedge e_{t'}}) = e_t \wedge e_{t'}$, $q(x) = \tau(xx^{\mathsf{T}})$ equals $\sum_t x_t^2 w_t + \sum_{t<t'} x_t x_{t'} e_t \wedge e_{t'}$, and $\Omega_{q(x)}$ has the off-diagonal entries $x_t x_{t'} + \sum_{t''} x_{t''}^2 (\Omega_{w_{t''}})_{tt'}$. In a product over a partition into pairs, a factor $x_{t''}^2$ gives a squared variable, so the coefficient of $x_1 \cdots x_n$ in $\operatorname{Pf} \Omega_{q(x)}$, a polynomial of degree n, is the number $(n-1)(n-3)\cdots 1$ of these partitions, which is odd. Lemma N.11 gives the identity, so some $\operatorname{Pf} \Omega_{q(1_S)}$ is nonzero, and $\Omega_{q(1_S)}$ is invertible.

Step 2 (odd n). For odd n let $\iota'\colon \Bbbk^{n-1} \to \Bbbk^n$ be the inclusion of the first $n-1$ coordinates and $\tau'(\mathsf{Z}) = \Lambda^2 \iota'^{\mathsf{T}} \tau(\iota' \mathsf{Z} \iota'^{\mathsf{T}})$. As $\iota' \Omega_w \iota'^{\mathsf{T}} = \Omega_{\Lambda^2 \iota' w}$ and $\Omega_{\Lambda^2 \iota'^{\mathsf{T}} w} = \iota'^{\mathsf{T}} \Omega_w \iota'$, the map τ' satisfies the hypothesis for $n-1$, and $\operatorname{rank} \Omega_{\tau'(\mathsf{Z})} \leq \operatorname{rank} \Omega_{\tau(\iota' \mathsf{Z} \iota'^{\mathsf{T}})}$; the even case for τ' and $\iota' 1_S 1_S^{\mathsf{T}} \iota'^{\mathsf{T}} = 1_S 1_S^{\mathsf{T}}$ give the last assertion. □

Proposition N.14 (Pfaffian identity)**.** *Let $n \geq 2$ be even and Ψ a multiplier lift on $\Bbbk^n$. Then (7.6) holds, and*

$$\sum_L \sum_{a,x} \sum_{b,y} \operatorname{Pf} \mathsf{M}_\Psi(a,b,x,y) = 1, \tag{N.3}$$

the sums over the $L \subseteq [n]$ with $|L| = n/2$, over the $a, x \in \{0,1\}^n$ supported in L, and over the $b, y \in \{0,1\}^n$ supported in L^c.

Proof. Step 1 (decomposition). For $t \neq t'$ put $\Psi^{\mathrm{cor}}_{tt'} = \Psi_{tt'} - \Psi^{<}_{tt'}$. By (7.5), $\Psi^{\mathrm{cor}}_{tt'} + \Psi^{\mathrm{cor}}_{t't} = \pi_{tt'} - \pi_{tt'} = 0$, so $\Psi^{\mathrm{cor}}_{tt'} = \Psi^{\mathrm{cor}}_{t't}$; write $\Psi^{\mathrm{cor}}_{\{t,t'\}}$ for it. For a 2-subset $\varepsilon = \{t,t'\}$ of $[n]$ with $t < t'$ put $e_\varepsilon = e_t \wedge e_{t'}$, and write $\Psi^{\mathrm{cor}}_{\{t,t'\}}[\varepsilon, \varepsilon']$ for the entry of $\Psi^{\mathrm{cor}}_{\{t,t'\}}$ in the row of e_ε and the column of $e_{\varepsilon'}$. Fix L with $|L| = n/2$. In the monomial $a_L b_{L^c} x_L y_{L^c}$ every a-variable has its index in L and every b-variable in L^c, so only the $\Psi_{tt'}$ with $t \in L$ and $t' \in L^c$, in particular $t \neq t'$, contribute. As the coordinate of $x \wedge y$ at $e_{\{u,u'\}}$ is $x_u y_{u'} + x_{u'} y_u$, for $t \neq t'$ and $u \neq u'$ the coefficient of $a_t b_{t'} x_u y_{u'}$ in the entry of M_Ψ at a 2-subset ε is

$$O_\varepsilon(t,t',u,u') + \Psi^{\mathrm{cor}}_{\{t,t'\}}[\varepsilon, \{u,u'\}], \tag{N.4}$$

where $O_\varepsilon(t,t',u,u') = 1$ if $t < t'$ and $\{t,t'\} = \{u,u'\} = \varepsilon$, and $O_\varepsilon(t,t',u,u') = 0$ otherwise.

Step 2 (configurations). Expanding the Pfaffian over the perfect matchings $\mathfrak{m}$ of $[n]$, whose elements we call edges, and each entry by (N.4), the coefficient of $a_L b_{L^c} x_L y_{L^c}$ in $\operatorname{Pf} \mathsf{M}_\Psi$ is the sum of the values of the *configurations* for L. A configuration consists of a perfect matching $\mathfrak{m}$, an assignment $\varepsilon \mapsto (t_\varepsilon, t'_\varepsilon, u_\varepsilon, u'_\varepsilon)$ on its edges such that each element of L occurs once among the t_ε and once among the u_ε, and each element of L^c once among the t'_ε and once among the u'_ε, and a choice at each edge of one of the two summands of (N.4), where the first summand may be chosen only if it equals 1. Its value is the product of the chosen summands. An edge is an *ordering edge* or a *correction edge* according to the chosen summand.

Step 3 (ordering edges). An ordering edge ε has $\{t_\varepsilon, t'_\varepsilon\} = \{u_\varepsilon, u'_\varepsilon\} = \varepsilon$ and $t_\varepsilon < t'_\varepsilon$. As $t_\varepsilon, u_\varepsilon \in L$ and $t'_\varepsilon, u'_\varepsilon \notin L$, this forces $t_\varepsilon = u_\varepsilon = \min \varepsilon \in L$ and $t'_\varepsilon = u'_\varepsilon = \max \varepsilon \notin L$, and its summand is 1.

Step 4 (correction data). Let $\varepsilon_1, \ldots, \varepsilon_m$ be the correction edges of a configuration, and put $J_i = \{t_{\varepsilon_i}, t'_{\varepsilon_i}\}$ and $J'_i = \{u_{\varepsilon_i}, u'_{\varepsilon_i}\}$. The value of the configuration is

$$\prod_{i=1}^{m} \Psi^{\mathrm{cor}}_{J_i}[\varepsilon_i, J'_i], \tag{N.5}$$

and the set $\Delta = \{(\varepsilon_i, J_i, J_i') : i \le m\}$ is its *correction data*. Each element of $[n]$ occurs exactly once among the $t_\varepsilon, t_\varepsilon'$, once among the $u_\varepsilon, u_\varepsilon'$ and once among the edges of $\mathfrak{m}$, and the ordering edges use the same two elements in all three roles (Step 3). So $J_1, \dots, J_m$, the sets $J_1', \dots, J_m'$ and the edges $\varepsilon_1, \dots, \varepsilon_m$ are three partitions of one set $U \subseteq [n]$, and the ordering edges form a perfect matching $\mathfrak{m}'$ of $[n] \setminus U$. Moreover $L \cap U$ meets each J_i and each J_i' in exactly one element.

Step 5 (counting). Fix a set $\Delta = \{(\varepsilon_i, J_i, J_i') : i \le m\}$, $m \ge 0$, of triples of 2-subsets in which the ε_i, the J_i and the J_i' partition the same set U, and call a subset of U meeting each J_i and each J_i' in exactly one element a *transversal* of Δ. The configurations, for all L together, with correction data Δ correspond bijectively to the pairs of a transversal L_U and a perfect matching $\mathfrak{m}'$ of $[n] \setminus U$: given the pair, put $L = L_U \cup \{\min \varepsilon : \varepsilon \in \mathfrak{m}'\}$ and $\mathfrak{m} = \{\varepsilon_1, \dots, \varepsilon_m\} \cup \mathfrak{m}'$, let t_{ε_i} and u_{ε_i} be the elements of J_i and J_i' in L_U and t'_{ε_i}, u'_{ε_i} the other ones, and use Step 3 on the edges of $\mathfrak{m}'$; by Steps 3 and 4 every configuration arises from exactly one pair. All of them have the value (N.5), and $[n] \setminus U$ has $(n-2m-1)(n-2m-3)\cdots 1$ perfect matchings (one if $U = [n]$), an odd number. The multigraph on U with the edges $J_1, \dots, J_m, J_1', \dots, J_m'$ has degree 2 at every vertex, with one edge of each kind, so it is a disjoint union of γ cycles of even length alternating between the kinds ($J_i = J_j'$ gives a cycle of length 2). A transversal contains exactly one endpoint of each of these edges, so on each cycle it is one of the two colour classes of its proper 2-colouring; there are 2^γ transversals, and $\gamma \ge 1$ if $m \ge 1$.

Step 6 (summation). Summing over L sums the values of all configurations. Grouped by their correction data, those with $m \ge 1$ contribute 2^γ times an odd number times (N.5), which is 0, and those with $m = 0$ are the pairs $(\emptyset, \mathfrak{m}')$, of value 1, contributing $(n-1)(n-3)\cdots 1 = 1$. This is (7.6).

Step 7 (0/1 inputs). Fix L and restrict $\operatorname{Pf} \mathsf{M}_\Psi$ to a, x supported in L and b, y supported in L^c. The result is a polynomial of degree at most $2n$ in the $2n$ variables a_t, x_t ($t \in L$) and $b_{t'}, y_{t'}$ ($t' \in L^c$). As in the proof of Lemma N.11, the sum of the values of such a polynomial over $\{0,1\}^{2n}$ is the coefficient of the product of all its variables, here of $a_L b_{L^c} x_L y_{L^c}$. Summing over L, (7.6) gives (N.3). □

Lemma N.15 (Pfaffian and determinant bounds)**.**

(1) *Let $n \ge 2$ be even. For z in $\Lambda^2_\pi c_0(n)$ or in its completion, $\|z\|_\pi \ge \frac{n}{2} |\operatorname{Pf} \Omega_z|^{2/n}$.*
(2) *Let $n \ge 1$. For Z in $c_0(n) \otimes_\pi c_0(n)$ or in its completion, $\|\mathsf{Z}\|_\pi \ge n\, |\det \mathsf{Z}|^{1/n}$.*

Proof. Step 1 (expansion). Let $z = \sum_{i \in I_0} x_i \wedge y_i$ with $x_i, y_i \in K^n$ and I_0 a finite ordered set. Then

$$\operatorname{Pf} \Omega_z = \sum_{I \subseteq I_0,\ |I| = n/2} \det(x_{i_1}, y_{i_1}, \dots, x_{i_{n/2}}, y_{i_{n/2}}), \qquad I = \{i_1 < \cdots < i_{n/2}\}. \tag{N.6}$$

Indeed, read with the Pfaffian of the alternating matrix of z over $\mathbb{Z}$, (N.6) is a polynomial identity with integer coefficients in the entries of the x_i, y_i. It holds over $\mathbb{Q}$: there $z^{\wedge n/2} = (n/2)!\operatorname{Pf}(\Omega_z)\, e_1 \wedge \cdots \wedge e_n$, the 2-forms $x_i \wedge y_i$ commute and have zero square, so $z^{\wedge n/2} = (n/2)! \sum_{|I| = n/2} \bigwedge_{i \in I} (x_i \wedge y_i)$, and $\bigwedge_{i \in I}(x_i \wedge y_i) = \det(x_{i_1}, y_{i_1}, \dots)\, e_1 \wedge \cdots \wedge e_n$. Hence it holds over $\mathbb{Z}$, and after reduction mod 2 over every commutative ring of characteristic 2.

Step 2 (bound). Put $\beta_i = \|x_i\|\, \|y_i\|$. A determinant is a sum of products containing one entry from each column, so by the ultrametric inequality each determinant in (N.6) has absolute value at most $\prod_{i \in I} \beta_i$. By the inequality of arithmetic and geometric means, $\prod_{i \in I} \beta_i \le \left(\frac{2}{n} \sum_{i \in I} \beta_i\right)^{n/2} \le \left(\frac{2}{n} \sum_{i \in I_0} \beta_i\right)^{n/2}$. By the ultrametric inequality again, $\sum_{i \in I_0} \beta_i \ge \frac{n}{2} |\operatorname{Pf} \Omega_z|^{2/n}$. Taking the infimum over the representations of z gives (1) on $\Lambda^2_\pi c_0(n)$. By (N.2) both sides are

continuous for the coordinate topology, in which $\Lambda^2 K^n$ is dense in the completion, so (1) holds on the completion.

Step 3 (item (2)). For $\mathsf{Z} = \sum_{i\in I_0} x_i y_i^{\mathsf{T}}$, the Cauchy–Binet formula gives

$$\det \mathsf{Z} = \sum_{I\subseteq I_0,\ |I|=n} \det(x_i : i\in I)\,\det(y_i : i\in I), \tag{N.7}$$

and Step 2 applies to (N.7) with $n/2$ replaced by n; the parity of n plays no role. □

Lemma N.16 (contractions)**.** *Let $n' \ge k \ge 1$, and let $l\colon K^{n'} \to K$ be linear with $|l(x)| \le \|x\|_\infty$ for all x. The contraction $l\lrcorner\colon \Lambda^k K^{n'} \to \Lambda^{k-1} K^{n'}$, $l \lrcorner (x_1 \wedge \cdots \wedge x_k) = \sum_j (-1)^{j-1} l(x_j)\, x_1 \wedge \cdots \wedge \widehat{x_j} \wedge \cdots \wedge x_k$, satisfies $\|l \lrcorner z\|_\pi \le \|z\|_\pi$ for $z \in \Lambda^k_\pi c_0(n')$, and so does its extension to the completions.*

Proof. It suffices to show $\|l\lrcorner\omega\|_\pi \le \prod_j \|x_j\|$ for $\omega = x_1\wedge\cdots\wedge x_k$ with all $x_j \ne 0$. If $l(x_j) = 0$ for all j, then $l \lrcorner \omega = 0$. Otherwise choose j_0 with $|l(x_{j_0})|/\|x_{j_0}\|$ maximal, so that $l(x_{j_0}) \ne 0$, and put $x'_j = x_j - \big(l(x_j)/l(x_{j_0})\big)x_{j_0}$ for $j \ne j_0$. Then $l(x'_j) = 0$, and by the ultrametric inequality and the choice of j_0, $\|x'_j\| \le \max\big(\|x_j\|, |l(x_j)|\,\|x_{j_0}\|/|l(x_{j_0})|\big) = \|x_j\|$. Replacing x_j by x'_j for $j \ne j_0$ does not change ω, so $l \lrcorner \omega = \pm l(x_{j_0}) \bigwedge_{j\ne j_0} x'_j$, whose projective norm is at most $|l(x_{j_0})| \prod_{j\ne j_0}\|x_j\| \le \prod_j \|x_j\|$. □

Proof of Theorem 7.20. Step 1 (item (1)). The first assertion is Proposition N.14, with $\det = \mathrm{Pf}^2$. The assertions on σ and τ are Propositions N.12 and N.13. The map $(x,y) \mapsto \mathcal{P}_<(xy^{\mathsf{T}} + yx^{\mathsf{T}})$ is bilinear and vanishes for $x = y$, so it induces $\sigma^<$. The matrix $\mathsf{Z} = \mathcal{P}_< \Omega_{x\wedge y}$ is strictly upper triangular, of rank at most $n-1$, and $\Omega_{\wedge \mathsf{Z}} = \mathsf{Z} + \mathsf{Z}^{\mathsf{T}} = \Omega_{x\wedge y}$, as $\Omega_{x\wedge y}$ has zero diagonal.

Step 2 (item (2): a multiplier lift on K^n). Let $\iota_n\colon K^n \to E$ be the inclusion of the first n coordinates, for $n' \ge 1$ let $\mathrm{res}_{n'}\colon E \to c_0(n')$ be the restriction to the first n' coordinates, and for $n \le n'$ let $\mathrm{res}^{n'}_n\colon c_0(n') \to c_0(n)$ be the restriction. The restrictions are contractive since $\|x\|_\infty \le \|x\|_E$, so their exterior powers (Proposition 3.1) and tensor powers are contractive; they extend to the completions, and the extensions are written with the same symbols. The 0/1 vectors 1_S, $S \subseteq [n]$, have $\|1_S\|_E \le \bar\varphi_E(n)$ and $\|1_S\|_\infty \le 1$.

Let $n' = n + k - 2$, let $1_{>n}$ be the indicator of $\{n+1,\dots,n'\}$ and $e_{>n} = e_{n+1}\wedge\cdots\wedge e_{n'}$, the empty product for $k = 2$. Let $\eta_{>n}\colon \Lambda^k K^{n'} \to \Lambda^2 K^{n'}$ be the composite of the contractions by the coordinate functionals of $n+1,\dots,n'$, in this order, and the identity for $k = 2$; then $\eta_{>n}(w \wedge e_{>n}) = w$ for $w \in \Lambda^2 K^n$. Put $\theta(w) = (\Lambda^2_\pi \iota_n w) \wedge e_{>n} \in \Lambda^k_\pi E$ for $w \in \Lambda^2 K^n$. As $\Lambda^k_\pi \mathrm{res}_{n'}(\theta w) = w \wedge e_{>n}$ in $\Lambda^k K^{n'}$,

$$\Lambda^2_\pi \mathrm{res}^{n'}_n \circ \eta_{>n} \circ \Lambda^k_\pi \mathrm{res}_{n'} \circ \theta = \mathrm{id} \qquad \text{on } \Lambda^2 K^n. \tag{N.8}$$

For $a, b \in K^n$, regarded in $c_{00}(\mathbb{N}, K)$, let $Q_2(a,b)$ be the sum, over the pairs of slots $j < j'$, of Q evaluated with a in slot j, b in slot j' and $1_{>n}$ in the other slots, and put

$$\Psi(a,b) = \Lambda^2_\pi \mathrm{res}^{n'}_n \circ \eta_{>n} \circ \Lambda^k_\pi \mathrm{res}_{n'} \circ Q_2(a,b) \circ \theta. \tag{N.9}$$

This is a K-linear map $\Lambda^2 K^n \to \Lambda^2 \widehat{K}^n$, K-bilinear in (a,b); extend it $\widehat{K}$-linearly in w and $\widehat{K}$-bilinearly in (a,b). For $\lambda \in K$, multilinearity gives $Q(a+\lambda 1_{>n}, \dots, a+\lambda 1_{>n}) = \sum_{j=0}^k \lambda^{k-j} q_j(a)$, where $q_j(a)$ is the sum, over the j-element sets of slots, of Q with a in these slots and $1_{>n}$ elsewhere; so $q_2(a) = Q_2(a,a)$. As $\mu_{a+\lambda 1_{>n}}$ multiplies e_t by a_t for $t \le n$ and by λ for $n < t \le n'$, $\Lambda^k_\pi \mu_{a+\lambda 1_{>n}}(\theta w) = \lambda^{k-2}\theta(\Lambda^2 \mu_a w)$. As K is infinite, comparing the coefficients of λ^{k-2} gives $Q_2(a,a)(\theta w) = \theta(\Lambda^2 \mu_a w)$, and (N.8) gives $\Psi(a,a) = \Lambda^2 \mu_a$ for $a \in K^n$. Taking $a = e_t$, $a = e_{t'}$ and $a = e_t + e_{t'}$, $t \ne t'$, (7.4) gives $\Psi(e_t, e_{t'}) + \Psi(e_{t'}, e_t) = \Lambda^2 \mu_{e_t + e_{t'}} - \Lambda^2 \mu_{e_t} - \Lambda^2 \mu_{e_{t'}} = \pi_{tt'}$, so Ψ is a multiplier lift on $\widehat{K}^n$.

Step 3 (item (2): the bound). Let $a,b,x,y\in\{0,1\}^n$. Each of the $k(k-1)/2$ terms of $Q_2(a,b)$ has norm at most $\|Q\|$; $\Lambda^2_\pi\operatorname{res}^{n'}_n$ and $\Lambda^k_\pi\operatorname{res}_{n'}$ are contractive; $\eta_{>n}$ is contractive by Lemma N.16; and $\theta(x\wedge y)$ is a single decomposable vector, so $\|\theta(x\wedge y)\|_\pi\le\|x\|_E\|y\|_E\prod_{t=n+1}^{n'}\|e_t\|_E\le\bar\varphi_E(n)^2$. Hence

$$\|\Psi(a,b)(x\wedge y)\|_\pi\le\tfrac{k(k-1)}{2}\,\|Q\|\,\bar\varphi_E(n)^2, \tag{N.10}$$

the norm being that of the completion of $\Lambda^2_\pi c_0(n)$. By (N.3) over $\widehat K$, a finite sum of the values $\operatorname{Pf}\mathsf M_\Psi(a,b,x,y)$ at 0/1 inputs is 1. As $\widehat K$ is nonarchimedean, one of them has absolute value at least 1, and Lemma N.15(1) gives $\|\Psi(a,b)(x\wedge y)\|_\pi\ge n/2$ for that input. With (N.10) this is (7.7).

Step 4 (item (3): sections). Let $\sigma=(\operatorname{res}_n\otimes\operatorname{res}_n)\circ s\circ\Lambda^2_\pi\iota_n$, extended $\widehat K$-linearly to $\Lambda^2\widehat K^n\to\widehat K^{n\times n}$. By naturality of $\wedge$, $\wedge\circ\sigma=\Lambda^2_\pi\operatorname{res}_n\circ\widehat\wedge{}^2_E\circ s\circ\Lambda^2_\pi\iota_n=\mathrm{id}$ on Λ^2K^n, hence on $\Lambda^2\widehat K^n$ by linearity. Proposition N.12 over $\widehat K$ gives $\sum_{S,T\subseteq[n]}\det\mathsf N(1_S,1_T)=1$ for $\mathsf N(x,y)=\sigma(x\wedge y)+xy^{\mathsf T}$, so some $|\det\mathsf N(1_S,1_T)|\ge1$, and Lemma N.15(2) gives $n\le\|\mathsf N(1_S,1_T)\|_\pi\le\|s\|\,\bar\varphi_E(n)^2+1$, so $\|s\|\ge(n-1)/\bar\varphi_E(n)^2$. For $c_0(n)$ itself take ι_n and res_n to be the identity here and in Steps 5 and 6.

Step 5 (item (3): retractions). Let n be even; for odd n apply the even case to $n-1$, as $\bar\varphi_E$ is nondecreasing. Let $\tau=\Lambda^2_\pi\operatorname{res}_n\circ\rho\circ(\iota_n\otimes\iota_n)$, extended $\widehat K$-linearly to $\widehat K^{n\times n}\to\Lambda^2\widehat K^n$. For $w\in\Lambda^2K^n$, $\operatorname{alt}^2w=\Omega_w$, so $\tau(\Omega_w)=\Lambda^2_\pi\operatorname{res}_n\circ\rho\circ\operatorname{alt}^2_E(\Lambda^2_\pi\iota_nw)=w$, and by linearity $\tau(\Omega_w)=w$ for $w\in\Lambda^2\widehat K^n$. Proposition N.13 over $\widehat K$ gives $\sum_{S\subseteq[n]}\operatorname{Pf}\Omega_{\tau(1_S1_S^{\mathsf T})}=1$, so some $|\operatorname{Pf}\Omega_{\tau(1_S1_S^{\mathsf T})}|\ge1$, and Lemma N.15(1) gives $n/2\le\|\tau(1_S1_S^{\mathsf T})\|_\pi\le\|\rho\|\,\bar\varphi_E(n)^2$.

Step 6 (item (3): projections). Let $\mathsf N(x,y)=(\operatorname{res}_n\otimes\operatorname{res}_n)R_2(\iota_nx\otimes\iota_ny)$, extended $\widehat K$-bilinearly. Its values are symmetric matrices, since $\widehat\Gamma^2_\pi E$ is the closed span of the pure squares and $\operatorname{res}_n\otimes\operatorname{res}_n$ maps pure squares to symmetric matrices, which form a closed subspace. As R_2 fixes pure squares, $\mathsf N(e_t,e_t)=e_te_t^{\mathsf T}$ and, polarizing $(e_t+e_{t'})^{\otimes2}$, $\mathsf N(e_t,e_{t'})+\mathsf N(e_{t'},e_t)=e_te_{t'}^{\mathsf T}+e_{t'}e_t^{\mathsf T}$; so $\mathsf N(x,x)=xx^{\mathsf T}$ for $x\in\widehat K^n$ by bilinear expansion. So $\mathsf N$ is an $[n]$-polarization over $\widehat K$ (Definition N.7), and Lemmas N.9, N.10 and N.11 give $\sum_{S,T}\det\mathsf N(1_S,1_T)=1$. As in Step 4, $n\le\|\mathsf N(1_S,1_T)\|_\pi\le\|R_2\|\,\bar\varphi_E(n)^2$.

Step 7 (item (4): goodness). Each listed E is a sequence space with $\|e_t\|_E=1$ and $\|\mu_a\|_{E\to E}\le\|a\|_\infty$ for finitely supported a, and $\sup_n n/\bar\varphi_E(n)^2=\infty$. For a symmetric sequence space, Definition 7.10 gives a norm that is monotone in $|x|$ and equal to 1 at the unit vectors; so $\bar\varphi_E=\varphi_E$, $\|x\|_\infty\le\|x\|_E$, and $|ax|\le|\lambda x|$ for a coordinate λ of a with $|\lambda|=\|a\|_\infty$ gives $\|\mu_a\|\le\|a\|_\infty$. As $\bar\varphi_E$ is nondecreasing, $(n-1)/\bar\varphi_E(n-1)^2\ge(n-1)/\bar\varphi_E(n)^2$, so $n/\bar\varphi_E(n)^2$ is also unbounded along even n. If (E,E) were good for Banach values in degree k, Lemma 3.4 would give a bounded k-linear $\widetilde Q\colon\mathcal L(E,E)^k\to\mathcal L(\Lambda^k_\pi E,\widehat\Lambda^k_\pi E)$ with $\widetilde Q(u,\dots,u)=\Lambda^k_\pi u$, and $Q(a_1,\dots,a_k)=\widetilde Q(\mu_{a_1},\dots,\mu_{a_k})$ would satisfy the hypotheses of item (2) with $\|Q\|\le\|\widetilde Q\|$, contradicting (7.7) for large even n. Lemma 3.4, Theorem 3.2(1) and the dichotomy of Proposition 2.5 then give the assertion on $A^k_{E,E;F}$, and Proposition 3.5(1),(2) the failure of $(\mathrm S^k_E)$ and $(\mathrm R^k_E)$. Item (3) gives the failure of (Γ^2_E).

Step 8 (item (4): nonarchimedean norms). Let $h\ge1$, $n=2h$, $n'=n+k-2$, and $z_h=\sum_{i=1}^h e_{2i-1}\wedge e_{2i}\wedge e_{>n}\in\Lambda^k_\pi E$, a sum of h decomposable vectors of norm at most 1. The contractive maps $\Lambda^k_\pi\operatorname{res}_{n'}$, $\eta_{>n}$ and $\Lambda^2_\pi\operatorname{res}^{n'}_n$ send it to $\sum_{i\le h}e_{2i-1}\wedge e_{2i}$, whose Pfaffian is 1, so $\|z_h\|_\pi\ge h$ by Lemma N.15(1). If $\|\cdot\|'$ were a nonarchimedean norm on $\widehat\Lambda^k_\pi E$ with $c\|\cdot\|_\pi\le\|\cdot\|'\le C\|\cdot\|_\pi$, then $ch\le\|z_h\|'\le C$ for all h, which is impossible.

Step 9 (item (4): pairs). Let $2 < r_1, r_2 \le \infty$, write ℓ^{r_i} for $\ell^{r_i}(\mathbb{N}, K)$, or for $c_0(\mathbb{N}, K)$ if $r_i = \infty$, and put $1/\infty = 0$. If (ℓ^{r_1}, ℓ^{r_2}) were good for Banach values in degree k, Lemma 3.4 would give a bounded k-linear $\widetilde{Q}\colon \mathcal{L}(\ell^{r_1}, \ell^{r_2})^k \to \mathcal{L}(\Lambda^k_\pi \ell^{r_1}, \widehat{\Lambda}^k_\pi \ell^{r_2})$ with $\widetilde{Q}(u, \dots, u) = \Lambda^k_\pi u$. Run Steps 2 and 3 with $Q(a_1, \dots, a_k) = \widetilde{Q}(\mu_{a_1}, \dots, \mu_{a_k})$, where μ_a is now a map $\ell^{r_1} \to \ell^{r_2}$, with $\theta_1(w) = (\Lambda^2_\pi \iota_n w) \wedge e_{>n}$ in $\Lambda^k_\pi \ell^{r_1}$ in place of θ on the right of (N.9) and the same expression $\theta_2(w)$ in $\Lambda^k_\pi \ell^{r_2}$ in the identity $\Lambda^k_\pi \mu_{a+\lambda 1_{>n}} \circ \theta_1 = \lambda^{k-2} \theta_2 \circ \Lambda^2 \mu_a$, and with $\mathrm{res}_{n'}$ defined on ℓ^{r_2}. Put $\delta = \max(0, 1/r_2 - 1/r_1)$. Hölder's inequality gives $\|\mu_a\| \le |\operatorname{supp} a|^\delta$ for 0/1 vectors a, and $\|1_S\|_{r_1} \le n^{1/r_1}$, so $n/2 \le \frac{k(k-1)}{2} \|\widetilde{Q}\| (k-2)^{(k-2)\delta} n^{2\delta + 2/r_1}$, where $(k-2)^{(k-2)\delta} = 1$ for $k = 2$. As $2\delta + 2/r_1 = 2/\min(r_1, r_2) < 1$, this fails for large even n. □

Appendix O. Proofs for familywise analyticity

This appendix proves the results of Section 11 not proved there, in its notation. We write $h^{\times n} = (h, \dots, h) \in P^n$, h_I for the tuple of the h_i with $i \in I$, and $\mathrm{Mult}^0(P; Y) = Y$.

O.1. **Power series and mixed exterior powers.**

Lemma O.1. (1) *If Y is a Banach space and $b_n \in \mathrm{Mult}^n(P; Y)$, $\|b_n\| \le CR^{-n}$, then $x \mapsto \sum_n b_n(x^{\times n})$ is analytic on the open ball of radius R about 0.*

(2) *Let $\Phi\colon Y_1 \times \dots \times Y_m \to Y$ be bounded m-linear, the spaces normed, and $f_a\colon U \to Y_a$ with $f_a(x_0 + h) = \sum_n b^a_n(h^{\times n})$ for $\|h\| < R$, $\|b^a_n\| \le CR^{-n}$. Then $\Phi(f_1(x_0 + h), \dots, f_m(x_0 + h)) = \sum_N \Phi_N(h^{\times N})$ for $\|h\| < R$, where $\Phi_N(h_1, \dots, h_N) = \sum_{n_1 + \dots + n_m = N} \Phi(b^1_{n_1}(h_{I_1}), \dots, b^m_{n_m}(h_{I_m}))$, I_a consecutive of sizes n_a, has norm at most $\|\Phi\| \binom{N+m-1}{m-1} C^m R^{-N}$. So the composite is analytic at x_0, and on U if the f_a are.*

Proof. (1) For $\|x_1\| < R$, the sum b'_j over $n \ge j$ and j-element $S \subseteq \{1, \dots, n\}$ of b_n with $h_1, \dots, h_j$ in the slots of S and x_1 elsewhere converges in $\mathrm{Mult}^j(P; Y)$ with a geometric bound, as $\sum_{n \ge j} \binom{n}{j} R^{-n} \|x_1\|^{n-j} = R(R - \|x_1\|)^{-j-1}$; for $\|h\| < R - \|x_1\|$ the majorant $\sum_n \|b_n\| (\|x_1\| + \|h\|)^n$ of the multilinear expansion gives $\sum_n b_n((x_1 + h)^{\times n}) = \sum_j b'_j(h^{\times j})$. (2) The bound counts compositions of N. For $\|h\| < R$ the terms $\Phi(b^1_{n_1}(h^{\times n_1}), \dots, b^m_{n_m}(h^{\times n_m}))$ have the majorant $\|\Phi\| \prod_a \sum_n C(\|h\|/R)^n$; their sums over $\max_a n_a \le L$ are values of Φ at partial sums of the f_a, and differ from their sums over $n_1 + \dots + n_m \le L$ by terms with $\max_a n_a > L/m$, whose majorant tends to 0. □

Lemma O.2. *Let G be a Banach space, $v_1, \dots, v_m \in \mathcal{L}(E, G)$, and $\mathbf{i} \in \mathbb{N}_0^m$ a* type, *$i_1 + \dots + i_m = k \ge 1$; a map $\sigma\colon \{1, \dots, k\} \to \{1, \dots, m\}$ has type $\mathbf{i}$ if $|\sigma^{-1}(l)| = i_l$ for all l, and there are $\binom{k}{\mathbf{i}} = k!/(i_1! \cdots i_m!)$ of them. Entries $i_l = 0$ use $\Lambda^0_\pi v = \mathrm{id}_K$ and the conventions stated with (6.3).*

(1) *There is a unique $\Lambda^{\mathbf{i}}_\pi(v_1, \dots, v_m) \in \mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi G)$ sending $x_1 \wedge \dots \wedge x_k$ to $\sum_\sigma v_{\sigma(1)} x_1 \wedge \dots \wedge v_{\sigma(k)} x_k$, over the σ of type $\mathbf{i}$; its norm is at most $\binom{k}{\mathbf{i}} \prod_l \|v_l\|^{i_l}$.*

(2) *The iterated shuffle $\mathrm{sh}^{\mathbf{i}}_E = (\mathrm{id} \otimes \mathrm{sh}^{(i_2, \dots, i_m)}_E) \circ \mathrm{sh}^{i_1, k-i_1}_E$, $\mathrm{sh}^{(k)}_E = \mathrm{id}$, into $\Lambda^{i_1}_\pi E \otimes_\pi \dots \otimes_\pi \Lambda^{i_m}_\pi E$ has norm at most $\binom{k}{\mathbf{i}}$, the iterated wedge product $\wedge^{\mathbf{i}}_G$ from $\widehat{\Lambda}^{i_1}_\pi G \mathbin{\widehat{\otimes}}_\pi \dots \mathbin{\widehat{\otimes}}_\pi \widehat{\Lambda}^{i_m}_\pi G$ to $\widehat{\Lambda}^k_\pi G$ is contractive, and $\Lambda^{\mathbf{i}}_\pi(v_1, \dots, v_m) = \wedge^{\mathbf{i}}_G \circ (\Lambda^{i_1}_\pi v_1 \otimes \dots \otimes \Lambda^{i_m}_\pi v_m) \circ \mathrm{sh}^{\mathbf{i}}_E$.*

Proof. Lemma 6.1, iterated and extended to completions, and $\|\mathrm{id} \otimes T\| \le \|T\|$ give the norm bounds of (2). By induction on m and (6.1), $\mathrm{sh}^{\mathbf{i}}_E(x_1 \wedge \dots \wedge x_k)$ is the sum of the $\epsilon(S_1, \dots, S_m)\, x_{S_1} \otimes \dots \otimes x_{S_m}$ over the ordered partitions with $|S_l| = i_l$, where $\epsilon(S_1, \dots, S_m)$, the sign of the listing of $S_1, \dots, S_m$ in turn, each increasingly, is $\epsilon(S_1)$ times the sign of the listing of $S_2, \dots, S_m$ relative to the increasing listing of the complement of S_1. With $\sigma(c) = l$

for $c \in S_l$ and $y_c = v_{\sigma(c)}x_c$, the composite in (2) sends $x_1 \wedge \cdots \wedge x_k$ to the sum of the $\epsilon(S_1,\dots,S_m)\, y_{S_1} \wedge \cdots \wedge y_{S_m} = y_1 \wedge \cdots \wedge y_k$, and ordered partitions correspond bijectively to maps of type **i**. So the composite is the operator of (1), unique as pure wedges span, and $\|\Lambda^{i_l}_\pi v_l\| \le \|v_l\|^{i_l}$ gives its bound. □

Lemma O.3. *Let G be a Banach space and $a, b \in \mathcal{L}(E,G)$. Then $\Lambda^k_\pi(a+\lambda b) = \sum_{j=0}^k \lambda^j \Lambda^{(k-j,j)}_\pi(a,b)$ for $\lambda \in K$, where $\Lambda^{(k-j,j)}_\pi(a,b) = \wedge^{k-j,j}_G \circ (\Lambda^{k-j}_\pi a \otimes \Lambda^j_\pi b) \circ \mathrm{sh}^{k-j,j}_E$ has norm at most $\binom{k}{j}\|a\|^{k-j}\|b\|^j$, and $\Lambda^{(0,k)}_\pi(a,b) = \Lambda^k_\pi b$. If $\lambda_0,\dots,\lambda_k \in K$ are distinct and (χ_{ji}) is the inverse of $(\lambda_i^j)_{0\le i,j\le k}$, then $\Lambda^k_\pi b = \sum_i \chi_{ki}\Lambda^k_\pi(a+\lambda_i b)$.*

Proof. Equation (6.3) with $w_1 = a$, $w_2 = \lambda b$ and $\Lambda^j_\pi(\lambda b) = \lambda^j \Lambda^j_\pi b$ gives the expansion, and Lemma 6.1 the bound. The Vandermonde matrix being invertible, the identities $\Lambda^k_\pi(a+\lambda_i b) = \sum_j \lambda_i^j \Lambda^{(k-j,j)}_\pi(a,b)$ give $\Lambda^{(k-j,j)}_\pi(a,b) = \sum_i \chi_{ji}\Lambda^k_\pi(a+\lambda_i b)$; take $j = k$. □

Lemma O.4. *Let H be a Banach space, $U' \subseteq P$ open and $w\colon U' \to \mathcal{L}(H)$ analytic with $\sup_{U'}\|w\| < 1$. Then $1+w$ is an invertible family on U'.*

Proof. Fix $x_1 \in U'$ and let $g_1 = 1 + w(x_1)$, so $\|g_1^{-1}\| \le (1 - \sup_{U'}\|w\|)^{-1}$ by the Neumann series. Write $w(x_1+h) - w(x_1) = \delta(h) = \sum_{n\ge1} b_n(h^{\times n})$, $\|b_n\| \le CR^{-n}$. With $C' = \|g_1^{-1}\|C$ and $t = \|h\|/R < 1/(1+C')$, $\|g_1^{-1}\delta(h)\| \le C't/(1-t) < 1$, so $(1+w(x_1+h))^{-1} = \sum_l (-g_1^{-1}\delta(h))^l g_1^{-1}$, whose multilinear expansion has the majorant $\|g_1^{-1}\|\sum_l (C't/(1-t))^l$; its degree-$N$ coefficient, the sum over the compositions $(n_1,\dots,n_l)$ of N of $(-1)^l g_1^{-1} b_{n_1}(h_{I_1}) g_1^{-1} \cdots b_{n_l}(h_{I_l}) g_1^{-1}$, has norm at most $\|g_1^{-1}\|(1+C')^N R^{-N}$. □

O.2. **Reductions and what bundles need.**

Proof of Proposition 11.2. Step 1 (item (1)). Let $u\colon U \to \mathcal{L}(H)$ be analytic and $x_0 \in U$. Choose an open $U_0 \subseteq U$ containing x_0 with $\|u\| \le C = \|u(x_0)\| + 1$ on U_0, an integer $m \ge 0$ with $|\vartheta|^m C < 1$, and the distinct $\lambda_i = \vartheta^{m+i}$, $0 \le i \le k$. By Lemma O.4 every $1+\lambda_i u$ is an invertible family on U_0, and Lemma O.3 with $G = H$, $a = 1$, $b = u(x)$ gives $\Lambda^k_\pi u = \sum_i \chi_{ki}\Lambda^k_\pi(1+\lambda_i u)$ on U_0.

Step 2 (item (2)). Let $\alpha_1\colon E_0 \to E$, $\alpha_2\colon D_0 \to D$, $\psi\colon P_0 \to P$ have bounded linear left inverses α_1', α_2', ψ'. For analytic $u_0\colon U' \to \mathcal{L}(E_0, \widehat{D}_0)$, $U' \subseteq P_0$ open, $u = \hat{\alpha}_2 u_0(\psi'\cdot)\alpha_1'$ is analytic on $\psi'^{-1}(U')$, hats denoting extensions to completions; hence so is $\Lambda^k_\pi u$, and $\Lambda^k_\pi u_0(y) = \widehat{\Lambda}^k_\pi \hat{\alpha}_2' \circ \Lambda^k_\pi u(\psi y) \circ \Lambda^k_\pi \alpha_1$ (Proposition 3.1). For $E \oplus E'$, "only if" is the retract case; conversely, for analytic $u\colon U \to \mathcal{L}(E\oplus E', \widehat{D})$, the injections ι_l and projections π_l of $E \oplus E'$, $u_l = u\iota_l$ and $k' \le k$, $u = u_1\pi_1 + u_2\pi_2$, (6.3) and Proposition 3.1 give

$$\Lambda^{k'}_\pi u = \sum_{j=0}^{k'} \wedge^{j,k'-j}_{\widehat{D}} \circ \big((\Lambda^j_\pi u_1 \circ \Lambda^j_\pi \pi_1) \otimes (\Lambda^{k'-j}_\pi u_2 \circ \Lambda^{k'-j}_\pi \pi_2)\big) \circ \mathrm{sh}^{j,k'-j}_{E\oplus E'},$$

bounded bilinear (linear for $j \in \{0,k'\}$) in analytic maps; apply Lemma O.1(2). The target slot is the same, with $u = \sum_l \iota_l'\pi_l' u$ for the injections ι_l' and projections π_l' of $\widehat{D} \oplus \widehat{D'}$ and $\Lambda^j_\pi(\iota_l'\pi_l' u) = \widehat{\Lambda}^j_\pi \iota_l' \circ \Lambda^j_\pi(\pi_l' u)$.

Step 3 (item (3)). Let g be as in Proposition 10.1, with $H = D \oplus E$, π_D the projection onto D and i_E the injection of E. For analytic $u\colon U \to \mathcal{L}(E,D)$, $x \mapsto g(u(x))$ is an invertible family, with inverse $g(-u(x))$, and $\pi_D g(u(x)) i_E = u(x)$ and Proposition 3.1 give

$$\Lambda^k_\pi u(x) = \widehat{\Lambda}^k_\pi \pi_D \circ \Lambda^k_\pi g(u(x)) \circ \Lambda^k_\pi i_E. \tag{O.1}$$

In degrees $j \le k$, analyticity along invertible families into $\mathcal{L}(H)$ gives goodness of $(P;H)$ by (1), hence of the four triples by (2); conversely, goodness of the four triples gives that of $(P;H)$ by (2) in both slots, and invertible families are analytic. For $\mathcal{C}$, goodness of $(P;G,G)$ for $G \in \mathcal{C}$ covers invertible families into $\mathcal{L}(G)$; conversely, for $E, D \in \mathcal{C}$, $D \oplus E \in \mathcal{C}$ and (O.1) applies. □

Proof of Corollary 11.3. Step 1 ((2) implies the last assertion and (1)). Let $E, D \in \mathcal{C}$, let F, F' be Banach spaces, and let $u\colon U \to \mathcal{L}(E,D)$ and $v\colon U \to \mathcal{L}(F,F')$ be analytic on an open $U \subseteq P$. By (2) and the paragraph after Definition 11.1, $x \mapsto A^k_{E,D;F}(u(x))$ is analytic. The joint action $B(u(x), v(x))$, which is $A^k_{E,D;F}(u(x))$ followed by postcomposition with $v(x)$, is bounded bilinear in the two families, hence analytic (Lemma O.1(2)). The last assertion of Theorem 2.1, for restricted categories of fibers as in the paragraph after it, gives the last assertion; (1) is its case of a trivial bundle with fiber F.

Step 2 ((1) implies (2)). Let $E, D \in \mathcal{C}$, let $u\colon U \to \mathcal{L}(E,D)$ be analytic on an open $U \subseteq P$, and let $H = D \oplus E \in \mathcal{C}$ and g be as in Proposition 10.1. The trivial bundle $U \times H$ with the trivializations id and $(x,z) \mapsto (x, g(u(x))z)$ is analytic, since its transitions $g(\pm u(x))$ are affine in $u(x)$. By (1), the induced transition $x \mapsto g(u(x))^*$ on $\mathrm{Alt}^k(H;F)$ is analytic for every Banach space F. The compression (10.1) makes $x \mapsto A^k_{E,D;F}(u(x))$ analytic, so $\Lambda^k_\pi u$ is analytic on U by Theorem 3.2(1) with $Y = \widehat{\Lambda}^k_\pi D$; as D is complete, this is (2). □

O.3. **Homogeneous families.** A *content* is a finitely supported $\mathbf{m}\colon \mathbb{N}_0 \to \mathbb{N}_0$ with $\sum_n \mathbf{m}(n) = k$; its degree is $N(\mathbf{m}) = \sum_n n\mathbf{m}(n)$, and its type is $(\mathbf{m}(n_1), \dots, \mathbf{m}(n_m))$ for its support $n_1 < \cdots < n_m$. The content of $(n'_1, \dots, n'_k) \in \mathbb{N}_0^k$ counts repeated values, so at most $\binom{N+k-1}{k-1}$ contents have degree N.

Lemma O.5. *Let $n \ge 0$, $j \ge 1$ and $\beta \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D}))$.*

1. *$c_j(\beta)$ depends only on u_β, and $c_j(\lambda\beta) = |\lambda|^j c_j(\beta)$ for $\lambda \in K^\times$.*
2. *$c_1(\beta) \le \|\beta\|$, and $c_j(\beta) \le (j!/|j!|)\|\beta\|^j$ if $j! \ne 0$ in K, the integer $j!$ in the numerator.*
3. *For $n \ge 1$, $h \mapsto \Lambda^j_\pi u_\beta(h)$ is analytic at some point of P if and only if $c_j(\beta) < \infty$, and it is then a continuous polynomial.*
4. *$c_j(b\beta a) \le \|b\|^j \|a\|^j c_j(\beta)$ for bounded linear $a\colon E' \to E$, $b\colon \widehat{D} \to \widehat{D'}$, with $(b\beta a)(h_1, \dots, h_n) = b\,\beta(h_1, \dots, h_n)\,a$.*

Proof. (1) The definition involves only $\Lambda^j_\pi u_\beta$, and $\lambda^j q$ lifts $u_{\lambda\beta}$ if q lifts u_β; apply this also to λ^{-1}. (2) β lifts u_β in degree 1. If $j! \ne 0$, the section $s = \mathrm{alt}^j_E / j!$ of $\wedge^j_E$ has norm at most $j!/|j!|$ (Proposition 4.1(1)), and $q(h_1, \dots, h_{jn}) = \widehat{\wedge}^j_D \circ (\beta(h_{I_1}) \otimes \cdots \otimes \beta(h_{I_j})) \circ s$, blocks of size n, has diagonal $\Lambda^j_\pi u_\beta(h) \wedge^j_E s = \Lambda^j_\pi u_\beta(h)$ by naturality of the wedge, and norm at most $(j!/|j!|)\|\beta\|^j$. (3) $T \mapsto T \circ \wedge^j_E$ is an isometry of $\mathcal{L}(\Lambda^j_\pi E, \widehat{\Lambda}^j_\pi D)$ onto the operators in $\mathcal{L}(\otimes^j_\pi E, \widehat{\Lambda}^j_\pi D)$ vanishing on $\ker \wedge^j_E$ (Lemma 3.7), a closed subspace, and $(h_1, \dots, h_{jn}) \mapsto \widehat{\wedge}^j_D \circ (\beta(h_{I_1}) \otimes \cdots \otimes \beta(h_{I_j}))$ is an ambient representative of degree $jn \ge 1$; apply Proposition 2.5. (4) If q lifts u_β, then $\widehat{\Lambda}^j_\pi b \circ q \circ \Lambda^j_\pi a$ lifts $u_{b\beta a}$ (Proposition 3.1). □

Lemma O.6. *Let $u(x_0 + h) = \sum_{n \ge 0} \beta_n(h^{\times n})$ for $\|h\| < R$, $\beta_n \in \mathrm{Mult}^n(P; \mathcal{L}(E, \widehat{D}))$, $\|\beta_n\| \le CR^{-n}$, $a = \beta_0$, and let $C_0, C_1 \ge 1$ and lifts $q_{n,j}$ of u_{β_n} in degree j with $\|q_{n,j}\| \le C_0 C_1^n$ be given for $j \le k$, $n \ge 1$. Then $\Lambda^k_\pi u$ is analytic at x_0, with Nth coefficient of norm at most $k!\binom{N+k-1}{k-1} \max(1, \|a\|)^k C_0^k C_1^N$.*

Proof. Step 1 (one content). For a content $\mathbf{m}$ of support $n_1 < \cdots < n_m$ and type $\mathbf{i}$ let $Q_l = q_{n_l, i_l}$ if $n_l \ge 1$, $Q_l = \Lambda^{i_l}_\pi a$ if $n_l = 0$, and, with consecutive blocks I_l of sizes $n_l i_l$,

$q_{\mathbf{m}}(h_1,\dots,h_{N(\mathbf{m})}) = \wedge^{\mathbf{i}}_{\widehat{D}} \circ (Q_1(h_{I_1}) \otimes \cdots \otimes Q_m(h_{I_m})) \circ \mathrm{sh}^{\mathbf{i}}_E$. By Lemma O.2(2), $q_{\mathbf{m}}(h^{\times N(\mathbf{m})}) = \Lambda^{\mathbf{i}}_\pi(u_{\beta_{n_1}}(h),\dots,u_{\beta_{nm}}(h))$ and $\|q_{\mathbf{m}}\| \le \binom{k}{\mathbf{i}} \prod_l \|Q_l\| \le k! \max(1,\|a\|)^k C_0^k C_1^{N(\mathbf{m})}$, as $m \le k$ and $C_0, C_1 \ge 1$. The sum q_N of the $q_{\mathbf{m}}$ of degree N obeys the stated bound, and $\sum_N \|q_N\| \|h\|^N < \infty$ for $\|h\| < 1/(2C_1)$.

Step 2 (the expansion). For $\|h\| < R$ and $y = x_1 \wedge \cdots \wedge x_k$, as in Lemma O.1(2), $\Lambda^k_\pi u(x_0+h)y = \omega_{\widehat{D}}(\sum_n u_{\beta_n}(h)x_1, \dots, \sum_n u_{\beta_n}(h)x_k)$ is the absolutely convergent sum of the $u_{\beta_{\sigma(1)}}(h)x_1 \wedge \cdots \wedge u_{\beta_{\sigma(k)}}(h)x_k$, $\sigma\colon \{1,\dots,k\} \to \mathbb{N}_0$. Grouped by the content of $(\sigma(1),\dots,\sigma(k))$ and then by degree, they give $\sum_N q_N(h^{\times N})y$ (Lemma O.2(1)); as pure wedges span, $\Lambda^k_\pi u(x_0+h) = \sum_N q_N(h^{\times N})$ for $\|h\| < \min(R, 1/(2C_1))$. □

Lemma O.7. (1) *If $(P;E,D)$ is good in degrees $a,b \ge 1$ and $\binom{a+b}{a} \ne 0$ in K, it is good in degree $a+b$.*

(2) *Let $k \ge 2$, with maximum norms on $E \oplus K$, $D \oplus K$. If $(P; E\oplus K, D \oplus K)$ is good in degree k, then $(P;E,D)$ is good in degree $k-1$.*

Proof. (1) Let $n = a+b$. For $v \in \mathcal{L}(E,\widehat{D})$, naturality of the wedge product and (6.2), extended to completions, give $\Lambda^n_\pi v = \binom{n}{a}^{-1} \wedge^{a,b}_{\widehat{D}} \circ (\Lambda^a_\pi v \otimes \Lambda^b_\pi v) \circ \mathrm{sh}^{a,b}_E$, bounded bilinear in $(\Lambda^a_\pi v, \Lambda^b_\pi v)$; apply Lemma O.1(2). (2) Let $u\colon U \to \mathcal{L}(E,\widehat{D})$ be analytic and F a Banach space. As $u \oplus 1$ is analytic into $\mathcal{L}(E\oplus K, \widehat{D}\oplus\widehat{K})$ and $\widehat{D}\oplus\widehat{K}$ completes $D\oplus K$, goodness and the prose after Definition 11.1 make $x \mapsto A^k_{E\oplus K,\widehat{D}\oplus\widehat{K};F}(u(x)\oplus 1)$ analytic, hence also $x \mapsto A^k_{E\oplus K,\widehat{D}\oplus K;F}(u(x)\oplus 1)$, restriction $\mathrm{Alt}^k(\widehat{D}\oplus\widehat{K};F) \to \mathrm{Alt}^k(\widehat{D}\oplus K;F)$ being an isometric isomorphism by density. The proof of Theorem 6.4, with $\widehat{D}$ for D, gives fixed bounded linear R_h, J_h with $A^{k-1}_{E,\widehat{D};F}(v) = R_h \circ A^k_{E\oplus K,\widehat{D}\oplus K;F}(v\oplus 1)\circ J_h$; so $x \mapsto A^{k-1}_{E,\widehat{D};F}(u(x))$ is analytic for every Banach F, and the paragraph after Definition 11.1 applies. □

Proof of Theorem 11.5. Step 1 (item (1)). Let (β_n) be as in (11.1). Each u_{β_n} is analytic (Proposition 2.5), so $c_j(\beta_n) < \infty$ by goodness and Lemma O.5(3). If $\limsup_n c_j(\beta_n)^{1/n} = \infty$, choose $n_1 < n_2 < \dots$ with $n_{i+1} > jn_i$ and $c_j(\beta_{n_i})^{1/n_i} \ge i$. By Lemma O.1(1) and goodness, $u(h) = \sum_i \beta_{n_i}(h^{\times n_i})$ is analytic on the open unit ball and $\Lambda^j_\pi u(h) = \sum_N q_N(h^{\times N})$ for $\|h\| < R'$, $\|q_N\| \le C' R'^{-N}$. As in Step 2 of the proof of Lemma O.6, $\Lambda^j_\pi u(h)(x_1\wedge\cdots\wedge x_j) = \sum_N s_N(h)$, with $s_N(h)$ the sum of the $u_{\beta_{\sigma(1)}}(h)x_1 \wedge \cdots \wedge u_{\beta_{\sigma(j)}}(h)x_j$ over the $\sigma\colon \{1,\dots,j\} \to \{n_1,n_2,\dots\}$ with $\sum_a \sigma(a) = N$, and Lemma A.1 on the lines $t \mapsto th$ gives $s_N(h) = q_N(h^{\times N})(x_1\wedge\cdots\wedge x_j)$. For $N = jn_i$ only $\sigma \equiv n_i$ qualifies, as $n_{i+1} > jn_i$ and $jn_{i-1} < n_i$; so q_{jn_i} lifts $u_{\beta_{n_i}}$, and $c_j(\beta_{n_i})^{1/n_i} \le C'^{1/n_i} R'^{-j}$ stays bounded, a contradiction.

Step 2 (item (2)). Let $u\colon U \to \mathcal{L}(E,\widehat{D})$ be analytic, $u(x_0+h) = \sum_n \beta_n(h^{\times n})$, $\|\beta_n\| \le CR^{-n}$. For integers $m, m' \ge 0$ with $|\vartheta|^{m'} < R$ and $|\vartheta|^{-m} \ge \max(C,1)$ and for $\lambda = \vartheta^{m'}$, $\|\vartheta^m \lambda^n \beta_n\| \le 1$, so (11.1) gives $C'_j, C''_j \ge 1$ with $c_j(\vartheta^m\lambda^n\beta_n) < C'_j C''^n_j$ for $n \ge 1$, $j \le k$, and Lemma O.5(1) gives lifts of u_{β_n} in degree j of norm at most $|\vartheta|^{-mj} C'_j (C''_j |\lambda|^{-j})^n$; apply Lemma O.6 with $C_0 = \max_j |\vartheta|^{-mj} C'_j$ and $C_1 = \max_j C''_j |\lambda|^{-j}$.

Step 3 (item (3)). The first assertion is Lemma O.5(2). In the second, "only if" is (1); conversely, we show goodness in degree $n \le k$ by induction. If $n < p$, apply (2) and the first assertion. If $n > p$ is not a power of p and p^l is the largest power of p below n, then $n = p^l + (n-p^l)$ is carry-free, so $\binom{n}{p^l} \ne 0$ in K by Lucas's theorem (proof of Corollary 6.3); apply Lemma O.7(1). If $n = p^i$, (G_j) holds for $j < n$ by induction and (1), and for $j = n$ by hypothesis; apply (2).

Step 4 (item (4)). The linearization $\tau_\beta\colon Y_n \to \widehat{D}$ of $(h_1,\dots,h_n,y)\mapsto \beta(h_1,\dots,h_n)y$ has norm $\|\beta\|$ and $\beta = \tau_\beta\circ\beta^{\mathrm{u}}_n$; apply Lemma O.5(4). If $(P;E,D_{\mathrm{u}})$ is good in degrees $j\le k$, then, with J_n the nth injection of D_{u}, (G_j) for $\beta_n = J_n\circ\beta^{\mathrm{u}}_n$, of norm at most 1, and $c_j(\beta^{\mathrm{u}}_n)\le c_j(\beta_n)$, by Lemma O.5(4) with the nth projection, make every $c_j(\beta^{\mathrm{u}}_n)$ finite with geometric growth; with the inequality, this gives (G_j) for every normed D, hence goodness in degrees $j\le k$ by (2). In characteristic p the degrees $p^i\le k$ give (G_{p^i}) for every D, and (3) applies. □

Proof of Corollary 11.6. (1) Degrees below p are good by Theorem 11.5(2),(3), carry-free sums by Lemma O.7(1) and Lucas's theorem; a least missing N that is not a power of p would be the carry-free sum $p^l+(N-p^l)$ of smaller elements, p^l the largest power of p below N. (2) By Proposition 11.2(2), $(P;E,D)$ and $(P;E\oplus K,D\oplus K)$ are good in the same degrees, so Lemma O.7(2) makes $\mathfrak{D}(P;E,D)$ closed under $k\mapsto k-1$, $k\ge 2$; apply (1). (3) is Proposition 11.11. (4) Lemma O.7(2), for all normed E, D at once, gives closure under $k\mapsto k-1$; by (1) the set contains $1,\dots,p-1$ and is closed under carry-free sums, so its least missing element is a power of p. □

O.4. **The familywise class.**

Lemma O.8. *Let $u(x_0+h)=\sum_n\beta_n(h^{\times n})$ for $\|h\|<R$, $\beta_n\in\mathrm{Mult}^n(P;\mathcal{L}(E,\widehat{D}))$, $\|\beta_n\|\le CR^{-n}$, and $\lambda\in K$, $0<|\lambda|<R$. Let $\beta^{\otimes}_n\colon \widehat{\otimes}^n_\pi P\to\mathcal{L}(E,\widehat{D})$ be the linearization of β_n for $n\ge 1$, and $\beta^{\otimes}_0(c)=c\beta_0$ for c in slot 0, $\widehat{K}$, by the $\widehat{K}$-vector space structure of $\mathcal{L}(E,\widehat{D})$. Then $\theta((w_n)_n)=\sum_n\lambda^n\beta^{\otimes}_n(w_n)$ is a linear family on $\mathcal{T}(P)$, $\|\theta\|\le C$, and $u(x_0+\lambda h)=\theta(\mathrm{ver}(h))$ for $\|h\|<1$. Conversely, $\theta\circ\mathrm{ver}$ is analytic on the open unit ball for every linear family θ on $\mathcal{T}(P)$.*

Proof. $\|\lambda^n\beta^{\otimes}_n(w_n)\|\le C(|\lambda|/R)^n\|w_n\|$, and $\theta(\mathrm{ver}(h))=\sum_n\beta_n((\lambda h)^{\times n})$. The converse is Lemma O.1(1), as $\|\theta\circ\mathrm{ver}_n\|\le\|\theta\|$. □

Lemma O.9. *Let Y be a Banach space, $0<R'\le R<1$, and $C_{k,R}=\sup_N\binom{N+k-1}{k-1}R^N<\infty$. For $\Phi\in\mathrm{Mult}^k(\mathcal{T}(P);Y)$, $\|\mathrm{ver}^\sharp\Phi\|_R\le C_{k,R}\|\Phi\|$ and $\Phi(\mathrm{ver}(h),\dots,\mathrm{ver}(h))=\sum_N(\mathrm{ver}^\sharp\Phi)_N(h^{\times N})$ for $\|h\|<1$; if Φ has zero diagonal, $\mathrm{ver}^\sharp\Phi\in\mathrm{Mult}^\omega_{R,0}(P;Y)$; and $\iota_{R'R}\circ\mathrm{ver}^\sharp$ is $\mathrm{ver}^\sharp$ with values in $\mathrm{Mult}^\omega_{R'}(P;Y)$. Hence vanishing after shrinking does not depend on R.*

Proof. There are at most $\binom{N+k-1}{k-1}$ compositions of N into k parts, and $\|\mathrm{ver}_n\|\le 1$; the identity is Lemma O.1(2) with radius 1. For zero diagonal, $\sum_N t^N(\mathrm{ver}^\sharp\Phi)_N(h^{\times N})=0$ for $|t|<1/\|h\|$; apply Lemma A.1. The inclusion keeps components, so $(\iota_{R'R})_*\,\mathrm{ob}^\omega_k(\theta;R)=\mathrm{ob}^\omega_k(\theta;R')$ by (11.2) and Corollary N.3(b), and pushouts of 0 vanish. □

Proof of Theorem 11.7. Step 1 (item (1)). The second and third conditions are equivalent: bounded linear maps into $\mathrm{Mult}^\omega_{R',0}(P;\widehat{\Lambda}^k_\pi D)$ are sequences of maps into the $\mathrm{Mult}^N_0(P;\widehat{\Lambda}^k_\pi D)$ with $\sup_N R'^N\|\cdot\|<\infty$, $\iota_{R'R}$ keeps components, and a pushout vanishes if and only if the map extends (Lemma 3.6(3)). Each $q_N\in\mathrm{Mult}^N(P;\mathcal{L}(\Lambda^k_\pi E,\widehat{\Lambda}^k_\pi D))$ gives $q^\sharp_N\colon\widehat{\Lambda}^k_\pi E\to\mathrm{Mult}^N(P;\widehat{\Lambda}^k_\pi D)$, $q^\sharp_N(y)(h_1,\dots,h_N)=q_N(h_1,\dots,h_N)y$, of the same norm, and conversely. If $\Lambda^k_\pi\theta(\mathrm{ver}(h))=\sum_N q_N(h^{\times N})$ for $\|h\|<R_0$, $\|q_N\|\le C'R_0^{-N}$, then for $z\in\widehat{\otimes}^k_\pi E$ the Nth terms of $\mathrm{ver}^\sharp\mathrm{ev}_\theta(z)$ and $q^\sharp_N\widehat{\wedge}^k_E z$ have the same diagonal, both giving the t^N-coefficient of $\mathrm{ev}_\theta(z)(\mathrm{ver}(th),\dots,\mathrm{ver}(th))=\widehat{\Lambda}^k_\pi\theta(\mathrm{ver}(th))(\widehat{\wedge}^k_E z)$ (Lemmas O.9 and A.1). So $\xi_N=\mathrm{pr}_N\,\mathrm{ver}^\sharp\mathrm{ev}_\theta-q^\sharp_N\widehat{\wedge}^k_E$ takes values in Mult^N_0, extends $(\mathrm{pr}_N\,\mathrm{ver}^\sharp\mathrm{ev}_\theta)|_{\mathcal{N}^k_E}$ and has norm at most $C_{k,R}\|\theta\|^kR^{-N}+C'R_0^{-N}$. Conversely, extensions ξ_N of norm at most CR'^{-N} make $\mathrm{pr}_N\,\mathrm{ver}^\sharp\mathrm{ev}_\theta-\xi_N$ vanish on $\mathcal{N}^k_E$, so it is $q^\sharp_N\widehat{\wedge}^k_E$

with $\|q_N^\sharp\| \le (C_{k,R}\|\theta\|^k + C)\min(R,R')^{-N}$ (Lemma 3.7), and $\sum_N q_N(h^{\times N}) = \Lambda_\pi^k\theta(\mathrm{ver}(h))$ for $\|h\| < \min(R,R')$ by Lemma O.9.

Step 2 (item (2)). For analytic u and x_0, Lemma O.8 gives λ, θ with $u(x_0+\lambda h) = \theta(\mathrm{ver}(h))$, and (1), with $h = \lambda^{-1}(x - x_0)$, gives analyticity of $\Lambda_\pi^k u$ at x_0. Conversely, $\theta \circ \mathrm{ver}$ is analytic on the open unit ball (Lemma O.8); apply goodness and (1).

Step 3 (item (3)). As $\theta(w) = \tau_\theta \circ \theta_{\mathrm{u}}(w)$, $\mathrm{ev}_\theta = (\widehat{\Lambda}_\pi^k\tau_\theta)_\flat\,\mathrm{ev}_{\theta_{\mathrm{u}}}$ on pure tensors. The map $(\widehat{\Lambda}_\pi^k\tau_\theta)_\flat$ commutes with $\mathrm{ver}^\sharp$ and $\iota_{R'R}$, preserves zero diagonals and has norm at most $\|\tau_\theta\|^k$; so Corollary N.3(b) gives the identity, vanishing after shrinking passes from θ_{u} to every θ, and (2) gives goodness for every D. Conversely, (2) for the Banach space $D = \mathcal{T}(P)\,\widehat{\otimes}_\pi\,E$ gives vanishing for θ_{u}. Finally $\theta_{\mathrm{u}} \circ \mathrm{ver} = \mathcal{U}$; apply (1). □

O.5. **Polarizable base models.**

Lemma O.10. *Let $n, i \ge 1$ and R_i a bounded projection of $\widehat{\otimes}_\pi^i P$ onto $\widehat{\Gamma}_\pi^i P$. Identify $\widehat{\otimes}_\pi^{ni} P$ with $(\widehat{\otimes}_\pi^i P)^{\widehat{\otimes} n}$ by grouping the positions $(a,1),\dots,(a,i)$ into the ath factor, and let* reg *be the isometry of $\widehat{\otimes}_\pi^{ni} P$ moving position (a,b) to (b,a), with target $(\widehat{\otimes}_\pi^n P)^{\widehat{\otimes} i}$. Then $q_{n,i}(h_1,\dots,h_{ni}) = \mathrm{reg} \circ R_i^{\otimes n}(h_1 \otimes \cdots \otimes h_{ni})$ is bounded ni-linear, $\|q_{n,i}\| \le \|R_i\|^n$, with values in $\widehat{\Gamma}_\pi^i(\widehat{\otimes}_\pi^n P)$, and $q_{n,i}(h^{\times ni}) = (h^{\otimes n})^{\otimes i}$.*

Proof. $R_i^{\otimes n}$ has norm at most $\|R_i\|^n$, takes values in the closed span of the $y_1^{\otimes i} \otimes \cdots \otimes y_n^{\otimes i}$, which reg sends to $(y_1 \otimes \cdots \otimes y_n)^{\otimes i}$, and fixes $(h^{\otimes i})^{\otimes n}$, which reg sends to $(h^{\otimes n})^{\otimes i}$. □

Lemma O.11. *Let G be a Banach space and $\mathbf{m}$ a content of type $\mathbf{i} = (i_1,\dots,i_m)$, let $\Omega_{\mathbf{m}}$ represent the left cosets in S_k of the subgroup preserving the consecutive blocks of sizes $i_1,\dots,i_m$, acting on $\widehat{\otimes}_\pi^k G$ by permuting factors, and $\mathrm{per}_{\mathbf{m}} = \sum_{\sigma\in\Omega_{\mathbf{m}}}\sigma$. Then $\|\mathrm{per}_{\mathbf{m}}\| \le \binom{k}{\mathbf{i}}$; on tensors invariant under that subgroup, $\mathrm{per}_{\mathbf{m}}$ does not depend on $\Omega_{\mathbf{m}}$; for $y_1,\dots,y_m \in G$, $\mathrm{per}_{\mathbf{m}}(y_1^{\otimes i_1} \otimes \cdots \otimes y_m^{\otimes i_m})$ is the coefficient of $t_1^{i_1}\cdots t_m^{i_m}$ in $(\sum_l t_l y_l)^{\otimes k}$ and lies in the span of the powers $y^{\otimes k}$, $y \in G$; and $\mathrm{per}_{\mathbf{m}}$ maps the closed span of such tensors into $\widehat{\Gamma}_\pi^k G$.*

Proof. The bound is the triangle inequality; on invariant tensors σ and $\sigma\sigma'$, σ' in the subgroup, agree. The cosets correspond bijectively to the words in which l occurs i_l times, whose sum is the coefficient. A linear functional on the span of the coefficients of $(\sum_l t_l y_l)^{\otimes k}$ vanishing on these powers vanishes on every coefficient, by Lemma B.6, as in Step 1 of the proof of Proposition 5.13. Continuity gives the last claim. □

Proof of Theorem 11.9. Step 1 (affine maps suffice). Affine maps are analytic. Conversely, fix $n \le k$, let $G = \widehat{K} \oplus \widehat{P}$ with the maximum norm, $\gamma(x) = (1,x)$, $Z = \widehat{\otimes}_\pi^n P = \widehat{\otimes}_\pi^n \widehat{P}$ and $W = \widehat{\Gamma}_\pi^n P = \widehat{\Gamma}_\pi^n \widehat{P}$ (density; paragraph before Theorem 3.16). Scalar multiplication on Z extends from K to $\widehat{K}$ by continuity and preserves the closed subspace W. The k-linear map $\Phi((\lambda_1,x_1),\dots,(\lambda_k,x_k)) = \lambda_{n+1}\cdots\lambda_k\, x_1 \otimes \cdots \otimes x_n$ has $\|\Phi\| \le 1$ and $\Phi((\lambda,x),\dots,(\lambda,x)) = \lambda^{k-n}x^{\otimes n} \in W$, so $x \mapsto x^{\otimes n} = \Phi(\gamma(x),\dots,\gamma(x))$ is analytic at 0 into W; Proposition 2.5, with the canonical map $P^n \to Z$, and Lemma 3.11 give polarizability in degree n.

Step 2 (the ambient sum). Let G, Z, W, Φ, U, γ be as in Definition 11.8, $x_0 \in U$ and $\gamma(x_0+h) = \sum_n \gamma_n(h^{\times n})$ for $\|h\| < R$, $\|\gamma_n\| \le CR^{-n}$. Let $\gamma_n^\otimes\colon \widehat{\otimes}_\pi^n P \to G$ and $\Phi^\otimes\colon \widehat{\otimes}_\pi^k G \to Z$ be the linearizations, with $\widehat{\otimes}_\pi^0 P = \widehat{K}$ and $\gamma_0^\otimes(c) = c\gamma_0$; as W is closed, $\Phi^\otimes(\widehat{\Gamma}_\pi^k G) \subseteq W$. By Lemma O.1(2), for $\|h\| < R$, $\Phi(\gamma(x_0+h),\dots,\gamma(x_0+h))$ is the absolutely convergent sum of the $\Phi(\gamma_{n_1'}(h^{\times n_1'}),\dots,\gamma_{n_k'}(h^{\times n_k'}))$, $(n_1',\dots,n_k') \in \mathbb{N}_0^k$, and by Lemma O.11 the terms of content $\mathbf{m}$, of support $n_1 < \cdots < n_m$ and type $\mathbf{i}$, sum to $\Phi^\otimes\,\mathrm{per}_{\mathbf{m}}(\gamma_{n_1}(h^{\times n_1})^{\otimes i_1} \otimes \cdots \otimes \gamma_{n_m}(h^{\times n_m})^{\otimes i_m})$.

Step 3 (lifting each content). Fix $\bar g$ as in (2) and projections R_i onto $\widehat\Gamma^i_\pi P$ with $\|R_i\| \le \bar g$, $i \le k$. Let $q_l = q_{n_l,i_l}$ (Lemma O.10) if $n_l \ge 1$, $q_l = 1 \in \widehat K$ if $n_l = 0$, and, with consecutive blocks I_l of sizes $n_l i_l$, $Q_{\mathbf m}(h_1,\dots,h_{N(\mathbf m)}) = \Phi^\otimes \operatorname{per}_{\mathbf m}(\bigotimes_l (\gamma^\otimes_{n_l})^{\otimes i_l} q_l(h_{I_l}))$. Then $\|Q_{\mathbf m}\| \le \|\Phi\| \binom{k}{\mathbf i} \prod_l \|\gamma_{n_l}\|^{i_l} \bar g^{n_l} \le \|\Phi\| k! C^k (\bar g/R)^{N(\mathbf m)}$, as $\bar g \ge 1$. Its values lie in W, as q_l takes values in $\widehat\Gamma^{i_l}_\pi(\widehat\otimes^{n_l}_\pi P)$, which $(\gamma^\otimes_{n_l})^{\otimes i_l}$ maps into $\widehat\Gamma^{i_l}_\pi G$, and Lemma O.11 applies. By Lemma O.10, $Q_{\mathbf m}(h^{\times N(\mathbf m)})$ is the content-$\mathbf m$ sum of Step 2.

Step 4 (item (1)). The sum q_N of the $Q_{\mathbf m}$ of degree N is bounded N-linear into W, with

$$\|q_N\| \le \|\Phi\|\, k!\, C^k \tbinom{N+k-1}{k-1} (\bar g/R)^N, \tag{O.2}$$

and for $\|h\| < R/\bar g$, $\sum_N q_N(h^{\times N})$ converges in the Banach space W to $\Phi(\gamma(x_0{+}h),\dots,\gamma(x_0{+}h))$, by Step 2.

Step 5 (items (2)–(4)). (2) For analytic $u\colon U \to \mathcal L(E,\widehat D)$ take $G = \mathcal L(E,\widehat D)$, $Z = \operatorname{Mult}^k(E;\widehat\Lambda^k_\pi D)$, W the image of $\mathcal L(\Lambda^k_\pi E,\widehat\Lambda^k_\pi D)$ under the closed isometry $T \mapsto T\circ\omega_E$ (Proposition 3.1), and $\Phi(u_1,\dots,u_k)(x_1,\dots,x_k) = u_1x_1 \wedge\dots\wedge u_kx_k$, of norm at most 1, with $\Phi(u,\dots,u) = \Lambda^k_\pi u\circ\omega_E$; apply (1) and (O.2). For bundles, $x \mapsto A^k_{E,\widehat D;F}(u(x))$ is analytic for Banach F (prose after Definition 11.1), and the last assertion of Theorem 2.1 applies as in the proof of Corollary 11.3. (3) If $n! \ne 0$ in K, P is polarizable in degree n (Proposition 4.1(2)); so (2) applies if $\operatorname{char} K = 0$ or $k < p$. Polarizability in degree p gives it in every degree (Theorem 6.10). A lower r-estimate, $r < 2$, bounds $\mathcal P_<$ (Theorem 5.6(1)), and Proposition 5.7(3) gives polarizability in every degree. (4) Projections with $\|R_n\| \le 2g_n(P)$ and Theorem G.3 give the first equivalence, and (1) the second. For nonarchimedean K, $g_n(\ell^1(\mathbb N,K)) = n!$ (Propositions 5.13(1) and G.4) is finite, while $(n!)^{1/n}$ is unbounded; apply (1) and (2). □

Proof of Theorem 11.10. Step 1 (item (1)). As the statement depends only on K as a normed field, Lemma 7.1 lets us assume K dense in $\kappa((X))$, as in the proof of Theorem 7.13. By Lemma 7.11(1) for P, $\|ay\|_P \le \|y\|_\infty \|a\|_P$ for $a \in P$ and $y \in E_0$, so θ_0 is a linear family, and $\|\theta_0(a)\| = \|\mu_a\|_{E_0\to P} = \|a\|_P$ for $a \in c_{00}(\mathbb N,K)$ (Lemma 7.11(2),(4)). The fifth condition gives the first, the first the second. If $\Lambda^k_\pi\theta_0$ is analytic at a point, θ_0 has a lift Q in degree k (prose after Definition 11.1). Its restriction to $c_{00}(\mathbb N,K)^k$ is a multiplier lift for (E_0,P) (Section K.4) satisfying (K.6) with $C_Q = \|Q\|$, and E_0, P are symmetric sequence spaces; so Steps 1–5 of the proof of Theorem 7.13 apply to (E_0,P), as in Step 2 of the proof of Proposition 7.16, and $x = 1_{\operatorname{supp} a}$, $y = 1_{\operatorname{supp} b}$ in (7.2) bound $\mathcal P_<$ on $P \otimes_\pi P$. This bound, item (3) of Theorem 8.1, gives polarizability in degree p (Proposition 5.7(3)), hence the fifth condition (Theorem 11.9(3)). The examples follow from Corollary 8.2 and Theorem 11.9(3), every $k \ge p$ being admissible in (L).

Step 2 (item (2)). Apply Theorem 7.18 with $E = c_{00}(\mathbb N,K)$, $E' = \ell^\infty(\mathbb N,\widehat K)$, $\psi = \mathrm{id}$, $\theta = \mathrm{id}$, $u_a = \mu_a = \mu_{w(i(a))}$, $F = \widehat\Lambda^k_\pi E'$, $\nu = \omega_{E'}$, $\theta_F = \mathrm{id}$: no bounded k-linear Q on $c_{00}(\mathbb N,K)^k$ has $Q(a^{\times k}) = \omega_{E'}\circ(\mu_a,\dots,\mu_a)$. Analyticity of $x \mapsto \Lambda^k_\pi\mu_{w(x)}$ at a point would give a lift Q_P of the linear family $x \mapsto \mu_{w(x)}$ (prose after Definition 11.1), and $(a_1,\dots,a_k) \mapsto Q_P(ia_1,\dots,ia_k)\circ\omega_E$ would be such a Q. For $c_0(\Sigma,K)$ and $\ell^\infty(\Sigma,K)$ take $\mathbb N \subseteq \Sigma$, i extension by zero and w restriction; for P with an equivalent nonarchimedean norm and $\dim_{\widehat K}\widehat P = \infty$, take $i(\xi) = \sum_n \xi_n e_n$ and $w = \theta_P|_P$ with e_n, θ_P from Lemma 7.17. Then $(\mathrm U^k_P)$ fails, and with it polarizability in degree p (Theorem 11.9(3)). If $\dim_{\widehat K}\widehat P = d < \infty$, a bounded $\widehat K$-linear isomorphism $T\colon \widehat P \to \ell^1(\{1,\dots,d\},\widehat K)$ exists as $\widehat K$ is complete, and $(T^{-1})^{\otimes n} R_n T^{\otimes n}$ transfers projections (Step 3 of the proof of Proposition 5.13); as passing to a dense subspace changes neither $\widehat\otimes^n_\pi$

nor $\widehat{\Gamma}^n_\pi$, Proposition 5.13(1) for $\ell^1(\{1,\dots,d\},K)$ makes P polarizable in every degree; apply Theorem 11.9(2). □

O.6. **Fibers, parameter-relative ordering and the staircase.**

Proof of Proposition 11.11. (1) For an analytic $u\colon U\to\mathcal{L}(E,\widehat{D})$, $\Lambda^k_\pi u(x)=Q(u(x),\dots,u(x))$ is analytic on U by Lemma O.1(2). For β as stated, $q(h_1,\dots,h_{kn})=Q(\beta(h_{I_1}),\dots,\beta(h_{I_k}))$, with $I_1,\dots,I_k$ the consecutive blocks of size n, is a lift of u_β in degree k of norm at most $\|Q\|\|\beta\|^k$. (2) The "if" direction is (1). Conversely, the identity of the open set $\mathcal{L}(E,\widehat{D})$ is an analytic family, so $u\mapsto\Lambda^k_\pi u$ is analytic at some point of $\mathcal{L}(E,\widehat{D})$; since $\widehat{\Lambda}^k_\pi\widehat{D}=\widehat{\Lambda}^k_\pi D$, Lemma 3.4 applies to $(E,\widehat{D})$. The second assertion follows from (1) and the first. □

Proof of Theorem 11.13. Step 1 (the map Π). Let G_r and δ_t be as in Section J.4; by density $\widehat{\otimes}^k_\pi G_r=\widehat{\otimes}^k_\pi\ell^r(\Sigma,\widehat{K})$ and $\widehat{\Gamma}^k_\pi G_r=\widehat{\Gamma}^k_\pi\ell^r(\Sigma,\widehat{K})$. For a composition $\mathbf{l}=(l_1,\dots,l_m)$ of k let $I_1,\dots,I_m$ be the consecutive blocks of sizes l_i and $\Omega_\mathbf{l}$ represent the left cosets of $S_{l_1}\times\cdots\times S_{l_m}$ in S_k, acting by permuting factors. For $a_1,\dots,a_k\in c_{00}(\Sigma,K)$ put

$$\Pi_\mathbf{l}(a_1,\dots,a_k)=\sum_{t_1<\cdots<t_m}\prod_{i=1}^m\prod_{j\in I_i}a_j(t_i)\,\delta_{t_1}^{\otimes l_1}\otimes\cdots\otimes\delta_{t_m}^{\otimes l_m},\qquad\Pi=\sum_{\mathbf{l}}\sum_{\sigma\in\Omega_\mathbf{l}}\sigma\circ\Pi_\mathbf{l}.$$

As $\sum_{\sigma\in\Omega_\mathbf{l}}\sigma\delta_\mathbf{t}=O_\mathbf{t}$ for nondecreasing $\mathbf{t}$ with constant blocks of sizes $\mathbf{l}$, Π is R_k of Proposition 5.7(3), for (δ_t), applied to $a_1\otimes\cdots\otimes a_k$. So $\Pi(a,\dots,a)=a^{\otimes k}$, and Π takes values in the span of the $O_\mathbf{t}$, which lie in the span of the powers by Step 4 of the proof of that proposition in Appendix J.

Step 2 (one composition). Let $\Delta=\{i:l_i\ge2\}$, j_i the element of I_i for $i\notin\Delta$, and $b_i=\prod_{j\in I_i}a_j$ for $i\in\Delta$, so $\|b_i\|_1\le\prod_{j\in I_i}\|a_j\|_2$ by Hölder's inequality and $\|\cdot\|_{l_i}\le\|\cdot\|_2$. For fixed strictly increasing $(t_i)_{i\in\Delta}$, the $i\notin\Delta$ form maximal runs of consecutive indices, and for a run the admissible t_i are the strictly increasing tuples in the open interval J of Σ between the neighboring fixed values. So $\Pi_\mathbf{l}(a_1,\dots,a_k)$ is the sum over $(t_i)_{i\in\Delta}$ of $\prod_{i\in\Delta}b_i(t_i)$ times the tensor product, in block order, of the $\delta_{t_i}^{\otimes l_i}$, $i\in\Delta$, of norm 1, and, for each run of length n, of $\mathcal{P}^{(n)}(\bigotimes_i a_{j_i}1_J)$, of norm at most $C'_n\prod_i\|a_{j_i}\|_2$ with $C'_n=\max(1,C_{n,r}|\vartheta|^{-n})$ (Theorem J.5). Hence $\|\Pi_\mathbf{l}(a_1,\dots,a_k)\|\le C_\mathbf{l}\prod_j\|a_j\|_2$, with $C_\mathbf{l}$ the product of the C'_n of the runs.

Step 3 (item (1)). So Π extends from the dense subspace $c_{00}(\Sigma,K)$ of $\ell^2(\Sigma,\widehat{K})$, with values in the closed subspace $\widehat{\Gamma}^k_\pi G_r$, and $\Pi(a,\dots,a)=a^{\otimes k}$ persists by continuity.

Step 4 (item (2), linear maps). Let $T\colon P\to G$ be bounded linear, and extend Φ to $\widehat{G}^k$; its diagonal stays in W. The coordinate map $V\to\ell^2(\Sigma,\widehat{K})$ extends to $w\colon P\to\ell^2(\Sigma,\widehat{K})$, $\|w\|\le C_2$; write $h(t)=(wh)(t)$ and $\iota\colon\ell^2\to\ell^r$ for the inclusion. As in Step 1 of the proof of Theorem 5.9, with P, G, T for E, D, u, the map $\xi\mapsto\sum_t\xi(t)T(e_t)$ extends from $c_{00}(\Sigma,K)$ to $\widetilde{T}\colon\ell^r(\Sigma,\widehat{K})\to\widehat{G}$ with $T=\widetilde{T}\circ\iota\circ w$. The linearization $\Phi^\otimes_r$ of $(\xi_1,\dots,\xi_k)\mapsto\Phi(\widetilde{T}\xi_1,\dots,\widetilde{T}\xi_k)$, extended to $\widehat{\otimes}^k_\pi G_r$, maps $\widehat{\Gamma}^k_\pi G_r$ into W, and $Q(a_1,\dots,a_k)=\Phi^\otimes_r\Pi(wa_1,\dots,wa_k)$ satisfies $Q(a,\dots,a)=\Phi^\otimes_r((\iota wa)^{\otimes k})=\Phi(Ta,\dots,Ta)$, by continuity from $c_{00}(\Sigma,K)$.

Step 5 (item (2), analytic maps). Let $\gamma(x_0+h)=\sum_n\gamma_n(h^{\times n})$ for $\|h\|<R$, $\|\gamma_n\|\le CR^{-n}$, let Σ_* be the disjoint union of the Σ^n, $n\ge0$, totally ordered in any way, and $\lambda\in K$ with $0<|\lambda|\le R/C_e$, $C_e=\sup_t\|e_t\|$. For $\|h\|<|\lambda|/C_2$ the vector $\xi(h)\in\ell^2(\Sigma_*,\widehat{K})$ with entries $\xi(h)_{(n,\mathbf{t})}=\lambda^{-n}h(t_1)\cdots h(t_n)$ has nth block of norm $(\|wh\|_2/|\lambda|)^n$, and ξ is analytic at 0 with nth coefficient of norm at most $(C_2/|\lambda|)^n$. As $\|\lambda^n\gamma_n(e_{t_1},\dots,e_{t_n})\|\le C$, (5.2) for $\widehat{G}$ (Proposition 5.8) extends $\delta_{(n,\mathbf{t})}\mapsto\lambda^n\gamma_n(e_{t_1},\dots,e_{t_n})$ to $T_*\colon\ell^r(\Sigma_*,\widehat{K})\to\widehat{G}$, and $T_*\iota\xi(h)=\sum_n\gamma_n(h^{\times n})=\gamma(x_0+h)$ for $\|h\|<\min(R,|\lambda|/C_2)$: first for $h\in V$, where the blocks of

$\xi(h)$ are finitely supported, then by continuity. Step 4 for $\ell^2(\Sigma_*, \widehat{K})$ with its unit vectors, $\widehat{G}$ and $T_* \circ \iota$ gives a bounded k-linear Q_{T_*} into W with $Q_{T_*}(\eta, \dots, \eta) = \Phi(T_*\iota\eta, \dots, T_*\iota\eta)$; so $\Phi(\gamma(x_0 + h), \dots, \gamma(x_0 + h)) = Q_{T_*}(\xi(h), \dots, \xi(h))$ near 0, analytic by Lemma O.1(2).

Step 6 (item (3)). The space $\widehat{D}$ satisfies (5.2): by Proposition 5.8 and continuity if D has an equivalent r-norm, and, if D has an equivalent nonarchimedean norm, as $\max_t|\xi_t| \le (\sum_t|\xi_t|^r)^{1/r}$, as in the proof of Corollary 5.10. Hence so does $G = \mathcal{L}(E, \widehat{D})$, as $\|\sum_t \xi_t u_t\| = \sup_{\|x\|\le 1}\|\sum_t \xi_t u_t x\|$, and G has an equivalent r-norm (Proposition 5.8). Let $Z = \mathcal{L}(\widehat{\otimes}^k_\pi E, \widehat{\Lambda}^k_\pi D)$, W its operators vanishing on $\mathcal{N}^k_E$, and $\Phi(u_1, \dots, u_k) = \widehat{\wedge}^k_D \circ (u_1 \otimes \cdots \otimes u_k)$, so $\Phi(u, \dots, u) = \widehat{\Lambda}^k_\pi u \circ \widehat{\wedge}^k_E$. For analytic $u\colon U \to G$, (2) makes $x \mapsto \widehat{\Lambda}^k_\pi u(x) \circ \widehat{\wedge}^k_E$ analytic into W; the inverse of the isometry $S \mapsto S \circ \widehat{\wedge}^k_E$ onto W (Lemma 3.7) and restriction to $\Lambda^k_\pi E$ give analyticity of $\Lambda^k_\pi u$. □

Proof of Theorem 11.14. Step 1 (the input from Theorem 7.7). Suppose such a Q exists. Step 1 of the proof of Theorem 7.7 in Appendix K uses only that Q is k-linear on $c_{00}(\mathbb{N}, K)^k$ with $Q(a, \dots, a) = \Lambda^k_\pi \mu_a$, and that $\|y\|_\infty \le \|y\|_E$, so that the determinant array of Section K.1 is defined on $\widehat{\Lambda}^k_\pi E$; it gives τ, slots $\gamma_1 \ne \gamma_2$ and $\delta \ne 0$, independent of m.

Step 2 (the norm bound for ζ). Fix $m \ge 1$, and let $a_1, \dots, a_k$ and $x_1, \dots, x_k$ be the inputs of Step 2 of that proof: their entries lie in $\{0, \pm 1\}$, a_{γ_1} and a_{γ_2} have $2m$ nonzero entries and the other a_γ two, and x_{γ_1}, x_{γ_2} have $4m$ and the other x_γ four. So $\zeta = Q(a_1, \dots, a_k)(x_1 \wedge \cdots \wedge x_k)$ has $\|\zeta\| \le \|Q\|\varphi^\pm_P(2)^{k-2}\varphi^\pm_E(4)^{k-2}\varphi^\pm_P(2m)^2\varphi^\pm_E(4m)^2$.

Step 3 (the rank bound). The slice of Lemma K.3 for the slots γ_1, γ_2 and $c_\gamma = C_\gamma(1)$, followed by restriction to the coordinates $A_n(1)$ and $B_{n'}(1)$, contractive as $\|y\|_\infty \le \|y\|_E$, gives z with $\operatorname{coeff}_0 z = \mathsf{Z}$ and $\|z\| \le k(k-1)\|\zeta\|$; so Lemma K.2(2) and $\operatorname{rank}\mathsf{Z} \ge m - 1$ give $m - 1 \le M_\varrho k(k-1)\|\zeta\|$ for every m, contradicting the hypothesis by Step 2.

Step 4 (the linear family and symmetric spaces). If $\Lambda^k_\pi T$ were analytic at a point, T would have a lift (prose after Definition 11.1), whose restriction to $c_{00}(\mathbb{N}, K)^k$ Steps 1–3 exclude. For a symmetric sequence space G and $y \in c_{00}(\mathbb{N}, K)$ with entries in $\{0, \pm 1\}$, $\|y\|_G = \|1_{\operatorname{supp} y}\|_G = \varphi_G(|\operatorname{supp} y|)$ by permutation invariance, and φ_G is nondecreasing, so $\varphi^\pm_G = \varphi_G$. □

Proof of Corollary 11.15. Step 1 (item (1)). For $r_1 < 2$, P has a lower r_1-estimate (example after Theorem 5.6); apply Theorem 11.9(3). For $E = \ell^{r_2}(\mathbb{N}, K)$, $r_2 < 2$, $(E, \widehat{E})$ is good for Banach values (Theorem 5.6(3)); apply Proposition 11.11. For $r_1 = 2$ apply Theorem 11.13(3) with the unit vectors of $\ell^2(\mathbb{N}, K)$, and with $r = r_2$ and the r_2-norm $\|\cdot\|_{r_2}$ of $\ell^{r_2}(\mathbb{N}, K)$ over the nonarchimedean K (paragraph after Corollary 5.10), or with any $r \in (2, \infty)$ and an equivalent nonarchimedean norm of E.

Step 2 (item (2)). For $E = \ell^{r_2}(\mathbb{N}, K)$ or $c_0(\mathbb{N}, K)$, $\|\mu_a\|_{E\to E} \le \|a\|_\infty \le \|a\|_P$, so $a \mapsto \mu_a$ extends from $c_{00}(\mathbb{N}, K)$ to a linear family $T\colon P \to \mathcal{L}(E, \widehat{E})$, and analyticity of $\Lambda^k_\pi T$ at a point would give a lift (prose after Definition 11.1). For $P = c_0(\mathbb{N}, K)$ and E equal to $\ell^{r_2}(\mathbb{N}, K)$, $r_2 \ge 2$, or $c_0(\mathbb{N}, K)$, its restriction to $c_{00}(\mathbb{N}, K)^k$ is bounded for the supremum norm, which Theorem 7.7 excludes. For $2 < r_1 < \infty$ and E equal to $c_0(\mathbb{N}, K)$ or $\ell^{r_2}(\mathbb{N}, K)$ with $1/r_1 + 1/r_2 < 1/2$, Theorem 11.14 applies with $\varphi_{\ell^{r_1}}(n) = n^{1/r_1}$ and $\varphi_E(n) = 1$ or n^{1/r_2}, as $(2m)^{2/r_1}(4m)^{2/r_2}/m \to 0$. If K is complete and E has an equivalent nonarchimedean norm, Lemma 7.17 gives bounded linear $\widehat{\psi}_E\colon c_0(\mathbb{N}, K) \to E$ and $\theta_E\colon E \to c_0(\mathbb{N}, K)$ with $\theta_E\widehat{\psi}_E = \mathrm{id}$, and $T_E(a) = \widehat{\psi}_E\mu_a\theta_E$ is a linear family on P; a lift Q_{T_E} of it gives the lift $\widehat{\Lambda}^k_\pi\theta_E \circ Q_{T_E}(a_1, \dots, a_k) \circ \Lambda^k_\pi\widehat{\psi}_E$ of $a \mapsto \Lambda^k_\pi\mu_a$ on $c_0(\mathbb{N}, K)$, which the case $E = c_0(\mathbb{N}, K)$ excludes.

Step 3 (item (3)). As $\|\mu_a\|_{E\to E} = \|a\|_\infty$ for $a \in c_{00}(\mathbb{N}, K)$ (Lemma 7.11(1),(3)), the argument of Step 1 of the proof of Theorem 11.10, with (E, E) for (E_0, P) and with $a = 1_{\operatorname{supp} x}$, $b = 1_{\operatorname{supp} y}$ in (7.2), shows that a lift in a degree $k \geq p$ bounds $\mathcal{P}_<$ on $E \otimes_\pi E$; Theorem 8.1, (3)⇒(7), then makes (E, E) and $(E, \widehat{E})$ good for Banach values in every degree. Conversely, if (E, E) is good for Banach values in a degree $k \geq p$, (1)⇒(7) of that theorem gives the same; then Proposition 11.11(1) makes $(c_0(\mathbb{N}, K); E)$ good in every degree, and the prose after Definition 11.1 makes the multiplier family liftable in every degree. □

O.7. **Tangent bundles and structure groups.**

Proof of Theorem 11.17. Step 1 (item (1), the shear chart). The factors P_1, D, E of P_0 are Banach spaces. As $(x, e) \mapsto u(x)e$ composes $(x, e) \mapsto (u(x), e)$ with the bounded bilinear evaluation, ψ and $(x, d, e) \mapsto (x, d-u(x)e, e)$ are analytic (Lemma O.1(2)); they map $U\times D\times E$ to itself, are mutually inverse, and have analytic derivatives (paragraph after Definition 11.16). By the product rule $D\psi(x, d, e)(h, b, c) = (h, b + (Du(x)h)e + u(x)c, c)$, so $\pi_D D\psi(x, d, e)i_E = u(x)$, and Proposition 3.1 gives the identity. Analyticity of $\Lambda^k_\pi D\psi$ at (x_0, d_0, e_0), composed with $x \mapsto (x, d_0, e_0)$ and $T \mapsto \widehat{\Lambda}^k_\pi \pi_D \circ T \circ \Lambda^k_\pi i_E$, would make $\Lambda^k_\pi u$ analytic at x_0; for ψ^{-1} use $\pi_D D(\psi^{-1})i_E = -u$.

Step 2 (item (1), the manifold M). The chart change from id_M to $J\circ\psi\circ J^{-1}$ is $J\circ\psi\circ J^{-1}$, with derivative $J\circ D\psi(J^{-1}y)\circ J^{-1}$ at $y \in M$; so the transitions of $\widehat{\Lambda}^k_\pi TM$ are $y \mapsto \widehat{\Lambda}^k_\pi J\circ\Lambda^k_\pi D\psi(J^{-1}y)\circ \Lambda^k_\pi J^{-1}$ and the same expression with ψ^{-1} (Proposition 3.1). They are analytic exactly where $\Lambda^k_\pi D\psi$ and $\Lambda^k_\pi D(\psi^{-1})$ are, so not at the points $J(x_0, d, e)$, by Step 1. For $\mathrm{Alt}^k(TM; \widehat{\Lambda}^k_\pi D)$ use $A^k_{E,D;F}(u) = A^k(i_E) \circ A^k(D\psi) \circ A^k(\pi_D)$, $A^k(J \circ D\psi \circ J^{-1}) = A^k(J^{-1}) \circ A^k(D\psi) \circ A^k(J)$ and the single target $F = \widehat{\Lambda}^k_\pi D$ of Theorem 3.2(1).

Step 3 (item (2)). If $(P_0; P_0)$ is good, then for f as in Definition 11.16, Df is an analytic $\mathcal{L}(P_0)$-valued family, so P_0 is tangent-good; tangent goodness gives the bundle statement (sentence after Definition 11.16); if the bundles are analytic, (1) with $P_1 = D = E = P_0$ and $P_0 \cong P_0^3$ makes $\Lambda^k_\pi u$ analytic for every analytic u into $\mathcal{L}(P_0)$.

Step 4 (item (3)). For $E = c_0(\mathbb{N}, K)$ and $E = \ell^r(\mathbb{N}, K)$, $4 < r < \infty$, $\|a \cdot e\|_E \leq \|a\|_\infty \|e\|_E$, so $(a, e) \mapsto a \cdot e$ is bounded bilinear, $D\psi(a, d, e)(h, b, c) = (h, b + h \cdot e + a \cdot c, c)$ and, with $i_3(c) = (0, 0, c)$ and $\pi_2(h, b, c) = b$, $\pi_2 D(\psi^{\pm 1})(a, d, e)i_3 = \pm\mu_a$. As in Step 1, analyticity at a point would make $a \mapsto \Lambda^k_\pi \mu_a$ analytic at a point of E, which Theorem 11.14 with $P = E$ excludes, as $\varphi_E(n)$ is 1 or $n^{1/r}$ and $(2m)^{2/r}(4m)^{2/r}/m \to 0$; interleaving coordinates gives $E \cong E^3$. If K is complete and P_0 is an infinite-dimensional Banach space with an equivalent nonarchimedean norm, Lemma 7.17 gives $\widehat{\psi}_{P_0}$, θ_{P_0} with $\theta_{P_0}\widehat{\psi}_{P_0} = \mathrm{id}$, so $P_0 \cong c_0 \oplus P_2$ with $c_0 = c_0(\mathbb{N}, K)$ and $P_2 = \ker \theta_{P_0}$, and $P_0 \cong c_0 \times c_0 \times (c_0 \oplus P_2)$; the chart $(a, d, e, z) \mapsto (a, d + a \cdot e, e, z)$ again has $\pi_2 D\psi i_3 = \mu_a$. □

Proof of Theorem 11.19. Step 1 (item (1)). Let g be an H-family and $x_0 \in U$. Left multiplication by $g(x_0)$ is a homeomorphism of H, so $g(U') \subseteq g(x_0)\mathsf{H}_0$ for an open $U' \ni x_0$; then $x \mapsto g(x_0)^{-1}g(x)$ is a V_0-family on U', and $\Lambda^k_\pi g(x) = \widehat{\Lambda}^k_\pi g(x_0) \circ \Lambda^k_\pi(g(x_0)^{-1}g(x))$ (Proposition 3.1). Conversely, for a V_0-family u and $x_0 \in U$ choose an open $U' \ni x_0$ and $N \geq 0$ with $|\vartheta|^N\|u\| < \varepsilon$ on U', and $\lambda_i = \vartheta^{N+i}$, $0 \leq i \leq k$. Each $1 + \lambda_i u$ is an H-family on U', with values in $1 + \mathsf{V}_0(\varepsilon)$, and Lemma O.3 with $G = \widehat{E}$, a the inclusion of E and $b = u(x)$ gives $\Lambda^k_\pi u = \sum_i \chi_{ki}\Lambda^k_\pi(1 + \lambda_i u)$. For Banach E take $\mathsf{H} = \mathsf{H}_0 = \mathrm{GL}(E)$, $\mathsf{V}_0 = \mathcal{L}(E)$ and $\varepsilon = 1$ (Neumann series).

Step 2 (item (2)). For $\|w\| < 1$, $\|(1 + w)x\| = \|x\|$ by the nonarchimedean inequality and $1 + w$ is invertible; apply (1) with $\mathsf{H} = \mathsf{H}_0 = \mathrm{Isom}(E)$, $\mathsf{V}_0 = \mathcal{L}(E)$, $\varepsilon = 1$.

Step 3 (item (3)). As $\sigma_\sharp\mu_a\sigma'_\sharp\mu_b = (\sigma\sigma')_\sharp\mu_{(a\circ\sigma')b}$ and $(\sigma_\sharp\mu_a)^{-1} = \mu_{1/a}\sigma_\sharp^{-1}$, H_m is a group. If $\sigma(t) \neq t$, the tth coordinate of $(\sigma_\sharp\mu_a - 1)e_t$ is -1, so $\|\sigma_\sharp\mu_a - 1\| \geq 1$; hence $\{h \in \mathsf{H}_\mathrm{m} : \|h-1\| < 1\} \subseteq \mathsf{H}_\mathrm{d}$, and its translates show that H_d is open in H_m. As $\|\mu_a\| = \|a\|_\infty$, the nonarchimedean inequality gives $1 + \mathsf{V}_\mathrm{d}(1) \subseteq \mathsf{H}_\mathrm{d}$; apply (1) with $\mathsf{H}_0 = \mathsf{H}_\mathrm{d}$, $\mathsf{V}_0 = \mathsf{V}_\mathrm{d}$, $\varepsilon = 1$, and H equal to H_m or H_d.

Step 4 (item (4)). By (3) it suffices that $\Lambda^k_\pi\mu_x$ is analytic at no point for the V_d-family $x \mapsto \mu_x$ on $c_0(\mathbb{N}, K)$, by Proposition 2.5 and Theorem 7.7 with $r = \infty$. □

Proof of Proposition 11.20. Step 1 (H is a discrete group). As $g(u)g(u') = g(u+u')$, H is a group. For $u \neq u'$ in U some $e_i^*((u-u')e_j)$ is a nonzero element of κ', so $\|g(u) - g(u')\| \geq \|u - u'\| \geq |e_i^*((u-u')e_j)| = 1$; hence H is discrete, and H-families and their exterior powers are locally constant.

Step 2 ($g(\mu_x)$ lies in the closed span). For $x \in c_0(\mathbb{N}, K)$, the expansions of the coordinates give $x = \sum_{n \geq n_0} \vartheta^n x_{[n]}$, $|\vartheta|^{n_0} \geq \|x\|_\infty$, with $x_{[n]} \in \kappa'^{\mathbb{N}}$ finitely supported as $x(t) \to 0$; so $\mu_{x_{[n]}} \in \mathsf{U}$, $\|\vartheta^n \mu_{x_{[n]}}\| \leq |\vartheta|^n$, and $g(\mu_x) - 1 = \sum_n \vartheta^n (g(\mu_{x_{[n]}}) - 1)$ lies in the closed span of H.

Step 3 (no point of analyticity). By (O.1), $\Lambda^k_\pi\mu_x = \widehat{\Lambda}^k_\pi\pi_D \circ \Lambda^k_\pi g(\mu_x) \circ \Lambda^k_\pi i_E$, so analyticity of $x \mapsto \Lambda^k_\pi g(\mu_x)$ at a point is excluded by Step 4 of the proof of Theorem 11.19. □

Proof of Proposition 11.21. Step 1 (item (1)). Suppose that Q is as in Theorem 11.14, with $\|Q(a_1, \dots, a_k)\| \leq \|Q\| \prod_i \|a_i\|_P$. Steps 2 and 3 of the proof of Theorem 7.20 apply to Q with $\|a_i\|_P$ in place of $\|a_i\|_\infty$. For 0/1 vectors a, b, x, y supported in $\{1, \dots, n\}$, $\|a\|_P, \|b\|_P \leq \varphi_P^\pm(n)$, $\|1_{>n}\|_P \leq \varphi_P^\pm(k-2)$, $\|x\|_E, \|y\|_E \leq \varphi_E^\pm(n)$ and $\|e_t\|_E \leq \varphi_E^\pm(1)$. So for every even $n \geq 2$

$$n \leq k(k-1)\, c_k\, \|Q\|\, \varphi_P^\pm(n)^2 \varphi_E^\pm(n)^2, \qquad c_k = \varphi_P^\pm(k-2)^{k-2} \varphi_E^\pm(1)^{k-2},$$

with $c_2 = 1$. As $\varphi_P^\pm$ and $\varphi_E^\pm$ are nondecreasing, $n/(\varphi_P^\pm(n)^2\varphi_E^\pm(n)^2)$ is unbounded along even n, which is a contradiction. The assertion on T follows as in Step 4 of the proof of Theorem 11.14.

Step 2 (items (2)–(4)). For $n \geq 1$, $\varphi^\pm_{c_0(\mathbb{N},K)}(n) = 1$ and $\varphi^\pm_{\ell^r(\mathbb{N},K)}(n) = n^{1/r}$. Item (2) follows from item (1) as in Step 2 of the proof of Corollary 11.15, since $n^{1-2/r_1-2/r_2} \to \infty$ when $1/r_1 + 1/r_2 < 1/2$, with $1/r_1 = 0$ for $P = c_0(\mathbb{N}, K)$ and $1/r_2 = 0$ for $E = c_0(\mathbb{N}, K)$. Item (3) follows as in Step 4 of the proof of Theorem 11.17, with item (1) for $P = E$ in place of Theorem 11.14, since $n^{1-4/r} \to \infty$ for $r > 4$. Item (4) follows as in Step 4 of the proof of Theorem 11.19, with item (1) for $P = E = c_0(\mathbb{N}, K)$ in place of Theorem 7.7. □

Appendix P. Deferred proofs

This appendix contains the proofs to which Sections 3–10 refer, in the order of their statements; the notation is that of the corresponding section.

P.1. **Proofs for Section 3.** The notation is that of Section 3.

Proof of Proposition 3.5. The maps Q are bounded and k-linear, and naturality gives $Q(u, \dots, u) = \Lambda^k_\pi u \circ \widehat{\wedge}^k_E \circ s = \Lambda^k_\pi u$ in (1) and $Q(u, \dots, u) = \rho \circ \mathrm{alt}^k_D \circ \Lambda^k_\pi u = \Lambda^k_\pi u$ in (2); Lemma 3.4 proves (1)–(3). For (4), given s, send $m \in \mathrm{Mult}^k(E; F)$ to the alternating map $m^\otimes \circ s \circ \omega_E$. If $m = m' \circ \omega_E$ with $m' \in \mathcal{L}(\Lambda^k_\pi E, F)$, then $m^\otimes = m' \circ \wedge^k_E$ and $m^\otimes \circ s = m'$; so this is a retraction, and it commutes with postcomposition since $(g \circ m)^\otimes = g \circ m^\otimes$. Conversely, the retraction for $F = \widehat{\otimes}^k_\pi E$ sends the canonical k-linear map to $s \circ \omega_E$ for some $s \in \mathcal{L}(\Lambda^k_\pi E, \widehat{\otimes}^k_\pi E)$. Postcomposition by $\widehat{\wedge}^k_E$ turns the canonical map into the alternating map ω_E, so $\widehat{\wedge}^k_E \circ s \circ \omega_E = \omega_E$ and $\widehat{\wedge}^k_E \circ s = \mathrm{id}$. Without completion use $F = \otimes^k_\pi E$. □

Proof of Lemma 3.6. (1) Given $\tilde\alpha_1$, $\alpha_2 - i'\tilde\alpha_1$ vanishes on $\ker q$, so $\tilde\alpha_3(qy) = (\alpha_2 - i'\tilde\alpha_1)y$ defines a linear map, bounded because q is open, with $q'\tilde\alpha_3 q = q'\alpha_2 = \alpha_3 q$. Given $\tilde\alpha_3$, $q'(\alpha_2 - \tilde\alpha_3 q) = 0$, so $\alpha_2 - \tilde\alpha_3 q = i'\tilde\alpha_1$ for a linear $\tilde\alpha_1$, bounded because i' is bounded below, and $i'\tilde\alpha_1 i = \alpha_2 i = i'\alpha_1$. (2) Apply (1) to $\mathcal{Y}' = \mathcal{Y}$ and the identities. (3) The space Y_α is a Banach space, and q_α is well defined, surjective and open, with kernel $\{[(y', iy)]\} = \{[(y' + \alpha y, 0)]\} = i_\alpha(Y')$; and $\|y'\| \le \|y' + \alpha y\| + C\|\alpha\|\|iy\|$ for all y shows that i_α is injective and bounded below. An extension $\tilde\alpha$ gives the left inverse $[(y', y)] \mapsto y' + \tilde\alpha y$ of i_α, and a left inverse λ gives the extension $y \mapsto \lambda[(0, y)]$, since $[(0, iy)] = [(\alpha y, 0)]$. Now apply (2). □

Proof of Lemma 3.7. Since $\wedge_E^k = \omega_E^\otimes$ is contractive, it suffices to find for y and $\delta > 0$ a preimage z with $\|z\| < \|y\| + 3\delta$. For $y \in \Lambda_\pi^k E$ take $z = \sum_j x_{j1} \otimes \cdots \otimes x_{jk}$ for a representation $y = \sum_j x_{j1} \wedge \cdots \wedge x_{jk}$ with $\sum_j \prod_a \|x_{ja}\| < \|y\| + \delta$. For $y \in \widehat\Lambda_\pi^k E$ write $y = \sum_{l\ge1} y_l$ with $y_l \in \Lambda_\pi^k E$, $\|y_1\| < \|y\| + \delta$ and $\sum_{l\ge2}\|y_l\| < \delta$, and choose preimages z_l with $\|z_l\| < \|y_l\| + 2^{-l}\delta$; the series $z = \sum_l z_l$ converges, $\widehat\wedge_E^k z = y$ and $\|z\| < \|y\| + 3\delta$. □

Proof of Lemma 3.10. (1)⇒(2): a retraction ρ, extended to $\widehat\otimes_\pi^k D$, satisfies $\rho\,\mathrm{alt}_D^k = \mathrm{id}$ on $\widehat\Lambda_\pi^k D$; so $\|y\| \le \|\rho\|\|\mathrm{alt}_D^k y\|$, ρ is a left inverse of the first map of $\mathcal{R}_D^k$, and Lemma 3.6(2) applies. (2)⇒(1): restrict a bounded left inverse to $\otimes_\pi^k D$. (1)⇔(3): $\mathrm{alt}_D^k \omega_D$ is the antisymmetrization of $(y_1, \dots, y_k) \mapsto y_1 \otimes \cdots \otimes y_k$. So $\eta(y_1, \dots, y_k) = \rho(y_1 \otimes \cdots \otimes y_k)$ gives $\mathcal{A}_D\eta = \rho\,\mathrm{alt}_D^k \omega_D = \omega_D$; conversely $\rho = \eta^\otimes$ satisfies $\rho\,\mathrm{alt}_D^k \omega_D = \mathcal{A}_D\eta = \omega_D$, and the wedges span $\Lambda_\pi^k D$. □

Proof of Lemma 3.11. (1)⇒(3): choose a projection R_n onto $\widehat\Gamma_\pi^n P$ with $\|R_n\| \le g_n(P) + \delta$. The linearization $\Phi^\otimes\colon \widehat\otimes_\pi^n P \to Z$ of Φ, of norm $\|\Phi\|$, maps each $h^{\otimes n}$, hence $\widehat\Gamma_\pi^n P$, into W. So $Q(h_1, \dots, h_n) = \Phi^\otimes R_n(h_1 \otimes \cdots \otimes h_n)$ takes values in W and has diagonal $\Phi^\otimes(h^{\otimes n})$. (3)⇒(2): take $W = \widehat\Gamma_\pi^n P$, $Z = \widehat\otimes_\pi^n P$ and $\Phi(h_1, \dots, h_n) = h_1 \otimes \cdots \otimes h_n$. (2)⇒(1): the linearization of the n-linear map maps $\widehat\otimes_\pi^n P$ into $\widehat\Gamma_\pi^n P$ and fixes every $h^{\otimes n}$, hence every point of $\widehat\Gamma_\pi^n P$. For the last assertion, for $\Phi \in \mathrm{Mult}^n(P; Y)$ the n-linear map $(h_1, \dots, h_n) \mapsto \Phi^\otimes R_n(h_1 \otimes \cdots \otimes h_n)$ has the diagonal of Φ, and it is 0 when Φ has zero diagonal, since $\Phi^\otimes$ then vanishes on $\widehat\Gamma_\pi^n P$. So it defines a right inverse of the quotient map, of norm at most $\|R_n\|$, that commutes with postcomposition. □

Proof of Proposition 3.13. Naturality of $\wedge^k$, extended to completions, gives $\widehat\wedge_G^k w^{\otimes k} = \widehat\Lambda_\pi^k w\,\widehat\wedge_E^k$. So $w^{\otimes k}(\mathcal{N}_E^k) \subseteq \mathcal{N}_G^k$, and $w^{\otimes k}|_{\mathcal{N}_E^k}$, $w^{\otimes k}$, $\widehat\Lambda_\pi^k w$ satisfy the hypotheses of Lemma 3.6(1) for $\mathcal{S}_E^k$ and $\mathcal{S}_G^k$. Now (a)⇔(b) because sections extend by continuity and restrict, (b)⇔(c) is Lemma 3.6(1), and (c)⇔(d) is Lemma 3.6(3). In (2), naturality of $\wedge^k$ and alt^k gives $\widehat\wedge_{G_1}^k v^{\otimes k} s\,\Lambda_\pi^k v' = \widehat\Lambda_\pi^k v\,\widehat\wedge_G^k s\,\Lambda_\pi^k v'$ and $\widehat\Lambda_\pi^k v\,\rho\,v'^{\otimes k}\,\mathrm{alt}_{G_0}^k = \widehat\Lambda_\pi^k v\,\rho\,\mathrm{alt}_E^k\,\Lambda_\pi^k v'$; in either case the result is $\widehat\Lambda_\pi^k v\,\Lambda_\pi^k w\,\Lambda_\pi^k v' = \Lambda_\pi^k(vwv')$. □

From here on, q and $\overline{\mathrm{ev}}$ are the maps defined after Lemma 3.14.

Proof of Theorem 3.16. In each case we give an extension as in Theorem 3.15(2), or a lift.

Step 1 (source). By Proposition 3.13(1), $w^{\otimes k}|_{\mathcal{N}_E^k}$ has a bounded linear extension $\xi_w\colon \widehat\otimes_\pi^k E \to \mathcal{N}_G^k$, and $\tau\xi_w$ extends $\mathrm{ev}\,|_{\mathcal{N}_E^k}$. For (S_E^k) take $w = \mathrm{id}_E$ and $\tau = \mathrm{ev}\,|_{\mathcal{N}_E^k}$. For the data (G, w, ε), the formula for τ in part (1) defines a bounded linear map on $\widehat\otimes_\pi^k G$ with values in $\widehat\Lambda_\pi^k \widehat D = \widehat\Lambda_\pi^k D$ (the paragraph before Theorem 3.16) and $\|\tau\| \le \|\varepsilon\|^k$. Naturality of $\wedge^k$ gives $\tau(\zeta)(u, \dots, u) = \widehat\Lambda_\pi^k \varepsilon(u)\,\widehat\wedge_G^k \zeta$, which vanishes for $\zeta \in \mathcal{N}_G^k$. On pure tensors, $\tau(w^{\otimes k}(x_1 \otimes \cdots \otimes x_k))(u_1, \dots, u_k) = \varepsilon(u_1)wx_1 \wedge \cdots \wedge \varepsilon(u_k)wx_k = u_1x_1 \wedge \cdots \wedge u_kx_k$, so $\tau w^{\otimes k} = \mathrm{ev}$ by (3.2) and continuity.

Step 2 (parameter). Let γ be a bounded linear right inverse of q. Then $\xi = \operatorname{ev} - \gamma\overline{\operatorname{ev}}\,\widehat{\wedge}^k_E$ equals ev on $\mathcal{N}^k_E$, and $q\xi = q\operatorname{ev} - \overline{\operatorname{ev}}\,\widehat{\wedge}^k_E = 0$. If $(\Gamma^k_{\mathcal{L}(E,D)})$ holds, $\mathcal{M}^k(\mathcal{L}(E,D);\widehat{\Lambda}^k_\pi D)$ splits by Lemma 3.11. In the general form, the operators vanishing on $\mathcal{N}^k_E$ form a closed subspace W of $Z = \mathcal{L}(\widehat{\otimes}^k_\pi E, \widehat{\Lambda}^k_\pi D)$, and $T \mapsto T\widehat{\wedge}^k_E$ is an isometry of $\mathcal{L}(\widehat{\Lambda}^k_\pi E, \widehat{\Lambda}^k_\pi D)$ onto W, by Lemma 3.7. Lemma 3.11(3) gives a bounded k-linear $Q'\colon P^k \to \mathcal{L}(\widehat{\Lambda}^k_\pi E, \widehat{\Lambda}^k_\pi D)$ with $Q'(h,\dots,h)\widehat{\wedge}^k_E = \Phi(h,\dots,h)$. Then $Q(u_1,\dots,u_k) = Q'(\psi u_1,\dots,\psi u_k)|_{\Lambda^k_\pi E}$ is bounded and k-linear with $Q(u,\dots,u) = \Lambda^k_\pi u$, and Lemma 3.4 applies.

Step 3 (target). The formula $\xi'(z)(u_1,\dots,u_k) = \operatorname{alt}^k_D\big(\operatorname{ev}(z)(u_1,\dots,u_k)\big) - (u_1\otimes\cdots\otimes u_k)\operatorname{alt}^k_E(\widehat{\wedge}^k_E z)$ defines a bounded linear $\xi'\colon \widehat{\otimes}^k_\pi E \to \operatorname{Mult}^k(\mathcal{L}(E,D);\widehat{\otimes}^k_\pi D)$ that equals $\operatorname{alt}_*\operatorname{ev}$ on $\mathcal{N}^k_E$. Its values have zero diagonal, since $\xi'(z)(u,\dots,u) = \operatorname{alt}^k_D \widehat{\Lambda}^k_\pi u\,\widehat{\wedge}^k_E z - u^{\otimes k}\operatorname{alt}^k_E \widehat{\wedge}^k_E z = 0$ by naturality of alt^k. Lemma 3.6(3) gives the first assertion. If ρ is a retraction of alt^k_D, extended to $\widehat{\otimes}^k_\pi D$, then $\rho\circ\xi'$ extends $\rho\operatorname{alt}^k_D\operatorname{ev}|_{\mathcal{N}^k_E} = \operatorname{ev}|_{\mathcal{N}^k_E}$.

Step 4 (relative target). Let $\rho'\colon \otimes^k_\pi G \to \widehat{\Lambda}^k_\pi \widehat{D} = \widehat{\Lambda}^k_\pi D$ satisfy $\rho'\operatorname{alt}^k_G = \Lambda^k_\pi w'$, and put

$$\xi(z)(u_1,\dots,u_k) = \operatorname{ev}(z)(u_1,\dots,u_k) - \rho'\big(\varepsilon(u_1)\otimes\cdots\otimes\varepsilon(u_k)\big)\operatorname{alt}^k_E(\widehat{\wedge}^k_E z).$$

This is a bounded linear map, of norm at most $1 + k!\|\rho'\|\|\varepsilon\|^k$, that equals ev on $\mathcal{N}^k_E$. By naturality of alt^k for $\varepsilon(u)\colon E \to G$, $\rho'\varepsilon(u)^{\otimes k}\operatorname{alt}^k_E = \rho'\operatorname{alt}^k_G \widehat{\Lambda}^k_\pi\varepsilon(u) = \widehat{\Lambda}^k_\pi w' \widehat{\Lambda}^k_\pi\varepsilon(u) = \widehat{\Lambda}^k_\pi u$, so $\xi(z)(u,\dots,u) = 0$. □

Proof of Theorem 3.18. Step 1 (part (1)). By (3.2) and (3.4), $\operatorname{ev}(j_\otimes e^\otimes_t) = Q^\sharp_<(e^\wedge_t)$ for $t \in \Sigma^k_<$, which is the first identity on a basis. Hence $q\circ Q^\sharp_< = \overline{\operatorname{ev}}\circ\widehat{\wedge}^k_E\circ j_\otimes\circ s_< = \overline{\operatorname{ev}}\circ j_\Lambda\circ\wedge_V\circ s_< = \overline{\operatorname{ev}}\circ j_\Lambda$.

Step 2 (part (2)). Since Q is a lift, $q\circ Q^\sharp = \overline{\operatorname{ev}}$, so $q\circ L = \overline{\operatorname{ev}}\circ j_\Lambda - q\circ Q^\sharp_< = 0$ by part (1). Thus L takes values in $\ker q = \operatorname{Mult}^k_0$, and Φ is k-linear with zero diagonal.

Step 3 (part (3)). Put $T = Q^\sharp_< + L$. Since $\|Ty\| \le \|T\|\|j_\Lambda y\|$, $j_\Lambda(\Lambda^k_K V)$ is dense and Mult^k is complete, $T = \widehat{T}\circ j_\Lambda$ for a bounded linear $\widehat{T}\colon \widehat{\Lambda}^k_\pi E \to \operatorname{Mult}^k$ with $\|\widehat{T}\| = \|T\|$. By part (1) and $q\circ L = 0$, $q\circ\widehat{T} = \overline{\operatorname{ev}}$ on a dense subspace, hence everywhere. So $\widehat{T} = Q^\sharp$ for a lift Q, with $\|Q\| = \|T\|$, and Q is unique by density. Conversely, a lift Q gives the correction L of part (2), with $\|Q^\sharp_< + L\| = \|Q^\sharp\circ j_\Lambda\| = \|Q\|$ by density. This proves the formula for $c_k(E,D)$ and the last assertion.

Step 4 (part (4)). Since $q\circ\operatorname{ev} = \overline{\operatorname{ev}}\circ\widehat{\wedge}^k_E$, part (1) gives $q(\operatorname{ev}(x_1\otimes\cdots\otimes x_k)) = \overline{\operatorname{ev}}(j_\Lambda(x_1\wedge\cdots\wedge x_k)) = q(Q^\sharp_<(x_1\wedge\cdots\wedge x_k))$, so $\Omega_<$ takes values in Mult^k_0. For $x \in V^k$,

$$\Omega_<(x) - L(x_1\wedge\cdots\wedge x_k) = \operatorname{ev}(x_1\otimes\cdots\otimes x_k) - (Q^\sharp_< + L)(x_1\wedge\cdots\wedge x_k);$$

since $\|\operatorname{ev}\| \le 1$ and $\|x_1\wedge\cdots\wedge x_k\|_\pi \le \prod_a\|x_a\|$, this gives the second inequality. For the first, let $\delta > 0$ and write $y \in \Lambda^k_K V$ as $\sum_i x_{i1}\wedge\cdots\wedge x_{ik}$ with $x_{ia} \in V$ and $\sum_i\prod_a\|x_{ia}\| < \|y\|_\pi + \delta$. Summing the identity for $(x_{i1},\dots,x_{ik})$ over i gives $\|(Q^\sharp_< + L)y\| \le (1 + \|\Omega_< - L\circ\wedge\|)(\|y\|_\pi + \delta)$; let $\delta \to 0$. The alternating k-linear maps $V^k \to \operatorname{Mult}^k_0$ are the maps $L\circ\wedge$, by the universal property of $\Lambda^k_K V$. So part (3) and the two inequalities give the remaining assertions. □

Proof of Lemma 3.19. Step 1 (algebra). Let $I_E \subseteq \otimes^k_K E$ be the span of the pure tensors with $x_a = x_{a+1}$ for some $a < k$. Modulo I_E, exchanging two adjacent entries changes the sign of a pure tensor, since $x\otimes y + y\otimes x = (x+y)\otimes(x+y) - x\otimes x - y\otimes y$. So I_E contains every pure tensor with two equal entries, and it is the kernel of the quotient map $\otimes^k_K E \to \Lambda^k_K E$, by the universal property of $\Lambda^k_K E$. The same holds for V.

Step 2 (closure). Let I be the image of I_E in $\otimes_\pi^k E$, let $z \in \otimes_\pi^k E$ be the class of $\tilde{z} \in \otimes_K^k E$, and let $\delta > 0$. Write the image of $\tilde{z}$ in $\Lambda_K^k E$ as $\sum_i x_{i1} \wedge \cdots \wedge x_{ik}$ with $\sum_i \prod_a \|x_{ia}\| < \|\wedge_E^k z\| + \delta$; by Step 1, $\tilde{z} - \sum_i x_{i1} \otimes \cdots \otimes x_{ik} \in I_E$, so the distance from z to I is at most $\|\wedge_E^k z\|$. If $z \in \mathcal{N}_E^k$ and $z_l \in \otimes_\pi^k E$ converge to z, then $\|\wedge_E^k z_l\| = \|\widehat{\wedge}_E^k (z_l - z)\| \to 0$, so z lies in the closure of I. As $I \subseteq \mathcal{N}_E^k$ and $\mathcal{N}_E^k$ is closed, $\mathcal{N}_E^k$ is the closure of I; with Step 1 this proves the first assertion.

Step 3 (entries in V). A pure tensor $x_1 \otimes \cdots \otimes x_k$ with $x_a = x_{a+1}$ is a limit of pure tensors with entries in V and equal entries in the slots a and $a+1$, since V is dense and $(x_1, \ldots, x_k) \mapsto x_1 \otimes \cdots \otimes x_k$ is continuous. With Step 2 this proves the last assertion. □

Proof of Corollary 3.20. By linearization and continuity, the bounded k-linear maps $\Psi \colon E^k \to \mathrm{Mult}_0^k$ correspond to the bounded linear maps $\xi \colon \widehat{\otimes}_\pi^k E \to \mathrm{Mult}_0^k$, by $\xi(x_1 \otimes \cdots \otimes x_k) = \Psi(x_1, \ldots, x_k)$, with $\|\xi\| = \|\Psi\|$. By (3.2) the displayed identity says $\xi(x_1 \otimes \cdots \otimes x_k) = \mathrm{ev}(x_1 \otimes \cdots \otimes x_k)$. If it holds whenever $x_a = x_{a+1}$ for some a, then $\xi = \mathrm{ev}$ on $\mathcal{N}_E^k$ by Lemma 3.19 and continuity, so it holds whenever two entries coincide; conversely, an extension of $\mathrm{ev}\big|_{\mathcal{N}_E^k}$ gives the identity on these tuples, whose pure tensors lie in $\mathcal{N}_E^k$. So the maps Ψ correspond to the extensions ξ of Theorem 3.15(2), and that theorem with its constants gives the corollary. □

Proof of Corollary 3.21. By Theorem 3.18(1), $q \circ \mathrm{ev} \circ \nu \circ j_\Lambda = \overline{\mathrm{ev}} \circ \widehat{\wedge}_E^k \circ \nu \circ j_\Lambda = \overline{\mathrm{ev}} \circ j_\Lambda = q \circ Q_<^\sharp$, so $\Omega_<^\nu$ takes values in Mult_0^k; it is homogeneous because ν is homogeneous and ev, j_Λ, $Q_<^\sharp$ are linear. Since $Q_<^\sharp$ is linear, $\Omega_<^\nu(\sum_i y_i) - \sum_i \Omega_<^\nu(y_i) = \mathrm{ev}\big(\nu(\sum_i j_\Lambda y_i) - \sum_i \nu(j_\Lambda y_i)\big)$; the argument of ev lies in $\mathcal{N}_E^k$ and has norm at most $2(1+\delta)\sum_i \|y_i\|_\pi$ by the bound on ν, and $\|\mathrm{ev}\| \le 1$. For a linear $L \colon \Lambda_K^k V \to \mathrm{Mult}_0^k$, $\Omega_<^\nu - L = \mathrm{ev} \circ \nu \circ j_\Lambda - (Q_<^\sharp + L)$ with $\|\mathrm{ev}(\nu(j_\Lambda y))\| \le (1+\delta)\|y\|_\pi$. So the least constant for $\Omega_<^\nu - L$ and $\|Q_<^\sharp + L\|$ each exceed the other by at most $1+\delta$, and Theorems 3.18(3) and 3.15 give the remaining assertions. □

Proof of Theorem 3.22. Put $C_\alpha = 2(c_+/c_-)\|\alpha\|(C_c C_e)^k$. For $z \in \otimes_K^k V$, $j_\otimes(z - s_<(\wedge_V z))$ lies in $\mathcal{N}_E^k$, since $\widehat{\wedge}_E^k \circ j_\otimes = j_\Lambda \circ \wedge_V$ and $\wedge_V \circ s_< = \mathrm{id}$; let $\alpha_0(z)$ be its image under α.

Step 1 (expansion). For $t \in \Sigma^k$ with a repeated entry put $\zeta_t = e_t^\otimes$. For $t \in \Sigma^k$ with distinct entries let $\sigma \in S_k$ be the permutation with $t \circ \sigma = (t_{\sigma(1)}, \ldots, t_{\sigma(k)}) \in \Sigma_<^k$, and put $\zeta_t = e_t^\otimes - \mathrm{sgn}(\sigma)\, e_{t\circ\sigma}^\otimes$. Since $e_{t_1} \wedge \cdots \wedge e_{t_k}$ is $\mathrm{sgn}(\sigma)\, e_{t\circ\sigma}^\wedge$ for distinct entries and 0 otherwise, $\wedge_V \zeta_t = 0$; and $\|j_\otimes \zeta_t\| \le 2C_e^k$. For $x_1, \ldots, x_k \in V$, expanding $x_l = \sum_t x_l(t) e_t$ gives the finite sum

$$x_1 \otimes \cdots \otimes x_k - s_<(x_1 \wedge \cdots \wedge x_k) = \sum_{t \in \Sigma^k} \prod_{l=1}^k x_l(t_l)\, \zeta_t.$$

Step 2 (bound). Since $|\!|\!|\cdot|\!|\!|_Y$ is a nonarchimedean norm, Step 1 gives

$$|\!|\!|\alpha_0(x_1 \otimes \cdots \otimes x_k)|\!|\!|_Y \le \max_t \prod_{l=1}^k |x_l(t_l)|\, |\!|\!|\alpha(j_\otimes \zeta_t)|\!|\!|_Y \le C_c^k \prod_{l=1}^k \|x_l\| \cdot \frac{2C_e^k \|\alpha\|}{c_-},$$

the maximum over the finitely many t with $\prod_l x_l(t_l) \neq 0$. As $\|\cdot\| \le c_+ |\!|\!|\cdot|\!|\!|_Y$, this gives $\|\alpha_0(x_1 \otimes \cdots \otimes x_k)\| \le C_\alpha \prod_l \|x_l\|$, and so $\|\alpha_0(z)\| \le C_\alpha \sum_i \prod_l \|x_{il}\|$ for every representation $z = \sum_i x_{i1} \otimes \cdots \otimes x_{ik}$ of $z \in \otimes_K^k V$ with entries in V. By the paragraph before Theorem 3.16, $\|\alpha_0(z)\| \le C_\alpha \|j_\otimes z\|$. Since $j_\otimes(\otimes_K^k V)$ is dense and Y is complete, there is a bounded linear $\tilde{\alpha} \colon \widehat{\otimes}_\pi^k E \to Y$ with $\tilde{\alpha} \circ j_\otimes = \alpha_0$ and $\|\tilde{\alpha}\| \le C_\alpha$.

Step 3 (agreement). If $z \in \otimes_K^k V$ and $\wedge_V z = 0$, then $\tilde{\alpha}(j_\otimes z) = \alpha_0(z) = \alpha(j_\otimes z)$. The pure tensors with entries in V and two equal adjacent entries lie in $\ker \wedge_V$, and their images span

a dense subspace of $\mathcal{N}_E^k$ by Lemma 3.19. So $\tilde{\alpha} = \alpha$ on $\mathcal{N}_E^k$ by continuity, and $\alpha_*[\mathcal{S}_E^k] = 0$ by Lemma 3.6(3).

Step 4 (the class). Let $\|\cdot\|_D$ be a nonarchimedean norm on $\widehat{\Lambda}_\pi^k D$ with $c_-\|\cdot\|_D \le \|\cdot\| \le c_+\|\cdot\|_D$. Take $Y = \mathrm{Mult}_0^k$ and, for $\Phi \in Y$, let $\|\Phi\|_Y$ be the least C with $\|\Phi(u_1,\dots,u_k)\|_D \le C\prod_l \|u_l\|$ for all $u_1,\dots,u_k \in \mathcal{L}(E,D)$. Then $\|\cdot\|_Y$ is a nonarchimedean norm on the Banach space Y, since $\|\cdot\|_D$ is one, and $c_-\|\cdot\|_Y \le \|\cdot\| \le c_+\|\cdot\|_Y$, since $\|\Phi\|$ is the least C with $\|\Phi(u_1,\dots,u_k)\| \le C\prod_l\|u_l\|$. The first part, with this Y and $\alpha = \mathrm{ev}\,|_{\mathcal{N}_E^k}$, gives $\mathrm{ob}_k(E,D) = 0$. □

P.2. **Proofs for Section 4.** The notation is that of Section 4.

Proof of Theorem 4.2. Since $k! = 0$ in K, $p = \mathrm{char}\,K$ satisfies $2 \le p \le k$. Let $\nu \in S_k$ be a cycle of length p and g the permutation matrix with $ge_i = e_{\nu(i)}$, which is monomial. Corollary 4.4(2) excludes (1)–(3) already for this g. For the algebraic powers, let H be the group generated by ν, acting on $I = [k]$, so that $V = K^k$. As in the proof of Corollary 4.4(2), a map equivariant for g is H-equivariant, and $\bar{\mathsf{H}}_{[k]} = \bar{\mathsf{H}}_{\mathbf{m}}$ has order p for the k-multiset $\mathbf{m}$ with value 1 on $[k]$; so Theorem 4.3(1)–(3) excludes such maps. □

Proof of Corollary 4.4. (1) We treat (1) and (2) of Theorem 4.3; for (3) replace T by $\operatorname{supp}\mathbf{m}$ and the stabilizers of T by those of $\mathbf{m}$. The first statement implies the second by restriction. If $h \in \mathsf{H}$ maps a k-subset T onto itself and induces on it a permutation of order $p = \mathrm{char}\,K$, then p divides $|\bar{C}_T|$ for the group C generated by h, and Theorem 4.3 excludes a C-equivariant map; so the second statement implies the third. If the first fails, Theorem 4.3 gives a k-subset T such that $p = \mathrm{char}\,K > 0$ divides $|\bar{\mathsf{H}}_T|$, and by Cauchy's theorem some element of H_T induces on T a permutation of order p; so the third fails. If $\mathrm{char}\,K = 0$, all $|\bar{\mathsf{H}}_T|$ and $|\bar{\mathsf{H}}_{\mathbf{m}}|$ are nonzero in K.

(2) Let H be the group generated by ν, acting on $I = [k]$ and on E_0 by $he_i = e_{h(i)}$; these operators are bounded since the e_i^* are, and satisfy (4.3) since $e_i^*(he_j) = \delta_{i,h(j)} = e_{h^{-1}(i)}^*(e_j)$. For $T = I$ and the k-multiset $\mathbf{m}$ with value 1 on I, $\mathsf{H}_T = \mathsf{H}_{\mathbf{m}} = \mathsf{H}$, and $\bar{\mathsf{H}}_T = \bar{\mathsf{H}}_{\mathbf{m}}$ has order p, which is 0 in K. An equivariance identity for g holds for its powers, since $(g^j)^{\otimes k} = (g^{\otimes k})^j$ and $\Lambda_\pi^k(g^j) = (\Lambda_\pi^k g)^j$, hence for all of H, which Theorem 4.3(4) excludes. □

Proof of Proposition 4.6. (1) This is Proposition B.8(1) applied to Q. With $u_S := \sum_{i\in S} u_i$, every k-linear Φ satisfies $\sum_{S\subseteq[k]}(-1)^{k-|S|}\Phi(u_S,\dots,u_S) = \sum_\sigma \Phi(u_{\sigma(1)},\dots,u_{\sigma(k)})$, as in (4.2). Apply this to $Q(\cdot)\circ\omega_E$ and to $(u_1,\dots,u_k) \mapsto \big((x_b) \mapsto u_1x_1 \wedge \cdots \wedge u_kx_k\big)$, which agree on the diagonal since $Q(u,\dots,u) = \Lambda_\pi^k u$: the right sides give $\sum_\sigma Q(u_{\sigma(1)},\dots,u_{\sigma(k)})\circ\omega_E = \Lambda_{\mathrm{sym}}(u_1,\dots,u_k)\circ\omega_E$, and pure wedges span $\Lambda_\pi^k E$. (2) A symmetric Q would give $\Lambda_{\mathrm{sym}} = k!\,Q = 0$ by (1). Let $f_1,\dots,f_k$ be the functionals, choose $x_1,\dots,x_k \in E$ with $f_i(x_j) = \delta_{ij}$ and, since $\Lambda_\pi^k D$ is dense in $\widehat{\Lambda}_\pi^k D \ne 0$, $d_1,\dots,d_k \in D$ with $d_1\wedge\cdots\wedge d_k \ne 0$. For $u_i = f_i(\cdot)\,d_i$ only $\sigma = \mathrm{id}$ contributes to (4.4), so $\Lambda_{\mathrm{sym}}(u_1,\dots,u_k)(x_1\wedge\cdots\wedge x_k) = d_1\wedge\cdots\wedge d_k \ne 0$. (3) By (4.1), $\wedge_D^k = \rho\circ\mathrm{alt}_D^k\circ\wedge_D^k = \sum_\sigma \mathrm{sgn}(\sigma)\rho\circ\tau_\sigma = k!\,\rho = 0$, so $\Lambda_\pi^k D = \wedge_D^k(\otimes_\pi^k D) = 0$. (4) For $w \in \Lambda_\pi^k E$, (4.1) gives $\mathrm{alt}_E^k(w) = \mathrm{alt}_E^k\,\widehat{\wedge}_E^k s(w) = \sum_\sigma \mathrm{sgn}(\sigma)\tau_\sigma s(w) = k!\,s(w) = 0$. With f_i and x_j as in (2), the bounded functional on $\widehat{\otimes}_\pi^k E$ linearizing $(y_a) \mapsto \prod_a f_a(y_a)$ takes the value $\sum_\sigma \mathrm{sgn}(\sigma)\prod_a f_a(x_{\sigma(a)}) = 1$ at $\mathrm{alt}_E^k(x_1\wedge\cdots\wedge x_k)$, a contradiction. □

P.3. **Proofs for Section 5.**

Proof of Theorem 5.2. Step 1 (simultaneous extension). Let Y be a nonarchimedean seminormed space, G a nonarchimedean spherically complete space, $T_\alpha\colon Y \to G$ a nonempty family of bounded operators and $r_\alpha \ge 0$ radii with $\|T_\alpha - T_\beta\| \le \max(r_\alpha, r_\beta)$, and T_0 a linear map on

a subspace Y_0 with $\|T_0 d - T_\alpha d\| \le r_\alpha \|d\|$ for all $d \in Y_0$ and all α. We extend T_0 to Y with all these bounds. For $x \notin Y_0$, the balls

$$\overline{B}\big(T_\alpha(d+x) - T_0 d,\ r_\alpha\|d+x\|\big), \qquad \alpha,\ d \in Y_0,$$

meet pairwise: for $r_\alpha \le r_\beta$, the centers for (α, d) and (β, e) differ by at most $\max(r_\alpha\|d + x\|, r_\beta\|e + x\|)$, by the hypotheses and the nonarchimedean inequality. A common point, given by spherical completeness, is a value at x extending T_0 to $Y_0 + Kx$ with all bounds, and Zorn's lemma extends T_0 to Y. With the single operator 0 and radius one, contractions extend to contractions: this is the extension mechanism of Ingleton's theorem [12].

Step 2 ($\mathrm{Alt}^k(E;F)$ *is spherically complete).* Let Y be the free vector space on the symbols $[x_1, \dots, x_k]$, $x \in E^k$, with the seminorm $\|\sum_x c_x[x]\| = \max_x |c_x| \prod_i \|x_i\|$, and let Y_0 be the span of the multilinearity and repeated-input relations. Alternating maps $E^k \to F$ correspond isometrically to bounded linear maps $Y \to F$ vanishing on Y_0: one direction uses the nonarchimedean inequality in F, the other evaluates on generators. Pairwise-meeting balls $\overline{B}(m_\alpha, r_\alpha)$ in $\mathrm{Alt}^k(E;F)$ satisfy $\|m_\alpha - m_\beta\| \le \max(r_\alpha, r_\beta)$, since this operator norm is nonarchimedean. So the maps T_α corresponding to the m_α, and $T_0 = 0$ on Y_0, satisfy the hypotheses of Step 1 with $G = F$; the extension of T_0 vanishes on Y_0, and its alternating map lies in every ball.

Step 3 (the retraction). The operator norm of $\mathrm{Mult}^k(E;F)$ is nonarchimedean. By Step 1 with $G = \mathrm{Alt}^k(E;F)$, spherically complete by Step 2, the inverse of the isometric inclusion $j_{E,F}$ extends to a contraction of $\mathrm{Mult}^k(E;F)$ onto $\mathrm{Alt}^k(E;F)$, which is the required retraction. Proposition 5.1 finishes the proof. In degree zero the alternating space is F. □

Proof of Theorem 5.4. Step 1 (ultrametric envelopes). As in the proof of Proposition I.1, with the nonarchimedean inequality of F in place of that of K, every bounded k-linear map m into F satisfies $\|m(x_1, \dots, x_k)\| \le \|m\| \prod_a q_Y(x_a)$. Hence $m \mapsto m \circ j_Y^{\times k}$ is an isometry of $\mathrm{Alt}^k(Y^u;F)$ onto $\mathrm{Alt}^k(Y;F)$, every $u \in \mathcal{L}(E,D)$ induces $u^u \in \mathcal{L}(E^u, D^u)$ with $\|u^u\| \le \|u\|$, linearly in u, and $A^k_{E,D;F}(u)$ corresponds to $A^k_{E^u,D^u;F}(u^u)$. A lift for (E^u, D^u) therefore gives one for (E,D) of no larger norm. Since $(Y^u)^u = Y^u$, we may assume $E = E^u$ and $D = D^u$.

Step 2 (the expansion on $c_0(I,K)$*).* Order I, and let (e_i) and (e_i^*) be the unit vectors and coordinate functionals of $c_0(I,K)$. For $m \in \mathrm{Alt}^k(c_0(I,K);F)$ and $x_1, \dots, x_k \in c_0(I,K)$, the family $\det(e_{i_a}^*(x_b))_{a,b}\, m(e_{i_1}, \dots, e_{i_k})$, $i_1 < \cdots < i_k$, is null: if a determinant has absolute value at least $\delta > 0$, every i_a lies in the finite set where some $|e_i^*(x_b)| \ge \delta / \max_b \|x_b\|^{k-1}$. Its sum is

$$m(x_1, \dots, x_k) = \sum_{i_1 < \cdots < i_k} \det\big(e_{i_a}^*(x_b)\big)_{a,b}\, m(e_{i_1}, \dots, e_{i_k}), \tag{P.1}$$

by sorting the multilinear expansion for finitely supported x_b, and by density in general, both sides being bounded and multilinear.

Step 3 (lifts). The maps

$$Q(u_1, \dots, u_k)(m)(x) = \sum_{i_1 < \cdots < i_k} \det\big(e_{i_a}^*(x_b)\big)_{a,b}\, m(u_1 e_{i_1}, \dots, u_k e_{i_k}),$$

for $u_a \in \mathcal{L}(c_0(I,K), D)$, and

$$Q'(u_1, \dots, u_k)(m)(x) = \sum_{i_1 < \cdots < i_k} \det\big(e_{i_a}^*(u_a x_b)\big)_{a,b}\, m(e_{i_1}, \dots, e_{i_k}),$$

for $u_a \in \mathcal{L}(E, c_0(I,K))$, are well defined by Step 2, with $u_a x_b$ in place of x_b in row a for Q'. They are alternating in x, of norm at most one, and have diagonal $A^k(u)(m)$ by (P.1). If Q_0 lifts $A^k_{c_0(I,K),D;F}$ and $\iota'\iota = \mathrm{id}_E$, then $Q_0(u_1\iota', \dots, u_k\iota')(m) \circ \iota^{\times k}$ lifts $A^k_{E,D;F}$; if Q_0 lifts $A^k_{E,c_0(I,K);F}$ and $\iota'\iota = \mathrm{id}_D$, then $Q_0(\iota u_1, \dots, \iota u_k)(m \circ \iota'^{\times k})$ does. Both have norm at most $(\|\iota\|\|\iota'\|)^k \|Q_0\|$.

Step 4 (countable type). If E is of countable type, so is E^u, which then has a t-orthogonal basis for every $0 < t < 1$ [20, Proposition 1.4]: nonzero vectors b_i with dense span and $\|\sum_i \lambda_i b_i\| \geq t \max_i |\lambda_i| \|b_i\|$ for finitely many scalars λ_i. The formulas for Q and (P.1), with this basis and its coordinate functionals in place of (e_i) and (e_i^*), give a lift of norm at most t^{-k}, as in the proof of Proposition I.1; similarly for D with Q'. □

Proof of Proposition 5.5. (1) The map $\Phi\colon \delta_i \mapsto e_{i_1} \wedge \cdots \wedge e_{i_k}$ is contractive on finitely supported vectors. The alternating map $\Omega(x_1, \ldots, x_k) = (\det(x_b(i_a))_{a,b})_{i \in J^k_<}$ satisfies $\|\Omega(x_1, \ldots, x_k)\|_1 \leq \sum_{i \in J^k_<} \sum_{\sigma \in S_k} \prod_a |x_{\sigma(a)}(i_a)| \leq \prod_b \|x_b\|_1$, since $(i, \sigma) \mapsto i \circ \sigma^{-1}$ is injective into J^k, tuples being maps $\{1, \ldots, k\} \to J$. By Proposition 3.1, Ω induces a contraction $\widetilde{\Omega}\colon \Lambda^k_\pi \ell^1(J, K) \to \ell^1(J^k_<, \widehat{K})$ with $\widetilde{\Omega}\Phi\delta_i = \delta_i$, because $(e_{i_b}(i'_a))_{a,b}$ is the identity for $i' = i$ and has a zero row for every other $i' \in J^k_<$. So Φ is isometric. Its range is dense, since wedges of finitely supported vectors span a dense subspace and expand into the $e_{i_1} \wedge \cdots \wedge e_{i_k}$, $i \in J^k_<$. Completion gives (1).

(2) By Lemma 3.7, $\widehat{\wedge}^k_E\colon \widehat{\otimes}^k_\pi E \to \widehat{\Lambda}^k_\pi E$ is a bounded linear surjection of Banach spaces. Projectivity gives a bounded linear s' with $\widehat{\wedge}^k_E \circ s' = \mathrm{id}$, whose restriction to $\Lambda^k_\pi E$ satisfies (S^k_E).

(3) Order J, and let $\iota\colon E \to \ell^1(J, K)$ and $\iota'\colon \ell^1(J, K) \to E$ be bounded linear with $\iota'\iota = \mathrm{id}_E$. The inverse of the isomorphism of (1), followed by $\delta_i \mapsto e_{i_1} \otimes \cdots \otimes e_{i_n}$, restricts to a contraction $s_0\colon \Lambda^n_\pi \ell^1(J, K) \to \widehat{\otimes}^n_\pi \ell^1(J, K)$ with $\widehat{\wedge}^n \circ s_0 = \mathrm{id}$, by the density shown in (1). By naturality of $\wedge^n$, the map $s = \iota'^{\otimes n} \circ s_0 \circ \Lambda^n_\pi \iota$ satisfies $\widehat{\wedge}^n_E \circ s = \widehat{\Lambda}^n_\pi \iota' \circ \Lambda^n_\pi \iota = \mathrm{id}$, and $\|s\| \leq (\|\iota\| \|\iota'\|)^n$. □

Proof of Corollary 5.10. Let ν be a nonarchimedean norm on D with $a\nu(y) \leq \|y\| \leq b\nu(y)$. Then $\|\sum_t \xi_t y_t\| \leq b \max_t |\xi_t| \nu(y_t) \leq (b/a)(\sum_t |\xi_t|^3)^{1/3} \max_t \|y_t\|$, so (5.2) holds with $r = 3$, D has an equivalent 3-norm by Proposition 5.8, and Theorem 5.9(2) gives the first assertion. The extension of the coordinate map is the map w of Theorem 5.9(1) for $r = 3$ followed by the contractive inclusion $\ell^3(\Sigma, \widehat{K}) \to c_0(\Sigma, \widehat{K})$, hence S^n-split by the ideal property in Proposition 3.13. □

Proof of Theorem 5.12. Step 1 (the space c_0). Order I and group the coordinate coefficients of each ambient b_n into nondecreasing words. Finite-coordinate uniqueness, as in the proof of Theorem 5.11, puts the grouped coefficients $c_{i_1, \ldots, i_n}$ in W, and the nonarchimedean inequality bounds their norms by $\|b_n\|$. The formula

$$q_n(x_1, \ldots, x_n) = \sum_{i_1 \leq \cdots \leq i_n} c_{i_1, \ldots, i_n} (x_1)_{i_1} \cdots (x_n)_{i_n}$$

converges, since outside finite sets of words some coordinate factor is arbitrarily small while the others are bounded; completeness of W supplies the sum, of norm at most $\|b_n\| \prod_a \|x_a\|$. The diagonals of q_n and b_n agree on finite supports, hence by density. Theorem 2.3 gives (1).

Step 2 (sorting in ℓ^1). For every set J, the sorting of Step 1 in degree d gives each bounded d-linear $B'\colon \ell^1(J, K)^d \to Z$ whose diagonal lies in W a lift into W with the same diagonal and of norm at most $d!\|B'\|$: W is complete, being closed in Z, each grouped coefficient has at most $d!$ terms, and the resulting series is absolutely summable, bounded by $d!\|B'\| \prod_a \|x_a\|_1$.

Step 3 (factorization of γ). Fix $x_0 \in U$, write $\gamma(x_0 + x) = \sum_n a_n(x, \ldots, x)$, and choose $\lambda \in K$ with $0 < |\lambda|$ within the radius of this expansion. Set $J = \coprod_{n \geq 0} I^n$ and

$$T(e_{n, i_1, \ldots, i_n}) = \lambda^n a_n(e_{i_1}, \ldots, e_{i_n}), \qquad g(x)_{n, i_1, \ldots, i_n} = \lambda^{-n} x_{i_1} \cdots x_{i_n}.$$

The values of T on unit vectors are uniformly bounded, so T extends by absolute summation to a bounded linear map $\ell^1(J, K) \to \widehat{H}$. The degree-$n$ multilinear coefficient of g has entries $\lambda^{-n}(x_1)_{i_1} \cdots (x_n)_{i_n}$ in its nth block and zero elsewhere, so its norm is at most $|\lambda|^{-n}$; its diagonal

block has sum norm $(\|x\|_1/|\lambda|)^n$. So g has a convergent multilinear expansion for $\|x\|_1 < |\lambda|$, without completeness of K, and $T(g(x)) = \gamma(x_0 + x)$ there.

Step 4 (conclusion of (2)). Extend B to $\widehat{H}^d$ by completeness of Z; its diagonal still lies in W, by closedness. By Step 2, the composite of this extension with $T^{\times d}$ has a bounded d-linear lift q into W with the same diagonal. Then $ja(\gamma(x_0 + x)) = jq(g(x), \dots, g(x))$ for $\|x\|_1 < |\lambda|$, so near x_0 the map $a\gamma$ is the diagonal of q, a bounded polynomial, composed with the analytic map g. Hence $a\gamma$ is analytic at every point x_0.

Step 5 (retracts). If $ri = \mathrm{id}_P$ for bounded linear $i\colon P \to V$ and $r\colon V \to P$, apply (1) or (2) to the family composed with r on $r^{-1}(U)$, and restrict along i; in (1) this multiplies the norms of the lifted coefficients by at most $(\|r\|\|i\|)^n$. □

Proof of Proposition 5.13. Step 1 (orbit sums). Let $L = \ell^1(J, K)$ with its unit vectors e_i, and order J. For nondecreasing $t \in J^n$, let O_t be the sum of the tensors $e_{\tilde{t}_1} \otimes \cdots \otimes e_{\tilde{t}_n}$ over the distinct rearrangements $\tilde{t}$ of t, let $b_1, \dots, b_m$ be the distinct entries of t, and for $\alpha \in \mathbb{N}^m$ with $\alpha_1 + \cdots + \alpha_m = n$ let $t(\alpha)$ be the nondecreasing tuple with α_i entries b_i. For $\lambda \in K^m$, expanding the tensor power gives $(\sum_i \lambda_i e_{b_i})^{\otimes n} = \sum_\alpha \lambda^\alpha O_{t(\alpha)}$. A linear functional φ on the span of the $O_{t(\alpha)}$ that vanishes on these pure powers makes the polynomial function $\lambda \mapsto \sum_\alpha \lambda^\alpha \varphi(O_{t(\alpha)})$ vanish on K^m, so $\varphi(O_{t(\alpha)}) = 0$ for all α, since K is infinite. Hence O_t lies in the span of the pure powers, and in $\widehat{\Gamma}^n_\pi L$. This argument is algebraic: it holds for arbitrary vectors $e_{b_1}, \dots, e_{b_m}$ of a vector space over K in place of the unit vectors.

Step 2 (the projection). As in the proof of Proposition 5.5(1), with the coordinate map $(x_1, \dots, x_n) \mapsto (x_1(t_1) \cdots x_n(t_n))_{t \in J^n}$ in place of Ω, the map $e_{t_1} \otimes \cdots \otimes e_{t_n} \mapsto \delta_t$ onto the unit vectors δ_t of $\ell^1(J^n, \widehat{K})$ extends to an isometric isomorphism $\widehat{\otimes}^n_\pi L \cong \ell^1(J^n, \widehat{K})$. Let $R_n \delta_t = O_t$ for nondecreasing t and $R_n \delta_t = 0$ otherwise. Then $\|R_n\| \le n!$, since O_t is a sum of at most $n!$ unit vectors, and R_n maps into the closed subspace $\widehat{\Gamma}^n_\pi L$ by Step 1. For finitely supported $h \in L$, the coefficients of $h^{\otimes n} = \sum_t h(t_1) \cdots h(t_n) \delta_t$ are constant on rearrangements of t, so $R_n h^{\otimes n} = h^{\otimes n}$. By continuity R_n fixes every pure power, hence $\widehat{\Gamma}^n_\pi L$. This proves (1).

Step 3 (finite coordinates). A normed space P with finite continuous coordinates has a bounded linear isomorphism T onto $\ell^1(\{1, \dots, d\}, K)$. The extension of $T^{\otimes n}$ to the completions is an isomorphism mapping $\widehat{\Gamma}^n_\pi P$ onto $\widehat{\Gamma}^n_\pi \ell^1(\{1, \dots, d\}, K)$, since $T^{\otimes n} h^{\otimes n} = (Th)^{\otimes n}$. So $(T^{-1})^{\otimes n} \circ R_n \circ T^{\otimes n}$, with R_n from Step 2 for $J = \{1, \dots, d\}$, is a projection onto $\widehat{\Gamma}^n_\pi P$. □

Proof of Proposition 5.14. Alternation is vanishing on repeated arguments, so $\Lambda^2_K P$ is the quotient of $P \otimes_K P$ by the span of the tensors $h \otimes h$. Hence $\widehat{\wedge}^2_P(h \otimes h) = 0$, and $\widehat{\Gamma}^2_\pi P \subseteq \mathcal{N}^2_P$ because $\mathcal{N}^2_P$ is closed. Conversely, let $z \in \otimes^2_\pi P$ with $\wedge^2_P z = 0$, represented by $\tilde{z} \in P \otimes_K P$. The image of $\tilde{z}$ in $\Lambda^2_K P$ has seminorm zero, so for $\epsilon > 0$ it equals $\sum_j x_j \wedge y_j$ with $\sum_j \|x_j\|\|y_j\| < \epsilon$; then $\tilde{z} - \sum_j x_j \otimes y_j$ lies in the span of the tensors $h \otimes h$, so z lies within ϵ of that span, and $z \in \widehat{\Gamma}^2_\pi P$. For $z \in \mathcal{N}^2_P$ choose $z_m \in \otimes^2_\pi P$ with $z_m \to z$. Then $\wedge^2_P z_m \to 0$, Lemma 3.7 gives $w_m \in \otimes^2_\pi P$ with $\wedge^2_P w_m = \wedge^2_P z_m$ and $\|w_m\| \le 2\|\wedge^2_P z_m\|$, and $z_m - w_m \in \widehat{\Gamma}^2_\pi P$ converges to z. The norm bounds are those of Lemma 3.9. □

P.4. **Proofs for Section 6.** The notation and the standing hypotheses are those of Section 6.

Proof of Lemma 6.1. Concatenating representations gives $\|z \wedge w\| \le \|z\|\|w\|$. The right side of (6.1) is n-linear in $(x_1, \dots, x_n)$, of norm at most $\binom{n}{a}$. It is alternating: if $x_i = x_j$ with $i \ne j$, the terms with i and j in one block vanish, and the others cancel in pairs by exchanging i and j between the blocks. Proposition 3.1 gives $\mathrm{sh}^{a,b}_E$; naturality is checked on pure wedges. Each of the $\binom{n}{a}$ terms of $\wedge^{a,b}_E \mathrm{sh}^{a,b}_E(x_1 \wedge \cdots \wedge x_n)$ is $\epsilon(S) x_S \wedge x_{S^c} = x_1 \wedge \cdots \wedge x_n$. The factorization of $\wedge^n_E$

holds on pure tensors; that of alt^n_E holds because every permutation of $\{1,\dots,n\}$ is uniquely a shuffle of two blocks followed by permutations within the blocks, and its sign is the product of their signs. □

Proof of Theorem 6.2. Identify $\mathrm{Alt}^j(Y;F')$ with $\mathcal{L}(\Lambda^j_\pi Y,F')$ by Proposition 3.1. For $T\in\mathrm{Alt}^n(D;F)$ define $T^\flat\in\mathrm{Alt}^a(D;G)$ by $T^\flat(z)(w)=T(z\wedge w)$; then $\|T^\flat\|\le\|T\|$. Let Q_a and Q_b be bounded lifts of $A^a_{E,D;G}$ and $A^b_{E,D;F}$ (Proposition 2.5). For $u_1,\dots,u_n\in\mathcal{L}(E,D)$ the bilinear map on $\Lambda^a_\pi E\times\Lambda^b_\pi E$ given by

$$(z,w)\longmapsto\big[Q_b(u_{a+1},\dots,u_n)\big(Q_a(u_1,\dots,u_a)(T^\flat)(z)\big)\big](w)$$

is bounded. Composing its linearization with $\binom{n}{a}^{-1}\mathrm{sh}^{a,b}_E$ gives $Q_n(u_1,\dots,u_n)(T)\in\mathrm{Alt}^n(E;F)$, bounded and n-linear in $(u_1,\dots,u_n)$, with values in $\mathcal{L}(\mathrm{Alt}^n(D;F),\mathrm{Alt}^n(E;F))$. On the diagonal the bilinear map is $(z,w)\mapsto T(\Lambda^a_\pi u\,z\wedge\Lambda^b_\pi u\,w)$, so naturality and (6.2) give $Q_n(u,\dots,u)(T)=T\circ\Lambda^n_\pi u=A^n_{E,D;F}(u)(T)$. For the last assertion take $a=1$: $A^1_{E,D;G}(u)(T)=T\circ u$ is linear in u. □

Proof of Corollary 6.3. By Lucas's theorem, $\binom{a+b}{a}\neq0$ in K exactly when the base-p addition of a and b has no carry: over $\mathbb{F}_p$, $(1+X)^{a+b}=\prod_i(1+X^{p^i})^{c_i}$, where the c_i are the base-p digits of $a+b$.

(1) Degrees below p belong to $\mathfrak{D}$ by Proposition 5.1(1). For a carry-free sum apply Theorem 6.2; if F is normed or Banach, so is $G=\mathrm{Alt}^b(D;F)$. For the scalar set, Theorem 3.2(2) shows that $A^j_{E,D;K}$ is analytic exactly when $A^j_{E,D;Y^*}$ is analytic for every normed Y; apply Theorem 6.2 with $F=Y^*$, for which $G=\mathrm{Alt}^b(D;Y^*)$ is the dual of $\Lambda^b_\pi D\otimes_\pi Y$.

(2) Let N be the least missing element; $N\ge p$ by (1). If N is not a power of p and p^j is the largest power of p below N, then $N=p^j+(N-p^j)$ is a carry-free sum with $0<N-p^j<N$, so $N\in\mathfrak{D}$, a contradiction. The second assertion follows.

(3) The pair (E,D) is not good in degree n_F; by (2), the least degree in which it is not good is p^i for some $i\ge1$, and $n_F\ge p^i$. If n_F had a nonzero digit in a position $j<i$, then (E,D) would be good in degree p^j, $A^{n_F-p^j}_{E,D;F}$ would be analytic, and $n_F=p^j+(n_F-p^j)$ would be carry-free, so Theorem 6.2 would make $A^{n_F}_{E,D;F}$ analytic. □

Proof of Theorem 6.4. Let $y=(0,1)\in E\oplus K$, and let pr_D and pr_K be the coordinate projections of $D\oplus K$ onto D and K. Define bounded linear maps $R\colon\mathrm{Alt}^k(E\oplus K;F)\to\mathrm{Alt}^{k-1}(E;F)$, of norm at most one, and $J\colon\mathrm{Alt}^{k-1}(D;F)\to\mathrm{Alt}^k(D\oplus K;F)$, of norm at most k, by $R(m)(x_1,\dots,x_{k-1})=m((x_1,0),\dots,(x_{k-1},0),y)$ and

$$J(m)(z_1,\dots,z_k)=\sum_{i=1}^{k}(-1)^{k-i}\,\mathrm{pr}_K(z_i)\,m(\mathrm{pr}_D\,z_1,\dots,\widehat{\mathrm{pr}_D\,z_i},\dots,\mathrm{pr}_D\,z_k).$$

The form $J(m)$ is alternating in every characteristic: if $z_a=z_b$ with $a<b$, only the terms $i=a$ and $i=b$ can be nonzero, their argument lists differ by moving one entry across $b-a-1$ others, and $(-1)^{k-a}+(-1)^{k-b}(-1)^{b-a-1}=0$. For $u\in\mathcal{L}(E,D)$, $(u\oplus1)(x_i,0)=(ux_i,0)$ and $(u\oplus1)y=(0,1)$, so only the term $i=k$ survives and

$$A^{k-1}_{E,D;F}(u)=R\circ A^k_{E\oplus K,D\oplus K;F}(u\oplus1)\circ J.$$

The right side is analytic in u, since $u\mapsto u\oplus1$ is affine and bounded. □

Proof of Theorem 6.13. Step 1 (pieces). Let $n\le k$. For a piece of source type and a section s_n for w in degree n, the map $Q_n(u_1,\dots,u_n)=\widehat{\wedge}^n_D\circ(w'\varepsilon(u_1)\otimes\cdots\otimes w'\varepsilon(u_n))\circ s_n$ has diagonal $\Lambda^n_\pi(w'\varepsilon(u))\circ\Lambda^n_\pi w$, as in the proof of Proposition 6.11(1). For a piece of target type and a

retraction ρ_n for w' in degree n, the map $Q_n(u_1,\dots,u_n)=\rho_n\circ(\varepsilon(u_1)w\otimes\cdots\otimes\varepsilon(u_n)w)\circ\mathrm{alt}_E^n$ has diagonal $\Lambda_\pi^n w'\circ\Lambda_\pi^n(\varepsilon(u)w)$, as in the proof of Proposition 6.11(2). Both diagonals equal $\Lambda_\pi^n(w'\varepsilon(u)w)$, which proves (1).

Step 2 (two maps). Let $\theta,\tilde\theta\colon\mathcal{L}(E,D)\to\mathcal{L}(E,\widehat D)$ be liftable in every degree $a\le k$, with lifts Q_a and $\tilde Q_a$, and put $Q_0=\tilde Q_0=\mathrm{id}_K$. For $n\le k$ the map

$$Q_n^+(u_1,\dots,u_n)=\sum_{a=0}^{n}\wedge_{\widehat D}^{a,n-a}\circ\big(Q_a(u_1,\dots,u_a)\otimes\tilde Q_{n-a}(u_{a+1},\dots,u_n)\big)\circ\mathrm{sh}_E^{a,n-a},$$

with $\wedge_{\widehat D}^{a,n-a}$ extended to the completions, is bounded and n-linear with diagonal $\Lambda_\pi^n(\theta(u)+\tilde\theta(u))$, by (6.3) for $G=\widehat D$.

Step 3 (conclusion). By Step 2 and induction on m, the map $u\mapsto\sum_i\theta_i(u)=u$ is liftable in degree k, and Lemma 3.4 gives (2); under the hypotheses of its last sentence, the formula of Step 2 keeps values in $\Lambda_\pi^n D$. Item (3) follows from (1) and (2) for decompositions into pieces of source and target type, with Step 1 for decompositions without completion, and from Proposition 6.11 for relative splittings. □

Proof of Proposition 6.15. Step 1 (the two bilinear maps). Given a decomposition (6.4) into pieces of source and target type, let b_S and b_R be the sums of the bounded bilinear maps $b_i(u,x)=w_i'\varepsilon_i(u)w_ix$ over the pieces of source type and of target type; then $b_S(u,x)+b_R(u,x)=ux$. For a piece of source type, the map $\xi_i\colon G_i\to\mathcal{L}(\mathcal{L}(E,D),\widehat D)$, $\xi_i(g)(u)=w_i'\varepsilon_i(u)g$, is bounded linear and $w_{b_i}=\xi_iw_i$. For a piece of target type, the linear map $\lambda_i\colon\mathcal{L}(E,D)\otimes_\pi E\to G_i'$ with $\lambda_i(u\otimes x)=\varepsilon_i(u)w_ix$ is bounded and $w_{b_i}'=w_i'\lambda_i$. By the paragraph on split operators in Section 6.1, $w_{b_S}=\sum w_{b_i}$ is $\mathrm{S}^{\le k}$-split and $w_{b_R}'=\sum w_{b_i}'$ is $\mathrm{R}^{\le k}$-split.

Step 2 (the two pieces). Conversely, given b_S and b_R, the piece $(\mathcal{L}(\mathcal{L}(E,D),\widehat D),\widehat D,w_{b_S},\mathrm{id}_{\widehat D},\varepsilon_S)$ with $\varepsilon_S(u)(T)=T(u)$ is of source type, the piece $(E,\mathcal{L}(E,D)\otimes_\pi E,\mathrm{id}_E,w_{b_R}',\varepsilon_R)$ with $\varepsilon_R(u)(x)=u\otimes x$ is of target type, and $\varepsilon_S(u)w_{b_S}x+w_{b_R}'\varepsilon_R(u)x=b_S(u,x)+b_R(u,x)=ux$. □

Proof of Lemma 6.16. (1) By (6.4), $v=\Pi\Phi(v)J=\sum_i\Pi w_i'\,\varepsilon_i(\Phi v)\,w_iJ$, and by the ideal property (Proposition 3.13(2)) the pieces $(G_i,G_i',w_iJ,\Pi w_i',\varepsilon_i\circ\Phi)$ have the types of the given pieces. A source-side relative splitting (G,w,ε) gives $(G,wJ,v\mapsto\Pi\varepsilon(\Phi v))$, and a target-side one (G,w',ε) gives $(G,\Pi w',v\mapsto\varepsilon(\Phi v)J)$. (2) Apply Proposition 6.14 with $G=E$, $G'=D$, $w=J$, $w'=\Pi|_D$ and $\varepsilon=\Phi$. □

Proof of Proposition 6.18. Step 1 (items (1) and (3)). Item (1) is Proposition 5.13. For (3), let $J\colon P_0\to P$ and $\Pi'\colon P\to P_0$ be bounded linear with $\Pi'J=\mathrm{id}_{P_0}$, and let R_n be a projection of $\widehat\otimes_\pi^nP$ onto $\widehat\Gamma_\pi^nP$. Then $\Pi'^{\otimes n}R_nJ^{\otimes n}$ takes values in $\widehat\Gamma_\pi^nP_0$, since $\Pi'^{\otimes n}$ maps pure powers to pure powers, and it fixes $h^{\otimes n}=\Pi'^{\otimes n}(Jh)^{\otimes n}$ for $h\in P_0$.

Step 2 (item (2)). Let $n\le k$, let J_1,J_2 and Π_1,Π_2 be the inclusions and projections of $P=P_1\oplus P_2$, and let $R_a^{(1)}$, $R_a^{(2)}$ be projections onto $\widehat\Gamma_\pi^aP_1$, $\widehat\Gamma_\pi^aP_2$ for $a\le n$, with $R_0^{(1)}=R_0^{(2)}=\mathrm{id}_K$. For $0\le a\le n$ let $\mathrm{sh}_a\colon\widehat\otimes_\pi^aP_1\,\widehat\otimes_\pi\,\widehat\otimes_\pi^{n-a}P_2\to\widehat\otimes_\pi^nP$ send $y\otimes z$ to the sum of the $\binom{n}{a}$ tensors obtained by distributing the factors of $J_1^{\otimes a}y$ and of $J_2^{\otimes(n-a)}z$ over the n slots, keeping their order, and put

$$R=\sum_{a=0}^{n}\mathrm{sh}_a\circ\big(R_a^{(1)}\otimes R_{n-a}^{(2)}\big)\circ\big(\Pi_1^{\otimes a}\otimes\Pi_2^{\otimes(n-a)}\big).$$

For $h=J_1h_1+J_2h_2$, $(\Pi_1^{\otimes a}\otimes\Pi_2^{\otimes(n-a)})(h^{\otimes n})=h_1^{\otimes a}\otimes h_2^{\otimes(n-a)}$ and $h^{\otimes n}=\sum_a\mathrm{sh}_a(h_1^{\otimes a}\otimes h_2^{\otimes(n-a)})$, so R fixes $h^{\otimes n}$, hence $\widehat\Gamma_\pi^nP$. The tensor $\mathrm{sh}_a(y^{\otimes a}\otimes z^{\otimes(n-a)})$ is the coefficient of λ^a in the

polynomial $\lambda \mapsto (\lambda J_1 y + J_2 z)^{\otimes n}$, hence a linear combination of its values, K being infinite. Since $R_a^{(1)} \otimes R_{n-a}^{(2)}$ takes values in the closed span of the tensors $y^{\otimes a} \otimes z^{\otimes (n-a)}$, the map R takes values in $\widehat{\Gamma}_\pi^n P$.

Step 3 (item (4)). Let $n \le k$ and let R_n be a projection onto $\widehat{\Gamma}_\pi^n P$. The operators that vanish on $\mathcal{N}_E^n$ form a closed subspace W of $Z = \mathcal{L}(\widehat{\otimes}_\pi^n E, \widehat{\Lambda}_\pi^n D)$, and $T \mapsto T \circ \widehat{\wedge}_E^n$ is an isometry of $\mathcal{L}(\widehat{\Lambda}_\pi^n E, \widehat{\Lambda}_\pi^n D)$ onto W (Lemma 3.7). Let $\Theta\colon \widehat{\otimes}_\pi^n P \to Z$ be the linearization of the bounded n-linear map $(y_1, \dots, y_n) \mapsto \widehat{\wedge}_D^n \circ \big(w'\varepsilon'(y_1)w \otimes \cdots \otimes w'\varepsilon'(y_n)w\big)$. By naturality of $\wedge^n$, $\Theta(y^{\otimes n}) = \widehat{\Lambda}_\pi^n(w'\varepsilon'(y)w) \circ \widehat{\wedge}_E^n$ lies in W, so Θ maps $\widehat{\Gamma}_\pi^n P$ into W. Hence $Q_n(u_1, \dots, u_n) = \Theta\big(R_n(\psi u_1 \otimes \cdots \otimes \psi u_n)\big)$, read in $\mathcal{L}(\widehat{\Lambda}_\pi^n E, \widehat{\Lambda}_\pi^n D)$ and restricted to $\Lambda_\pi^n E$, is bounded and n-linear, and its diagonal is $\Lambda_\pi^n(w'\varepsilon(u)w)$, since R_n fixes $(\psi u)^{\otimes n}$. The last assertion follows from Theorem 6.13(1)–(3). □

Proof of Corollary 6.24. (1) Extend w' to $\widehat{G}'$. Then $\theta_{E,D}(u) = w' \circ \theta_{G,G'}(\varepsilon(u)) \circ w$ for $u \in \mathcal{L}(E, D)$. Proposition 6.23(b), (c) and (a), applied in this order to w, w' and $\psi = \varepsilon$, give $\mathrm{ob}_k(E, D) = (\widehat{\Lambda}_\pi^k w)^*((\widehat{\Lambda}_\pi^k w')_\flat)_*(\varepsilon^\sharp)_* \,\mathrm{ob}_k(G, G')$, and pullbacks commute with pushouts (Corollary N.3). (2) The family ε on $\mathcal{L}(E, D)$ satisfies $\theta_{E,D} = \varepsilon_w$, so Proposition 6.23(b) gives the formula. If w is S^k-split, then $\mathrm{sp}_k(w) = 0$ by Proposition 3.13(1), and the pushout of a split sequence splits by Lemma 3.6(2) and (3). □

Proof of Proposition 6.27. Step 1 (the alternating map). Fix $u^1, \dots, u^k \in \mathcal{U}$, and let $E_{\mathrm{fin}} \subseteq E$ be the subspace of vectors with finitely many nonzero blocks. For $\nu = (\nu_1, \dots, \nu_k) \in I^k$ let $i_1 < \cdots < i_m$ be the distinct entries of ν, let $S_l = \{a : \nu_a = i_l\}$ and $n_l = |S_l|$, and let $\epsilon(S_1, \dots, S_m)$ be the sign of the permutation of $\{1, \dots, k\}$ that lists S_1, then S_2, and so on, each in increasing order. For $x_1, \dots, x_k \in E_{\mathrm{fin}}$ put

$$\Phi(x_1, \dots, x_k) = \sum_{\nu \in I^k} \epsilon(S_1, \dots, S_m) \bigwedge_{l=1}^{m} \widehat{\Lambda}_\pi^{n_l} J'_{i_l} \, Q_{n_l}^{i_l}\big(u_{i_l}^{(l)}\big)\Big(\bigwedge_{a \in S_l} \Pi_{i_l} x_a \Big),$$

a finite sum in $\widehat{\Lambda}_\pi^k D$. Here $u_{i_l}^{(l)}$ is the tuple of the $u_{i_l}^j$ with $n_1 + \cdots + n_{l-1} < j \le n_1 + \cdots + n_l$, in increasing order of j; the wedges over S_l are taken in increasing order; and the outer wedge is the iterated wedge product, extended to completions. Each u^j enters one slot of one map $Q_{n_l}^{i_l}$, determined by the sizes n_l, so Φ is k-linear in $(x_1, \dots, x_k)$ and in $(u^1, \dots, u^k)$. It is alternating: let $x_a = x_b$ with $a \ne b$. The terms with $\nu_a = \nu_b$ vanish, since one block wedge contains $\Pi_{\nu_a} x_a$ twice. The term of $\nu \circ (a\ b)$ is minus the term of ν with x_a and x_b exchanged: the two tuples have the same entries and block sizes, the permutations defining their signs differ by the transposition $(a\ b)$ followed by permutations inside the blocks, and each block factor is alternating in the vectors of its block; so the remaining terms cancel in pairs.

Step 2 (the bound). The wedge product is contractive and $\|\widehat{\Lambda}_\pi^n J'_i\| \le 1$, so the term of ν has norm at most $C^m \prod_j (c\|u^j\|) \prod_a \|\Pi_{\nu_a} x_a\| \le C^k c^k \prod_j \|u^j\| \prod_a \|\Pi_{\nu_a} x_a\|$. Summing over ν gives $\|\Phi(x_1, \dots, x_k)\| \le C^k c^k \prod_j \|u^j\| \prod_a \|x_a\|$, since $\sum_\nu \prod_a \|\Pi_{\nu_a} x_a\| = \prod_a \|x_a\|$. Hence Φ extends by continuity to a bounded k-linear map on E^k, alternating because a tuple in E^k with two equal entries is a limit of such tuples in E_{fin}^k, and Proposition 3.1 gives $Q(u^1, \dots, u^k) \in \mathcal{L}(\Lambda_\pi^k E, \widehat{\Lambda}_\pi^k D)$, k-linear in $(u^1, \dots, u^k)$, with $\|Q\| \le C^k c^k$.

Step 3 (the diagonal). Let $u^j = u$ for every j. The l-th block factor of the term of ν is then $\bigwedge_{a \in S_l} J'_{i_l} u_{i_l} \Pi_{i_l} x_a$, so by the choice of the sign the term is $\bigwedge_{a=1}^k J'_{\nu_a} u_{\nu_a} \Pi_{\nu_a} x_a$. Summing over ν gives $\bigwedge_a \big(\sum_i J'_i u_i \Pi_i x_a\big) = u x_1 \wedge \cdots \wedge u x_k$ for $x_1, \dots, x_k \in E_{\mathrm{fin}}$. Wedges of vectors of E_{fin} span a dense subspace of $\Lambda_\pi^k E$, so $Q(u, \dots, u) = \Lambda_\pi^k u$. The last assertion is Lemma 3.4. □

Proof of Proposition 6.28. Step 1 (retracts and sums). Let $J\colon Y_0 \to Y$ and $\Pi'\colon Y \to Y_0$ be bounded linear with $\Pi' J = \mathrm{id}_{Y_0}$. Since $\mathrm{id}_{Y_0} = \Pi' \mathrm{id}_Y J$, the conditions (S^k) and (R^k) pass to Y_0 by Proposition 3.13(2), and polarizability by Proposition 6.18(3). For pairs, apply Lemma 6.16(2) in the situation of its second sentence. In (2), the conditions pass from the sum to Y_1 and Y_2 by (1). Conversely, $\mathrm{id}_{Y_1 \oplus Y_2}$ is the sum of the inclusion-projection composites of the two summands, each $\mathrm{S}^{\leq k}$-split, respectively $\mathrm{R}^{\leq k}$-split, by the ideal property, so their sum is split in the same degrees by the paragraph on split operators in Section 6.1; polarizability is Proposition 6.18(2). For the pairs, let J_a, Π_a be the inclusion and projection of E_a in $E_1 \oplus E_2$, and J'_b, Π'_b those of D_b in $D_1 \oplus D_2$. Every $u \in \mathcal{L}(E_1 \oplus E_2, D_1 \oplus D_2)$ is $\sum_{a,b} J'_b(\Pi'_b u J_a)\Pi_a$. For each (a,b) the map $u \mapsto J'_b(\Pi'_b u J_a)\Pi_a$ is liftable in every degree $n \leq k$, by the formula of Proposition 6.14 for the pair (E_a, D_b), with $w = \Pi_a$, $w' = J'_b$ and $\varepsilon(u) = \Pi'_b u J_a$. Theorem 6.13(2), applied in each degree $n \leq k$, concludes.

Step 2 (completion of a space). By the paragraph before Theorem 3.16, applied to the dense subspace Y of $\widehat{Y}$, the inclusion induces isometries $\otimes^k_\pi Y \to \otimes^k_\pi \widehat{Y}$ and $\Lambda^k_\pi Y \to \Lambda^k_\pi \widehat{Y}$ with dense images. So the completed powers of Y and $\widehat{Y}$ agree, as do the maps $\widehat{\wedge}^k$ and alt^k on them, and $\widehat{\Gamma}^k_\pi Y = \widehat{\Gamma}^k_\pi \widehat{Y}$, since $h \mapsto h^{\otimes k}$ is continuous. Sections and retractions for id_Y extend by continuity to sections and retractions for $\mathrm{id}_{\widehat{Y}}$, since both sides of the defining identities are continuous, and those for $\mathrm{id}_{\widehat{Y}}$ restrict to ones for id_Y. The projections onto $\widehat{\Gamma}^k_\pi Y = \widehat{\Gamma}^k_\pi \widehat{Y}$ are the same.

Step 3 (completion of a pair). Let D be complete. Restriction is an isometry of $\mathcal{L}(\widehat{E}, D)$ onto $\mathcal{L}(E, D)$; write $\widehat{u}$ for the extension of $u \in \mathcal{L}(E, D)$. By Step 2, $\Lambda^k_\pi E$ is a dense subspace of $\Lambda^k_\pi \widehat{E}$ with the induced norm. A lift $\widehat{Q}$ for $(\widehat{E}, D)$ gives the lift $Q(u_1, \dots, u_k) = \widehat{Q}(\widehat{u}_1, \dots, \widehat{u}_k)|_{\Lambda^k_\pi E}$ for (E, D). A lift Q for (E, D) gives the lift for $(\widehat{E}, D)$ whose value at $(v_1, \dots, v_k)$ is the extension by continuity of $Q(v_1|_E, \dots, v_k|_E)$; its diagonal is $\Lambda^k_\pi v$, since $\Lambda^k_\pi v$ extends $\Lambda^k_\pi(v|_E)$. Finally $\mathcal{L}(E, D) \subseteq \mathcal{L}(E, \widehat{D})$ isometrically for any normed D, and $\widehat{\Lambda}^k_\pi \widehat{D} = \widehat{\Lambda}^k_\pi D$, so a lift for $(E, \widehat{D})$ restricts to a lift for (E, D). Lemma 3.4 concludes. □

P.5. **Proofs for Section 7.** The notation is that of Section 7.

Proof of Lemma 7.1. Step 1 (value group and residue field). The field $\widehat{K}$ is closed in the complete field $\kappa((X))$, hence complete. Put $\mathcal{O} = \{a \in \widehat{K} : |a| \leq 1\}$ and $\mathfrak{m} = \{a \in \widehat{K} : |a| < 1\}$. For $a \in \widehat{K}^\times$ there is $a' \in K$ with $|a - a'| < |a|$, hence $|a'| = |a|$; so $|\widehat{K}^\times| = |K^\times| = |\vartheta|^{\mathbb{Z}}$, and $|a| \leq |\vartheta|$ for every $a \in \mathfrak{m}$. Since $\mathcal{O} = \widehat{K} \cap \kappa[[X]]$ and $\mathfrak{m} = \widehat{K} \cap X\kappa[[X]]$, the map coeff_0 restricts to a ring homomorphism $\mathcal{O} \to \kappa$ with kernel $\mathfrak{m}$. So $\mathcal{O}/\mathfrak{m}$ is isomorphic to a subfield κ_1 of κ, and $|\kappa_1|$ is a power of p.

Step 2 (Teichmüller representatives). Since $\widehat{K}$ has characteristic p, $\mathrm{Fr}(a) = a^{|\kappa_1|}$ is a continuous ring endomorphism of $\mathcal{O}$; let κ' be its subring of fixed points. A nonzero $b \in \kappa'$ satisfies $b^{|\kappa_1|-1} = 1$, since $\mathcal{O}$ is a domain; so κ' is a field, $|b| = 1$, and the reduction map $\kappa' \to \mathcal{O}/\mathfrak{m}$ is injective. For surjectivity let $a \in \mathcal{O}$ and $a_n = \mathrm{Fr}^n(a)$. Since $y^{|\kappa_1|} = y$ on κ_1, $w = \mathrm{Fr}(a) - a$ lies in $\mathfrak{m}$, and $|a_{n+1} - a_n| = |w^{|\kappa_1|^n}| \leq |\vartheta|^{|\kappa_1|^n}$. Hence $a_n \to b \in \mathcal{O}$ with $\mathrm{Fr}(b) = b$ and $|b - a| < 1$. So $\kappa' \cong \kappa_1$ is a finite field of characteristic p.

Step 3 (expansions). Let $a \in \mathcal{O}$. By Step 2 there is $b_0 \in \kappa'$ with $a - b_0 \in \mathfrak{m}$, and $(a - b_0)/\vartheta \in \mathcal{O}$ by Step 1. Repeating this gives $b_0, b_1, \dots \in \kappa'$ with $a - \sum_{i<n} b_i \vartheta^i \in \vartheta^n \mathcal{O}$ for every n, so $a = \sum_{i \geq 0} b_i \vartheta^i$. If all b_i lie in κ' and $b_n \neq 0$, then $|\sum_{i \geq n} b_i \vartheta^i| = |\vartheta|^n$, since $|b_n \vartheta^n| = |\vartheta|^n$ and the later terms have absolute value at most $|\vartheta|^{n+1}$. Applied to the difference of two expansions, this shows that the expansion is unique.

Step 4 (the isomorphism). For $g = \sum_{i \geq m} b_i X^i$ in $\kappa'((X))$ put $\sigma(g) = \sum_{i \geq m} b_i \vartheta^i$, which converges since $|b_i \vartheta^i| \leq |\vartheta|^i$. The map σ is additive, and it is multiplicative since in the

complete nonarchimedean field $\widehat{K}$ the product of two convergent series is the sum of their Cauchy product, whose coefficients $\sum_{i+j=n} b_i b'_j$ lie in κ'. If $g \neq 0$ has order m, Step 3 applied to $\vartheta^{-m}\sigma(g)$ gives $|\sigma(g)| = |\vartheta|^m$; so σ is isometric, hence injective. For $a \in \widehat{K}^\times$ choose n with $|a| = |\vartheta|^n$ (Step 1); then $\vartheta^{-n}a = \sum_{i\geq 0} b_i \vartheta^i$ by Step 3, and $a = \sigma(\sum_i b_i X^{i+n})$. So σ is an isometric field isomorphism with $\sigma(X) = \vartheta$. It restricts to an isometric isomorphism of the dense subfield $\sigma^{-1}(K)$ of $\kappa'((X))$ onto K, and every statement that depends only on K as a normed field transfers along this isomorphism. □

Proof of Lemma 7.11. (1) For $x \in c_{00}(\mathbb{N}, K)$ and $t \in \mathbb{N}$, $|x(t)|e_t \leq |x|$ pointwise, so $|x(t)| \leq \|x\|_E$ by monotonicity and $\|e_t\|_E = 1$. For $x \in E$ and $t \in [n]$ this gives $|x(t)| \leq \|x1_{[n]}\|_E \leq \|x\|_E$. If b is finitely supported with $|b| \leq 1$ and $[n]$ contains its support, then $bx = b(x1_{[n]})$ and $|bx| \leq |x1_{[n]}|$, so $\|bx\|_E \leq \|x1_{[n]}\|_E \leq \|x\|_E$.

(2) For $x \in E$, ax is finitely supported and $\|ax\|_D \leq \sum_{t\in\operatorname{supp} a} |a(t)|\|e_t\|_D\|x\|_E$ by (1); so μ_a is bounded. Moreover $ax = a(x1_{\operatorname{supp} a})$, where $x1_{\operatorname{supp} a}$ is supported in $\operatorname{supp} a$ and has norm at most $\|x\|_E$ by (1). This gives the formula.

(3) The map $a \mapsto \mu_a$ is linear, and $\|\mu_a e_t\|_D = |a(t)|$ gives $\|\mu_a\|_{E\to D} \geq \|a\|_\infty$; so $a \mapsto \|\mu_a\|_{E\to D}$ is a norm. Since $\mu_{e_t}x = x(t)e_t$, (1) gives $\|\mu_{e_t}\|_{E\to D} = 1$. For an injective map $\gamma\colon \mathbb{N} \to \mathbb{N}$ and finitely supported y, let $y^\gamma(\gamma(t)) = y(t)$ and $y^\gamma = 0$ off $\gamma(\mathbb{N})$. Then $\|y^\gamma\| = \|y\|$ in E and in D, since γ restricted to $\operatorname{supp} y$ extends to a permutation of $\mathbb{N}$. The vectors supported in $\operatorname{supp} a^\gamma$ are the x^γ with $\operatorname{supp} x \subseteq \operatorname{supp} a$, and $a^\gamma x^\gamma = (ax)^\gamma$; so (2) gives $\|\mu_{a^\gamma}\|_{E\to D} = \|\mu_a\|_{E\to D}$. Multiplication of a by a sequence of absolute value 1 does not change $|ax|$. If $|a| \leq |b|$, let $c = a/b$ on $\operatorname{supp} b$ and $c = 0$ elsewhere; then $|c| \leq 1$, $ax = b(cx)$ and $\|cx\|_E \leq \|x\|_E$ by (1).

(4) For $x \in c_{00}(\mathbb{N}, K)$, $|ax| \leq \|a\|_\infty |x|$ pointwise, so $\|ax\|_D \leq C\|ax\|_E \leq C\|a\|_\infty\|x\|_E$. If D is symmetric and $\|x\|_\infty \leq 1$, then $|ax| \leq |a|$, so $\|ax\|_D \leq \|a\|_D$, with equality at $x = 1_{\operatorname{supp} a}$. Finally $\|ax\|_\infty \leq \|a\|_\infty\|x\|_\infty \leq \|a\|_\infty\|x\|_E$ by (1), and (3) gives the reverse inequality. □

P.6. **Proofs for Section 9.**

Proof of Proposition 9.1. Step 1 (the lift). Each $u \in \mathcal{L}(E, D)$ is λu_0 for a unique $\lambda \in K$, and $|\lambda| = \|u\|/\|u_0\|$. Hence Q is a well-defined k-linear map $\mathcal{L}(E, D)^k \to \mathcal{L}(\Lambda^k_\pi E, \Lambda^k_\pi D)$ of norm at most $\|\Lambda^k_\pi u_0\|/\|u_0\|^k \leq 1$. Its diagonal is $\lambda^k \Lambda^k_\pi u_0 = \Lambda^k_\pi(\lambda u_0)$, so Lemma 3.4 gives (1).

Step 2 (sufficiency in (2)). Put $\varepsilon(\lambda u_0) = \lambda \operatorname{id}_D$ and $\varepsilon'(\lambda u_0) = \lambda \operatorname{id}_E$. Then $\varepsilon(u) \circ u_0 = u = u_0 \circ \varepsilon'(u)$ for every $u \in \mathcal{L}(E, D)$. If u_0 is S^k-split, (D, u_0, ε) is a source-side relative splitting; if u_0 is R^k-split, (E, u_0, ε') is a target-side relative splitting. Either is a splitting decomposition of degree k (Definition 6.12).

Step 3 (necessity in (2), one piece). A splitting decomposition with one piece writes u_0 as a composite of bounded operators one of which is S^k-split or R^k-split (Definition 6.12). By Proposition 3.13(2), u_0 is S^k-split or R^k-split.

Step 4 (necessity in (2), several pieces). For a decomposition into at least two pieces of source and target type, Proposition 6.15 gives, in its notation, bounded bilinear maps b_S, b_R with $b_S(u, x) + b_R(u, x) = ux$, w_{b_S} $\mathrm{S}^{\leq k}$-split and w'_{b_R} $\mathrm{R}^{\leq k}$-split; their values lie in D, since D is complete. Put $u_S = b_S(u_0, \cdot)$ and $u_R = b_R(u_0, \cdot)$. Since u_S is w_{b_S} followed by evaluation at u_0, and u_R is $x \mapsto u_0 \otimes x$ followed by w'_{b_R}, Proposition 3.13(2) makes u_S S^k-split and u_R R^k-split. Both lie in $K\,u_0$ and $u_S + u_R = u_0$, so there is $a \in K$ with $u_S = a\,u_0$ and $u_R = (1 - a)\,u_0$. If $a \neq 0$, then $u_0 = a^{-1}u_S$ is S^k-split; otherwise $u_0 = u_R$ is R^k-split.

Step 5 (the dual). If f is a nonzero bounded linear functional on E, the operators $x \mapsto f(x)\,d$ with $d \in D$ form a subspace of $\mathcal{L}(E, D)$ of dimension $\dim D \geq 2$, contrary to $\mathcal{L}(E, D) = K\,u_0$. □

P.7. **Proofs for Section 10.**

Proof of Proposition 10.3. Choose E, D from Theorem 7.2(2), so that $A = A^k_{E,D;K}$ is analytic at no point.

Step 1 (assertion (1)). Put $P = \mathcal{L}(E,D) \times D \times E$ and use the charts id_P and $\psi(u,d,e) = (u, d+ue, e)$. Operator evaluation is bounded bilinear, so ψ and $\psi^{-1}(u,d,e) = (u, d-ue, e)$ are polynomial analytic, with $D\psi(u,d,e)(v,a,b) = (v, a+ve+ub, b)$. Let J extend forms from D along the projection $\pi_D\colon P \to D$, and let R restrict forms on P along $i_E(b) = (0,0,b)$. These bounded linear maps give

$$R \circ A^k_{P,P;K}(D\psi(u,d,e)) \circ J = A(u).$$

If the transition were analytic at (u_0, d_0, e_0), restriction to the affine slice $u \mapsto (u, d_0, e_0)$ would make A analytic at u_0. The reverse transition is treated in the same way with $-u$.

Step 2 (assertion (2)). Let $U = \mathrm{Alt}^k(D;K)$, $Y = U \times D$ and $N = \mathcal{L}(E,D) \times U \times E$. Define the bounded linear map $\omega_0(m,d) = m \circ \pi_D^{\times k}$, with $\pi_D\colon Y \to D$ the projection, and the polynomial map $h(u,m,e) = (m, ue)$. Since $Dh(u,m,e)(v,a,b) = (a, ve+ub)$, restricting the values of $h^*\omega_0$ to the E-directions gives $f(u,m) = A(u)(m)$. If $h^*\omega_0$ were analytic as an $\mathrm{Alt}^k(N;K)$-valued map at (u_0, m_0, e_0), fixing $e = e_0$ would make f analytic near (u_0, m_0); since Banach-valued power series differentiate termwise on a smaller ball, $\partial_m f(u,m) = A(u)$ would be analytic near u_0, a contradiction. In $\mathrm{Mult}^k(N;K)$ the pullback is a bounded multilinear expression in Dh and $\omega_0 \circ h$, hence analytic. □

Proof of Proposition 10.4. Throughout, for Banach spaces E and D' put $P = \mathcal{L}(E,D') \times D' \times E$, with the maximum norm, and use the charts id_P and $\psi(u,d,e) = (u, d+ue, e)$ of Step 1 of the proof of Proposition 10.3. Both charts are polynomial analytic, and $g = D\psi(u,d,e)$ is $g(v,a,b) = (v, a+ve+ub, b)$, with $g^{-1}(v,a,b) = (v, a-ve-ub, b)$.

Step 1 (values in F). For (1), take E, F from Theorem 7.2(1) and $D' = E$. The maps J, R of Step 1 of the proof of Proposition 10.3, with values in F, give

$$R \circ A^k_{P,P;F}(D\psi(w,d,e)) \circ J = A^k_{E,E;F}(w) \qquad (w \in \mathcal{L}(E,E)),$$

which is analytic at no point; restriction to affine slices, as in that step, shows that neither transition is analytic at any point. Only bounded linear compression and affine slices are used, so K need not be complete, and F has no equivalent nonarchimedean norm by Theorem 7.2(1). For $K = \mathbb{F}_q((X))$ with its X-adic absolute value, Remark E.2 allows $E = \ell^\infty(\mathbb{N}, K)$ and $F = F_1$, which has no equivalent nonarchimedean norm by Proposition E.1. Then P is nonarchimedean, since operator norms into a nonarchimedean space and maximum norms of products are nonarchimedean.

Step 2 (tangent values). Suppose that $A^k_{E,D;F}$ is analytic at no point, and put $D' = D \times F$. As functions of the point (u,d,e), the induced transitions on tangent-valued forms are $m \mapsto g \circ m \circ (g^{-1}, \dots, g^{-1})$ and $m \mapsto g^{-1} \circ m \circ (g, \dots, g)$. Let $\pi_D\colon P \to D$ and $\pi_F\colon P \to F$ be the components of the middle factor, and put $i_F(y) = (0, (0,y), 0)$ and $i_E(b) = (0,0,b)$. The bounded linear map

$$\Phi(T)(m) = \pi_F \circ T\big(i_F \circ m \circ \pi_D^{\times k}\big) \circ i_E^{\times k}$$

sends operators T on $\mathrm{Alt}^k(P;P)$ to operators $\mathrm{Alt}^k(D;F) \to \mathrm{Alt}^k(E;F)$, and it sends $m \mapsto h \circ m \circ (f, \dots, f)$, for $f, h \in \mathcal{L}(P,P)$, to $m \mapsto (\pi_F h i_F) \circ m \circ (\pi_D f i_E, \dots, \pi_D f i_E)$. Since $\pi_F g^{\pm 1} i_F = \mathrm{id}_F$ and $\pi_D g^{\pm 1} i_E = \pm u_D$, where $u_D \in \mathcal{L}(E,D)$ is the D-component of u, Φ sends the two transitions to $A^k_{E,D;F}(\mp u_D) = (\mp 1)^k A^k_{E,D;F}(u_D)$. If either transition were analytic at (u_0, d_0, e_0), restriction to the affine slice $w \mapsto (u_0 + (w,0), d_0, e_0)$, $w \in \mathcal{L}(E,D)$, would make $A^k_{E,D;F}$ analytic at the D-component of u_0.

Step 3 (the models). For (2), apply Step 2 with E, F from Theorem 7.2(1) and $D = E$; then P contains the isometric copy $i_F(F)$ of F, so it admits no equivalent nonarchimedean norm. For (3), apply Step 2 with E, D from Theorem 7.2(2) and $F = K$; then P is a nonarchimedean Banach space, as in Step 1. □

Proof of Theorem 10.7. Step 1 (sheaf and pullback). Locality of analytic coefficient maps gives the sheaf property. Addition and multiplication by an analytic function are bounded multilinear operations on the coefficient spaces. For an analytic map h, the formula

$$h^*\eta = (\eta \circ h) \circ (Dh, \dots, Dh)$$

is analytic in the ambient space and has alternating values. The chain rule gives identity and composition laws for pullback.

Step 2 (the product). The shuffle formula (10.2) has $\binom{k+l}{k}$ terms, each of norm at most $\|\eta\|\|\zeta\|$, so it is a bounded bilinear operation on multilinear coefficient spaces, of norm at most $\binom{k+l}{k}$. For alternating inputs it is alternating: terms with two repeated arguments in one block vanish, and the remaining terms cancel in pairs by exchanging those arguments between blocks; this argument includes characteristic two. Formula (10.2) commutes with simultaneous precomposition, hence glues in charts and commutes with pullback. Complementing S proves graded commutativity. Expanding either bracketing of a triple product yields the same signed sum over ordered three-block partitions, proving associativity.

Step 3 ($d\eta$ is a form). On an open subset of a model space P, the operation (10.3) applied to $D\eta$ is bounded linear with norm at most $k+1$, so $d\eta$ is ambient analytic. Each $D\eta(y)(v_i)$ is alternating: difference quotients have alternating values, and that subspace is closed. If $v_a = v_b$ with $a < b$, the summands other than $i = a, b$ vanish; alternation moves the remaining repeated argument across $b - a - 1$ positions, making those two terms cancel. Hence $d\eta$ is a form.

Step 4 ($d^2 = 0$). The second derivative of an analytic map is symmetric in every characteristic: if c_2 is the bilinear coefficient of degree two of a power series, the second derivative at its center is $(v, w) \mapsto c_2(v, w) + c_2(w, v)$. In the expansion of $d(d\eta)$, pair the terms differentiated in directions (v_a, v_b) and (v_b, v_a) for $a < b$. Their values agree and their signs are $(-1)^{a+b-1}$ and $(-1)^{a+b}$, so $d^2\eta = 0$.

Step 5 (naturality and gluing of d). Differentiate the pullback formula. Terms differentiating η give $h^*(d\eta)$ after the signed sum (10.3). The other terms insert $D^2h(v_a, v_b)$ into one argument of η, and symmetry of D^2h and alternation of η cancel them in pairs. Therefore $d(h^*\eta) = h^*(d\eta)$; applied to chart changes, this identity shows that the local derivatives glue to $d\colon \Omega^k_{\mathrm{amb}}(M) \to \Omega^{k+1}_{\mathrm{amb}}(M)$.

Step 6 (Leibniz rule). Differentiating the shuffle formula (10.2) gives the Leibniz rule: placing the derivative index in the second block moves it past the first k arguments and contributes $(-1)^k$. Every identity in Steps 1–6 uses finite sums and cancellation, with no division by a factorial. □

Proof of Proposition 10.9. Clearly (2) implies (1). If (1) holds, put $T = D\eta(y)$ and $\beta(v_0, \dots, v_{p-1}) = T(v_0)(v_1, \dots, v_{p-1})$. Grouping the permutations by the image of 0 and using alternation of each $T(v_i)$,

$$\mathcal{A}_Y\beta(v_0, \dots, v_{p-1}) = (p-1)!\sum_{i=0}^{p-1}(-1)^i T(v_i)(v_0, \dots, \widehat{v_i}, \dots, v_{p-1}) = (p-1)!\,\omega(v_0, \dots, v_{p-1}),$$

and $(p-1)! = -1$ in K by Wilson's theorem, so $\omega = \mathcal{A}_Y(-\beta)$. If (3) holds, the bounded linear map $\eta(y)(v_1, \dots, v_{p-1}) = \sum_\sigma \operatorname{sgn}(\sigma)\,\beta(y, v_{\sigma(1)}, \dots, v_{\sigma(p-1)})$, summed over the permutations σ of $\{1, \dots, p-1\}$, satisfies $D\eta(y) = \eta$, and the same grouping gives $d\eta = \mathcal{A}_Y\beta = \omega$. For

$F = \widehat{\Lambda}^p_\pi Y$ the equivalence with (R^p_Y) is Lemma 3.10, whose proof also gives the uncompleted case, and Proposition 3.5 gives goodness. □

Proof of Theorem 10.11. Step 1 (nodes). For an integer $a \geq 0$ with binary digits $a_l \in \{0,1\}$, so that $a = \sum_l a_l 2^l$, put $\xi_a = \sum_l a_l \varpi^l \in K$. Then $|\xi_a| \leq 1$, and for $a \neq b$ the least l with $a_l \neq b_l$ is the exponent $v(a-b)$ of 2 in $a-b$, so $|\xi_a - \xi_b| = |\varpi|^{v(a-b)}$ by the ultrametric inequality. For $0 \leq a \leq i$ the numbers $|a-b|$, $b \leq i$, $b \neq a$, are $1, \dots, a$ and $1, \dots, i-a$, and the exponent of 2 in a factorial is at most its argument; so

$$\prod_{b \leq i,\ b \neq a} |\xi_a - \xi_b|^{-1} = |\varpi|^{-v(a!)-v((i-a)!)} \leq |\varpi|^{-i}.$$

Hence the divided difference $\sum_{a \leq i} \psi(\xi_a) \prod_{b \leq i,\, b \neq a} (\xi_a - \xi_b)^{-1}$ of order i of a map ψ from K to a normed space is a combination of the $i+1$ values $\psi(\xi_a)$, $a \leq i$, with coefficients of absolute value at most $|\varpi|^{-i}$.

Step 2 (item (1): the bound). If $m = 0$, b is constant and the bound is clear. Arguments w_c with $\alpha_c = 0$ do not change $b^{[\alpha]}(w)$, so we may assume $\alpha_c \geq 1$ for all c; then $r \leq m$. If $r = 1$, then $b^{[\alpha]}(w_1) = b(w_1)$. Let $r \geq 2$ and $N = r-1$. The map $\Phi(s) = b(s_1 w_1 + \dots + s_N w_N + w_r)$ on K^N is a polynomial of total degree at most m with coefficients in Y, and $b^{[\alpha]}(w)$ is its coefficient of $s_1^{\alpha_1} \cdots s_N^{\alpha_N}$. Newton's interpolation formula on the nodes $\xi_0, \dots, \xi_m$ in each variable gives

$$\Phi(s) = \sum_{i \in \{0,\dots,m\}^N} \delta_i \prod_{c=1}^{N} q_{i_c}(s_c), \qquad q_i(s) = \prod_{b<i} (s - \xi_b),$$

where $\delta_i \in Y$ is the divided difference of Φ of order i_c in the variable s_c, on the nodes $\xi_0, \dots, \xi_{i_c}$, for $c = 1, \dots, N$. A divided difference of order i of a polynomial of degree less than i vanishes, and the monomials of Φ have total degree at most m, so $\delta_i = 0$ unless $|i| \leq m$. By Step 1, δ_i is a combination of at most $\prod_c (i_c + 1) \leq 2^{|i|}$ values of Φ at points with coordinates ξ_a, with coefficients of absolute value at most $|\varpi|^{-|i|}$, and each of these values has norm at most $\|b\|(\|w_1\| + \dots + \|w_r\|)^m$, since $|\xi_a| \leq 1$. The polynomials q_i have coefficients of absolute value at most 1. So the coefficient of $s_1^{\alpha_1} \cdots s_N^{\alpha_N}$ is a sum of at most $\binom{m+N}{N} \leq 4^m$ terms, one for each i with $|i| \leq m$, each of norm at most $(2/|\varpi|)^m \|b\|(\|w_1\| + \dots + \|w_r\|)^m$.

Step 3 (item (1): expansions). Let $x, y \in W$ and $0 \leq i \leq m$. If $x, y \neq 0$, choose $\lambda, \mu \in K$ with $|\varpi| < |\lambda|\|x\| \leq 1$ and $|\varpi| < |\mu|\|y\| \leq 1$. As $b^{[(m-i,i)]}(\lambda x, \mu y) = \lambda^{m-i} \mu^i b^{[(m-i,i)]}(x,y)$, Step 2 gives

$$\|b^{[(m-i,i)]}(x,y)\| \leq (16/|\varpi|^2)^m \|b\|\, \|x\|^{m-i} \|y\|^i, \tag{P.2}$$

which holds trivially if $x = 0$ or $y = 0$. Since $b(x+y) = \sum_{i \leq m} b^{[(m-i,i)]}(x,y)$, b is continuous. For fixed x, the map $y \mapsto b^{[(m-i,i)]}(x,y)$ is a bounded polynomial law of degree i: on the span of finitely many vectors it is a coefficient of a polynomial with coefficients in Y, and (P.2) bounds it. So $b(x+y) = \sum_{i \leq m} b^{[(m-i,i)]}(x,y)$ is a finite expansion at x, and b is DP on W.

Step 4 (item (2)). Let $f(x_0 + h) = \sum_n a_n(h)$ for $\|h\| < R$ and $g(f(x_0) + w) = \sum_m b_m(w)$ for $\|w\| < R'$, with $\|a_n\| \leq MR^{-n}$ and $\|b_m\| \leq M'R'^{-m}$, where $M, M' > 0$. Put $\sigma = \min(R/2, |\varpi|RR'/(32M))$ and $\Delta(h) = \sum_{n \geq 1} a_n(h) = f(x_0 + h) - f(x_0)$. For $\|h\| \leq \sigma$, $\sum_{n \geq 1} \|a_n(h)\| \leq 2M\sigma/R \leq |\varpi|R'/16$. For a finitely supported sequence $\alpha = (\alpha_n)_{n \geq 1}$ of nonnegative integers put $|\alpha| = \sum_n \alpha_n$, $\deg \alpha = \sum_n n\alpha_n$ and $\phi_\alpha(h) = b^{[\alpha]}_{|\alpha|}(a_1(h), a_2(h), \dots)$, computed with $a_1(h), \dots, a_L(h)$ for any L at least the largest index in the support of α. On the span of finitely many vectors the a_n are polynomials whose coefficients span a finite-dimensional subspace of W, on which $b^{[\alpha]}_{|\alpha|}$ is a polynomial in the coordinates of its arguments with coefficients

in Y; and $\phi_\alpha(\lambda h) = \lambda^{\deg\alpha}\phi_\alpha(h)$. With (1), ϕ_α is a bounded polynomial law of degree $\deg\alpha$, and for $\|h\| \le \sigma$

$$\|\phi_\alpha(h)\| \le (8/|\varpi|)^{|\alpha|} M' R'^{-|\alpha|} (2M\sigma/R)^{|\alpha|} \le M' 2^{-|\alpha|}.$$

By homogeneity, $\|\phi_\alpha(h)\| \le M'(\|h\|/(|\varpi|\sigma))^{\deg\alpha}$ for all h. At most 2^{d-1} sequences α have $\deg\alpha = d \ge 1$, one for each partition of d. So for $\|h\| \le \tau = |\varpi|\sigma/4$ the family $(\phi_\alpha(h))_\alpha$ is absolutely summable in the completion $\widehat{Y}$. For each L, $b_m(\sum_{n\le L} a_n(h))$ is the sum of the $\phi_\alpha(h)$ with $|\alpha| = m$ and support in $\{1,\dots,L\}$; as $L \to \infty$ the left side tends to $b_m(\Delta(h))$, by continuity of b_m. Summing over m, the family has sum $\sum_m b_m(\Delta(h)) = g(f(x_0+h))$, since $\|\Delta(h)\| < R'$. Grouped by degree instead, it has the finite sums $\theta_d(h) = \sum_{\deg\alpha = d} \phi_\alpha(h)$, bounded polynomial laws of degree d with values in Y and $\|\theta_d\| \le 2^{d-1} M' (|\varpi|\sigma)^{-d}$ for $d \ge 1$. So $\sum_d \theta_d(h)$ converges in $\widehat{Y}$ to $g(f(x_0+h)) \in Y$ for $\|h\| \le \tau$; its partial sums lie in Y, so it converges in Y, and $g \circ f$ is DP at x_0.

Step 5 (item (3)). Let $ja(h) = \beta(h,\dots,h)$ as in Definition 10.10(2). For $h_1,\dots,h_r \in P$, the map $s \mapsto \beta(\sum_c s_c h_c, \dots, \sum_c s_c h_c)$ is a homogeneous polynomial of degree n with coefficients in Z. As K is infinite, its coefficients are linear combinations of its values, by interpolation in each variable, and these values lie in $j(W)$. So a is a bounded polynomial law of degree n, with $\|a\| \le \|\beta\|$; in particular analytic maps are DP. Now let $T\colon K^r \to P$ be a bounded linear isomorphism, K^r with the maximum norm and unit vectors $e_1,\dots,e_r$, and let $a\colon P \to W$ be a bounded polynomial law of degree n. Then $a(Tz) = \sum_{|\gamma|=n} v_\gamma z^\gamma$ with $v_\gamma = (a\circ T)^{[\gamma]}(e_1,\dots,e_r)$, and $\|v_\gamma\| \le (8r/|\varpi|)^n \|T\|^n \|a\|$ by (1). The n-linear map

$$\beta_a(h^1,\dots,h^n) = \sum_{t_1 \le \dots \le t_n} v_{\gamma(t)} \prod_{c=1}^{n} (T^{-1}h^c)_{t_c},$$

summed over the nondecreasing words t in $\{1,\dots,r\}$, where $\gamma(t)$ counts the letters of t, has diagonal a, since each monomial z^γ comes from exactly one such word, and $\|\beta_a\| \le \binom{n+r-1}{n}(8r\|T\|\|T^{-1}\|/|\varpi|)^n\|a\| \le 2^{r-1}(16r\|T\|\|T^{-1}\|/|\varpi|)^n\|a\|$. Applied to the coefficients of a DP expansion, this bound, geometric in n, gives an analytic expansion with a smaller positive radius.

Step 6 (item (4)). By Proposition 2.2 and its proof, the restriction of B to alternating maps is the diagonal of a bounded $(k+1)$-linear map of norm at most 1 with values in $\mathcal{L}(\mathrm{Alt}^k(D;F), \mathrm{Mult}^k(E;F'))$, and its values lie in $\mathcal{L}(\mathrm{Alt}^k(D;F), \mathrm{Alt}^k(E;F'))$, which postcomposition by $j_{E,F'}$ embeds isometrically; the same holds for $A^k_{E,D;F}(u) = B(u, \mathrm{id}_F)$ in degree k. By Lemma 3.7, $\Theta \mapsto \Theta \circ \wedge^k_E$ is a linear isometry of $\mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$ into $\mathcal{L}(\otimes^k_\pi E, \widehat{\Lambda}^k_\pi D)$, and restriction to the dense subspace $\Lambda^k_\pi E$ is an isometric isomorphism of $\mathcal{L}(\widehat{\Lambda}^k_\pi E, \widehat{\Lambda}^k_\pi D)$ onto $\mathcal{L}(\Lambda^k_\pi E, \widehat{\Lambda}^k_\pi D)$, as $\widehat{\Lambda}^k_\pi D$ is complete. These isometries map $\Lambda^k_\pi u$ and $\widehat{\Lambda}^k_\pi u$ to $\wedge^k_D \circ (u \otimes \dots \otimes u)$, the diagonal of the k-linear map $(u_1,\dots,u_k) \mapsto \wedge^k_D \circ (u_1 \otimes \dots \otimes u_k)$ of norm at most 1. Item (3) bounds the four norms by 1. □

For the proof of Proposition 10.13 we use the following notation. For a field $\Bbbk$ and integers $m, n \ge 1$, read a tensor $t \in (\Bbbk^m)^{\otimes n}$ as the function $t(i_1,\dots,i_n)$ on $[m]^n$, where $[m] = \{1,\dots,m\}$, and for linear forms $\phi_1,\dots,\phi_n$ on $\Bbbk^m$ let $t(\phi_1,\dots,\phi_n)$ be the value of $\phi_1 \otimes \dots \otimes \phi_n$ at t. Let $\Gamma^n(\Bbbk^m)$ be the space of symmetric tensors, let $\mathrm{diag}(t) \in \Bbbk^m$ have the entries $t(i,\dots,i)$, and let the *flattening rank* $\mathrm{frk}(t)$ be the rank of the $m \times m^{n-1}$ matrix of the entries $t(i_1,\dots,i_n)$, with rows indexed by i_1; it is subadditive and does not change under extension of scalars. A *projection onto the symmetric tensors* is a linear map Ψ of $(\Bbbk^m)^{\otimes n}$ into $\Gamma^n(\Bbbk^m)$ with $\Psi(t) = t$ for $t \in \Gamma^n(\Bbbk^m)$. Symmetric tensors with entries in a commutative $\Bbbk$-algebra are combinations of

symmetric tensors over $\Bbbk$, so the linear extension of Ψ to such entries maps tensors with these entries to symmetric ones and fixes the symmetric ones.

Lemma P.1 (projections onto symmetric tensors)**.** *Let $m \geq 1$.*

1. *Let $\Bbbk$ be a field, $a \geq 1$, $x = (x_1, \dots, x_m)^{\mathsf{T}}$ a column of indeterminates, and $\mathsf{M}(x)$ an $m \times m$ matrix over $\Bbbk[x_1, \dots, x_m]$ with $\mathsf{M}(x)x = (x_1^a, \dots, x_m^a)^{\mathsf{T}}$. Then $\det \mathsf{M}(x)$ has the coefficient 1 at $x_1^{a-1} \cdots x_m^{a-1}$.*
2. *Let $\Bbbk$ have characteristic $p > 0$, let $C \geq 0$ be an integer with $|\Bbbk| \geq 2C + 2$, and let $\mathcal{X} \subseteq \Gamma^p(\Bbbk^m)$ be a linear space whose elements have flattening rank at most C. Then $\dim \operatorname{diag}(\mathcal{X}) \leq C$.*
3. *Let f be a function from an abelian group into $[0, \infty)$ with $f(z + z') \leq f(z) + f(z')$, and let Φ be a map from $(\mathbb{F}_p^m)^p$ into this group that is additive in each argument. Then the average of $f \circ \Phi$ over $(\mathbb{F}_p^m)^p$ is at least 2^{-p} times its value at every point.*
4. *Let $\Bbbk$ be a field of characteristic $p > 0$, and let Ψ be a projection of $(\Bbbk^m)^{\otimes p}$ onto $\Gamma^p(\Bbbk^m)$. Then the average of $\operatorname{frk} \Psi(y^1 \otimes \cdots \otimes y^p)$ over $(y^1, \dots, y^p) \in (\mathbb{F}_p^m)^p$ is at least $m/\big(2^p(1 + \lfloor \log_p m \rfloor)\big)$.*

Proof. Step 1 (item (1)). Let D be the diagonal matrix with the entries $x_1^{a-1}, \dots, x_m^{a-1}$, and put $\mathsf{N} = \mathsf{M} - \mathsf{D}$, so $\mathsf{N}x = 0$. As the determinant is linear in each row, $\det \mathsf{M} = \sum_{S \subseteq [m]} \det \mathsf{M}_S$, where M_S has the rows of N in the positions in S and those of D elsewhere, and $\det \mathsf{M}_\emptyset = x_1^{a-1} \cdots x_m^{a-1}$. Let $S \neq \emptyset$, and let J_S be the ideal generated by the x_i^a, $i \notin S$. The vector $\mathsf{M}_S x$ has the entries 0 in the positions in S and x_i^a elsewhere, so multiplication by the adjugate of M_S gives $\det(\mathsf{M}_S) x_j \in J_S$ for every j. For $j \in S$, x_j is a nonzerodivisor modulo J_S, which is generated by monomials in the other variables; so $\det \mathsf{M}_S \in J_S$. Every monomial of an element of J_S is divisible by some x_i^a, so $x_1^{a-1} \cdots x_m^{a-1}$ does not occur in $\det \mathsf{M}_S$.

Step 2 (item (2): a common kernel). For $t \in \Gamma^p(\Bbbk^m)$ and a linear form ϕ on $\Bbbk^m$ let $t^\flat(\phi) \in (\Bbbk^m)^{\otimes(p-1)}$ have the entries $\sum_i \phi(e_i)\, t(i, i_2, \dots, i_p)$, e_i the unit vectors; the linear map $t^\flat$ has rank $\operatorname{frk}(t)$, and $t^\flat(\phi)(\phi, \dots, \phi) = t(\phi, \dots, \phi)$. Let $t_0 \in \mathcal{X}$ have the largest flattening rank $r_0 \leq C$, let $\mathcal{U}$ be the kernel of $t_0^\flat$, of codimension r_0, with a complement $\mathcal{U}'$, and let $\mathcal{I}$ be the image of $t_0^\flat$, with a complement $\mathcal{I}'$. For $t \in \mathcal{X}$ write $t^\flat$ in blocks $\mathsf{B}_{11}(t) \colon \mathcal{U}' \to \mathcal{I}$, $\mathsf{B}_{12}(t) \colon \mathcal{U} \to \mathcal{I}$, $\mathsf{B}_{21}(t) \colon \mathcal{U}' \to \mathcal{I}'$ and $\mathsf{B}_{22}(t) \colon \mathcal{U} \to \mathcal{I}'$; then $\mathsf{B}_{11}(t_0)$ is invertible and the other blocks of $t_0^\flat$ vanish. Put $\mathsf{A}_\mu = \mathsf{B}_{11}(t_0) + \mu \mathsf{B}_{11}(t)$ for $\mu \in \Bbbk$. If $\mu \neq 0$ and $\det \mathsf{A}_\mu \neq 0$, the Schur complement of A_μ in $(t_0 + \mu t)^\flat$ vanishes, since $\operatorname{frk}(t_0 + \mu t) \leq r_0$; that is,

$$\det(\mathsf{A}_\mu)\, \mathsf{B}_{22}(t) = \mu\, \mathsf{B}_{21}(t) \operatorname{adj}(\mathsf{A}_\mu) \mathsf{B}_{12}(t). \tag{P.3}$$

Both sides are polynomials in μ of degree at most r_0, and $\det \mathsf{A}_\mu$ has at most r_0 roots, since it is nonzero at $\mu = 0$. So (P.3) holds for all μ outside a set of at most $r_0 + 1$ elements, hence, as $|\Bbbk| \geq 2r_0 + 2$, for at least $r_0 + 1$ values of μ, and so identically. At $\mu = 0$, (P.3) gives $\mathsf{B}_{22}(t) = 0$: $t^\flat(\phi) \in \mathcal{I}$ for $\phi \in \mathcal{U}$.

Step 3 (item (2): diagonal forms). The elements of $\mathcal{I}$ vanish when one of their arguments lies in $\mathcal{U}$, as t_0 is symmetric. So $\delta_t(\phi) = t(\phi, \dots, \phi) = t^\flat(\phi)(\phi, \dots, \phi)$ vanishes for $\phi \in \mathcal{U}$ and $t \in \mathcal{X}$. As t is symmetric and the binomial coefficients $\binom{p}{1}, \dots, \binom{p}{p-1}$ vanish in $\Bbbk$, δ_t is additive, and $\delta_t(c\phi) = c^p \delta_t(\phi)$ for $c \in \Bbbk$. Let $\phi_1, \dots, \phi_{r_0}$ be a basis of $\mathcal{U}'$, and write the coordinate forms as $e_i^* = \psi_i + \sum_j c_{ij} \phi_j$ with $\psi_i \in \mathcal{U}$. Then $\operatorname{diag}(t)$ has the entries $\delta_t(e_i^*) = \sum_j c_{ij}^p \delta_t(\phi_j)$, so it is the image of the vector $(\delta_t(\phi_j))_j \in \Bbbk^{r_0}$, which depends linearly on t, under a matrix that does not depend on t. Hence $\dim \operatorname{diag}(\mathcal{X}) \leq r_0 \leq C$.

Step 4 (item (3)). For z in a finite subgroup B of the group, the average of f over B is at least $f(z)/2$, since $f(b) + f(z - b) \geq f(z)$ and $b \mapsto z - b$ permutes B. For fixed other arguments,

$y^j \mapsto \Phi(y^1, \dots, y^p)$ is additive on $\mathbb{F}_p^m$, so as y^j runs through $\mathbb{F}_p^m$ its value runs equally often through each element of its image, a finite subgroup. Hence the average of $f \circ \Phi$ over y^j is at least half its value at any fixed y^j. Applying this in the slots $j = 1, \dots, p$ in turn gives (3).

Step 5 (item (4): an invertible diagonal). Let $x = (x_1, \dots, x_m)$ be a vector of indeterminates, extend Ψ linearly to entries in $\Bbbk[x_1, \dots, x_m]$, and let $\mathsf{M}(x)$ be the matrix over $\Bbbk[x_1, \dots, x_m]$ of the linear map $v \mapsto \operatorname{diag} \Psi(x^{\otimes(p-1)} \otimes v)$, whose columns are $\operatorname{diag} \Psi(x^{\otimes(p-1)} \otimes e_i)$. Its value at a point $\xi \in \Bbbk^m$ is the matrix of the same map with ξ in place of x. As Ψ fixes the symmetric tensor $x^{\otimes p}$, $\mathsf{M}(x)x = \operatorname{diag}(x^{\otimes p}) = (x_1^p, \dots, x_m^p)^{\mathsf{T}}$. By (1), the form $\det \mathsf{M}(x)$ of degree $m(p-1)$ has the coefficient 1 at $x_1^{p-1} \cdots x_m^{p-1}$. The sum $\sum_{c \in \mathbb{F}_p} c^b$ is -1 if $b > 0$ and $p-1$ divides b, and 0 otherwise, and $x_1^{p-1} \cdots x_m^{p-1}$ is the only monomial of degree $m(p-1)$ whose exponents are positive multiples of $p-1$; so $\sum_{\xi \in \mathbb{F}_p^m} \det \mathsf{M}(\xi) = (-1)^m$, and $\mathsf{M}(\xi_0)$ is invertible for some $\xi_0 \in \mathbb{F}_p^m$.

Step 6 (item (4): a value of large rank). Let $\Bbbk'' \supseteq \Bbbk$ be a field with $|\Bbbk''| \geq 2m$, with $\Bbbk'' = \Bbbk$ if $\Bbbk$ is infinite, and let Ψ'' be the linear extension of Ψ to entries in $\Bbbk''$. The tensors $\Psi''(\xi_0^{\otimes(p-1)} \otimes u)$, $u \in \Bbbk''^m$, form a linear space whose diagonals form $\mathsf{M}(\xi_0)\Bbbk''^m = \Bbbk''^m$, so by (2) with $C = m-1$ one of them has flattening rank m. Hence some $m \times m$ minor $\mathsf{P}(u)$ of the matrix of entries of $\Psi''(\xi_0^{\otimes(p-1)} \otimes u)$, a polynomial of degree at most m in each coordinate of u, is not the zero polynomial. Let $d = 1 + \lfloor \log_p m \rfloor$, the least integer with $p^d > m$. As $|\Bbbk''| > m$, and $|\Bbbk''|$ is a power of p if $\Bbbk''$ is finite, $\Bbbk''$ contains an $\mathbb{F}_p$-subspace $\mathcal{W}$ of dimension d, so $|\mathcal{W}| = p^d > m$. A nonzero polynomial in r variables of degree at most m in each variable does not vanish on all of $\mathcal{W}^r$: for $r = 1$ it has at most m roots, and for $r > 1$ write it as $\sum_{e=0}^m \mathsf{P}_e u_r^e$ with polynomials P_e in the other variables; some P_e is nonzero, by induction it does not vanish at some point of $\mathcal{W}^{r-1}$, and there the polynomial is a nonzero polynomial in u_r of degree at most m. With $r = m$ and the minor P, there is $v \in \mathcal{W}^m$ with $\operatorname{frk} \Psi''(\xi_0^{\otimes(p-1)} \otimes v) = m$. Write $v = \sum_{i=1}^d \omega_i v_i$ with an $\mathbb{F}_p$-basis $\omega_1, \dots, \omega_d$ of $\mathcal{W}$ and $v_i \in \mathbb{F}_p^m$. Then $\Psi''(\xi_0^{\otimes(p-1)} \otimes v) = \sum_i \omega_i \Psi(\xi_0^{\otimes(p-1)} \otimes v_i)$, and subadditivity gives $\operatorname{frk} \Psi(\xi_0^{\otimes(p-1)} \otimes v_{i_0}) \geq m/d$ for some i_0.

Step 7 (item (4): averaging). The flattening rank is subadditive, and $(y^1, \dots, y^p) \mapsto \Psi(y^1 \otimes \cdots \otimes y^p)$ is additive in each argument, so by (3) the average in (4) is at least 2^{-p} times the flattening rank of the value at $(\xi_0, \dots, \xi_0, v_{i_0})$, hence at least $m/(2^p d)$. □

Proof of Proposition 10.13. Step 0 (the field under (a)). Assume (a). The completion $\widehat{K}$ has characteristic p, so its valuation ring is a complete discrete valuation ring containing the prime field $\mathbb{F}_p$, with ϑ as a uniformizer. By Cohen's structure theorem for complete regular local rings containing a field [6], this ring is isomorphic to the power series ring $\kappa[[X]]$ over its residue field κ, by an isomorphism taking ϑ to X; it extends to an isomorphism of $\widehat{K}$ onto $\kappa((X))$, which is isometric for the absolute value $|a| = |\vartheta|^{\operatorname{ord} a}$ on $\kappa((X))$. All objects in Proposition 10.13 are defined from K as a normed field, so under (a) we may assume that K is a subfield of $\kappa((X))$ with this absolute value.

Step 1 (item (1): t_m). Let (f_k) be a basis of Q_m with dual basis (f_k^*), and let e_l be the unit vector of $\ell^1(I_m, K)$ at l, so that ε_l is its image in G_m. Then $\sum_l \bar{y}_l \otimes e_l = \sum_k f_k \otimes (f_k^*(\bar{y}_l))_l$ lies in $Q_m \otimes N_m$, so $\sum_l \bar{y}_l \otimes \varepsilon_l = 0$ in $Q_m \otimes G_m$. Hence t_m lies in the kernel of the map $z \otimes g \mapsto \bar{z} \otimes g$ of $(K^m)^{\otimes p} \otimes G_m$ onto $Q_m \otimes G_m$, which is $\Gamma^p(K^m) \otimes G_m$. As K is infinite, $\Gamma^p(K^m)$ is spanned by the powers $u^{\otimes p}$, so $t_m \in V$. The vectors $y_l^1, \dots, y_l^p$ have entries in $\mathbb{F}_p$, hence norm at most 1, and $\|\varepsilon_l\| \leq 1$, so $\pi(t_m) \leq |I_m| = p^{pm}$.

Step 2 (a representation gives a projection). Let $t_m = \sum_i c_i u_i^{\otimes p} \otimes g_i$ with $c_i \in K$, $u_i \in c_0$, $g_i \in G$, and let $\delta > 0$. The coordinate projection of c_0 onto K^m and the projection of G

onto G_m are contractive; the p-th tensor power of the first, tensored with the second, fixes t_m and does not increase $\sum_i |c_i| \|u_i\|^p \|g_i\|$, so we may assume $u_i \in K^m$ and $g_i \in G_m$. Choose $\hat{g}_i \in \ell^1(I_m, K)$ with image g_i and $\|\hat{g}_i\| \leq (1+\delta)\|g_i\|$, and put $\sigma_l = \sum_i c_i \hat{g}_i(l)\, u_i^{\otimes p} \in \Gamma^p(K^m)$ for $l \in I_m$. With π_m the projective norm of the p-fold tensor power of K^m with the maximum norm,

$$\sum_{l \in I_m} \pi_m(\sigma_l) \leq \sum_i |c_i| \|u_i\|^p \|\hat{g}_i\| \leq (1+\delta) \sum_i |c_i| \|u_i\|^p \|g_i\|. \tag{P.4}$$

The tensors $\sum_l \sigma_l \otimes e_l = \sum_i c_i u_i^{\otimes p} \otimes \hat{g}_i$ and $\sum_l y_l \otimes e_l$ both map to t_m in $(K^m)^{\otimes p} \otimes G_m$, so their difference lies in $(K^m)^{\otimes p} \otimes N_m$. The vectors $(f_k^*(\bar{y}_l))_l$ span N_m, so the difference is $\sum_k \tau_k \otimes (f_k^*(\bar{y}_l))_l$ for some $\tau_k \in (K^m)^{\otimes p}$. So $\sigma_l = y_l + \sum_k f_k^*(\bar{y}_l)\tau_k$ for $l \in I_m$. Put $\Psi_K(z) = z + \sum_k f_k^*(\bar{z})\tau_k$ for $z \in (K^m)^{\otimes p}$; then $\Psi_K(y_l) = \sigma_l$. The linear map $\Theta(q) = q + \sum_k f_k^*(q)\bar{\tau}_k$ of Q_m vanishes at every $\bar{y}_l$, since $\bar{\sigma}_l = 0$, and these span Q_m, as I_m contains the tuples of unit vectors; so $\Theta = 0$, and the image of $\Psi_K(z)$ in Q_m is $\Theta(\bar{z}) = 0$. As $\Psi_K(z) = z$ when $\bar{z} = 0$, Ψ_K is a projection of $(K^m)^{\otimes p}$ onto $\Gamma^p(K^m)$.

Step 3 (item (1) under (a): constant coefficients). Let $K \subseteq \kappa((X))$ be as in Step 0. By the paragraph before Lemma P.1, with $\Bbbk = K$, the linear extension $\widetilde{\Psi}$ of Ψ_K to entries in $\kappa((X))$ is a projection of $(\kappa((X))^m)^{\otimes p}$ onto $\Gamma^p(\kappa((X))^m)$. For $z \in (\kappa^m)^{\otimes p}$ put $\Psi(z) = \operatorname{coeff}_0 \widetilde{\Psi}(z)$. Then Ψ is κ-linear with symmetric values and fixes the symmetric tensors over κ, so it is a projection of $(\kappa^m)^{\otimes p}$ onto $\Gamma^p(\kappa^m)$. As the y_l have entries in $\mathbb{F}_p \subseteq K \cap \kappa$, $\widetilde{\Psi}(y_l) = \Psi_K(y_l) = \sigma_l$, and $\Psi(y_l) = \operatorname{coeff}_0 \sigma_l$. The matrix of the entries of σ_l, with rows indexed by the first index, read in the completion of $\ell^\infty([m], K) \otimes_\pi \ell^\infty([m]^{p-1}, K)$, has norm at most $\pi_m(\sigma_l)$, since an elementary tensor of $p-1$ vectors of K^m, read as a function on $[m]^{p-1}$, has as maximum norm the product of their maximum norms. So Lemma K.2(2), which holds for every field κ of characteristic p (see the beginning of Appendix K), with $|\vartheta|$ in place of ϱ and the index sets $[m]$ and $[m]^{p-1}$, gives $\operatorname{frk} \Psi(y_l) \leq M_{|\vartheta|} \pi_m(\sigma_l)$, and by (P.4)

$$\sum_{l \in I_m} \operatorname{frk} \Psi(y_l) \leq (1+\delta) M_{|\vartheta|} \sum_i |c_i| \|u_i\|^p \|g_i\|. \tag{P.5}$$

Step 4 (item (1) under (a)). The tensors y_l, $l \in I_m$, are the $y^1 \otimes \cdots \otimes y^p$ with $(y^1, \ldots, y^p) \in (\mathbb{F}_p^m)^p$, so by Lemma P.1(4) the left side of (P.5) is at least $mp^{pm}/\big(2^p(1 + \lfloor \log_p m \rfloor)\big)$. As the representation of t_m and $\delta > 0$ are arbitrary, $\nu(t_m) \geq mp^{pm}/\big(2^p M_{|\vartheta|}(1 + \lfloor \log_p m \rfloor)\big)$. With Step 1 this is (1) under (a).

Step 5 (item (1) under (b)). Let $p = 2$. For $S \subseteq [m]$ let $1_S \in \mathbb{F}_2^m$ have the entries 1 at the elements of S and 0 elsewhere, so that I_m consists of the pairs $l = (1_S, 1_T)$, $S, T \subseteq [m]$. Identify $(K^m)^{\otimes 2}$ with $K^{m \times m}$ by $a \otimes b \mapsto ab^{\mathsf{T}}$; then $\Gamma^2(K^m)$ is the space of symmetric matrices and $y_l = 1_S 1_T^{\mathsf{T}}$ for $l = (1_S, 1_T)$. First, $|\det \sigma_{l_0}| \geq 1$ for some $l_0 \in I_m$. If $m = 1$, then $\Psi_K = \mathrm{id}$ and $\sigma_{l_0} = 1$ for $l_0 = (1_{\{1\}}, 1_{\{1\}})$. Let $m \geq 2$, and let $\wedge\colon K^{m \times m} \to \Lambda^2 K^m$ be the linear map of Theorem 7.20(1), with m in place of its n; by its formula there, it is onto, its kernel is the space of symmetric matrices, and $\wedge(ab^{\mathsf{T}}) = a \wedge b$. As $\Psi_K - \mathrm{id}$ vanishes on the symmetric matrices, it is $\chi \circ \wedge$ for a linear map $\chi\colon \Lambda^2 K^m \to K^{m \times m}$, and $\wedge \circ \chi = \mathrm{id}$, since Ψ_K takes symmetric values and $-1 = 1$ in K. So $\Psi_K(ab^{\mathsf{T}}) = \chi(a \wedge b) + ab^{\mathsf{T}}$, and Proposition N.12, applied over the field K to χ, with m in place of its n, gives $\sum_{S,T \subseteq [m]} \det \Psi_K(1_S 1_T^{\mathsf{T}}) = 1$. As K is nonarchimedean, some term has absolute value at least 1.

By Lemma N.15(2), $\pi_m(\sigma_{l_0}) \geq m$. By Lemma P.1(3), applied to the norm π_m and to the map $(y^1, y^2) \mapsto \Psi_K(y^1 \otimes y^2)$, the average of $\pi_m(\sigma_l)$ over the 4^m elements $l \in I_m$ is at least

$\pi_m(\sigma_{l_0})/4$, so $\sum_{l\in I_m}\pi_m(\sigma_l)\ge m4^{m-1}$. As the representation of t_m and $\delta>0$ are arbitrary, (P.4) gives $\nu(t_m)\ge m4^{m-1}$. With Step 1 this is (1) under (b).

Step 6 (item (2)). On the span of finitely many vectors of H, R_n is a homogeneous polynomial of degree n with coefficients in V: the coefficients of $x^{\otimes p}$, as a polynomial in the coordinates of x in a finite-dimensional subspace of c_0, are symmetric tensors, and symmetric tensors in $c_0^{\otimes p}$ are combinations of powers, as K is infinite. Moreover $\nu(R_n(\lambda,x,g))\le|\lambda|^{n-p-1}\|x\|^p\|g\|\le\|(\lambda,x,g)\|^n$. So R_n is a bounded polynomial law of degree n with $\|R_n\|\le1$. Suppose $jR_n(h)=\beta(h,\dots,h)$ for a linear isometry j of V, or of its completion, into a normed space Z and a bounded n-linear $\beta\colon H^n\to Z$, and let $\beta_\otimes\colon\otimes^n_\pi H\to Z$ be the linear map with $\beta_\otimes(h_1\otimes\cdots\otimes h_n)=\beta(h_1,\dots,h_n)$, of norm $\|\beta\|$. Put $\mathbf{1}=(1,0,0)\in H$ and $d_n=n!/(p!\,(n{-}p{-}1)!)$. For $t\in(K^m)^{\otimes p}\otimes G$, read in $\otimes^{p+1}H$, let $\iota(t)\in\otimes^nH$ be the sum, over the d_n words $w\in\{1,2,3\}^n$ with p letters 1 and one letter 2, of the tensor obtained from $t\otimes\mathbf{1}^{\otimes(n-p-1)}$ by moving the first p factors, in order, to the positions of the letters 1 of w, the $(p+1)$-st factor to the position of the letter 2, and the factors $\mathbf{1}$ to the positions of the letters 3. Since the inclusions of c_0 and G into H are isometric, permutations of the factors are isometries of $\otimes^n_\pi H$ and $\|\mathbf{1}\|=1$, $\iota(t)$ has norm at most $d_n\pi(t)$ in $\otimes^n_\pi H$, with $\pi(t)$ the projective norm of t in $c_0^{\otimes p}\otimes G$. For $x\in K^m$, $g\in G$ and $s=(s_1,s_2,s_3)\in K^3$ put $h_s=(s_3,s_1x,s_2g)$. Then $\beta_\otimes(h_s^{\otimes n})=\beta(h_s,\dots,h_s)=jR_n(h_s)=s_1^ps_2s_3^{n-p-1}\,j(x^{\otimes p}\otimes g)$, and $\iota(x^{\otimes p}\otimes g)$ is the coefficient of $s_1^ps_2s_3^{n-p-1}$ in the polynomial $h_s^{\otimes n}$. As K is infinite, comparing coefficients gives $\beta_\otimes(\iota(x^{\otimes p}\otimes g))=j(x^{\otimes p}\otimes g)$. So $\beta_\otimes(\iota(t))=j(t)$ for t in the span of the $x^{\otimes p}\otimes g$, $x\in K^m$, $g\in G$, which contains t_m by Step 1. Hence $\nu(t_m)=\|j(t_m)\|\le d_n\|\beta\|\pi(t_m)\le d_n\|\beta\|p^{pm}$, and (1) gives $m\le2^pd_nM_{|\vartheta|}\|\beta\|(1+\lfloor\log_p m\rfloor)$ under (a) and $m\le4d_n\|\beta\|$ under (b), for every m, which is impossible. □

For the proof of Proposition 10.14 we extend the notation before Lemma P.1 from $\Bbbk^m$ to $\Bbbk^{\mathcal J}$, for a field $\Bbbk$ and a finite set $\mathcal J$. A tensor $t\in(\Bbbk^{\mathcal J})^{\otimes d}$ is read as a function on the words $w=(w_1,\dots,w_d)\in\mathcal J^d$, and $\Gamma^d(\Bbbk^{\mathcal J})$ is the space of symmetric tensors, with $\Gamma^0(\Bbbk^{\mathcal J})=\Bbbk$. Let $\mathbb N^{\mathcal J}_d$ be the set of $\alpha\in\mathbb N^{\mathcal J}$ with $|\alpha|=d$; for $\mathcal J=[m]$ it is the set $\mathbb N^m_d$ of Section 10.4. The *content* of a word $w\in\mathcal J^d$ is the $\alpha\in\mathbb N^{\mathcal J}_d$ whose entry at each element of $\mathcal J$ is the number of letters of w equal to it. Let $O_\alpha\in\Gamma^d(\Bbbk^{\mathcal J})$ be the sum, over the words w of content α, of the tensor product of the unit vectors of $\Bbbk^{\mathcal J}$ at the letters $w_1,\dots,w_d$. The O_α, $\alpha\in\mathbb N^{\mathcal J}_d$, form a basis of $\Gamma^d(\Bbbk^{\mathcal J})$, and $t=\sum_\alpha t_\alpha O_\alpha$, where t_α is the value of t at any word of content α.

The *product* of $t\in\Gamma^d(\Bbbk^{\mathcal J})$ and $t'\in\Gamma^{d'}(\Bbbk^{\mathcal J})$ is the sum $t\cdot t'\in\Gamma^{d+d'}(\Bbbk^{\mathcal J})$ of the $\binom{d+d'}{d}$ tensors obtained from $t\otimes t'$ by the permutations of the $d+d'$ factors that keep the order of the first d factors and the order of the last d'; a scalar in $\Gamma^0(\Bbbk^{\mathcal J})$ multiplies. For $\Bbbk=K$ this is the shuffle product of Section L.3. As tensor powers of linear maps commute with permutations of the factors, $q^{\otimes(d+d')}(t\cdot t')=q^{\otimes d}(t)\cdot q^{\otimes d'}(t')$ for every linear map q of $\Bbbk^{\mathcal J}$ into some $\Bbbk^{\mathcal J'}$. Let $\mathcal J=[m]$, $\alpha\in\mathbb N^m_d$ and $\alpha'\in\mathbb N^m_{d'}$. The coefficient of $O_\alpha\cdot O_{\alpha'}$ at a word of content $\alpha+\alpha'$ is the number of ways to choose the d of its letters that come from the first factor, which must contain α_i of the $\alpha_i+\alpha'_i$ letters equal to i, for each i; so

$$O_\alpha\cdot O_{\alpha'}=\prod_{i=1}^m\binom{\alpha_i+\alpha'_i}{\alpha_i}\,O_{\alpha+\alpha'},\tag{P.6}$$

and the product is commutative and associative on the spaces $\Gamma^d(\Bbbk^m)$, $d\ge0$. For $t\in\Gamma^{d+d'}(\Bbbk^m)$ and $\gamma\in\mathbb N^m_d$ let $\partial^\gamma t\in\Gamma^{d'}(\Bbbk^m)$ have the coefficients $(\partial^\gamma t)_\alpha=t_{\alpha+\gamma}$, $\alpha\in\mathbb N^m_{d'}$; for $d=p-1$ and $d'=1$ this is the slice of Section 10.4. Then $\partial^\gamma\partial^{\gamma'}=\partial^{\gamma+\gamma'}$, and $\partial^\gamma O_\alpha$ is $O_{\alpha-\gamma}$ if $\gamma\le\alpha$ entrywise and 0 otherwise. With (P.6) and Vandermonde's identity $\sum_\gamma\prod_i\binom{\alpha_i}{\gamma_i}=\binom{d+d'}{d}$ for

$\alpha \in \mathbb{N}^m_{d+d'}$, the sum over the $\gamma \in \mathbb{N}^m_d$ with $\gamma \le \alpha$, this gives

$$\sum_{\gamma \in \mathbb{N}^m_d} (\partial^\gamma t)\cdot O_\gamma = \binom{d+d'}{d} t \qquad (t \in \Gamma^{d+d'}(\Bbbk^m)). \tag{P.7}$$

Lemma P.2 (an identity for symmetric tensors)**.** *Let $\Bbbk$ be a field of characteristic $p>0$ and $m \ge 1$, let L be a matrix over $\Bbbk$ with rows and columns indexed by $\mathbb{N}^m_{p-1}$, and let $\mathsf{L}^\sharp$ be the linear map of $\Gamma^p(\Bbbk^m)$ with*

$$\mathsf{L}^\sharp(t) = \sum_{\gamma,\gamma' \in \mathbb{N}^m_{p-1}} \mathsf{L}_{\gamma\gamma'}\,(\partial^\gamma t)\cdot O_{\gamma'}.$$

(1) $t = \sum_{\gamma\in\mathbb{N}^m_{p-1}} \big((\mathrm{id}-\mathsf{L}^\sharp)(\partial^\gamma t)\big)\cdot O_\gamma$ *for every* $t \in \Gamma^{2p-1}(\Bbbk^m)$.

(2) $\operatorname{rank}(\mathrm{id}-\mathsf{L}^\sharp) \ge \binom{m+2p-2}{2p-1} \big/ \binom{m+p-2}{p-1} \ge (p-1)!\,m^p/(2p-1)!$.

Proof. Step 1 (the identity part). By (P.7) with $d=p-1$ and $d'=p$, $\sum_\gamma (\partial^\gamma t)\cdot O_\gamma = \binom{2p-1}{p-1} t$, the sum over $\mathbb{N}^m_{p-1}$. The integer $\binom{2p-1}{p-1}(p-1)!$ is the product $(p+1)(p+2)\cdots(2p-1)$, which is congruent to $(p-1)!$ modulo p, and $(p-1)!$ is prime to p; so $\binom{2p-1}{p-1}=1$ in $\Bbbk$, and $\sum_\gamma (\partial^\gamma t)\cdot O_\gamma = t$.

Step 2 (the part of $\mathsf{L}^\sharp$). For $\gamma \in \mathbb{N}^m_{p-1}$, $\mathsf{L}^\sharp(\partial^\gamma t) = \sum_{\gamma',\gamma''} \mathsf{L}_{\gamma'\gamma''}\,(\partial^{\gamma'}\partial^\gamma t)\cdot O_{\gamma''}$. As the product is commutative and associative and $\partial^{\gamma'}\partial^\gamma = \partial^\gamma\partial^{\gamma'}$,

$$\sum_\gamma \mathsf{L}^\sharp(\partial^\gamma t)\cdot O_\gamma = \sum_{\gamma',\gamma''} \mathsf{L}_{\gamma'\gamma''}\, O_{\gamma''} \cdot \sum_\gamma \big(\partial^\gamma(\partial^{\gamma'} t)\big)\cdot O_\gamma,$$

all sums over $\mathbb{N}^m_{p-1}$. By (P.7) with $d=p-1$ and $d'=1$, applied to $\partial^{\gamma'} t \in \Gamma^p(\Bbbk^m)$, the inner sum is $p\,\partial^{\gamma'} t = 0$. With Step 1 this is (1).

Step 3 (item (2)). By (1), the linear map $t \mapsto \big((\mathrm{id}-\mathsf{L}^\sharp)(\partial^\gamma t)\big)_{\gamma\in\mathbb{N}^m_{p-1}}$ from $\Gamma^{2p-1}(\Bbbk^m)$ into the product of $|\mathbb{N}^m_{p-1}|$ copies of the image of $\mathrm{id}-\mathsf{L}^\sharp$ is injective. As $\dim \Gamma^d(\Bbbk^m) = |\mathbb{N}^m_d| = \binom{m+d-1}{d}$, this gives $\binom{m+2p-2}{2p-1} \le \binom{m+p-2}{p-1}\operatorname{rank}(\mathrm{id}-\mathsf{L}^\sharp)$. The quotient of the two binomial coefficients is $(p-1)!/(2p-1)!$ times the product of the p numbers $m+p-1,\dots,m+2p-2$, hence at least $(p-1)!\,m^p/(2p-1)!$. □

Proof of Proposition 10.14. Let $m \ge 1$, and let $\mathcal{J}_m$ be the disjoint union of $[m]$ and $\mathbb{N}^m_{p-1}$. For a field $\Bbbk$ containing $\mathbb{F}_p$ write e_i, $i\in[m]$, and e_γ, $\gamma \in \mathbb{N}^m_{p-1}$, for the unit vectors of $\Bbbk^{\mathcal{J}_m}$, and read $\Bbbk^m$ as the span of the e_i. For $T \in \Gamma^p(\mathbb{F}_p^m)$ let $q_T\colon \Bbbk^{\mathcal{J}_m} \to \Bbbk^m$ be the linear map with $q_T(e_i)=e_i$ and $q_T(e_\gamma) = \partial^\gamma T$; its matrix has entries in $\mathbb{F}_p$. For a real number $\tau \ge 1$ and $z = \sum_i z_i e_i + \sum_\gamma z_\gamma e_\gamma$ in $K^{\mathcal{J}_m}$ put $\|z\|_\tau = \sum_i |z_i| + \tau\sum_\gamma |z_\gamma|$; by (10.5), $\|h\|_{T,\tau} = \inf\{\|z\|_\tau : q_T(z)=h\}$. Call the elements $\mathbf{a}$ of $\mathbb{N}_p^{\mathcal{J}_m}$ *positions*, let $|\mathbf{a}|_\partial$ be the sum of the entries of $\mathbf{a}$ at the elements of $\mathbb{N}^m_{p-1}$, and call $\mathbf{a}$ *nonlinear* if $|\mathbf{a}|_\partial \ge 2$. For a word $w \in \mathcal{J}_m^p$ let $|w|_\partial$ be the number of its letters in $\mathbb{N}^m_{p-1}$, and for $S \in \Gamma^p(\Bbbk^{\mathcal{J}_m})$ let $\operatorname{supp} S$ be the set of positions $\mathbf{a}$ with $S_{\mathbf{a}} \ne 0$. Put $\mathfrak{b}_m = (p-1)!\,m^p/(2p-1)!$.

Step 1 (norms). If $h = q_T(z)$, the i-th entry of h is $z_i + \sum_\gamma z_\gamma T_{\gamma+e_i}$, of absolute value at most $\|z\|_\tau$, as $|T_\alpha| \le 1$ and $\tau \ge 1$. So no entry of $h \in \mathcal{E}_{T,\tau}$ has absolute value above $\|h\|_{T,\tau}$: the coordinate forms of $\mathcal{E}_{T,\tau}$ are bounded and separate points, so π is a norm on $\mathcal{E}_{T,\tau}^{\otimes p}$, and so is $\|\cdot\|_\Gamma$ on $\Gamma^p(\mathcal{E}_{T,\tau})$. A representation $t = \sum_l c_l h_l^{\otimes p}$ in $\Gamma^p(P)$, for a normed space P, gives $\pi(t) \le \sum_l |c_l|\|h_l\|^p$; so $\|t\|_\Gamma \ge \pi(t)$, by (10.4).

Step 2 (the projective norm). In $\mathcal{E}_{T,\tau}$, $\|e_i\| \le 1$ and $\|\partial^\gamma T\| \le \tau$, by (10.5). For a word $(i_1,\dots,i_{p-1}) \in [m]^{p-1}$ let $\gamma(i_1,\dots,i_{p-1})$ be its content. The value of T at $(i_1,\dots,i_{p-1},i)$ is the

i-th entry of $\partial^{\gamma(i_1,\dots,i_{p-1})}T$, so

$$T = \sum_{(i_1,\dots,i_{p-1})\in[m]^{p-1}} e_{i_1}\otimes\cdots\otimes e_{i_{p-1}}\otimes\partial^{\gamma(i_1,\dots,i_{p-1})}T,$$

a sum of m^{p-1} elementary tensors whose factors have norms with product at most τ. Hence $\pi(T)\le\tau m^{p-1}$ in $\mathcal{E}_{T,\tau}^{\otimes p}$.

Step 3 (a representation gives a symmetric tensor over $\mathbb{F}_p$). The elements of K of absolute value below 1 form an $\mathbb{F}_p$-linear subspace that does not contain 1, so there is an $\mathbb{F}_p$-linear map $\rho\colon K\to\mathbb{F}_p$ with $\rho(1)=1$ that vanishes on it; then ρ fixes $\mathbb{F}_p$, and $|x|\ge 1$ for every $x\in K$ with $\rho(x)\ne 0$. Let $\delta>0$ and $T=\sum_l c_l h_l^{\otimes p}$ with $c_l\in K$, $h_l\in\mathcal{E}_{T,\tau}$ and $\sum_l|c_l|\|h_l\|_{T,\tau}^p\le\|T\|_\Gamma+\delta$. Choose $\hat h_l\in K^{\mathcal{J}_m}$ with $q_T(\hat h_l)=h_l$ and $\|\hat h_l\|_\tau\le(1+\delta)\|h_l\|_{T,\tau}$. The tensor $\widetilde S=\sum_l c_l\hat h_l^{\otimes p}$ is symmetric, and $q_T^{\otimes p}(\widetilde S)=T$. For $z\in K^{\mathcal{J}_m}$ the sum of the numbers $\tau^{|w|_\partial}|z^{\otimes p}(w)|$, $w\in\mathcal{J}_m^p$, is $\|z\|_\tau^p$; so, by the triangle inequality,

$$\sum_{w\in\mathcal{J}_m^p}\tau^{|w|_\partial}|\widetilde S(w)|\le\sum_l|c_l|\|\hat h_l\|_\tau^p\le(1+\delta)^p(\|T\|_\Gamma+\delta). \tag{P.8}$$

Let $S\in\Gamma^p(\mathbb{F}_p^{\mathcal{J}_m})$ have the values $S(w)=\rho(\widetilde S(w))$. As the matrix of q_T has entries in $\mathbb{F}_p$, each value of $q_T^{\otimes p}(S)$ is the image under ρ of the corresponding value of $q_T^{\otimes p}(\widetilde S)=T$, which lies in $\mathbb{F}_p$; so $q_T^{\otimes p}(S)=T$. For $\mathbf{a}\in\operatorname{supp}S$ and a word w of content $\mathbf{a}$, $\rho(\widetilde S(w))=S_{\mathbf{a}}\ne 0$, so $|\widetilde S(w)|\ge 1$; and $|w|_\partial=|\mathbf{a}|_\partial$. Taking one such word for each $\mathbf{a}\in\operatorname{supp}S$, (P.8) gives $\sum_{\mathbf{a}\in\operatorname{supp}S}\tau^{|\mathbf{a}|_\partial}\le(1+\delta)^p(\|T\|_\Gamma+\delta)$. As $\Gamma^p(\mathbb{F}_p^{\mathcal{J}_m})$ is finite, one S occurs for arbitrarily small δ; so

$$\|T\|_\Gamma\ge\sum_{\mathbf{a}\in\operatorname{supp}S}\tau^{|\mathbf{a}|_\partial}\quad\text{for some } S\in\Gamma^p(\mathbb{F}_p^{\mathcal{J}_m})\text{ with } q_T^{\otimes p}(S)=T. \tag{P.9}$$

Step 4 (supports with few nonlinear positions). Let $S\in\Gamma^p(\mathbb{F}_p^{\mathcal{J}_m})$, and let $\mathcal{K}_S\subseteq\mathbb{N}_p^m$ be the set of the $\gamma+e_i$, $i\in[m]$, for the $\gamma\in\mathbb{N}_{p-1}^m$ at which some nonlinear $\mathbf{a}\in\operatorname{supp}S$ has a nonzero entry. A position has nonzero entries at no more than p elements of $\mathbb{N}_{p-1}^m$, so $|\mathcal{K}_S|$ is at most pm times the number of nonlinear positions in $\operatorname{supp}S$. We show that at most $p^{|\mathcal{K}_S|+\binom{m+p-1}{p}-\mathfrak{b}_m}$ tensors $T'\in\Gamma^p(\mathbb{F}_p^m)$ satisfy $q_{T'}^{\otimes p}(S)=T'$. Write $S=S^{(0)}+S^{(1)}+S^{(2)}$, where $S^{(0)}$, $S^{(1)}$ and $S^{(2)}$ are supported in the positions $\mathbf{a}$ with $|\mathbf{a}|_\partial=0$, $|\mathbf{a}|_\partial=1$ and $|\mathbf{a}|_\partial\ge 2$. The positions with $|\mathbf{a}|_\partial=0$ are the $\alpha\in\mathbb{N}_p^m$, and $q_{T'}$ fixes the e_i, so $q_{T'}^{\otimes p}(S^{(0)})=\sum_{\alpha\in\mathbb{N}_p^m}S_\alpha O_\alpha$ does not depend on T'. For $\gamma,\gamma'\in\mathbb{N}_{p-1}^m$ let $\mathbf{a}(\gamma,\gamma')$ be the position with the entry 1 at γ, the entries of γ' on $[m]$ and 0 elsewhere. These are the positions with $|\mathbf{a}|_\partial=1$, and $O_{\mathbf{a}(\gamma,\gamma')}=e_\gamma\cdot O_{\gamma'}$, as each word of content $\mathbf{a}(\gamma,\gamma')$ arises from exactly one of the p permutations that define this product, applied to exactly one tensor $e_\gamma\otimes e_{i_1}\otimes\cdots\otimes e_{i_{p-1}}$ with $(i_1,\dots,i_{p-1})$ of content γ'. As the product commutes with $q_{T'}$,

$$q_{T'}^{\otimes p}(S^{(1)})=\sum_{\gamma,\gamma'\in\mathbb{N}_{p-1}^m}S_{\mathbf{a}(\gamma,\gamma')}\,(\partial^\gamma T')\cdot O_{\gamma'}=\mathsf{L}^\sharp(T'),$$

with $\mathsf{L}_{\gamma\gamma'}=S_{\mathbf{a}(\gamma,\gamma')}$ in Lemma P.2. Finally, $q_{T'}^{\otimes p}(S^{(2)})$ depends only on the vectors $q_{T'}(e_\gamma)=\partial^\gamma T'$ for the γ at which some $\mathbf{a}\in\operatorname{supp}S^{(2)}$ has a nonzero entry, that is, on the coefficients T'_α, $\alpha\in\mathcal{K}_S$. So if $q_{T'}^{\otimes p}(S)=T'$, then $(\mathrm{id}-\mathsf{L}^\sharp)(T')$ equals $q_{T'}^{\otimes p}(S^{(0)})+q_{T'}^{\otimes p}(S^{(2)})$, which depends only on $(T'_\alpha)_{\alpha\in\mathcal{K}_S}$. For each of the $p^{|\mathcal{K}_S|}$ choices of these coefficients the solutions T' lie in one

coset of the kernel of $\mathrm{id} - \mathsf{L}^\sharp$, which has at most $p^{\binom{m+p-1}{p}-\mathfrak{b}_m}$ elements by Lemma P.2(2), as $\dim \Gamma^p(\mathbb{F}_p^m) = \binom{m+p-1}{p}$.

Step 5 (item (1)). Put $\mathfrak{n}_1 = \mathfrak{b}_m/(3p^2(1+\log_p m))$, $\mathfrak{n}_2 = \mathfrak{b}_m/(3pm)$ and $\tau = \max\{1, (m/(p(1+\log_p m)))^{1/2}\}$. There are $\binom{|\mathcal{J}_m|+p-1}{p} \le |\mathcal{J}_m|^p$ positions, and $|\mathcal{J}_m| = m + \binom{m+p-2}{p-1} \le m + m^{p-1} \le 2m^{p-1}$; so p times the number of positions is at most $p\,2^p m^{p(p-1)} \le p^{p^2} m^{p^2} = p^{p^2(1+\log_p m)}$, as $p\,2^p \le p^{p^2}$. Every $S \in \Gamma^p(\mathbb{F}_p^{\mathcal{J}_m})$ that is nonzero at fewer than $\mathfrak{n}_1$ positions is the sum of $\lceil \mathfrak{n}_1 \rceil - 1$ tensors $O_{\mathbf{a}}$, $\mathbf{a}$ a position, each multiplied by an element of $\mathbb{F}_p$; so there are at most $p^{\mathfrak{n}_1 p^2(1+\log_p m)} = p^{\mathfrak{b}_m/3}$ of them. If moreover fewer than $\mathfrak{n}_2$ of its positions are nonlinear, then $|\mathcal{K}_S| \le pm\mathfrak{n}_2 = \mathfrak{b}_m/3$, and by Step 4 at most $p^{\binom{m+p-1}{p}-2\mathfrak{b}_m/3}$ tensors T' satisfy $q_{T'}^{\otimes p}(S) = T'$. So at most $p^{\binom{m+p-1}{p}-\mathfrak{b}_m/3}$ of the $p^{\binom{m+p-1}{p}}$ elements T' of $\Gamma^p(\mathbb{F}_p^m)$ satisfy $q_{T'}^{\otimes p}(S) = T'$ for such an S. Choose $T \in \Gamma^p(\mathbb{F}_p^m)$ outside them: every S with $q_T^{\otimes p}(S) = T$ is nonzero at no fewer than $\mathfrak{n}_1$ positions, or at no fewer than $\mathfrak{n}_2$ nonlinear positions. The weights $\tau^{|\mathbf{a}|_\partial}$ in (P.9) are at least 1, and at least τ^2 for nonlinear $\mathbf{a}$, and $\tau^2\mathfrak{n}_2 \ge \mathfrak{n}_1$, as $\tau^2 \ge m/(p(1+\log_p m))$; so $\|T\|_\Gamma \ge \mathfrak{n}_1 > 0$ in $\Gamma^p(\mathcal{E}_{T,\tau})$, and $T \ne 0$. If $\tau > 1$, Step 2 gives

$$\frac{\|T\|_\Gamma}{\pi(T)} \ge \frac{\mathfrak{n}_1}{\tau m^{p-1}} = \frac{(p-1)!}{3p^{3/2}(2p-1)!}\Big(\frac{m}{1+\log_p m}\Big)^{1/2}.$$

If $\tau = 1$, then $m \le p(1+\log_p m)$, so the right side of (10.6) is at most $(p-1)!\,\pi(T)/(3p(2p-1)!) < \pi(T) \le \|T\|_\Gamma$, by Step 1.

Step 6 (item (2): the law). If K is complete, the finite-dimensional spaces $\mathcal{E}_m$ are complete, and so is their ℓ^1-sum $\mathcal{E}$. By Step 1 the coordinate forms of the spaces $\mathcal{E}_m$, composed with the projections of $\mathcal{E}$ onto them, are bounded; they separate points, so π is a norm on $\mathcal{E}^{\otimes p}$ and $\|\cdot\|_\Gamma \ge \pi$ is a norm on $\Gamma^p(\mathcal{E})$. For finitely many vectors (λ_l, h_l) of $K \oplus \mathcal{E}$ and scalars s_l, the value of the map at $\sum_l s_l(\lambda_l, h_l)$ is $(\sum_l s_l\lambda_l)^{n-p}(\sum_l s_l h_l)^{\otimes p}$, a homogeneous polynomial of degree n in the s_l whose coefficients are combinations of symmetric tensors of the span of the h_l, which lie in $\Gamma^p(\mathcal{E})$ as K is infinite. Moreover $\|\lambda^{n-p}h^{\otimes p}\|_\Gamma \le |\lambda|^{n-p}\|h\|^p \le \|(\lambda,h)\|^n$. So the map is a bounded polynomial law of degree n and norm at most 1.

Step 7 (item (2): the ratio on $\Gamma^p(\mathcal{E})$). The inclusion of $\mathcal{E}_m$ in $\mathcal{E}$ is isometric, and the projection of $\mathcal{E}$ onto $\mathcal{E}_m$ is a contractive left inverse. So, for $t \in \Gamma^p(\mathcal{E}_m)$, $\pi(t)$ computed in $\mathcal{E}^{\otimes p}$ is at most $\pi(t)$ computed in $\mathcal{E}_m^{\otimes p}$, and $\|t\|_\Gamma$ computed in $\Gamma^p(\mathcal{E})$ is at least $\|t\|_\Gamma$ computed in $\Gamma^p(\mathcal{E}_m)$, as the p-th tensor power of the projection fixes t and does not increase the cost $\sum_l |c_l|\|h_l\|^p$ of a representation. By (1), applied to $t = T^{(m)} \ne 0$, $\|\cdot\|_\Gamma/\pi$ is unbounded on the nonzero elements of $\Gamma^p(\mathcal{E})$.

Step 8 (item (2): comparing coefficients). Suppose that $j(\lambda^{n-p}h^{\otimes p}) = \beta((\lambda,h),\dots,(\lambda,h))$ for all $(\lambda,h) \in K \oplus \mathcal{E}$, for a linear isometry j of $\Gamma^p(\mathcal{E})$, or of its completion, into a normed space Z and a bounded n-linear map $\beta\colon (K\oplus\mathcal{E})^n \to Z$. Put $\mathbf{1} = (1,0)$ and $h^+ = (0,h)$ for $h \in \mathcal{E}$. For $h_1,\dots,h_p \in \mathcal{E}$ let $\beta'(h_1,\dots,h_p)$ be the sum, over the $\binom{n}{p}$ sets of p of the n slots, of the value of β at the n-tuple with $h_1^+,\dots,h_p^+$ in these slots, in this order, and $\mathbf{1}$ in the others. Then β' is p-linear, and $\|\beta'\| \le \binom{n}{p}\|\beta\|$, as $\|\mathbf{1}\| = 1$ and $\|h^+\| = \|h\|$. For $h \in \mathcal{E}$ and $s = (s_1,s_2) \in K^2$, the value $\beta((s_2,s_1h),\dots,(s_2,s_1h))$, with $(s_2,s_1h) = s_2\mathbf{1} + s_1h^+$, is a homogeneous polynomial of degree n in s whose coefficient of $s_1^p s_2^{n-p}$ is $\beta'(h,\dots,h)$, while $j(s_2^{n-p}(s_1h)^{\otimes p}) = s_1^p s_2^{n-p} j(h^{\otimes p})$. As K is infinite, comparing coefficients, as in Step 6 of the proof of Proposition 10.13, gives $j(h^{\otimes p}) = \beta'(h,\dots,h)$. Let $\beta'_\otimes\colon \otimes_\pi^p \mathcal{E} \to Z$ be the linear map with $\beta'_\otimes(h_1\otimes\cdots\otimes h_p) = \beta'(h_1,\dots,h_p)$, of norm $\|\beta'\|$. As $\Gamma^p(\mathcal{E})$ is spanned by powers, $j(t) = \beta'_\otimes(t)$ for $t \in \Gamma^p(\mathcal{E})$, so $\|t\|_\Gamma = \|j(t)\| \le \binom{n}{p}\|\beta\|\,\pi(t)$ for $t \in \Gamma^p(\mathcal{E})$, against Step 7. □

For the proof of Theorem 10.15 let $p = \operatorname{char} K > 0$, with $\Gamma^p(P)$, $\|\cdot\|_\Gamma$, $\|\cdot\|_\mathrm{F}$, π, the hollow tensors and the sets $\mathbb{N}_p^r$ as in Section 10.4. For a subspace P_0 of P we read $P_0^{\otimes p}$ as a subspace of $P^{\otimes p}$, as over every field. For vectors $h_1, \dots, h_r \in P$ and $\alpha \in \mathbb{N}_p^r$ let $O_\alpha(h_1, \dots, h_r)$ be the sum of the tensors $h_{w_1} \otimes \cdots \otimes h_{w_p}$ over the words $w \in [r]^p$ whose content, defined before Lemma P.2, is α; the same sum with the unit vectors of K^r in place of $h_1, \dots, h_r$ is the tensor O_α there, for $\Bbbk = K$ and $\mathcal{J} = [r]$. Expanding the power gives

$$(s_1h_1 + \cdots + s_rh_r)^{\otimes p} = \sum_{\alpha \in \mathbb{N}_p^r} s_1^{\alpha_1} \cdots s_r^{\alpha_r}\, O_\alpha(h_1, \dots, h_r) \qquad (s_1, \dots, s_r \in K), \tag{P.10}$$

and $O_\alpha(h_1, \dots, h_r)$ lies in $\Gamma^p(P)$, since the coefficients of this polynomial are linear combinations of its values, by interpolation in each variable, as in Step 5 of the proof of Theorem 10.11.

Lemma P.3 (linearization)**.** *Let $a\colon P \to W$ be a bounded polynomial law of degree p.*

1. *There is a unique linear map $L_a\colon \Gamma^p(P) \to W$ with $L_a(h^{\otimes p}) = a(h)$ for $h \in P$, and $\|L_a(t)\| \le \|a\|\|t\|_\Gamma$ for $t \in \Gamma^p(P)$. If a is a Frobenius law, L_a vanishes on the hollow tensors, and $\|L_a(t)\| \le \|a\|\|t\|_\mathrm{F}$.*
2. *If $M \ge 0$ and $\|L_a(t)\| \le M\pi(t)$ for $t \in \Gamma^p(P)$, then a is ambient of degree p, with a p-linear map β of norm at most M in Definition 10.10(2).*
3. *If P satisfies (F_P) with constant C_F and (N_P) with constant C_N, then $\|t\|_\Gamma \le \big(C_\mathrm{F} + (1 + C_\mathrm{F})C_\mathrm{N}\big)\pi(t)$ for $t \in \Gamma^p(P)$.*

Proof. Step 1 (item (1): the map L_a). Let $\sum_i \lambda_i h_i^{\otimes p} = 0$, a finite sum with $\lambda_i \in K$ and $h_i \in P$. Let $b_1, \dots, b_m$ be a basis of the span of the h_i, write $h_i = \sum_l s_{i,l} b_l$, and put $s_i^\alpha = s_{i,1}^{\alpha_1} \cdots s_{i,m}^{\alpha_m}$ for $\alpha \in \mathbb{N}_p^m$. By (P.10) and Definition 10.10(1), $h_i^{\otimes p} = \sum_\alpha s_i^\alpha O_\alpha(b_1, \dots, b_m)$ and $a(h_i) = \sum_\alpha s_i^\alpha a^{[\alpha]}(b_1, \dots, b_m)$, the sums over $\mathbb{N}_p^m$. The tensors $O_\alpha(b_1, \dots, b_m)$ are sums over disjoint sets of words of the linearly independent tensors $b_{w_1} \otimes \cdots \otimes b_{w_p}$, hence linearly independent, so $\sum_i \lambda_i s_i^\alpha = 0$ for every α, and $\sum_i \lambda_i a(h_i) = 0$. As the powers span $\Gamma^p(P)$, $L_a(\sum_i \lambda_i h_i^{\otimes p}) = \sum_i \lambda_i a(h_i)$ defines the unique such linear map, and $\|L_a(t)\| \le \sum_i |\lambda_i| \|a(h_i)\| \le \|a\| \sum_i |\lambda_i| \|h_i\|^p$ for every representation $t = \sum_i \lambda_i h_i^{\otimes p}$; the infimum over these is the bound, by (10.4).

Step 2 (item (1): Frobenius laws). Let a be a Frobenius law and $h_1, \dots, h_p \in P$. By (P.10), for $s_1, \dots, s_p \in K$,

$$\sum_{\alpha \in \mathbb{N}_p^p} s_1^{\alpha_1} \cdots s_p^{\alpha_p} L_a\big(O_\alpha(h_1, \dots, h_p)\big) = a(s_1h_1 + \cdots + s_ph_p) = s_1^p a(h_1) + \cdots + s_p^p a(h_p).$$

As K is infinite and $p \ge 2$, comparing the coefficients of $s_1 \cdots s_p$ gives $L_a(O_{(1,\dots,1)}(h_1, \dots, h_p)) = 0$. The words of content $(1, \dots, 1)$ are the orderings of $[p]$, so $O_{(1,\dots,1)}(h_1, \dots, h_p)$ is one of the tensors that span the hollow tensors, and L_a vanishes on the hollow tensors. If $t - t'$ is hollow, then $\|L_a(t)\| = \|L_a(t')\| \le \|a\|\|t'\|_\Gamma$; the infimum over t' is the bound.

Step 3 (item (2)). For $v \in W$ and $y \in P^{\otimes p}$ let $\|(v, y)\|_Z$ be the infimum of $\|v + L_a(t)\| + M\pi(y - t)$ over $t \in \Gamma^p(P)$, the quotient seminorm of $\|v\| + M\pi(y)$ on $W \oplus P^{\otimes p}$ modulo the subspace of the pairs $(L_a(t), -t)$. Let Z be the quotient of $W \oplus P^{\otimes p}$ by the kernel of $\|\cdot\|_Z$, with the norm $\|\cdot\|_Z$, and $[v, y]$ the class of (v, y). The linear map $j(v) = [v, 0]$ is an isometry: $t = 0$ gives $\|[v, 0]\|_Z \le \|v\|$, and $\|v + L_a(t)\| + M\pi(-t) \ge \|v\| - \|L_a(t)\| + M\pi(t) \ge \|v\|$ for every t, by the hypothesis on L_a. The map $\beta(h_1, \dots, h_p) = [0, h_1 \otimes \cdots \otimes h_p]$ is p-linear with $\|\beta(h_1, \dots, h_p)\|_Z \le M\pi(h_1 \otimes \cdots \otimes h_p) \le M\|h_1\| \cdots \|h_p\|$, and $\beta(h, \dots, h) - ja(h) = [-a(h), h^{\otimes p}] = 0$, as $t = h^{\otimes p}$ shows.

Step 4 (item (3)). Let $t \in \Gamma^p(P)$ and $\delta > 0$. By (F_P) there is $t' \in \Gamma^p(P)$ with $t - t'$ hollow and $\|t'\|_\Gamma \le C_\mathrm{F}\pi(t) + \delta$. Then $\pi(t - t') \le \pi(t) + \|t'\|_\Gamma \le (1 + C_\mathrm{F})\pi(t) + \delta$, as $\pi \le \|\cdot\|_\Gamma$, and by (N_P), $\|t\|_\Gamma \le \|t'\|_\Gamma + C_\mathrm{N}\pi(t - t') \le C_\mathrm{F}\pi(t) + \delta + C_\mathrm{N}\big((1 + C_\mathrm{F})\pi(t) + \delta\big)$. Let $\delta \to 0$. □

Lemma P.4 (bases of representations)**.** *Let P_0 be a finite-dimensional subspace of P and $t \in \Gamma^p(P) \cap P_0^{\otimes p}$, and for linear forms $\phi_1, \dots, \phi_p$ on P_0 let $t(\phi_1, \dots, \phi_p)$ be the value of $\phi_1 \otimes \cdots \otimes \phi_p$ at t, as before Lemma P.1.*

(1) *Let $b_1, \dots, b_m$ be a basis of P_0, with dual basis $b_1^*, \dots, b_m^*$. Then $t - \sum_{i=1}^m t(b_i^*, \dots, b_i^*)\, b_i^{\otimes p}$ is hollow.*

(2) *Let $t = \sum_{k=1}^r x_{k,1} \otimes \cdots \otimes x_{k,p}$ with nonzero vectors $x_{k,c}$ that span P_0. There is a set $\mathcal{B} \subseteq [r] \times [p]$ such that the vectors $x_{k,c}$, $(k,c) \in \mathcal{B}$, form a basis of P_0 whose dual basis $(x_{k,c}^*)_{(k,c)\in\mathcal{B}}$ satisfies $|t(x_{k,c}^*, \dots, x_{k,c}^*)|\, \|x_{k,c}\|^p \le \|x_{k,1}\| \cdots \|x_{k,p}\|$ for every $(k,c) \in \mathcal{B}$.*

Proof. Step 1 (item (1)). Let q be a linear projection of P onto P_0. As $q^{\otimes p}$ fixes $P_0^{\otimes p}$ and maps powers to powers, $t = q^{\otimes p}(t)$ is a combination of powers $h^{\otimes p}$, $h \in P_0$. For a word $w \in [m]^p$ put $b_w = b_{w_1} \otimes \cdots \otimes b_{w_p}$; these tensors form a basis of $P_0^{\otimes p}$, and $t = \sum_w t(b_{w_1}^*, \dots, b_{w_p}^*)\, b_w$. For a power $h^{\otimes p}$, $h \in P_0$, the coefficient $b_{w_1}^*(h) \cdots b_{w_p}^*(h)$ of b_w depends only on the content $\alpha \in \mathbb{N}_p^m$ of w; hence so does that of t, and t is a combination of the tensors $O_\alpha(b_1, \dots, b_m)$. If every $\alpha_i < p$, then $\alpha_1! \cdots \alpha_m!$ is invertible in K, and for a word w of content α the tensor $O_{(1,\dots,1)}(b_{w_1}, \dots, b_{w_p})$, hollow as in Step 2 of the proof of Lemma P.3, is $\alpha_1! \cdots \alpha_m!\, O_\alpha(b_1, \dots, b_m)$, as the reorderings of the letters of w give each word of content α exactly $\alpha_1! \cdots \alpha_m!$ times; so $O_\alpha(b_1, \dots, b_m)$ is hollow. Otherwise $\alpha_i = p$ for one i, the tensor $O_\alpha(b_1, \dots, b_m)$ is $b_i^{\otimes p}$, and its coefficient is $t(b_i^*, \dots, b_i^*)$.

Step 2 (item (2): the choice of $\mathcal{B}$). Call $\mathcal{B} \subseteq [r] \times [p]$ a *basis set* if the vectors $x_{k,c}$, $(k,c) \in \mathcal{B}$, form a basis of P_0; basis sets exist, as these vectors span P_0. Fix a basis of P_0, and for a basis set $\mathcal{B}$ let $\det_\mathcal{B}$ be the determinant of the coordinates of the vectors $x_{k,c}$, $(k,c) \in \mathcal{B}$, in a fixed order of $[r] \times [p]$. Put $\upsilon_{k,c} = (\|x_{k,1}\| \cdots \|x_{k,p}\|)^{1/p} / \|x_{k,c}\| > 0$, so that $\upsilon_{k,1} \cdots \upsilon_{k,p} = 1$ for each k, and choose, among the finitely many basis sets, a basis set $\mathcal{B}$ with the largest value of $|\det_\mathcal{B}| \prod_{(k,c)\in\mathcal{B}} \upsilon_{k,c}$. Then

$$|x_{k,c}^*(x_{k',c'})|\, \upsilon_{k',c'} \le \upsilon_{k,c} \qquad \big((k,c) \in \mathcal{B},\ (k',c') \in [r] \times [p]\big). \tag{P.11}$$

This is clear if $(k',c') \in \mathcal{B}$ or $x_{k,c}^*(x_{k',c'}) = 0$. Otherwise $x_{k',c'} = \sum_{(k'',c'')\in\mathcal{B}} x_{k'',c''}^*(x_{k',c'})\, x_{k'',c''}$, whose coefficient at (k,c) is $x_{k,c}^*(x_{k',c'}) \ne 0$, so replacing (k,c) by (k',c') gives a basis set $\mathcal{B}'$, and $\det_{\mathcal{B}'} = \pm x_{k,c}^*(x_{k',c'}) \det_\mathcal{B}$ by multilinearity and alternation of the determinant, the sign coming from the order of $[r] \times [p]$. The choice of $\mathcal{B}$ gives $|x_{k,c}^*(x_{k',c'})|\, \upsilon_{k',c'} / \upsilon_{k,c} \le 1$.

Step 3 (item (2): the bound). Let $(k,c) \in \mathcal{B}$. Then $t(x_{k,c}^*, \dots, x_{k,c}^*) = \sum_{k'=1}^r \prod_{c'=1}^p x_{k,c}^*(x_{k',c'})$, so by the ultrametric inequality and (P.11)

$$|t(x_{k,c}^*, \dots, x_{k,c}^*)| \le \max_{k'} \prod_{c'=1}^p \frac{\upsilon_{k,c}}{\upsilon_{k',c'}} = \upsilon_{k,c}^p = \frac{\|x_{k,1}\| \cdots \|x_{k,p}\|}{\|x_{k,c}\|^p}.$$

□

Proof of Theorem 10.15. Step 1 (item (1)). Let $t \in \Gamma^p(P)$ and $\delta > 0$; we may assume $t \ne 0$. Choose a representation $t = \sum_{k=1}^r x_{k,1} \otimes \cdots \otimes x_{k,p}$ with $\sum_k \|x_{k,1}\| \cdots \|x_{k,p}\| \le \pi(t) + \delta$; dropping the terms with a zero factor, all $x_{k,c}$ are nonzero. Let P_0 be their span, so $\dim P_0 \le \dim P$ and $t \in P_0^{\otimes p}$, and take $\mathcal{B}$ from Lemma P.4(2). By Lemma P.4(1), $t - t'$ is hollow for $t' =$

$\sum_{(k,c)\in\mathcal{B}} t(x^*_{k,c},\dots,x^*_{k,c})\, x_{k,c}^{\otimes p}$, so

$$\|t\|_{\mathrm{F}} \le \|t'\|_\Gamma \le \sum_{(k,c)\in\mathcal{B}} |t(x^*_{k,c},\dots,x^*_{k,c})|\,\|x_{k,c}\|^p$$
$$\le \sum_{k=1}^{r} |\mathcal{B}\cap(\{k\}\times[p])|\,\|x_{k,1}\|\cdots\|x_{k,p}\| \le \min(p,\dim P_0)\,(\pi(t)+\delta),$$

since $\mathcal{B}$ has $\dim P_0$ elements. Let $\delta\to 0$.

Step 2 (item (2)). By Lemma P.3(1) and item (1), $\|L_a(t)\| \le \|a\|\|t\|_{\mathrm{F}} \le p\|a\|\,\pi(t)$ for $t\in\Gamma^p(P)$; apply Lemma P.3(2) with $M = p\|a\|$.

Step 3 (item (3): the bound). By item (1) and Lemma P.3(3) with $C_{\mathrm{F}} = p$ and $C_{\mathrm{N}} = C$, $\|t\|_\Gamma \le (p+(p+1)C)\pi(t)$ for $t\in\Gamma^p(P)$. So $\|L_a(t)\| \le (p+(p+1)C)\|a\|\,\pi(t)$ by Lemma P.3(1); apply Lemma P.3(2) with $M = (p+(p+1)C)\|a\|$.

Step 4 (item (3): the converse). Let W_0 be the quotient of $\Gamma^p(P)$ by the kernel of $\|\cdot\|_\Gamma$, normed by $\|\cdot\|_\Gamma$, and $a_\Gamma(h)\in W_0$ the class of $h^{\otimes p}$. By (P.10) and (10.4), a_Γ is a bounded polynomial law of degree p with $\|a_\Gamma\|\le 1$; by hypothesis $ja_\Gamma(h) = \beta(h,\dots,h)$ for a linear isometry j of W_0 into a normed space Z and a bounded p-linear $\beta\colon P^p\to Z$. Let $\beta_\otimes\colon P^{\otimes p}\to Z$ be the linear map with $\beta_\otimes(h_1\otimes\cdots\otimes h_p) = \beta(h_1,\dots,h_p)$; then $\|\beta_\otimes(y)\|\le\|\beta\|\pi(y)$ for $y\in P^{\otimes p}$. As $\beta_\otimes(h^{\otimes p}) = ja_\Gamma(h)$ and the powers span $\Gamma^p(P)$, $\beta_\otimes$ agrees on $\Gamma^p(P)$ with j composed with the quotient map, so $\|t\|_\Gamma\le\|\beta\|\pi(t)$ for $t\in\Gamma^p(P)$, in particular for hollow t. □

For the proof of Proposition 10.16 let $n\ge 1$, with $\Gamma^n(P)$, $\|\cdot\|_\Gamma$, π, $\mathfrak{a}_n(P)$, $[m]$, $c_0([m],K)$ and $\mathfrak{c}_K$ as in Section 10.4. The statements of the paragraph before Lemma P.3, the first assertion of Lemma P.3(1) and Lemma P.3(2) hold with n in place of p, with the same proofs, which do not use that p is the characteristic of K. In particular (P.10) holds with n in place of p, with $O_\alpha(h_1,\dots,h_r)\in\Gamma^n(P)$ for $\alpha\in\mathbb{N}_n^r$. As in Step 4 of the proof of Theorem 10.15, let W_0 be the quotient of $\Gamma^n(P)$ by the kernel of $\|\cdot\|_\Gamma$, normed by $\|\cdot\|_\Gamma$, and let $a_\Gamma(h)\in W_0$ be the class of $h^{\otimes n}$. As after (10.4), $\pi\le\|\cdot\|_\Gamma$ on $\Gamma^n(P)$. When two spaces or two fields occur, we say in which space and over which field π, $\|\cdot\|_\Gamma$ and $\mathfrak{a}_n$ are computed. The completion $\widehat{P}$ of P is a normed space over $\widehat{K}$, the scalar multiplication extending by continuity, and it is nonarchimedean if P is.

Lemma P.5 (properties of the ambience constants)**.** *Let P' be a normed space.*

1. *a_Γ is a bounded polynomial law of degree n with $\|a_\Gamma\|\le 1$. If $ja_\Gamma(h) = \beta(h,\dots,h)$ for all $h\in P$, for a linear isometry j of W_0 into a normed space and a bounded n-linear map β on P^n, then $\mathfrak{a}_n(P)\le\|\beta\|$.*
2. *If Ξ is a bounded linear bijection of P onto P' with bounded inverse, then $\mathfrak{a}_n(P')\le(\|\Xi\|\|\Xi^{-1}\|)^n\mathfrak{a}_n(P)$.*
3. *If P' is a subspace of P, with the restricted norm, and some linear projection of P onto P' has norm at most 1, then $\mathfrak{a}_n(P')\le\mathfrak{a}_n(P)$.*
4. *$\mathfrak{a}_n(P)$ is at most the supremum of $\mathfrak{a}_n(P_0)$ over the finite-dimensional subspaces P_0 of P, with the restricted norm.*

Proof. Step 1 (item (1)). This is Step 4 of the proof of Theorem 10.15, with n in place of p; it does not use that p is the characteristic of K.

Step 2 (item (2)). We may assume that $\mathfrak{a}_n(P)$ is finite. Let $t'\in\Gamma^n(P')$ and $t = (\Xi^{-1})^{\otimes n}(t')$. The maps $\Xi^{\otimes n}$ and $(\Xi^{-1})^{\otimes n}$ carry powers to powers and elementary tensors to elementary tensors, so $t\in\Gamma^n(P)$, and they multiply the cost of a representation by at most $\|\Xi\|^n$ and $\|\Xi^{-1}\|^n$, respectively. Hence $\|t'\|_\Gamma\le\|\Xi\|^n\|t\|_\Gamma\le\|\Xi\|^n\mathfrak{a}_n(P)\pi(t)\le(\|\Xi\|\|\Xi^{-1}\|)^n\mathfrak{a}_n(P)\pi(t')$.

Step 3 (item (3)). We may assume that $\mathfrak{a}_n(P)$ is finite. Let Π be such a projection and $t \in \Gamma^n(P')$. Then $\Pi^{\otimes n}$ fixes t and carries a representation of t in $\Gamma^n(P)$ to one in $\Gamma^n(P')$ of no larger cost, and representations of t in $P'^{\otimes n}$ are representations in $P^{\otimes n}$. So $\|t\|_\Gamma$ computed in $\Gamma^n(P')$ is at most $\|t\|_\Gamma$ computed in $\Gamma^n(P)$, which is at most $\mathfrak{a}_n(P)\pi(t)$ with $\pi(t)$ computed in $P^{\otimes n}$, hence at most $\mathfrak{a}_n(P)\pi(t)$ with $\pi(t)$ computed in $P'^{\otimes n}$.

Step 4 (item (4)). Let $t \in \Gamma^n(P)$ and $\delta > 0$, and let $t = \sum_k x_{k,1} \otimes \cdots \otimes x_{k,n}$ with $\sum_k \|x_{k,1}\| \cdots \|x_{k,n}\| \le \pi(t) + \delta$. Let P_0 be the span of the $x_{k,c}$. As in Step 1 of the proof of Lemma P.4, $t = q^{\otimes n}(t) \in \Gamma^n(P_0)$ for a linear projection q of P onto P_0. Then $\|t\|_\Gamma$ computed in $\Gamma^n(P)$ is at most $\|t\|_\Gamma$ computed in $\Gamma^n(P_0)$, hence at most $\mathfrak{a}_n(P_0)(\pi(t) + \delta)$. Let $\delta \to 0$. □

Lemma P.6 (orthogonal bases)**.** *Let K be complete, and let P be nonarchimedean, of finite dimension $m \ge 1$. Then $\mathfrak{a}_n(P) \le \mathfrak{c}_K^{-n}\mathfrak{a}_n(c_0([m], K))$.*

Proof. Step 1 (distances). Let P_1 be a subspace of P, $x \in P \setminus P_1$ and $\operatorname{dist}(x, P_1) = \inf_{y \in P_1} \|x - y\|$. By Lemma A.3, P_1 is complete, hence closed in P, so $\operatorname{dist}(x, P_1) > 0$. Let K be discretely valued. If nonzero vectors $u_1, \dots, u_k \in P$ have norms in distinct cosets of the subgroup $|\vartheta|^{\mathbb{Z}}$ of the multiplicative group $(0, \infty)$, then the nonzero numbers among $|\lambda_i|\|u_i\|$, for $\lambda_1, \dots, \lambda_k \in K$, are distinct, so the nonarchimedean norm gives $\|\lambda_1 u_1 + \cdots + \lambda_k u_k\| = \max_i |\lambda_i|\|u_i\|$; hence the u_i are linearly independent, and $k \le m$. So the nonzero values of the norm of P lie in at most m cosets of $|\vartheta|^{\mathbb{Z}}$, and only finitely many of them lie in the interval $[\operatorname{dist}(x, P_1), \|x\|]$. Hence $\|x - y\| = \operatorname{dist}(x, P_1)$ for some $y \in P_1$.

Step 2 (orthogonal bases). Let $0 < \mathfrak{o} \le 1$, with $\mathfrak{o} < 1$ unless K is discretely valued. We show by induction on m that P has a basis $b_1, \dots, b_m$ with

$$\|\lambda_1 b_1 + \cdots + \lambda_m b_m\| \ge \mathfrak{o} \max_i |\lambda_i|\|b_i\| \qquad (\lambda_1, \dots, \lambda_m \in K). \tag{P.12}$$

For $m = 1$ every nonzero vector will do. Let $m \ge 2$, let P_1 be a hyperplane of P, and let $b_1, \dots, b_{m-1}$ be a basis of P_1 with (P.12) for $\mathfrak{o}^{1/2}$ in place of $\mathfrak{o}$. Let $x \in P \setminus P_1$. By Step 1 there is $y \in P_1$ with $\|x - y\| \le \mathfrak{o}^{-1/2} \operatorname{dist}(x, P_1)$: if $\mathfrak{o} < 1$ since $\operatorname{dist}(x, P_1) > 0$, and if $\mathfrak{o} = 1$ by its last sentence. Put $b_m = x - y$, so that $\operatorname{dist}(b_m, P_1) = \operatorname{dist}(x, P_1) \ge \mathfrak{o}^{1/2}\|b_m\|$. Let $v = v' + \lambda_m b_m$ with $v' = \sum_{i<m} \lambda_i b_i$. As $-v' \in P_1$, $\|v\| \ge |\lambda_m| \operatorname{dist}(b_m, P_1) \ge \mathfrak{o}^{1/2}|\lambda_m|\|b_m\|$, and so $\|v'\| \le \max(\|v\|, |\lambda_m|\|b_m\|) \le \mathfrak{o}^{-1/2}\|v\|$. Hence $\mathfrak{o} \max_{i<m} |\lambda_i|\|b_i\| \le \mathfrak{o}^{1/2}\|v'\| \le \|v\|$ and $\mathfrak{o}|\lambda_m|\|b_m\| \le \|v\|$, which is (P.12).

Step 3 (the bound). Multiplying the b_i by nonzero scalars keeps (P.12). If K is discretely valued, take $\mathfrak{o} = 1$ in Step 2 and scale the b_i by powers of ϑ so that $|\vartheta| < \|b_i\| \le 1$, and put $\mathfrak{o}' = |\vartheta|$. Otherwise $|K^\times|$ is a subgroup of $(0, \infty)$ that contains $|\varpi|$ and is not of the form $|\vartheta|^{\mathbb{Z}}$, hence dense, since a subgroup of $(0, \infty)$ that contains a number other than 1 and is not dense is generated by its largest element below 1; for $0 < \mathfrak{o} < 1$ take the b_i of Step 2 and scale them so that $\mathfrak{o} < \|b_i\| \le 1$, and put $\mathfrak{o}' = \mathfrak{o}$. The linear bijection $\Xi(\lambda) = \sum_i \lambda_i b_i$, $\lambda = (\lambda_1, \dots, \lambda_m)$, of $c_0([m], K)$ onto P satisfies $\|\Xi(\lambda)\| \le \max_i |\lambda_i|\|b_i\| \le \|\lambda\|$, as P is nonarchimedean, and $\|\Xi(\lambda)\| \ge \mathfrak{o} \max_i |\lambda_i|\|b_i\| \ge \mathfrak{o}\mathfrak{o}'\|\lambda\|$, by (P.12). Lemma P.5(2) gives $\mathfrak{a}_n(P) \le (\mathfrak{o}\mathfrak{o}')^{-n}\mathfrak{a}_n(c_0([m], K))$. This is the claim if K is discretely valued; otherwise let $\mathfrak{o} \to 1$. □

Lemma P.7 (completion)**.**

(1) *$\mathfrak{a}_n(P)$ is at most $\mathfrak{a}_n(\widehat{P})$ computed over $\widehat{K}$.*
(2) *For $m \ge 1$, $\mathfrak{a}_n(c_0([m], \widehat{K}))$ computed over $\widehat{K}$ is at most $\mathfrak{a}_n(c_0([m], K))$.*

Proof. Step 1 (continuity). Let $a\colon P \to W$ be a bounded polynomial law of degree n, and let $R > 0$. For $x, y \in P$ with $\|x\|, \|y\| \le R$, $a(x + y) - a(x) = \sum_{i=1}^n a^{[(n-i,i)]}(x, y)$, and (P.2) gives

$$\|a(x + y) - a(x)\| \le n(16/|\varpi|^2)^n \|a\| \, R^{n-1}\|y\|. \tag{P.13}$$

If $\|x\| \le R/2$ and $\|x+y\| \le R/2$, then $\|y\| \le R$; so, by (P.13), a is Lipschitz on the set of vectors of P of norm at most $R/2$, for every $R > 0$. Hence a is uniformly continuous on every bounded subset of P, and it extends uniquely to a continuous map $\hat{a}\colon \widehat{P} \to \widehat{W}$, which satisfies $\|\hat{a}(x)\| \le \|a\|\|x\|^n$ by continuity.

Step 2 (laws extend). Let a and $\hat{a}$ be as in Step 1, $r \ge 1$ and $\alpha \in \mathbb{N}_n^r$. By interpolation in each variable on $n+1$ distinct elements of K, as in Step 5 of the proof of Theorem 10.11, there are finitely many scalars λ_l and $\xi_{l,1}, \dots, \xi_{l,r}$ in K, depending only on r and α, with $a^{[\alpha]}(h_1, \dots, h_r) = \sum_l \lambda_l a(\xi_{l,1}h_1 + \cdots + \xi_{l,r}h_r)$ for all $h_1, \dots, h_r \in P$. Define $\hat{a}^{[\alpha]}(h_1, \dots, h_r) = \sum_l \lambda_l \hat{a}(\xi_{l,1}h_1 + \cdots + \xi_{l,r}h_r)$ for $h_1, \dots, h_r \in \widehat{P}$. The identity

$$\hat{a}(s_1h_1 + \cdots + s_rh_r) = \sum_{\alpha \in \mathbb{N}_n^r} s_1^{\alpha_1} \cdots s_r^{\alpha_r}\, \hat{a}^{[\alpha]}(h_1, \dots, h_r)$$

holds for $s \in K^r$ and $h_1, \dots, h_r \in P$, by Definition 10.10(1) and the choice of the λ_l and $\xi_{l,i}$; both sides are continuous in $(s, h_1, \dots, h_r) \in \widehat{K}^r \times \widehat{P}^r$, so it holds there. Hence $\hat{a}\colon \widehat{P} \to \widehat{W}$ is a bounded polynomial law of degree n over $\widehat{K}$, with $\|\hat{a}\| \le \|a\|$.

Step 3 (item (1)). Let $C = \mathfrak{a}_n(\widehat{P})$, computed over $\widehat{K}$, be finite. By Step 2, a_Γ extends to a bounded polynomial law $\hat{a}_\Gamma\colon \widehat{P} \to \widehat{W_0}$ of degree n over $\widehat{K}$, of norm at most 1. As $\|\hat{a}_\Gamma\| \le 1$ and $\|t\|_\Gamma \le C\pi(t)$ on $\Gamma^n(\widehat{P})$, the first assertion of Lemma P.3(1) and Lemma P.3(2), with n in place of p, over $\widehat{K}$ and with $M = C$, give a $\widehat{K}$-linear isometry $\hat{\jmath}$ of $\widehat{W_0}$ into a normed space Z over $\widehat{K}$ and a bounded n-linear map $\hat{\beta}$ on $\widehat{P}^n$ over $\widehat{K}$, of norm at most C, with $\hat{\jmath}\hat{a}_\Gamma(h) = \hat{\beta}(h, \dots, h)$ for $h \in \widehat{P}$. The restrictions of $\hat{\jmath}$ to W_0 and of $\hat{\beta}$ to P^n are a linear isometry of W_0 into Z, read as a normed space over K, and a bounded n-linear map of norm at most C, and $\hat{\jmath}(a_\Gamma(h)) = \hat{\beta}(h, \dots, h)$ for $h \in P$. Lemma P.5(1) gives $\mathfrak{a}_n(P) \le C$.

Step 4 (projective norms in $c_0([m], K)$). Read $K^m \subseteq \widehat{K}^m$, so that the tensors $e_{w_1} \otimes \cdots \otimes e_{w_n}$, $w \in [m]^n$, are a basis of $(K^m)^{\otimes n}$ over K and of $(\widehat{K}^m)^{\otimes n}$ over $\widehat{K}$; call the coefficients of a tensor in this basis its coefficients. For $z \in (K^m)^{\otimes n}$ we show that $\pi(z)$ is the same over K and over $\widehat{K}$, the spaces being $c_0([m], K)$ and $c_0([m], \widehat{K})$. Over $\widehat{K}$ it is at most its value over K. Over K, $\pi(z)$ is at most the sum of the absolute values of the coefficients of z. Let $z = \sum_k x_{k,1} \otimes \cdots \otimes x_{k,n}$ with $x_{k,c} \in \widehat{K}^m$, let $\mathfrak{r} = \sum_k \|x_{k,1}\| \cdots \|x_{k,n}\|$, and let $0 < \delta \le 1$. Choose $x'_{k,c} \in K^m$ with $\|x'_{k,c} - x_{k,c}\| \le \delta\|x_{k,c}\|$; then $\|x'_{k,c}\| \le \|x_{k,c}\|$. Every coefficient of

$$x_{k,1} \otimes \cdots \otimes x_{k,n} - x'_{k,1} \otimes \cdots \otimes x'_{k,n} = \sum_{c=1}^{n} x'_{k,1} \otimes \cdots \otimes x'_{k,c-1} \otimes (x_{k,c} - x'_{k,c}) \otimes x_{k,c+1} \otimes \cdots \otimes x_{k,n}$$

has absolute value at most $\delta\|x_{k,1}\| \cdots \|x_{k,n}\|$. So $z' = \sum_k x'_{k,1} \otimes \cdots \otimes x'_{k,n}$ lies in $(K^m)^{\otimes n}$, $\pi(z') \le \mathfrak{r}$ over K, and $z - z'$ has coefficients of absolute value at most $\delta\mathfrak{r}$. Over K this gives $\pi(z) \le \mathfrak{r} + m^n\delta\mathfrak{r}$. Let $\delta \to 0$ and take the infimum over the representations of z over $\widehat{K}$.

Step 5 (item (2)). Let $C = \mathfrak{a}_n(c_0([m], K))$ be finite, let $t \in \Gamma^n(c_0([m], \widehat{K}))$, computed over $\widehat{K}$, and let $\delta > 0$. Write $O_\alpha = O_\alpha(e_1, \dots, e_m)$ for $\alpha \in \mathbb{N}_n^m$. By the paragraph before Lemma P.2, with $\Bbbk = \widehat{K}$ and $\mathcal{J} = [m]$, $t = \sum_\alpha t_\alpha O_\alpha$ with $t_\alpha \in \widehat{K}$, as t is symmetric. As noted before Lemma P.5, the O_α lie in $\Gamma^n(c_0([m], K))$, computed over K; let M_0 be the sum of their seminorms $\|O_\alpha\|_\Gamma$ there. Choose $t'_\alpha \in K$ with $|t_\alpha - t'_\alpha| \le \delta$ and put $t' = \sum_\alpha t'_\alpha O_\alpha$. Representations over K are representations over $\widehat{K}$, so, over $\widehat{K}$, $\|t - t'\|_\Gamma \le \delta M_0$, hence $\pi(t - t') \le \delta M_0$ and $\pi(t') \le \pi(t) + \delta M_0$. With Step 4,

$$\|t\|_\Gamma \le \|t'\|_\Gamma + \delta M_0 \le C\pi(t') + \delta M_0 \le C(\pi(t) + \delta M_0) + \delta M_0,$$

where $\|t\|_\Gamma$ and $\pi(t)$ are computed over $\widehat{K}$, $\|t'\|_\Gamma$ over K, which is at least its value over $\widehat{K}$, and $\pi(t')$ over either field. Let $\delta \to 0$. □

Lemma P.8 (degree two)**.** *For every $m \geq 1$, $\mathfrak{a}_2(c_0([m], K)) \leq 3/2$.*

Proof. Identify $(K^m)^{\otimes 2}$ with $K^{m\times m}$ by $x \otimes x' \mapsto xx'^{\mathsf{T}}$, as in Step 5 of the proof of Proposition 10.13. Then the elements of $\Gamma^2(c_0([m], K))$ are symmetric matrices, $\|\mathsf{S}\|_\Gamma$ is the infimum of $\sum_l |c_l| \|u_l\|^2$ over the representations $\mathsf{S} = \sum_l c_l u_l u_l^{\mathsf{T}}$, and $\pi(\mathsf{S})$ is the infimum of $\sum_l \|x_l\| \|x'_l\|$ over the representations $\mathsf{S} = \sum_l x_l x_l'^{\mathsf{T}}$. A matrix is *integral* if its entries have absolute value at most 1. For a matrix X and $k \geq 1$ let $\mathfrak{d}_k(\mathsf{X})$ be the largest absolute value of a $k \times k$ minor of X, or 0 if there is none, and $\mathfrak{d}_0(\mathsf{X}) = 1$. A determinant is a sum of products with one entry from each column, so by the ultrametric inequality the minors of an integral matrix have absolute value at most 1, and by the Cauchy–Binet formula $\mathfrak{d}_k(\mathsf{X})$ does not increase when X is multiplied on either side by an integral matrix.

Step 1 (a pivot). Let $\mathsf{S} \in K^{m\times m}$ be symmetric and nonzero, with entries S_{ij}, and let $\mathfrak{f}$ be the largest absolute value of an entry. If $|\mathsf{S}_{ii}| = \mathfrak{f}$ for some i, put $\mathcal{V} = \{i\}$; otherwise put $\mathcal{V} = \{i, j\}$ for some $i \neq j$ with $|\mathsf{S}_{ij}| = \mathfrak{f}$. Let $\mathsf{S}_\mathcal{V}$ be the submatrix of S with rows and columns in $\mathcal{V}$, and $\mathsf{S}_{[m]\mathcal{V}}$ the one with columns in $\mathcal{V}$. Then $|\det \mathsf{S}_\mathcal{V}| = \mathfrak{f}^{|\mathcal{V}|}$; in the second case $\det \mathsf{S}_\mathcal{V} = \mathsf{S}_{ii}\mathsf{S}_{jj} - \mathsf{S}_{ij}^2$ with $|\mathsf{S}_{ii}\mathsf{S}_{jj}| < \mathfrak{f}^2$. So $\mathsf{S}_\mathcal{V}^{-1}$, its adjugate divided by its determinant, has entries of absolute value at most $\mathfrak{f}^{-1}$, and $\mathsf{G} = \mathsf{S}_{[m]\mathcal{V}}\mathsf{S}_\mathcal{V}^{-1}$ is integral, with the identity matrix as its rows in $\mathcal{V}$. Put $\mathsf{S}' = \mathsf{S} - \mathsf{G}\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}}$. It is symmetric with entries of absolute value at most $\mathfrak{f}$, and its rows and columns in $\mathcal{V}$ vanish, as the rows in $\mathcal{V}$ of $\mathsf{G}\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}}$ form $\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}} = \mathsf{S}_{[m]\mathcal{V}}^{\mathsf{T}}$. If $\mathcal{V} = \{i\}$, then $\mathsf{G}\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}} = \mathsf{S}_{ii}uu^{\mathsf{T}}$ for the column u of G. If $\mathcal{V} = \{i, j\}$ and u_1, u_2 are the columns of G at i and j, then

$$\mathsf{G}\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}} = (\mathsf{S}_{ii} - \mathsf{S}_{ij})u_1u_1^{\mathsf{T}} + (\mathsf{S}_{jj} - \mathsf{S}_{ij})u_2u_2^{\mathsf{T}} + \mathsf{S}_{ij}(u_1 + u_2)(u_1 + u_2)^{\mathsf{T}},$$

three terms with coefficients of absolute value at most $\mathfrak{f}$ and vectors of norm at most 1. In both cases

$$\|\mathsf{G}\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}}\|_\Gamma \leq \tfrac{3}{2}|\mathcal{V}|\mathfrak{f}. \tag{P.14}$$

Let $\mathsf{U} \in K^{m\times m}$ have the columns of G at the indices in $\mathcal{V}$ and the columns of the identity matrix id elsewhere, and let $\mathsf{E} \in K^{m\times m}$ agree with S at the positions in $\mathcal{V} \times \mathcal{V}$ and vanish elsewhere. The nonzero columns of $\mathsf{U} - \mathrm{id}$ lie at indices in $\mathcal{V}$, and its rows in $\mathcal{V}$ vanish; so $(\mathsf{U} - \mathrm{id})^2 = 0$, $\mathsf{U}^{-1} = 2\mathrm{id} - \mathsf{U}$ is integral, and $\mathsf{U}\mathsf{S}' = \mathsf{S}' = \mathsf{S}'\mathsf{U}^{\mathsf{T}}$, since the rows and columns of S' in $\mathcal{V}$ vanish. As $\mathsf{U}\mathsf{E}\mathsf{U}^{\mathsf{T}} = \mathsf{G}\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}}$, this gives $\mathsf{S} = \mathsf{U}(\mathsf{E} + \mathsf{S}')\mathsf{U}^{\mathsf{T}}$, so $\mathfrak{d}_k(\mathsf{S}) \geq \mathfrak{d}_k(\mathsf{E} + \mathsf{S}')$. For $0 \leq k_1 \leq \min(k, |\mathcal{V}|)$, a $k_1 \times k_1$ minor of $\mathsf{S}_\mathcal{V}$ and a $(k - k_1) \times (k - k_1)$ minor of S' with rows and columns outside $\mathcal{V}$ together form a $k \times k$ minor of $\mathsf{E} + \mathsf{S}'$ whose matrix is, after reordering its rows and columns, block diagonal. As the nonzero minors of S' have rows and columns outside $\mathcal{V}$, $\mathfrak{d}_k(\mathsf{S}) \geq \mathfrak{d}_{k_1}(\mathsf{S}_\mathcal{V})\,\mathfrak{d}_{k-k_1}(\mathsf{S}')$; here $\mathfrak{d}_1(\mathsf{S}_\mathcal{V}) = \mathfrak{f}$, and $\mathfrak{d}_2(\mathsf{S}_\mathcal{V}) = \mathfrak{f}^2$ if $|\mathcal{V}| = 2$.

Step 2 (the pivot sequence). Apply Step 1 to S, then to S' if it is nonzero, and so on. A row that vanishes stays zero, as the row of G at its index, and so that of $\mathsf{G}\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}}$, vanishes; and the rows in $\mathcal{V}$, which contain an entry of absolute value $\mathfrak{f}$, vanish after the step. So after at most m steps the matrix is zero, and S is the sum of the matrices $\mathsf{G}\mathsf{S}_\mathcal{V}\mathsf{G}^{\mathsf{T}}$ of the steps. List the numbers $\mathfrak{f}$ of the steps, each once if $|\mathcal{V}| = 1$ and twice if $|\mathcal{V}| = 2$, as $\mathfrak{f}_1, \dots, \mathfrak{f}_r$. As the entries of S' have absolute value at most $\mathfrak{f}$, $\mathfrak{f}_1 \geq \cdots \geq \mathfrak{f}_r > 0$. By (P.14), $\|\mathsf{S}\|_\Gamma \leq \frac{3}{2}(\mathfrak{f}_1 + \cdots + \mathfrak{f}_r)$. By the last sentence of Step 1 and induction on the number of steps, applied to S', whose sequence is $\mathfrak{f}_{|\mathcal{V}|+1}, \dots, \mathfrak{f}_r$,

$$\mathfrak{d}_k(\mathsf{S}) \geq \mathfrak{f}_1 \cdots \mathfrak{f}_k \qquad (1 \leq k \leq r)\,: \tag{P.15}$$

for $k \leq |\mathcal{V}|$ take $k_1 = k$, and for $k > |\mathcal{V}|$ take $k_1 = |\mathcal{V}|$.

Step 3 (products and sums). Let $\mathfrak{s}_1, \dots, \mathfrak{s}_r > 0$ with $\mathfrak{f}_1 \cdots \mathfrak{f}_k \le \mathfrak{s}_1 \cdots \mathfrak{s}_k$ for $1 \le k \le r$. Then $\mathfrak{f}_1 + \cdots + \mathfrak{f}_r \le \mathfrak{s}_1 + \cdots + \mathfrak{s}_r$. Indeed, $\log(\mathfrak{s}_k/\mathfrak{f}_k) \le \mathfrak{s}_k/\mathfrak{f}_k - 1$, so $\mathfrak{s}_k - \mathfrak{f}_k \ge \mathfrak{f}_k \log(\mathfrak{s}_k/\mathfrak{f}_k)$. Put $\mathfrak{l}_k = \sum_{i \le k} \log(\mathfrak{s}_i/\mathfrak{f}_i) \ge 0$, $\mathfrak{l}_0 = 0$ and $\mathfrak{f}_{r+1} = 0$; summation by parts gives

$$\sum_{k=1}^{r} \mathfrak{f}_k \log(\mathfrak{s}_k/\mathfrak{f}_k) = \sum_{k=1}^{r} \mathfrak{f}_k(\mathfrak{l}_k - \mathfrak{l}_{k-1}) = \sum_{k=1}^{r} (\mathfrak{f}_k - \mathfrak{f}_{k+1})\mathfrak{l}_k \ge 0.$$

Step 4 (the projective norm). Let $\mathsf{S} = \sum_{l=1}^{r'} x_l x_l'^{\mathsf{T}}$, and let $\mathfrak{s}_1 \ge \mathfrak{s}_2 \ge \cdots \ge \mathfrak{s}_{r'}$ be the numbers $\|x_l\|\|x_l'\|$ in nonincreasing order. By the Cauchy–Binet formula, a $k \times k$ minor of S is the sum, over the sets of k indices l, of the products of the corresponding $k \times k$ minors of the matrices with the columns x_l and x_l', l in the set; by the ultrametric inequality such a minor of the first matrix has absolute value at most the product of the $\|x_l\|$, and likewise for the second. So $\mathfrak{d}_k(\mathsf{S}) \le \mathfrak{s}_1 \cdots \mathfrak{s}_k$ if $k \le r'$, and $\mathfrak{d}_k(\mathsf{S}) = 0$ if $k > r'$. By (P.15), $r' \ge r$ and $\mathfrak{f}_1 \cdots \mathfrak{f}_k \le \mathfrak{s}_1 \cdots \mathfrak{s}_k$ for $k \le r$, so $\mathfrak{s}_1, \dots, \mathfrak{s}_r > 0$, and Step 3 gives $\mathfrak{f}_1 + \cdots + \mathfrak{f}_r \le \mathfrak{s}_1 + \cdots + \mathfrak{s}_r \le \sum_l \|x_l\|\|x_l'\|$. Taking the infimum over the representations, $\mathfrak{f}_1 + \cdots + \mathfrak{f}_r \le \pi(\mathsf{S})$, and Step 2 gives $\|\mathsf{S}\|_\Gamma \le \frac{3}{2}\pi(\mathsf{S})$, also for $\mathsf{S} = 0$. □

Proof of Proposition 10.16. Step 1 (item (1)). Let $\mathfrak{a}_n(P) < \infty$ and $a\colon P \to W$ be a bounded polynomial law of degree n. By the first assertion of Lemma P.3(1), with n in place of p, $\|L_a(t)\| \le \|a\|\|t\|_\Gamma \le \mathfrak{a}_n(P)\|a\|\,\pi(t)$ for $t \in \Gamma^n(P)$; apply Lemma P.3(2), with n in place of p and $M = \mathfrak{a}_n(P)\|a\|$. Conversely, if every bounded polynomial law of degree n on P is ambient of degree n, so is a_Γ, and Lemma P.5(1) gives $\mathfrak{a}_n(P) < \infty$.

Step 2 (item (2)). We may assume that $C = \mathfrak{a}_{n+1}(K \oplus P)$ is finite. The map $a_+(\lambda, h) = \lambda a_\Gamma(h)$ from $K \oplus P$ to W_0 is a bounded polynomial law of degree $n+1$ with $\|a_+\| \le 1$: for finitely many $(\lambda_l, h_l) \in K \oplus P$ and scalars s_l, $a_+(\sum_l s_l(\lambda_l, h_l)) = (\sum_l s_l\lambda_l)\, a_\Gamma(\sum_l s_l h_l)$, and $\|a_+(\lambda, h)\| \le |\lambda|\|h\|^n \le \|(\lambda, h)\|^{n+1}$. By (1), $ja_+(\lambda, h) = \beta((\lambda, h), \dots, (\lambda, h))$ for a linear isometry j of W_0 into a normed space Z and a bounded $(n+1)$-linear map $\beta\colon (K \oplus P)^{n+1} \to Z$ with $\|\beta\| \le C$. Put $\mathbf{1} = (1, 0)$ and $h^+ = (0, h)$ for $h \in P$. For $h_1, \dots, h_n \in P$ let $\beta'(h_1, \dots, h_n)$ be the sum, over the $n+1$ slots, of the value of β at the $(n+1)$-tuple with $\mathbf{1}$ in this slot and $h_1^+, \dots, h_n^+$ in the others, in this order. Then β' is n-linear, and $\|\beta'\| \le (n+1)\|\beta\|$, as $\|\mathbf{1}\| = 1$ and $\|h^+\| = \|h\|$. For $h \in P$ and $s \in K$, $\beta((1, sh), \dots, (1, sh))$, with $(1, sh) = \mathbf{1} + sh^+$, is a polynomial in s whose coefficient of s^n is $\beta'(h, \dots, h)$, while $ja_+(1, sh) = s^n ja_\Gamma(h)$. As K is infinite, comparing coefficients gives $ja_\Gamma(h) = \beta'(h, \dots, h)$, and Lemma P.5(1) gives $\mathfrak{a}_n(P) \le \|\beta'\| \le (n+1)C$.

Step 3 (item (3)). The second inequality follows from Lemma P.5(3), applied to the coordinate projection of $c_0(\mathbb{N}, K)$ onto $c_0([m], K)$. For the first, by Lemma P.5(4), and as $\mathfrak{a}_n(\{0\}) = 0$, it suffices to show $\mathfrak{a}_n(P_0) \le \mathfrak{c}_K^{-n} \sup_{m \ge 1} \mathfrak{a}_n(c_0([m], K))$ for every finite-dimensional subspace $P_0 \ne \{0\}$ of P. Its completion $\widehat{P}_0$ is a nonarchimedean normed space over $\widehat{K}$. The $\widehat{K}$-span of a basis of P_0 in $\widehat{P}_0$ is complete by Lemma A.3, hence closed, and it contains the dense subspace P_0; so it is $\widehat{P}_0$, whose dimension m' over $\widehat{K}$ satisfies $1 \le m' < \infty$. If $\lambda_k \to \lambda$ in $\widehat{K}$ with $\lambda_k \in K$ and $\lambda \ne 0$, then $|\lambda_k - \lambda| < |\lambda|$ and so $|\lambda_k| = |\lambda|$ for large k; hence $|\widehat{K}^\times| = |K^\times|$, and $\mathfrak{c}_{\widehat{K}} = \mathfrak{c}_K$. Lemma P.7(1), Lemma P.6 over $\widehat{K}$ and Lemma P.7(2) give

$$\mathfrak{a}_n(P_0) \le \mathfrak{a}_n(\widehat{P}_0) \le \mathfrak{c}_K^{-n}\, \mathfrak{a}_n(c_0([m'], \widehat{K})) \le \mathfrak{c}_K^{-n}\, \mathfrak{a}_n(c_0([m'], K)),$$

where the second and third constants are computed over $\widehat{K}$.

Step 4 (item (4)). Apply item (3) with $n = 2$ and Lemma P.8. □

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